INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Retail

What the industry is, how it runs, and how to model it — for architects who have never worked in it

A working guide for an enterprise architect walking into a retailer for the first time. The SKU, the store and the shelf, and why everything else is downstream of them. Assortment, space and the buying calendar. The supply chain that fills the shelf and the forecast that decides what goes on it. Omnichannel, and what it actually costs. Then the part that matters most: how to turn all of it into a model somebody on the trading floor would recognise as true.

South African lens, with North American practice flagged where it differs. Sourced throughout. October 2026.

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CHAPTER 1

The industry in one chapter

A retailer buys for less and sells for more, and the whole discipline is the inventory in between. This chapter is the book compressed: enough to follow the conversation in a retailer this afternoon, and a map of where the rest of the detail lives.

Two ordinary mornings

A shopper in Mitchells Plain fills a trolley on the first Saturday of the month. Forty-one items, R1,186, paid in cash. She swipes a loyalty card at the till and eleven of the forty-one items ring at a lower price than the shelf edge shows, because the member price is applied inside the sale, not afterwards. Two items are weighed at the counter and carry a barcode that was printed in that store and means nothing anywhere else. One is out of stock, so she buys a different brand, and the retailer will never know that it lost the sale it wanted and won the one it did not.

Eleven kilometres away, in the same group, a pharmacist hands over a repeat prescription. The price of the medicine was not set by the retailer and cannot be discounted, promoted or matched against a competitor: it is a nationally gazetted figure, and the pharmacy's income on the line is a fee calculated by a formula with four bands [1]. The claim goes to a medical scheme and is adjudicated while the patient waits. The item has two identifiers — one for scanning it at a till, one for claiming it from the scheme — maintained by two different bodies on two different cadences, and the mapping between them is the retailer's own problem and nobody else's.

Same group, same morning, same general ledger. Almost nothing else in common.

That is the first thing to take from this book: retail is not one business, and the parts of it that look most similar from the outside — they all have tills, they all have shelves — are the parts that diverge most once you get below the till. This chapter is the whole book compressed: enough to follow the conversation in a retailer this afternoon, and a map of where the rest of the detail lives.

Throughout the book the worked example is **Milkwood Retail Group**, an invented South African multi-format retailer: roughly 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm and an online business. Nothing about Milkwood is real. It exists because grocery, pharmacy and apparel genuinely need different architectures, and one example that carries all three is worth more than three that carry one each.

Buy for less, sell for more — and the whole discipline is the inventory in between

A retailer buys goods it did not make and sells them to people who could have bought them elsewhere. It performs no physical transformation. Its entire economic contribution is four things: deciding what to stock, having it present when the customer arrives, buying in quantities no individual could, and putting it within reach. Assortment, availability, aggregation, access.

Everything else follows from one fact. Between the purchase and the sale, the retailer **owns the inventory**. Inventory is cash that has been converted into a physical asset which is perishable, stealable, fashion-dated and space-consuming. Every discipline in retail is a discipline of that asset:

- **Buying** decides what inventory to own and at what cost.
- **Planning** decides how much and when, in money.
- **Allocation and replenishment** decide where it physically sits.
- **Space planning** decides how much shelf it gets.
- **Pricing and promotion** decide at what price it leaves.
- **Markdown** is the admission that the price was wrong.
- **Shrink** is the inventory that left without being sold.
- **Supply chain** moves it.
- **Store operations** presents it and takes the money.

Read that list again as a systems list and the shape of a retail estate becomes obvious. It is not one core with satellites. It is a **merchandise system of record** holding the item, the location, the cost, the price and the on-hand quantity, surrounded by a dozen specialist engines that read from it and write back to it. There is no single system in retail that occupies the position a deposit core occupies in a bank or a policy administration system occupies in an insurer.

> The integrity question in retail architecture is always the same sentence: **what does this system believe we own, where, at what value, as at when?** Retail IT failures are overwhelmingly failures of that one answer. Keep the sentence. You will use it in every workshop.

Why a retailer is not a factory with a till

Six differences, each of which turns into a constraint you will have to design around.

Demand is not an order book. A manufacturer builds to order, or to forecast with a lead time it controls. A retailer discovers demand only at the till, after it has already bought. Forecasting is therefore not a planning nicety. It is the mechanism by which the business commits capital.

The unit of decision and the unit of record differ. Buyers commit in *money, for a season, for a category*. The stock ledger records *units of a SKU at a location*. Reconciling those two grains is where most merchandise-planning architecture lives, and where most of it fails.

Price is not a property of the product. It is a property of the product at a location on a date under a promotion for a customer segment. A price file is a time series of effective-dated rules, not an attribute on an item record.

The physical store is a system component. Till uptime, shelf execution, cold-chain temperature and on-shelf availability are operational IT problems with direct revenue consequences, in thousands of locations, many of them with unreliable power and poor connectivity. Store systems must trade offline and reconcile later. That is the normal operating condition, not the disaster case.

Labour is the second-largest cost and it is scheduled against a forecast. Store rosters are built from a forecast of transactions by hour. A bad forecast buys queues or idle payroll. In South Africa the floor under that cost is statutory: the national minimum wage rose to R30.23 an ordinary hour from 1 March 2026, from R28.79 [2].

The customer is anonymous by default. Unless the shopper identifies themselves — a loyalty card, a card token, an online account — the retailer holds a basket with no customer attached. The loyalty identification rate therefore decides whether customer analytics is possible at all, and South Africa is unusually fortunate here, which we come to below.

The three grains everything is downstream of

Three objects define the grain of all retail data. Get them right and most modelling decisions follow. Get them wrong and the data platform gets rebuilt.

The SKU — stock-keeping unit. The smallest thing the retailer distinguishes for stock purposes: a specific product in a specific size, colour and pack. A 400g tin of a named brand of baked beans is one SKU; the 410g tin is another. A shirt in navy, size M is one SKU; the same shirt in navy, size L is another. On-hand, cost, price and sales are all held at this grain.

The store. The second grain, and a more slippery object than it looks. A group with 2,839 South African supermarkets [3] has 2,839 assortments, 2,839 potential price positions, 2,839 stock ledgers and 2,839 cost structures. The store is also a *legal entity* question, because in a franchise or a voluntary-trading estate the store may not belong to the group at all.

The shelf. Where the SKU meets the store. It is expressed as a **planogram**: a drawing of a defined length of shelving — a **bay**, a **gondola** run, an **end cap** — showing which SKU sits in which position, at how many **facings**, how many units deep. The planogram is the join between the merchandise hierarchy and the physical store. It is also the only object that makes space productivity measurable, and replenishment ultimately computes against a shelf capacity derived from it.

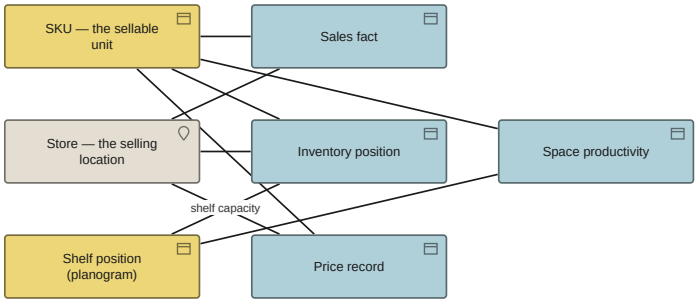


Figure 1.1 The three grains, and the four facts that are projections of them. Sales are SKU × store × time; inventory is SKU × location; price is SKU × zone × date; space is SKU × store × fixture. A model that does not make item, location and time its primary dimensions will be rebuilt.

Why these three and nothing else? Because sales are SKU × store × time. Inventory is SKU × location. Price is SKU × price zone × date. Space is SKU × store × fixture. Every retail analytic, forecast, allocation and margin number is a projection of one of those four. A retail data model that does not make item, location and time its primary dimensions will be rebuilt, and you will be the one explaining why.

Two warnings about the third grain, because it is the one that gets left out of the model. First, the join between the article and the bay exists only inside the space-planning system, and in most estates it is re-derived rather than mastered — which is why planogram compliance is so hard to measure and why "the shelf says we have space for four facings" is an assertion nobody can audit. Second, the measurements decay. Trading area per store, per department and per fixture is among the most reliably wrong data items in retail, because nobody owns it after the store opens and every refit changes it quietly. Both facts matter the moment anybody asks for sales per square metre or sales per linear metre by category, which they will, in your first month.

The arithmetic that governs everything

Retail margins are thin and the thinness is structural, not a sign of a badly run business. Work the numbers once and you will never again ask why a retailer is so anxious about a forecast.

South African grocery, latest published. Shoprite's FY2026, the 52 weeks to 28 June 2026: group sales R270.8bn, gross margin 24.5%, trading margin 6.0%; its Supermarkets RSA segment at a 26.1% gross margin and a 6.6% trading margin [3]. Pick n Pay's group FY2026: gross margin 18.8%, trading margin 1.4% — and the Pick n Pay segment on its own at a 17.0% gross margin and a trading margin of **minus 1.3%** [4]. Boxer at 21.6% gross and 5.7% trading [5]. Cashbuild, in building materials, at a 25.3% gross margin and a 2.6% normalised operating margin [6]. SPAR's group gross margin is 10.8% at a 2.1% operating margin, but SPAR is mostly a wholesaler and the figure is not comparable [7].

A rand of sales, and the cost of a 1% forecast error

Grocery margins as published; the error arithmetic illustrative, at 25% gross and 5% trading

A rand of grocery sales	**R1.00**
less cost of goods sold	−R0.755
Gross margin (Shoprite group FY2026, 24.5%)	R0.245
less store labour, rent, power, distribution, shrink, depreciation, head office	−R0.185
Trading profit (6.0%)	**R0.060**
Same rand at Cashbuild: 25.3% gross →	R0.026
Same rand in the Pick n Pay segment: 17.0% gross →	−R0.013
Over-forecast 1% of a R100bn purchase book	**R1.0bn of stock**
cleared at a 30% average markdown	−R300m margin
equals the trading profit on	**R6.0bn of sales**
Under-forecast and lose 1% of sales	**−R1.0bn turnover**
trading profit forgone at 5%	−R50m
plus an unmeasured share of the abandoned basket	not counted

In grocery, out-of-stock is the worse error; in apparel, over-stock is, because the item has no second season.
Forecast bias should be deliberately different in the two businesses.

Figure 1.2 A rand of grocery sales, worked down to trading profit, and what a 1% forecast error costs against it. Published South African gross and trading margins [3][4][2][6]; the forecast-error arithmetic is mine, on an illustrative R100bn purchase book.

Read the gap in that figure slowly, because it is the whole economics of the industry. Shoprite converts 24.5 cents of gross margin on a rand of sales into 6.0 cents of trading profit. Cashbuild converts 25.3 into 2.6. Pick n Pay's supermarket business converts 17.0 into less than nothing. **Between gross margin and trading profit sits the entire cost of being a retailer** — store labour, rent, electricity and diesel, distribution, shrink, waste, depreciation and head office. Shoprite's employee costs alone were R10.8bn in one half-year, up 8.6% [8].

Now the arithmetic that matters to you, worked out, because it is the single best argument you will ever have for spending money on data quality.

Take a grocery retailer at a 25% gross margin and a 5% trading margin. A rand of sales produces five cents of trading profit. So a rand of inventory bought and never sold at full price destroys the margin on four rands of sales. Concretely, on a R100bn purchase book:

- **Over-forecast by 1%** and you are holding R1bn of unwanted stock. Clear it at an average 30% markdown and the margin loss is R300m — the trading profit on R6bn of sales, or 6% of turnover.
- **Under-forecast and go out of stock on 1% of sales** and you lose R1bn of turnover and R50m of trading profit, plus an unmeasured share of the basket, because the shopper who cannot find the item frequently abandons the trip.

The asymmetry is the point, and it reverses by sector. In **grocery**, out-of-stock is worse than over-stock: the item was going to sell anyway, and the shopper substitutes the store rather than the product. In **apparel**, over-stock is worse: the item has no second season and the markdown is permanent. Forecast bias should therefore be *deliberately different* in the two businesses, and the systems should let it be. A single group-wide forecast-accuracy target across grocery and apparel is not a stretch goal; it is a category error.

> A 3% net margin cannot absorb a forecasting error, and nothing in a retailer's accounting absorbs it for you. A bank with a 20% return on equity and a 180 basis-point credit loss ratio has a pricing structure that *expects* loss and prices for it. A retailer has no provision line for being wrong about demand. The error lands in gross margin, in the same period, with no smoothing of any kind.

One more piece of arithmetic that trips up newcomers, and it is worth fixing now. **Gross margin** is (sales − cost of sales) ÷ **sales**. **Markup** is (price − cost) ÷ **cost** [9]. Different denominators, different numbers, and in a conversation where a buyer says "we're on forty" and a finance manager says "we're on twenty-nine", both may be describing the same item. Ask which denominator. Every time.

The cash conversion cycle, and why supplier terms are a strategy

The **cash conversion cycle** is days inventory outstanding plus days sales outstanding minus days payables outstanding. For a retailer paid at the till, days sales outstanding is near zero. So the cycle collapses to inventory days minus payable days — and when payable days exceed inventory days, the retailer's working capital is *negative*: the supplier funds the stock.

Shoprite's own balance sheet at 28 December 2025 shows the mechanism. Inventories R33,738m, trade and other receivables R6,816m, trade and other payables R43,939m [10]. Payables exceed inventory by R10.2bn. The company discloses inventories at 12.9% of trailing sales, and notes explicitly that the period-end cash balance of R15,076m is *before* creditor payments made after the cut-off [10] — in other words, the reported cash position is partly a function of where the payment run fell relative to the balance-sheet date.

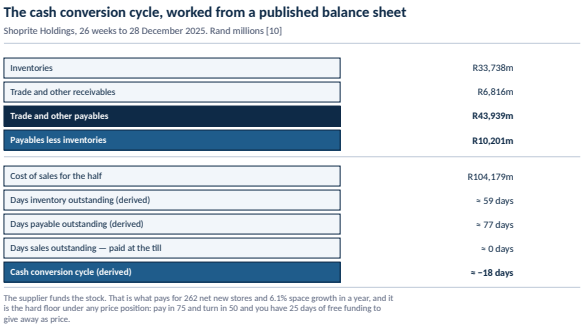


Figure 1.3 Negative working capital, from one published balance sheet at 28 December 2025 [10]. The day counts are my calculation on half-year cost of sales over 182 days, not a disclosure, and payables include non-trade items — an order of magnitude, not a reported figure.

Those derived day counts are my calculation from the published balance sheet, not a disclosure, and payables include non-trade items, so treat them as an order of magnitude. The direction is not in doubt.

Why this is a strategy rather than an accounts-payable detail, in four steps:

It funds expansion. Shoprite added 262 net new Supermarkets RSA stores and grew space 6.1% in FY2026 [3]. With negative working capital, each new store's opening stock is substantially supplier-funded, so growth generates cash instead of consuming it.

It is the hard floor under a price position. A retailer that pays in 30 days and turns stock in 60 must fund 30 days of stock from equity or debt and must earn a margin that services it. A retailer that pays in 75 and turns in 50 has 25 days of free funding available to give away as price. That is a large part of why scale players can undercut independents on price without being better at anything else.

It is a competition matter. The Competition Commission's Grocery Retail Market Inquiry found that national chains obtain more favourable trading terms than wholesalers and buyer groups *even where those groups buy larger volumes*, and that rebates for in-store functions are generally unavailable to independents [11]. Terms are therefore a regulated subject, not a treasury one.

It breaks on perishables and on imports. Fresh produce turns in days and is often paid faster. Imported apparel is paid on shipment and sells twelve weeks later, so an importing apparel retailer has a *positive* cycle and funds its own stock — which is part of why apparel gross margins run 41–57% and grocery's run 17–26% [12][13][14].

The architectural consequence is specific. Payment terms are per-supplier, sometimes per-category, sometimes per-purchase-order data, with effective dates, settlement discounts and rebate interactions. They live in the supplier master and the purchasing terms engine, they drive the payment proposal run, and they are an input to landed cost. A retailer whose terms live in a contract PDF cannot compute a true cost of goods, and cannot answer the Commission's question about differential terms.

The South African market in a page

Stats SA publishes retail trade sales monthly as release P6242.1, and it is the authoritative measure of *formal* retail. May 2026, preliminary [15]:

Type of retailer	Share of total	Year on year, current prices	Year on year, constant 2019 prices
General dealers	46.3%	2.9%	1.0%
Textiles, clothing, footwear and leather	16.6%	4.2%	3.7%
Food, beverages and tobacco in specialised stores	9.2%	3.5%	−0.4%
All other retailers	9.2%	5.8%	6.7%
Pharmaceuticals, medical goods, cosmetics and toiletries	8.0%	4.5%	2.1%
Hardware, paint and glass	7.5%	2.2%	−0.4%
Household furniture, appliances and equipment	3.1%	3.6%	8.1%
Total	**R128,615m in the month**	**3.5%**	**2.3%**

Three readings. "General dealers" is the supermarket groups, and it is nearly half of formal retail — and the category where constant-price growth is weakest, which means the sector is growing on price rather than volume. **Apparel is 16.6% of formal retail**, far larger than outsiders expect, and growing faster in real terms than food. And furniture and appliances is only 3.1% of current-price retail but grew 8.1% in real terms, which means its prices are falling.

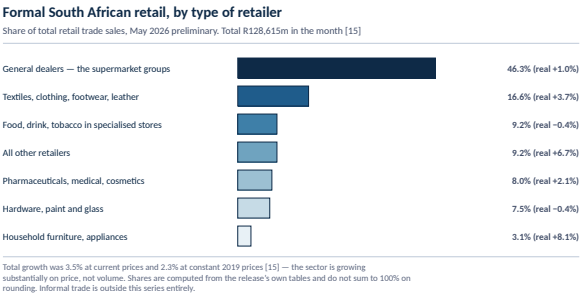


Figure 1.4 Formal South African retail by type of retailer, May 2026 [15]. Apparel is far larger than outsiders expect; furniture's real growth exceeds its nominal growth, which means its prices are falling.

The listed groups, with their latest reported figures, give the shape of the sector in a single table. Note that the period ends differ, so there is no common "latest" date:

Group	Period	Turnover	Gross margin	Trading or operating margin	Stores
Shoprite Holdings	52wk to 28 Jun 2026	R270.8bn, +7.2%	24.5%	6.0% trading	2,839 Supermarkets RSA [3]
Pick n Pay Group	52wk to 1 Mar 2026	R120.3bn, +3.4%	18.8%	1.4% trading	2,261 including franchise and Boxer [4]
SPAR Group	52wk comparable to 26 Sep 2025	R132.4bn, +1.6%	10.8%	2.1% operating	1,846 South African stores [7]
Woolworths Holdings	52wk to 28 Jun 2026	R84.5bn, +4.3%	Food 24.9%; fashion, beauty and home 46.0%	adjusted EBIT R5.3bn	not disclosed [12]
Pepkor Holdings	10m to 31 Jul 2026	R89.2bn, +11.9%	not disclosed	—	6,677 [13]
Mr Price Group	52wk to 28 Mar 2026	R41.1bn retail sales, +4.3%	41.2%	14.2% operating	3,182 [14]
Clicks Group	6m to 28 Feb 2026	R24.9bn, +7.4%	not disclosed	9.1% trading	1,003 stores, 795 pharmacies [22]
Massmart	delisted 2023	no public South African results	—	—	about 122 Game stores in South Africa [35]

Massmart is the structural exception and a useful fact in its own right. Walmart took 51% in 2011, bought the remainder and delisted the group from the JSE in late 2022, so the second-largest general-merchandise estate in the country publishes no South African segment results at all [35]. Any market-share model that needs it is relying on estimates, and should say so.

The concentration is well documented. The Commission put the top five at approximately 64% of the formal grocery market in 2019 [11]. Independent estimates since put Shoprite alone at 30–35% of food retail, with modern retail at about 80% of grocery sales and a 2024 grocery Herfindahl-Hirschman index of 1,766.8 out of 10,000 [16][17]. Supermarket density in 2023 was 4.9 per 10,000 people, roughly four times the sub-Saharan African average [17].

Three market facts that are easy to miss and expensive to miss.

The informal trade is large and mostly invisible in the data. Trade Intelligence valued the informal grocery sector, spaza shops and mobile traders included, at R184bn [18]; about 150,000 spaza shops operate, mostly run by foreign nationals [18][19]. The Commission's own final report gives no figure at all for the size of the informal market [11]. For an architect the consequence is practical: the informal retailer is a **wholesale customer, not a store**. It needs cash-and-carry, buying groups, van sales and route-to-market systems, not planograms. And demand models built only on formal till data systematically understate volume in exactly the low-income areas where the discount formats compete.

Cash is the majority of transactions. The Reserve Bank's 2023 Payments Study, on a diary of 210,207 payments by 4,624 participants, put cash at 56% of payment volume and 21% of value, at an average payment of R208.44, against debit cards at 34% of volume and 55% of value [20]. Separately, **over 80% more transactions occur on the first of the month than on the 22nd**, with peak volume on the 25th [21]. Cash handling, offline till operation and a monthly payday amplitude are first-order South African design concerns. An architecture that treats cash as a declining legacy tender is wrong for this market.

Loyalty identification is unusually high. South Africa has one of the world's highest loyalty participation rates: 85% of economically active South Africans use loyalty programmes, at an average of 10.4 programmes each, up from 4.6 in 2015 [22]. The large grocery programmes report identification on roughly 80–84% of sales [22][23][4]. Two consequences follow. South African grocers have customer-level visibility most global retailers do not. And the dominant mechanic is an **instant member price at the till**, not points in an account — which means the promotion engine has to evaluate member status *inside the authorisation path of the sale*, with a latency budget in milliseconds, at a reported rate of more than 2,700 loyalty swipes a minute on the largest programme [22].

And one structural fact that changes the meaning of every system diagram in the rest of this book.

A voluntary-trading or franchise group does not master a price, an assortment or an inventory position for much of its turnover. SPAR is a *wholesaler*, not a chain: stores are owned by independent retailers who hold membership of the group and a separate Guild, there is no royalty on turnover, SPAR earns its margin on wholesale supply, and members keep pricing discretion, product-mix flexibility and the right to source elsewhere [24][25]. SPAR reports "retailer loyalty" of 78.6% as a headline operating metric [7] — the share of its members' purchases that comes through its own distribution centres. Pick n Pay runs 620 franchise stores alongside 992 company-owned ones [23]. In those estates, replenishment, range, pricing and loyalty are *offered services*, not central controls, and the integration is business-to-business rather than intra-enterprise.

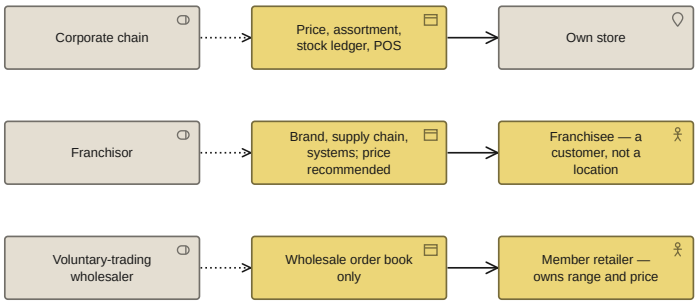


Figure 1.5 What the group actually masters under each of the three ownership models that coexist in South African retail. In a voluntary-trading estate there is no group price file, no group assortment and no group stock ledger for the member business — and every reference architecture in part III assumes there is.

Hold that picture. When somebody draws you a retail reference architecture with a single item master at the top and a price distribution path running down to every till, ask which share of the group's turnover that diagram actually governs. In a voluntary-trading estate the honest answer may be a third of it.

Two further South African specifics for the list, both covered properly later. Store power and connectivity: load-shedding turned offline store operation into the normal design case, and the disclosed costs were real — an additional R560m of diesel across one group's South African stores during stages five and six over six months in late 2022, about R93m a month, and R500m in the six months to 31 December 2023 [26][27]; a second group spent an additional R346m year on year over ten months [26]. **Those are historic peak figures from 2022 to 2024, not a run rate**, and the operational position has improved materially since. And the landlord: South African retail is overwhelmingly shopping-centre retail, with the Clur index putting national annualised trading density at R43,612 per square metre and base rent at 6.6% of sales in the second quarter of 2026 [28]. A retailer at a 25% gross margin and a 6% trading margin that pays 7% of sales in base rent has given away more than its entire trading margin before paying a single wage.

What breaks, predictably

This is the part of the chapter to bookmark. Four failures recur in retail estates with enough regularity that you should walk in expecting them and go looking in week one.

The article that exists three times

Six words — article, style, option, variant, SKU, product — with overlapping meanings, and the overlap causes more integration defects than anything else in a retail estate. Four distinct concepts are involved: the thing a buyer buys and plans, the differentiated offer the customer chooses before size, the sellable barcoded unit, and the physical grouping it arrives in. Grocery collapses the first three into one object. Apparel never does: one shirt is one planning object, four colourways and perhaps twenty-eight sellable units.

So a grocery-born merchandising system that models the article as a flat object is correct for grocery and wrong for apparel, and a group running both either runs two merchandising systems or runs one with the apparel business torturing a flat model into a style-colour-size shape through naming conventions. The vendors formalise this differently and the differences are real: SAP distinguishes a single article, a generic article with variants, and a structured article [29]; Oracle's merchandise hierarchy is a fixed six levels from company down to subclass, with items attaching at subclass [30]. Neither is wrong. They are not the same model, and an interface between them needs a mapping that somebody owns.

The symptom you will find is three records for one physical product: one created by a buyer in the merchandising system, one created by a content editor in the online catalogue because the merchandising record had no photograph, and one created at a store because the case arrived with a barcode nobody recognised. Each is "correct" in its own system. None is the same object.

Two specific defects in this family are worth memorising now, because both are cheap to prevent and expensive to find later. The first is the **dropped leading zero**. A GTIN-13 carried in a case-level barcode field needs a leading zero, because that field is fourteen digits [34]; one system stores twelve digits, another thirteen, another fourteen, and a fourth stores the digits as a numeric type that silently drops the zero. Normalise every identifier to a fourteen-digit, right-aligned, zero-padded *string* at every boundary and the whole class of defect disappears. The second is the **wrong case quantity** — the conversion factor between the traded unit the retailer orders and the consumer unit the store sells. A case quantity wrong by a factor of ten multiplies or divides every order, every receipt variance, every stock position and every forecast consumption calculation by the same wrong number, and stays nearly invisible until a store receives ten pallets of something.

And a third, which is not a defect at all but surprises everybody: a material share of a grocer's turnover passes through identifiers that are valid only inside that retailer. Loose produce, butchery, deli and bakery items are weighed on an in-store scale that prints a barcode encoding either the price or the weight, under a restricted-circulation prefix reserved for internal use [34]. Those items cannot be matched to a supplier's article and they break every standards-based integration. Architects routinely discover this in week six. Chapter 3 settles the vocabulary; chapters 9 and 10 build the model.

The price that three systems each think they own

The archetypal retail finding, and it is worth setting out exactly, because each system is right.

The **merchandising system** holds regular retail by article, zone and effective date, because the margin calculation and the stock ledger need it there [30]. The **promotions and pricing engine** holds promotional and markdown prices and the basket-level rules — "three for R50", "spend R500 get R50 off" — because those cannot be expressed as a price on an article at all. The **point of sale** holds the price it will actually charge, in a local file, because it has to be able to trade when the line is down. Add the **electronic shelf label controller** and the **website** and there are five stores of one concept, with five update paths and five latencies.

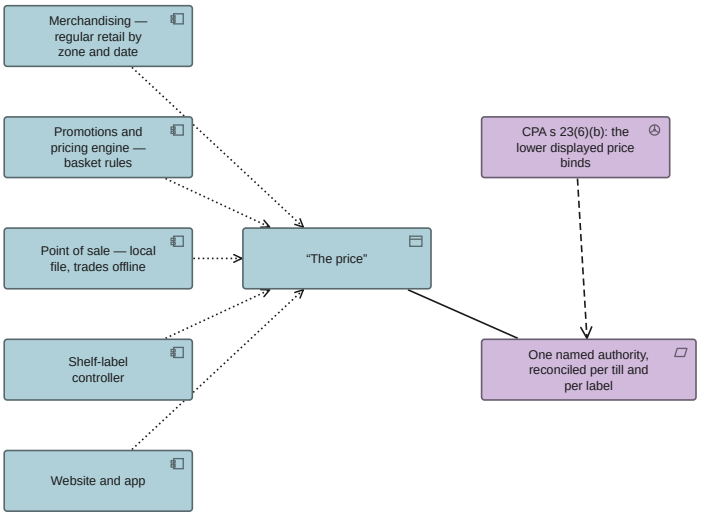


Figure 1.6 Five systems, one concept, five update paths and five latencies — and each was correct in isolation. The failure is that none is declared authoritative and no reconciliation exists, while section 23(6)(b) of the Consumer Protection Act binds the retailer to the lower of whatever two of them display [31].

Each was correct in isolation. The failure is that none is declared authoritative and no reconciliation exists, so the question "what price will this article ring at, at this store, tomorrow" has no systemic answer.

The shelf label that silently fails to update

This one is a legal exposure, and architects consistently file it as a data-quality issue.

Section 23(6)(b) of the Consumer Protection Act 68 of 2008 provides that where more than one price is concurrently displayed for goods, the supplier **must not charge more than the lower or lowest of those prices** [31]. So a shelf label that disagrees with the till is not an operational irritation. It is a statutory obligation to sell at the lower number.

The escape is narrow. Section 23(9) relieves the supplier of a displayed price containing an "inadvertent and obvious error" only once the supplier has *both* corrected the display *and* taken reasonable steps to inform consumers who may have seen it. The Consumer Goods and Services Ombud's Advisory Note 2 of 1 March 2021 sets out what reasonable steps look like — withdraw the product, fix the label, then resell; for in-store notices, put them up at the affected stores and keep dated photographic proof — and notes that a consumer who has already acted as if buying, for example by taking the item to the till, may still pay the error price [32].

> Read that as a set of requirements, because that is what it is. The retailer needs **a provable record of what price was displayed, where, and when**; a correction workflow that produces evidence; and a reconciliation between the price in the point of sale, the price on the shelf edge and the price in the advertising. A weekly manual shelf-label audit does not satisfy any of the three. Electronic shelf labels turn this into an integration problem with confirmation per label, which is better — and they make the label-to-article binding a controlled object, which is a new risk in its own right.

The overnight batch that did not finish

The overnight cycle is the estate's heartbeat, and architects arriving from banking or insurance consistently under-weight it. The dependency chain is the point: the store's day has to close before sales can be polled; sales have to post before the forecast can run; the forecast has to finish before replenishment can run; replenishment has to finish before orders can go out against supplier cut-offs that are contractual; and the price file has to be built, distributed and *applied* before the stores open.

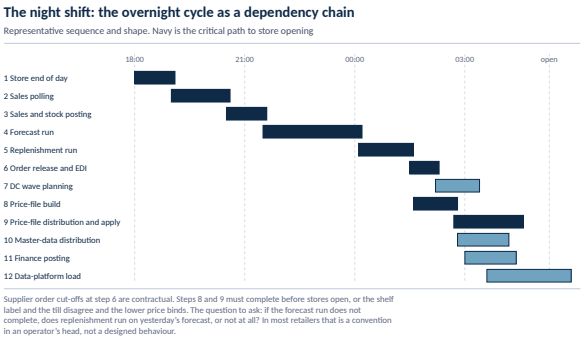


Figure 1.7 The overnight cycle as a dependency chain, with the critical path in navy. The question worth asking in your first fortnight is the last one in the note: nobody will volunteer the answer.

Four questions to ask in your first fortnight, in this order. What is the cycle's total duration and how much slack is there to store opening? Which jobs are on the critical path and which are merely slow? What is the recovery procedure for each — re-run, skip, or run degraded? And what happens if the forecast run does not complete: does replenishment run on yesterday's forecast, or not run at all?

In most retailers the answer to that last question is a convention held in an operator's head rather than a designed behaviour. Finding that out is one of the more valuable things you will do in your first month.

The map of the book

Twenty-two items, in five parts and four appendices. One line each.

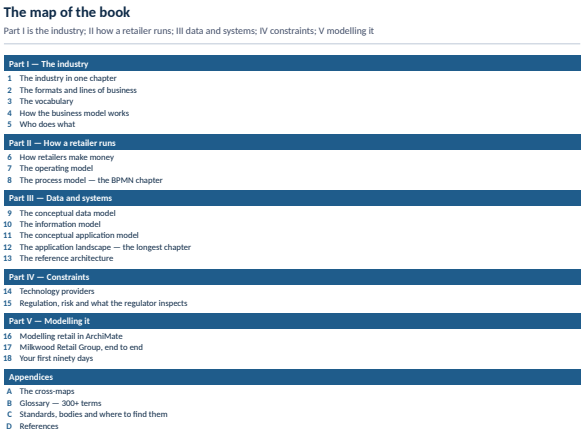


Figure 1.8 The twenty-two items, in five parts and four appendices.

Part I — the industry. Chapter 2 takes the sixteen formats one at a time and makes the case that a pharmacy and an apparel chain share almost nothing below the till. Chapter 3 is the vocabulary, in themed runs, with the arguments attached. Chapter 4 works the business model: where the money comes from and what the retailer sells to whom. Chapter 5 names the jobs — about forty-five roles in eleven families — because an architect who cannot name the jobs cannot run a workshop.

Part II — how a retailer runs. Chapter 6 is the economics in detail: margin mix, markdown, shrink, supplier income, promotional funding, omnichannel unit costs, returns, and the choice between retail and cost inventory accounting. Chapter 7 is the operating model. Chapter 8 is the process model, and it is the one chapter drawn in BPMN rather than ArchiMate, because in a process model the hand-off is the point.

Part III — data and systems. Chapter 9 is the conceptual data model: the twenty-five things the business talks about. Chapter 10 is the information model, including time, effective dating and the restatement problem. Chapter 11 is the conceptual application model — the applications a retailer *needs*, named by what they do. Chapter 12 is the real application estate, class by class, and it is the longest chapter in the book. Chapter 13 assembles a reference architecture and is honest about what retail does not have.

Part IV — constraints. Chapter 14 is the technology providers, the standards bodies and the ownership changes that commentary routinely gets wrong. Chapter 15 is regulation, risk and what each regulator actually inspects.

Part V — modelling it. Chapter 16 maps retail onto ArchiMate 4 element by element. Chapter 17 takes Milkwood end to end. Chapter 18 is your first ninety days. Then four appendices: the cross-maps, a glossary of more than three hundred terms, the standards bodies and where to find them, and the references.

Three things that will surprise you

If you have come from banking or insurance — and most architects working this market have — three things will land differently, and they are the three worth carrying into your first week.

There is no system of record for the thing you sell. In a bank the balance is in the core and the core is right by definition; a disagreement is a defect to be fixed against an authority. In a retailer the authority for "how much of this do we have on this shelf" is the shelf, the system is a derivation, and the two disagree continuously and legitimately — because stock was stolen, broken, mis-received, mis-scanned, sold as the wrong item or written off without a record. Ask a retailer how much stock it has of a SKU in a store and the honest answer is a question: who is asking, and what for? Chapter 3 gives nine different correct answers. The architectural response is not a better quantity column. It is to make the **inventory movement** the durable fact — append-only, signed, typed, located, timestamped, reason-coded — and to treat every stock position as a derivation over movements. A cached quantity column is acceptable as a derivation with a documented recomputation path. It is not acceptable as the system of record, because the moment two channels disagree there is nothing to adjudicate between them.

The store has to keep trading when nothing else works. A branch that cannot transact sends the customer to the app. A store that cannot transact sends the customer to the competitor 400 metres away, and perishables degrade while the queue abandons. So the design rule is unusual and non-negotiable: **the store's ability to complete a sale must not depend on any system outside the store.** Everything else may degrade. Completing the sale may not. "Offline mode" turns out to be four separable capabilities — local item lookup and pricing, local transaction logging, offline tender acceptance, and offline loyalty and gift-card handling — and most estates have two of them. The hard one is card tender, because an authorisation needs a round trip; the mechanisms available are floor limits, stand-in authorisation at the terminal or the store controller, and store-and-forward of the request, and each of them trades fraud risk for availability. Those parameters are a commercial decision wearing technical clothing. Test the capability specifically rather than accepting "we have offline mode", and ask who set the floor limit and when.

Time is a trading calendar, not a clock. Retailers do not report in calendar months. They report in retail weeks grouped four-five-four, so that comparable months hold the same number of weekends and holidays stay aligned — a convention the National Retail Federation publishes as a voluntary guide, developed informally in the 1930s and widely adopted in the 1940s [33]. A 52-week year is 364 days, so a 53rd week is needed roughly every five or six years on no fixed schedule [33]. The consequence shows up everywhere: Pick n Pay's FY2026 was 52 weeks against a 53-week FY2025, and the same prior-year turnover appears in one results pack as both R116.3bn on a pro-forma 52-week basis and R119.0bn on the statutory 53 weeks [4]. South African retailers also have different year ends — Shoprite in June, Pick n Pay in late February or early March, SPAR in September, Clicks in August — so there is no common "latest" date, which is a reporting-layer problem before it is anything else.

And the calendar has one more consequence that will affect you directly: the **peak trading freeze**. A retailer freezes production change for the duration of peak, and the freeze is wider than software releases — it covers infrastructure, master-data structural change, range change, planogram resets and store refits. For a South African grocer it commonly runs from about mid-November to mid-January. Two things follow. The delivery year is compressed into roughly nine months, so a programme plan that assumes twelve is wrong by a quarter. And the freeze *accumulates* change, which makes the January and February releases the largest and riskiest of the year, landing immediately after the period in which the estate was least exercised.

Nobody will tell you any of this in the kick-off meeting. Ask.

THE CHAPTER IN TEN LINES

- Retail adds no physical transformation. Its contribution is assortment, availability, aggregation and access — and between purchase and sale it owns the inventory, which is what every discipline in the business is about.
- There is no core system in retail analogous to a deposit core or a policy administration system. There is a merchandise system of record and a dozen specialist engines reading and writing to it.
- Three grains govern everything: the SKU, the store and the shelf. Sales, inventory, price and space are projections of them, and a model that does not make item, location and time primary will be rebuilt.
- A 3% net margin cannot absorb a forecasting error. Over-forecast 1% of a R100bn purchase book and a 30% markdown destroys the trading profit on R6bn of sales; there is no provision line that smooths it.
- Gross margin divides by sales and markup divides by cost. Ask which denominator, every time.
- The retailer’s working capital is negative because the supplier funds the stock, so payment terms are a strategy, a competition-law object, and per-supplier effective-dated data — not an accounts-payable detail.
- South African specifics that change designs: cash at 56% of payment volume, 80%+ more transactions on the first of the month than the 22nd, loyalty identification on roughly 80–84% of grocery sales, and an informal channel outside the official series.
- A voluntary-trading or franchise group does not master a price, an assortment or an inventory position for much of its turnover. Ask which share of the group a reference architecture actually governs.
- Four things break predictably: the article that exists three times, the price that five systems each think they own, the shelf label that fails silently — a legal exposure under CPA s 23(6)(b), not a data-quality issue — and the overnight batch that did not finish.
- The store’s ability to complete a sale must not depend on any system outside the store. Offline mode is four separable capabilities and most estates have two.
- Time in retail is a trading calendar, not a clock: 4-5-4 periods, a 53rd week roughly every five or six years, different year ends across the sector, and a peak freeze that compresses the delivery year into about nine months.

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CHAPTER 2

The formats and lines of business

A retail group is a dozen businesses sharing a till, a ledger and a customer master. This chapter is the taxonomy: what each format sells, what it consumes, what it needs, and why a pharmacy and an apparel chain share almost nothing below the point of sale.

A format is an economic machine, not a shop size

A **format** is a defined store type with a size, a range, a price envelope and a shopper mission. That sounds like a marketing construct. It is not. A format is the unit at which the economics of retail are actually decided, because the format fixes the gross margin available, the stock turn achievable, the capital per store, the labour model and — this is the part architects miss — which systems the business cannot trade without.

Sixteen formats follow. For each: what it sells, who buys, the margin and turn profile, the capital it consumes, the systems it needs, and where it genuinely diverges from its neighbours. Published margins are the latest reported; turn figures are derived or qualitative where no disclosure exists, and they are labelled as such.

Read it with one question in mind, because the whole chapter is an argument for one claim: **a pharmacy and an apparel chain share almost nothing below the point of sale.** The shared layer in a multi-format group is thinner than most reference architectures admit, and the cost of pretending otherwise is a decade of workarounds.

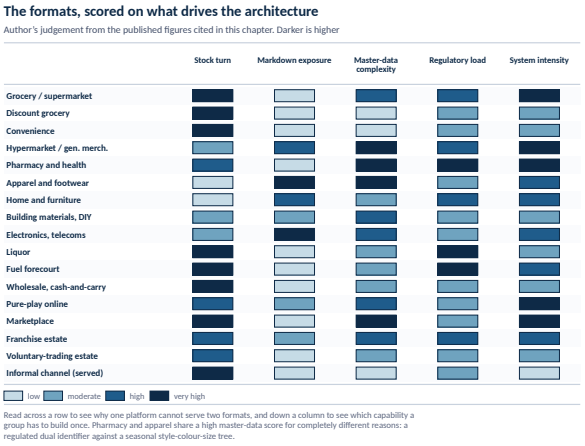


Figure 2.1 Sixteen formats plus the informal channel, scored on the five dimensions that decide the architecture. The scores are my judgement from the published margin, turn and estate data in this chapter, not a sourced index — use them to order a conversation, not to settle one.

The grocery family

Grocery and supermarket

Sells food — fresh produce, meat, bakery, deli, dairy — ambient groceries, frozen, household and personal care, with liquor and pharmacy frequently attached as separately licensed departments. **Buys** everyone, weekly or more often.

Margin and turn. Gross margin runs 17–26%: the Pick n Pay segment at 17.0%, Boxer at 21.6%, Shoprite group at 24.5%, its Supermarkets RSA segment at 26.1%, Woolworths Food at 24.9% [1][2][3][4]. Trading margins span minus 1.3% to 6.6% in the same market in the same year [1][3]. The turn is the fastest in retail: Shoprite discloses inventories at 12.9% of trailing sales [5]. Volumes are enormous — 1.3 billion customer visits and 9.3 billion items in one group's financial year [3].

Capital. Moderate per store and heavy in aggregate, because the format needs refrigeration, distribution centres and a vehicle fleet. The working capital is negative, so growth generates cash (chapter 1).

Systems it cannot trade without. This is the highest-volume, most time-critical estate in retail. Daily or multi-daily **replenishment** per SKU per store on min-max or forecast-driven order generation. **Fresh and perishable management** with shelf life, markdown-to-clear and waste write-off. **Weighed items** priced at an in-store scale with their own barcode symbology. **Recipe and own-manufacture** for bakery, butchery and deli, with yield and traceability. **Cold-chain telemetry.** A **promotion engine evaluated inside the sale path** at loyalty-member level. And **cash handling at scale**. Pricing is high-low and promotion-intensive: promotions were 40% of one group's Supermarkets RSA turnover in the half to December 2025, up from 37.9% [6]. At 40% of turnover the promotion engine is not a marketing tool. It is a primary pricing system in the authorisation path of every sale.

Where it diverges. From everything else, on cadence. The grocery day is a daily cycle with a monthly payday amplitude on top of it. Nothing else in retail reorders every item in every store every night.

Discount and limited-assortment grocery

A sub-format worth separating, because its architecture is deliberately *simpler* and that is the commercial point. It sells a narrow range — typically a few thousand SKUs against twenty thousand or more in a full supermarket — heavy in staples and bulk packs, in a low-cost box, to price-driven shoppers.

The economics invert what newcomers expect. Boxer runs a **lower** gross margin than a full-range supermarket, 21.6%, at a **higher** trading margin, 5.7% [2]. It achieved internal selling-price *deflation* of 1.2% while growing turnover 12.3% and growing like-for-like volume [2]. Range discipline is the mechanism: a narrow range makes the forecast accurate, the turn fast, the distribution-centre handling simple and the store labour low.

The architectural instruction is unusual: the system must prevent range creep. Fewer SKUs means case- and pallet-level replenishment rather than unit picking, so the distribution profile is cross-dock and full case. Every exception a buyer wins erodes the model.

Convenience

Sells a top-up basket — bread, milk, snacks, drinks, cigarettes, ready-to-eat — at a price premium for proximity and hours. **Buys** the immediate-need shopper. Higher gross margin than a supermarket, very high turn on a small range, and a very high fixed cost per square metre.

South African convenience is largely attached to someone else's site: around 190 forecourt stores under one grocery brand's franchise, a second grocer's express format inside a fuel brand's network, a third operating more than 330 stores with another fuel brand, and a container format built for township trade [7][8][9].

Systems. Small store, often franchised, often inside another business's premises. So: head-office-driven ordering with **automated replenishment and no store intervention**; a limited range on national pricing; a till that handles fuel, airtime, bill payment and lottery alongside grocery; and integration with the host fuel brand's systems. The back office is minimal by design, which means every capability the store does not have, head office must.

Hypermarket and general merchandise

Sells full grocery plus apparel, homeware, electronics, toys, garden, do-it-yourself and seasonal goods under one roof, typically eight to fifteen thousand square metres. **Buys** the monthly bulk shopper and the destination shopper. Margin is blended — food at grocery rates, general merchandise at 30–45% — with much slower turn on the general merchandise and heavy seasonal and markdown exposure.

This is the hardest format to run, and the reason is architectural. It is several businesses sharing one stock ledger: fresh food replenishment next to seasonal apparel allocation next to big-ticket electronics with serial numbers and warranties. It needs grocery replenishment **and** apparel-style allocation and markdown, **plus** bulky-goods delivery, assembly and home-delivery scheduling. And the allocation of space between categories is a continuous commercial argument that the space-planning system has to arbitrate rather than merely record.

Pharmacy and health — a regulated business inside a retail business

This is the format where an architect most reliably gets it wrong, because the dispensary is not a department. It is a separately licensed, separately regulated, separately priced professional practice that happens to be inside a shop. It gets the extended treatment here because almost every instinct you bring from the rest of the estate is wrong inside its walls.

Three businesses, three economics

Prescription medicines, Schedules 2 to 6, where the price is fixed nationally and the pharmacy earns a regulated fee. **Pharmacist-initiated and over-the-counter medicines**, Schedules 0 and 1, where the pharmacy earns a margin. And **front shop** — beauty, personal care, baby, healthcare devices, vitamins — which is ordinary retail at retail margins.

The blended result is the highest-return business in South African retail: one pharmacy-led group reported a 9.1% trading margin and a **45.7% return on equity** in the half to February 2026, with retail pharmacy market share of 24.9% and 795 pharmacies across 1,003 stores [10]. The other reported a 22.2% gross margin and a 31.1% total income margin, with the dispensary at 38.2% of retail turnover and a 24.7% retail pharmacy share [11]. The strategic logic is one sentence: **the dispensary brings the footfall and the front shop brings the margin.** Private label at 31% of one chain's front-shop sales is where a good deal of that margin comes from [12].

How the price works, and why the retailer does not set it

South African medicine pricing is the single biggest divergence in this book, and it is worth getting exactly right [13].

The **Single Exit Price** is the only price at which a manufacturer may sell a medicine to anyone other than the state. Unlike a factory-gate price, the SEP is the price **the final seller charges the patient or the scheme**, and every buyer pays the same SEP regardless of volume. Wholesalers add no mark-up at all; the manufacturer pays the wholesaler a **logistics fee** out of the exit price, which means a seller that owns its own wholesaler can earn logistics income the buyer never sees.

The pharmacy's income on a prescription is a **regulated dispensing fee**: a flat component plus a percentage of the SEP, across four price bands, with the percentage falling as the SEP rises. In the September 2025 gazette's top band, where the SEP exceeds R1,530.73, a pharmacist's fee may not exceed R270.54 plus 5% of the SEP [13]. The other three bands are not in the public material I could source, so I will not invent them. Final sellers must disclose both the SEP and the dispensing fee charged, and may not exceed the annual maximum [13].

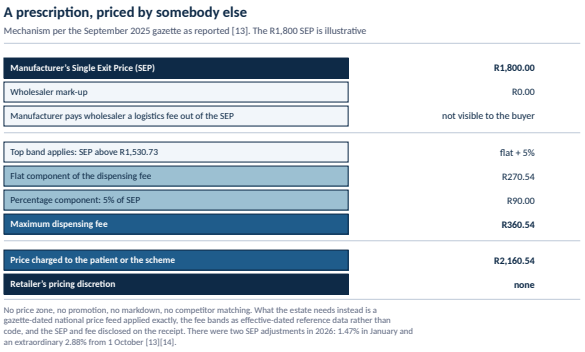


Figure 2.2 A dispensed prescription, priced. The Single Exit Price is set nationally and the pharmacy's income is a banded formula, so the retailer has no pricing lever at all. The top band is the only one in the sourced material; the three lower bands were not obtainable and are not invented here [13].

The annual adjustment is gazetted. There were **two adjustments in 2026**: 1.47% effective January, and an extraordinary second adjustment of 2.88% effective 1 October 2026 on medicines listed on the national pricing database as at 30 September [13][14]. Any source that gives you "the 2026 SEP increase" as a single figure is wrong, and if your price-file ingestion assumes one annual event, it broke in October.

The legal basis is the Medicines and Related Substances Act 101 of 1965 as amended in 1997, with the pricing provisions taking effect in 2004. SAHPRA registers medicines and maintains the database but **does not regulate prices** [13]. The Act also prohibits bonus, rebate and incentive schemes on medicines, with the section paused for Schedule 0 products but still applying to the others [13].

> The pharmacy retailer **cannot price medicines**. There is no price zone, no promotion, no markdown, no competitor matching and no margin optimisation on a scheduled line. What exists instead is a national price feed that must be ingested on a published schedule and applied exactly, a fee calculated by a banded formula whose parameters change by Government Gazette, and a disclosure obligation on the receipt. The pricing engine must be **bypassed entirely** for Schedule 1 to 6 lines, and the article master must carry the schedule as a controlling attribute. A design that treats medicines as articles with a price is wrong at the first SEP adjustment.

Two identifier schemes for the same product

This is the best single fact in the pharmacy section and it is one most architects have never heard.

A South African retail pharmacy's item master carries **two identifiers for the same physical product**. The **GTIN** is how the product is received, scanned and sold. The **NAPPI code** is how it is claimed and reimbursed: a unique identifier for pharmaceutical, medicinal, surgical, medical appliance and healthcare consumable products, used in South African electronic healthcare reimbursement since the early 1990s, with the trademark owned and the file managed day to day by MediKredit; the public-domain file is free and carries code, description, strength, pack size and manufacturer, with fuller product information available by subscription [15].

The two schemes are maintained by different bodies, on different cadences, under different allocation rules, and **they do not map one to one** — pack-size and formulation variation resolves differently in each. So the GTIN-to-NAPPI mapping is a master-data asset with a compliance consequence, and the retailer owns it. Nobody else will maintain it. Chapter 3 draws the two authorities side by side.

The wall, and it is specified in the rules

The separation between the dispensary and the shop around it is physical as well as logical, and the Good Pharmacy Practice rules published under section 35A of the Pharmacy Act 53 of 1974 specify it [16]. A pharmacy located inside another business must be clearly identified and demarcated with **permanent, floor-to-ceiling barriers** enclosing the waiting, clinic, semi-private and private areas; it must have a single compliant entry and exit point; and the pharmacist must have unfettered 24-hour access. Another business operating inside a pharmacy needs Council approval. **Only registered pharmacy personnel may have direct access to medicine**, and keys or access devices must remain on the person of the responsible pharmacist or another pharmacist [16].

The confidentiality rules are what constrain the data architecture. Access to patient information is limited to those who need it for the patient's care or in the public interest; disclosure requires the patient's **written** consent and must itself be recorded on the patient's record; and anonymised patient data may be collated for research or commercial purposes only with names, addresses, telephone numbers and postal codes removed [16]. Put that beside POPIA, under which health information is special personal information with a stricter processing basis, and a hard boundary falls out:

- Patient and prescription data is access-controlled to pharmacy personnel. Not store staff, not head-office merchandising, not marketing, not the customer data platform.
- The loyalty programme may know that a basket included a dispensary transaction and its value. It must not know what was dispensed, absent a separate consented basis.
- The data platform receives dispensary data either pseudonymised or into a segregated zone with its own access model, and the rules name the specific identifiers to strip [16].
- Audit logging on the patient medication record is a professional-conduct requirement, not an IT nicety.

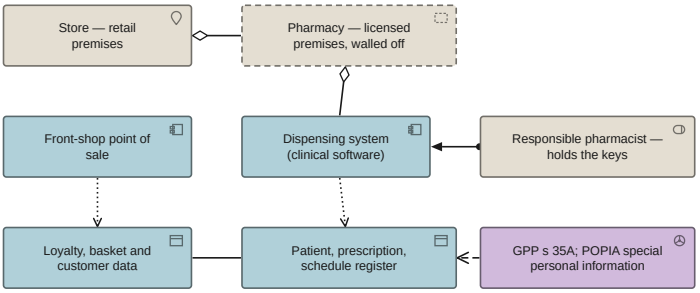


Figure 2.3 The dispensary is not a department. Good Pharmacy Practice requires permanent, floor-to-ceiling demarcation and restricts direct access to medicine and to patient information to registered pharmacy personnel [16]; POPIA treats health information as special personal information. Drawn as a wall rather than a lane, because that is what it is.

One more rule with a sharp architectural edge. Good Pharmacy Practice rule 1.2.11.4(f) requires that software used for warnings or interaction checking be **the latest version available** [16]. That turns a content database into a regulated currency obligation sitting on the compliance critical path — and it has real consequences for a store that has been offline for two days, or for a distributed estate where content updates are pushed on a best-effort schedule.

What the dispensing system actually is

Patient registration; prescriber capture and validation; prescription capture with item, strength, dose, quantity, directions and repeats; clinical checking for interaction, duplication, dose range, allergy and contra-indication; generic substitution offer and recording; label generation; the scheduled-substance register; repeat management; medical-scheme eligibility and claim submission with real-time adjudication at the counter; the patient medication record; dispensary stock control held separately from front-shop stock; and the statutory reports and registers.

It must master the patient, the prescription, the dispensing event and the scheduled-substance register **alone**. No other system in the estate may be authoritative for any of them. In practice this is clinical software with a retail attachment, not retail software with a clinical module — and that distinction decides the procurement.

Scale, for context: the South African Pharmacy Council register at 16 October 2026 showed 4,367 community pharmacies, 326 private and 627 public institutional pharmacies, 330 manufacturing pharmacies, 175 private and 11 public wholesale pharmacies and 11 academic institutions — 5,851 registered pharmacies in all [17].

> **North American aside.** The Canadian model is a different machine and a single global template fits neither. There is **no national exit price**; dispensing fees and ingredient mark-ups are set province by province for public drug plans, with pharmacies charging "usual and customary" fees to private plans and the uninsured; billable **professional services** — vaccination, medication review, prescribing — are a revenue line worth $634m nationally in 2024 with no South African equivalent; and generic manufacturer rebates, estimated at 25–50% of generic purchase value, are confidential [18]. Ownership differs too: about 71% of Ontario community pharmacies are pharmacist-owned, and Quebec law permits only pharmacists to own a pharmacy [18]. The architectural summary is worth keeping: **a Canadian dispensing system is a claims-adjudication system with a clinical front end; a South African one is a price-compliance system with a clinical front end.**

The apparel and big-ticket family

Apparel and footwear

Sells fashion with a season, a style, a colour and a size curve. **Buys** everyone, discretionary, and highly income-elastic.

Margin and turn. The highest gross margins in retail and the slowest turn, with much of the margin given back in markdown. Mr Price group at 41.2%, TFG Africa at 41.6%, Lewis at 43.7%, Woolworths fashion, beauty and home at 46.0%, Truworths group at 51.3% with its Africa business at 54.0%, Country Road Group at 57.7% [19][20][21][4][22]. Apparel is 16.6% of formal South African retail sales [23].

Credit mix is the dividing line *within* the sector, and it changes the architecture more than the clothes do: Mr Price at 10.6% credit sales, Pepkor at 18.0%, TFG Africa at 25.8%, Truworths at 71% account sales, Lewis at 69.4% [19][24][20][22][21]. Bad debt tracks it — Truworths' net bad debt and expected credit loss at 11.9% of account sales with an allowance of 21.7% of gross active receivables, Mr Price's provision at 13.1% of receivables on a 21.9% approval rate [22][19].

Systems, and five genuine divergences from grocery.

The item does not persist. A new assortment is created each season, so the item master is a high-churn object and historical analysis has to work across SKUs that no longer exist. Grocery systems that assume a stable item master break here, not gradually but immediately.

Style, colour, size and the size curve. Planning happens at option level, allocation at SKU level, and the **size curve** — the planned distribution of units across sizes — is a first-class, measured, store-clustered object that grocery has no equivalent of. A store that has sold out of size M has a *broken* assortment while still holding stock, and the system has to be able to say so.

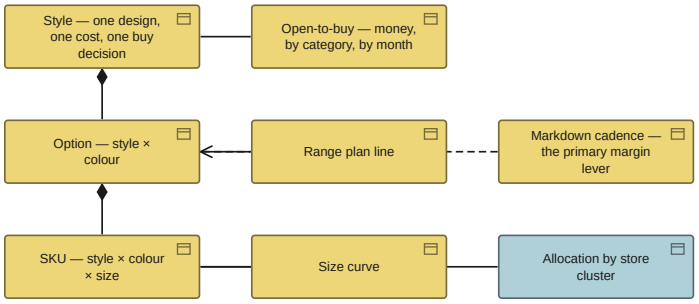


Figure 2.4 Why an apparel merchandising system is a different shape from a grocery one. The buyer commits to the style against an open-to-buy held in money; the planner works at option level; the store sells the SKU; and the size curve is a first-class planning object that grocery has no equivalent of.

Pre-season commitment. The buy is committed months ahead against an **open-to-buy** budget, usually with imported lead times. In season the only levers left are allocation, transfer and markdown.

Markdown is the primary margin lever, and it runs on a cadence: full price for the first weeks, a first markdown on slow options at the first review, a second mid-season, a clearance in the exit weeks, then terminal stock disposal. Each decision — which options, in which stores, at what depth, in which week — is irreversible, because a marked-down price cannot be raised. Mr Price's gross margin *rose* 70 basis points to 41.2% while comparable sales grew 1.1% and units grew 0.5% [19]: a margin improvement achieved by not marking down, which is only possible if the buy was accurate.

Clustering, not store by store. Thousands of stores are grouped by size, demographics and climate, and assortments are planned per cluster. Local versus imported lead time is the strategic variable, because a short local lead time allows an in-season repeat buy and a long import lead time does not.

Home, furniture and appliances

Sells big-ticket, low-frequency, often credit-funded goods, usually delivered. **Buys** households at a life event. Gross margins of 42.8% to 43.7% [19][21] on the slowest turn in retail. Lewis's debtors' book of R9.2bn **exceeds its annual merchandise sales of R5.5bn** [21], which tells you what business it is really in.

The sale is not the delivery, and that is the whole system requirement. Order management has to hold a sold-but-undelivered position — simultaneously a liability and a stock reservation — schedule delivery and assembly, handle part-deliveries and multi-item orders, and manage extended warranties and service. And note what happens in one till transaction: **the credit agreement, the insurance policy and the delivery order are three contracts created at once**, with the credit agreement carrying prescribed National Credit Act disclosures, an affordability assessment and cooling-off rights. Lewis reports "satisfactory paying customers" at 82.6% as a headline metric [21], which tells you where management attention sits.

Building materials and do-it-yourself

Sells cement, timber, boards, roofing, plumbing, hardware, paint, tiles and sanitaryware, in units that are heavy, bulky, sold by length, area or mass, and frequently delivered. **Buys** small builders, contractors and homeowners. Cashbuild at a 25.3% gross margin, a 2.6% operating margin and an average basket of R734 [25]; hardware, paint and glass is 7.5% of formal retail and shrank 0.4% in real terms in May 2026 [23].

Units of measure are the architecture. Product sold per metre, per square metre, per bag, per pallet, cut to length, with conversion factors and minimum order quantities. **Quotations** that convert to orders. **Trade accounts** with credit limits and project references. Delivery with vehicle and crane constraints. And returns of cut or mixed product that cannot be resold.

One live capability most groups forget: Cashbuild's store count *fell* from 252 to 242 with its P&L Hardware chain halving from 34 to 17 [25]. A contracting estate makes store-closure data flows — stock liquidation, transfer, lease exit — an operational capability rather than a project.

Electronics, appliances and telecoms

Sells high-value, serial-numbered, rapidly obsolescing goods, plus airtime, data, handsets and contracts. The thinnest gross margins in non-food retail: one group's telecoms segment at 20.7% on R1.5bn of sales [19].

Systems. Serial-number tracking from receipt to sale to warranty claim. IMEI and SIM activation integrated with mobile networks. **Airtime and data as virtual stock** — no physical inventory, real cost, real fraud exposure. Price-protection and stock-protection claims against vendors when prices fall. Extended warranty as a separate insurance-like product. And the shop-in-shop model, with its own stock-ownership and settlement question: one group runs 86 standalone telecoms stores plus 482 store-in-store concepts [19].

The licensed and site-bound family

Liquor

Sells beer, wine, spirits and ready-to-drink from licensed premises with statutory trading hours. High turn, moderate margin, very high volume growth — one group's liquor format grew 12.1% to R21.3bn in FY2026 across 886 stores under two banners [3][6]; a voluntary-trading group's liquor banner did 50 million transactions with an average basket up 0.70% [26].

The licence is the constraint and it is geographic. Liquor retail is licensed provincially, with trading days and hours set by provincial acts and municipal by-laws, and national chains hold longer liquor trading hours than small traders [27][28]. That means **per-store, per-day, per-hour trading rules**

enforced at the till, age verification, separately licensed premises — which is why liquor is a physically adjacent store rather than an aisle — excise-relevant product data, and separate stock and cash control. A single national "store open hours" field cannot express any of it.

Fuel forecourt

Sells regulated-price fuel plus a convenience shop, coffee, car wash and increasingly a quick-service restaurant. Fuel prices are regulated, so operators work on fixed margins per litre regardless of price level, and **fuel is typically 80–90% of turnover but a small share of profit** [29]. Meanwhile forecourt convenience sales rose about 4% to R40bn in 2024 from about 3,000 stores, roughly 15% of South Africa's primary convenience channels, forecast to about R48bn by 2028 [7]. And **46% of forecourt store customers are not buying fuel** [7].

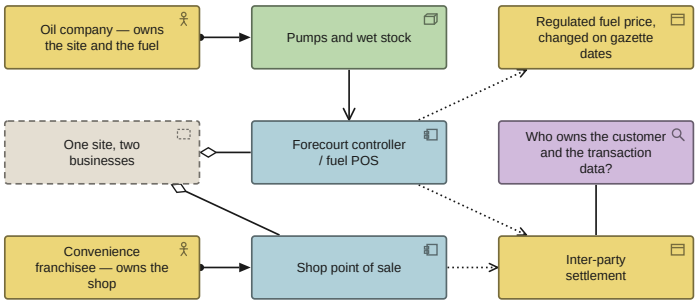


Figure 2.5 The forecourt is two businesses with two owners on one site, and the settlement between them — together with the question of who owns the customer and the transaction data — is the central architectural question of the format. Forty-six per cent of forecourt store customers are not buying fuel [7].

The forecourt is two businesses with two owners on one site, and that is the architectural issue. Fuel needs wet-stock reconciliation — dip readings against pump throughput against deliveries, with variance thresholds that flag leaks and theft — pump-to-POS integration, and regulated price changes on gazette dates, often running on the oil company's systems rather than the retailer's. The shop is a franchised convenience store with its own supplier, range and loyalty. **The settlement between the two, and the question of who owns the customer and the transaction data, is the central architectural question of the format.**

Wholesale and cash-and-carry

Sells cases and pallets to independent retailers, spaza shops, caterers and small businesses, for cash or on account. Very low gross margin, very high turn: a wholesale-weighted group's blended gross margin is 10.8% at a 2.1% operating margin [26]; a pharmacy group's wholesale total income margin is 8.0% against its retail's 30.8% [11].

The customer is a business, so: trade accounts with credit limits, terms and ageing; customer-specific and volume-break pricing, which the Commission has asked be made uniform and objectively justified [27]; case and pallet units of measure; **VAT invoicing rather than till slips**; route-to-market and van sales for delivered wholesale; and buying-group and rebate administration.

One number worth staring at. One pharmacy group's wholesale arm books R13.3bn of **internal** sales to its own stores against R3.2bn external [11]. Intercompany elimination is therefore a first-order reporting requirement, not an accounting afterthought, and the group's R21.3bn turnover is not R18.1bn plus R16.8bn.

The digital family

Pure-play online

Sells its own inventory, other sellers' inventory, or both, to urban, banked, connected shoppers. South African online retail was about R130bn in 2025, forecast to about R159bn in 2026, from R30.2bn in 2020 — roughly 10.7% of total retail on a constant-price basis, with the sources disagreeing on basis and saying so [30]. The detail worth noticing: adults shopping online **fell** to 34.2% in 2025 from 36.6% while internet access reached 79.1% [30]. The market is growing on spend per shopper, not on shopper count.

The economics are brutal and public. The benchmark pure-play's core e-commerce unit runs at about a 1% margin, and the group reached its first full-year profit before tax — R181m on R16.4bn of revenue — in the year to mid-2026 after a R214m prior loss [30][31]. The structural reason is in chapter 6: an online order costs picking labour, packing materials and either a delivery or a collection operation, against the **same shelf price**.

Systems. Catalogue, cart and checkout are the easy part. The hard parts are a single **available-to-promise** view that reconciles store on-hand to a promise and discounts it for phantom inventory; pick-list generation and in-store picking tools with substitution rules and customer approval; **slot capacity management** per store per window; route planning; weight-variable repricing, because the picked 1.3 kg chicken is not the ordered 1.2 kg; age-restricted handling at handover; and payment authorisation followed by final capture **at a different amount** — a reconciliation path most retail finance systems were never built for.

Marketplace

Sells other people's inventory on your platform. It looks like a channel and it is a different business. Marketplace is 60% of the benchmark platform's gross merchandise value, with 15,000 active sellers [30].

1P and 3P are two businesses in one platform. First-party needs buying, inventory, pricing and fulfilment. Third-party needs seller onboarding, catalogue ingestion and matching, listing quality control, order routing to the seller, seller performance management, commission and **take rate** calculation, payout and settlement, and liability allocation for returns and defects. And fulfilment-as-a-service is a third business again: the platform's fulfilment arm grew revenue 93.5% with over half of its top forty sellers using it [30].

> **GMV is not revenue.** On a third-party sale the platform books commission, not the sale. That is why the benchmark platform's revenue of R16.4bn and its gross merchandise value of about R30bn in 2025 differ by roughly a factor of two [31][30]. A dashboard that mixes the two produces a growth rate nobody can reconcile, and a board that believes it has doubled when it has not.

Two lines of business that are not formats

A format has stores. These two have neither stores nor stock, and both sit inside South African retailers at material scale. Miss them and your application model has a hole in it.

Retail credit and retail financial services

Sells a store card, a revolving account, an instalment agreement, insurance and increasingly transactional banking, to the retailer's own customers. The numbers are not small. Lewis earns R10.3bn of revenue on R5.5bn of merchandise sales [21]. One apparel group's financial services arm earns R947m of other revenue with no retail sales at all and charges R744m of interest and fees on debtors [19]. Woolworths Financial Services contributed R228m of profit after tax at a 7.0% annualised impairment rate [4]. Pepkor's financial services revenue was R4,897m, up 28.6% [24]. TFG Africa's debtors' book is R9.4bn [20].

This is a regulated credit provider inside a retailer: National Credit Act registration, prescribed affordability assessment, bureau enquiry, prescribed disclosures, statutory maximum pricing, credit life insurance with a statutory cap, collections, debt review and write-off. The architecture is a lending stack adjacent to a retail stack, joined at the customer and at the till. Two consequences architects miss. First, **the credit decision sits in the sale path** at the point of sale, with a latency budget. Second, the retail margin and the credit margin are separately managed and they can conflict — a looser approval rate grows sales and worsens bad debt — so the only correct view is contribution after credit loss, by cohort. Neither business can be managed on the other's reporting.

And one mechanic with no North American equivalent at scale: **lay-by**. The customer pays in instalments and takes the goods on final payment, so the retailer holds both the cash and the stock. It is not credit, it is not yet a sale, it is regulated under the Consumer Protection Act, and under section 62 each instalment remains the consumer's property until delivery with the goods at the supplier's risk. So it needs its own ledger, its own stock reservation and its own cancellation and refund rules — and it is routinely bolted onto a point of sale that was not designed for it. Pepkor reports lay-by sales as a separate line [24].

Retail media

Sells advertising and promotional space — in store, on shelf, on the receipt, in the app, on the website, in the loyalty communication — to the retailer's own suppliers, priced off the retailer's first-party data. Near-pure margin, because the inventory is space and audience the retailer already owns. One discount grocer launched a retail media division in FY2026 [2]; the asset underneath such a business is a loyalty base of tens of millions of members and more than a billion transactions a year [39]. No South African retailer publishes retail media revenue, so there is no figure to give.

Systems: ad inventory management, audience segmentation from loyalty data, campaign measurement with incrementality testing, and a billing relationship with the same supplier the buying team is negotiating cost price with. That last point is the governance problem. Retail media income and supplier income are economically adjacent and must be contractually and accounting-separate, or the retailer has recreated the supplier-income transparency problem in a brand-new line of business.

The ownership models, and what each does to the architecture

Three ownership models coexist inside single South African groups, and they must be modelled as different businesses. This is the section to read twice, because it relocates the authority for price, assortment and stock — and every system diagram in part III is drawn against one of these three assumptions.

Corporate. The group owns the store, the stock and the price. Everything in the estate is the group's system of record. This is the model every retail reference architecture assumes.

Franchise. The franchisee owns the business; the group owns the brand, the supply chain and usually the systems. One group ran 620 franchise outlets against 992 company-owned at 1 March 2026, with franchisees sourcing inventory through the group's distribution network, paying a royalty for brand, systems, supply chain and digital capability, and funding their own capital, operating costs, staffing and management [32]. Architecturally the franchise store is **a customer, not a location** — which is exactly why Oracle's model allows a "store" to represent a customer location such as a franchisee [33]. Price recommendation versus price mandate becomes a contractual question. Stock is *sold* to the franchisee rather than transferred. And the group's visibility of the franchisee's shelf depends entirely on what the franchise agreement compels.

The performance divergence inside one brand is real: franchise like-for-like growth of 0.9% against company-owned like-for-like of 3.9% in FY2026, a 300 basis-point gap [1].

Voluntary trading. The strongest form of devolution, a top-five South African grocery model, and the one most general retail reference material gets wrong. SPAR Southern Africa is **a wholesaler, not a chain**. Stores are owned by independent retailers who hold membership of the group and of a separate non-profit **Guild**, retaining legal ownership of their businesses. There is **no franchise fee and no royalty on turnover**; the group earns its margin on the wholesale supply of goods through its distribution centres. Members commit to buy a defined share of stock through those centres —

which is why the group reports **"retailer loyalty" of 78.6%** as a headline operating metric — and they keep pricing discretion, product-mix flexibility and the right to source locally [26][34][35].

> **In a voluntary-trading estate the group does not have an item master, a price file or a stock ledger to master for the member business.** It can recommend, list, promote and fund. It cannot mandate. There is no group price file because members set shelf prices. There is no group stock ledger because the group knows what it shipped, not what the member holds, unless the member shares point-of-sale and inventory data by agreement, in heterogeneous formats, with gaps. The point-of-sale estate is not mandated, so a corporate chain rolls out one till platform and a voluntary-trading group integrates many. Credit risk points the other way: the receivable is *from* the members. And store ownership changes without group consent, which gives store master data an ownership dimension with its own lifecycle.

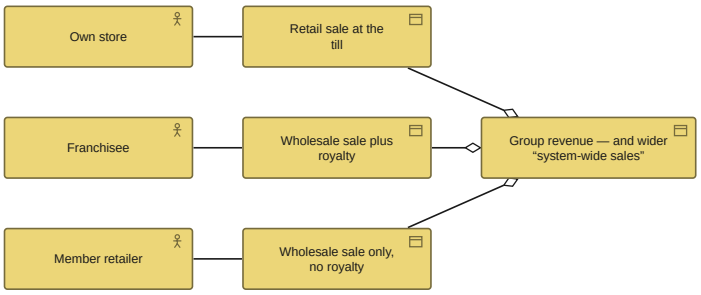


Figure 2.6 Three ownership models, three revenue recognition patterns, inside one group. Different timing, different VAT treatment and different credit exposure — which is why a group reporting "turnover" has to say which of the three it means.

Three further consequences follow in any group that runs more than one model.

Three revenue recognition patterns in one group. Own-store retail sales, wholesale sales to franchisees and members, and royalty income — with different timing, different VAT treatment and different credit exposure. Italtile reports **system-wide turnover of R11.3bn** against a group trading profit of R1.8bn [36], a figure that includes franchisee sales the group never books as revenue. A reporting layer that does not carry the distinction will produce two "turnover" numbers and have to explain the difference to an audit committee.

Entitlement, not control. A franchisee gets a subset of group systems with a contractual, auditable boundary. That is an access-management requirement with a commercial contract behind it, not a role in a directory.

Data rights must be explicit. Who owns franchise point-of-sale data, loyalty data and customer data, and on what lawful basis does it cross the boundary? That is a POPIA question and it needs an answer before the integration is built, not after.

The informal channel

Not a format the group operates, but a channel it serves, and it is large: the informal grocery sector including spaza shops and mobile traders has been valued at R184bn, across about 150,000 shops [37][38]. Consumers spend less than 40% of their grocery budget there, concentrated in day-to-day items [27]. The chains are entering it with container formats, and a discount clothing group's explicitly named "informal market platform" moved R58.4bn of throughput in ten months [24].

The architecture is a wholesale architecture, not a retail one. Cash-and-carry, buying groups, van sales, route to market. And the integration point is **payments and value-added services** — airtime, electricity, bill payment, money transfer — because that is how formal platforms reach informal retailers. The credit question is the whole opportunity: an informal retailer with no financial statements and no fixed address is an underwriting problem solvable, if at all, from the platform's own transaction data.

Where the architectures genuinely diverge

The honest answer is that the shared layer is thinner than most reference architectures admit.

Genuinely shared across every format: the point-of-sale transaction log; tender and settlement; the customer and loyalty identity; the general ledger, tax and statutory reporting; human resources, payroll and rostering; the store and property master; and the network, device and in-store technology estate.

Genuinely divergent:

Concern	Grocery	Apparel	Pharmacy	Building materials
Item lifecycle	years, stable	one season, disposable	national medicine code, regulated	years, stable
Planning grain	category money plus SKU/store units	option money plus size curve	dispensing volume plus front shop	branch plus unit of measure
Replenishment	daily forecast or min-max	pre-season allocation plus in-season transfer	usage-based with scheduled-stock controls	branch order plus direct to site
Price authority	the retailer	the retailer	**the state: SEP plus fee formula**	the retailer plus quotation
Primary margin lever	promotion funding	**markdown**	front-shop mix and private label	mix and trade terms
Dominant loss	**waste and shrink**	**markdown and returns**	expiry and scheduled-stock discrepancy	breakage and cut-to-length
Regulatory core	food safety, cold chain, liquor hours	labelling, country of origin	**Medicines Act, Pharmacy Act, scheme claims**	product standards, delivery safety
Third-party money	supplier income	supplier income, lower	**medical scheme reimbursement**	trade credit

What the format mix does to the application estate

Everything above lands on one design rule, and it is the conclusion of the chapter.

Build one shared commerce and customer layer, one shared finance layer, and separate merchandising and operations stacks per format family. Groups that have tried to run a single merchandise system across grocery, apparel and pharmacy have ended up with a grocery system plus two sets of workarounds. The published evidence is indirect but consistent: Woolworths reports Food and fashion, beauty and home as separate segments with separate margins, separate space and separate online shares [4]; Clicks keeps retail and its distribution business as distinct businesses [10].

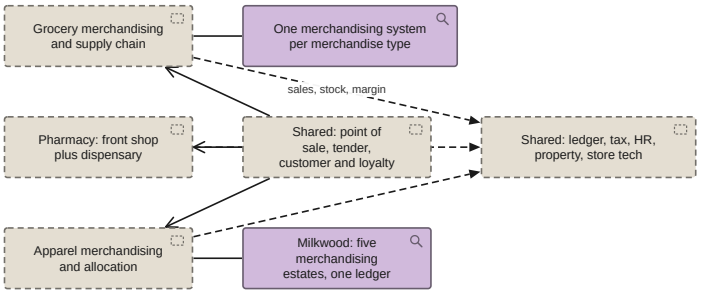


Figure 2.7 The design rule the chapter argues for: one shared commerce and customer layer, one shared finance layer, and separate merchandising and operations stacks per format family. Expect a merchandising system per merchandise type, not per banner.

So expect a merchandising system **per merchandise type, not per banner** — because food and fashion master-data models are genuinely incompatible, not merely differently configured. Expect multiple instances of merchandising, store inventory, point of sale and e-commerce, by banner, by country and by acquired business. Expect one instance of the general ledger. And expect the merchandise financial plan to exist, in part, as a spreadsheet, because the planning function is widely not systematised.

Milkwood, our invented group, therefore has five merchandising estates rather than one: grocery, pharmacy front shop, pharmacy dispensary, apparel, and cash-and-carry. Its shared layer is the customer, the ledger, the property master and the store technology. When somebody proposes "consolidating onto one platform", the question to ask is not whether it is technically possible. It is which of the four divergent concerns in the table above the single platform will be made to handle badly, and whether the business has agreed to that trade in writing.

THE CHAPTER IN TEN LINES

- A format is an economic machine, not a shop size: it fixes the margin available, the turn achievable, the capital per store, the labour model and the systems the business cannot trade without.
- Grocery is the highest-volume, most time-critical estate in retail, and at promotions of 40% of turnover its promotion engine is a primary pricing system inside the authorisation path of every sale.
- Discount grocery runs a lower gross margin at a higher trading margin, and the mechanism is range discipline — so the architectural instruction is unusual: the system must prevent range creep.
- The pharmacy dispensary is a separately licensed professional practice inside a shop. The retailer cannot price medicines: the Single Exit Price is national and the dispensing fee is a four-band formula changed by gazette.
- A South African pharmacy's item master carries two identifiers for one product — GTIN for receiving, scanning and selling, NAPPI for claiming and reimbursement — maintained by different bodies, not mapping one to one, with the retailer owning a mapping nobody else maintains.
- Good Pharmacy Practice specifies a physical wall and restricts access to medicine and to patient data to registered pharmacy personnel, which makes the dispensary a hard boundary in the data architecture, not a department with a permission set.
- Apparel diverges from grocery on five counts: the item does not persist, planning is at option level, the size curve is a first-class object, markdown is the primary margin lever, and assortments are planned per cluster.
- Retail credit and retail media are lines of business with no stores and no stock, and both sit inside South African retailers at material scale.
- Three ownership models coexist and they relocate the authority for price, assortment and stock. In a voluntary-trading estate the group has a wholesale order book and whatever the member chooses to share.
- One group runs three revenue recognition patterns — own-store retail, wholesale to franchisees and members, and royalty — with different timing, VAT treatment and credit exposure.
- The design rule: one shared commerce and customer layer, one shared finance layer, separate merchandising and operations stacks per format family. Expect a merchandising system per merchandise type, not per banner.

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CHAPTER 3

The vocabulary

Retail runs on a few hundred ordinary words used with precise and conflicting meanings. This chapter defines them in themed runs with the arguments attached, and ends with the five words that mean different things in different parts of the same retailer.

Four sentences you cannot yet read

> *"Option's selling through at a hundred and ten against curve but the size curve's broken in cluster three, so I'm transferring rather than taking the first markdown."*

> *"GMROI on the category's fine, it's the tail dragging cover — forty per cent of the SKUs, six per cent of the sales, and nobody'll delist because of the supplier income."*

> *"Promo's funded but the uplift was pantry loading, so the incrementality's negative once you net the cannibalisation on the KVI."*

> *"Fifty-two scripts rejected on the switch this morning — SEP's moved and the NAPPI mapping's stale on the new pack size."*

None of that is jargon for its own sake. Every phrase is compressed and carries real content, and until you can decompress it you will nod along in a trading meeting and miss the thing you were brought in to catch.

This chapter is the decompression. It runs in ten themed runs rather than alphabetically, because the words only make sense in clusters — the alphabetical list is appendix B. Each run gives a table of terms and then prose on the two or three that are routinely misused. The chapter ends with the five ordinary English words that mean different things in different parts of the same retailer, which is an architecture problem rather than a language problem.

One instruction before you start. The first run is the most important thing in this book. Spend longer on it than feels proportionate.

Run one: the merchandise words

Term	What it means
SKU	Stock-keeping unit: the smallest thing distinguished for stock purposes. In North American usage, the sellable barcoded unit
Article	The European ERP term, SAP's especially, for the merchandise-managed item. Synonymous with SKU in most grocery usage
Product	Ambiguous. In e-commerce, the *navigation* object: a multipack and a single are one product and two articles
Style	An apparel design: one cut, one cost base, one buy decision. Not stockable, not sellable
Option	Style plus one differentiating axis the customer chooses before size — normally colour. The unit of range planning
Variant	Any axis of difference below a parent: size, colour, flavour, pack size, scent
Size curve	The planned distribution of units across sizes for an option
GTIN	Global Trade Item Number: the GS1 product identifier. A concept, not a format
EAN / EAN-13	Legacy name for the 13-digit GTIN. The rest-of-world retail barcode
UPC / UPC-A	Legacy name for the 12-digit GTIN. North American
ITF-14	The case-level symbology, tolerant of poor print on corrugated board
GLN	Global Location Number: identifies a party or a place. The external key for "which store, which supplier site"
Case / inner / outer / traded unit	The packaging levels above the consumer unit, each with its own GTIN and quantity
Pack / prepack / ratio pack	A pre-assembled mixed group: one order line, several SKUs inside
Private label / own brand	Sold under the retailer's own brand, with the retailer taking the brand owner's role
Core range / tail	The lines every store must carry; the slow lines that are most of the item count and little of the sales

Every system in the estate has its own answer to "what is a SKU"

This is the single most argued-about piece of retail modelling, and the argument is not pedantry: it decides whether the retailer can forecast, allocate, report margin by range, or build its online catalogue at all.

The vocabulary collides because four traditions meet in one estate. European ERP says *article*. North American merchandising says *SKU*. Apparel says *style*, *option*, *colour-size*. E-commerce says *product* and *variant*. Hold it as **four distinct concepts that different trades name differently**:

Concept	What it is	Grocery name	Apparel name	SAP name	Identified by
The planning object	What a buyer buys and a planner plans: one design, one cost, one negotiation	article, product	**style**	**generic article**	internal style number
The differentiated offer	The style in one differentiating attribute, normally colour	rarely needed	**option**, colourway	part of the variant key	style plus colour code
The sellable unit	The thing with its own barcode, its own stock position and its own till price	**article**, SKU	**SKU**	**variant**	GTIN
The physical grouping	How the sellable unit arrives: inner, case, pallet, prepack	**case**, outer, traded unit	**prepack**, ratio pack	**structured article**	GTIN-14 or SSCC

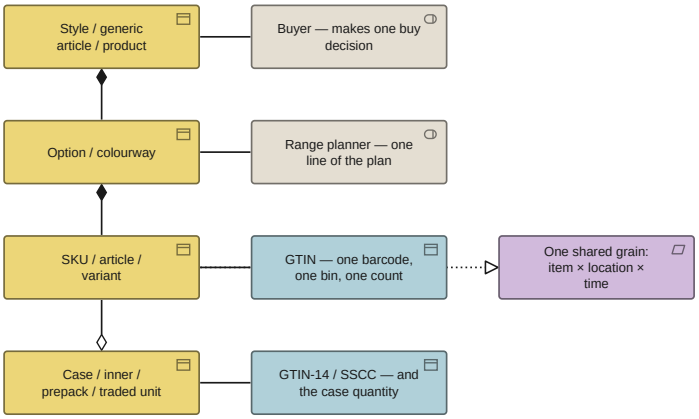


Figure 3.1 Four concepts, four vocabularies, one estate. Grocery collapses the top three into a single object and is right to; apparel never can. An interface between a system that collapses them and a system that does not needs a mapping somebody owns.

The structural reason the words collide. In food, the planning object and the sellable unit are usually the same thing: a two-litre bottle of milk is one article, one barcode, one stock position. In fashion they never are: a shirt is one planning object, four colourways and perhaps twenty-eight sellable units, and the buyer commits to the planning object while the store sells the sellable unit. Grocery *degenerates* the model cleanly — one style, one option, one SKU — which is why grocery-born systems cannot be retrofitted to apparel and apparel-born systems are gratuitous in grocery.

The vendors formalise it differently and the differences are operational, not cosmetic. SAP distinguishes a **single article**, a **generic article with variants** and a **structured article**; and — read this twice — characteristics assigned to hierarchy levels and merchandise categories are inherited downwards but are **informational only**, supporting analysis with no operational effect in the wider system [1]. A characteristic that looks like a product attribute may be a reporting tag with nothing behind it, and a project that assumes otherwise builds logic on sand. SAP also supplies two objects with no Oracle equivalent: a **reference article** used as a creation template, never sent to the point of sale and therefore never operational; and a **value-only article**, supporting inventory management on a value basis at category level [1].

Two settling questions that resolve almost every argument in the room:

1. *Is its stock counted separately, and can it be sold on its own?* Both yes, it is a SKU. If its stock is counted with its siblings, it is an option or a style. If it can be sold but never counted — a weighed loose item, a service, a deposit — it is a **sellable non-stock item**, which needs its own type rather than a fake SKU with the inventory flag switched off.
2. *Would the buyer make one buy decision for it?* That is the style. *Would the range planner put it on one line of the plan?* That is the option.

Five mistakes you will meet

One **product table with a level column**, which survives until a query must mean "sales by style this week against last" and "stock by SKU by store tonight" at once. **Size as a variant axis equal to colour** — colour drives photography, breadth and markdown, size drives the size curve, allocation ratios and lost-sale analysis, and one undifferentiated axis destroys both. **The barcode as the key**, when a SKU may legitimately carry several GTINs and a GTIN may be reused after a sunset period. **The e-commerce product as master**, when online grouping is a navigation decision that disagrees with the merchandise hierarchy. And **a new style created when the supplier changes**, when the design did not change and supplier belongs on a dated supply arrangement.

The identifiers, and the two defects that cause most of the pain

GTIN is the concept; the number of digits is the format. GTIN-8 for very small packs, GTIN-12 for the United States and Canada, GTIN-13 for everywhere else, GTIN-14 for packages containing several units — built from the inner GTIN-13 without its check digit, plus an indicator digit, plus a new check digit [2].

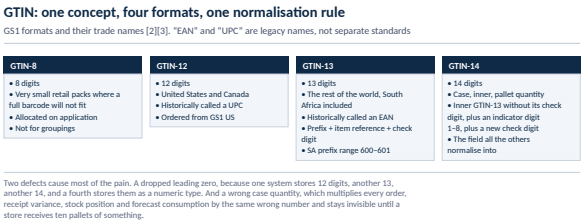


Figure 3.2 The GTIN formats and the one discipline that removes most retail integration defects. Normalise everything to fourteen digits, right-aligned, zero-padded, stored as a string — at every boundary [2][3].

Two legitimate strategies exist for packaging levels and the retailer cannot assume which a supplier used: a separate GTIN-13 per level, or a GTIN-13 on the consumer unit with GTIN-14s derived from it [3]. The article master must hold either.

The single most common retail data defect is a GTIN that lost a leading zero in transit. The instruction is explicit: when a GTIN-13 travels in a GS1-128 or ITF-14 barcode it gains a leading zero, because those symbologies use a fourteen-digit field [3]. Normalise to GTIN-14, right-aligned, zero-padded, stored as a *string*, at every boundary, and the class of defect disappears.

The single highest-impact article-data error is a wrong case quantity. It multiplies or divides every order, every receipt variance, every stock position and every forecast consumption calculation by the same wrong number, and it stays invisible until a store receives ten pallets of something.

One South African specific. GS1 South Africa trades as the Consumer Goods Council of South Africa and the local prefix range is 600–601 [4]. The Council has had to publicly correct the belief that a prefix shows where a product was *made*: it shows where the company is *registered* [4]. **The prefix tells you nothing you can act on**, so country of origin, local-content reporting and duty classification each need their own attribute.

Finally, the thing coming at the industry. GS1's 2D barcode programme points at the end of 2027 for retailers to be able to scan and process GS1-compliant 2D barcodes at the point of sale. It is a **readiness ambition, not a mandate with a legal date** — the GS1 US material gives no day or month, runs a voluntary crawl-walk-run model and says explicitly that it is not necessary to implement all 2D capabilities by that date, and the global and South African positions were not sourceable [5]. Never write that GS1 requires 2D by 2027. The durable content is the consequence: the scanner estate needs upgrading in every lane; the point of sale must parse a Digital Link URI and extract the GTIN from it; the item master becomes the authority for batch and expiry, which previously lived only on packaging; and because the Digital Link resolves to the brand owner's content, **the scan becomes a consumer touchpoint the retailer does not control.**

Run two: the hierarchy words

Term	What it means
Merchandise hierarchy	The tree from department down to the item. The most load-bearing reference data in the estate
Department	The top commercial level. Not a label: a bundle of policy, as below
Category	A *shopper*-defined grouping of substitutable products; the unit of commercial management
Merchandise category	SAP's lowest node; articles attach only here, never to a hierarchy level
Reclassification	Moving an item between hierarchy nodes. A dangerous act, as below
Organisational hierarchy	Company, chain, area, region, district, store. Retail operations' structure
Price zone	A set of locations sharing a price. A hierarchy over locations that matches no other
Cluster	A group of stores planned together on size, demographics or climate
Virtual location	A non-physical selling or stock-holding location: a website, a franchisee, a committed pool

The two canonical shapes differ and you will meet both. Oracle's is **fixed**: company, division, group, department, class, subclass, with items attaching at subclass; all six levels are required and six is the maximum [6]. SAP's is **deliberately variable**: a top node with unlimited levels beneath it, numbered bottom-up, branches of differing depth, and two node types — merchandise categories, which hold articles, and hierarchy levels, which never do [1].

A department is not a label. In Oracle's model the department record carries the profit calculation type, the open-to-buy calculation type, the purchase type, the markup calculation type and percentages, expected annual sales, outlier tolerances, and buyer assignment; VAT regions attach at department level and default to the items in it [6]. A department is a bundle of **accounting policy, tax default, buying authority, budget and statistical parameter**. Move an article between departments and you move all of it.

> **Reclassification is the trap.** Oracle states that when an item is reclassified, "its sales history, purchase orders, inventory, and planning information are reclassified too" [6]. Read the consequence. A reclassification that *does* carry history rewrites last year's numbers so that this year's comparison works — and last year's published report no longer reconciles. A reclassification that *does not* carry history preserves the published report and destroys the comparison. Both are defensible. The retailer must choose, must choose the same way every time, and in practice different systems in the same estate choose differently. Finding out which systems do which is a real deliverable.

The six hierarchies that disagree

A retailer runs at least six overlapping classification structures, each with a legitimate owner and a legitimate reason to differ.

Hierarchy	Owner	Purpose	Why it differs
Merchandise	Trading and buying	Buying authority, open-to-buy, tax default, margin policy	Organised around how the retailer *buys* — by supplier market and buyer role
Financial	Finance	Statutory and management reporting, segment disclosure	Organised around legal entity and reportable segment, which cut across buying
Reporting / category	Category management, marketing	The consumer decision tree — how shoppers think	Organised around *shopper* behaviour; "breakfast" is not a merchandise department
Space	Space planning	Planogram, bay, module, fixture	Organised around physical adjacency on the shelf
Organisational	Retail operations	Store management, labour, district reporting	Organised around span of control, which changes with every restructure
Price zone	Pricing	Where one price applies	Organised around competitive geography and format

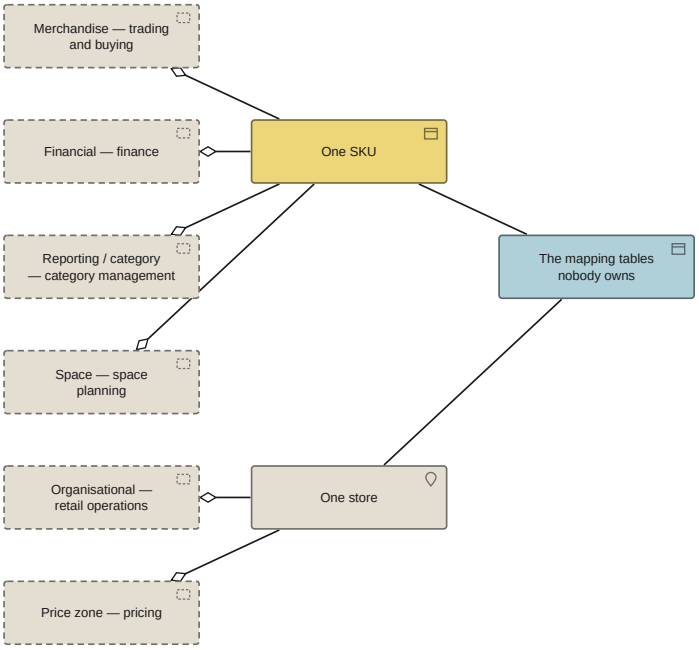


Figure 3.3 Six classification structures over the same item and the same store, each with a legitimate owner and a legitimate reason to differ. Do not try to collapse them; master the mappings, version them, and publish one page saying which hierarchy answers which question.

The honest standards response is the ARTS Operational Data Model's: model a *merchandise hierarchy function* so that different business processes get different hierarchies over the same items, with their own named levels in their own order, joined at item level [7]. Most real estates instead pick one hierarchy, declare it canonical, and maintain the other five as mapping tables of varying age — in the warehouse, in a spreadsheet, or in the heads of three analysts.

Four consequences to expect, and to state early:

- **Totals that do not tie.** Sales by merchandise department and by reporting category both sum to total sales, but neither sums to the other at any intermediate level, because the mappings are many-to-many. Both are right, and every executive reads whichever flatters.
- **Reclassification asymmetry.** The merchandise hierarchy restates history; the warehouse's reporting hierarchy usually does not, because it is a slowly changing dimension. A reclassified article's history moves in one system and stays in the other, permanently.
- **The space hierarchy has no shared key**, as above.
- **The organisational hierarchy changes fastest.** A regional restructure re-parents 300 stores, and every report, target, labour model and price zone built on last quarter's structure is now wrong — and the price zone was *not* re-parented, because it is maintained separately.

Do not try to collapse the six into one; that project has failed repeatedly. Instead: make item and location the only mandatory shared grains, require every hierarchy to be versioned and effective-dated with a stated restatement policy, master the mappings as first-class data with an owner rather than as warehouse code, and publish one page saying which hierarchy answers which question. That prescription is my judgement; what is sourced is that ARTS models multiple hierarchies per business function [7] and that Oracle restates history on reclassification [6].

Run three: the buying and planning words

Term	What it means
Buying	Selecting suppliers, range and cost price
Planning	Determining how much to buy, in money, by period
Range	The set of lines the retailer offers; in apparel also a seasonal collection
Assortment	The range carried by a specific store or cluster
Listing / delisting	Admitting a line to the range, or removing it. Delisting is a contractual event
Open-to-buy (OTB)	Planned receipts less on-order commitments: the buying budget, in money
Allocation	Deciding which units of a fixed buy go to which store
Replenishment	Automatically reordering to maintain availability
Sell-through	Units sold as a percentage of opening units on hand
Size curve	The planned distribution of units across sizes for an option

Buying and planning are different jobs, and that is the distinction outsiders always get wrong. The buyer decides *what* and *at what cost*. The planner decides *how much*, in money, by period. Open-to-buy is held **in money, by category, by month or week — not in units** [8], which is precisely why the two are separate roles. Every merchandise-planning architecture has to reconcile the money plan to the unit plan and almost none does so cleanly; when somebody says the planning tool "doesn't tie", that is the gap being described.

Sell-through is the core apparel in-season metric and the one most often misapplied. Units sold divided by beginning-of-period units on hand [8]: of what I had, how much went? Measured at option and store-cluster level, weekly, against a planned curve. Above plan in week three is a repeat-buy signal if lead times allow; below plan is a markdown signal. **Grocery barely uses it**, because grocery replenishes rather than selling through a finite buy — a clean illustration of why one planning system cannot serve both.

Cluster is not a region. Stores cluster on size, demographic and climate, cutting across the organisational hierarchy and across price zones. A retailer that allocates by region rather than cluster sends winter coats to Durban because Durban shares a district with the Midlands.

Run four: the space words

Term	What it means
Planogram	A drawing specifying which SKU sits where on a defined piece of shelving
Facings	How many units of a SKU are presented to the shopper across the shelf front
Bay	A standard unit of shelving width
Gondola	A double-sided free-standing shelving run
End cap	The display at the end of a gondola. Premium promotional space, and usually paid for
Shelf-edge label (SEL)	The price label on the shelf front. A legal object, see run seven
Pick face	The shelf position stock is taken from; its capacity drives replenishment
Space plan / floor plan	Selling space allocated between categories; the layout of fixtures and aisles
Trading area	The floor area used for selling. The denominator for trading density
Dark store / MFC / CFC	Online-only fulfilment facilities: a store-format one, a small automated unit in a back room, a dedicated warehouse

Three of these are routinely misused. **Facings** gets confused with units: a SKU with three facings and four units deep holds twelve units on the shelf, and replenishment computes against that capacity, not the facing count. **Bay** gets used as a synonym for gondola, which makes planogram counts meaningless — a gondola run is several bays. And **trading area** gets confused with lettable area, which is the leased figure and is larger; rent is charged on one and trading density measured on the other, so quoting a density against the wrong denominator flatters a store.

And the point from chapter 1 bears repeating: the planogram is the only object joining the merchandise hierarchy to the physical store, and in most estates that join is re-derived rather than mastered.

Run five: the inventory words

Term	What it means
Cover / weeks of supply	On-hand divided by average weekly unit sales. Grocery says days, apparel says weeks
Forward cover	The same, computed on *forecast* rather than historic sales
Stock turn	Cost of sales divided by average inventory
Safety stock	Buffer held against demand and lead-time variability
Reorder point / order-up-to level	The min and the max in min-max replenishment
Variance / adjustment	Counted less book; the posted correction, which should always carry a reason code
Phantom inventory	Stock the system holds that does not physically exist
Negative stock	A system on-hand below zero. Always a process defect, never a state
Available to promise (ATP)	The position the retailer is willing to sell online

Stock on hand has nine different answers

Ask a retailer how much stock it has of a SKU in a store and the honest response is a question: who is asking, and what for? There are at least nine correct answers, and they are not variants of one number.

Position	What it is	Who asks
Physical / book stock	What the system believes is bodily present	Store manager, stock take, finance
Available to sell / ATP	Physical less reservations, less unsellable, optionally plus inbound inside a promise horizon	Website, call centre, click and collect
On order	Ordered from the supplier, not yet shipped	Replenishment, open-to-buy
In transit	Shipped, not received. Owned, not available, and at somebody's risk	Finance — it is on the balance sheet — and supply chain
Reserved / allocated	Physically present but committed to a customer order, a pick, a transfer out or a lay-by	Fulfilment
Damaged / quarantined / expired	Present, owned, not sellable. Awaiting write-off, return or recall	Loss prevention, quality, finance
In a customer's basket	Soft-held by an online session. Not a stock position in most systems, and the commonest cause of oversell	Digital
Customer-owned, retailer-held	Paid for, not collected: lay-by, click-and-collect awaiting pickup, repair	Finance, legal
Consignment / supplier-owned	On the shelf, not owned. Concessions, magazines, some pharmacy lines	Finance

A tenth appears in pharmacy: **bulk stock under dispensing** — a thousand-tablet container partially dispensed, whose remaining contents are a stock position measured in a different unit from the purchase unit.

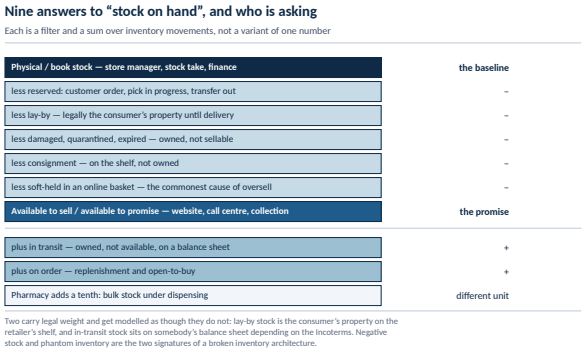


Figure 3.4 The nine answers to “how much stock do we have”, with the two that carry legal weight marked. Every one of them is a derivation over inventory movements, which is why the movement log and not the quantity column is the system of record.

Two carry legal weight and get modelled as though they do not. Under the Consumer Protection Act each lay-by instalment remains the consumer's property until delivery, with the goods at the supplier's risk — so lay-by stock on the retailer's shelf is legally *not* the retailer's stock. And in-transit stock sits on somebody's balance sheet depending on the Incoterms, so counting it at both ends is a misstatement, not a rounding problem.

> **"Stock" is not a column.** The durable facts are **inventory movements** — append-only, signed, typed, located, timestamped, with a reason code and a document reference — plus **inventory status buckets**. Every position in the table above is a derivation: a filter and a sum over movements, optionally plus a pending-document set. A cached quantity is acceptable as a derivation with a recomputation path. It is not acceptable as the system of record, because the moment two channels disagree there is nothing to adjudicate between them.

Phantom inventory and negative stock are the two signatures of a broken inventory architecture. Negative stock means movements are being posted out of order, or receipts are missing. Phantom inventory — stock the system has and the shelf does not — is what causes online pick failures and false "available" promises, and it is the reason an available-to-promise figure should always be discounted rather than taken at face value.

Run six: the money words

Term	What it means
Gross margin	(Net sales − cost of sales) ÷ **net sales**
Markup	(Price − cost) ÷ **cost**. A different denominator
Maintained markup (MMU)	Ticket price less cost less allowances. The margin actually achieved
Markdown	A *permanent* price reduction on unsold stock
Discount	A *temporary* reduction applied at the point of sale
GMROI	Gross margin rand divided by average inventory at cost
Shrink	Book stock less counted stock: the unexplained loss
Supplier income / back margin	Money from suppliers that is not an invoice-price reduction
Rebate / overrider	A retrospective payment based on purchase volume
Trading profit / trading margin	The South African primary operating measure, before capital items

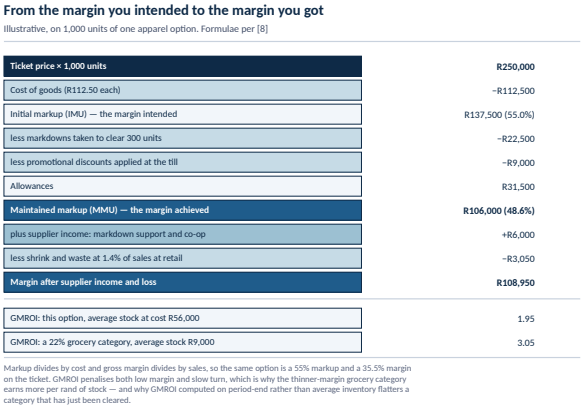


Figure 3.5 The money words as one arithmetic walk, on an illustrative apparel option. Initial markup is what you intended; maintained markup is what you got; allowances are the bridge [8]. The GMROI comparison underneath is why a high-margin category can still be the wrong place to put the money.

Gross margin against markup is the arithmetic error that most often derails a meeting. Gross margin divides by sales, markup divides by cost [8]. An item bought at R60 and sold at R100 carries a 40% gross margin and a 66.7% markup. Both numbers are correct. Ask which denominator, every time.

Markdown against discount decides whether a retailer can explain its own margin. A markdown is permanent, retailer-funded, and under the retail inventory method it reduces inventory value the moment it is applied, with no physical movement at all. A discount is temporary, often supplier-funded, and the base price survives underneath it. Systems that model both as "a price record with dates" lose the funding attribution and the accounting treatment, and the retailer cannot then answer the most important question in merchandise performance: how much of our margin erosion was promotional and how much was markdown?

GMROI is the best one-number test of whether a category earns its stock investment, because it penalises both low margin and slow turn [8]. It is also most often computed on the wrong inventory base — period-end rather than average — which flatters a category that has just been cleared.

Shrink has four causes with sharply different detection economics: external theft, internal theft, administrative and process error, and waste. The asymmetry to internalise is that external theft is most discussed and most expensive to reduce, while **administrative error is usually the largest recoverable share and costs almost nothing to find**, because it is already visible in the retailer's own data — negative on-hands, receipts without purchase orders, counts out by a whole case. The first shrink project should be a reconciliation project.

> **There is no published South African shrink figure.** South African retailers do not disclose shrink separately in results, and no local industry survey was sourceable. The only available benchmark is American: the United States National Retail Federation's security survey put shrink at 1.6% of sales in fiscal 2022, $112.1bn in total, split 36.15% external theft, 28.85% internal theft and 27.29% process error, across 177 brands and more than 97,000 locations [9]. **That is a US figure. Say so whenever you use it.** A South African retailer cannot benchmark its own shrink against anything, which is a genuine strategic disadvantage and the strongest available argument for building internal measurement properly.

Supplier income is the most argued-over number in a retailer's profit and loss, for four reasons that are all architecture problems. *Timing*: rebates earned on purchases can be recognised before the goods are sold, pulling profit forward, and the correct treatment defers the rebate to the stock still on hand — which means attributing it to specific stock, and most systems attribute it to nothing. *Classification*: cost-of-goods reduction or other income, which decides gross margin, inventory value and segment reporting. *Estimation*: tiered annual rebates accrue monthly on a judgement about year-end volume. And it is a *competition-law object* — the Commission found rebates for in-store functions generally unavailable to wholesalers and buyer groups, some with no efficiency justification, and back-end margins lacking transparency [10]. A retailer whose supplier income lives in a spreadsheet and a credit-note queue cannot pass an audit, cannot answer the regulator, and cannot tell you its true gross margin.

Run seven: the price words

Term	What it means
Price zone	A group of stores sharing a price position
Known value item (KVI)	A line whose price shoppers remember and compare
EDLP	Every day low price: stable low prices with little promotion
High-low	Higher base prices with frequent deep promotions
Effective dating	Recording the date a change applies from, independently of when it was captured
Internal selling-price inflation	The retailer's own weighted shelf-price movement on a fixed basket
Competitor price index	Prices relative to a competitor's basket. In a pricing meeting "CPI" may mean this, not consumer inflation
Multibuy / mix-and-match / BOGOF / threshold	Three for R50; the same across a set; buy one get one free; spend R500 get R50 off
Funded / unfunded promotion	One the supplier contributes to; one the retailer pays for out of its own margin
Baseline / uplift / incrementality	What would have sold anyway; sales above it; uplift net of cannibalisation, halo and pantry loading

EDLP against high-low is a strategy choice with architectural consequences, not a marketing preference. An every-day-low-price retailer runs few price changes and can distribute a price file weekly. A high-low retailer runs thousands a week, concentrated into a cycle, every one needing to land at every till and every shelf label before the promotion starts — which is what puts the price-file build on the overnight critical path. South African grocery is firmly high-low: promotions at 40% of one group's supermarket turnover [11].

Effective dating is a legal requirement here, not a convenience. A price record is not "the price"; it is "the price from this date, in this zone, on this channel, for this customer segment". A change entered today for next Monday must not affect today's selling. A retroactive change must be either refused or accompanied by an accounting adjustment. Two changes for the same article, zone and date are a conflict that must resolve deterministically. And "what was the price of this article at this store at 14:37 last Tuesday" must be answerable, because a customer complaint, a margin investigation and a regulator may each ask it.

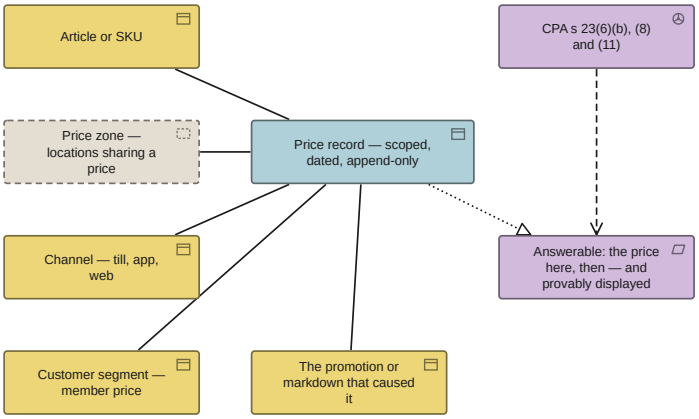


Figure 3.6 Why price is an entity and not an attribute. Four scoping axes, an append-only version history, and a statutory requirement to prove what was displayed where and when [12][13]. A column called “price” on the article record is wrong within a week.

That last point has a South African sharpness newcomers miss. Section 23(6)(b) of the Consumer Protection Act provides that where more than one price is concurrently displayed, the supplier may not charge more than the lower or lowest of them [12]. A fully obscuring second price becomes the

displayed price under section 23(8), and a displayed reduction placard makes the displayed price the shown price less the reduction under section 23(11) [12]. So the retailer must be able to **prove what was displayed, where, when, and what it did about an error** — and section 23(9)'s escape for an "inadvertent and obvious" error applies only once the display has been corrected *and* reasonable steps taken to inform affected consumers [13].

One board-level number is pure data architecture. **Internal selling-price inflation** is a weighted index of the retailer's own shelf prices on a fixed basket, reconstructed from price history, quoted to one decimal place in results announcements — one group reported 0.8%, a seven-year low, against official food inflation of 3.9%, with 11,500 products in deflation [11]; another reported internal *deflation* of 1.2% [14]. **A retailer that cannot reproduce its own price file as at an arbitrary past date cannot compute it.** Effective-dated, bi-temporal price data is a reporting requirement, not a design preference.

Run eight: the supply chain words

Term	What it means
Replenishment	Automatic reordering to maintain availability
Min-max	A reorder point and an order-up-to level per SKU per location
Cross-dock / flow-through	Receiving and despatching without put-away; pre-allocated stock passing through without storage
DSD	Direct store delivery: supplier to store, bypassing the distribution centre
Three-way match	Reconciling purchase order, receipt and invoice before payment
Fill rate	The proportion of ordered lines or units actually supplied
OTIF	On time, in full. The combined supplier delivery reliability measure
Cold chain / temperature excursion	The temperature-controlled path from supplier to shelf; a breach of its range
MAPE	Mean absolute percentage error. The standard forecast-accuracy measure
Bias	Persistent directional forecast error. Worse than random error

DSD is not a logistics footnote. A direct-store-delivery line never passes through the distribution centre, so the receipt happens at the store, the invoice match happens centrally, and the stock position and the order lifecycle have a different shape. Bread, soft drinks, dairy and newspapers are commonly DSD. Modelling direct-to-store deliveries as though they went through the distribution centre is one of the more damaging data-model errors in grocery.

OTIF and fill rate are not the same, and suppliers quote whichever flatters. Fill rate is *how much* of what was ordered arrived. OTIF requires it to arrive *and* on the agreed day. A supplier at a 98% fill rate and a 71% OTIF is costing the retailer availability while reporting a good number.

MAPE and bias are not the same either, and bias is the dangerous one. MAPE measures the size of the error; bias measures its direction. A forecast 15% out in both directions and averaging correct is manageable with safety stock. A forecast 6% out but always low starves the shelf every week, and the safety stock absorbs it until it does not.

Run nine: the channel words

Term	What it means
Omnichannel	Trading across store, online and app on shared stock and price
Click and collect / BOPIS	Buy online, pick up in store
Ship from store	Fulfilling an online order from store stock, for delivery
Dark store	A store-format facility used only for online fulfilment
Last mile / drop / drop density	The final delivery leg; one delivery; drops per route area
1P / 3P	First-party (platform-owned) and third-party (seller-owned) inventory
GMV	Gross merchandise value: total value transacted, not the platform's revenue
Take rate	Platform commission as a percentage of GMV
Identification rate	The share of sales linked to an identified customer
Instant savings / member price	A loyalty discount applied at the till in real time

Three of these are misused in ways that cost money.

BOPIS is a margin statistic, not a convenience statistic. Click and collect is structurally cheaper than delivery because the last mile is free: the customer provides the vehicle. The picking cost is identical; the delivery cost disappears. So one apparel group's figure of more than 55% of online orders collected in store [15] is telling you about unit economics, not preferences.

GMV against revenue, repeated from chapter 2 because it keeps happening: on a third-party sale the platform books commission, not the sale. Mixing the two in one dashboard produces a growth rate nobody can reconcile.

Ship from store makes the store three things at once — a selling location, a fulfilment node and the origin of a delivery — and its stock is partly available to walk-in customers, partly committed to online orders, partly being picked. That needs at minimum a committed/available split and arguably a virtual location per channel. The symptom when it is missing is familiar: an online order accepted against stock a walk-in customer has just bought.

Run ten: the pharmacy words

Term	What it means
NAPPI code	The South African identifier for medicinal, surgical and healthcare consumable products, used for claiming and reimbursement
Single Exit Price (SEP)	The only price at which a manufacturer may sell privately, and the price the final seller charges
SEP adjustment	The gazetted maximum annual increase to existing SEPs
Dispensing fee	The regulated flat-plus-percentage fee per prescription, across four SEP bands
Schedule / scheduling	Classification from Schedule 0 to 6, deciding who may sell, who may prescribe, whether a prescription is needed and what registers are mandatory
Adjudication	Real-time assessment of a scheme claim at the counter: eligibility, formulary, co-payment, levy, reject code

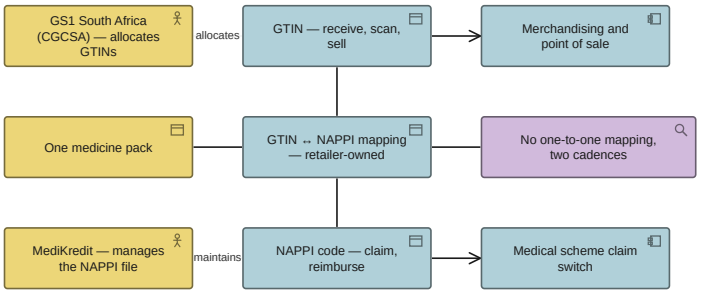


Figure 3.7 Two identifier authorities, two cadences, one product — and a mapping the retailer owns because nobody else maintains it [17]. The two schemes do not map one to one: pack-size and formulation variation resolves differently in each.

Three notes on this run, because it is the run most likely to be got wrong in a workshop.

Scheduling is a controlling attribute, not a classification. The schedule decides whether a line can be sold at a till at all, who may sell it, whether a register entry is mandatory, and whether a prescription and a repeat authorisation are required. It belongs on the article master as an attribute that *changes system behaviour*, not in a reporting hierarchy.

The dispensing fee is a formula, not a price. It is a flat component plus a percentage of the SEP, falling as the SEP rises, across four bands, with the parameters set annually by gazette [16]. Hard-code the bands and a gazette becomes a software release. Hold them as effective-dated reference data and it becomes a data load.

NAPPI and GTIN are two authorities, not two fields. The GTIN is how the product is received, scanned and sold; the NAPPI code is how it is claimed and reimbursed [17]. Different bodies, different cadences, different allocation rules, and no one-to-one mapping. The retailer owns the mapping and nobody else maintains it.

The words that mean different things in the same retailer

Five ordinary English words. Each one means something different in two parts of the same business, and in every case the ambiguity is an architecture problem rather than a language problem — because two systems agree on the word and disagree on the referent, and the integration passes testing.

Article

In the merchandising system, the sellable unit with a cost, a price and a stock position. In the apparel buying office, often the *style* — the thing being bought. In SAP, a term covering single, generic and structured articles, which are three different shapes. In the online catalogue, usually not used at all, because the catalogue says "product" and means the navigation grouping. Ask anybody who says "article" whether theirs has a barcode and whether its stock is counted on its own. That settles it in ten seconds.

Stock

Nine answers, set out in run five. The failure is that a dashboard labelled "stock on hand" can show any of them, and three dashboards in one retailer commonly show three. Make the position name part of the metric name; "stock" alone is not a metric.

Price

At least nine types, set out in run seven, each scoped by location or zone, by channel, by customer segment and by date. "What is the price of this" has no answer without four qualifiers — and the system that answers it fastest is usually the point of sale, which holds the one number that is legally binding and the one that is hardest to query historically.

Store

Five things, and they do not share a boundary.

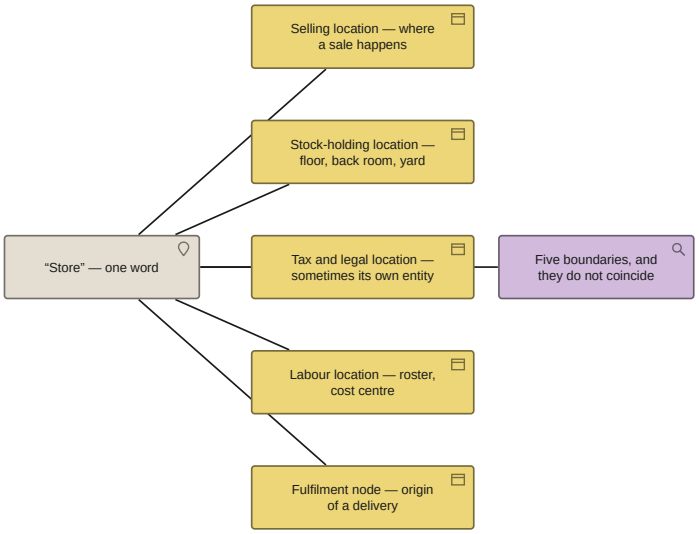


Figure 3.8 One word, five senses, five different boundaries. A concession inside a store, or a forecourt shop trading under a fuel retailer's licence, differs on every one of them — which is why Oracle's model lets a "store" be a physical location, a virtual one such as a website, or a customer such as a franchisee [18].

A **selling location** — where a transaction happens. A **stock-holding location** — which may include a back room, a yard and a channel-committed pool. A **tax and legal location** — a VAT jurisdiction and, for a franchise or member business, a separate legal entity. A **labour location** — a roster, a cost centre and a management span. And a **fulfilment node** — the origin of a delivery, with its own slot capacity. A concession inside a store, or a forecourt shop trading under a fuel retailer's licence, has a different boundary for each of the five. Oracle's model handles the hardest case by letting a "store" represent a physical selling location, a *virtual* selling location such as an e-commerce site, or a *customer* location such as a franchisee [18] — the right instinct, and also an admission that one word is doing too much work.

Customer

Six things at least. A **party** — a person or organisation the retailer knows about. A **customer** — a role: a party with at least one purchase, derived and channel-specific. A **loyalty member** — a party enrolled in a programme, with a number, a tier, a balance, a consent record and a terms version: a contractual relationship. An **account holder** — a party with a credit facility, a regulated relationship under the National Credit Act. A **patient** — in pharmacy, a regulated health-information relationship. And a **card or identifier** — the till token, of which there may be several per membership.

Collapsing party and loyalty member is the most expensive modelling decision in retail customer data. A shopper can transact without being identified, and in South African grocery the unidentified basket is a large minority, so the transaction cannot require a customer key. Meanwhile a member can enrol and never shop, one household holds one membership and three people, and one party has two memberships because she enrolled twice.

POPIA makes the distinction legal rather than stylistic. The loyalty member has given a *purpose-bounded* consent, and using loyalty data outside that purpose is further processing; the Information Regulator's guidance states that using loyalty data for marketing beyond the original purpose requires written consent, with sharing a member's data with group companies for marketing as its own example [19]. So purpose is an attribute of a consent, and consent is an entity with a dated scope. Consent as a boolean on a customer record is not a simplification. It is a compliance gap.

What to do with this chapter

Three things, in your first fortnight.

Run the four settling questions. Is its stock counted separately and can it be sold alone? Would the buyer make one buy decision for it? Could two different numbers both be correct at the same moment? If this number were wrong, who would notice and what would break? Those four resolve more modelling arguments than any amount of standards reading.

Write down which hierarchy answers which question, on one page, and get somebody senior to disagree with it in writing. The disagreement is the valuable part.

Make the position name part of the metric name, for stock, for price and for sales. "Stock", "price" and "turnover" are not metrics in a retailer. They are families of metrics that different parts of the business compute differently, correctly, and without telling each other.

THE CHAPTER IN TEN LINES

- The merchandise run is the most important vocabulary in the book, because every system in the estate has its own answer to what a SKU is — four concepts, four trade vocabularies, one estate.
- Grocery collapses style, option and SKU into one object and is right to. Apparel never can, which is why one merchandising system cannot serve both without one business working around it.
- Two settling questions resolve most arguments: is its stock counted separately and can it be sold alone, and would the buyer make one buy decision for it?
- Normalise every trade identifier to a fourteen-digit, zero-padded string at every boundary. The dropped leading zero and the wrong case quantity cause most retail integration defects.
- GS1's 2027 date is a readiness ambition, not a mandate. The durable consequences are the scanner estate, Digital Link parsing at the till, batch and expiry moving into the item master, and a scan that resolves to content the retailer does not control.
- A department is a bundle of accounting policy, tax default, buying authority, budget and statistical parameter — so reclassification either rewrites published history or destroys the comparison, and the retailer must choose one and keep choosing it.
- Six hierarchies disagree legitimately. Make item and location the only mandatory shared grains, version every hierarchy, master the mappings, and publish one page saying which hierarchy answers which question.
- "Stock on hand" has nine correct answers. The durable fact is the inventory movement; every position is a derivation, and a cached quantity is a cache, not a system of record.
- Gross margin divides by sales and markup divides by cost. Markdown is permanent and retailer-funded; a discount is temporary and often supplier-funded; a system that models both as a dated price record cannot explain its own margin.
- There is no published South African shrink figure. The NRF benchmark is a United States figure and must be labelled as one; an SA retailer cannot benchmark its own shrink at all.
- Price is an entity, not an attribute: typed, scoped by location, channel, segment and date, append-only, and legally binding at the shelf under CPA s 23(6)(b).
- Five ordinary words — article, stock, price, store and customer — mean different things in different parts of the same retailer, and in every case two systems agree on the word, disagree on the referent, and pass testing.

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CHAPTER 4

How the business model works

An architect who cannot draw a retailer's business model on one page cannot tell a system that is merely old from a system that is load-bearing. Two instruments fix that: the nine-block canvas, and the customer journey with the systems named and the failures marked.

The question you will be asked in week three

Somebody will put a system in front of you and ask whether it matters. It arrives as a budget question. Do we replace the merchandise system. Do we keep paying the licence on the space-planning tool. Do we really need electronic shelf labels in 420 stores.

You cannot answer any of those from a technical assessment. A system matters in proportion to the part of the business model it carries and the part of the shopper's experience it decides. Retail makes that harder than banking does, because no single system in a retailer corresponds to a bank's deposit core. There is a merchandise system of record holding the item, the location, the cost, the retail price and the quantity on hand, and a dozen specialist engines reading from it and writing back into it. The integrity question is always the same one: *what does this estate believe we own, where, at what value, as at when?* Most expensive retail IT failures are failures of that one answer.

This chapter gives you two instruments that fit on a page each. The first is the business model on nine blocks. The second is the customer journey, done properly, with the systems named and the failure points marked. Between them you can say, in a sentence a chief financial officer will accept: *this is the only place the authorised retail price is decided, it feeds the till, the label, the scale and the app, and the law says the lowest displayed price binds us — so it is not a licence renewal.*

Neither instrument is retail-specific. The nine-block canvas is Osterwalder and Pigneur's [1]; journey mapping comes from service design. What makes them worth doing here is that retail punishes the generic version of both. A canvas whose customer-segment block says "shoppers" and whose value proposition says "great products at great prices" is worse than no canvas, because somebody will believe it and design against it. A multi-format South African group has at least six segments that need different operating models, and two of them are not shoppers at all.

There is a second reason to do this early, specific to retail. In most industries the business model describes how value is created; in a retailer it describes a bet. The retailer buys goods weeks or months before demand is observed and discovers whether it was right only at the till. A grocer at a 24% gross margin and a 5% trading margin turns a rand of sales into five cents of trading profit, so a rand of stock cleared at a 30% markdown destroys the profit on six rands of sales, and there is no provision line for being wrong about demand. Every instrument in this chapter has to survive that fact, and the plain canvas does not — which is why the section on what the canvas leaves out matters more than the walk through the blocks.

A note on the worked example. **Milkwood Retail Group is invented.** It is a mid-tier South African multi-format retailer: roughly 420 supermarkets, a 180-store pharmacy and health chain, an apparel format of about 260 stores, 22 cash-and-carry branches serving independent traders, and an online business. Nothing in it is a case study of any real retailer. The multi-format shape is deliberate: grocery, pharmacy and apparel have genuinely different architectures below the point of sale. Milkwood's own figures are illustrative and internally consistent; the market figures around them are sourced.

The canvas for a multi-format retail group

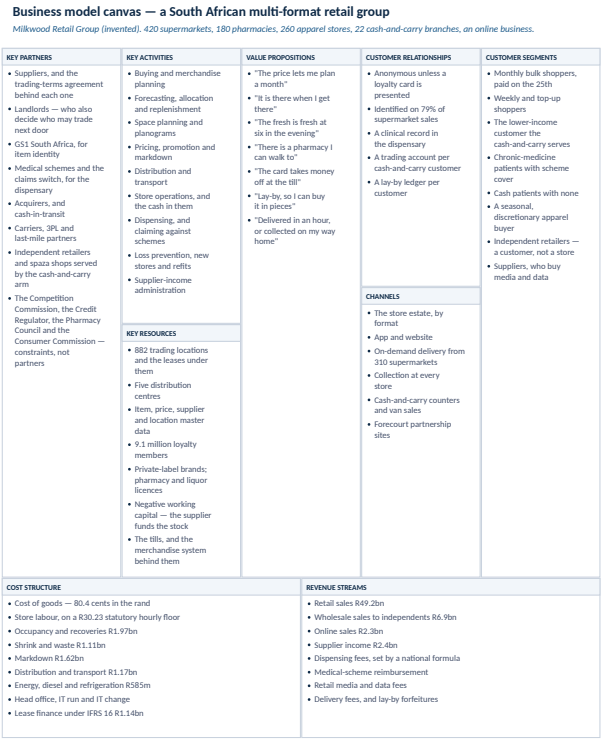


Figure 4.1 The nine-block business model canvas for Milkwood Retail Group, an invented South African multi-format retailer. Not a case study of any real group; the shape is typical, and the shape is the point. Milkwood's own figures are illustrative.

Walk the blocks. Each implies a capability, and most imply a system you will have to find and form a view about.

Customer segments

Milkwood serves monthly bulk shoppers, weekly and top-up shoppers, lower-income customers in townships and rural towns, chronic-medicine patients with scheme cover, cash patients with none, and a discretionary apparel buyer. Those are six businesses in one group. They differ in basket size by two orders of magnitude, in trip frequency, in what good service means and in whether price or range decides the trip.

Two segments outsiders miss. The first is **the independent retailer**. Milkwood's cash-and-carry arm sells to spaza shops, independent grocers and small caterers; South Africa's informal grocery trade was valued at about R184bn across roughly 150,000 spaza shops [2], and the Competition Commission found consumers generally spend less than 40% of their grocery budget there [3]. An independent retailer is a *customer*, not a store: no planogram, no loyalty card, no till of the group's, but a credit limit, a trading account, a delivery schedule and a route. The second is **the supplier**, who buys shelf space, promotional slots, leaflet pages, data and in-app advertising. Supplier income is a material revenue stream (Chapter 6), so the supplier appears in two blocks of the same canvas.

Architecture implication. Segments this far apart cannot share a cost to serve, and in retail they cannot share a merchandising stack either. The dispensary's item is identified by a national medicine code and priced by the state; the apparel item exists for one season and is planned on a size curve; the grocery item lives for years and is replenished daily. One merchandise system configured for all three becomes a grocery system with two sets of workarounds. That is the honest reason for the multi-stack estate you will walk into; acquisitions are the other, and the two get blamed for each other.

Value propositions

Write these as the shopper would say them afterwards. *The price lets me plan a month. It is there when I get there. The fresh is still fresh at six in the evening. There is a pharmacy I can walk to. The card takes money off at the till. Lay-by, so I can buy it in pieces.*

Two of those are measured publicly. One listed pharmacy group reports 53.6% of households within 5 km of one of its pharmacies [4]; one grocery loyalty programme reported R16.5bn of till savings to members in a year [5]. Both are capability statements with a system behind them.

Architecture implication. Availability is a promise made by a forecast and kept by a replenishment run, against a store stock figure that is wrong by the amount of the store's shrink. Freshness is kept by a production plan, a shelf-life rule and a markdown decision taken by a department manager at four in the afternoon. Price stability is kept by a price file that reaches every till, every shelf-edge label and every scale before the store opens. None of those three is kept by the system a vendor would sell you as the answer.

Channels

The store estate by format, the app, the website, on-demand delivery picked in stores, collection at every store, cash-and-carry counters, van sales, and forecourt partnership sites, which are a real South African channel in their own right [8]. The scale of store-fulfilled online in South Africa is the striking

part: one grocer serves on-demand delivery from 997 stores and reached R25.5bn, 11.1% of its South African supermarket sales, at 94.0% on-time delivery and 96.8% order fulfilment [6]; another serves from more than 620 [7].

Architecture implication. Channels decide latency and they decide where data is created. A till must price a basket with no network at all. A shelf label must update before the store opens or the retailer is obliged to sell at the old price. An online availability promise is made in a few hundred milliseconds from a stock figure refreshed overnight. A dispensary claim is adjudicated in real time with the patient at the counter. Four channels, four latency budgets, one item master underneath. The honest conclusion is one price authority, one promotion definition and one available-to-promise service behind many front ends. The estate you will find has a price in the merchandise system, a price in the promotion engine and a price in the point of sale, with no stated authority between them.

Customer relationships

The retail default is **anonymity**. Unless the shopper identifies themselves, the retailer has a basket with no customer attached, so the whole of customer analytics rests on one operational number: the identification rate. South African programmes publish it — one reports 83.7% of sales on a 12.9 million-member card [4], and the loyalty research finds 64% of economically active consumers prefer swiping a card to scanning an app [5]. At 2,700 card swipes a minute on a large programme [5], a one-second latency increase is an operational event, not a performance ticket.

Architecture implication. Identity in a retailer fragments four ways: the loyalty member, the online account, the credit or lay-by account, and the card token. Joining them is the work, and the dispensary adds a fifth — a clinical patient record that must not be casually merged with a marketing profile, because the lawful basis for processing it is different. An architect who treats customer identity as a single-system problem has not met the pharmacy.

Revenue streams

Milkwood's R58.4bn of turnover is not one revenue stream but several, with different recognition, timing and owners: retail sales of R49.2bn, wholesale sales to independents of R6.9bn, online sales of R2.3bn, **supplier income of R2.4bn**, regulated dispensing fees, medical-scheme reimbursement, retail media and data fees, delivery fees, and lay-by forfeitures.

Three of those behave unlike retail sales. **Supplier income** is money from suppliers that is not a reduction in the invoice price — rebates, listing fees, co-op advertising, space payments, markdown support. It is the most argued-over number in a retailer's P&L and it has an accounting-standards problem attached (Chapter 6). **The dispensing fee** is set by a national formula in four price bands and adjusted by gazette; the retailer cannot price it and cannot promote on it [9]. **Lay-by** is a goods-reservation contract: the customer pays in instalments and takes the goods on final payment, so the retailer holds both the cash and the stock. It is not credit and not a sale, it is regulated under the Consumer Protection Act, and it is routinely bolted onto a point of sale never designed for it. One listed group reports lay-by sales as a separate line [10].

Architecture implication. Each stream needs its own recognition path and its own reconciliation. Supplier income needs effective-dated trading-terms agreements, attribution to a supplier, a category, a SKU set and a period, and then an accrual, a claim, a credit note and a reconciliation of the difference. Scheme reimbursement needs a receivables sub-ledger. Lay-by needs its own ledger, its own stock reservation and its own cancellation rules. A retailer that books all of this through the sales interface will restate something.

Key resources

Eight hundred and eighty-two trading locations and the leases under them. Five distribution centres. The item, price, supplier and location master data. A loyalty database of 9.1 million members. Private-label brands. Pharmacy and liquor licences. Negative working capital. And the tills.

Two are easy to under-rate. **The lease book** is not property-team reference data; under IFRS 16 every store lease is a right-of-use asset and a lease liability, and for a large South African grocer it can run to several times the debt book — one discloses lease liabilities of R50.4bn against borrowings of R6.7bn [11]. Lease term, escalation, options, break dates, area and turnover clauses are accounting-critical master data. **Negative working capital** is the second: a retailer that pays suppliers in 75 days and turns stock in 57 is funded by its suppliers, which is how new stores get stocked without consuming cash. Payment terms are therefore per supplier, sometimes per purchase order, effective-dated, and interacting with settlement discounts and rebates — so a retailer whose terms live in a contract PDF cannot compute a true cost of goods.

Key activities

Buying and merchandise planning; forecasting, allocation and replenishment; space planning; pricing, promotion and markdown; distribution; store operations; dispensing and scheme claiming; loss prevention; new stores; and supplier-income administration.

Architecture implication. Notice what that list is: a sequence of decisions about one asset. Buying decides what stock to own and at what cost, planning how much in money and when, allocation and replenishment where it sits, space planning how much shelf it gets, pricing at what price it leaves. Markdown is the admission that the price was wrong and shrink is the stock that left without being sold. Every activity block in a retail canvas is a decision about inventory — which is the clue that the canvas has no block for the thing the whole model turns on.

Key partners

Suppliers and their trading terms. Landlords — who in South African shopping centres also decide, through exclusivity clauses, whether a competitor may trade next door; the Grocery Retail Market Inquiry made exclusive leases a remedy with deadlines running to November 2024 [3]. GS1 South Africa, for item identity. Medical schemes and the claims switch. Acquirers and cash-in-transit. Carriers. And the independent retailers served by the cash-and-carry arm, who are partner, customer and debtor at once.

And then the bodies that are not partners at all but constraints: the Competition Commission, the National Credit Regulator, the South African Pharmacy Council and the National Consumer Commission. Putting a regulator in the partner block is the commonest mistake on a retail canvas, and it makes the constraint read as negotiable when it is not.

Cost structure

Cost of goods at about 80 cents in the rand. Store labour, against a statutory floor that rose to R30.23 per ordinary hour from 1 March 2026 [12]. Occupancy and recoveries. Shrink and waste. Markdown. Distribution. Energy, diesel and refrigeration. Head office, IT run and IT change. Lease finance under IFRS 16.

Two South African lines deserve naming. **Energy**: during the load-shedding years retailers spent heavily on diesel to keep trading and the cold chain intact. The disclosed figures are **historic peak, not a run rate** — about R500m in a single half-year at one grocer and R346m over ten months at another, all in the 2022 to 2024 period [13][14]. Load-shedding was suspended for long stretches from 2024, but the design positions it forced are the right ones anyway. **Shrink** is the other: a cost line nobody can benchmark, because there is no published South African shrink figure at all (Chapter 6).

What the canvas leaves out

Five things, and each drives more retail architecture than anything printed on the canvas.

The inventory. There is no block for *the business is a bet on demand placed weeks or months before demand is observed*. That one sentence determines every merchandising system in the estate: forecasting, planning, allocation, replenishment, markdown and the stock ledger exist because of it. The canvas can show that buying is a key activity. It cannot show that the firm owns a perishable, stealable, fashion-dated, space-consuming asset, and that the discipline of retail is the discipline of that asset.

The shelf. Space is finite, contested and rented. The South African measure is trading density; the Clur Shopping Centre Index gives national annualised trading density of R43,612 per square metre, with rent at 6.6% of sales nationally [15]. Shelf space is arbitrated between suppliers by sales per linear metre and recorded in a planogram, and the planogram is the only object joining the merchandise hierarchy to the physical store. No canvas block holds it, and no system owns it properly either: product dimension data arrives wrong from suppliers, fixture data decays after every refit, and store compliance with the plan is mostly unmeasured.

The trading calendar. Price, cost, terms, promotions, planograms, assortments and leases are all effective-dated. Retailers report in 52- and 53-week years, and a 53rd week is needed roughly every five or six years on no fixed schedule [16]. Every like-for-like comparison, every seasonal index in the forecast, every period in the merchandise plan and every labour model is expressed in retail periods — and three major South African retailers have three different year ends. Add a change freeze across peak trading that compresses the delivery year into about nine months. **Effective dating is the architecture**, and the calendar is the dimension that makes it computable.

The supplier. The partner block invites a list of logos. What it cannot express is that the supplier is a counterparty in a continuous negotiation whose terms are structured data: cost price by effective date, case pack, lead time, minimum order quantity, settlement discount, rebate tier, promotional funding, markdown support, logistics allowance. The Commission found that national chains obtain more favourable terms than wholesalers and buyer groups even where those groups buy larger volumes, that some rebates have no efficiency justification, and that back-end margins lack transparency; its recommended remedy is uniform, objectively justified trade terms not based on volume [3]. That turns supplier terms from a procurement artefact into reportable structured data, which almost no retailer holds.

The regulator, and the clocks it sets. Four examples, each a hard date in a system. The Single Exit Price file changes on a gazetted date across every pharmacy at once — two adjustments in 2026, 1.47% in January and an extraordinary 2.88% effective 1 October on medicines listed as at 30 September [17]. Section 56 of the Consumer Protection Act gives a consumer six months from delivery to return defective goods, which is a clock per sale line. Section 23(6)(b) makes the lower of two displayed prices binding [18][19]. Good Pharmacy Practice requires the latest version of interaction-checking software, which puts a content vendor's release cadence on the retailer's compliance critical path [20]. None of those is a policy engine. They are statutory clocks, and the system either evidences them or does not.

> If your canvas has a regulator in the key-partners block, move it. A constraint drawn as a partner reads as negotiable, and the first programme decision taken against that drawing will be wrong.

Two more canvases, and the second is the interesting one

A pure-play online retailer

The same canvas for a pure-play online retailer

What changes, and the changes are structural rather than cosmetic.

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
• Third-party sellers — 60% of marketplace GMV at the SA benchmark • Payment gateways and card schemes • Couriers and pickup-point networks • Cloud	• Seller onboarding and catalogue ingestion • Listing quality and product matching • Order routing and fulfilment • Seller performance, payout and settlement • Dispute and liability allocation • Paid customer acquisition **KEY RESOURCES** • Fulfilment centres • The catalogue and its match quality • The seller base • The recommendation and search stack	• Range nobody can stock • "It arrives tomorrow" • Price transparency • A subscription that removes the minimum order	• Always identified — there is no anonymous order • Account-based, with a full behavioural history • Subscription as the retention mechanic **CHANNELS** • App and website only • Delivery and pickup points • Seller-fulfilled despatch	• Urban, banked, delivery-addressable shoppers • Sellers, who are the second customer • Brands buying on-platform media

COST STRUCTURE	REVENUE STREAMS
• Cost of goods on the first-party book only • Fulfilment and last mile — the line that replaces rent • Paid acquisition • Payment acceptance and fraud • Technology, as a build cost rather than a run cost	• Marketplace commission — a take rate, not a margin • First-party retail sales • Fulfilment-as-a-service fees • Delivery fees and subscription • On-platform advertising

Figure 4.2 The same canvas for a pure-play online retailer. The structural changes are the revenue model (commission, not margin), the activity set (sellers, not stores) and the absence of any offline-till requirement.

Four changes, all structural. **There is no store estate**, so no rent to sales and no trading density — and no free last mile either, which is the trade; the cost line that replaces rent is fulfilment and delivery. **Revenue is substantially commission rather than sales**: marketplace is about 60% of the South African benchmark platform's gross merchandise value across some 15,000 active sellers, so GMV and revenue diverge by roughly a factor of two and the P&L is a take-rate business with a thin first-party book attached [6][21]. **The key activities invert** — seller onboarding, catalogue ingestion and matching, listing quality, order routing, performance management, payout and settlement, and fulfilment as a service. **Margins are structurally thin**: the core South African e-commerce unit runs at about 1%, on a first full-year profit before tax of R181m [6][21].

The architectural simplification is worth naming, because it is the largest one in this book: **a pure-play has no offline-capable till requirement.** No store means no store server, no local price file, no floor limits, no store-and-forward journal, no cash handling, no scale, no shelf label and no load prioritisation when the power goes. An architect moving between a pure-play and a store retailer will mis-estimate the work by an order of magnitude, in whichever direction they are travelling.

A voluntary-trading or franchise group

The same canvas for a voluntary-trading or franchise group

The interesting one. The shopper is not the customer, and the group masters neither the price, the assortment nor the stock.

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
• The member or franchisee — partner, customer and debtor at once • A members' guild or franchise council, which must consent • Suppliers, negotiated centrally for terms the member may decline • Landlords, often of the member rather than the group	• Wholesale buying and distribution • Recommending a range, a price and a promotion • Brand, advertising and the leaflet • Recruiting, financing and retaining members • Measuring how much of the member's purchasing is captured **KEY RESOURCES** • The distribution centres • The banner and the brand • Private label • The membership agreement • A receivable from the members	• To the member: buying power, brand, logistics, systems if wanted • To the shopper: whatever the member decides it is	• Contractual with the member, not the shopper • Measured as loyalty — the captured share of member purchasing • No group-wide shopper identity unless members share data **CHANNELS** • The members' stores, which the group does not own • Distribution centres and dropshipment • A group app the member may or may not support	• Members and prospective members • The shopper, reached only through the member

COST STRUCTURE	REVENUE STREAMS
• Cost of goods bought for resale to members • Distribution — the dominant own cost • Member credit losses and guarantees • Brand and advertising • Group systems offered, not imposed	• Wholesale supply margin — in the low teens, not the mid-twenties • Franchise royalty where the model is franchise, not voluntary • Marketing levy • Rebates earned on central buying

Figure 4.3 The same canvas for a voluntary-trading or franchise group. The customer is the retailer, the group earns a wholesale margin rather than a retail one, and the group masters neither the price, the assortment nor the stock for much of its turnover.

This is the one that changes the meaning of everything downstream, and most reference material ignores it because the model is rare outside Europe and southern Africa.

In a **voluntary-trading group** the stores are owned by independent retailers who hold membership of the group and of a members' guild, and who retain legal ownership of their businesses. There is no franchise fee and no royalty on turnover: the group earns its margin on the wholesale supply of goods through its distribution centres, and members keep pricing discretion, product-mix flexibility and the right to source locally [22]. South Africa's largest example of the model reports **retailer loyalty of 78.6%** as a headline operating metric [23] — the share of its members' purchasing it captures. A business whose central metric is how much of its customers' buying it managed to win is not a chain.

In a **franchise group** the franchisee owns the business and the group owns the brand, the supply chain and usually the systems, charging a royalty. One South African grocer runs 620 franchise outlets alongside 992 company-owned stores, with franchisees sourcing inventory through the group's supply chain and funding their own capital, staffing and operating costs [7]. In the same year franchise like-for-like growth ran at 0.9% against company-owned at 3.9% [24] — a 300 basis-point divergence inside one fascia.

Here is the sentence that matters, and it belongs in week one:

> In a voluntary-trading or heavily franchised estate, **the group does not have a price, an assortment or an inventory position to master** for much of its turnover. It has a wholesale order book and whatever the member chooses to share.

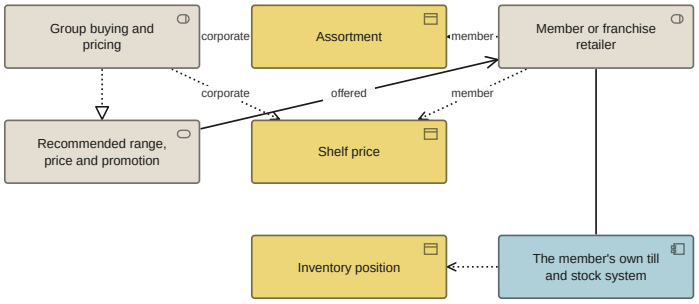


Figure 4.4 Where authority over the assortment, the shelf price and the inventory position actually sits. In a corporate store the left-hand role holds all three. In a voluntary-trading store the group's only relationship is a service it offers, and the member decides.

Read the figure as three overlays on one diagram. In a corporate store, group buying and pricing holds the assortment and the shelf price, the group's merchandise system holds the inventory position, and every system is the group's system of record. In a franchise store the group usually mandates price and range by contract, the stock is *sold* to the franchisee rather than transferred, and visibility of the shelf depends on what the agreement compels. In a voluntary-trading store, group buying can recommend, list, promote and fund — but the member decides, and the group's relationship to the assortment and the price is a *service it offers* rather than a control it exercises.

Everything downstream inherits that. There is no group item master, because the retailer owns the range decision, so any "single source of truth for assortment" is advisory data at the member boundary. There is no group price file, so national price compliance is a reporting and persuasion problem rather than a configuration push. There is no group stock ledger: the group knows what it shipped, not what the member has on hand, unless the member shares point-of-sale and inventory data — by agreement, in heterogeneous formats, with gaps. The till estate is not mandated, so the group integrates many rather than rolling out one. Credit risk points the wrong way, because the group's receivable is from its members. Governance is a two-party negotiation with a guild that must consent. And store ownership can change without the group's consent, so store master data carries an ownership dimension with its own lifecycle [22][23].

	Corporate group	Pure-play online	Voluntary-trading or franchise group
Who the customer is	The shopper	The shopper, and the seller	**The retailer**, and the shopper only through them
Who owns the store	The group	No store	The member or franchisee
Who sets the shelf price	The group	The platform	**The member** (voluntary) or the group by contract (franchise)
Who owns the assortment	The group	The seller, within policy	**The member**
Who holds the stock position	The group, per store	The platform and the seller	The group knows what it shipped, and no more
Revenue recognition	Retail sale	Commission plus a thin 1P book	Wholesale sale, plus royalty in franchise
The group's control mechanism	Configuration	Policy and platform rules	**Contract, persuasion and funding**
What you must model extra	Nothing	Seller, listing, payout	Member entity, membership, consent, credit, data rights

The right-hand column is the one that breaks reference architectures. Replenishment, range, pricing and loyalty all become *offered services* rather than central controls, and the integration is business-to-business rather than intra-enterprise. Such a group also has three revenue layers that are routinely conflated — group turnover, system-wide sales and member retail sales.

Four customer journeys, with the breaks marked

A customer journey map earns its place only if it names the systems and marks the failures. Four cover most of what a South African multi-format group does. Read the right-hand column of each figure first.

A weekly grocery shop with a loyalty card

Eleven steps. Nine of them touch the price file.

STEP	WHO AND WHAT SYSTEM	WHERE IT BREAKS
1 **Trip decision**	The shopper, and nothing of the retailer's. Payday governs it: more than 80% more transactions on the 1st than the 22nd	▲ The forecast and the roster are built on a weekly average, so the 1st is under-staffed and the 22nd is over-staffed
2 **The shelf price is read**	Shelf-edge label or electronic shelf label, bound to an article and driven from the price file	▲ The label did not update. Where two prices are displayed, the lower binds — so a silent label failure is a legal exposure, not a data-quality issue
3 **Basket assembled**	The shopper, against the planogram and the promotional signage	▲ The planned item is not there. The shopper substitutes the store, not the product, so the loss is the trip and not the line
4 **Loyalty card presented**	Point of sale plus the loyalty service; a card beats an app for most economically active consumers	Not presented: the basket is anonymous, the instant saving is lost, and so is the only record that this shopper exists
5 **Each item scanned, price resolved**	Point of sale, against a price file held locally on the till	▲ The till and the shelf disagree. The till wins operationally; the label wins legally; the shopper notices either way
6 **Promotions evaluated**	Promotion engine inside the till, with member status as an input	▲ A multibuy, a member price and a spend threshold on one basket can combine in several orders with different totals, and the precedence rules are rarely written down or consistent across channels
7 **Weighed items priced**	The scale, with its own price file and its own distribution path	A scale mis-key is a margin loss with no audit trail, and the scale's price file is a second distribution channel nobody tests
8 **Age-restricted items challenged**	Till prompt, cashier judgement, and the store's own licence conditions	Liquor trading days and hours are per premises, per day, per hour — and are usually hard-coded rather than held as licence data
9 **Tender**	Cash, at 56% of national payment volume and 21% of value; card; wallet	▲ Line down. Cash is trivial; a card authorisation is a round trip. Floor limits and stand-in authorisation are a commercial risk decision disguised as a technical setting
10 **Accrual and savings printed**	Loyalty service, deferred to a queue if the till is offline	Offline accrual is accepted on risk; offline redemption is declined, because the balance is unknown — which the shopper experiences as the card not working
11 **Posting, and tonight's order**	Till journal → sales polling → stock posting → forecast → replenishment → order to the DC and the supplier	▲ A store that does not close its trading day blocks every downstream job for that store, and the gap has to be filled tomorrow without double-counting

Figure 4.5 A weekly grocery shop with a loyalty card. The triangles mark the breaks that cost the retailer a basket, a trip or a statutory exposure.

Four failures deserve expansion.

Price file divergence. The shelf edge, the till, the scale and the app can all disagree, and each has its own distribution path. The commercial consequence is margin; the legal consequence is sharper. Section 23(6)(b) of the Consumer Protection Act provides that where more than one price is displayed at the same time, the supplier must not charge more than the lowest of them [18][19]. The escape is narrow: section 23(9) relieves a supplier of a displayed price containing an inadvertent and obvious error only once the supplier has both corrected the display and taken reasonable steps to inform affected consumers, and the Consumer Goods and Services Ombud's guidance on what reasonable steps look like includes withdrawing the product, fixing the label and keeping dated photographic proof [18]. So a shelf label that silently fails to update is not a data-quality defect. It is a statutory obligation to sell at the lower number, and the evidence that you corrected it is itself a data requirement.

Promotion precedence. A basket can simultaneously qualify for a multibuy, a member-only price and a spend threshold. Those combine in several orders with different totals, and the precedence rules are rarely documented and frequently differ between the till, the app and the online basket. Promotions ran at 40% of one major grocer's South African supermarket turnover in a recent half-year, up from 37.9% [25]. At that share the promotion engine is not a marketing tool; it is a primary pricing system in the sale path, evaluated at the till in milliseconds with loyalty member status as an input.

Offline. Cash is 56% of South African payment volume and 21% of value, at an average cash payment of R208.44 [26], and more than 80% more transactions happen on the 1st of the month than on the 22nd, peaking on the 25th [27]. Together those mean a South African store must keep selling with no network on the busiest day of the month. Offline capability is four separable things — local item lookup and pricing, local transaction logging, offline tender acceptance, and offline loyalty — and most estates have the first two. Test the third specifically rather than accepting "we have offline mode", because the first two without the third means a card basket cannot be tendered.



A prescription dispensed at a pharmacy counter

The cleanest role separation in retail, and the only price the retailer does not set.

STEP	WHO AND WHAT SYSTEM	WHERE IT BREAKS
1 Prescription or repeat presented	Dispensary counter; the dispensing system, which is clinical software	A repeat with no authorisations left, discovered with the patient standing there
2 Patient identified, clinical record retrieved	The dispensing system's own patient record	▲ The dispensary's patient and the loyalty member are two records for one person, and nothing joins them
3 Prescriber validated	A prescriber register inside the dispensing system	Maintained by hand, so it goes stale and the validation becomes a formality
4 Item matched to the national medicine database	The medicine identity and status feed, which SAHPRA maintains	▲ The item master carries two identifier schemes for one product — GTIN to receive, scan and sell; NAPPI to claim and reimburse — on different cadences, not mapping one to one, with the retailer owning a mapping nobody else maintains
5 Generic substitution offered	Pharmacist's professional judgement, with the scheme formulary alongside	The system may propose a substitution. It may not decide one
6 Schedule checked, professional authorisation applied	Role-based authorisation in the dispensing system, under the Pharmacy Act	▲ Authorisation bypassed under counter pressure. The dispensing act is personal and statutory; a system that lets an assistant complete it is a compliance failure, not a usability improvement
7 Interaction and allergy screening	Licensed clinical content, which Good Pharmacy Practice requires to be the latest version available	▲ That one rule turns a content subscription into a regulated currency obligation on the compliance critical path, and an offline store is running last month's version
8 Price computed	Single Exit Price from the national file, plus a dispensing fee set by a four-band formula	▲ A stale price file is a regulatory breach and an unrecoverable margin error. There were two adjustments in 2026 — 1.47% in January and an extraordinary 2.88% from 1 October
9 Scheme claim submitted and adjudicated	A real-time claim to the scheme through a claims switch	▲ A reject at the counter carrying a code the pharmacy cannot resolve. The fallback is the patient paying cash and claiming themselves
10 Patient pays the balance	The point of sale, under a second price regime in the same shop	Front shop and dispensary are two margin models, two price authorities and two tax treatments on one receipt
11 Label printed, checked, counselled, dispensed	The pharmacist, personally	Cold-chain items handed over without a maintained chain, and no record that they were
12 Registers, stock, receivable	Scheduled-substance register; stock decrement; the scheme receivable	▲ Weeks later the paid amount differs from the claimed amount and nobody can reconstruct the adjudication that produced it

Figure 4.6 A prescription dispensed at a pharmacy counter. The price is not the retailer's, the authorisation is personal and statutory, and the same physical product carries two identifier schemes.

The pharmacy is the format where an architect most reliably gets it wrong, because the dispensary is not a department. It is a separately licensed, separately regulated, separately priced business that happens to be inside a shop. At 16 October 2026 the South African Pharmacy Council register showed 5,851 registered pharmacies, of which 4,367 were community pharmacies [28].

The price is not the retailer's. The Single Exit Price is the only price at which a manufacturer may sell a medicine to anyone other than the state, and unlike a factory-gate price it is the price the final seller charges the patient or the scheme — every buyer pays the same SEP regardless of volume. Wholesalers add no mark-up; the manufacturer pays a logistics fee out of the exit price. The pharmacy earns a regulated dispensing fee: a flat component plus a percentage of the SEP across four bands, the percentage falling as the SEP rises. The Act also prohibits bonus, rebate and incentive schemes on medicines [9]. So the dispensary has no price lever, no promotional lever and no rebate lever, and its gross margin is capped by a formula rather than negotiated. The front shop is ordinary retail at retail margins, which is the whole strategic logic of pharmacy retail: the dispensary brings the footfall and the front shop brings the margin.

Two identifier schemes for one product, and this is almost never modelled. A South African retail pharmacy's item master carries a **GTIN**, which is how the product is received, scanned and sold, and a **NAPPI code**, which is how it is claimed and reimbursed. NAPPI identifies pharmaceutical, medicinal, surgical and healthcare-consumable products for electronic reimbursement and has been the South African standard since the early 1990s; the trademark and the file are MediKredit's, the public file is free, and allocation of a code is not an endorsement [29]. The two schemes are maintained by different bodies on different cadences under different allocation rules, and they do not map one to one — pack-size and formulation variation resolves differently in each. **The GTIN-to-NAPPI mapping is a master-data asset with a compliance consequence, the retailer owns it, and nobody else will maintain it.**

Professional authorisation is a control the system may not override. A responsible pharmacist is statutorily accountable for the pharmacy's compliance with the Pharmacy Act, the Good Pharmacy Practice rules and the premises licence, and that accountability is personal and cannot be delegated to a manager or a system. The dispensing act cannot be performed by a till operator, and a pharmacist's assistant works within a scope that excludes acts reserved to a pharmacist. Role-based authorisation in a dispensing system is therefore a statutory control, not an IT preference [20][28]. And Good Pharmacy Practice rule 1.2.11.4(f) requires that where software is relied on for warnings or interaction checks, it must be the latest version available [20]. That one clause converts a content subscription into a regulated currency obligation: the drug-database vendor's release cadence sits on the retailer's compliance critical path, and a store that has not taken a content update in weeks is non-compliant on the plain words of the rule.

The claim is adjudicated with the patient standing there. Eligibility, formulary, co-payment and levy are resolved in real time, and a reject the pharmacy cannot resolve leaves the patient paying cash and claiming themselves. Weeks later the paid amount differs from the claimed amount, and if the adjudication cannot be reconstructed the receivable cannot be worked. That is a claims system living inside a point of sale, with its own receivables sub-ledger — which is why the dispensing system should be integrated to the point of sale rather than merged with it.

> **NORTH AMERICA DIVERGES MOST SHARPLY HERE**
>
> South Africa has one national Single Exit Price and one national four-band dispensing-fee formula, set by a Pricing Committee and adjusted by gazette [9]. Canada has no national exit price: dispensing fees and ingredient mark-ups are set province by province for public drug plans, pharmacies charge "usual and customary" fees to private plans, and billable professional services — vaccination, medication review, prescribing — were worth about $634m nationally in 2024 [30]. A Canadian pharmacy system needs per-province fee and mark-up engines, per-plan adjudication, multiple formularies, preferred pharmacy networks and a professional-services billing capability. A South African one needs a national price-file ingestion and a formula. Do not port one to the other.

An online order picked in a store and collected

An online order picked in a store and collected

The store is the warehouse, the picker is the shelf-filler, and the stock figure is a belief.

STEP	WHO AND WHAT SYSTEM	WHERE IT BREAKS
1 **Customer browses**	App or site, against an available-to-promise view derived from store on-hand	▲ That on-hand is book stock, which is wrong by the amount of the store's shrink — and nobody has told the website by how much
2 **Basket built**	E-commerce platform and catalogue	Weight-variable items are ordered as a quantity and will be picked as a weight
3 **Collection slot chosen**	Slot capacity per store, per time window	▲ Slots are sold against a notional capacity rather than against picker hours, so the app promises labour the store does not have
4 **Payment authorised, not captured**	Payment service, holding an authorisation	An authorisation has a life. A slot two days out can outlive it
5 **Order routed to the store**	Order management, selecting a fulfilling location	The store manager's roster was built yesterday on a forecast that did not include this order
6 **Pick list generated**	Picking application on a handheld, sequenced in pick-path order	The pick path is derived from a planogram the store has not implemented, so the sequence sends the picker back and forth
7 **Picker walks the store**	A store-based picker, working alongside trade	▲ Phantom inventory — the system has it, the shelf does not. This is the dominant failure mode of store-picked online, and it is a shrink-architecture problem wearing a digital costume
8 **Substitutions resolved**	Picking app, with a message to the customer and a reply expected	▲ The customer declines. The basket shrinks and the fixed pick and handover cost is now spread over less revenue
9 **Weight-variable items captured**	A scale in the pick area, repricing the line	▲ The picked 1.3 kg chicken is not the ordered 1.2 kg, so the order total moves after the card was authorised
10 **Packed, segregated, staged**	Chilled and frozen separated; a staging area at the collection point	▲ A chilled order waiting outside temperature discipline is a food-safety event, not an inconvenience, and the clock on it is nobody's task
11 **Customer arrives and is identified**	Handover at the collection point, with age-restricted items verified there	Verification has moved from a till with a prompt to a handover desk that may have neither the prompt nor the authority
12 **Final capture and posting**	Payment capture at the adjusted amount; inventory and sales posted; the sale allocated to a channel	▲ The captured amount differs from the authorised one. And then the real argument: whose sale, whose margin, whose like-for-like, and whose shrink when the pick failed

Figure 4.7 An online order picked in a store and collected. The dominant failure is phantom inventory, and the closing argument is always about whose sale it was.

Phantom inventory is the dominant failure mode, and it is a shrink-architecture problem wearing a digital costume. The website promises from the store's book stock, which differs from the shelf by the store's unmeasured shrink. That is why a grocer publishes order fulfilment — 96.8% at one South African operation [6] — as a headline metric alongside on-time delivery. The fix is not better pickers. It is an available-to-promise view that discounts store on-hand by a measured confidence factor per category, which first requires the retailer to measure the discrepancy per category.

Slot capacity is a labour constraint pretending to be a configuration setting. Slots are sold against a notional number per store per window rather than against picker hours rostered. The picker and the shelf-filler are the same payroll, so an online order at five o'clock on the 1st competes directly with store trade.

Payment is authorised at one amount and captured at another. Substitutions, unpicked lines and weight-variable items all move the total after authorisation; the picked 1.3 kg chicken is not the ordered 1.2 kg. The design needs authorisation and capture with an amount tolerance, a rule for what happens outside it, and a reconciliation path — and most retail finance systems were never built for a sale whose value changes between promise and settlement.

And then the accounting argument. Whose sale, whose margin, whose like-for-like and whose shrink when the pick failed? The only safe answer is a **channel dimension on every sale and every stock movement from the outset**, because retrofitting one is a multi-year data project.

A garment bought online and returned in store

A garment bought online and returned in store

Where the Consumer Protection Act gives the election to the consumer, and the systems give it to nobody.

STEP	WHO AND WHAT SYSTEM	WHERE IT BREAKS
1 **Ordered online, possibly in two sizes**	E-commerce platform. Ordering two sizes to keep one is deliberate and common	The second size was never going to be kept, and the forecast counted both as demand
2 **Shipped from a DC or from a store**	Order management selects the source; a carrier moves it	A store-fulfilled online sale leaves the store's stock and puts the store's like-for-like in dispute
3 **Delivered**	Carrier, with proof of delivery	▲ The delivery date starts the six-month statutory clock, and the only system that knows that date belongs to the carrier
4 **Customer decides to return**	The app, or they walk into a store without telling anyone	A change of mind and a defect are different legal animals, and the app offers one button for both
5 **The store looks for the original sale**	The point of sale, searching for an order it did not create	▲ Separate order and point-of-sale systems with no shared sale-line identity. This is the commonest defect in cross-channel retail: without it the refund is a goodwill decision and the stock movement is untraceable
6 **Eligibility confirmed**	Retailer return policy — or section 56 of the Consumer Protection Act if the goods are defective	▲ Section 56 gives the consumer six months from delivery, with the consumer electing repair, replacement or refund, at the supplier's risk and expense. Treating a statutory return as a policy return either over-refunds or breaches the Act
7 **Return recorded with a reason code**	Returns function in the point of sale	Reason codes chosen by whoever wants the queue to move, so the reason data is noise
8 **Refunded to the original tender**	Tender routing in the till	▲ The online payment was a card token or a wallet the till cannot refund to, so the customer gets a different card or a voucher — a dispute and a refund-fraud vector in one
9 **Loyalty accrual reversed**	Loyalty service, on an event from the return	Missed, leaving the member holding points for a purchase they did not keep
10 **Graded and dispositioned**	Resaleable, repack, mark down, or write off	▲ Not recorded, so a garment that is terminal stock re-enters inventory at full retail and distorts both the stock value and the sell-through
11 **Restocked, or returned to the DC**	Store inventory, or reverse logistics and return to vendor	▲ Received and never restocked is shrink, and because it was never on a count it surfaces months later with no cause attached
12 **Margin restated, sell-through recomputed**	Finance and merchandise planning	▲ An online sale fulfilled from a DC and returned to a franchise store creates an inter-entity settlement nobody designed; on a marketplace item the liability is the seller's and the platform has to allocate it

Figure 4.8 A garment bought online and returned in a store. Section 56 of the Consumer Protection Act gives the election to the consumer, which makes this a statutory clock per sale line rather than a policy engine.

Returns differ by channel and category by an order of magnitude: grocery returns are negligible, store apparel returns modest, and online apparel returns the dominant cost, because the customer cannot try the garment on and frequently orders two sizes deliberately. No reliable South African return-rate figure could be sourced, and the global figures in circulation come from low-quality aggregators, so this book prints none. The cost of a return is never the refund, and Chapter 6 does that arithmetic.

And then the South African divergence, which is the whole point of this one. Under the Consumer Protection Act 68 of 2008, section 55 sets implied quality standards for goods and section 56(1) makes producers, importers, distributors **and retailers** each implicitly warrant that goods meet them — so the retailer is liable irrespective of who made the product. Section 56(2) gives the consumer **six months from delivery** to return goods that fail those standards, "without penalty and at the supplier's risk and expense", and the remedy is repair, replacement or refund **at the consumer's direction**. The National Consumer Commission's position is that the consumer elects: a supplier may offer a repair after the consumer has elected a refund, but the consumer is not bound by it, and the Tribunal has held that a consumer with a defective vehicle could cancel and claim a refund within six months without first allowing a repair [31].

Read that as a systems requirement rather than as law. It is **a statutory clock per sale line**, running from the delivery date, independent of whatever return policy the retailer publishes. The system must know, for any item brought back: what was paid, what tender was used, when it was delivered, whether it is defective or a change of mind, and which remedy the consumer has elected. A retailer that conflates the statutory return with the policy return will either over-refund systematically or breach the Act, and will not know which. In North America the equivalent problem is a configurable policy engine plus state or provincial consumer-protection law. **The South African system needs a clock; the North American system needs a policy engine.** They are not the same component.

Two more failures here recur everywhere. **The store cannot find the online sale** — separate order and point-of-sale systems with no shared sale-line identity, which makes the refund a goodwill decision and the stock movement untraceable. And **refund to the original tender** — the online payment was a card token or a wallet the till cannot refund to, so the customer receives a different card refund or a voucher, which is both a dispute and the largest internal-fraud vector at a till.

The moments that decide the relationship

The moments that decide whether the shopper comes back

Where a view is formed, and the system that forms it. The shaded rows are the ones nobody forgives.

THE MOMENT	WHAT THE SHOPPER CONCLUDES	THE SYSTEM THAT DECIDES IT
The item is on the shelf, or is not	Either this shop works, or I will drive the 400 metres to the other one	Forecasting and replenishment, against a store stock position that is wrong by the amount of the store's shrink
The shelf price and the till price agree	They are straight with me, or they are not	The price file and its distribution to the till, the label, the scale and the app — four paths, one number
The queue at half past five on the 1st	I will come earlier next month, or I will go elsewhere	Workforce management, scheduled against a forecast of transactions by hour
The card goes through	Nothing — until it does not	Payment acceptance, floor limits and stand-in authorisation
The scheme pays, or the patient does	This pharmacy understood my cover	Real-time claim adjudication at the counter, and the reject codes behind it
The substitution the picker chose	They knew what I meant, or they sent me tinned peas	Picking tooling and the substitution rule set, with the customer in the loop
The refund, in a store, for a thing bought online	This is one shop, or it is two shops wearing one name	A sale-line identity that survives crossing a channel
The saving printed at the foot of the receipt	The card is worth carrying	The loyalty service and the promotion engine, resolved at the till in milliseconds
The fresh counter at six in the evening	They run out, or they throw it away	Production planning, waste markdown and the department manager's discretion

Figure 4.9 The moments at which a shopper forms a view, and the system that forms it. Three of these decide whether the trip is repeated, and two of the three are decided by the price file.

Use that table in a budget conversation. Asked whether the price-file distribution mechanism matters, the answer is not "it is old". It is that the mechanism decides two of the three moments a shopper does not forgive, and one of them carries a statutory consequence. Five failure classes recur across all four customer journeys; learn to spot them and you will diagnose a retail estate faster than the people who work in it.

Failure class	What it looks like	Where it shows up
One sale, two identities	The sale line has no identity that survives crossing a channel, so a return, a refund or a loyalty reversal becomes a goodwill decision	Return in store for an online purchase; online order picked in a store; marketplace items
The stock position is a belief, not a fact	Phantom inventory, negative stock, adjustments without reason codes, shrink discovered at a count months later	Online pick failures; false availability promises; a replenishment run ordering nothing because the system thinks the shelf is full
Price exists in several places with no single authority	The merchandise system, the promotion engine, the point of sale, the label and the scale each hold a number, with no stated authority	Shelf and till disagree; the advert runs before the price lands; promotion precedence differs by channel
Statutory clocks nobody models	A legal deadline implemented as a batch job, a policy or a person's diary	Section 56's six months from delivery; the gazetted SEP date; liquor trading hours; interaction-content currency
Fixed cost per order against a variable basket	Pick, pack and drop costs are largely fixed per order while the basket is not, so small orders destroy margin invisibly	Every omnichannel economics argument you will be asked to settle (Chapter 6)

Capability first, or system first

You now have two ways to answer the week-three question, and they are not equivalent.

System first starts from the application portfolio and produces a list of systems with ages, costs, vendors and support statuses. That is useful for licence and lifecycle decisions, but it can only see what exists. A system-first assessment of Milkwood's online operation would record an e-commerce platform,

an order management system, a picking application and a payment service, note that all four are current and supported, and conclude the estate is healthy — while the business loses margin on every order because nothing measures how wrong the store's stock figure is.

Capability first starts from the value stream, asks what the business must be able to do, and only then asks what realises it. The missing box is the whole value of the method.

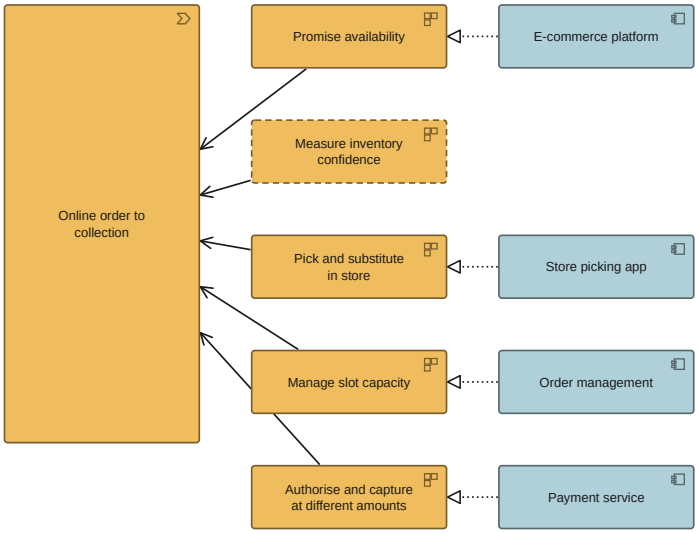


Figure 4.10 Capability first, drawn properly. Four capabilities have an application behind them. The dashed one — measuring how far the book stock departs from the shelf — has nothing, and no vendor sells it. That is the finding a portfolio assessment cannot produce.

The dashed box is the finding. *Measure inventory confidence* — knowing, per category and per store, how far the book stock departs from the shelf, and discounting the online promise accordingly — is a capability the value stream needs and nothing realises. It will never appear in a portfolio assessment, because there is no system to assess. It will never appear in a vendor evaluation, because no vendor sells it as a product. It appears only when you draw the value stream, list the capabilities it needs, and find one with nothing under it.

That is the practical difference. A system-first review tells you what you have; a capability-first review tells you what you are missing, and in retail what you are missing is usually a measurement rather than a system. The three most valuable gaps in a typical retail estate are all measurements: inventory confidence, promotional incrementality net of cannibalisation and pantry loading, and the true cost to serve by channel and basket size. None has a vendor. All three decide whether the business is making money.

So do both, in this order. Draw the canvas and argue about it with the trading director. Draw the four customer journeys and argue about the right-hand column with store operations and the pharmacy. Derive the capabilities from the value streams those journeys sit in. *Then* open the portfolio, map each capability to whatever realises it, and mark the empty ones. The empty ones are your programme; the full ones are a licence conversation, and you can now tell the difference in a sentence.

THE CHAPTER IN TEN LINES

- A system matters in proportion to the part of the business model it carries and the part of the shopper's experience it decides, not in proportion to its age.
- There is no deposit-core equivalent in retail: there is a merchandise system of record and a dozen engines around it, and the integrity question is always what we own, where, at what value, as at when.
- The canvas has no block for the inventory, the shelf, the trading calendar, the supplier's terms or the regulator's clocks — and those five drive more retail architecture than anything on it.
- A pure-play online retailer earns a take rate rather than a margin, and its largest architectural simplification is that it has no offline-capable till requirement at all.
- In a voluntary-trading or heavily franchised estate the group has no price, no assortment and no inventory position to master for much of its turnover, which turns replenishment, range, pricing and loyalty into offered services.
- Where two prices are displayed the lower binds, so a shelf label that silently fails to update is a statutory exposure rather than a data-quality defect.
- A pharmacy's price is set by the state, its item carries both a GTIN and a NAPPI code on different cadences, and its professional authorisation is a control the system may not override.
- Store-picked online fails mainly on phantom inventory, and the fix is a measured confidence discount on store on-hand rather than better pickers.
- Section 56 of the Consumer Protection Act gives the consumer six months from delivery and the election of remedy, which is a statutory clock per sale line and not a policy engine.
- Five failure classes recur: one sale with two identities, a stock position that is a belief, price with no single authority, statutory clocks nobody models, and a fixed cost per order against a variable basket.

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CHAPTER 5

Who does what

Every role in a retailer: what the person does all day, the decision they own, the data they need, the system they live in, and what they complain about. This is the chapter that lets you run a discovery workshop without being found out.

Why the org chart is useless

Somebody will hand you an organisation chart in your first week. Take it, because it tells you who has a budget and who to copy on an email. Then put it away, because it will not tell you a single thing you need.

Here is why. A retailer's org chart is a chart of *functions* — trading, supply chain, store operations, finance, technology — and almost nothing a retailer does happens inside one of them. A promotion is agreed by a buyer, funded by a supplier, priced by a pricing analyst, forecast by a replenishment analyst, ordered by a supply chain planner, picked by a distribution centre, filled and labelled in store, rung up by a till, and claimed back from the supplier nine weeks later. Ten hands, five functions, one object. The chart shows the five boxes and none of the ten hands.

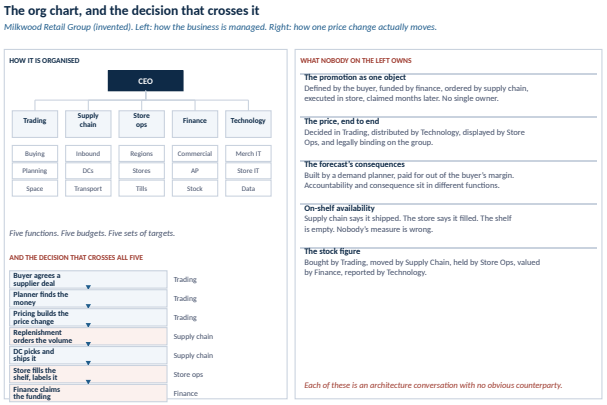


Figure 5.1 The org chart of Milkwood Retail Group, invented, against one decision moving through it. The right-hand panel lists what nothing on the left owns end to end — which is where the architecture work is.

The right-hand panel of that figure is the useful half. Each of those five things is something the business depends on completely and nobody owns end to end. That is not a criticism of the retailer; it is the shape of the industry. Retail is a sequence of decisions about one asset, taken by different people at different times with different information, and the handover between them is where the architecture work is.

There is a second reason the chart misleads, specific to retail: **the people who decide are not the people accountable for the consequence.** A demand planner builds a forecast. Wrong on the high side, the margin loss appears as markdown in the buyer's category P&L; wrong on the low side, the lost sale appears as an availability failure in store operations' scorecard. The planner's own measure is forecast accuracy, which is on neither report. **Wherever a decision and its consequence sit in different functions you will find a system nobody has asked to improve**, because improving it does not help whoever owns it.

A third reason, and the one architects miss longest. In a franchise or voluntary-trading estate, **some of the people in the value chain do not work for the retailer at all.** The member retailer sets the price, the franchisee hires the cashier, and the independent trader buying from the cash-and-carry arm is a customer with a credit limit. The org chart has no box for them, and they hold decisions the group's systems are designed to make.

So the instrument is not the org chart. It is a decision map.

> A note on the worked example. **Milkwood Retail Group is invented**: roughly 420 supermarkets, a 180-store pharmacy and health chain, 260 apparel stores, 22 cash-and-carry branches and an online business. The role names below are the ones the industry uses. The job descriptions are what the person actually does all day, and the last column of each table is where your work comes from.

The decision map

The decisions, and who actually owns each one

Ask "who decides?" rather than "who reports to whom?". The third column is where the argument is.

VALUE-CHAIN STAGE	THE DECISION	WHO OWNS IT	WHO THINKS THEY DO
Range	What is listed and delisted	Category buyer	The supplier, and the store manager who loses a facing
Cost	What we pay, and on what terms	Category buyer	Procurement, in groups that have one
Money	How much may be bought, by when	Merchandise planner, through open-to-buy	The buyer
Forecast	How many units, by SKU and store	Demand planner or replenishment analyst	Nobody — and the loss lands in the buyer's margin
Space	Which SKU gets which facing	Space planner	The buyer who negotiated the listing
Price	The shelf price, and its zones	Pricing manager	The buyer, and in a franchise estate the franchisee
Promotion	Mechanic, scope, period, funding	Promotions planner	Marketing, who owns the leaflet
Markdown	Timing, depth and scope	Markdown manager or buyer	Finance, once the margin lands
Allocation	Which store gets which units	Allocator	The area manager who wants more
Execution	Whether the plan reached the shelf	Store manager	Head office, which pushed the task
The stock figure	What we own, where, at what value	Nobody, in most retailers	Everybody, differently — which is the finding

Figure 5.2 The decision map. Run it as a workshop: the disagreement between the third and fourth columns is the finding, and the shaded bottom row is the one nobody owns.

Run that figure as a workshop. Put the eleven stages on a wall, ask who decides each one, and write down every name you are given. You will get two or three names per row, and the disagreement is the finding — not a problem to resolve politely but the reason the data in that part of the estate is unreliable.

The bottom row is the one to watch. In most retailers nobody owns *what we own, where, at what value, as at when*. Trading bought it, supply chain moved it, store operations holds it, finance values it, technology reports it, and each number is defensible from where its owner sits. The architect's job is not to pick a winner. It is to establish which system is the record for the quantity, which for the value, and what reconciliation exists between them — then to say so in writing, because in most retailers nobody ever has.

Buying and merchandising

The distinction outsiders always collapse

This is the single most consequential misunderstanding an architect can bring to a retailer, so it gets the long treatment.

Buying decides *what*. Planning decides *how much, in money, and when*.

They are different jobs with different mandates, different systems, different measures and frequent open conflict. An outsider hears "merchandising" and pictures one person who chooses products and orders them. In any retailer above a few dozen stores those are two organisations.

The **buyer** — often titled category manager — selects suppliers, negotiates the cost price and the trading terms, chooses the range, agrees the promotional plan and its funding, manages the supplier relationship and develops own-brand lines. What the buyer owns is *what is in the range and what it costs*. The buyer is measured on category sales and margin rate, and on supplier income earned. The buyer's instincts run toward range and volume, because both are how a buyer grows a category.

The **merchandise planner** builds the sales, margin, markdown, inventory and receipt plan by category by week or by retail period, and manages **open-to-buy**. What the planner owns is *the money*: how much may be bought, and when. The planner is measured on stock turn, terminal stock, markdown against plan, and whether the category's inventory investment earned its keep. The planner's instincts run toward caution, because the planner carries the consequence of excess.

Now the mechanism, because this is where the architecture lives. Open-to-buy is arithmetic:

> $$\textit{planned receipts} = \textit{planned sales} + \textit{planned closing inventory} + \textit{planned markdowns} - \textit{opening inventory}$$

>

> $$\textit{open-to-buy} = \textit{planned receipts} - \textit{commitments already on order}$$

Open-to-buy is held in money, by category, by period — not in units. The stock ledger, meanwhile, records *units of a SKU at a location*. Those are two different grains, and reconciling them is where most merchandise-planning architecture lives and where most of it fails. A planner who approves R13m of open-to-buy for a category has not said which SKUs, which stores or which sizes. A buyer who places a purchase order for 9,000 units of an option has not said what it does to the category's money plan until somebody values it at cost, adds the landed-cost up-charges, and posts it against the commitment.

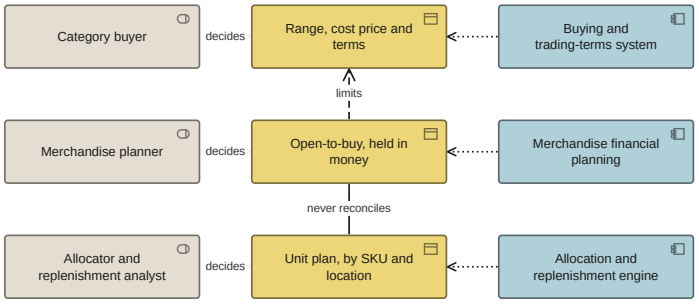


Figure 5.3 Buying, planning and allocation — three roles, three decisions, three systems. Open-to-buy limits the range decision and is held in money; the unit plan is held in units at a location. The association between them is the reconciliation most merchandise estates never close.

Three consequences follow, and all three are testable in week two.

One. A system design that merges the two roles has mis-modelled the organisation. If the merchandise system lets a buyer raise a purchase order without checking it against open-to-buy, the money plan is advisory and the business will over-buy. If the planning system holds the money but has no reliable feed of what is already on order, the open-to-buy number is wrong in the other direction and the business will under-buy going into a season. Ask to see the on-order feed. Ask when it last reconciled. In most retailers the answer involves a weekly extract and a spreadsheet.

Two. A buyer who plans over-buys and a planner who buys buys safe. That is an incentive, not a character judgement. The separation of the two roles is a *control*, and a system that collapses it has removed a control without anybody recording it.

Three. The two jobs need genuinely different software, and in grocery and apparel they need different software again. Apparel plans an **option** — a style in a colour — against a **size curve**, and manages a finite buy through a season to a terminal date. Grocery plans a category's money and replenishes a near-permanent SKU against a daily forecast. Apparel measures **sell-through**: units sold divided by units on hand at the start of the period. Grocery barely uses sell-through at all, because grocery replenishes rather than selling through a finite buy. That one difference is the cleanest illustration in this book of why one planning system cannot serve both, and why a group that runs supermarkets and clothing has two merchandise estates whatever the target architecture slide says.

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Buyer / category manager	Supplier meetings, cost negotiations, range reviews, own-brand development, promotional plans	What is in the range and what it costs	Supplier cost and terms history, category sales and margin, competitor prices, supplier income earned, range gaps	A buying and trading-terms system, a category review pack, spreadsheets	No single view of supplier income by supplier; terms held in contracts rather than data; range reviews rebuilt each cycle; no reliable landed cost
Merchandise planner	Builds and re-forecasts the sales, margin, markdown and receipt plan; manages open-to-buy	How much may be bought, and when	Plan against actual at category and option level, forward cover, sell-through curves, markdown against plan, on-order	A merchandise planning system, and Excel	The unit plan and the money plan never reconcile; no version control on plans; on-order data is unreliable, so open-to-buy is wrong
Range / assortment planner	Decides the assortment per store cluster, including localisation and the tail	The cluster assortment	Cluster definitions, rate of sale by cluster, duplication and cannibalisation, space available	An assortment planning tool, or the merchandise system	Clusters defined once and never revisited; no way to measure what the long tail costs

Allocation and space

Allocation and replenishment both get stock to a location and they are not the same thing. **Allocation** distributes a *finite buy* already committed — a seasonal apparel order, a promotional volume, a launch quantity. **Replenishment** keeps a *continuing* line in stock against a forecast. An allocator pushes; a replenishment engine pulls. Any multi-format group needs both, and the two engines will disagree about the same store on the same day.

Space is the third, and the one with no good owner. The planogram joins the merchandise hierarchy to the physical store — which SKU, which bay, how many facings, how deep — and it is also an input replenishment needs, because the order-up-to level derives from a **pick face** capacity. Get the planogram wrong and replenishment over-orders into a shelf that cannot hold it.

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Allocator / distributor	Decides which store gets which units, in what size curve, on what date; initiates in-season transfers	Where the stock physically goes	Store-cluster performance, size-curve sell-through, on-hand and in-transit, store capacity and planogram minimums	An allocation engine	Allocation rules cannot express "give the strong store the good sizes"; no visibility of what the store put on the shelf; transfers cost more than they recover
Replenishment analyst	Maintains forecast parameters, minimum and maximum levels, safety stock, order multiples and promotional uplift across hundreds of thousands of SKU-location pairs	Availability against stockholding	Forecast accuracy by SKU-location, out-of-stock measurement, lead-time reliability by supplier, promotional uplift, waste	The forecasting and replenishment engine	Promoted volume not ordered — the commonest availability failure there is; the forecast cannot be overridden without breaking the next cycle
Space planner	Builds planograms, reconciles range to physical space, issues store-specific plans	The shelf	SKU dimensions and case configuration, fixture surveys, sales per linear metre, pick-face capacity	A space-planning tool plus a store-survey database	Dimension data is wrong, so planograms do not fit; stores do not implement the plan and nobody measures compliance; fixture data decays after every refit

Supply chain and logistics

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Supply chain planner (inbound)	Converts the receipt plan into purchase orders; manages supplier lead times, book-in slots and inbound capacity	The order	Lead time and on-time-in-full reliability by supplier, distribution centre capacity, order multiples	The purchase order system and a supplier portal	Suppliers deliver short and the system has no memory of it
Demand planner	Owns the forecast: baseline, seasonality, event and promotional effects, new-line forecasting, accuracy measurement	The forecast	History at the right grain, event calendar, promotional history net of cannibalisation, bias by segment	The forecasting engine	Accuracy measured at the wrong grain to be actionable; no accountability for a bad forecast, because the loss lands in the buyer's P&L
Distribution centre manager	Runs receiving, put-away, slotting, picking waves, despatch and the yard	Today's throughput and tomorrow's labour	Inbound schedule, order profile by wave, pick productivity, dock utilisation	A warehouse management system	Store orders arrive in an unworkable profile; no advance notice of promotional volume; slotting not re-optimised as velocity changes
Transport / fleet planner	Builds routes and loads, manages drivers, backhaul and third-party carriers	The route plan and the cost per drop	Order volumes and weights by store, vehicle and driver availability, delivery windows, fuel and toll costs	A transport management system	Stores cannot receive in the window the route needs; cost per drop invisible at store level
Cold-chain / food-safety manager	Owns temperature discipline from supplier to shelf, with monitoring, excursion response and audit evidence	Whether the chain held	Continuous temperature telemetry with excursion history, joined to stock	A monitoring platform, and paper	Telemetry exists but is not joined to stock, so a write-off cannot be evidenced to an excursion

Notice that four of those five complaints are the same complaint: **a fact was known somewhere and never joined to the thing it explains.** The short delivery was recorded and never attributed to the supplier. The temperature excursion was logged and never attributed to the write-off. The forecast error was measured and never attributed to the markdown. That pattern is the most reliable source of high-value, low-cost architecture work in a retailer, and it is nearly always a data-model problem rather than a new system.

Store operations, from the manager to the till

The store is where the plan either happens or does not. Everything upstream is a proposal.

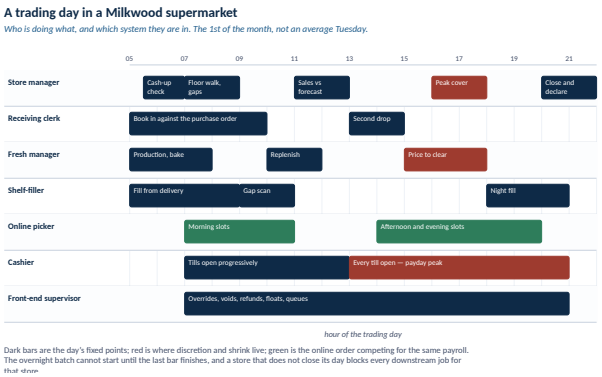


Figure 5.4 A trading day in a Milkwood supermarket on the 1st of the month. The online picker and the shelf-filler are the same payroll, and the batch cycle cannot start until the last bar finishes.

Three features of that day shape the architecture. **It is hour-shaped**, so labour is scheduled against a forecast of transactions by hour and a bad forecast produces either queues or idle payroll on a statutory minimum wage of R30.23 per ordinary hour from 1 March 2026 [1]. **It has fixed points that gate everything downstream**: the trading day must be closed and sealed before the night's batch can run, and a store that does not close blocks every job for that store. And **it is where discretion lives** — the override, the void, the refund, the waste markdown, the manual price change. Each is a legitimate operational necessity and a shrink vector at once, which is why the front-end supervisor is the most important internal control point in a retailer.

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Regional / area manager	Runs a portfolio of stores on sales, availability, shrink, labour cost and standards	Store-manager performance and local exception approval	Like-for-like by store, labour against forecast, shrink and waste, audit findings	A reporting portal and email	Reports arrive too late to act on; targets set centrally without regard to local trade
Store manager	Runs the store P&L, the roster, the cash, the stock, the standards and the team	Execution — whether the plan happened on the shelf	Sales against forecast by hour, availability and gaps, waste, labour hours used, cash position, outstanding tasks	A store back office and a task list	A dozen systems for one job; tasks pushed from head office with no visibility of total load; the system blocks sensible local decisions
Department manager (fresh, butchery, bakery, deli)	Orders, prepares, prices to clear, manages waste and hygiene records	Today's waste and availability	Shelf life, waste history, markdown-to-clear guidance, production plans and yields	A store ordering and production module	The ordering system does not understand production; waste recording is manual and therefore unreliable, which corrupts shrink reporting
Replenisher / shelf-filler	Fills shelves, faces up, reports gaps, handles deliveries	Whether the shelf has stock on it	The planogram, the delivery manifest, gap-scanning tooling	A handheld	The planogram does not match the fixture; back-stock cannot be located
Cashier / till operator	Scans, resolves price queries, takes tender, handles voids and refunds, verifies age restrictions, enrols loyalty	The integrity of the transaction	A fast till, current prices, clear promotion messaging, override rules	The point of sale	Price disputes they cannot resolve; the till is slow at peak; refund rules change without notice
Front-end supervisor	Authorises overrides, voids and refunds; manages queues, cash-ups and floats	The override decision	Exception history by operator, float and variance position	The point of sale and a cash office system	Authority rules are a laminated sheet rather than a configuration
Receiving clerk	Books in deliveries against the purchase order, records shortages and damages	The accuracy of the inbound stock record	Expected deliveries, purchase order detail, claim status	A back-office receiving screen	No time to check at the dock; claims against suppliers not followed up, so short deliveries quietly become shrink
Stock controller	Runs cycle counts and stock takes, investigates variances, posts adjustments	Whether the perpetual inventory is true	Variance history by SKU and location, movement history, reason-coded adjustments	Store inventory management	Adjustments posted without reasons, so variance analysis is impossible
Online picker (store-based)	Picks and packs online orders in the store alongside trade, resolving substitutions	Pick accuracy and substitution judgement	Pick list in path sequence, substitution rules, customer response	A picking app on a handheld	The system sends them for stock that is not there; pick targets conflict with helping customers

Pharmacy roles and their statutory position

The pharmacy is the one place in a retailer where a role's authority comes from statute rather than from a job grade, and that changes what a system is allowed to do.

At 16 October 2026 the South African Pharmacy Council register showed 5,851 registered pharmacies, 4,367 of them community pharmacies [2]. Each licensed premises has a **responsible pharmacist** who is statutorily accountable for its compliance with the Pharmacy Act, the Good Pharmacy Practice rules made under section 35A, and the premises licence. That accountability is personal. It cannot be delegated to a store manager, a regional manager or a system.

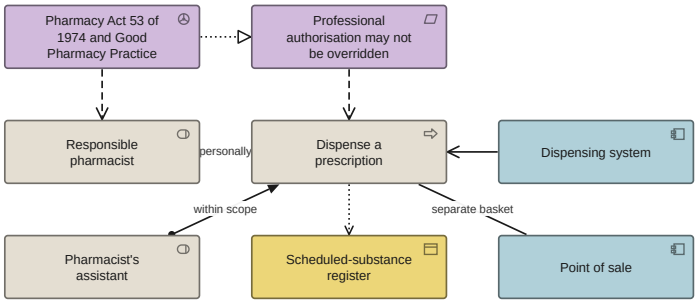


Figure 5.5 The pharmacy boundary drawn as a wall. The requirement comes from statute, the responsible pharmacist is personally assigned, the assistant is assigned only within scope, and the point of sale is associated with the dispense rather than part of it.

Read the figure as a wall rather than a boundary. Two things sit on the far side of it and must not be moved.

The dispensing act is reserved. A pharmacist interprets and dispenses the prescription, screens interactions, counsels the patient and authorises generic substitution. A pharmacist's assistant — basic or post-basic — works within a defined scope that excludes acts reserved to a pharmacist. So **role-based**

authorisation in a dispensing system is a statutory control, not an IT preference, and a configuration that lets a front-shop supervisor complete a dispense to clear a queue is a compliance failure dressed as a usability improvement. The corollary is the one architects get wrong: you may not simplify the authorisation model to match the rest of the estate. The dispensary's authorisation model is the regulator's, and the rest of the estate has to live with it.

The pharmacist's professional judgement overrides the system, and the system may not override the pharmacist. The software may propose a generic substitution, flag an interaction or price a line. It may not decide the clinical question. That sounds obvious until you meet a design that blocks a dispense the pharmacist has judged appropriate, or one that completes a dispense the pharmacist has not seen. Both happen. And Good Pharmacy Practice rule 1.2.11.4(f) requires that where software is relied on for warnings or interaction checks it must be **the latest version available** [3] — which turns a content subscription into a regulated currency obligation with a monitored distribution mechanism attached.

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Responsible pharmacist	Keeps the pharmacy lawfully operating: registers, procedures, staff registration, inspection readiness	Whether the pharmacy may lawfully trade	Registers, licence status, inspection findings, staff registration status, content currency	The dispensing system and a lever-arch file	Compliance evidence is manual; the group's IT release calendar does not recognise a statutory obligation
Dispensing pharmacist	Interprets and dispenses prescriptions, screens interactions, counsels patients, authorises substitution, vaccinates where permitted	The clinical appropriateness of the dispense	Patient medication history, prescriber detail, interaction and allergy data, formulary and scheme rules, stock	The dispensing system	Scheme adjudication rejects at the counter with codes they cannot resolve; the point of sale and the dispensing system hold different views of the same person
Pharmacist's assistant	Works under supervision within a defined scope: pre-packing, basic dispensing support, front-shop medicine advice	Nothing reserved to a pharmacist	A system-enforced scope boundary	The dispensing system	Scope boundaries enforced by convention rather than by the system
Pharmacy / front-shop manager	Runs the commercial business around the dispensary: front-shop range, promotions, labour, shrink	The margin, where the dispensary owns the footfall	Front-shop category performance, private-label penetration, dispensary footfall conversion	The retail back office	Two systems, two item masters, one customer standing in front of them
Medical-scheme claims clerk	Reconciles submitted claims to scheme remittances, works rejections and short payments	The scheme receivable	Claim, adjudication response, remittance, reason codes	A claims reconciliation tool, or a spreadsheet	Cannot reconstruct the adjudication that produced the paid amount

E-commerce and digital

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
E-commerce trading manager	Runs the online range, pricing, merchandising, search and promotion	Online sales and conversion	Funnel analytics, search-term and zero-result reports, availability by channel, basket and margin by order	The e-commerce platform's admin	Availability data is not trustworthy; the online price must match the store price but the systems are separate
Digital product owner	Owns the app and web experience, the release cadence and adoption	The experience	Adoption, retention, release telemetry, abandonment points	A backlog and an analytics tool	The backend cannot support the experience; store-side constraints such as slots and picking are invisible in the app
Fulfilment operations manager	Owns pick, pack, slot capacity and on-time delivery	The promise the app makes	Slot capacity against rostered picker hours, pick rates, on-time and fulfilment rates	An order management system	Capacity promised by the app exceeds the labour available
Marketplace / seller operations manager	Onboards sellers, polices catalogue and listing quality, manages performance, commission and payout, arbitrates disputes	The seller base and the take rate	Seller cancellation, late-despatch and defect rates, catalogue match quality, commission and settlement accuracy	A marketplace management console	Catalogue matching is manual; liability allocation between platform and seller is decided case by case

Marketing, loyalty and CRM

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Loyalty programme manager	Owns membership growth, identification rate, earn-and-burn mechanics, cashback liability and partner integrations	The mechanic and its cost	Identification rate by store, redemption and breakage, incremental spend by cohort, liability balance	The loyalty platform	Can measure participation but not incrementality
CRM / customer analytics manager	Segments the base, builds personalised offers, measures campaign incrementality	Who gets which offer	Customer-level transaction history at line level, consent status, a stable identity across channels	A customer data platform and a campaign tool	Customer identity fragments across loyalty, online account, credit account and card token
Trade marketing / retail media manager	Sells and executes supplier-funded activity and on-site media	The media and funding calendar	Inventory of media slots, supplier commitments, delivery evidence	An ad server and a spreadsheet	The boundary against supplier income is undefined, so the same rand is claimed twice in two places

Pricing and promotions

Where pricing sits in the organisation changes the governance of the price file, and you should establish it in week one. In some retailers pricing is inside merchandising and reports to the trading director; in others it is a separate function with its own head. In the first case the buyer effectively controls the shelf price. In the second there is a genuine second opinion before a price moves.

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Pricing manager	Sets and maintains the price file: architecture, zones, rounding, price-ending rules, competitor response	The shelf price, and the internal selling-price inflation the board reports	A competitor price index, elasticity estimates, cost-price changes, margin impact simulation, zone definitions	A price management system	No simulation before committing; cost increases arrive faster than the price file can respond; shelf labels lag the till
Promotions planner	Builds the promotional calendar, mechanics, funding and forecast uplift; locks the leaflet	The promotional calendar and its funding status	Historic promotional performance net of cannibalisation, funding agreed by supplier, forecast uplift for replenishment	A promotion engine and the leaflet plan	The calendar locks before funding is confirmed; promoted volume not ordered
Markdown / clearance manager	Decides markdown timing, depth and scope against exit targets	The terminal margin	Sell-through by option by cluster, forward cover, remaining weeks, prior markdown response	A markdown module, or Excel	Markdown taken as a promotion, so the inventory revaluation never happens

That last complaint is worth stopping on, because it is a data-model defect with a financial consequence. A **markdown** is a permanent price reduction that revalues the stock; a **promotion** is a temporary reduction at the point of sale that does not. Execute a markdown as a promotion and the inventory is never revalued, so the margin hit arrives later as a stock write-down and nobody can explain the difference. The two must be separate objects, and in a surprising number of estates they are the same object with a flag.

Finance, and the supplier-income team

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Commercial / category finance analyst	Partners the buyer and planner on margin, terms and promotional economics	The margin number everybody argues about	Gross margin decomposed into cost, price, mix, markdown, shrink and supplier income	The general ledger, and Excel	Gross margin cannot be decomposed, so a 120 basis-point movement cannot be acted on
Supplier income accountant	Accrues, claims, settles and reconciles rebates, listing fees, co-op advertising and markdown support; evidences the conditions	The number most likely to cause a restatement	Effective-dated trading-terms agreements, purchase and sales volumes by supplier and SKU, promotional records, claim status	A rebate module if lucky, a spreadsheet and a credit-note queue if not	Terms are in contracts rather than data, claims are disputed with no shared evidence, and the audit requires reconstruction
Accounts payable	Matches invoice to order to receipt, applies terms and settlement discounts, deducts claims, runs the payment proposal	When suppliers get paid — the negative working capital the business runs on	Three-way match status, terms by supplier, claim and deduction detail	The ERP	Match failures caused by receiving errors; deductions disputed because the claim evidence is somewhere else
Inventory accountant	Values stock under the retail or the cost method, books shrink and markdown, reconciles the stock ledger to the general ledger	The stock value in the accounts	Movement history, reason-coded adjustments, count results against a book position as at the count date	The stock ledger and the general ledger	Adjustments without reason codes; counts not reconciled to a book position as at the count date
Store financial controller	Produces the store P&L, allocates central and distribution costs, calculates turnover rent	The store P&L	Store sales and costs, allocation rules, lease terms including turnover clauses	A reporting layer	Allocations are arbitrary, so no store manager believes their own P&L
Internal audit	Tests store controls, cash, refunds, voids, receiving, stock takes and supplier claims	What gets tested	Exception reports by store and operator	Whatever exists	No exception reporting, so testing is sampling

The supplier-income accountant deserves attention out of all proportion to the size of the team. For Milkwood, supplier income of R2.4bn sits against trading profit of R2.9bn, so an error of a tenth in the accrual moves trading profit by 8%. Chapter 6 does the arithmetic; the organisational point is that the number is produced by a small team working from contracts that are not data, and nobody in the business has an incentive to make it harder to claim.

Loss prevention

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Head of loss prevention	Owns total shrink and its four causes: external theft, internal theft, administrative error, waste	The shrink reduction plan	Shrink by store, category and cause; exception reports; count variance history	A loss-prevention analytics tool	Shrink is only measured at stock take, so causes are months cold
Shrink analyst	Finds the pattern: the store, the SKU, the till, the operator, the supplier, the process	Where to point an investigation	Transaction-level till data, adjustment history, receiving variance, waste records	Exception-based reporting	**This role is the cheapest return in retail and the most under-tooled**
Investigator (internal)	Works till exception, refund and receiving cases to conclusion, including disciplinary and criminal referral	The case outcome	Case evidence joined across till, stock and roster data	A case management tool, or a notebook	Evidence lives in four systems and no two agree on a timestamp

Property and new stores

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Property / leasing manager	Finds sites, negotiates leases, manages renewals, break options, exclusivity and usage clauses	The lease	Catchment and competitor data, trading density benchmarks, rent-to-sales affordability, lease expiry profile	A lease management system	Lease data needed by finance, operations and the auditor lives in three places
Store development manager	Takes a site from signature to opening: design, fit-out, fixtures, systems installation, stock-up, hiring, training	Opening on the planned date	Project plan, equipment lead times, systems provisioning status	A project tool	Store systems provisioning is the last thing anybody thinks of and the first thing that delays an opening
Format / concept developer	Designs the format: adjacency, flow, fixture, size and range envelope	What a store of this format is	Format performance by catchment, range envelope by size band	Drawings and a space tool	New formats are designed before anybody checks the merchandise system can represent them
Facilities and energy manager	Owns refrigeration, air conditioning, generators, inverters, solar and the energy bill	Whether the store can trade when the power fails	Asset register, telemetry, outage history, energy consumption by site	A building management and monitoring platform	Telemetry is per site and per vendor, so the estate view has to be rebuilt each time

The store development manager's complaint is the systemic one. A retailer opening a few hundred stores a year — and South African grocers do open at that rate — is running a continuous provisioning pipeline: a location in the location master, a price zone, an assortment cluster, a planogram set, tills, scales, label printers, handhelds, a network link, a payment estate and a loyalty configuration. Treated as a project each time it is slow and error-prone. Treated as a product — store opening as a repeatable, automated capability — it is one of the highest-return pieces of work available in a growing retailer.

Technology

Role	What they do all day	The decision they own	The data they need	The system they live in	What they complain about
Head of merchandising systems	Owns the merchandise, planning, pricing and replenishment estate	The merchandising roadmap	Portfolio, release calendar, incident and change history	A service management tool	Every business function wants its own master data
Store systems manager	Owns tills, scanners, scales, label printers, handhelds, networks, UPS and payment devices across every site	Whether stores can trade	Device telemetry, uptime by store, offline incidence, patch status	A device management platform	Releases cannot be tested at estate scale; offline reconciliation failures surface days later
Master data manager	Owns the golden record for item, supplier, price and location, and the stewardship workflow around each	What a valid record is	Data quality rules, rejection rates, source system behaviour	A master data or product information management platform	GTIN and dimension data supplied by vendors is wrong and there is no gate to stop it entering
Integration architect	Owns the flow of sale, stock, price, promotion and order events between store, DC, head office, online and partners	The contract between systems	Message volumes, failure and replay rates, reconciliation breaks	An integration platform and a lot of file transfer	Nothing owns the reconciliation when a flow fails, so breaks are found by the business
Data and analytics lead	Owns the sales, stock and customer data layer and the derived measures that end up in the results announcement	The definition of the number	Conformed history, dimension versions, measure definitions	A data platform	The number in the board pack and the number in the campaign tool differ, and both are defended
Payments and acquiring manager	Owns tender acceptance, acquiring relationships, tokenisation and authentication	How money is taken	Approval and decline rates by tender and channel, acquirer fees, fraud rates	A payment switch console	Decline rates are owned by nobody and blamed on everybody
Information security and privacy officer	Owns cardholder data scope across thousands of terminals and the privacy obligations attached to a loyalty base of millions	What is in scope	Terminal inventory, network segmentation, consent records, retention rules	A compliance platform	Scope is defined by where card data happens to flow rather than by design

The trading calendar governs every diary in the building

Every role above works to the same clock, and an architect from banking or insurance will under-weight it. In banking the calendar is month-end, quarter-end and year-end. In retail the calendar *is* the operating model.

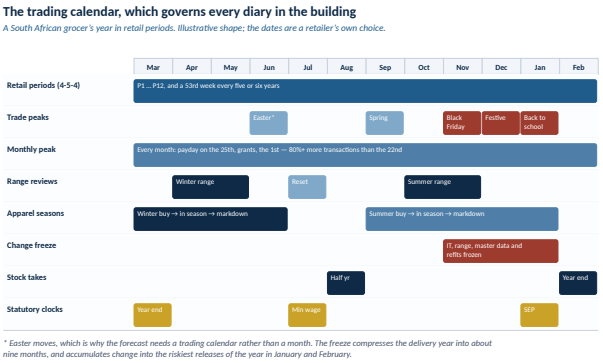


Figure 5.6 The trading calendar. Illustrative shape for a South African grocer; the specific dates are each retailer's own choice and the change freeze is practitioner consensus rather than a published fact.

Four things in that figure are worth stating plainly.

The retail year is not the calendar year. The National Retail Federation's **4-5-4 calendar** is a voluntary industry guide dividing the year into months on a four-week, five-week, four-week pattern, so that comparable months contain the same number of Saturdays and Sundays. It came into wide use in the 1940s, because as weekends grew as a share of sales the varying number of weekends per calendar month made month-on-month comparison meaningless. A 52-week year is 364 days, so a **53rd week** is needed roughly every five or six years on no fixed schedule, added for sales reporting only [4]. South Africa adds a wrinkle: major retailers have different year ends, so sector comparison requires period alignment and a data platform that can hold more than one retail year definition.

That makes the calendar an **architectural object**, not a reporting convention. The calendar dimension must carry the retail week number, the period with its 4-5-4 length, the quarter, the year, the 53-week flag and the comparable prior-period mapping including the restatement convention. Build it once, centrally, and make every system read it. Most estates let three systems derive it independently — and the fashion business uses a Sunday-start week while the food business uses a Monday-start week, so Monday's sales land in different weeks in different systems.

The monthly peak is a South African first-order concern. Payday on the 25th, social grants, and the 1st of the month produce a monthly amplitude sharper than in most markets: more than 80% more transactions happen on the 1st than on the 22nd [5]. That shapes the roster, the cash position, the delivery schedule, the till count and the network capacity — monthly, not seasonally.

Peak is not the shape a UK or US architect expects. The festive peak runs from Black Friday into early January, with December dominating. **Back to school** in January is a second genuine peak in stationery, uniform and school shoes, and because the South African school year starts in January it collides directly with the post-festive trough in food. Easter moves, which is precisely why a forecast needs a trading calendar rather than a month.

And then the freeze. A retailer freezes change to the production estate across peak, and the freeze is wider than IT change: it typically covers software releases, infrastructure change, structural master-data change — hierarchy reclassification especially — range change, planogram resets and store refits. No South African retailer publishes its change calendar, so the specific windows are practitioner consensus rather than a sourced fact, and you should confirm them in your first week. The duration is the point: a freeze from mid-November to mid-January **compresses the delivery year into about nine months**, so a programme plan that assumes twelve is wrong by a quarter. It also accumulates change, which makes the January and February releases the largest and riskiest of the year, immediately after the period in which the estate was least exercised.

Your real counterparties

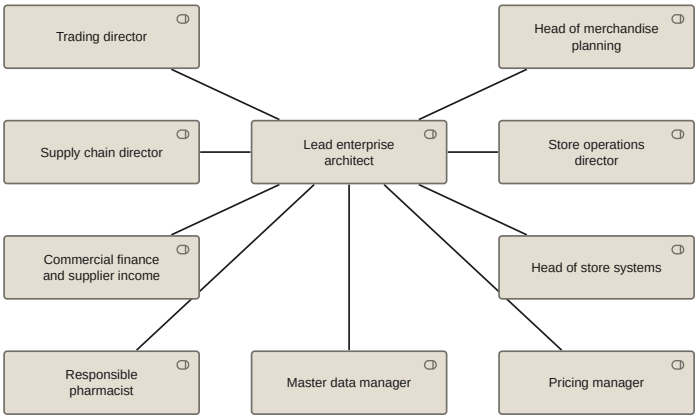


Figure 5.7 The architect's real counterparties. Nine named people, not nine functions. The chief information officer is deliberately absent: the decisions you need are commercial and operational.

Nine people. Not nine functions — nine named individuals, and you should be able to write their names next to those boxes by the end of your second week.

Two are easy to miss and both are load-bearing. The **responsible pharmacist** is there because nobody else in the group can tell you which of your designs is unlawful, and the answer is not negotiable. The **master data manager** is there because they are the only person who knows what is actually in the item, supplier, price and location masters as opposed to what the data dictionary says — and almost every retail programme fails on master data rather than on function.

The one deliberately absent is the chief information officer. You will report to them or near them, but they are not a counterparty for this work: the decisions you need are commercial and operational, and the people who hold them sit in trading, supply chain and store operations.

Who decides what, on one page

The decision map tells you who owns each decision. A RACI tells you who else has to be in the room, and it is the artefact that turns a role list into something a programme can use.

Who does what, from a range review to the shelf

Eleven steps, eight roles. Fill this in with the retailer's own names in week two and watch the arguments start.

	THE STEP	Category buyer	Merchandise planner	Space planner	Pricing manager	Supply chain planner	Store operations	Commercial finance	Master data
1	Category strategy and the review brief	R	C	C	I	I	I	C	
2	Range decision — list and delist	A	C	C	I	I	C	I	R
3	Cost price and trading terms	A	I		C	I		R	C
4	Promotional plan and funding	A	C	I	R	C	I	C	
5	Money plan and open-to-buy	C	A		I	C		R	
6	Planogram and space allocation	C	I	A		I	C		R
7	The buy — purchase orders placed	A	C	I	I	R		I	C
8	Allocation and replenishment parameters	C	C	C		A	I		R
9	Price file and shelf-edge labels	C	I	C	A	I	R	I	C
10	Execution on the shelf	I	I	C	I	C	A		
11	Review, and the supplier claim	C	C	I	I	I	I	A	I

A accountable (the one neck). R responsible (does the work). C consulted. I informed. Blank means nobody — and every blank in a real retailer's version of this grid is a defect waiting to be found.

Figure 5.8 Eleven steps from a range review to the shelf, against eight roles. Every blank cell in a real retailer's version of this grid is a defect waiting to be found.

Do not hand that out as a finding. Hand it out half-filled, with the retailer's own job titles in the column headings, and let the people in the room argue. Two patterns emerge within an hour. **Rows with two accountables** are a genuine governance gap, and the price file row is the usual offender: pricing is accountable for the number, store operations for it appearing on the shelf, and when a label fails neither thinks it was theirs. **Blank cells** are worse — a blank in the supplier-claim row means a promotion ran, somebody agreed the funding, and no function is consulted when the claim goes in nine weeks later. Every blank is a defect waiting to be found, and finding them costs one workshop.

Four questions

Ask these four in the first fortnight, of the people above, and write down the answers verbatim. They are chosen because each one has a single short answer that the business thinks is obvious and that the estate frequently contradicts.

FOUR QUESTIONS, ASKED OF EVERYBODY

Who decides the shelf price for this SKU in this store on this date — and which

system is the authority when two of them disagree?

What does the business believe we own, where, at what value, as at when — and

which system is the system of record for the quantity, and which for the value?

If the forecast run does not complete tonight, does replenishment run on

yesterday's forecast, or not run at all? Who decided that, and where is it written down?

When does change freeze, what exactly does the freeze cover, and who may lift it?

The first finds the price-authority problem, the commonest expensive defect in retail and the one with a statutory consequence in South Africa. The second finds the inventory-integrity problem, which is the root of most of the rest. The third tests whether the overnight cycle has designed behaviour or a convention held in an operator's head — in most retailers it is the latter, which makes discovering it one of the more valuable things you will do in your first month. The fourth tells you how long your delivery year actually is.

THE CHAPTER IN TEN LINES

- The org chart shows five functions; nothing a retailer does happens inside one of them, so the instrument you want is a decision map rather than a reporting line.
- Wherever a decision and its consequence sit in different functions you will find a system nobody has asked to improve.
- Buying decides what; planning decides how much, in money, and when — and a design that merges the two has removed a control without recording that it did.
- Open-to-buy is held in money by category and period while the stock ledger is held in units at a location, and reconciling those two grains is where most merchandise-planning architecture fails.
- Allocation pushes a finite buy and replenishment pulls a continuing line; a multi-format group needs both and the two engines will disagree.
- Most supply chain complaints are one complaint: a fact was known somewhere and never joined to the thing it explains.
- The front-end supervisor who authorises overrides, voids and refunds is the single most important internal control point in a retailer.
- In the pharmacy, role-based authorisation is a statutory control rather than an IT preference, and the pharmacist's professional judgement is a control the system may not override.
- The 4-5-4 trading calendar with its 53rd week is an architectural object that every system should read from one place, and the peak change freeze compresses the delivery year to about nine months.
- Four questions settle most of it: who decides the price, what do we own where and as at when, what happens if the forecast run fails, and when does change freeze.

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- **2.** South African Pharmacy Council, Registered pharmacies by sector (as at 16 October 2026) https://interns.pharma.mm3.co.za/Statistics/PharmaciesBySector
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- **4.** National Retail Federation, "4-5-4 Calendar" https://nrf.com/resources/4-5-4-calendar
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CHAPTER 6

How retailers make money

A retailer buys goods it did not make, owns them until somebody buys them, and earns a few cents on the rand for the trouble. An architect who cannot read that arithmetic will fund the wrong programme.

The arithmetic, properly

An architect who cannot read a retailer's profit and loss account will build the wrong thing. Not a slightly sub-optimal thing — the wrong thing. You will be asked to fund a data platform, and the honest business case for it is that gross margin cannot be decomposed into its six causes, so a 120 basis-point movement cannot be acted on. You will be asked to replace a planning system, and the honest case is that the money plan and the unit plan do not reconcile, so open-to-buy is wrong in one direction or the other all year. Both of those are arithmetic arguments. Neither survives being made in the language of capability maturity.

So this chapter is the arithmetic. It is not accountancy and you will not have to produce any of these numbers yourself. You will have to **read** them, recognise when one of them is being computed wrongly, and know which system produces each.

Three properties of retail arithmetic are worth fixing in your mind before the numbers start.

The denominator moves. Gross margin percentage divides by net sales; **markup** percentage divides by cost. They are different numbers and confusing them is the commonest arithmetic error in retail conversation [1]. Similarly, stock turn uses *average* inventory and almost everybody computes it on the period-end balance.

Every figure needs a period basis attached. Retailers report in 52- and 53-week years. One South African grocer's prior-year turnover appears in the same results pack as both R116.3bn on a pro-forma 52-week basis and R119.0bn on the actual 53-week basis [2]. Every growth rate must declare its week basis, and an architect who drops that qualifier ships reconciliation defects into every reporting layer they build.

The margins are not comparable across formats, and sometimes not within a group. Published South African gross margins, latest reported, run from a wholesaler's 10.8% through grocery at 17.0% to 26.1%, to apparel and homeware at 41.2% to 57.7% [2][3][4][5][6][7]. A wholesale margin on a wholesale cost base is not a retail margin, and comparing them tells you nothing except which business you are looking at.

Milkwood Retail Group — the P&L, read as a retailer reads it

Invented, illustrative and internally consistent. A full year, R millions. Use these numbers; they recur through the book.

THE LINE	R MILLION	% OF SALES
Turnover (net of VAT and returns)	**58,400**	**100.0**
Cost of goods sold, at invoice cost	(46,940)	(80.4)
Gross profit before supplier income	**11,460**	**19.6**
Supplier income recognised in cost of goods	2,448	4.2
Gross profit	**13,908**	**23.8**
of which: markdown taken	(1,620)	(2.8)
of which: shrink and waste	(1,110)	(1.9)
THE COST OF BEING A RETAILER		
Store labour	(4,380)	(7.5)
Occupancy, rates and recoveries	(1,969)	(3.4)
Energy, diesel and refrigeration	(585)	(1.0)
Distribution and transport	(1,170)	(2.0)
Other store operating cost	(777)	(1.3)
Head office, support and IT run	(1,110)	(1.9)
Depreciation on owned assets	(1,019)	(1.7)
Trading profit — the number management runs on	**2,898**	**5.0**
Lease finance cost under IFRS 16	(1,142)	(2.0)
Net other finance cost	(188)	(0.3)
Profit before tax	**1,568**	**2.7**
Taxation at 27%	(423)	(0.7)
Profit after tax	**1,145**	**2.0**

Read the two shaded totals together. Twenty-three point eight cents of gross margin on a rand of sales becomes five cents of trading profit and two cents of profit after tax. Between them sits the entire cost of being a retailer — and there is no provision line anywhere for being wrong about demand.

Figure 6.1 Milkwood Retail Group's profit and loss account, invented and illustrative. The two shaded totals are the shape of retail: 23.8 cents of gross margin becomes five cents of trading profit, and there is no provision line for being wrong about demand.

> **Milkwood Retail Group is invented**, and so are its financial figures. They are internally consistent and chosen to be typical of a mid-tier South African multi-format group: 420 supermarkets, 180 pharmacy and health stores, 260 apparel stores, 22 cash-and-carry branches and an online business. Use them as worked examples; the market figures cited around them are sourced.

Read the figure once more and notice the shape. Gross margin of 23.8% becomes trading profit of 5.0% and profit after tax of 2.0%. Between the first and the second sits the entire cost of being a retailer: labour, rent, energy, distribution, the stores themselves. And nowhere in that structure is there a provision line for being wrong about demand. A bank at a 20% return on equity prices for expected credit loss. A retailer at a 5% trading margin has nothing equivalent. The error lands in gross margin, in the same period, with no smoothing at all.

Gross margin and the margin mix

Gross margin % = (net sales − cost of goods sold) ÷ net sales [1]. Easy. The hard part is that it moves without anybody deciding anything.

The same group, by segment — and why the mix moves the number

Milkwood Retail Group, invented. Five businesses with gross margins 32 points apart sharing one reported figure.

SEGMENT	TURNOVER RM	SHARE	GROSS MARGIN	TRADING MARGIN	WHY IT IS WHERE IT IS
Supermarkets (420)	38,400	65.8%	24.2%	5.4%	Fresh and ambient sit 10–15 points apart; promotions run near 40% of turnover
Pharmacy and health (180)	7,200	12.3%	26.8%	7.1%	Dispensary margin is capped by the state; the front shop carries the number
Apparel (260)	3,600	6.2%	42.5%	6.2%	The highest gross margin and the slowest turn, with much of it given back in markdown
Cash-and-carry (22)	6,900	11.8%	10.4%	1.9%	A wholesale margin on a wholesale cost base — not comparable to a retail one
Online (Milkwood Now)	2,300	3.9%	19.0%	(1.8)%	Same shelf price, plus the pick and the drop
Group	58,400	100.0%	23.8%	5.0%	A weighted average nobody decided

Figure 6.2 The same group by segment. Five businesses with gross margins 32 points apart share one reported number, and growing the lowest-margin one moves the group figure without a single price or cost changing.

Four mix effects do it, and each needs its own attribution in reporting.

Category mix. Fresh and ambient grocery sit 10 to 15 points apart; a pharmacy's dispensary and front shop sit more than 20 points apart. Sell more fresh, or more front shop, and the blended number moves.

Format mix. Milkwood's cash-and-carry arm carries a 10.4% gross margin. Grow it 10% with nothing else changing and group gross margin falls from 23.82% to 23.66% — a 16 basis-point decline caused by growth. In the real market the same effect appears as one grocer's non-South African supermarkets trading at a 3.0% trading margin against 6.6% domestically [2].

Channel mix. An online order carries the pick and the drop against the same shelf price. Milkwood Now runs at a 19.0% gross margin against the supermarket's 24.2%, and a negative trading margin.

Promotional mix. Promotions at 40% of one grocer's South African supermarket turnover, up from 37.9% a year earlier [8], move blended margin without a single cost price changing.

Here is the requirement that follows, and it is the single most common data gap in a retailer:

> Any movement in gross margin must decompose into **cost price, retail price, mix, markdown, shrink and waste, supplier income, and promotional funding** — at category level, for the period, reconciling to the reported number. A retailer that can only report "gross margin was down 120 basis points" cannot act on it, and one large listed group reports exactly that, at group level [5].

That decomposition is not a report. It is a data model: cost and price with effective dates, movement history with reason codes, supplier income attributed to stock, and promotion records reconstructable as at any past date. Build the model and the report is trivial. Build the report on top of a model that cannot support it and you will rebuild it every year.

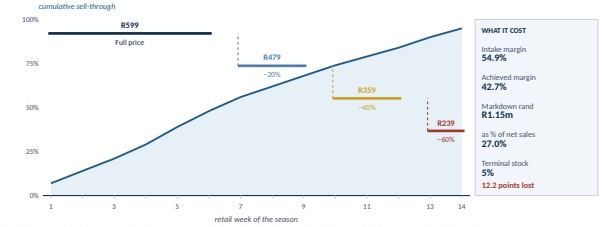
Markdown, and the markdown cadence

A **markdown** is a *permanent* price reduction taken because the item will not sell at its current price. A **discount** is a temporary reduction applied at the point of sale [1]. They are different objects and treating them as one object with a flag is a real and common defect, because a markdown revalues the stock and a discount does not.

Markdown % = markdown rand ÷ net sales [1]. And **maintained markup = ticket price − cost of goods − allowances**, where allowances are discounts plus markdowns [1] — which is to say, markdown is the bridge between the margin you intended and the margin you got.

The markdown cadence — one apparel option, fourteen weeks

Milkwood, illustrative. 9,000 units bought at a R270 cost on a R599 ticket. Four decisions, none of them reversible.



The decisions are which options, in which stores, at what depth, in which week — and each is irreversible, because a marked-down price cannot be raised. The system has to hold a markdown plan inside the open-to-buy, execute dated price changes to hundreds of stores, revalue the inventory where the retail method applies, and report actual against plan. Grocery's equivalent is the daily waste markdown on short-dated perishables: same mechanism, daily cadence, store-level discretion, and a direct interaction with shrink.

Figure 6.3 The markdown cadence for one apparel option. Four price decisions across fourteen weeks, each irreversible, costing 12.2 points of margin against the intake. The decision is about the future and the data is about the past.

Now the sentence that matters most in this section. **A markdown is a decision about the future taken with data about the past.** In week three of a fourteen-week season you know cumulative sell-through, the planned curve, and nothing at all about the weather in week nine. You have to choose a depth now because the remaining weeks are finite and the exit date is fixed. Mark down too early and you give away margin on units that would have sold at full price. Mark down too late and you clear at 60% instead of 20%, or you do not clear at all. There is no version of this decision that can be

deferred until the information arrives, and that is why markdown optimisation tools are bought and so often not trusted: the tool is right on average and the buyer remembers the three times it was wrong.

The architectural consequence is cadence, not optimisation. The system must hold a **markdown plan** — planned markdown rand by week by category, sitting inside the open-to-buy — execute dated price changes to hundreds or thousands of stores, **revalue the inventory at the new retail** where the retail method applies, and report actual against plan. Four jobs, usually spread over three systems.

Grocery's equivalent is the **waste markdown** on short-dated perishables: the same mechanism at a daily cadence, decided by a department manager at four in the afternoon, with direct interaction with shrink. Milkwood's R1,620m of markdown is not evenly spread: apparel took 11.4% of its own sales in markdown, supermarkets 2.2%, pharmacy 1.9% and online 1.5%. And a margin *improvement* is sometimes just the absence of markdown — one listed apparel group's gross margin rose 70 basis points while comparable sales grew 1.1% and units 0.5% on 3.8% selling-price inflation [4], which is a margin gain achieved by not marking down, and only possible if the buy was accurate in the first place.

Shrink

Shrink is the difference between the stock the system says you own and the stock a physical count finds. Four causes, and the detection economics differ sharply.

Shrink, by cause — what each costs to find, and where to look first

Milkwood's own measured split, invented and illustrative: R1,110m, 1.9% of sales. There is no published South African shrink figure to compare it with.

CAUSE	RM	SHARE	HOW IT IS DETECTED	COST TO DETECT	WHAT IT IS REALLY
Waste and spoilage	455	41%	Recorded write-offs at the point of waste	Low, if staff record it	A forecasting and ordering failure in fresh, measurable at source
Administrative and process error	266	24%	Negative on-hands, receipts without a purchase order, counts out by a whole case	Lowest of all — visible in data you already own	A data-quality problem in your own systems, and the first project
External theft	256	23%	Observation, tagging, cameras, till exception reporting	Highest, and mostly fixed	The cause most discussed and the dearest to reduce
Internal theft	133	12%	Exception reporting on voids, refunds, no-sales, discounts, cash variance	Cheap per case once built	An authority and audit problem at the front end
Total	1,110	100%	A stock take, months later	The real cost is the delay	A number you cannot benchmark in South Africa

Figure 6.4 Shrink by cause, with the detection economics. Milkwood's split is invented; the shaded rows are the asymmetry that matters — the cheapest cause to find is the largest recoverable one, and the total cannot be benchmarked in South Africa at all.

The asymmetry in that figure is the thing to internalise, because it inverts the usual priority. **External theft is the cause most discussed and the most expensive to reduce.** Administrative and process error is usually the largest recoverable share and costs almost nothing to find, because it is visible in the data you already own: negative on-hands, receipts without a purchase order, counts that disagree by a whole case, transfers posted one way and never the other, unrecorded waste. **The first shrink project in any retailer should be a reconciliation project**, not a camera project.

And now the honest statement, which matters more than any number in this chapter.

> **THERE IS NO PUBLISHED SOUTH AFRICAN SHRINK FIGURE**
>
> South African retailers do not disclose shrink separately in their results, and no industry survey figure could be sourced. **The only usable benchmark is American**: the United States National Retail Federation's National Retail Security Survey put shrink at **1.6% of sales in fiscal 2022**, up from 1.4%, totalling $112.1bn, split external theft 36.15%, internal and employee theft 28.85% and process and control errors 27.29%, across 177 brands representing $1.6tn of sales and more than 97,000 locations [9]. **That is a United States figure and must be labelled as one.** Apply the split to a South African retailer only as a hypothesis: this market's crime profile, cash intensity and informal-market adjacency make the external-theft share plausibly higher. A South African retailer **cannot benchmark its own shrink**, which is a genuine strategic disadvantage and the best argument there is for building internal measurement properly.

What the systems must do follows directly. Perpetual inventory per SKU per location. **Cycle counting** on a risk-weighted schedule as well as full stock takes. A **reason-coded adjustment** ledger with no unexplained adjustments, ever. Negative-stock prevention. Receiving controlled against the purchase order. Return to vendor as a tracked flow. Till exception analytics.

Two signatures tell you the shrink architecture is broken, and you can test for both in an afternoon. **Negative stock** means movements are being posted out of order or receipts are missing. **Phantom inventory** — stock the system has and the shelf does not — is what causes online pick failures and false availability promises, and it is the reason an online grocery operation publishes order fulfilment as a headline metric.

Stock turn, cover and GMROI

Four measures, and they are about the same thing from different angles: whether the money tied up in stock is earning anything.

- **Inventory turn = cost of goods sold ÷ average inventory**, at cost or in units [1].
- **Weeks of supply = on-hand units ÷ average weekly unit sales**; *forward* weeks of supply uses the forecast rather than history [1]. Practitioners say "cover" and "weeks of supply" interchangeably; grocery says days, apparel says weeks.
- **Stock-to-sales = opening inventory ÷ prior-period sales** [1].
- **GMROI = gross margin rand ÷ average inventory at cost** [1].

Stock turn, margin and GMROI — four categories, worked

Milkwood, illustrative. The highest-margin category has the worst return on its stock investment, which is the entire point of the measure.

CATEGORY	SALES RM	GROSS MARGIN	GM RM	AVERAGE STOCK AT COST RM	TURN	COVER	GMROI
Fresh produce	2,600	28.0%	728	54	34.7×	11 days	13.48
Ambient grocery	4,200	19.0%	798	420	8.1×	45 days	1.90
Health supplements	640	38.0%	243	148	2.7×	136 days	1.64
Ladies' outerwear	410	46.0%	189	118	1.9×	195 days	1.60

Figure 6.5 Turn, cover and GMROI worked for four categories. The shaded rows are the lesson: the highest gross margin in the table earns the worst return on its stock investment, and the lowest earns the best.

GMROI is the single best one-number test of whether a category earns its stock investment, because it penalises both low margin and slow turn. The figure makes the point better than any argument: **fresh produce at a 28% margin returns R13.48 of gross margin for every rand of stock, and ladies' outerwear at a 46% margin returns R1.60.** A buyer arguing for more space on margin rate alone is arguing for the wrong answer.

Two traps. First, GMROI is most often computed on the wrong inventory base — period-end rather than average — which flatters a category that was out of stock at the cut-off. Second, both turn and cover need a consistent valuation base, so a retailer running the retail method in one department and the cost method in another (which happens, sometimes inside one group) cannot compare them without a conversion nobody documents.

At group level the arithmetic is coarser and still useful. Milkwood's inventory of R7,310m against R46,940m of cost of goods implies about **6.4 times** turn and **57 days** of cover. For scale, one South African grocer's half-year balance sheet implies roughly 6.2 times and 59 days on the same basis, and the company's own disclosed metric is inventories at 12.9% of trailing sales [10] — my calculation from its figures, not its disclosure.

The cash conversion cycle, which is why supplier terms are a strategy

One more piece of inventory arithmetic, and it is the one that explains how a retailer funds itself.

Cash conversion cycle = days inventory outstanding + days sales outstanding − days payables outstanding. For a cash-paying retailer days sales outstanding is near zero, so the cycle is roughly inventory days minus payable days — and whenever payable days exceed inventory days the working capital is **negative**. The supplier funds the stock.

Milkwood's illustrative position: inventory R7,310m, trade and other receivables R1,180m, trade and other payables R9,640m. On cost of goods of R46,940m that is about 57 inventory days and 75 payable days, so a cycle of about **minus eleven days**. One listed South African grocer shows the same mechanism at scale: inventories of R33,738m against trade and other payables of R43,939m, with payables exceeding inventory by R10.2bn, and the company notes explicitly that its reported period-end cash balance is *before* creditor payments made after the cut-off [10]. That last point is worth reading twice: the reported cash position is partly a function of where the payment run fell relative to the balance-sheet date.

Three consequences. **It funds expansion**, because each new store's opening stock is substantially supplier-funded, so growth generates cash rather than consuming it. **It is the hard floor under a price position** — a retailer that pays in 75 days and turns in 50 has 25 days of free funding it can give away as price, and one that pays in 30 and turns in 60 cannot. And **it breaks on imports**: imported apparel is paid on shipment and sells twelve weeks later, so an importing apparel retailer has a *positive* cycle and funds its own stock, which is part of why apparel gross margins are 41% to 57% and grocery's are 18% to 26% [4][5][7].

The architectural consequence is dull and expensive to get wrong. Payment terms are **per supplier, sometimes per category, sometimes per purchase order**, with effective dates, settlement discounts and rebate interactions. They live in the supplier master and the purchasing terms engine, they drive the payment proposal run, and they are an input to landed cost. A retailer whose terms live in a procurement contract PDF cannot compute a true cost of goods — and cannot answer the Commission's question about differential terms either.

Open-to-buy, the control that actually governs the buying organisation

Everything in Chapter 5 about buying and planning being different jobs comes down to this one piece of arithmetic.

Open-to-buy, worked — one apparel category, one retail period

The control that actually governs a buying organisation. Held in money, not in units. R millions.

THE LINE	R MILLION
Planned sales for the period	82
Planned closing inventory	196
Planned markdowns	9
Less opening inventory	(210)
Planned receipts	**77**
Less commitments already on order	(64)
Open-to-buy — what the buyer may still commit	**13**

A buyer who finds a R20m opportunity has three choices and no fourth: cancel R7m of what is already on order, accept a higher closing inventory and the markdown that follows it, or raise the markdown plan now and tell finance. A system that lets the purchase order through without testing it against this number has removed the control.

Figure 6.6 Open-to-buy, worked. Held in money by category and period, against a stock ledger held in units by SKU and location — and the reconciliation between those two grains is where merchandise planning architecture usually fails.

Open-to-buy is held in money, by category, by period. The stock ledger is held in units, by SKU, by location. Those are different grains and the reconciliation between them is where most merchandise-planning architecture lives and where most of it fails [1].

Three tests you can run in week two.

1. **Ask to see the on-order feed into the planning system.** Open-to-buy is planned receipts *minus commitments already on order*. If the on-order figure comes from a weekly extract, the number is stale by up to a week in a business where a week is a trading period. If it comes from a spreadsheet, it is not a control.

2. **Ask whether a purchase order can be raised that breaches open-to-buy.** If yes, the money plan is advisory, and the business will over-buy. If no, ask what the override process is, because there will be one.
3. **Ask how the money plan and the unit plan reconcile.** The honest answer in most retailers is that they do not, and that a planner reconciles them by hand at period end.

A retailer whose open-to-buy is right has a buying organisation under control. One whose open-to-buy is wrong has either excess stock heading for markdown or empty shelves heading into a season, and in both cases the system was complicit.

Sell-through

Sell-through % = units sold ÷ units on hand at the beginning of the period [1]. It is the core apparel in-season measure: of what I had, how much went? It is measured at option and store-cluster level, weekly, against a planned curve. Above plan in week three is a repeat-buy signal if lead times allow; below plan is a markdown signal.

Grocery barely uses it, because grocery replenishes rather than selling through a finite buy. That single difference is the cleanest demonstration in this book of why one planning system cannot serve a supermarket chain and a clothing chain, and why a multi-format group has two merchandise estates whatever the target architecture slide says.

Like-for-like, and why the definition is a data-model requirement

Like-for-like growth — comparable-store sales, "comps" — is this year's sales over last year's sales for the same stores, including only stores trading in both periods [1][11]. Every retail investor relations deck leads with it. **No accounting standard governs the inclusion rules.**

The contest is entirely in those rules, and there are seven of them:

- **Qualifying period.** How long must a store have traded? Twelve months, thirteen, fourteen? A new store's honeymoon sales inflate the base if included too early.
- **Refurbishment.** A store closed for six weeks and reopened with 40% more space — comparable or not? Most retailers exclude during closure and re-include after a waiting period, and practice varies.
- **Relocation.** Moved 400 metres into a bigger unit is usually new. Moved 40 metres is usually comparable.
- **Space change.** A store that added 20% selling space without closing is still "the same store" under most definitions, which is why like-for-like and space growth must be read together.
- **Online.** Is a store-fulfilled online sale inside that store's comp? One South African grocer fulfils on-demand delivery from 997 stores and another from more than 620 [2][12]; whether those sales sit in the comp changes the number materially.
- **The 53rd week.** One group's financial year was 52 weeks against a prior 53-week year, and the pack presents prior-year turnover on both bases [2].
- **Closures.** Closing loss-making stores *raises* like-for-like mechanically.

And the evidence that the rules bite. One grocer's segment turnover **fell 1.6%** while its like-for-like was **+3.1%** [2] — a 4.7-point gap explained by closures. Another's turnover **rose 12.3%** with like-for-like at 4.5% and new stores contributing 7.8% [13]. A third reported retail sales up 2.4% with like-for-like also 2.4% [6], meaning no net space contribution at all. Milkwood's own illustrative bridge: turnover +4.1%, like-for-like +1.9%, new space +3.2%, closures −1.0%.

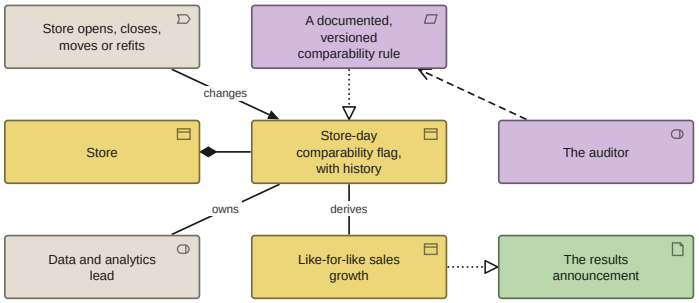


Figure 6.7 Like-for-like growth as a data requirement rather than a report. A store event changes the comparability flag; a versioned rule set realises it; the auditor has an interest in the rule set; and the number ends up in the results announcement.

So like-for-like is not a report. It is a **derived figure requiring a per-store, per-day comparability flag, with history, against a documented and versioned rule set, maintained as stores open, close, move and refit.** It is audited. It appears in the results announcement. And in most retailers it is maintained in a spreadsheet by one person, which is an audit finding waiting to happen and one of the easiest high-credibility wins available to an architect in the first ninety days.

Space productivity

Sales per square metre = net sales ÷ selling area [1]. The South African market measure is **trading density**, and the Clur Shopping Centre Index gives national annualised trading density of **R43,612 per square metre**, with super-regional centres at R53,643 and community and smaller centres at R49,523; rent-to-sales runs 6.6% nationally, 7.2% super-regional and 4.7% community [14].

Milkwood's illustrative position shows why one group number is useless. Its supermarkets trade at about R38,900 per square metre on 2,350 square metres average, at a 2.9% rent-to-sales. Its pharmacies trade at about R83,300 per square metre on 480 square metres, at 6.4%. **The pharmacy's density is more than twice the supermarket's and it pays more than twice the rent percentage** — so the two formats compete for the same shopping-centre space on completely different arithmetic, and the group's blended figure describes neither.

Two further measures matter. **Sales per linear metre** of shelf frontage is the category and planogram measure, and the one used to arbitrate shelf space between suppliers. **Sales per store** is the format and property measure.

And one trap. **Space growth is a growth lever that dilutes productivity.** A retailer growing space faster than sales is reducing trading density by arithmetic, and South African grocers have grown space at 6% to 7% a year [2][13]. The architectural requirement is unglamorous and almost always unmet: trading area per store, per department and per fixture, versioned by refit date, reconciled to the lease's lettable area. That data decays immediately and nobody owns it.

Basket, units per transaction and conversion

Three definitions and one decomposition that is the whole of store performance management [1].

- **Average transaction value = net sales ÷ transactions.**
- **Units per transaction = units ÷ transactions.**
- **Conversion rate = transactions ÷ foot traffic.**

And then: **sales = traffic × conversion × average transaction value**, where **average transaction value = units per transaction × average unit retail**.

That matters because a sales decline is one of four completely different problems with four completely different responses. Fewer people came — a catchment, marketing or competitor problem. The same people came and did not buy — a range, availability, price or service problem. They bought fewer things — a basket and promotion problem. They bought the same things cheaper — a mix and price problem. An executive team that cannot split a sales movement into those four is guessing, and the systems usually cannot split it because nobody counts the traffic.

The South African evidence shows how different the shapes are. One grocer grew supermarket transactions 5.6% with basket size up only 1.7% [8] — growth driven by frequency. A voluntary-trading group grew transactions 0.54% to nearly 300 million with average basket up 2.29% [6] — the opposite. One apparel group grew units sold 0.5% on selling-price inflation of 3.8% [4] — all of the growth from price. Those are three different businesses with three different problems, and the single word "sales" describes none of them.

Two architecture notes. **Conversion requires footfall counting**, which requires door counters, which most South African grocers do not have or do not publish. And **online conversion is measurable natively and is an order of magnitude lower** than store conversion, so the two must never be compared without stating the denominator — a mistake that appears in roughly every omnichannel board paper.

Supplier income

This is the most argued-over number in a retailer's profit and loss account, an accounting-standards minefield, and a data problem nobody owns. It deserves more of your attention than its size suggests, and its size is already large.

What it is: money received from suppliers that is *not* a reduction in the invoice price of goods.

Supplier income — R2,448m, and 84% of trading profit

Milkwood, invented. Six types, six different accounting answers, and one data problem nobody owns.

TYPE	RM	SHARE	THE ACCOUNTING QUESTION	THE DATA IT NEEDS
Volume rebate / overrider	1,102	45%	A reduction in cost of goods, deferred to the stock still on hand	Purchases by supplier by period against a tier, attributed to specific stock
Promotional funding and co-op advertising	710	29%	Cost reduction if tied to goods; income if it is a genuine service	A key from the funding back to the promotion that earned it
Listing fees and new-line support	196	8%	Generally a cost reduction, spread over the listing period	The listing event, the SKU set and the period
Settlement discount	171	7%	Economically interest; frequently booked as margin	Payment date against the agreed terms
Markdown support and price protection	147	6%	A cost reduction against the marked-down stock	The markdown event, its units and its depth
Space and display payments	122	5%	Usually a service fee, so income	The planogram position and the period it was held
Total	2,448	100%	Four questions: timing, classification, estimation, and whether the terms are lawful	An effective-dated trading-terms agreement per supplier — which is what almost nobody has

Figure 6.8 Supplier income by type for Milkwood, invented: 4.2% of sales and 84% of trading profit. Six types, six accounting answers, and one requirement — an effective-dated trading-terms agreement per supplier — that almost nobody has.

Milkwood's R2,448m of supplier income is 4.2% of sales and **84% of trading profit**. Sit with that for a moment. An error of one tenth in the accrual moves trading profit by 8%. A classification change between cost reduction and other income moves gross margin, inventory value and segment reporting simultaneously. This is not a finance back-office matter; it is a line big enough to be the difference between a profit and a loss.

Four reasons it is contested, and all four are architecture problems.

Timing. Rebates earned on *purchases* can be recognised before the goods are *sold*, which pulls profit forward. The correct treatment defers the rebate attributable to stock still on hand — which requires the rebate to be attributed to specific stock. Most systems do not attribute it to anything.

Classification. Is it a reduction in cost of goods, and therefore in inventory value, or is it other income? Volume rebates are generally a cost reduction; a genuine service fee is income. The distinction determines gross margin, inventory value and segment reporting.

Estimation. Tiered rebates on annual volumes must be accrued monthly on an estimate of year-end volume, and the estimate is the retailer's judgement. A buyer who believes the tier will be reached and a finance team that does not have a genuine disagreement with a profit consequence.

It is a competition-law object. The Competition Commission's Grocery Retail Market Inquiry found that **rebates for in-store functions are generally unavailable to wholesalers and buyer groups**, that **some rebates have no efficiency justification**, that the evidence on pass-through to consumer prices was mixed, and that **back-end margins lack transparency**. Its recommended remedy is **uniform, objectively justified trade terms not based on volume** [15]. That makes supplier income reportable by type and by customer, which is a data requirement, not a policy.

> **NORTH AMERICA REGULATES THE CONDUCT; SOUTH AFRICA RECOMMENDS**
>
> South Africa has a Commission *recommendation* for uniform terms, with voluntary compliance sought by 25 May 2020 [15]. Canada has a registered code: the **Canada Grocery Code took effect 1 January 2026**, registered by Costco, Empire, Loblaw, Metro and Walmart Canada plus more than 150 other retailers and suppliers, with a Dispute Resolution Management Process from the same date, administered by the Office of the Grocery Sector Code of Conduct and funded by membership fees scaled to company size and market share, addressing unexpected fees, cost increases imposed without notice, and late payments [16]. Same commercial problem, different enforceability — and different data obligations.

So the requirements. Supplier income must be **agreement-based**: an effective-dated trading-terms agreement per supplier, carrying type, basis, rate, tier, period and condition. **Attributable**: to a supplier, a category, a SKU set, a promotion and a period. **Accrued and settled separately**: accrual, claim, credit note, deduction from payment, and reconciliation of the difference. **Auditable**: the evidence that the condition was met. And **reportable by type and by customer**, for the uniform-terms question.

> *A retailer whose supplier income lives in a buyer's spreadsheet and a credit-note queue cannot pass an audit, cannot answer the regulator, and cannot tell you its true gross margin.*

Private label economics

Private label is product sold under the retailer's own brand, sourced from a contract manufacturer or an own factory. The retailer takes the brand owner's role: specification, packaging, quality, liability and demand risk.

The economics look simple and contain one trap. Private label carries a higher *percentage* gross margin than the equivalent branded product — typically five to fifteen points — at a lower shelf price, because the brand owner's marketing and margin are removed. **But it earns no supplier income.** No rebate, no listing fee, no co-op advertising. So the comparison against a branded line must be **margin after supplier income**, not gross margin, and a retailer that cannot compute margin after supplier income by SKU cannot make the private-label decision rationally. Given the state of most supplier-income data, very few can.

It also moves working capital and risk onto the retailer: minimum order quantities are factory-sized, lead times are longer, and obsolete packaging is the retailer's loss. South African penetration is material where it is pursued — one pharmacy group's private label is 31% of front-shop sales, and one voluntary-trading group reports 23% penetration across 5,169 SKUs [17][6].

The architectural requirements are a product-development lifecycle that no merchandise system has: brief, specification, sample, approval, artwork, packaging copy, nutritional and regulatory data, allergen declaration and country of origin; supplier quality management and audit; own-brand GTIN allocation under the retailer's own GS1 prefix; **recall to batch level**; and the liability that comes with being the responsible party under the Consumer Protection Act.

Promotional economics, funded and unfunded

A promotion is an effective-dated price or offer rule with a mechanic, a scope, a period and a funding arrangement. The mechanics are a finite list: simple price reduction, multibuy, mix-and-match across a set, percentage off, buy-one-get-one, spend threshold, member-only price, bundle, coupon and cashback.

Funded against unfunded is the only question that matters commercially.

	Funded	Unfunded
Who pays	The supplier, through a lower cost price for the period, a lump sum, a per-unit rebate on promoted volume, or markdown support	The retailer, out of its own margin
Margin effect	Rate may be maintained or improved	Rate falls by construction
Justification	The commercial deal	Incremental volume, basket uplift, traffic, or competitive necessity
What it needs	A supplier agreement, evidence, and a key from the funding to the promotion	A measured incrementality case
How it fails	The calendar locks before funding is confirmed, so the promotion runs unfunded and nobody notices until the claim is refused	Uplift is measured instead of incrementality, so the promotion looks successful and was not

That last cell is the measurement problem and it is universal. Promotional performance requires **incrementality**, not uplift: sales during the promotion, minus the baseline that would have occurred anyway, minus **cannibalisation** of substitute lines, plus **halo** on complements, net of **pantry loading** — the shopper buying forward and depressing later weeks. Most retailers measure uplift because uplift is easy and incrementality is hard, and therefore systematically overstate promotional effectiveness.

Scale makes this urgent rather than academic. Promotions were **40% of one South African grocer's supermarket turnover** in a recent half-year, up from 37.9% [8], and one voluntary-trading group attributed its margin pressure explicitly to promotions and rising costs [18]. **At 40% of turnover the promotion engine is not a marketing tool; it is a primary pricing system in the sale path.** It must be defined once and executed identically in store,

online, in the app and in the loyalty message; evaluated at the till in milliseconds with member status as an input; **priced into the forecast so the promoted volume is actually ordered** — the single commonest cause of promotional out-of-stock; funded against a supplier agreement; and reconstructable as at any past date, because internal selling-price inflation, margin attribution and supplier claims all depend on it.

Omnichannel economics

The structural fact is short. **An online grocery order costs the retailer money the store order does not** — picking labour, packing materials, and either a delivery or a collection operation — all against the same shelf price.

What an online grocery order actually earns

Milkwood Now, invented and illustrative, per order in rand. Same shelf price, same gross margin, different cost to serve.

PER ORDER	IN STORE, R420 BASKET	DELIVERED, R420	DELIVERED, R980	COLLECTED, R980
Sales	420	420	980	980
Gross margin at 24.2%	102	102	237	237
Store cost to present and sell, at 18.8% of sales	(79)	(79)	(184)	(184)
Pick and pack	—	(35)	(55)	(55)
Last mile	—	(48)	(48)	—
Payment and order handling	(2)	(12)	(18)	(18)
Delivery or collection fee earned	—	29	29	—
Result	21	(43)	(39)	(20)
As a percentage of the basket	5.0%	(10.2)%	(4.0)%	(2.0)%

Figure 6.9 What an online grocery order actually earns, per order, illustrative. The shaded rows are the answer: basket size and collection both help, and neither closes the gap on picking labour alone.

Five conclusions follow, and the figure contains four of them.

Store picking is the most expensive model per unit and the cheapest to start. It needs no capital, uses existing stock and existing labour, and reaches the whole estate immediately. That is exactly why one South African grocer serves on-demand delivery from 997 stores [2] and another from more than 620 [12]. It is also why neither can be profitable at scale on picking labour alone.

Click-and-collect is structurally cheaper than delivery, because the last mile is free: the customer provides the vehicle. The picking cost is identical and the drop cost disappears. So one apparel retailer's **more than 55% in-store collection rate on online orders** [4] is not a convenience statistic. It is a margin statistic.

Delivery economics are a density function, not a volume function. Cost per drop falls with drops per route in a smaller area, so a city becomes profitable before a suburb and a suburb before a town. A national roll-out plan that ignores drop density is a plan to lose money in a defined sequence.

Basket size is the lever, because pick and drop costs are largely fixed per order. Compare the second and third columns of the figure: the same delivered model loses R43 on a R420 basket and R39 on a R980 basket, which is the whole case for minimum order values, and for the subscription that removes them — Amazon Prime in South Africa at R59 a month or R399 a year removes minimum order thresholds [12], which is the only mechanism that makes small baskets viable.

And dark stores and automated fulfilment centres trade capital for unit cost, and only pay back above a volume threshold in a defined catchment.

> **THE CONVENTION IS THE ARGUMENT**
>
> Allocating full store cost to an online order, as the figure does, is one of two defensible conventions. The other treats the store's cost as sunk — the building and the shelf are there anyway — and gives a positive contribution on every line. Both are defensible and they produce opposite answers, which is exactly why the two camps inside a retailer never agree. **The channel P&L convention is a modelled decision that must be made explicitly, written down, and applied consistently in every system that reports a channel margin.** If it is not, the e-commerce director and the finance director will each be right.

The only well-quantified public cost model for online grocery fulfilment is North American, labour-only, in US dollars, with no analysis date stated and figures that do not internally reconcile: in-store valet picking at 114 labour seconds per unit against 12 seconds for a customer shopping the store themselves, an automated micro-fulfilment centre at 64 seconds, a manual dark store at 76 and an automated customer fulfilment centre at 59; home delivery at $11 to $12 per drop in suburban and metropolitan areas, with a range of $10 to $25, falling with drop density [19]. **Treat those as a United States shape, not a South African cost.** The South African evidence is thinner and more useful: one grocer states that its 25.3% supermarket gross margin is *after* absorbing its on-demand delivery costs [8][12] — the online cost sits inside the reported grocery margin, not below it — and its on-demand business reached R25.5bn, 11.1% of supermarket sales, with 94.0% on-time delivery and 96.8% order fulfilment [2][12]. Online shares elsewhere: one group's food business at 7.3% and its fashion and homeware at 6.3% [7]; one apparel group at 14.8% of group and 8.2% in Africa [5]; the benchmark pure-play's core e-commerce unit at about a 1% margin [12].

Finally, the requirements the arithmetic implies. A **single available-to-promise** view that reconciles store on-hand to a promise, discounted for phantom inventory. Pick-list generation and picking tooling with substitution rules and customer approval. **Slot capacity as a labour-constrained resource**, not a configuration number. Route planning. **Weight-variable item repricing** — the picked 1.3 kg chicken is not the ordered 1.2 kg. Age-restricted handling at handover. And **authorisation and capture at different amounts**, with a tolerance and a reconciliation path, which is a flow most retail finance systems were never built for.

Returns

Return rate % = return units ÷ sales units [1]. Grocery returns are negligible, store apparel returns are modest, and online apparel returns are the dominant cost, because the customer cannot try the garment on and frequently orders two sizes deliberately. No reliable South African return-rate figure could be sourced and the global figures in circulation come from low-quality aggregators, so this book prints none.

The cost of a return is not the refund. It is inbound carriage; receipt and inspection; grading into resaleable, repack, markdown or write-off; restocking to the right location; the lost margin where the item is now out of season; the payment processing cost on both legs; and the customer-service handling. A garment that arrives back in week 11 of a 14-week season is terminal stock at whatever clearance price exists — so the return has not cost the refund, it has cost the refund plus the difference between the full price and the clearance price.

And then the South African clock. Section 56(2) of the Consumer Protection Act 68 of 2008 gives the consumer **six months from delivery** to return goods that fail the quality standards in section 55, "without penalty and at the supplier's risk and expense", with the remedy — repair, replacement or refund — **at the consumer's election** [20]. That is a statutory entitlement per sale line, running from the delivery date, independent of retailer policy, and the retailer is liable irrespective of who made the product. So the return architecture needs a **statutory clock per sale line**, a link from the return to the original sale line carrying price paid, tender and loyalty accrual, grading and disposition with reason codes, return to vendor where the terms allow, and refund controls — because refunds are the largest single internal-fraud vector at a till.

Two accounting choices that change the data model

Both of these are decided by finance, usually before an architect arrives, and both determine what the data model has to be. Getting either wrong at design time is not recoverable by configuration.

The retail method against the cost method. Under the **retail method**, inventory is tracked at *selling price* all year and converted to cost only for the financial statements and tax, using a cumulative markup. Markdowns are highly visible and flag problems early, shrink is easy to identify because a count at retail compares directly to a book retail position, and conscientious markdown handling takes care of lower-of-cost-or-market — but it demands strict record-keeping. Under the **cost method**, inventory is tracked at cost by specific identification, average cost, FIFO or LIFO. Markdowns are not visible in the same way, because a price reduction does not change average cost; shrink requires item-level control; and year-end valuation needs extra methods that are more open to dispute. About **70% of retailers use the retail method** — mostly department stores, fashion and many hard-goods retailers — with the remainder, mostly grocers, on cost [21].

Why an architect must care: the method determines **whether the stock ledger's primary value is retail or cost, whether a price change is an inventory revaluation event, how shrink is computed, and which package configurations are viable at all.** A markdown under the retail method reduces inventory value the moment it is applied, with no physical movement whatsoever. Under the cost method it does not. That is the same business event producing two different ledgers, and a group that runs both — which happens, sometimes by department inside one group — cannot compare gross margin across them without a conversion nobody documents. Expect an imported package to assume the method you do not use.

IFRS 16, and why "trading profit" is the number management uses. Every store lease is a right-of-use asset and a lease liability, so rent splits into depreciation and finance cost. Milkwood's R2,898m of trading profit becomes R1,568m before tax largely because of R1,142m of lease finance. At scale the effect is larger than the debt book: one grocer carried lease liabilities of R50,372m against borrowings of R6,746m [10], and one group sold and leased back three distribution centres for about R2bn on fifteen-year triple-net leases [22], converting an asset into a liability and a rental stream.

Three consequences. Retail "trading profit" is reported *before* lease finance precisely because management treats rent as an operating cost while IFRS 16 does not — **so there are two defensible store profit and loss accounts and both appear in the same results pack**. Store-closure decisions carry a lease-exit cost that dominates the merchandise economics. And lease data — term, escalation, options, break dates, area, turnover clauses — is **accounting-critical master data**, not property-team reference data.

The four numbers a retail executive watches

Strip away everything above and a retail executive team watches four numbers, weekly.

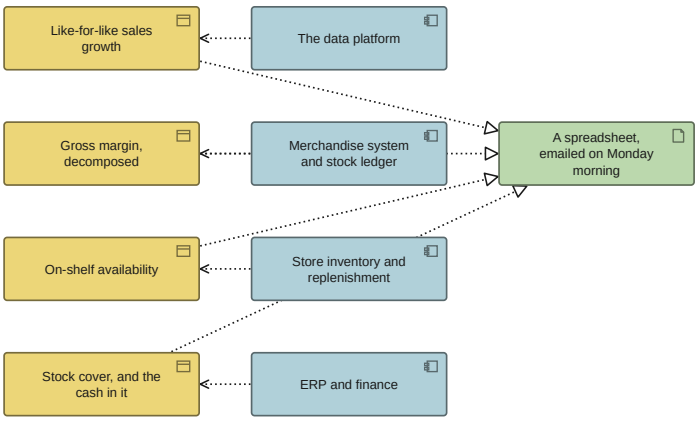


Figure 6.10 The four numbers a retail executive watches, the system that produces each, and the artefact they all actually arrive in. That spreadsheet is the most important application in the business and it is not in the portfolio.

Like-for-like sales growth, because it is the only growth figure that is not flattered by opening stores. Produced by the data platform, from a comparability flag that is maintained by hand.

Gross margin, and its decomposition, because margin is where every error in the business eventually lands. Produced by the merchandise system and the stock ledger, and decomposable in very few retailers.

On-shelf availability, because it is the promise the whole model rests on and the one measure the customer experiences directly. Produced by store inventory and replenishment, against a store stock figure that is wrong by the amount of the store's shrink.

Stock cover, and the cash inside it, because inventory is the asset and the working capital at the same time. Produced by ERP and finance, on a valuation base that may differ by department.

And now the honest point, which is the reason that figure has a third column.

> In most retailers all four numbers arrive in a spreadsheet. Someone extracts from three systems, adjusts for the things the systems get wrong, applies the definitions nobody has written down, and emails it on Monday morning. That spreadsheet is the most important application in the business and it is not in the application portfolio.

Do not sneer at it, and do not propose replacing it in month one. It exists because the systems cannot produce the number, and the adjustments in it are a precise specification of what is wrong with the estate. Get a copy. Read the formulas. Each manual adjustment is a defect with a name, an owner and a quantified cost — which is the best requirements document you will be given, and the only one that was written by somebody with nothing to sell you.

THE CHAPTER IN TEN LINES

- Gross margin divides by sales and markup divides by cost; confusing them is the commonest arithmetic error in retail, and every growth rate needs its week basis attached.
- Gross margin moves without anyone deciding anything, so any movement must decompose into cost, price, mix, markdown, shrink, supplier income and promotional funding — which is a data model, not a report.
- A markdown is permanent and revalues stock; a discount is temporary and does not. A markdown is also a decision about the future taken with data about the past, so the architecture is cadence rather than optimisation.
- Administrative error is the largest recoverable share of shrink and the cheapest to find, so the first shrink project should be a reconciliation project — and there is no published South African shrink figure to benchmark the total against, only a United States one.
- GMROI penalises both low margin and slow turn, which is why the highest-margin category in a retailer is frequently the worst investment of stock.
- Open-to-buy is held in money by category and period while the stock ledger is held in units by SKU and location, and a purchase order that can breach it means the control is not a control.
- Like-for-like has no governing standard, so it is a derived figure needing a per-store, per-day comparability flag with history against a versioned rule set.
- Supplier income is 4.2% of sales and 84% of trading profit in the worked example, and it needs effective-dated trading-terms agreements, attribution to stock, separate accrual and settlement, and reporting by type and by customer.
- Store-picked online fails on fixed cost per order against a variable basket, and whether a channel looks profitable depends on a store-cost allocation convention that must be chosen explicitly.
- All four of the numbers an executive watches arrive in a spreadsheet, and the manual adjustments in it are the best requirements document you will be given.

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CHAPTER 7

The operating model

A retailer is a few hundred shops that each have to keep selling when everything else fails, coordinated by a centre that may or may not be allowed to tell them what to do. This chapter is how one actually runs: the value chain, who owns the store, what each channel costs, the trading calendar, the overnight clock, and the four questions to ask of any change.

Four hundred and twenty shops and one price list

A bank has branches. An insurer has brokers. A retailer has shops, and the difference is not one of scale. A shop holds stock the group paid for, charges prices the group set, employs people the group rosters, and is the only place in the whole organisation where money is actually collected. Four hundred and twenty of them, each with its own power supply, its own network link and its own opinion about what is on its own shelves.

That is the operating model problem in one line. Almost everything a retailer decides centrally arrives at a shop as an instruction, and almost everything it knows comes back from one.

Milkwood Retail Group, which is invented, carries the worked examples in this book. It is a South African multi-format group: about 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm selling to independent traders, and an online business picked out of the supermarkets. Nothing about Milkwood is real. The shape is real, and the shape is the point — grocery, pharmacy and apparel have genuinely different architectures below the till, and a group that runs all three has to decide, repeatedly and usually badly, what is shared.

Two questions decide a retailer's operating model, and an architect should ask both in the first week.

Who owns the shop? If the group owns it, the group's systems are systems of record. If a franchisee or an independent member owns it, they are systems of *offer*, and the centre persuades rather than instructs.

What time is it? Not the clock on the wall. The retail week, the retail period, the position in the trading year, and how many hours remain before the first store opens tomorrow. Retail time is its own object, and it governs everything.

The value chain is six clocks, not six steps



Figure 7.1 The retail value chain: six groups of work, each a pair of processes. Read it as what the business decides, not as a sequence — all six run at once, on six different clocks.

Read the figure as what the business does, grouped by what each group decides. The six groups are stable across formats; what changes between a grocer and an apparel chain is the speed each one runs at.

Range and plan decides what the retailer will sell and how much money it may spend doing so. Two separate jobs: the buyer decides *what*, the merchandise planner decides *how much, in money, and when*. They have separate systems and frequent arguments, and any design that merges the two roles has mis-modelled the organisation. The outputs are a range — the list of articles authorised for a cluster of stores for a period — and an open-to-buy, which is the constraint that stops the business committing more than it can sell.

Buy turns a decision into a commitment. Cost price, trading terms, supplier income, and then the purchase order with its surprisingly elaborate state machine: created, approved, transmitted, acknowledged, amended, part-received, fully received, closed, cancelled. Note what lives here and nowhere

else: goods *for resale* run through the merchandising system and never touch ERP procurement, which handles shelving, uniforms and IT on a different approval model with a different supplier master. Architects routinely scope "procurement" and describe the smaller half.

Supply moves it. Receive at the distribution centre, hold, pick a store order, load, route, deliver. Three distribution-centre archetypes behave differently enough to be different system configurations: the stockholding ambient DC, the cross-dock that has no buffer and therefore no tolerance for a late inbound, and the cold-chain DC that picks first-expired-first-out and carries a traceability obligation.

Execute is the part head office forgets it depends on. The shelf filled, the planogram followed, the label right, the promotional display actually built. A promotion's forecast is partly an *execution* forecast, because the uplift depends on whether somebody built the end-of-aisle display. The single biggest failure in space planning is that the physical store does not match the planogram and nobody measures the gap.

Sell is the till, the online order and the dispensary. It is the only group with a hard latency budget and the only one that must keep working when every other system is unreachable.

Settle pays the supplier, claims the supplier income, values the stock and produces the numbers the board reports. It is also where the retail method of accounting makes a price change into an inventory revaluation event, which is covered in Chapter 6 and matters here because it means **Settle is affected by decisions made in Execute**, not only by movement.

Now the part that is easy to miss. These six are not a pipeline that an article flows down. They are six clocks running at different speeds over the same article at the same time.

Group	Its clock	What that does to the architecture
Range and plan	Annual to seasonal; a category review cycle measured in months	Needs comparable history, so a hierarchy reclassification is destructive
Buy	Weekly for local grocery; twenty to thirty weeks for imported apparel	Two genuinely different buying systems, not one with a parameter
Supply	Nightly, against contractual supplier cut-offs	Sits on the overnight critical path; a late order is a lost delivery day
Execute	Daily, store by store, with a shift pattern	Task compliance is the measure, and it is measured by sampling
Sell	Sub-second at the till; sixty minutes on an on-demand online order	Rules out a batch interface anywhere on that path
Settle	Period, quarter, year, under the 4-5-4 calendar	Needs every other clock expressed in retail periods

An architect who treats the value chain as a sequence will design interfaces that assume each stage finishes before the next begins. They do not. At eight o'clock on a Tuesday evening Milkwood is simultaneously reviewing next winter's range, placing tomorrow's store orders, filling tonight's shelves, selling, and closing last period's stock ledger. The interfaces have to carry all six at once, which is why the integration estate is the sediment described in Chapter 12 rather than a layer.

Who owns the shop

This is the question that most changes what an architect may design, and South Africa has all three answers in its top five grocers.

Corporate-owned. The group owns the store, the stock and the price. Every system in the estate is a system of record, replenishment is a central control, and a price change is a configuration push. This is the model every imported reference architecture assumes.

Franchised. The franchisee owns the business; the group owns the brand, usually the supply chain, and whatever systems the agreement compels. Pick n Pay reported **992 company-owned and 620 franchise stores** at 1 March 2026, with franchisees sourcing inventory through the group's central distribution network, paying a royalty for brand, systems, supply chain and digital capability, and funding their own capital, staff and operating costs [2]. Shoprite runs OK Franchise with 614 stores, where the group's revenue is the wholesale sale rather than the till [24]. Franchise like-for-like growth at Pick n Pay was 0.9% against 3.9% for company-owned stores in FY2026 — a 300 basis point divergence inside one brand [2].

Voluntary trading. The strongest form of devolution, rare outside Europe and southern Africa, and a top-five South African grocery model. SPAR stores are independently owned; SPAR describes the model as explicitly *not* a franchise; SPAR acts as wholesaler and distributor and earns its income from selling goods to the stores; retailers join by signing a membership agreement with a non-profit Guild and agree to its standards; and **retailers may stock their stores from any supplier**, so each store has its own range [4]. There is no royalty on turnover [5]. The business is literally measured by how much of its members' purchasing it captures: SPAR reports "retailer loyalty" of 78.6% as a headline operating metric [6].

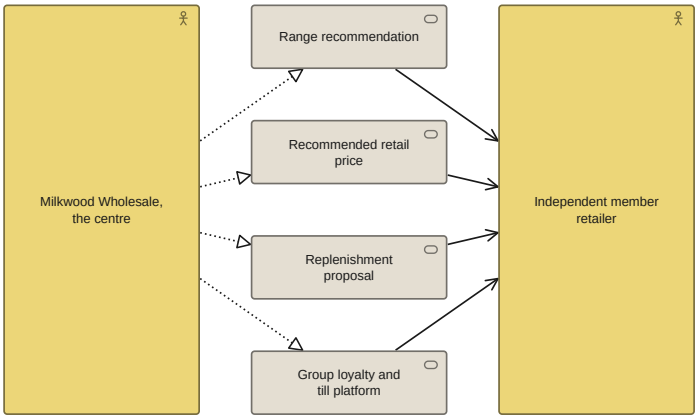


Figure 7.2 A voluntary-trading boundary, drawn honestly. The centre realises four services; every relationship crossing to the member is Serving, never Assignment, because the centre does not perform the member's work and cannot compel it. The member owns the right-hand box and everything inside it.

Look at the relationships in that figure, because the relationship type is the whole finding. Nothing crosses the member boundary as an instruction. Every arrow is a **Serving** relationship from a service the centre realises to a process the member owns. In ArchiMate terms the centre cannot be assigned to the member's process, because it does not perform it and cannot compel it.

> **WHAT DEVOLUTION REMOVES**
>
> In a voluntary-trading or heavily franchised estate the group **does not have an inventory position, a price or an assortment to master** for a large part of its turnover. It has a wholesale order book and whatever the member chooses to share.
>
> - **No group item master.** The member owns the range decision. "Single source of truth for assortment" is advisory data at the member boundary.
> - **No group price file.** Members set shelf prices. National price compliance is a reporting and persuasion problem, not a configuration push [4][5].
> - **No group stock ledger.** The centre knows what it shipped, not what the member holds, unless the member shares point-of-sale and inventory data — by agreement, in heterogeneous formats, with gaps.
> - **No mandated till.** A corporate chain rolls out one point-of-sale platform. A voluntary-trading group integrates many [5].

Three consequences follow for the architecture, and they are structural rather than inconvenient.

Entitlement replaces control. A franchisee or member gets a defined subset of group systems across a contractual, auditable boundary. That is an identity and entitlement problem with a commercial contract behind it, and the contract — not the role model — is the authority. Data rights have to be explicit: who owns the member's point-of-sale data, the loyalty data and the customer data, and on what basis any of it crosses the boundary. In South Africa that last part is a Protection of Personal Information Act question before it is an integration question.

The store becomes a customer, not a location. This sounds like modelling pedantry and is not. Oracle's merchandising location model allows a "store" to represent a physical selling location, a virtual selling location such as a website, *or a customer location such as a franchisee* [7]. Stock is sold to the franchisee rather than transferred to it, which means a sales transaction, a receivable, a credit limit and an ageing — none of which exist for a corporate store. The group now carries member credit risk a corporate chain does not have.

Three revenue recognition patterns live in one group. Own-store retail sales, wholesale sales to franchisees and members, and royalty income, each with different timing, different VAT treatment and different credit exposure. Milkwood has all three: corporate supermarkets, a franchised pharmacy tail, and a cash-and-carry arm whose customers are independent traders and spaza shops. One general ledger, three revenue models, and a finance team that can tell you which is which long before the data model can.

> **PUT THE OWNERSHIP MIX ON PAGE ONE OF ANY RETAIL ARCHITECTURE**
>
> Not in an appendix. The first question a reviewer should be able to answer from your document is what share of turnover the group can actually control. If you have modelled a central pricing capability for an estate where 40% of stores set their own prices, you have modelled a product the business cannot buy.

Channels, formats, and what each costs to serve

A **format** is a kind of shop: a hypermarket, a supermarket, a discounter, a convenience store, a pharmacy, a forecourt shop, a cash-and-carry. A **channel** is a way of buying: walk in, order online for delivery, order online and collect, order through a marketplace, buy on account from a wholesale counter. Retailers run many of both, and the architecture below the till diverges by format far more than it does by channel.

Concern	Grocery	Apparel	Pharmacy	Cash-and-carry
Item lifecycle	Years, stable	One season, disposable	National medicine code, regulated	Years, stable
Planning grain	Category money, plus unit by article and store	Option money, plus size curve	Dispensing volume, plus front shop	Branch, plus unit of measure
Replenishment	Nightly forecast and order	Pre-season allocation, in-season transfer	Usage-based with scheduled-stock controls	Branch order, direct to trader
Who sets the price	The retailer	The retailer	**The state** — single exit price plus a gazetted fee formula	The retailer, plus quotation
Main margin lever	Promotion funding	**Markdown**	Front-shop mix and private label	Mix and trading terms
Dominant loss	**Waste and shrink**	**Markdown and returns**	Expiry and scheduled-stock discrepancy	Breakage and credit
Third-party money	Supplier income	Supplier income, lower	**Medical scheme reimbursement**	Trade credit

The honest reading: a pharmacy and an apparel chain share almost nothing below the point of sale. What they genuinely share is the transaction log, tender and settlement, customer and loyalty identity, the general ledger and tax, payroll and rostering, the store and property master, and the in-store technology estate. Everything else is a format decision. The design rule that follows is one shared commerce and customer layer, one shared finance layer, and **separate merchandising and operations stacks per format family**. Groups that have tried to run one merchandising system across grocery, apparel and pharmacy have ended up with a grocery system plus two sets of workarounds.

Channels are where the cost-to-serve argument happens, and where an architect should be careful.

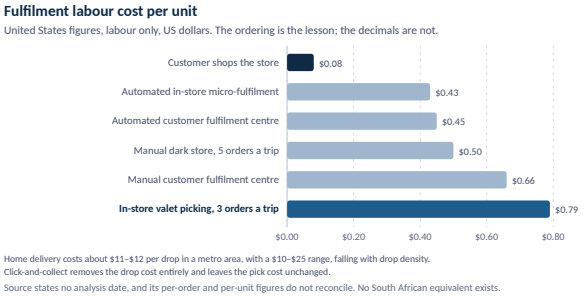


Figure 7.3 Fulfilment labour cost per unit, United States figures in US dollars, labour only [8]. The source gives no analysis date and its per-order and per-unit numbers do not reconcile. Read the ordering, not the decimals — and note that no South African equivalent exists.

Those are United States labour-only figures, in dollars, from a model that does not state its analysis date and whose per-order and per-unit numbers do not reconcile [8]. They are in the book because they are the only publicly available cost model of online grocery fulfilment this research could find, and because the *ordering* is more useful than the values. Read four conclusions from it and ignore the decimals.

Store picking is the most expensive model per unit and the cheapest to start. It needs no capital, uses stock and labour that already exist, and reaches the whole estate at once. That is exactly why South African grocery went that way: Shoprite serves Checkers Sixty60 from 997 stores and Pick n Pay fulfils asap! orders from more than 620 [2][23]. In-store fulfilment being the dominant South African model has a direct architectural consequence — **store systems, store labour and store inventory accuracy are the binding constraints on online growth, not the website.**

Click-and-collect is structurally cheaper than delivery, because the customer supplies the last mile. The picking cost is identical and the drop cost disappears. Mr Price reported more than 55% of online orders collected in store in FY2026 [9]; that is a margin statistic dressed as a convenience statistic. It is also a different order-management flow with a different failure mode — the uncollected order decaying on a shelf with no system expiry.

Delivery economics are a density function, not a volume function. Cost per drop falls with drops per route in a small area, so a city becomes profitable before a suburb and a suburb before a town. A capacity model that treats delivery as a national service will be wrong in both directions at once.

Basket size is the lever that makes any of it work, because pick and drop costs are largely fixed per order. Shoprite states its 25.3% Supermarkets RSA gross margin is reported *after* absorbing Sixty60 delivery costs [9][23] — the online cost sits inside the grocery margin rather than below it, which is worth knowing before anyone asks you to model online as a separate business.

One more number belongs here, because it sets the ceiling on all of the above. South African retail is overwhelmingly shopping-centre retail, and the Clur Shopping Centre Index for the second quarter of 2026 puts annualised trading density at **R43,612 per square metre** nationally with base rent at **6.6% of sales** — 7.2% in super-regional centres, 4.7% in community centres [10]. A retailer on a 25% gross margin and a 6% trading margin that pays 7% of sales in base rent has given away more than its whole trading margin before paying a single wage. That is why the format mix keeps shifting to smaller, cheaper, closer shops, and why trading area per store is a data item worth checking: nobody owns it after the store opens, and it is reliably wrong.

> **DO NOT QUOTE A COST TO SERVE YOU CANNOT SOURCE**
>
> There is no published South African cost-to-serve per channel. If a programme business case needs one, get the number the finance team will defend in writing, label it as theirs, and date it. An unsourced cost per order repeated in three decks becomes a fact in six months, and then somebody closes a channel with it.

The trading calendar

Retail time is not calendar time. This is the section architects arriving from banking or insurance under-weight most, and it is the cheapest thing to get right.

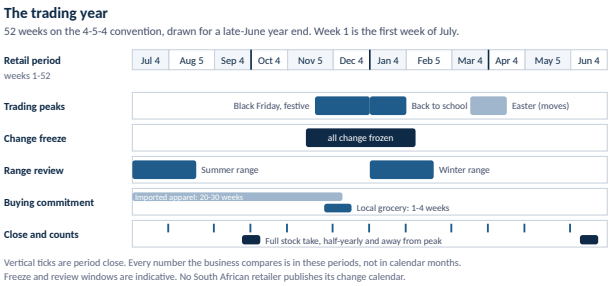


Figure 7.4 The trading year on the 4-5-4 convention [1], drawn for a late-June year end. Peak, the change freeze, the review windows and the import lead time are the four bands that will constrain your programme plan. Freeze and review windows are indicative; no South African retailer publishes its change calendar.

The 4-5-4 convention. The National Retail Federation publishes the 4-5-4 calendar as a **voluntary guide**: the year divides into months on a four-week, five-week, four-week pattern so that comparable months contain the same number of Saturdays and Sundays and holidays stay aligned [1]. It came out of informal industry discussion in the 1930s and into wide use in the 1940s, because as weekends grew as a share of sales a varying number of weekends per calendar month made month-on-month comparison meaningless [1].

The trading week. Monday to Sunday, or Sunday to Saturday, depending on the chain — and in a multi-format group, depending on the format. Food businesses commonly run Monday to Sunday; fashion businesses commonly run Sunday to Saturday. In a group that has both, Monday's sales land in different weeks in different systems, and every comparison built across the two is quietly wrong by one day.

The 53rd week. Fifty-two weeks is 364 days, so a day accumulates every year and a leap year adds another. A 53rd week is needed roughly every five to six years on no fixed schedule. The Federation's rule: if four or more days remain in January after the 52-week layout, a 53rd week is added **for sales reporting only**. 2016 had three leftover days and no 53rd week; 2017 had four and got one. 2006, 2012, 2017 and 2023 were 53-week years [1]. For comparability the following year either restates by shifting each week back one and ignoring the first, or ignores the 53rd week entirely [1].

Period end is the close of a 4-5-4 month: stock ledger reconciled, supplier income accrued, margin reported, plan compared to actual. It is not the calendar month end, which means the finance close and the statutory calendar disagree by up to a fortnight, and the mapping between them is reference data somebody has to own.

Three South African year ends, three definitions. Pick n Pay's FY26 was a 52-week period ended 1 March 2026, and the group reported online growth on a "52/52-week comparable basis" because the prior year was not comparable [2]. Shoprite's year ended 28 June 2026 [3]. Clicks reports to the end of August [11]. Any sector comparison needs period alignment, and a data platform that benchmarks has to support more than one retail year definition.

> **BUILD THE CALENDAR ONCE, CENTRALLY, AND MAKE EVERY SYSTEM READ IT**
>
> The calendar dimension carries: retail week number, retail period with its 4-, 5- or 4-week length, retail quarter, retail year, the 53-week flag, the week-start day *per format*, and the comparable-prior-period mapping including the restatement convention [1]. Most estates instead let three systems derive it independently and then spend a quarter reconciling like-for-like. This is the single cheapest piece of master data in retail and the one most often absent.

The buying calendar runs on lead time, and the lead times differ by a factor of twenty. A local grocery line is ordered weekly or nightly against a one- to four-week supplier lead time. An imported apparel range is committed twenty to thirty weeks before it reaches a shelf, through letters of credit or documentary collection, customs entry, duty, freight apportionment, container consolidation and partial shipment, with an expected landed cost that must be trued up to actual. The range review that authorises it sits further back still. So the apparel buyer is working on next winter while the grocery buyer is working on next week, and both report to the same trading director on the same weekly cycle.

Peak trading in South Africa is not the United Kingdom or United States shape. The festive peak runs from Black Friday, now comprehensively adopted, through to early January, with December dominating the year. **Back to school** in January is a second genuine peak, concentrated in stationery, uniform and school shoes, and because the South African school year starts in January it collides directly with the post-festive trough in food. Easter moves. And month-end produces a *monthly* peak that is sharper here than in most markets: platform data for the first quarter of 2026 shows over 80% more transactions on the 1st of the month than on the 22nd, with peak volume on the 25th [12]. That is payments data rather than grocery footfall, but the pattern it reflects — salary and social-grant dates concentrating demand into a few days — is well known in South African retail. The consequence is that **the day-of-month profile matters as much as the day-of-week profile**, and most imported forecasting configurations do not model it. Lane count, cash-handling capacity and queue management are sized for the 25th, not the 12th.

The change freeze is the architectural consequence of peak. A retailer freezes change to the production estate for the duration, and the freeze is wider than IT: software releases, infrastructure change, structural master-data change — hierarchy reclassification especially — range change, planogram resets and store refits. For a South African grocer the window commonly spans mid-November to mid-January; for a general merchandiser it may run from before Black Friday to the end of the January sale. No South African retailer publishes its change calendar, so treat the specific dates as practitioner consensus and confirm them in your first week; the existence of the practice is not in doubt.

Two second-order effects are worth more than the freeze itself.

The freeze compresses the delivery year into about nine months. A programme plan that assumes twelve is wrong by a quarter, and the error compounds because the plan was probably built by someone whose last industry had no freeze.

The freeze accumulates change. January and February carry the largest and riskiest releases of the year, landing immediately after the period in which the estate was least exercised and most patched around. If you are choosing when to put a new system into a retailer, that is the window to avoid and the one that will be offered to you.

The batch clock

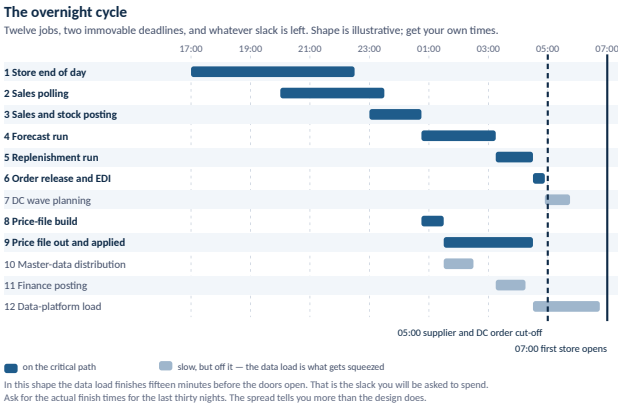


Figure 7.5 The overnight cycle as a timing diagram rather than a swimlane, because it has no hand-offs and no decisions — only dependencies and deadlines. The shape is illustrative. The exercise is to draw your own, with the last thirty nights' actual finish times on it.

This is the most useful picture in the book for an architect arriving from banking or insurance, and it is drawn as a timing diagram rather than a swimlane deliberately. There are no decisions in it and no hand-offs between people. There are twelve jobs, a set of dependencies, two immovable deadlines and a quantity of slack that nobody has measured.

The twelve, in dependency order, with the dependency being the point:

1. **Store end of day.** Tills closed and reconciled, journals finalised, cash declared, the trading day sealed. Must complete *per store*, and a store that does not close blocks everything downstream for that store.
2. **Sales polling.** Transaction data pulled from every store to the centre. The volume-dominant job. A store with a failed line is polled late or not at all, and the gap must be filled next cycle without double counting.
3. **Sales and stock posting.** Transactions applied to inventory and to the stock ledger; store stock positions published.
4. **Forecast run.** On the refreshed history. Usually the longest single job in the cycle.
5. **Replenishment run.** On the refreshed forecast and stock position, per store, on that store's order calendar. Must finish before the order cut-off.
6. **Order release and transmission.** To distribution centres and to suppliers, against supplier cut-off times that are *contractual*.
7. **Distribution-centre wave planning.** The day's picking plan, which depends on the orders.
8. **Price-file build.** The day's effective price and promotion changes, assembled per store or per price zone.
9. **Price-file distribution and application.** Pushed to every till and every electronic shelf label controller, with confirmation. Must complete before stores open.
10. **Master-data distribution.** New and changed articles to stores, tills, website and labels.
11. **Finance posting.** Sales journals, stock-ledger entries, the general ledger interface.
12. **Data-platform load.** The warehouse's daily load, usually the last dependent job and the one that gets squeezed.

In banking the overnight batch is a constraint on one system, and if it is late the customer sees a stale balance in an app. In retail the overnight cycle is the heartbeat of the whole estate, the deadline is a shop door opening, and the failure is visible at a shelf edge with a legal consequence attached.

Four questions establish the critical path, and all four are answerable in a morning.

What is the cycle's total duration, and its slack to store opening? Ask for the actual finish times for the last thirty nights, not the design. The distribution of finish times tells you more than the design document.

Which jobs are on the critical path and which are merely slow? The data-platform load is usually slow and off the path. The forecast run is usually on it.

What is the recovery procedure for each — re-run, skip, or run degraded? Written down, per job, with an owner.

What happens if the forecast run does not complete: does replenishment run on yesterday's forecast, or not run at all? In most retailers the answer to that is a convention held in an operator's head rather than a designed behaviour. Finding it out is one of the more valuable things an architect does in a first month, and writing it down is the first thing in the estate you will have improved.

Above the nightly clock sit the weekly and periodic ones: the weekly trading review and the weekly promotion changeover; the periodic close with its stock-ledger and general-ledger reconciliation; the quarterly and annual statutory close; and the stock-take weekend, which freezes movement and runs a different, heavier batch. A design that satisfies the nightly cycle and breaks the stock-take weekend has failed in a way that surfaces once a year, in front of the auditors.

The store has to keep selling

A store that cannot take money is closed. The cost is not the lost basket. It is the queue abandoning, the perishables degrading, the labour paid for no output, and — where a competitor is 400 metres away — the customer who does not come back. There is no banking equivalent: a branch that cannot transact sends the customer to the app. A shop that cannot transact sends the customer to the shop across the road.

So the design rule is unusual and not negotiable. **The store's ability to complete a sale must not depend on any system outside the store.** Replenishment, loyalty accrual, reporting and price updates may all degrade. Completing a sale may not.

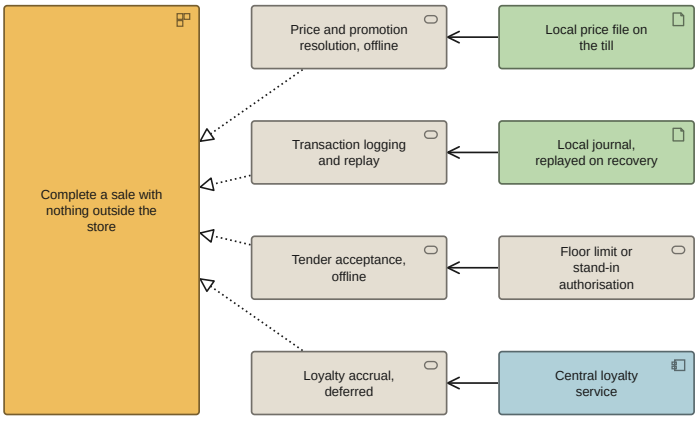


Figure 7.6 The one capability a store may never lose, and the four services that realise it. Most South African estates have the first two and the fourth. The third — taking a card with the line down — is the one to test specifically, because without it the store cannot trade.

"Offline mode" is four separable capabilities, and retailers routinely have two of them.

1. **Offline item lookup and pricing.** The till holds a local price file and promotion rule set and can price any basket without a call. Most estates do have this, and it is why the price file is distributed to every till rather than queried.
2. **Offline transaction logging.** The till writes its journal locally and forwards when the link returns, with duplicate suppression on recovery. Usually present.
3. **Offline tender acceptance.** The hard one. Cash is trivial; card is not, because an authorisation needs a round trip to the acquirer. The mechanisms are floor limits, stand-in authorisation at the terminal or store controller, and store-and-forward of the authorisation request. Each trades fraud risk for availability, and the parameters are a commercial decision disguised as a technical one.
4. **Offline loyalty and gift card.** Accrual can be deferred. *Redemption* cannot be, safely, because the balance is unknown. Most retailers accept the risk on accrual and decline redemption offline.

A store with (1) and (2) but not (3) cannot trade when the line is down, because in a card-heavy basket mix most baskets cannot be tendered. Test that specifically. "We have offline mode" is not an answer to the question.

The classical design put a **store server** in the back office: it held the price file and master data, distributed to tills, aggregated the journal, ran the store's inventory application and was the single integration point with the centre. It made the store autonomous and it made the estate 2,000 servers to patch, back up, replace and secure, in rooms with variable power, variable temperature and no IT staff. The direction of travel is to remove it — tills cache enough to trade standalone, and the store's integration becomes a direct connection to a central or cloud service, with till-to-till coordination handled by a lightweight local service, by one till acting as primary, or by the cloud with local fallback. Vendors market this as resilient cloud point of sale [13]. The trade is explicit: **fewer things to manage, more dependence on the link.** In South Africa that is precisely the dependence you do not want, and a retailer removing store servers has to over-invest in till-level autonomy and in dual connectivity to compensate.

What load-shedding did

Between roughly 2019 and 2024 South Africa ran systematic rotational power cuts, administered in stages. Every figure that follows is **historic peak, drawn from the 2022 to 2024 period, and not a run rate** — load-shedding was suspended for extended stretches from 2024 onward [14]. Dates are given because an undated load-shedding number in a business case is worse than none.

What	Figure	As at
Shoprite diesel spend to keep stores trading	R1.3bn	Reported September 2023 [15]
Shoprite diesel spend	R754m	Reported August 2024 [16]
Checkers business diesel spend	About R1bn a year	Trade press during the crisis; the article carries no date in the source set [17]
Pick n Pay half-year loss	Load-shedding named as a cause	Trade press during the crisis [18]
National diesel saving once load-shedding stopped	R17.25bn year on year	Eskom, after the suspension from 2024 [14]

The four system-level consequences are the part that outlives the crisis.

Cold chain. Refrigeration and freezing cannot tolerate rotational outages without either generation or loss. Generators changed the store's technical estate: fuel logistics, automatic transfer switching, load prioritisation so refrigeration and tills come before lighting and air conditioning, and a monitoring obligation — because a generator that fails during a scheduled outage destroys a store's perishable stock in hours. Temperature monitoring and logging moved from a quality nicety to an operational control, and the reading has to reach somebody who can act inside the window.

The payment estate. A store on a generator still needs the acquirer, which needs the network, which needs mobile and fixed-line infrastructure that was itself being shed. ICASA's inquiry into the electricity crisis recorded operator submissions that network availability fell below 93% at Stage 2 for some operators; that most backup systems need nine to twelve hours to recharge and above Stage 4 did not get it; that tower companies recorded a 250% increase in generator loss during Stage 6 in 2022; and that battery lifespan fell to two to three years [19]. So card authorisation failed for reasons entirely outside the retailer's estate, repeatedly and predictably. The responses were dual connectivity — fixed plus multiple mobile access points, in some cases satellite — longer-duration uninterruptible power on the till and network estate rather than only on the server, and revisited floor limits and stand-in parameters.

Store systems hardware. Repeated hard power loss destroys disks, corrupts databases and shortens the life of every device in the store. Uninterruptible power on the till and network layer became standard rather than optional, and the case for removing the store server strengthened: a cached till survives a power event better than a back-office database does.

Theft of the mitigation. Generators, batteries and cabling became theft targets, as the tower-company generator losses show [19]. The security posture of the store's technical estate became a real design input rather than a facilities footnote.

> **THE POSITIONS LOAD-SHEDDING FORCED ARE THE RIGHT ONES ANYWAY**
>
> The store trades on local capability. The link is an optimisation, not a dependency. Power and network are both assumed to fail. And the mitigation is itself an asset that needs monitoring and protection. Write those four into the architecture principles, because somebody will propose designing them back out on cost grounds within two years.

Suppliers, and the portal

A retailer's suppliers are not vendors in the IT sense. They are the other half of the operating model: they fund promotions, they pay for shelf access, they ship against contractual cut-offs, and in a grocer they are collectively the largest single source of profit after the gross margin itself.

What moves between them and the retailer is an order, a despatch advice, a receipt advice, an invoice and a remittance — and the honest answer is that it moves more slowly and in flatter files than anyone expects. Two messages carry most of the architectural weight. The **despatch advice** (DESADV in the EDIFACT world dominant in South Africa, the 856 advance ship notice in the North American X12 world) is what makes receipt-by-scanning possible: if the supplier sends a hierarchical despatch advice keyed on serial shipping container codes, the distribution centre scans a pallet label and receives the whole content in one event, with discrepancies raised by exception [20][21]. Without it, receipt is a line-by-line count, and the difference in labour cost and receipt accuracy is large. The **price catalogue** message (PRICAT) is meant to solve article-data onboarding and is the least adopted of the set, because it requires the supplier to maintain data to the retailer's standard. Most retailers run a portal or a spreadsheet template instead.

Electronic data interchange in a large retailer is a minority of the traffic. The majority is a spreadsheet emailed to a buyer's assistant, a delimited file dropped on an agreed path at an agreed time, a fixed-width file from a 1990s interface nobody has touched, a portal where a supplier types data in by hand, and a small but growing set of interfaces. Each has its own error handling, and the usual error handling is "the file did not arrive and nobody noticed until Thursday".

The **supplier portal** carries order visibility and acknowledgement, despatch-advice creation for suppliers without electronic data interchange, claim and self-billing visibility, performance scorecards on fill rate and on-time-in-full, new-line submission forms, and increasingly sales and stock data sharing. That last one has an architectural tail worth watching: Pick n Pay disclosed launching Boxer B-Inside in FY26 to extend supplier insight and create a new revenue stream [2]. **Supplier data sharing is becoming a product**, which means the data platform acquires an external, commercial, contractually committed consumer, with the availability and accuracy obligations that implies.

Two South African constraints sit on top. The Grocery Retail Market Inquiry's final report recommended that fast-moving consumer goods suppliers be offered **uniform, objectively justified trading terms not based on volume**, and found that rebates for in-store functions such as merchandising, shelf access and promotions were generally unavailable to wholesalers and buying groups [22]. That makes trading-terms data comparable-by-obligation rather than free text, and it makes the link from a listing decision to its trading term something that has to survive an inquiry. And the Inquiry's remedies on long-term exclusive lease provisions mean **lease terms are structured data with competition-law consequences** — exclusivity clauses, usage restrictions, escalations, renewal options and dates — rather than scanned documents in a property folder [22].

> ***North American aside.*** *Large North American retailers operate formal vendor compliance manuals with monetary chargebacks for despatch-advice errors, label non-conformance, late delivery and carton-content mismatch. A South African supplier-facing design that assumes performance management means a scorecard conversation will be under-built for a North American trading relationship, where it means an automated deduction against the invoice and a dispute process [20].*

Which boundaries are real

Most of the boundaries an architect inherits in a retailer are accidents. A few are not, and telling them apart is most of the value you add in the first ninety days.

Real, because something outside the organisation enforces them. The **store** is a legal, tax, labour, stock-holding and selling boundary at once, and those five do not always have the same edge — a concession inside a store, or a forecourt shop under a fuel retailer's licence. The **dispensary** is a separately licensed professional practice with floor-to-ceiling physical separation, and Chapter 8 draws it as a wall rather than a lane for that reason. The **member or franchisee** boundary is a contract, and the contract is the authority. The **price zone** is the unit at which a price legally exists. The **trading period** is the

unit in which every number is comparable. And the **till** is the unit of cash accountability, which is why operator-level exception data is a loss-prevention asset.

Accidents, usually of acquisition or of a single programme. Banner boundaries that survive because two merchandising systems were never merged. The store server, which exists because wide-area links used to be expensive. "Online" as a separate business with its own catalogue, price and stock — a boundary that was correct in 2015 and is now the reason the site and the shelf disagree. And the split between a merchandising system and a planning system that is really a split between a system and a spreadsheet.

One boundary gets misclassified in both directions, so it is worth naming: **the split of merchandising by merchandise type is usually real, not an accident.** Food and fashion master data models are genuinely incompatible — one is a stable article with a case quantity and a shelf life, the other is a style with options, variants and a size curve that dies in twelve weeks. A group running supermarket, pharmacy and apparel banners will have a merchandising system per merchandise type rather than per banner, and consolidating them is a much larger piece of work than the application inventory suggests.

FOUR QUESTIONS TO ASK OF ANY PROPOSED CHANGE

Ask these of every initiative, every business case and every vendor proposal in a retailer. They take ten minutes and they will catch most of what goes wrong.

1. **Whose shop is it?** Does the centre instruct or persuade here, and for what share of turnover? A capability that assumes control in a devolved estate is a capability nobody can use.
2. **What does this do to the overnight cycle?** Does it add a job, lengthen one, or move something onto the critical path — and what is the remaining slack to the first store opening?
3. **When in the trading year can it land?** Is the target date inside the freeze, inside peak, or in the January pile-up immediately after it?
4. **Can the store still complete a sale when the link is down?** And specifically: can it still take a card?

THE CHAPTER IN TEN LINES

- Two questions settle a retailer's operating model: who owns the shop, and what time it is in the trading year. Ask both in week one.
- The value chain is six groups of work — range and plan, buy, supply, execute, sell, settle — running simultaneously on six clocks that differ by a factor of a thousand.
- South Africa has corporate, franchise and voluntary-trading estates in its top five grocers. In a devolved estate the centre has no price, no assortment and no inventory position to master, and every central capability becomes an offered service.
- A franchised store is a customer, not a location: stock is sold to it, which creates a receivable, a credit limit and an ageing that a corporate store never has.
- A pharmacy and an apparel chain share almost nothing below the till. Build one shared commerce and customer layer, one shared finance layer, and separate merchandising stacks per format family.
- Store picking is the most expensive fulfilment model per unit and the cheapest to start, which is why South African grocery chose it — and why store systems, store labour and store inventory accuracy are the binding constraints on online growth.
- Build the retail calendar once, centrally, with the week-start day held per format. Three systems deriving it independently is the most common and most expensive avoidable error in retail data.
- Peak and the change freeze compress the delivery year to about nine months and pile the riskiest releases into January and February. Plan for it or lose a quarter.
- The overnight cycle is twelve dependent jobs against a deadline that is a shop door opening. Find out what happens when the forecast run does not finish; the answer is usually in an operator's head.
- A store must be able to complete a sale with nothing outside it working. Offline mode is four separable capabilities, and the one estates lack is taking a card.
- Load-shedding figures are historic peak from 2022 to 2024, not a run rate — date every one of them. The positions the crisis forced are the right ones for any distributed estate anyway.

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CHAPTER 8

The process model

A process architecture is the inventory of what a retailer does; a process model is the inside of one of them. Here are both — twelve end-to-end processes, and all twelve drawn in BPMN with the roles in lanes, because in a retailer the question that matters is always who is holding the work right now and what they are waiting for.

The binder, and the thing that is not in it

Ask a retailer for its process model and you will be handed a binder. In the invented Milkwood Retail Group it runs to about 500 pages, was last edited for a point-of-sale upgrade four years ago, and is mostly screenshots of the back-office system with the fields annotated. It explains, in forensic detail, how to key a stock adjustment. It does not explain how a promotion booked in August ends up on a shelf label in November, which is the part that crosses six systems, four roles and a legal deadline.

That is the normal starting position, and it is a particular kind of failure rather than laziness. The retailer documented its procedures and called the result a process architecture. Nobody wrote down the list of its processes.

> A **process architecture** is the inventory: which end-to-end processes exist, what triggers each, what outcome ends it, who owns it, and how they fit together. A **process model** is the inside of one of them: steps, roles, decisions, data, hand-offs. The first is a map of the country. The second is a street plan of one town. A thousand street plans still will not tell you how many towns there are.

An architect needs both, for reasons that have nothing to do with documentation.

A process is where a capability turns into work with a cost and a clock. "Pricing" is a capability and has no cycle time. "Change a price and take a promotion live" is a process: it has a lead time, a distribution success rate, a label compliance rate and a statutory exposure when it fails. Capabilities are for deciding what matters. Processes are for deciding what to change.

Processes are the only artefact the business will argue with. No trading director has ever disputed a capability map, because nothing is at stake in one. Put a swimlane model of the overnight replenishment run on the wall and the supply chain manager will dispute it inside ninety seconds. A model that provokes no argument has told you nothing.

A process model forces you to name who decides. The moment you draw lanes you have to answer: who may approve a price below margin floor, who may override a system-generated store order, who may refund without a receipt, who may refuse to dispense. In retail those answers matter unusually much, because the gap between the authority a person has and the authority the system enforces is where most shrink lives.

Systems are bought along process boundaries. When a retailer buys a pricing platform it is buying a span of process. If the inventory says pricing ends at the price-file build, then distribution, application and per-label confirmation fall outside the procurement — and those are the parts with the legal consequence. A wrong boundary at L1 becomes a wrong contract at signature.

Why this chapter leaves ArchiMate

Everything else in this book is drawn in ArchiMate 4, and that is right for relating a process to a capability, an actor, a business object and an application service. It is a poor notation for *sequence*. ArchiMate has no gateway. Its triggering relationship carries no condition. It cannot show a hand-over between two performers without a convention you have to teach the room first. And it has no swimlanes, which is fatal here, because in a retailer the hand-over is usually where the work stops.

BPMN 2.0 [1] is a sequence notation with a lane per performer. It answers the operations question — who is holding this right now, and what are they waiting for — and the architect's question — where does the work cross an organisational or system boundary, because that is where it fails. This chapter is drawn in BPMN, and it is the only chapter in the book that is.

Saying so plainly matters, because the two notations have to stay reconcilable.

> THE MAPPING BACK, AND IT IS MECHANICAL
>
> Each **pool** in this chapter is one **Process** element in the ArchiMate model, and it is one of the twelve in the L1 map below. Each **lane that is a person** is a **Role**, related to that Process by **Assignment**. Each **lane that is a system** is an **Application Component**, related to the Process by **Serving**. Each **data store or data object** in a figure is a **Business Object**, related by **Access**. Keep the names identical in both notations and the models stay reconcilable; let them drift and you have two process architectures and a meeting about which one is true.

Two consequences. Do not model in BPMN what you can state in ArchiMate: if the question is which applications serve a process, draw it once in ArchiMate and stop. And do not let the BPMN become the system of record for structure — the pool name, the lane list and the data stores belong in the repository as ArchiMate elements.

The lanes in this chapter use the job titles from Chapter 5. A reminder of the ones that recur, because an architect who cannot name the jobs cannot run the workshop.

Role in the lanes	What the person actually does all day
Category manager / buyer	Chooses the range, negotiates cost and trading terms, agrees promotional plans and their funding
Merchandise planner	Owns the money: the sales, margin, markdown and intake plan, and the open-to-buy
Allocator	Decides which store gets which units in which size, and initiates in-season transfers
Pricing manager	Owns the shelf price: zones, rounding, price endings, competitor response, the price file
Master data steward	Works a queue of article and supplier submissions and decides whether the data passes
Store manager	Owns execution — whether the plan happened on the shelf — plus cash, labour, shrink and tasks
Responsible pharmacist	Personally and statutorily accountable for whether the dispensary may lawfully operate [8]
Online picker	Picks customer orders in a trading store and makes the substitution judgements

L0, L1, L2, L3 — and the warning

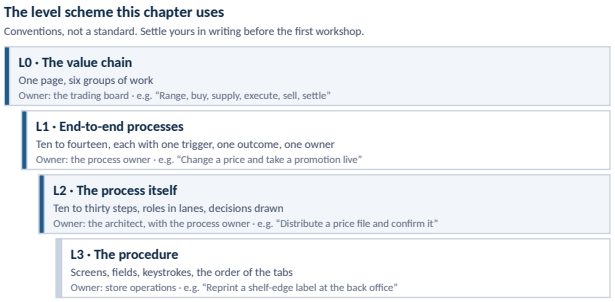


Figure 8.1 The level scheme this chapter uses. They are conventions, not a standard — APQC numbers its levels differently [2] and retail has no BIAN to adopt [4]. Settle yours in writing before the first workshop.

Those are the definitions this chapter uses. They are conventions. They are not authoritative, and nobody else's are either.

APQC publishes a Retail Process Classification Framework; its Excel release documented at version 6.1.1, published 28 April 2023, describes twelve enterprise-level categories with process groups and more than a thousand processes and activities, listed as public content [3]. That is a useful starting inventory, and it numbers its levels as category, process group, process, activity and task — which makes APQC's "level 2" a process *group* and shifts everything in this chapter down a number [2]. Two cautions. APQC's industry index shows other industries at version 8.0 with 2026 dates while the Retail entry shows no version, so **do not quote a Retail PCF version without re-checking it** [2]. And its licensing terms for non-member commercial use are not stated on that page, which is the question practitioners most often fail to ask.

The deeper point is that **retail has no BIAN.** There is no retail equivalent of a maintained service landscape with control records, semantic APIs and a certification scheme. The nearest industry body, the Object Management Group's retail programme, runs an active point-of-sale device and digital-receipt standards effort on top of data models absent from the formal specification catalogue [4]. A retail process architecture is something you build — from a vendor model, from APQC's taxonomy, or from interviews — not something you adopt. Saying so early saves a programme three months looking for the thing that does not exist.

> **SETTLE THE LEVELS IN WRITING BEFORE THE FIRST WORKSHOP**
>
> This sounds like pedantry. It is the most expensive pedantry you will skip. A programme with two definitions of "level 2" produces two populations of models at different grain, discovers the mismatch at integration, and reconciles by hand at the worst possible moment. Write the definitions on one page, get the process owners to initial it, and put it in the modelling conventions. Half an hour, week one.

Two more things about levels.

They are not a decomposition of equal depth, and should not be. It is normal for a retailer to have twelve L1 processes, nine modelled at L2 because they carry the volume or the pain, two at L3 because they are being automated this year, and one left at L1 because nothing is changing there. A programme that insists on uniform depth is producing documents, not architecture.

Below L2, stop. L3 belongs to operations. It changes weekly, it is tied to a screen layout, and in a 420-store estate it varies by store. Model L1 and L2, point at the operating procedures, and do not own them.

The L1 process map

Here is a defensible L1 inventory for a multi-format retailer. Twelve processes in four groups.



Figure 8.2 An L1 process inventory for a multi-format retailer: twelve end-to-end processes grouped by what they are for rather than by which department performs them. Each one maps to a single ArchiMate Process element.

And here is what makes it an architecture rather than a list: each one has a trigger, an outcome, an owner, and an honest note.

#	Process	Trigger	Outcome	Honest about it
1	Review the range and authorise an assortment	The category's slot in the review calendar	An authorised range per cluster, with listings and delists dated	A **cycle**, not a triggered flow. Its output decays within a month of sign-off
2	Onboard a supplier and list an article	A new supplier, or a new line from an existing one	A supplier that can be ordered from and paid, and an article that scans	Nine hand-offs. The commonest failure in all of retail
3	Plan the money and hold open-to-buy	A planning period	A sales, margin, markdown and intake plan, and a spend constraint	A **cycle**, and usually a spreadsheet. Left unmodelled here, deliberately
4	Order goods, receive them and pay for them	A proposal, an allocation, or a manual buy	Goods on hand, a matched invoice, a payable	Goods for resale only. Goods *not* for resale is a different process in ERP
5	Forecast and replenish	The nightly run, on the store's order calendar	A released order and stock on a shelf	A **nightly cycle** with a contractual cut-off. The highest-volume decision engine in the estate
6	Allocate a fixed buy	A fixed quantity received or planned	A store-level quantity plan and the movements that execute it	Not replenishment. Supply is fixed and the decision is distributive
7	Change a price and take a promotion live	A booked promotion, a cost change, a competitor move	The new price charged at every till and shown on every label, on time, with evidence	Six systems, four roles, and a statutory consequence when it fails [5][6]
8	Execute a markdown	Terminal stock, end of season, slow sell-through, a delist	Stock sold through at a reduced price, inventory revalued, margin recorded	Not a promotion. The difference is an accounting entry [20]
9	Serve a customer at the point of sale	A customer presents a basket	A completed sale, a settled tender, depleted stock, a compliant receipt	In a pharmacy retailer this has a separately licensed sibling: dispense a prescription
10	Fulfil a customer order across channels	A customer checks out	Goods in the customer's hands and a settled payment, or a documented failure	Needs an order management system most South African estates do not have
11	Take a return, refund and dispose	A customer brings goods back, in store or by courier	A refund, a disposition executed, inventory correctly stated	The statutory floor is section 56, not the retailer's policy [6][7]
12	Count the stock and state the inventory	The count calendar, or the stock-take weekend	A corrected position, a measured variance and a shrink number	A **cycle**. Also, quietly, the mechanism that corrects a year of interface failures

Four rules make an inventory like this usable.

One trigger, one outcome, one owner. A candidate with two triggers is two processes.

Nothing between the trigger and the outcome goes on the list. "Validate a GTIN" is a step inside several processes, not a thirteenth process. This rule keeps an inventory at twelve items rather than forty.

Cross-cutting work is a service, not a process. Article master maintenance, price-file distribution, customer notification, tax calculation and task issue are consumed by several processes and owned by none. Put them on the application side of the model.

Be honest about which items are not processes. Four of the twelve are cycles rather than triggered flows: the review cycle in 1, the planning cycle in 3, the nightly cycle in 5, and the count calendar in 12. Leave them in, because removing them makes the business think you have missed the work. Mark them, because forcing a swimlane onto a cycle produces neither a process model nor a schedule.

FIVE PROCESSES ALMOST ALWAYS MISSING FROM A RETAILER'S OWN INVENTORY

Ask for the retailer's list and compare. Five things are nearly always absent, and each for the same reason — the owner is too junior to have a seat at the architecture table. **Dispense a prescription.** Filed as "pharmacy", which is a department, not a process. It is a separately licensed professional practice with its own statutory records [8][9]. **Agree, accrue, claim and settle supplier income.** The number most likely to cause a restatement, owned by an accountant with a spreadsheet and a credit-note queue. **Maintain the article master and the hierarchies.** Treated as data administration. It is the process that decides whether every other process can run. **Trade the day at a store, open to close.** Everybody knows it happens; nobody owns it end to end, which is why the end-of-day run is skipped twice a year and discovered at period close. **Manage the trading calendar and the change freeze.** A real process with a real owner in two retailers out of ten, and an email from somebody in November in the other eight.

What follows models twelve of the twelve at L2, in fourteen figures, with two processes split at a natural seam. They are drawn in order of how much an architect learns from them rather than in value-chain order, because the first three are worth a week each and the rest are worth a day.

1. A price change and a promotion going live

Start here. This process crosses six systems and four roles, it has a hard external deadline — the advert runs whether you are ready or not — and the failure mode has a statutory consequence that most retailers treat as a data-quality issue.

It splits at the price-file build, because everything to the left is a decision taken in head office over weeks and everything to the right is a distributed systems problem with hours of slack.

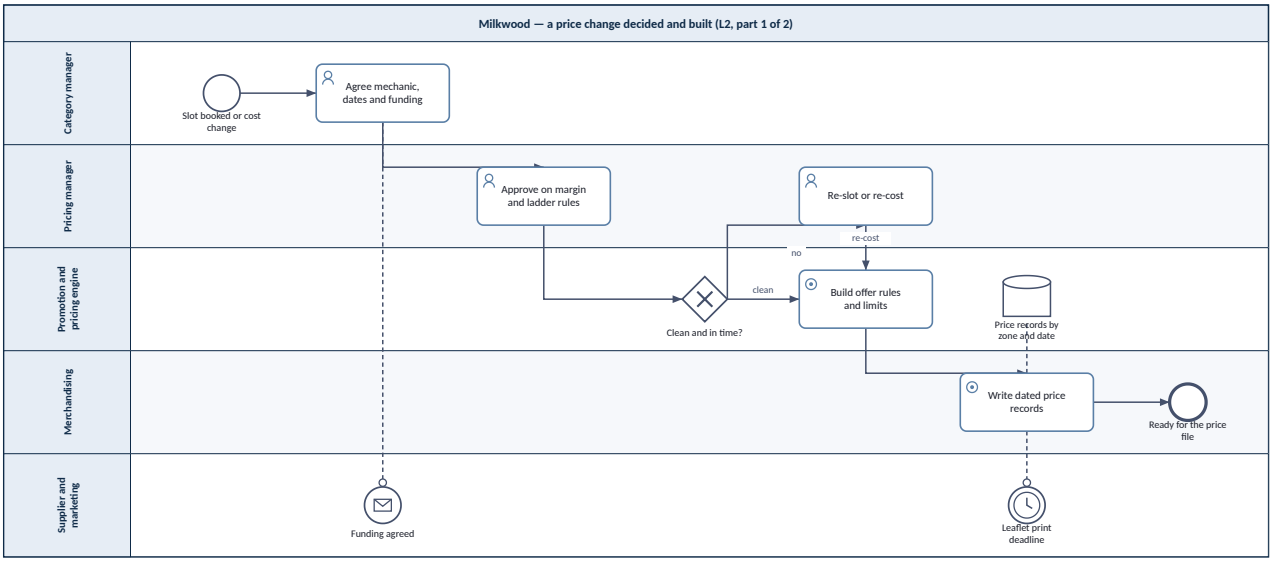


Figure 8.3 A price change decided and built. One gateway stands for two checks with different owners — does this offer conflict with another, and can the file still reach every store before the advert runs. Undo that simplification in the retailer's own model.

The decision points. One gateway, standing for two checks that are usually two systems: does this offer conflict with another on the same article, and can the price file still reach every store before the advert runs? Conflict detection has to be machine-checked, because humans will not catch overlaps at the volume a grocer runs — thousands of price changes a week, concentrated into a weekly cycle. The lead-time check is the one most often absent, and its absence is why adverts run before prices land.

Data created or changed. The promotion definition with its mechanic, funding, period, participating articles and locations. The offer rules, which are *not* prices — "three for R50" and "spend R500, get R50 off" are basket-level rules evaluated against the whole basket, with a priority and an exclusivity, which is why a promotions engine is a separate system class. And the price records: article, zone, effective date. A price record is not "the price"; it is "the price from this date", and two records for the same article, zone and date are a conflict that must resolve deterministically.

Systems and measures. Promotion and pricing engine, merchandising, the deals system for supplier funding, marketing and the leaflet path. Measure the proportion of promotions whose funding was confirmed before the calendar locked, the conflict rate caught by the engine against the rate caught by a human, and the lead time from approval to price-file readiness.

Where it goes wrong. Funding recorded in the deals system with no key back to the promotion, so the single most important question in merchandise performance — how much of our margin erosion was promotion and how much was markdown — has no answer. A promotion built on an article being delisted. And mechanics implemented in point-of-sale code rather than as data, so the catalogue is bounded by what somebody coded in 2008 and a new mechanic means a release across 420 stores.

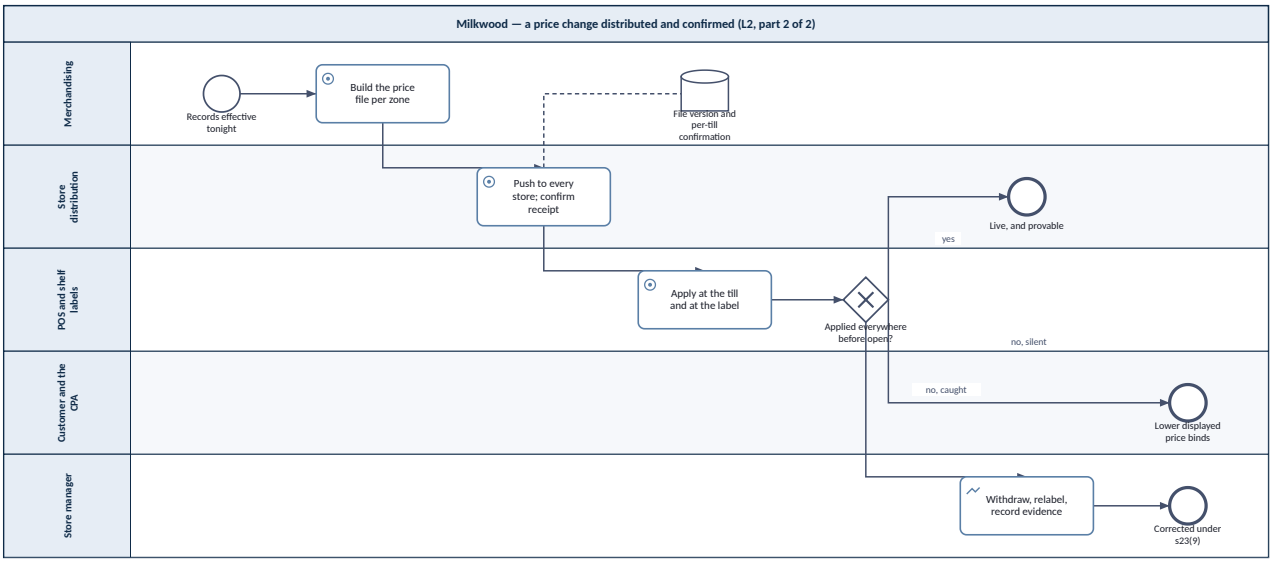


Figure 8.4 A price change distributed and confirmed. The gateway has three exits on purpose: applied, failed and caught, and failed and not caught. The third is the one no retailer draws and every retailer has, and it ends in a statutory obligation to sell at the lower displayed price [5][6].

Now the half with the law in it.

The decision points. One gateway, and it has three exits, which is the whole lesson. *Applied everywhere before the store opens?* Yes, and the price is live and provable. No, and somebody noticed — then the correction workflow runs. No, and nobody noticed — and that third exit is the one no retailer draws and every retailer has.

Here is why it matters. Section 23(6)(b) of the Consumer Protection Act provides that where more than one price is displayed at the same time, the supplier must not charge more than the lower of them [5][6]. A shelf label that silently fails to update is therefore not an operational irritation. It is a statutory obligation to sell at the lower number, at every shelf that failed, for as long as it displays. The escape is narrow: section 23(9) relieves the supplier of a displayed price containing an "inadvertent and obvious error" only once the supplier has *both* corrected the display *and* taken reasonable steps to tell affected consumers of the error and the correct price [5]. The Consumer Goods and Services Ombud's Advisory Note 2 of 1 March 2021 sets out what reasonable steps look like — withdraw the product, fix the label, then resell; for in-store notices, put them up at the affected stores and keep dated photographic proof — and notes that a consumer who has already acted as if buying, for example by carrying the item to the till, may still pay the error price [5]. Section 23(10) relieves the supplier where an unauthorised person altered, defaced, covered or removed the price [5].

The architectural consequence is direct. The retailer needs a provable record of what was displayed, where and when; a correction workflow that produces evidence rather than just a fix; and a reconciliation between the price in the till, the price on the shelf and the price in the advertising. That is why the data store in this figure is drawn at all. Per-till and per-label confirmation is not telemetry. It is the evidence.

Systems and measures. Price-file build, store distribution, the tills, the electronic shelf label estate or the shelf-edge label printer, and task management for the manual confirmations. Measure distribution success rate by store, label compliance rate, price-integrity audit pass rate, and the number of pricing complaints with their outcomes. Then ask the question nobody volunteers: how many labels in the estate are currently showing a price the till will not charge? If the answer is unknown, that is the finding.

Where it goes wrong. Distribution engineered for best effort rather than confirmed delivery, so "we sent it" is mistaken for "it arrived". Electronic shelf labels updated from a different source than the till price file, so the two can disagree by design. Label-to-article bindings maintained by hand, so a planogram reset orphans hundreds of labels at every store that got the same update. And a correction that fixes the shelf and produces no evidence, which under section 23(9) is not a correction at all [5].

2. A prescription dispensed

The cleanest role separation in the book, the most explicit regulatory gates, and a price the retailer has no discretion over whatsoever.

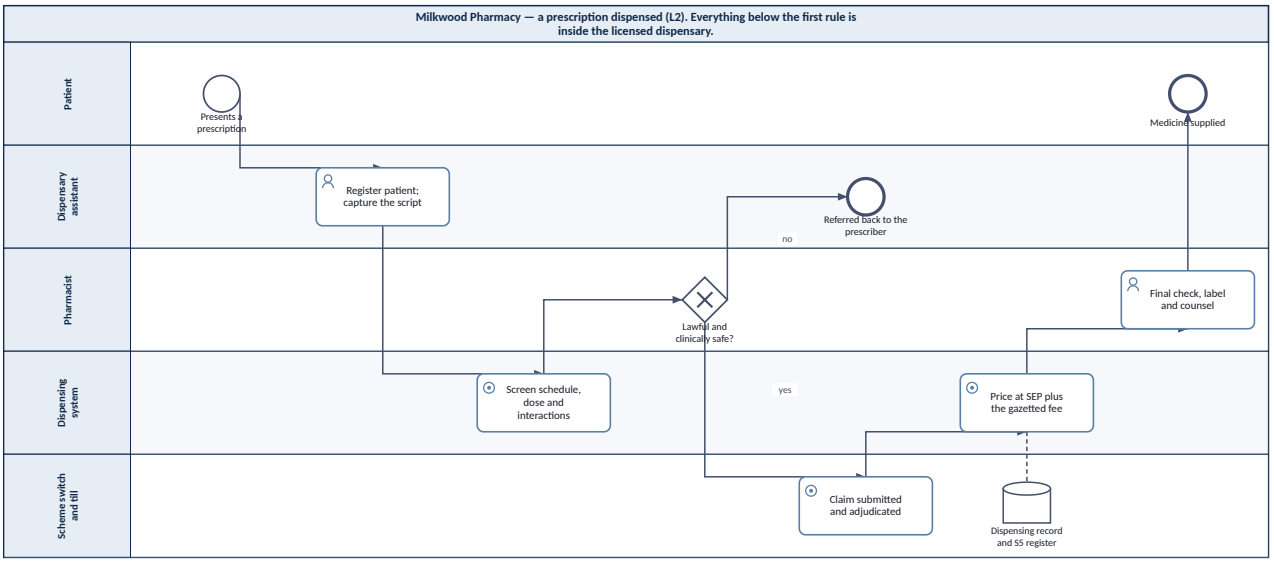


Figure 8.5 A prescription dispensed. The line between the patient lane and everything below it is not an organisational boundary — Good Pharmacy Practice requires a floor-to-ceiling physical barrier, a single compliant entrance and pharmacist-only access to medicine [8]. Draw it as a wall.

Notice what the figure does *not* draw as a lane. The boundary between the patient and everything below is not an organisational line. Good Pharmacy Practice requires a pharmacy located inside another business to be demarcated with **permanent, floor-to-ceiling barriers**, with a single compliant entry and exit, and with the pharmacist holding unfettered twenty-four-hour access; only registered pharmacy personnel may have direct access to medicine, and the keys or access devices stay on the person of a pharmacist [8]. It is a wall. Draw it as one, and resist every request to model the dispensary as a department of the store.

The decision points. One gateway carries two obligations an architect must keep apart. The *legal* check is the schedule: the Medicines and Related Substances Act and its General Regulations classify medicines into schedules that determine who may sell, who may prescribe, whether a prescription is required, whether it may be repeated, and what register entries are mandatory [9]. The *clinical* check is interaction, duplication, dose range and allergy — and Good Pharmacy Practice requires the software used for it to be the latest version available [8]. That turns a content database into a regulated currency obligation on the compliance critical path, with a sharp consequence for a store whose link is down.

Data created or changed. Patient, prescriber, prescription with item, strength, quantity, directions and repeats; the substitution offer and the patient's recorded decision; the scheme claim and its adjudication; the dispensing record; the scheduled-substance register entry; and the patient medication record. The dispensing system must master all of these *alone*.

The price is not a price. Medicines are sold at the **single exit price** plus a capped dispensing fee, both set nationally and both disclosed [10][12]. The pharmacy retailer cannot price them: there is no zone, no promotion, no markdown, no competitor match, no margin optimisation. What there is instead is a national price feed to ingest on a published schedule, and a banded fee formula whose parameters change by Government Gazette. And the cadence is not annual in practice — **2026 had two adjustments, a 1.47% increase effective in January and an extraordinary 2.88% increase effective 1 October 2026 on medicines listed as at 30 September 2026** [10][11]. Any design that treats the SEP as a yearly file load is already wrong, and any source quoting "the 2026 SEP increase" as a single figure is wrong too.

The identifier problem, which is the best fact in this chapter. A South African retail pharmacy's item master carries **two identifier schemes for the same physical product**. The GTIN is how it is received, scanned and sold. The **NAPPI code** is how it is claimed and reimbursed — a unique identifier for pharmaceutical, medicinal, surgical and healthcare consumable products, in use in South African electronic healthcare reimbursement since the early 1990s, with the trademark owned and the file managed day to day by MediKredit, a free public file and a subscription file beyond it [24]. They are maintained by different bodies, on different cadences, with different allocation rules, and they do not map one to one, because pack-size and formulation variation resolves differently in each. **The GTIN-to-NAPPI mapping is a master data asset with a compliance consequence, and nobody outside the retailer will maintain it.**

Measures. Scripts per day, near-miss rate, time to dispense, claim rejection rate by reason, register audit findings, and repeat adherence — Clicks reported 53% of scripts on its repeat service in 2025, across 780 pharmacies in 990 stores [13].

Where it goes wrong. The SEP file loaded by hand and late, so the price is wrong and the retailer is non-compliant. The dispensing-fee bands hard-coded, so a gazette change becomes a software release. The register kept on paper in parallel because nobody is sure the electronic version satisfies the regulation — a genuinely contested point in South African pharmacy that an architect should settle in writing rather than assume. Patient data flowing into the group warehouse because a generic extract was pointed at the whole store database, when Good Pharmacy Practice limits access to those who need it for the patient's care, requires written consent for disclosure, requires the disclosure to be recorded on the patient's record, and names the identifiers to strip before any data is collated for research or commercial use [8]. And interaction-checking software that is not on the latest version, which is an explicit breach [8].

3. An online order to collection or delivery

The third of the three worth a week. It is also the process where the missing system is most visible.

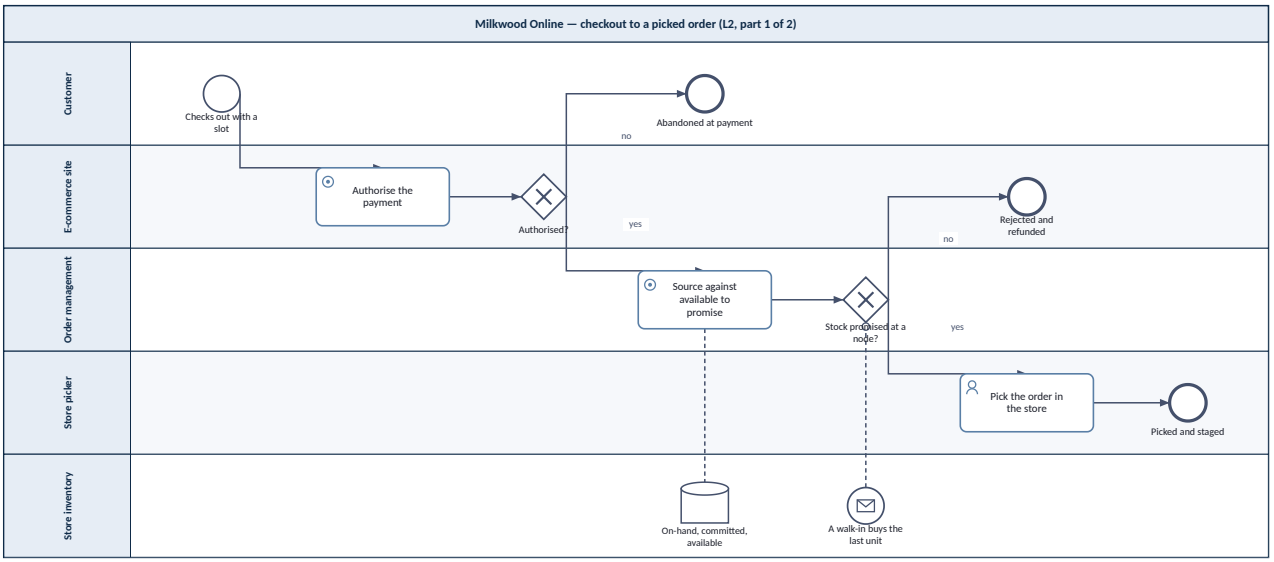


Figure 8.6 An online order from checkout to a picked basket. Lane three is the order management system, which most South African estates do not have; its absence shows up as a website promising stock from a nightly feed. The message event at the bottom is the walk-in customer buying the unit that was already promised.

The lane most South African estates do not have is lane three. An **order management system** owns the customer order as an object with a life — created, paid, sourced, allocated, picked, packed, dispatched, delivered, returned — and it makes the sourcing decision against available-to-promise inventory, node capacity, cost to serve and promise. Many retailers have an e-commerce platform and a courier integration and nothing in between, and the symptom is an e-commerce platform doing a bad job of promising stock against a nightly feed. If you take one thing from this figure, take the habit of asking which system holds the order's state. If the answer is "the website, and also the courier's portal", there is no order management system.

The decision points. Two. *Is the payment authorised?* — and it fails more often than a design assumes: platform data for 2026 reports nearly one in four online card attempts not succeeding, with invalid card details about 27% of failures, withdrawal limits about 15% and 3D Secure abandonment about 5% [17]. That is the strongest available argument for alternative tenders at checkout. *Is the stock actually promised at a node?* — which is the hard one.

Available-to-promise is the contention this figure exists to show. The message event in the bottom lane is not decoration. A store that fulfils online orders is simultaneously a selling location, a fulfilment node and the origin of a delivery, and the stock in it is partly available to a walk-in customer, partly committed to an online order and partly being picked. Promising it needs a near-real-time store position, which is structurally inaccurate; a safety buffer that is per-article and per-store rather than a flat percentage; and a substitution policy for when the pick fails. South African grocery has solved this operationally, with substitution and refund, rather than architecturally, with accuracy. That is a legitimate design choice and it should be written down as one rather than discovered.

In-store fulfilment is the dominant South African model. Shoprite serves Checkers Sixty60 from 997 stores [15]; Pick n Pay fulfils asap! orders from more than 620, offering over 30,000 products with delivery inside sixty minutes [16]. A sixty-minute promise is a systems promise: order-to-pick-start has to be minutes, which rules out a batch interface anywhere on that path.

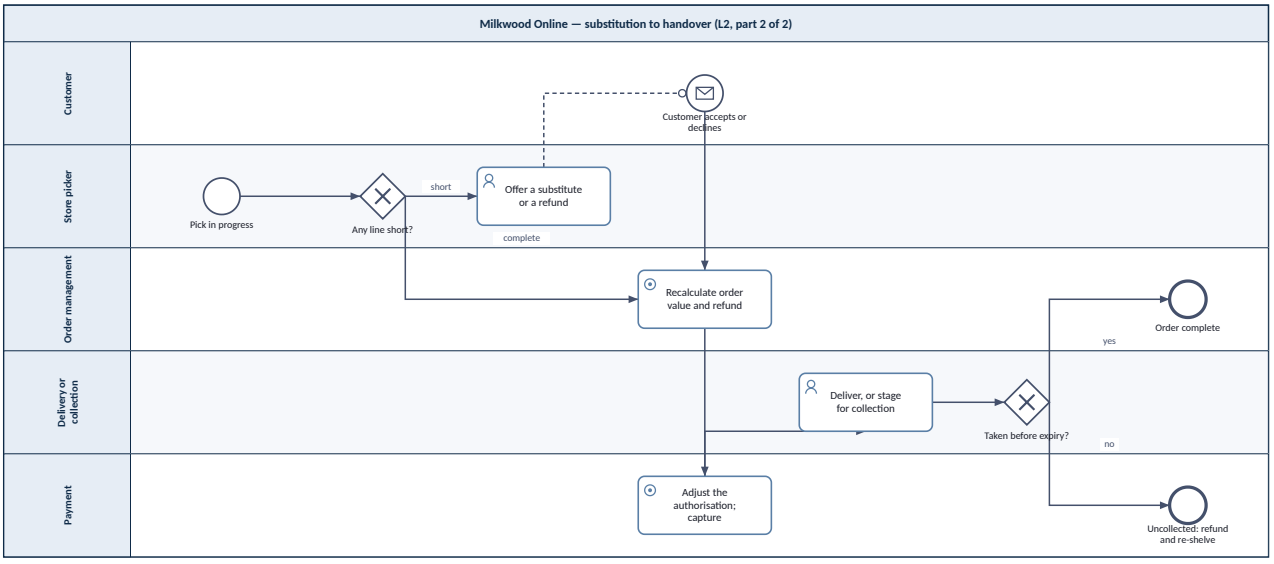


Figure 8.7 The second half of an online order: the substitution loop, the handover and the payment that is captured at a different amount from the one authorised. The expiry branch matters because click-and-collect is more than half of online orders in South African non-food [14].

The second half is the part that gets built last and costs the most.

The decision points. *Is any line short?* — which starts the substitution loop, and the loop is where the money leaks. *Was the order taken before the collection window expired?* — which matters more than it sounds, because click-and-collect is material in South African non-food: Mr Price reported more than 55% of online orders collected in store in FY2026 [14].

The payment path is the one most retail finance systems were never built for. The authorisation was taken at checkout for one amount. The customer is charged a different amount, after substitutions, shorts and weight-variable items — the picked 1.3 kilogram chicken is not the ordered 1.2 kilogram one. So the architecture needs authorisation at one value, capture at another, and a reconciliation between them that survives a refund leg. Get the sequence wrong and the refund is calculated twice: once in the picking application and once in the order management system, which is the single most common defect in South African grocery e-commerce.

Where it goes wrong. Substitution rules held in the picking application rather than the order management system. Uncollected orders decaying on a shelf with no system expiry. Age-restricted lines sold online with no verification at the handover point. Returns of online orders arriving at a store with no system path back to the online order. And the sixty-minute promise broken by a nightly interface that nobody realised was on the path.

4. Onboard a supplier and list an article

The commonest process failure in retail, and the one with the most hand-offs.

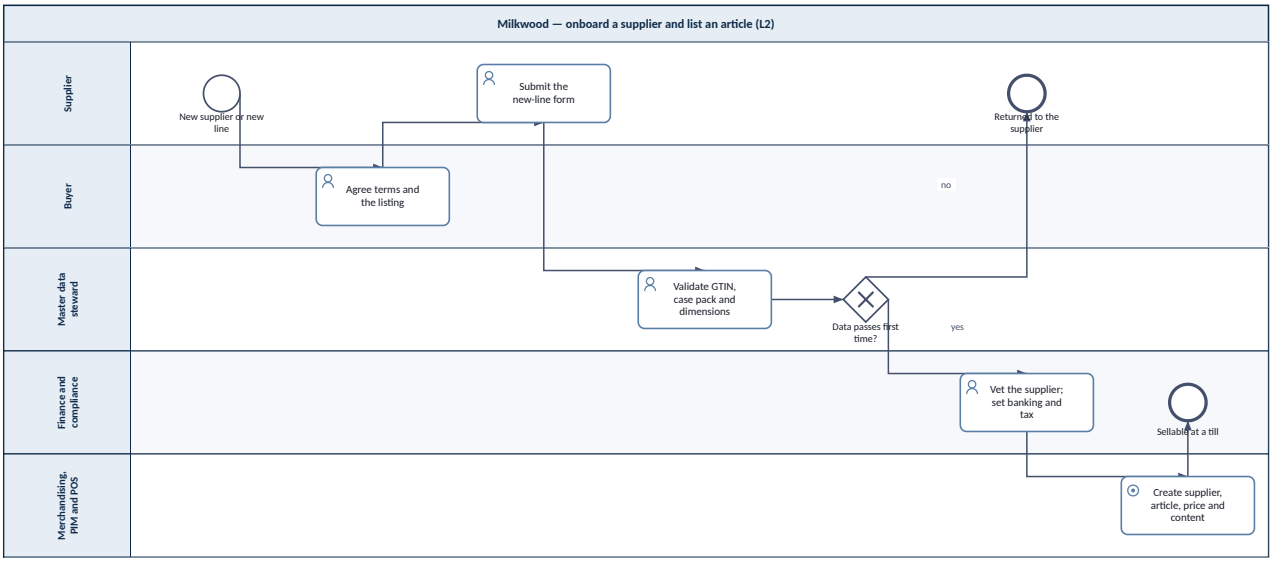


Figure 8.8 Onboard a supplier and list an article. Nine hand-offs between a commercial agreement and an article that scans at a till. The measure that matters is the percentage of new lines sellable on their planned launch date, and almost nobody collects it.

The decision point. *Does the data pass first time?* The honest version of this gateway has a loop counter on it, because the interesting number is the number of round trips, and that is what sets the lead time.

Data created or changed. Supplier legal entity, sites, banking, tax registration and classification; the article record with GTINs at every packaging level, case quantity, dimensions, hierarchy, cost and price; trading terms; and product content for the website. Three of those are routinely wrong in ways that cost real money: a GTIN entered with a dropped leading zero, a case quantity wrong by a factor of ten — which produces a purchase order ten times too large — and article dimensions that are guesses, which makes every planogram built on them fiction.

Controls worth insisting on. Segregation between the person who requests a supplier and the person who approves its banking detail, because supplier banking change is the standing social-engineering target in every retailer. GTIN validity and uniqueness checking at the boundary. Case-quantity verification against a physical sample. And, in South African grocery, an audit trail on the listing decision that survives an inquiry: the Grocery Retail Market Inquiry found rebates for in-store functions such as merchandising, shelf access and promotions generally unavailable to wholesalers and buying groups, and recommended uniform, objectively justified trading terms not based on volume [22]. The link from a listing to its trading term has to be data, not an understanding.

Where it goes wrong. The single most common process failure in retail is this: **the article is listed, the promotion is booked, the advert is printed, and the article is not sellable at the till on day one**, because one of nine hand-offs slipped. Measure the percentage of new lines sellable on their planned launch date. It is the most diagnostic single number in a retail master data programme, and almost nobody collects it.

5. Review the range and authorise an assortment

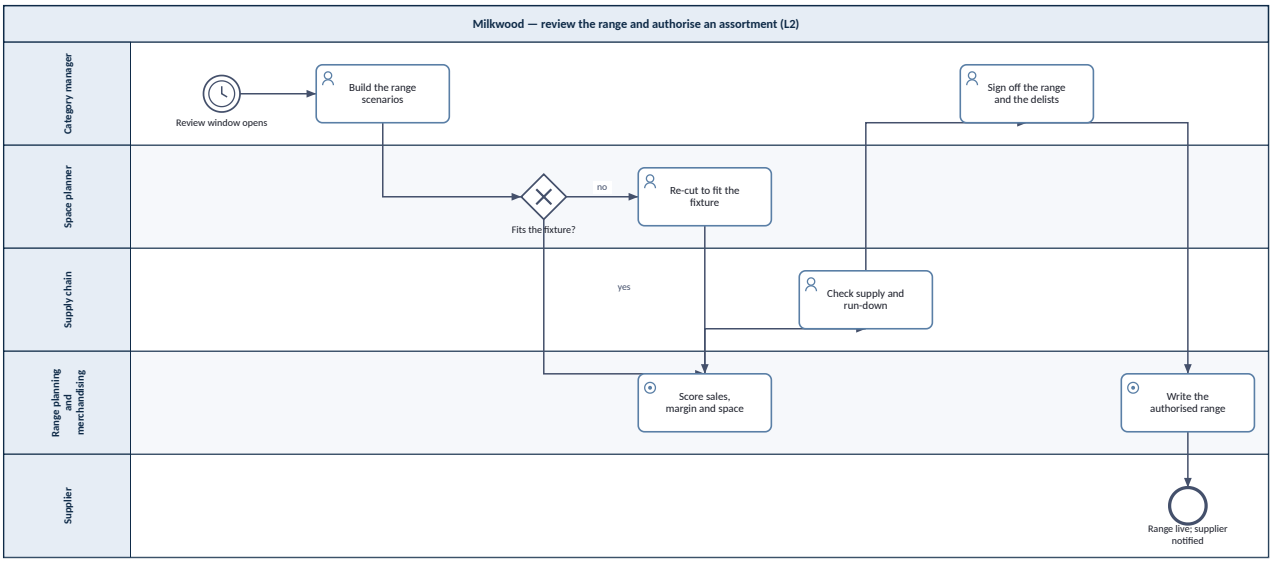


Figure 8.9 Review the range and authorise an assortment. Space feasibility is drawn before commercial sign-off deliberately; most retailers discover it afterwards, when the only remaining options are expensive.

The decision point. *Does it fit the fixture?* Space feasibility is discovered after commercial sign-off in most retailers, which is the wrong order, and the figure puts it first on purpose. The space data lives in a different system at a different grain: the planogram knows bays, facings and shelf capacity; the merchandising system knows articles and locations; and the join between them is the most fragile piece of reference data in store operations.

Data created or changed. The range authorisation by cluster, the listing decision, the delist decision with its run-down date, and the new-article requests that feed process 2. The authorisation object is the one the rest of the estate needs and most often cannot get: replenishment must not order an unranged article into a store, the website must not offer it for collection there, and the planogram must not include it. In practice the range is held as a status flag in merchandising *and* as a plan in the range-planning tool, and the two diverge within a month of the review ending.

Where it goes wrong. The decision taken in a spreadsheet whose output is a slide deck, re-keyed by hand. Cluster definitions set once in 2017. A delist with no run-down plan, so the stock is stranded and the only tool left is markdown. And no feedback loop from performance into the next review.

6. Order goods, receive them and pay for them

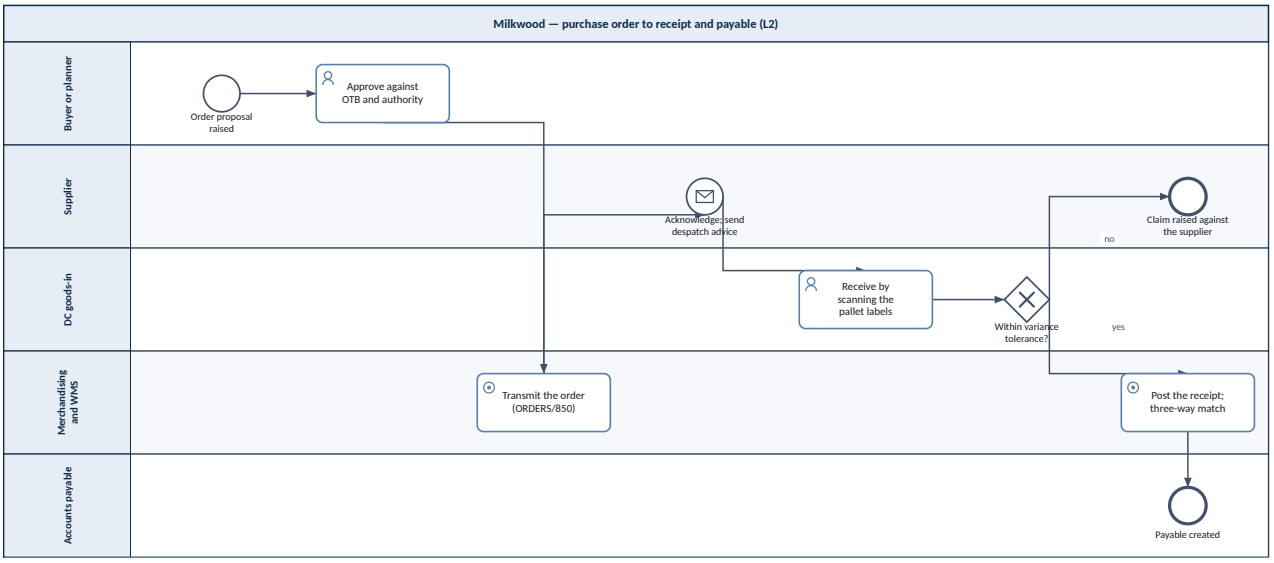


Figure 8.10 Purchase order to receipt and payable. The whole shape of the receipt step depends on whether the supplier sends a hierarchical despatch advice keyed on serial shipping container codes [18][19]. Without one, receipt is a line-by-line count.

The decision point. *Is the receipt within variance tolerance?* Tolerances are usually set too wide in one place and too narrow in another, so errors pass at the dock and accounts payable drowns in exceptions.

The one message that changes this process. A hierarchical despatch advice keyed on serial shipping container codes lets a distribution centre or a store scan a pallet label and receive the whole content in one event, with discrepancies raised by exception [18][19]. Without it, receipt is a line-by-line count,

and the difference in labour cost and accuracy is large. Ask whether the retailer mandates it, what its compliance rate is, and what happens to the suppliers who do not comply.

Where it goes wrong. Orders amended by telephone and not in the system, so the receipt cannot match. Deals applied at invoice rather than at order, so order cost and invoice cost differ systematically and intake margin cannot be reconciled. Goods received against the wrong location, so the stock exists nowhere. Short deliveries accepted because the driver was waiting. And claims raised against suppliers and never chased, which is where a measurable amount of money is lost outright.

7. Forecast and replenish

The highest-volume decision engine in the estate: article by location by day, every night, against a contractual cut-off.

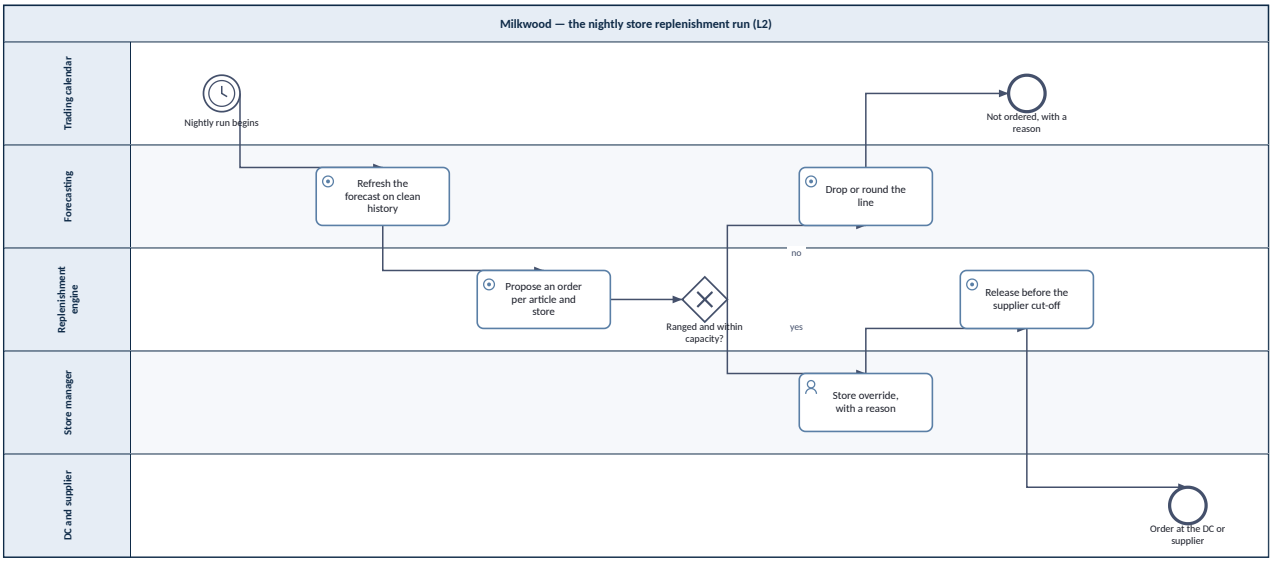


Figure 8.11 The nightly replenishment run. One gateway carries three constraints owned by three different systems — range, shelf capacity and case multiple. The override belongs in the model because it must be applied to the forecast, not to the order.

The decision point. *Is the line ranged, within shelf capacity and a whole number of cases?* Three constraints in one gateway, and each is a different system's data — range from assortment, capacity from the planogram, case multiple from the supplier-article record.

The step architects undervalue is the override. It belongs in the figure because it belongs in the architecture: the override must be captured with an owner, a reason and a magnitude, and it must be applied to the *forecast* rather than to the order. An override applied to the order means the model never learns, and the engine's performance can never be assessed.

Where it goes wrong. Sales treated as demand, with no correction for out-of-stocks, so stock-outs become self-perpetuating. Promotional uplift held as a flat category multiplier set once. Case rounding that over-orders slow lines into small stores. Presentation stock as a guessed constant because the space interface was never built. Promotional pre-build triggered by the promotion's start date rather than the delivery lead time, so stock arrives after the advert. And the override used routinely rather than exceptionally.

One South African specific belongs in the parameters. The monthly pay and grant cycle produces a demand spike sharper than most markets: platform data for the first quarter of 2026 shows over 80% more transactions on the 1st of the month than on the 22nd, with peak volume on the 25th [17]. The day-of-month profile matters as much as the day-of-week profile, and most imported forecasting configurations do not model it.

8. Allocate a fixed buy

Replenishment answers "the shelf is low, send more" and assumes supply is repeatable. Allocation answers "we bought 40,000 of these and will never buy more — how many to each store?" and assumes supply is fixed. They are different processes and different system classes, and conflating them is a scoping error that survives into a contract.

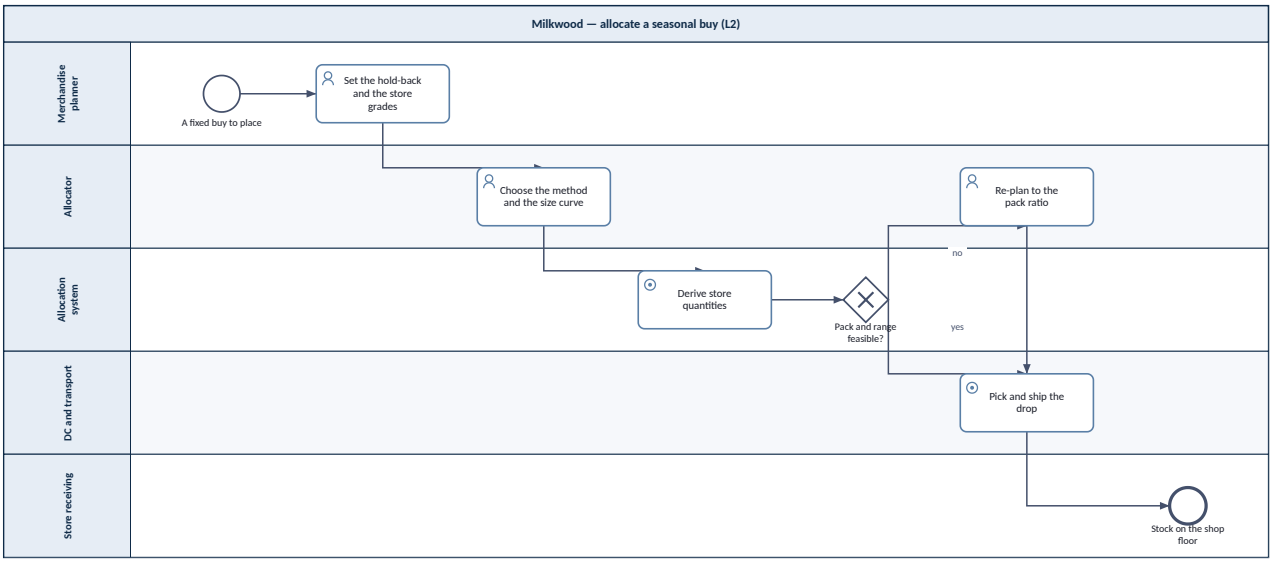


Figure 8.12 Allocate a seasonal buy. Not replenishment: supply is fixed and the decision is distributive. The feasibility gateway is where prepack ratios turn a proportional calculation into a combinatorial one [21].

The decision point. *Is the plan feasible in packs and in range?* In apparel and footwear the decision is not how many but how many of each size, and the size profile differs by store. If the buy arrives in supplier-assembled prepacks containing a fixed size ratio, the allocation is constrained to integer multiples of that ratio, so the optimisation becomes combinatorial rather than proportional, and the gap between the ideal size curve and the achievable pack mix is a real and measurable margin cost [21].

Where it goes wrong. National size curves applied to a store with a different size profile. The whole buy pushed on day one with no hold-back, so there is nothing left to correct with. Allocation to a store that has not opened or is closing. And the allocation engine unaware of the range, so stores receive lines they cannot sell.

9. Serve a customer at the point of sale

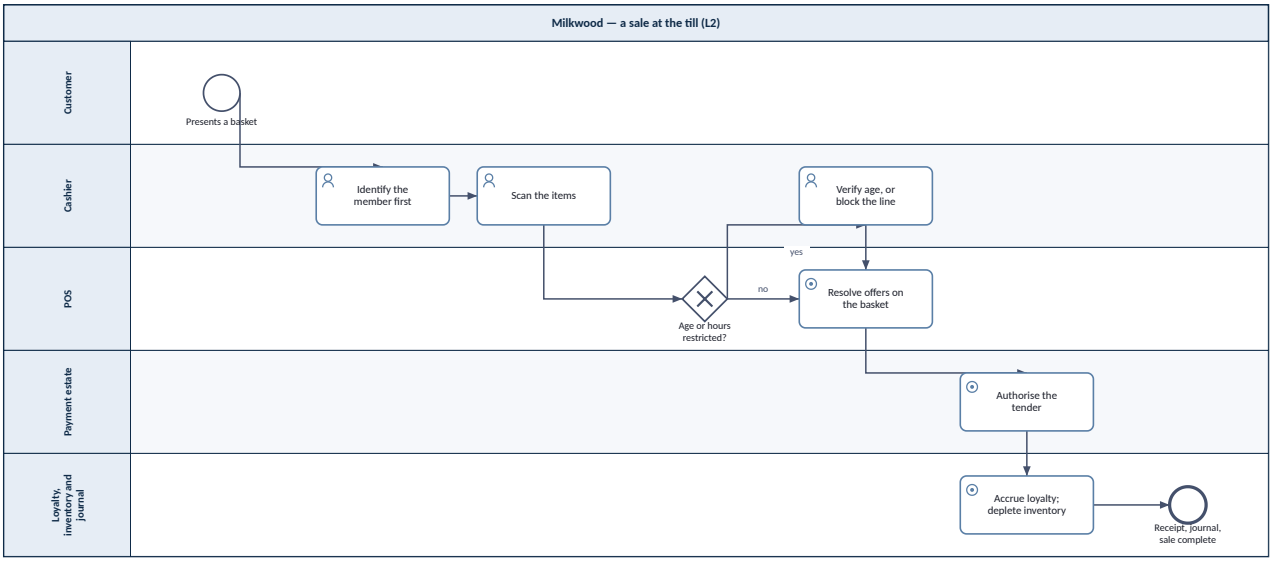


Figure 8.13 A sale at the till. Loyalty identification is drawn first because personalised offers must apply to the basket. The restriction gateway is a time-and-place block rather than an age prompt, and its rule set is provincial and municipal reference data [23].

The decision point. *Is the line age-restricted or hours-restricted?* This is one of the few genuinely South-Africa-specific point-of-sale requirements, and it is routinely under-built. Tobacco sale to persons under 18 is prohibited under the Tobacco Products Control Act 83 of 1993 as amended [23]. Liquor trading days and hours are set provincially and by municipal by-law. The consequence is not an age prompt; it is a **time-and-place-dependent sale block** — the same article must be sellable at one store at 19:00 and blocked at another — and the rule set is reference data with many owners, which has to be maintained and distributed like a price file.

Sequence matters, and the first step is the one retailers get wrong. Loyalty identification belongs *before* the first scan, because personalised offers have to apply to the basket. Identify the member after the items are scanned and the offers do not apply, the customer complains, and the cashier overrides.

Where it goes wrong. Promotion resolution wrong because two offers overlap and nothing detects the conflict — and in South Africa, if two prices end up displayed, section 23(6)(b) may require the customer to get the better of them anyway [5][6]. The age prompt dismissed habitually, which is visible in operator-level exception data that nobody analyses. The restricted-line block absent because the municipal hours were never loaded. A card declined with no alternative tender configured. And an offline mode that works for cash and not for card, which means the store cannot actually trade.

10. Take a return, refund and dispose

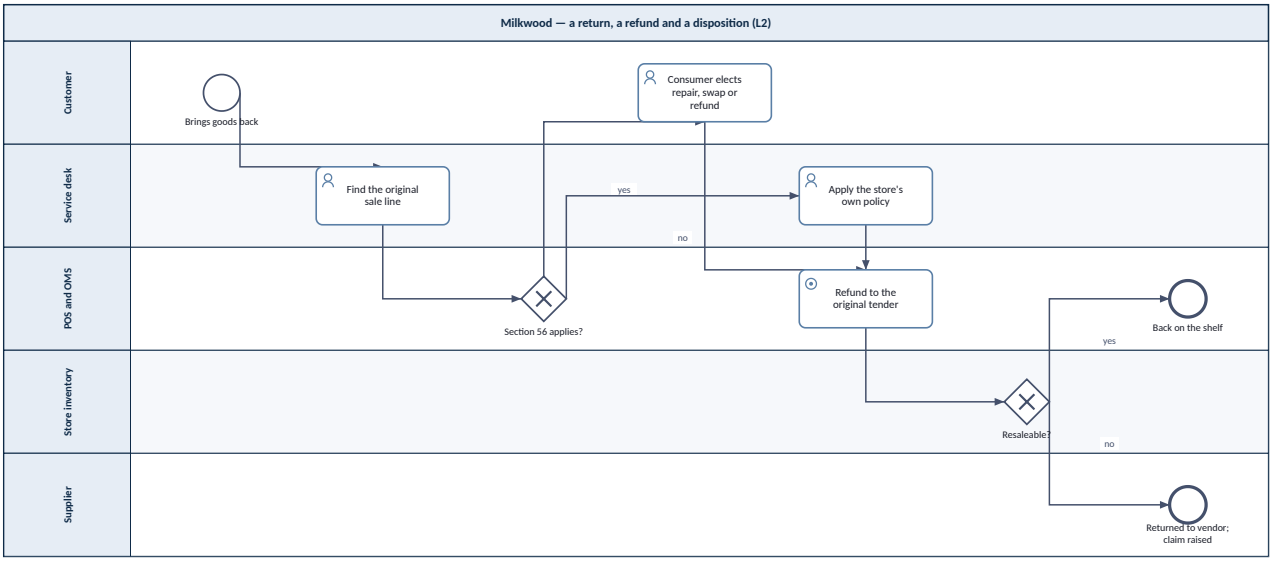


Figure 8.14 A return and a refund. The first gateway is a statute, not a policy: section 56 gives the consumer six months from delivery and lets the consumer elect repair, replacement or refund, at the supplier's risk and expense [6][7]. The second gateway is where the money is.

The decision point is a statute, not a policy. Under the Consumer Protection Act, section 56(1) makes producers, importers, distributors and retailers each implicitly warrant that goods meet the quality standards in section 55, so the retailer is liable irrespective of who made the product. Section 56(2) gives the consumer **six months from delivery** to return goods that fail that standard, "without penalty and at the supplier's risk and expense" — the supplier bears the cost of the return — and the remedy is repair, replacement or refund **at the consumer's election** [6][7]. The National Consumer Commission's position is that the consumer elects; a supplier may offer a repair after the consumer has elected a refund, but the consumer is not bound by it [7].

So the architecture needs a per-sale-line statutory clock, not a configurable policy engine. The system must be able to answer, for any item presented at a service desk, when it was delivered, what was paid, on what tender, against which loyalty accrual, and whether six months have elapsed. That requires a durable link from the return to the original sale line, across channels — bought online, returned in store, refunded to the original tender. North American practice is the mirror image: a configurable policy engine on top of a shorter statutory floor. A single global template fits neither.

The second decision has the money in it. A returned item put straight back on the shelf is worth full retail; one sent to a returns centre is worth retail minus handling; one returned to the vendor is worth cost; one destroyed is worth nothing. Retailers that do not make this decision systematically default to "back on the shelf" and carry unsellable stock as good. In pharmacy, returned medicines generally cannot re-enter stock at all.

Where it goes wrong. Refund issued before inspection. Reason codes too coarse to feed the range review. Inventory reinstated at the store when the goods went to a returns centre. Online returns accepted in store with no link to the online order, so the refund and the inventory movement reconcile nowhere.

11. Count the stock and state the inventory

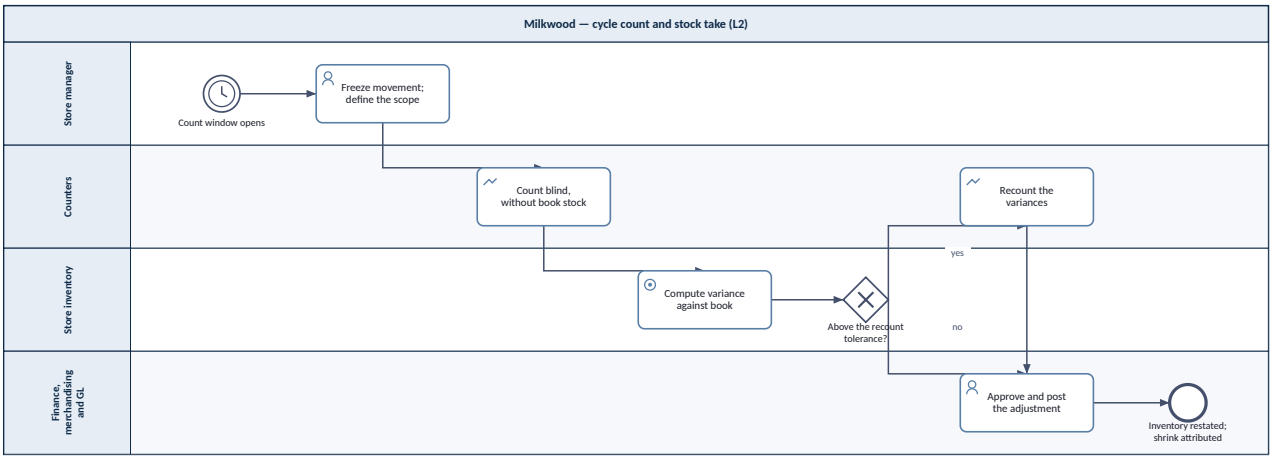


Figure 8.15 Cycle count and stock take. Blind counting and the movement freeze are what make the variance mean anything. Read the variance as a measure of integration quality as well as of theft.

The decision point. *Is the variance above the recount tolerance?* Variances below the threshold are ignored, which is correct per item and wrong in aggregate, because systematic error accumulates in exactly the band nobody looks at.

Three controls make the count worth running. Blind counting, so the counter does not see book stock. The movement freeze, because counting while trading continues puts same-day sales into the variance. And segregation between the person who counts and the person who approves.

Here is the finding every architect should look for, and almost nobody does. The full count is the mechanism that quietly corrects a year of accumulated interface failures. Which means the stock-take variance is partly a measure of integration quality — and nobody reads it that way. Ask for the variance by store, then ask which stores had the most interface exceptions. If the two lists correlate, you have just turned a finance number into an architecture argument.

One honest caveat. **There is no published South African shrink figure.** The United States National Retail Federation benchmark exists and may be quoted only as a United States figure. A South African retailer cannot benchmark its own shrink against anything, which is a real strategic disadvantage and the best available argument for building the internal measurement properly.

12. Execute a markdown

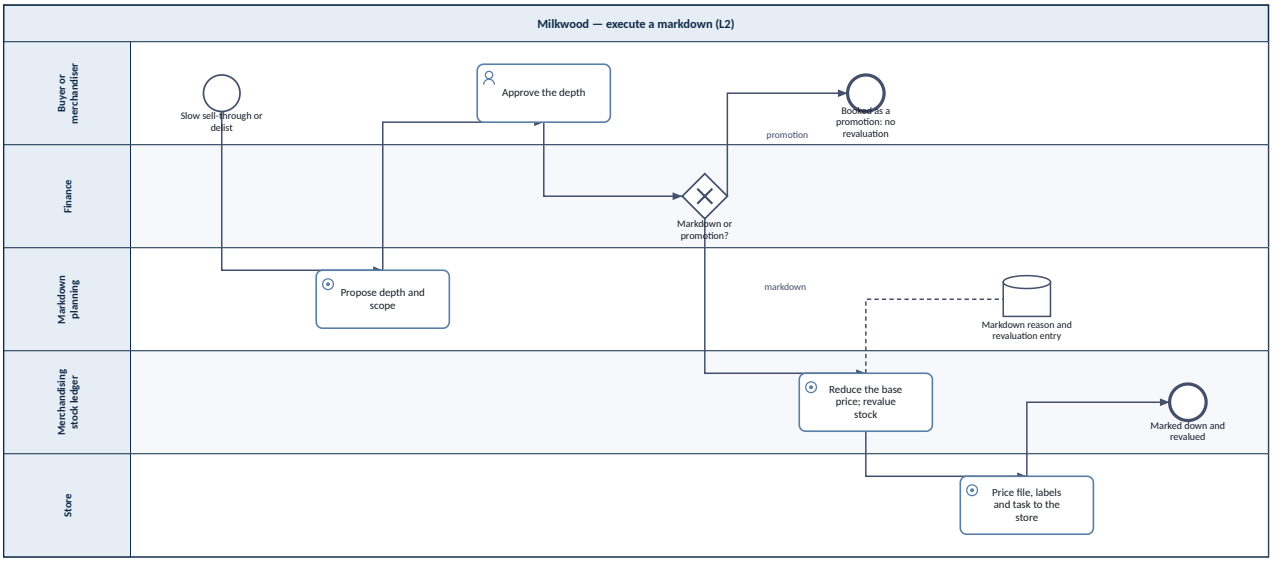


Figure 8.16 Execute a markdown. The gateway is an accounting decision disguised as a commercial one: a markdown revalues inventory under the retail method [20], a promotion does not. The upper branch is the failure every retailer has booked at least once.

The decision point is an accounting one disguised as a commercial one. A **promotion** is a time-boxed offer with a defined end, usually funded in whole or part by the supplier, with the base price unchanged underneath it. A **markdown** is a permanent reduction to the base price, funded by the retailer. Under the retail inventory method a markdown reduces inventory value the moment it is applied, with no physical movement at all; under the cost method it does not. Which method applies is configured at department level in the merchandising system, alongside the open-to-buy calculation type [20], so two departments in the same retailer can mean different things by the same event.

The failure branch in the figure is the real one. A markdown booked as a promotion never triggers the revaluation, so the margin hit does not land in the period it belongs to — it arrives later as a stock write-down, in somebody else's reporting cycle, with no traceable cause. Systems that model both as "a price record with dates" lose the funding attribution and the accounting treatment together.

Where it goes wrong. The price reduced at the till but not on the label, or the reverse — and under section 23(6)(b) the lower one wins [5][6]. Markdowns executed centrally for stock that is not in the stores that got the label. And the markdown's reason never captured, so the business cannot tell a buying error from a weather event, and repeats both.

Moving to a target process architecture

Everything above is current state. The reason to draw it is to change it, and three honest things follow.

Measure before you redesign. A target process built on opinion is a preference. Three sources of evidence sit in every retailer and are routinely unused. **Event timestamps** — order released at, file distributed at, label confirmed at — which give you the real cycle time and its distribution. **Reason-code distributions** — adjustment, override, return, variance — which say where the process actually fails rather than where people think it does. And **thirty randomly chosen closed cases traced by hand**, which is a week of unglamorous work and the best investment an architect makes in the first ninety days.

Split at a seam, not at a page break. Two processes here are drawn as two figures each, and in both cases the split is a real change of medium. The price change splits where a head-office decision becomes a distributed systems problem. The online order splits where the order stops being data and becomes a physical object somebody is carrying. If a process will not fit, cut where the character of the work changes; if there is no such point, the model is too detailed.

Know what the change costs the night. Every process here touches the overnight cycle from Chapter 7. Before committing to a target, answer what the change does to the critical path and to the slack before stores open. A target process that is better by every operational measure and adds forty minutes to the batch is not better.

The trap: modelling the process you were told about

The last thing, and the most important.

Every process model handed to you is an idealisation, and it is an idealisation in predictable ways. The person describing it is describing the version they are accountable for, which is the version that works.

Six tells that you have been handed an idealisation

Check every model you are given against these before you build anything on it.

THE TELL	WHAT IT MEANS
No rework loop anywhere	Nobody drew the third of the volume that goes round twice
Every gateway resolves one way	The other branch is being done in somebody's spreadsheet
Cycle times are round numbers	They are service-level targets, not measurements
A step with no named owner	It is done by whoever is free that morning, or not at all
No failure end event	The process as drawn has no way to lose, so nobody owns losing
The store does what head office says	Execution compliance is sampled by a walk-round, not known

None of these means the model is wrong. All of them mean it is incomplete in a way the business already knows about.

Figure 8.17 Six tells that the model you have been handed is the version that works rather than the version that runs. None of them means the model is wrong; all of them mean it is incomplete in a way the business already knows about.

Three habits catch most of it.

Find the spreadsheet. It sits exactly where the process crosses a boundary no system spans, and it is a free, accurate map of the missing part of your architecture. In retail there are always at least three: the merchandise plan, the allocation, and whatever the supplier income accountant uses to track claims. Inventory the ones that make commitments and establish which are load-bearing.

Ask the person two levels down. The process owner describes the design. The person doing it describes the process. Go to a store — not the flagship, and not with the regional manager — and watch an end-of-day run, a price change and a delivery receipt. Two days in two stores will correct more of your model than two weeks of workshops.

Model the failure path. A process drawn with only its happy path is a process with no architecture requirements, because every requirement worth having comes from what happens when something does not work. The three best figures in this chapter are the three with a drawn failure: the silent label, the referral back to the prescriber, and the markdown booked as a promotion. Each of those branches is where the money, the risk and the design work actually are.

THE CHAPTER IN TEN LINES

- A process architecture is the inventory — trigger, outcome, owner. A process model is the inside of one. Most retailers have hundreds of the second and none of the first.
- This is the one chapter that leaves ArchiMate, because ArchiMate has no gateway and no swimlane. The mapping back is mechanical: pool to Process, human lane to Role by Assignment, system lane to Application Component by Serving, store to Business Object by Access.
- Retail has no BIAN. APQC's Retail PCF is the nearest usable inventory, its level numbering differs from everyone else's, and its version and licence both need checking before you lean on it.
- Twelve L1 processes in four groups is a defensible inventory. Four of them are cycles rather than triggered flows — say so rather than forcing a swimlane onto them.
- Five processes are almost always missing from a retailer's own list: dispense a prescription, supplier income, article master maintenance, the trading day at a store, and the trading calendar itself.
- The price change is the process worth a week of your time: six systems, four roles, a deadline set by an advert, and a gateway whose third exit is a statutory obligation to sell at the lower displayed price.
- Draw the pharmacy boundary as a wall, not a lane. Good Pharmacy Practice makes it a floor-to-ceiling physical barrier with pharmacist-only access, and the dispensary must master patient, prescription and register alone.
- A pharmacy item master carries two identifiers for the same product — GTIN to sell it, NAPPI to claim it — maintained by different bodies and not mapping one to one. The retailer owns a mapping nobody else maintains.
- The order management system is the lane most South African estates lack, and available-to-promise is the contention between an online order and the walk-in customer buying the last unit.
- Section 56 is a per-sale-line statutory clock, not a policy engine: six months from delivery, the consumer elects the remedy, at the supplier's risk and expense.
- Split a process at a change of medium, not at a page break. The price change splits where a decision becomes a distributed systems problem; the online order splits where data becomes a physical object.
- Every handed-over model is an idealisation. No rework loops, gateways that resolve one way, round-number cycle times and a step with no owner are the tells. Find the spreadsheet, go to a store, and model the failure path.

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CHAPTER 9

The conceptual data model

Fifteen boxes, drawn in an afternoon, that will outlive every system in the estate — and the one argument about what an article is that the whole model stands or falls on.

The afternoon that produces one

Milkwood Retail Group's lead architect drew the group's conceptual data model on a wall in her third week. It took an afternoon. In the room: the general manager of grocery trading, a buying controller from apparel, the head of supply chain, a space planner, the responsible pharmacist from the health division, and the financial controller. Fifteen boxes went up. Two of them were still being argued about when everyone left.

Milkwood is invented — a South African multi-format group of roughly 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm and an online business. Nothing about it is a case study. It is here so the abstractions have somewhere to land, and its multi-format shape is deliberate, because grocery, pharmacy and apparel have genuinely different architectures underneath an identical-looking till.

Three years later that photograph is the first thing a new architect at Milkwood is shown. Below it, the information model has been rewritten once and the physical schema three times. The photograph has not changed, because the business has not changed. A retailer in 1975 had articles, suppliers, locations, prices, stock movements and transactions. So does a retailer in 2026, and so will one in 2050, whatever is in the database by then.

That is the whole case for the artefact. It is the cheapest thing in the data architecture to produce and the most expensive to buy back once a merchandising replacement has started without one.

What it is not

Five prohibitions, and each will be argued about in the first hour.

No attributes. Not even the obvious ones. No `description` on Article, no `quantity` on Inventory Movement. The moment an attribute appears, the meeting becomes a negotiation about attributes and the people who came to settle vocabulary go back to their desks.

No keys. Not internal, not surrogate, and above all not the barcode. A SKU may legitimately carry several GTINs and a GTIN may be reissued after a sunset period [1], so the barcode identifies a trade item, not a merchandise record. Chapter 10 spends a page on why.

No resolved many-to-many. One delivery carries lines from three orders. Leave that as it is. Resolving it is a design decision and this is not a design.

No system names. Not the merchandising system, not the promotions engine, not the warehouse system. If the model changes when a system is replaced, it was never a conceptual model.

No time, tax or consent structure. No effective-from, no VAT code, no POPIA consent flag. All three are mandatory, all three are the subject of chapter 10, and none of them belongs in this room.

The prohibitions are about audience, not minimalism. Watch the rule break once and you will enforce it for ever. Somebody writes `stock` on Article, because obviously an article has stock. Within ten minutes the online manager is explaining that available-to-promise is a different number, the controller is explaining that in-transit stock is on the balance sheet and on nobody's shelf, and the pharmacist is explaining that a part-dispensed container of a thousand tablets is stock in a unit nobody else uses. All three are right, and none of it belongs here. The afternoon becomes four sessions and produces an information model nobody owns.

Three tests for whether something is an entity

1. **Can you define it in one sentence in words the business already uses**, without saying "a record of" or "a table holding"?
2. **Does the business count it, name it or argue about it?** "How many SKUs do we range in a convenience store" is a question people fight over. "How many article-location junction rows" is not.
3. **Would it survive replacing every system tomorrow?** An inventory movement would. An "article header" would not.

The third test is the sharpest and it kills most candidates. Twelve to twenty boxes is the right range. Below twelve you have collapsed something — usually Article. Above twenty-five you have started the information model and lost the room. Fifteen is a good number for a retailer and it is the number this chapter builds to.

Article, style, option, SKU: the argument the model lives or dies on

This is the single most argued-about piece of retail modelling, and it is not pedantry. It decides whether the group can forecast, whether it can allocate, whether it can report margin by range, and whether its online catalogue can be built at all.

The vocabulary collides because four trades meet in one estate.

Word	Where it comes from	What it usually means there
Article	European ERP, SAP IS-Retail	The merchandise-managed thing: what the buyer buys, the system prices, the shelf holds. Generic article plus variants is the SAP shape [2]
SKU	North American merchandising	Stock keeping unit — the finest grain counted separately. Size-and-colour-specific in apparel, identical to the article in grocery
Style	Apparel and footwear	The design: one cut, one construction, one cost base, one buy decision. Not sellable, not stockable
Option	UK and South African apparel	Style plus colourway. The unit of range planning and of the photoshoot
Variant	Generic and e-commerce	Any axis of difference below the parent: size, colour, flavour, pack size, scent
Colour-size	Apparel operations	Style plus colour plus size. The thing with a barcode and a bin

Every system in the estate answers differently, and each answer is correct inside its own box. Oracle's merchandising documentation gives a fixed six-level merchandise hierarchy with items attaching at subclass [3]. SAP builds a merchandise category hierarchy with an unlimited number of levels, numbered bottom-up, where branches may differ in depth and articles attach only to merchandise categories and never to hierarchy levels [2]. It also ships two objects Oracle has no equivalent of: a *reference article* used as a creation template, which is never sent to the till, and a *value-only article*, which manages inventory by value at category level [2]. The ARTS Operational Data Model refused to name the levels at all, modelling a merchandise hierarchy *function* so that different business processes get different hierarchies over the same items [4]. Three vendors, three shapes, one estate.

The three levels that actually exist

Underneath the six words there are three concepts, and a fourth that is about packaging rather than merchandise.

- **Style** — a design or recipe identity carrying cost structure, composition, country of origin, lifecycle and the buy decision. Not stockable. Not sellable.
- **Option** — style plus one axis the customer chooses *before* size: colour, flavour, scent, pack format. The unit of range planning, of merchandise financial planning in fashion, and of online product-page identity.
- **SKU** — the stockable, sellable, barcoded atom. One bin, one count, one price point, one row on the sale line item.
- **Trade unit** — how the SKU arrives: inner, case, pallet, prepack. A grouping of the atom, not a level above it.

One style, five options, eight sizes each, forty SKUs. Grocery degenerates this cleanly: one style, one option, one SKU. That is exactly why grocery-born systems cannot be retrofitted to apparel, and why apparel-born systems are gratuitous in grocery — and why a group running both runs two merchandising systems, or runs one with the apparel business torturing a flat model into a style-colour-size shape through naming conventions.

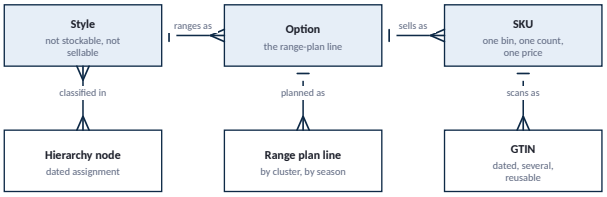


Figure 9.1 The article argument in crow's-foot notation. Three levels, one trade-item identifier family, and the two things that hang off the parents. **ArchiMate carries no notation for multiplicity**, and multiplicity is the argument here, so this figure is drawn in crow's foot and the structural views are drawn in ArchiMate.

The rule that settles it

Two questions, asked in this order, end the argument in every room I have been in.

> **Is its stock counted separately, and can it be sold on its own?** Both yes: it is a SKU. "Its stock is counted with its siblings": it is an option or a style. "It can be sold but never counted" — a weighed loose item, a service, a deposit: it is a **sellable non-stock item**, which needs its own type rather than a fake SKU with the inventory flag switched off.
>
> Then, for the parents: **would the buyer make one buy decision for it?** That is the style. **Would the range planner put it on one line of the plan?** That is the option.

The rule works because it asks about two things the business already does — counting and deciding — rather than about levels. Ask "is this a SKU or an article" and you get a vocabulary fight. Ask "is its stock counted separately" and you get an answer from the stock-taker.

Five mistakes, in the order you will meet them

1. **One `PRODUCT` table with a level column.** It survives until a query must mean "sales by style this week against last" and "stock by SKU by store tonight" at the same time, and the self-join answers differently depending on which level was populated.
2. **Making size a variant axis equal to colour.** Colour drives photography, assortment breadth and markdown behaviour. Size drives the size curve, allocation ratios and lost-sale analysis. An out-of-stock size is a lost sale on the *option*, because the customer may buy the next size up. One undifferentiated axis destroys the size-curve forecast and the broken-size markdown trigger at once.
3. **Using the barcode as the key.** Covered above, and it is the commonest of the five.
4. **Letting the e-commerce "product" be the master.** Online grouping is a navigation decision that changes with merchandising fashion. A multipack and a single are one navigation product and two SKUs.
5. **Creating a new style when a supplier changes.** The design did not change. Supplier belongs to a dated supply arrangement, or like-for-like at style level breaks at every re-source.



Figure 9.2 The same three levels in three formats. Grocery collapses them, fashion needs all three, and pharmacy keeps one axis — pack size — that grocery does not need and that carries a reimbursement consequence.

The six hierarchies, all of them correct

A retailer does not have *a* product hierarchy. It has at least six, they disagree, and each is right for its purpose. The architectural error is to pick one and declare it canonical.



Figure 9.3 Six hierarchies over one SKU, each with a legitimate owner and a legitimate reason to differ. Picking one and declaring it canonical is the error; mastering the mappings is the work.

They disagree legitimately. Merchandise hierarchies are strictly one-parent and get restructured when buying teams reorganise. Online navigation is many-parent and gets re-cut every season. Financial hierarchies cannot be restructured retrospectively without restating the stock ledger, so finance resists exactly the change merchandise wants. Space hierarchies follow how the customer shops the aisle: "breakfast" spans four merchandise classes. And regulated classification follows statute, which lines up with no commercial category at all — brown bread is zero-rated for VAT, brown bread rolls are not [5].

The conceptual answer is three boxes, not one: a **hierarchy** with a type, a **hierarchy node** with a parent, and an **article-to-node assignment**. All three dated, which is chapter 10's problem. One `category_id` serving six purposes is the failure, and the symptom is that two correct reports never tie at any intermediate level.

> **THE SETTLING QUESTION**
>
> If two people disagree about which category something belongs in, are they disagreeing about a fact or about a purpose? About a purpose means two hierarchies, and the useful deliverable is one page saying which hierarchy answers which question.

Supplier, vendor and supply source are three things

"Supplier" and "vendor" are used interchangeably in most retailers and should not be. Four distinct things hide behind them, and the split is the difference between a buying system that works and one that cannot say where an open order is going.

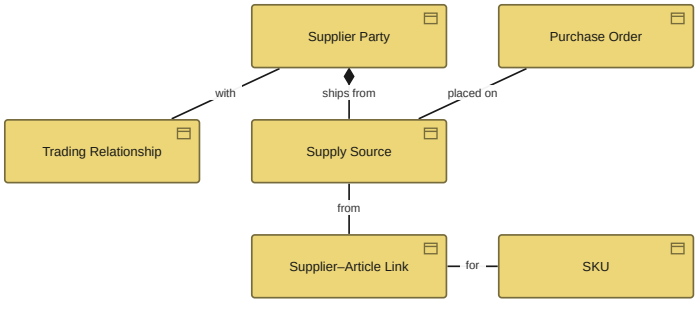


Figure 9.4 Supplier as four business objects rather than one. The purchase order is placed on a supply source, not on a legal entity — which is what lets a dual-sourced SKU say where an open order is going.

Supplier party is the legal entity: registered name, CIPC number, VAT number, B-BBEE status, banking details, country. **Trading relationship** is the dated commercial arrangement with that party: terms, payment days, rebate structure, currency, Incoterms, returns policy. **Supply source** is the specific place goods leave from — a factory, a depot, an agent's warehouse — carrying lead time, minimum order quantity, order multiple and its own GLN. **Supplier-article link** is the dated join to a SKU, carrying cost price, pack configuration, case GTIN and the primary-source flag.

One legal entity supplies through three factories with different lead times. One SKU is dual-sourced and the split changes weekly. A single `VENDOR` row with a `lead_time` column carries none of it. And in South Africa the supplier record is a compliance object, not an administrative one: the Grocery Retail Market Inquiry recommended that fast-moving consumer-goods suppliers offer trade terms **uniformly** to all retailers, wholesalers and buyer groups, each with an objective justification and not qualified for by volume [6]. A comparison across suppliers is the compliance artefact, so trading terms have to be structured rather than asserted in a contract PDF.

The settling question: *if this changed, would the open purchase orders need re-pointing?* Lead time and pack configuration, yes — supply source. VAT number, no — supplier party.

The buying chain, which is not a sequence

The chain from order to claim looks sequential and is not. That is the whole difficulty.

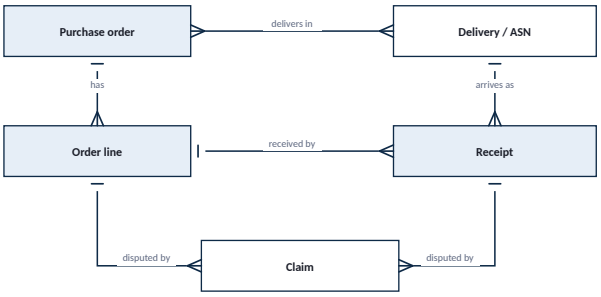


Figure 9.5 The buying chain, with the multiplicity that makes it not a sequence. One order, four deliveries; one delivery, three orders' lines; one line, two claims. Shaded boxes carry a quantity that is a different number in each.

The relationships are many-to-many in every direction. One order delivers in four drops. One delivery carries lines from three orders. One receipt covers part of one line. One line generates two claims — a short delivery and a price difference. Model this as `order_line.received_qty` and you destroy the ability to answer *when* stock arrived, which is the input to lead-time calculation, to service-level measurement and to the whole of replenishment.

Three quantities are not the same number: **ordered**, **expected** after the supplier's confirmation and advance ship notice, and **received**. A fourth — **invoiced** — belongs to accounts payable and routinely disagrees with all three. The three-way match is the control, and it exists only if the three are separate facts.

And pack mathematics is a first-class problem, not a unit-of-measure footnote. Orders are placed in cases, delivered on pallets, received in cases, sold in eaches and counted in whichever unit the stock-taker used. A wrong case quantity multiplies or divides every order, every receipt variance, every stock position and every forecast consumption figure by the same wrong number, and stays invisible until a store receives ten pallets of something.

The settling question for the whole chain: *could this happen more than once for one order line?* Receipts and claims can. The order line cannot. That is the test for a child entity rather than a column.

Nine answers to "how much stock do we have?"

Ask a retailer how much stock it holds of a SKU in a store and the honest reply is a question: who is asking, and for what?

POSITION	OWNED	PRESENT	SELLABLE	WHO ASKS
Physical / book stock	●	●	●	Store manager, finance
Available to promise	●	●	●	Website, contact centre
On order	●	○	○	Replenishment, open-to-buy
In transit	●	○	○	Finance, supply chain
Reserved / allocated	●	●	○	Fulfilment
Damaged / quarantined	●	●	○	Loss prevention, quality
In a customer's basket	●	●	◑	Digital
Customer-owned, held	○	●	○	Lay-by, click-and-collect
Consignment, supplier-owned	○	●	●	Concession, magazines
Bulk under dispensing	●	●	◑	Dispensary — the tenth

● yes ◗ qualified ○ no

Figure 9.6 Nine answers to "how much stock do we have", and a tenth in pharmacy. Owned, present and sellable are three independent facts, which is why none of these is a variant of the others and why none of them is a column.

Nine positions, and a tenth in pharmacy. Each has a different owner and each breaks something different when it is wrong, and two of them are not the retailer's stock at all. Under section 62 of the Consumer Protection Act each lay-by instalment remains the consumer's property until delivery, and the

goods remain at the supplier's risk [7] — so a lay-by on the retailer's shelf is the customer's goods in the retailer's custody. Consignment stock in a concession is the supplier's.

The structural consequence is the most important single line in this chapter. **Stock is not a column.** The durable fact is the **inventory movement**: append-only, signed, typed, located, timestamped, with a reason code and a document reference. Every position above is a derivation — a filter and a sum over movements, optionally plus a set of pending documents. A `qty_on_hand` column is acceptable as a cached derivation with a recomputation path. It is not acceptable as the system of record, because the moment two channels disagree there is nothing to adjudicate with.

> Silent negative stock is a signal, not a state. A system that tolerates it has told you that its movement log and its position column have drifted apart and nobody is reconciling them.

The settling question: *if this number were wrong, who would notice and what would break?* Each distinct answer is a distinct named position.

Location, down to the bin

Location is where retailers get away with a bad model longest, and it costs them during any format or channel change. Seven levels earn a place, and the model needs them as specialisations of one thing rather than as a flat `STORE` table with a type column.

Legal entity, which owns the stock and holds the VAT registration. **Site** — store, distribution centre, dark store, depot, pharmacy, concession, kiosk, franchise store — with a number, a GLN, an address, a municipality, a format and dated opening and closing dates. **Selling area**, because the dispensary, the liquor section, the bakery and the butchery are separately licensed and separately counted. **Fixture**, the bay or gondola that holds the goods. **Till**, which records the transaction, holds a cash float and sits inside or outside the card-data boundary. **Warehouse bin**, the distribution centre's storage address. And **virtual location** — in transit, quarantine, web stock, a third-party logistics pool, a supplier's consignment pool.

Virtual locations are mandatory, not tidy. Every inventory movement must have both ends. Without them, receipts come from nowhere, write-offs go nowhere, and stock in transit is counted at both ends.

Site is also a regulatory entity. A pharmacy licence is issued per premises, with conditions, one for each premises [8]. Liquor retail is licensed provincially, with hours set by province and municipality. So the site carries a dated set of licences, and the till has to refuse a sale the licence does not permit at that hour. Trading hours as one pair of times cannot express Sunday liquor restrictions, public holidays or after-hours pharmacy, and one national configuration produces unlawful sales in one province and blocks lawful ones in another.

The settling questions: *can stock be at it, and move to and from it?* Then it is a location, physical or virtual. *Does it have its own licence, lease or trading hours?* Then it is a site.

Price is an entity, not an attribute

A price is a dated, scoped, typed, sourced number with a reason. As a column on the article it is wrong within a week.

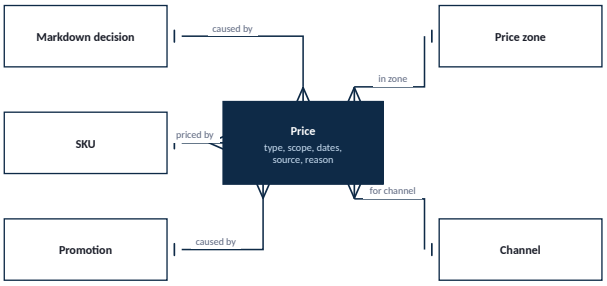


Figure 9.7 Price as an entity with a type, a scope and dates. Two numbers can be correct at the same instant in different zones or channels, and the Consumer Protection Act binds the retailer to the lower of any two displayed concurrently.

At least nine kinds of price coexist, and more than one can be correct at the same instant. **Cost** — and there are several: standard, last, weighted average, landed, and net of supplier income. **Regular retail**, VAT-inclusive in South Africa. **Promotional**, which is temporary and leaves the regular price alive underneath it. **Markdown**, which is permanent, replaces the regular price and revalues stock. **Price zone**, the set of locations that share a price. **Recommended retail price**, which is the supplier's suggestion and which it is unlawful to enforce. **Member price**, which is party-based rather than location-based. **Unit price** per kilogram or litre, which legal metrology prescribes the format of. And **regulated price** — in South African pharmacy the Single Exit Price is the only price at which a manufacturer may sell to anyone other than the state, with a capped dispensing fee layered on top [9]. The retailer does not set it, cannot promote it and cannot mark it down.

Effective dating here is a legal requirement rather than a convenience. Under section 23 of the Consumer Protection Act a supplier may not charge more than the displayed price, and **where more than one price is displayed concurrently the supplier may not charge more than the lower or lowest of them** [10]. The escape is narrow: the error must be inadvertent and obvious, *and* the supplier must have corrected the display, *and* taken reasonable steps to inform consumers who may have seen it [10]. All of that is a data obligation. The retailer must be able to prove what price was displayed, where, when, and what it did about an error — which means price records are append-only versions, never updates.

The settling question: *could two different numbers both be correct at the same moment?* Different zone, channel, customer or date: yes. Then it is an entity with a scope and dates.

Promotion, offer, and what actually happened in the basket

A promotion is a funded, dated, mechanic-driven intent. An offer is the condition-and-reward rule a basket can satisfy. A **discount application** is what actually fired. Retailers that model one entity for all three cannot explain their own margin.

The reason the third box is not optional: offers interact. A basket satisfying three offers can be discounted in several valid orders producing different totals; the engine picks one; and finance needs to know which offer cost what, because the supplier has to be billed for its share. Store only the net line price and none of that is recoverable. Multi-buys, mix-and-match and spend thresholds are not prices at all — they are basket-level rules with a priority and an exclusivity, which is why they cannot be collapsed into the price box.

One South African wrinkle worth a line. A promotional competition attached to a promotion is a legally distinct animal: the regulations prescribe the competition rules, cap the cost of an electronic entry at R1.50 including all subsequent electronic communication about that entry, require independent oversight by an accountant, auditor, attorney or advocate, and require records to be kept for at least three years and produced to the Commission at the promoter's own expense [11]. A competition treated as marketing rather than as a records obligation is a conceptual modelling error with a compliance tail.

The transaction

The retail transaction is the highest-volume entity in the estate and the most abused.

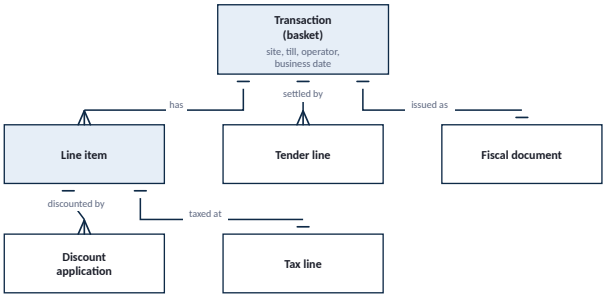


Figure 9.8 The basket, and the four children that are routinely collapsed into columns. Tender is many per basket, tax is per line because a grocery basket mixes VAT rates, and the fiscal document is a prescribed artefact rather than a report.

The invariant that the model must be able to assert, per basket: **sum of line nets, plus tax, less discounts, equals sum of tenders, plus change, plus rounding.** If it cannot, the cash-up will not reconcile and nobody will be able to say why.

Tender is many per basket, not a column — a split across three cards is three tender lines. Tax is per line, not per basket at a blended rate, because a single South African grocery basket mixes standard-rated and zero-rated items [5]. And the till slip is a document rather than a report: a full tax invoice is required above R5,000 and an abridged one at R5,000 or less, with prescribed particulars including the supplier's VAT registration number and, for the full form, the recipient's details [12]. A point of sale that cannot produce a compliant full invoice blocks exactly the trade a cash-and-carry arm depends on.

The settling questions: *would a customer dispute need it?* Then it is part of the transaction record and immutable. *Would finance need to restate it?* Then it needs an adjustment event, not an edit.

Customer, loyalty member, patient

They are not one entity, and collapsing them is the most expensive modelling decision in retail customer data.

- **Party** — a person or organisation the retailer knows about. The identity.
- **Customer** — a *role*: a party with at least one purchase. Derived and dated.
- **Loyalty membership** — a contractual relationship: membership number, tier, points balance, enrolment date, consent record, programme terms version.
- **Account holder** — a party with a credit facility, which is a regulated relationship.
- **Patient** — a party with a dispensing history, which is a regulated health relationship.
- **Identifier** — the card or token used at the till. Many per membership, and replaceable.

A shopper can transact without being identified, and in South African grocery that is the dominant case. So **the transaction cannot require a customer key**, and an anonymous basket has to reconcile like any other. A loyalty member can enrol and never shop. One household holds one membership and three people. One party holds two memberships because she enrolled twice.

POPIA makes the distinction legal rather than stylistic. Using loyalty data for a purpose outside the one it was collected for is further processing, and the Information Regulator's guidance says plainly that sharing a loyalty member's data with group companies for marketing requires the customer's written consent [13]. So *purpose* is an attribute of a consent, consent is an entity with a dated scope, and a single customer view built by joining banner datasets without that consent is a breach dressed as an architecture decision. Chapter 10 takes this apart properly.

The settling question: *would the retailer still need to know this person exists if they left the programme?* Yes: party. *Does it depend on programme terms?* Membership.

The second ring, briskly

Seven more families earn a mention here and a box only sometimes. Each is real; the question is whether it is an entity or a typed event over one of the boxes already on the page.

Space. Fixture, store layout, planogram, facing, planogram assignment, compliance observation. A planogram is usefully defined as a dated SKU-to-position assignment with a derived capacity [14], and shelf capacity is the number the rest of the estate needs: it caps the replenishment order and is the denominator of on-shelf availability.

Forecast and demand. Sales are what happened; demand is what customers wanted. They differ by lost sales, substitution and promotional pull-forward, and a forecast trained on sales under-forecasts everything that was out of stock — the self-reinforcing error that quietly destroys availability. A forecast is an append-only versioned fact with a production timestamp, because it can only be assessed against what was known when it was made. And the **override** — who adjusted it, when, why, by how much — is the most valuable and least-stored fact in retail planning.

Allocation. The one-time push of a finite quantity to many locations, with a method, a constraint and a size curve. It must not share an entity with replenishment: allocation is capacity-constrained distribution under fixed supply, replenishment is demand-driven pull against open supply. Fashion allocates, grocery replenishes, most South African general merchandisers do both. *Is the quantity fixed before the destinations are chosen?* Then it is an allocation.

Markdown. A permanent price reduction that revalues stock, which makes it an accounting event as well as a pricing one. *Does the regular price come back?* No: markdown. I model it as a typed, reasoned, authorised price event rather than as a separate entity, with one exception — the forward-looking **markdown provision** is a finance object that must reconcile to the decision pipeline, and it does deserve a box in a retailer whose margin story is a markdown story.

Shrink event. The gap between book and counted stock, and the architectural mistake is to treat it as a number. It is a decomposition: external theft, internal theft, unclaimed supplier shortage, administrative error, scan error, waste and spoilage, expiry, damage, process loss in the bakery and the butchery, and the honest residual. Shrinkage is conventionally measured as inventory loss from theft, damage, administrative error, waste and supplier fraud as a percentage of sales [15]. Known loss and unknown loss behave differently, respond to different interventions and belong to different owners; one `adjustment` reason with a free-text comment separates none of it.

Return. Not a negative sale. A distinct event with its own eligibility, its own stock disposition and its own **legal basis** — and the basis must be stored, because each one carries a different clock. Section 56 of the Consumer Protection Act gives the consumer six months from delivery with the *consumer* electing repair, replacement or refund, at the supplier's risk and expense, whatever the retailer's policy says [10]. Section 44 of the Electronic Communications and Transactions Act gives a seven-day cancellation without reason after receipt, with only the direct cost of returning the goods chargeable and a refund due within thirty days [16]. A goodwill return is neither, and should be labelled as such, because it is a cost decision rather than a compliance one.

Supplier income agreement. Rebates, listing fees, promotional funding, settlement discounts, volume bonuses. The most audit-sensitive area in grocery retail, because it determines reported gross margin and is accrued on an estimate made by the team whose performance it flatters. *Does the entitlement depend on something the retailer must prove it did?* Then the proof is part of the model — the promotion that ran, the facings given, the volume achieved.

The dispensing record, which is unlike anything else

Nothing else in a retail estate resembles this. A dispensing record is at the same time a sale line, a clinical record, a controlled-substance ledger entry, a reimbursement claim and a statutory record whose retention clock has nothing to do with the transaction's. Most retail books skip it. A South African group with a pharmacy chain cannot.

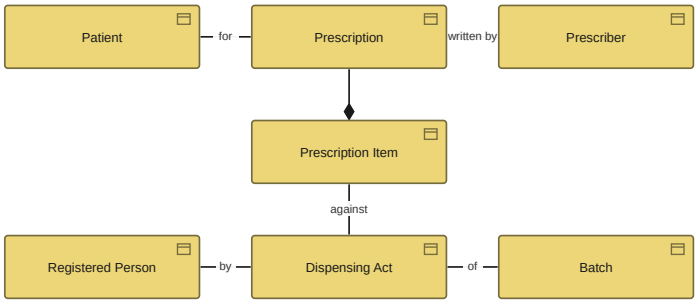


Figure 9.9 The dispensing cluster. The prescription item and the dispensing act reference different articles whenever a generic is substituted, and the batch and the registered person sit on the act because that is what a recall and an inspection ask for.

Seven things make it structurally different from a sale line.

1. **The dispensed product is not necessarily the prescribed product.** Generic substitution means the prescription item and the dispensing act reference different SKUs, with the substitution decision, the patient's agreement and the prescriber's prohibition all recorded.
2. **Batch traceability is mandatory.** A recall must reach the individual patient, which means the batch is on the dispensing act, not just on the receipt.
3. **The actor is legally significant.** The act is attributable to a named registered person and to the responsible pharmacist on duty — which is why a credential, dated and verifiable, is a runtime control rather than an HR report. The till must be able to refuse an action for which no qualified person is present.

4. **The schedule is a dated attribute of the article, not a constant.** Substances are scheduled 0 to 8, the schedule decides who may supply on what authority, and the schedules are amended by gazette repeatedly [17].
5. **The price is regulated** and the retailer does not set it [9].
6. **It is special personal information** under POPIA, which means a different lawful basis, tighter access control and a different retention logic from loyalty data.
7. **It cannot be deleted on request.** A deletion request collides with a statutory retention obligation and the statute wins — but only if the obligation is recorded against the record. Good Pharmacy Practice also limits access to patient information to those who need it for the patient's care, requires written consent for disclosure, requires the disclosure itself to be recorded on the patient's record, and names the identifiers to strip before patient data may be collated for commercial purposes [18].

The settling question is a list: if a regulator asked who did this, to whom, with which batch, on whose authority, and what was said — could the system answer? That list *is* the entity.

The fifteen on one page

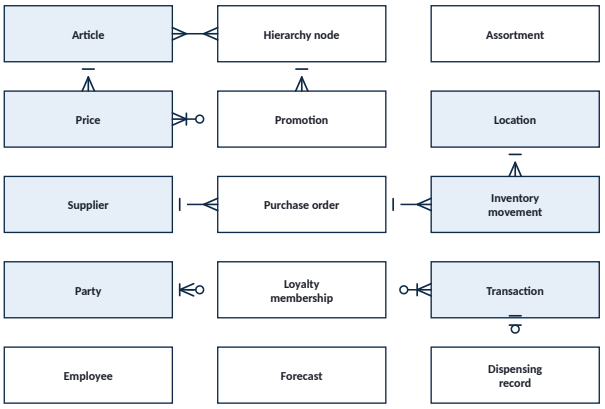


Figure 9.10 The fifteen. Photograph this one. Shaded boxes are the spine; multiplicity is drawn only where a room has argued about it. Markdown, return, allocation, replenishment and shrink are deliberately absent — they are typed events over these boxes.

Photograph this one. The shaded boxes are the spine: Article, Price, Location, Supplier, Party, Inventory Movement and Transaction. Multiplicity is drawn only where a room has argued about it. Everything else on the page hangs off the spine, and the second-ring families from the previous section are deliberately absent — they are typed events over these boxes, and the model is better for saying so out loud.

Entity	The question it answers
Article	What do we sell, at which of the three levels?
Hierarchy node	Whose number is this, and under which structure?
Assortment	Should this store carry this, and is it ranged, planogrammed or orderable?
Price	What may we charge, where, through which channel, from when?
Promotion	What did we offer, who funded it, and what fired in the basket?
Location	Where is it, who owns the stock there, and what may be sold there?
Supplier	Who do we buy from, on what terms, out of which door?
Purchase order	What have we committed to, and has it arrived?
Inventory movement	Why did the position change, and who says so?
Party	Who is this person or organisation, whatever they hold?
Loyalty membership	What did they agree to, and what do we owe them?
Transaction	What happened at the till, and does it reconcile?
Employee	Who did this, and were they qualified to?
Forecast	What did we expect, as at when, and who changed it?
Dispensing record	Who supplied what to whom, on whose authority?

The entities that should not be there

Every first draft carries some of these.

ASKED FOR		WHAT IT ACTUALLY IS
Stock	→	a derivation over movements
Sale	→	a type of transaction
Barcode	→	a dated identifier of the SKU
Store	→	one subtype of location
Category	→	a node in one of six hierarchies
Customer	→	a role a party plays
Vendor	→	three things wearing one word
Online product	→	a navigation grouping, not a thing
Markdown	→	a typed, reasoned price event
Return	→	a transaction with a legal basis
Shrink	→	the residual after decomposition
Master data	→	a lifecycle state, not an entity

Figure 9.11 Twelve things that arrive as boxes and are not entities. Each is a derivation, a role, a type or an identifier, and each one admitted costs a later argument about which copy is right.

Three that are borderline, and when to let them in

Planogram. Let it in when space is a binding commercial constraint and the group runs range reviews with a space tool — which is most grocers. Leave it off when assortment is decided on money and the shelf follows, which is most cash-and-carry. The test is whether anybody argues about facings in a trading meeting.

Supplier income agreement. Let it in when back-margin is material to reported gross margin, which in grocery it is. The Inquiry found a lack of transparency around back-end margins and recommended structured, comparable trade terms [6]; a group whose income lives in spreadsheets cannot answer the Commission and cannot answer its own auditor. Leave it off in a pure own-brand or apparel business where there is no back margin to misstate.

Shrink event. The purist position is that it is an inventory movement with a reason code and does not deserve a box. The practical position is that in a South African retailer shrink has its own investigation, its own evidence, its own owner and its own CCTV footage, and a model that hides it behind a reason code loses the argument with loss prevention in the first meeting. I let it in, and I say in the notes that it is a classification rather than a new fact. Worth knowing before that meeting: **there is no published South African shrink figure**, so a group cannot benchmark its own number, and the honest thing is to say so rather than import a United States retail-federation figure and present it as local.

The same fifteen, three different models

The page is the same. The weight is not.

Grocery. Article degenerates to one level and the levels argument disappears. Inventory movement and Transaction carry everything, because volume is the architecture. Price is zoned and changes weekly. Assortment is a cluster decision. Supplier income is material. Forecast is per SKU per store per day and mostly zeroes. Planogram matters because space is the constraint.

Fashion. Article needs all three levels and the size curve besides. Allocation matters and replenishment barely exists. Markdown is the margin story, so the markdown provision earns a box. Forecast is at option level by week. Supplier income is thin or absent, and price is national and seasonal rather than zoned and weekly.

Pharmacy. Article collapses to one level but pack size becomes load-bearing: a 30-tablet and a 90-tablet pack of the same molecule are separate SKUs and separate reimbursement lines. Price is largely not the retailer's to set [9]. Location gains the dispensary as a separately licensed selling area [8]. Employee gains a credential that is a runtime control. And Dispensing Record, Prescription, Patient and Prescriber arrive as a cluster with their own access model — one that must not flow into the marketing profile.

One group, three formats, three different models over one conceptual page. That is the argument for drawing the page at group level and the variants beneath it, rather than arguing about whose model is the real one.

From here to the information model

The conceptual model says what exists. It is deliberately silent on everything that makes it work: the attributes, the keys, the codes, the owners and above all the time. Chapter 10 picks each box up and asks what the business must *know* about it, and the honest answer is that the time problem in a retailer is harder than the time problem in a bank.

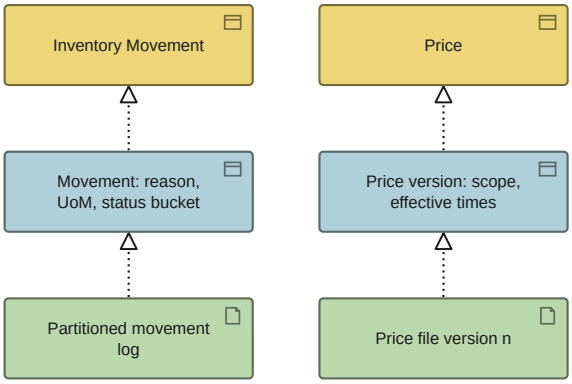


Figure 9.12 The realisation ladder out of this chapter. The business object is stable for a decade, the data object changes when the industry changes, the artefact changes when a vendor ships. Three pages, not one.

Three layers, realised upward. The business object is stable for a decade. The data object changes when the industry changes. The artefact changes when a vendor ships. Draw them on separate pages and a replacement programme becomes scopeable. Draw them on one page and you have produced something nobody can use.

> Run the afternoon with the people who argue about the words, not with the people who own the systems. Fifteen boxes, business language, multiplicity only where it is contested, and the article rule written on the page where everyone can see it. Then stop, photograph it, and let them change it. Everything below is yours.

THE CHAPTER IN TEN LINES

- A conceptual data model is fifteen business entities in business words: no attributes, no keys, no system names and no dates.
- Three tests decide an entity — a one-sentence business definition, the business counts or argues about it, and it survives replacing every system.
- Style, option and SKU are the three levels that exist; article, variant and colour-size are trade names for them, and every system in the estate answers differently.
- The rule that settles it: if its stock is counted separately and it can be sold on its own, it is a SKU — otherwise it is a parent or a sellable non-stock item.
- A retailer has six hierarchies over one SKU and all six are correct; master the mappings instead of picking a winner.
- Supplier, trading relationship, supply source and supplier-article link are four things, and the purchase order is placed on the third.
- The buying chain is many-to-many in every direction, and ordered, expected, received and invoiced are four different numbers.
- There are nine answers to “stock on hand” and a tenth in pharmacy, so stock is a derivation over an append-only inventory movement log, never a column.
- Price is a dated, scoped, typed entity, because the Consumer Protection Act binds the retailer to the lower of two concurrently displayed prices.
- The dispensing record is a sale line, a clinical record, a controlled-substance entry, a claim and a statutory record at once — and nothing else in retail is like it.

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- **10.** Consumer Protection Act 68 of 2008, section 23 (disclosure of price of goods or services) https://www.acts.co.za/consumer-protection/23_disclosure_of_price_of_goods_or_services
- **11.** Consumer Protection Act Regulations, regulation 11 (promotional competitions) https://www.acts.co.za/consumer-protection/r293_11__promotional_competitions
- **12.** SARS, Tax Invoices https://www.sars.gov.za/businesses-and-employers/tax-invoices/
- **13.** Information Regulator (South Africa), Guidance Note on Direct Marketing in terms of POPIA, paras 9.6.1–9.6.2 https://inforegulator.org.za/wp-content/uploads/2020/07/GUIDANCE-NOTE-ON-DIRECT-MARKETING-IN-TERMS-OF-THE-PROTECTION-OF-PERSONAL-INFORMATION-ACT-4-OF-2013-POPIA.pdf
- **14.** Wikipedia, "Planogram" https://en.wikipedia.org/wiki/Planogram
- **15.** TROC Global, "What is retail shrinkage?" (retail glossary) https://trocglobal.com/resources/definitions/what-is-retail-shrinkage/
- **16.** Electronic Communications and Transactions Act 25 of 2002, section 44 (cooling-off period) https://source.acts.co.za/electronic-communications-and-transactions-act-2002/44__cooling_off_period.php
- **17.** SAHPRA, Consolidated Schedules, August 2025 https://www.sahpra.org.za/document/consolidated-schedules-august-2025/
- **18.** Rules relating to Good Pharmacy Practice, published under section 35A of the Pharmacy Act 53 of 1974 (consolidated) https://www.saflii.org/za/legis/consol_reg/rrtgpp362

CHAPTER 10

The information model

What the business has to know about each of the fifteen boxes — the attributes, the identifiers, the codes, the owners, and the calendar that does not have twelve months in it.

The layer where the arguments get expensive

Chapter 9 produced fifteen boxes that a trading director would recognise. This chapter asks what the business has to *know* about each of them, and that is where the cost moves from an afternoon to a programme.

The information model is the layer below the conceptual one and above the physical schema. It holds attributes that earn their place, the identifiers that let two systems agree they are talking about the same thing, the code lists that decide what is lawful, the owner of each fact, the trail that proves where it came from, and — the part a retailer finds hardest — time.

Milkwood Retail Group is invented: a South African multi-format group of about 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm and an online business. Its information model took eleven months and two restarts. The first restart happened when somebody asked which of three cost figures the margin report used. The second happened when the group could not reproduce the prior-year comparative in its own trading statement after a departmental reorganisation.

Neither failure was technical. Both were failures of this layer.

> A retailer's time problem is harder than a bank's. A bank reconstructs a balance as at a date. A retailer has to do that *and* run a calendar that does not align to months, *and* compare two periods that are not the same length, *and* tolerate several hundred stores reporting yesterday's trade at different times today.

Attributes, where they earn their place

The conceptual model banned attributes. This one needs them, and the discipline that replaces the ban is a test.

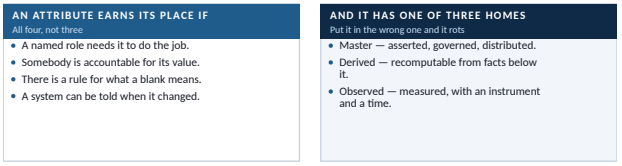


Figure 10.1 Four questions, and three homes. The fourth question is the one that gets skipped, and an attribute nobody can date cannot be audited, restated or reproduced.

The fourth question is the one that is skipped. An attribute nobody can date is an attribute that cannot be audited, cannot be restated and cannot be used in a reproduction of last Tuesday's price. The three homes matter just as much. **Master** attributes are asserted by somebody who is accountable: cost, retail price, VAT code, schedule. **Derived** attributes are recomputable from facts below them: stock on hand, gross margin, available to promise. **Observed** attributes are measured with an instrument at a time: shelf temperature, planogram compliance, a counted quantity. Put a derived attribute in the master layer and it goes stale silently; put an observed one there and you lose the instrument and the time, which is the only part that makes it evidence.

The article master shows the problem at full size. It carries, at minimum, seven groups of attribute plus a lifecycle status with dates. Six of the groups have an obvious owner. The seventh does not, and it is the one that cannot be back-filled.

Group	What is in it	Who actually knows
Identity	Internal number, level, GTINs, supplier reference, legacy keys from prior systems	Merchandising
Descriptive	Till-display description (character-limited by the hardware), long description, brand, own-label flag, copy, images	Marketing and the website
Classification	A node per hierarchy, GPC brick, tax code, regulated classification	Six different owners
Physical	Dimensions, weight, pack configuration, case and pallet quantities, temperature class, shelf life	The supplier
Commercial	Cost in its several flavours, retail, margin target, price family, VAT rate, deposit	Buying and finance
Operational	Orderable, stockable, sellable, weighed, serialised, batch-tracked, age-restricted, scale item, PLU	Supply chain and store operations
Regulatory	VAT zero-rating, medicine schedule, liquor and tobacco class, allergens and nutrients, packaging materials and masses, country of origin, declared quantity	Nobody, until an inspection

The till-display character limit in the second row is not a detail. It is a point-of-sale hardware constraint that propagates upward into the master, and it is the reason the description a customer sees on a slip is not the description on the website.

The regulatory group is the one people forget. Packaging material and mass, for the extended-producer-responsibility return [1]. Nutrient values per 100g, without which the group cannot assess its exposure to a front-of-pack labelling regime. Allergens. Country of origin, which is a label requirement and therefore has to agree with the label. VAT zero-rating [2]. Medicine schedule [3]. Liquor and tobacco classification. The declared quantity that legal metrology prescribes the format of [4]. A retailer that did not capture packaging material and mass at article creation cannot produce the EPR return without a project; one that did not capture nutrients cannot answer a labelling question without going back to every supplier.

Age restriction is a control, not a label. The article must carry the restriction *and* the check type, the till must enforce it, and the enforcement must be recorded. A free-text "18+" in the description is a note, not a control, and it will be discovered during an inspection rather than during testing.

Keys, and why the barcode is not one

Two systems agree about a product when they share an identifier. A retailer has at least six identifier namespaces in play, they overlap, and only one of them is internal.

FORM	DIGITS	WHERE YOU MEET IT
GTIN-8	8	Very small packs. Not for groupings
GTIN-12	12	North America — the old "UPC"
GTIN-13	13	The rest of the world — the old "EAN"
GTIN-14	14	Case, inner, pallet. Indicator digit 1–8
Variable measure	13	In-store, weighed. Never leaves the estate
GS1-128 / DataMatrix	—	Carries batch, expiry, serial, weight

THE DEFECT THAT COSTS THE MOST

0 0 6 0 0 1 2 3 4 5 6 7 8

A GTIN-13 stored as a number loses its leading zero in transit.
Normalise to fourteen digits, right-aligned, zero-padded, as a string.

Figure 10.2 The GTIN families a retailer meets, and the defect that costs the most. Normalise every identifier to fourteen digits as a string at every boundary and a whole class of integration defect disappears.

Four facts carry most of the weight.

The GS1 prefix is not a country of origin. South Africa is allocated prefixes 600 and 601, and GS1 states plainly that prefixes do not identify a product's country of origin, because a company can manufacture anywhere [5]. GS1 South Africa, which trades as the Consumer Goods Council of South Africa, has had to say so publicly, because consumers read a non-600 prefix as counterfeit [6]. Retailers nonetheless build origin reports off the prefix and are wrong. Country of origin is a declared, separately sourced, separately governed attribute.

A SKU may legitimately carry several GTINs, and a GTIN may be reissued. Supplier change, promotional pack, multipack, relabelled import and re-issue after a sunset period all produce a many-to-one or a temporally ambiguous relationship. So the GTIN is a *dated identifier entity* with a type, a primary flag and a validity period — not the primary key of the article. The internal article number is the key, and it is the one thing in the estate that must never be reused.

Case quantity is the highest-impact number on the record. The each scans at the till, the case scans at goods receiving, the pallet scans on the dock. GS1 UK sets out two legitimate strategies — a separate GTIN-13 for every packaging level, or a GTIN-13 on the consumer unit with GTIN-14s derived for everything above it [7] — and the retailer cannot assume which one a supplier used. A wrong case quantity multiplies or divides every order, every receipt variance, every stock position and every forecast consumption figure by the same wrong number.

Variable-measure codes must never leave the estate. A weighed item's barcode is printed in-store and embeds a price-lookup code with either a weight or a price, under a prefix reserved for restricted distribution inside one company. A material share of a grocer's turnover passes through identifiers that are valid nowhere else, cannot be matched to a supplier's article, and break every GS1-based exchange. Keep them in their own field, with their own type, and never publish them.

Three more namespaces round it out. **GLN** identifies parties and locations and is how trading partners address a store or a distribution centre — so the GLN-to-internal-location mapping is reference data with an owner, and when it is wrong goods arrive at the wrong place with paperwork for the right one. **SSCC** identifies *this* pallet rather than a class of item, which is what makes receipt-by-scanning possible [8]. And **PLU** is the scale's own namespace, distinct from the GTIN, distributed down a fifth channel nobody remembers until it breaks.

Two schemes for one product, and nobody owns the bridge

The cleanest example of a master-data problem with no owner sits in the pharmacy chain.

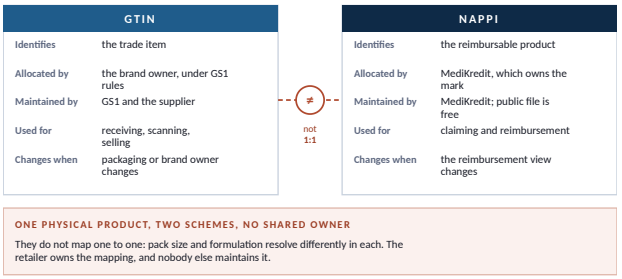


Figure 10.3 One physical product, two identifier schemes, two custodians, two cadences — and a mapping the retailer owns because nobody else will. This is the clearest master-data problem with no owner in a South African retail estate.

A NAPPI code is a unique identifier for pharmaceutical, medicinal, surgical and healthcare consumable products, used to move product information electronically across the healthcare chain. MediKredit owns the NAPPI trademark and manages the file day to day; the code has been the South African electronic reimbursement standard since the early 1990s; the public file is free and includes the code, description, strength, pack size and manufacturer, with the fuller data set available by subscription; and allocation of a code does not endorse the product [9].

So a South African retail pharmacy's item master carries **two identifier schemes for the same physical thing**. The GTIN is how the box is received, scanned and sold. The NAPPI is how it is claimed and reimbursed. They are allocated by different bodies, under different rules, on different cadences, for different purposes — and they do not map one to one, because pack-size and formulation variation resolves differently in each.

The architectural consequence is the part worth remembering. **The GTIN-to-NAPPI mapping is a master-data asset with a compliance consequence, and the retailer owns it.** GS1 will not maintain it. MediKredit will not maintain it. The scheme will not maintain it. It is a table in the retailer's estate, it decides whether a dispensed line can be claimed, and in most pharmacy retailers it is maintained by one person who learned it from the person before them.

Treat it the way you would treat any other unowned, load-bearing mapping: give it an owner, a change process, a reconciliation against both source files, and an exception report for every dispensed line whose NAPPI could not be resolved. I could not source NAPPI's code structure, check-digit rules or allocation process — MediKredit's public page does not describe them — so confirm those directly before designing validation.

Classification: the codes that decide what is lawful

Classification in a retailer is not taxonomy. Several of the codes decide whether a sale is legal, and they are owned outside the business.

VAT. The standard rate is 15%, and a list of basic foodstuffs is zero-rated under Part A of Schedule 2 of the VAT Act [2]. The boundaries are finer than any commercial category: brown bread is zero-rated, brown bread rolls are not; raw vegetables are, prepared salads are not; milk is, flavoured milk is not. **There is no hierarchy under which the VAT rate is inheritable**, which is why chapter 9 put regulated classification on its own hierarchy. The VAT code is a governed attribute with a change history and an approval trail, because a wrongly standard-rated staple over-collects from the poorest customers and a wrongly zero-rated line under-declares to SARS.

Medicine schedule. Substances are scheduled 0 to 8, and the schedule decides who may supply, on what authority, whether a prescription is required and whether a register entry is generated. SAHPRA publishes consolidated schedules — most recently August 2025 — and amends them by gazette repeatedly [3]. Schedule is therefore a *dated* attribute with an ingestion process behind it, because a gazette amendment silently changes what a till may sell.

Licence class. Liquor is licensed per premises by the province, with hours set provincially and municipally; tobacco carries its own restrictions and, since September 2024, may not be sold in a pharmacy at all. So the article's licence class resolves against the *site's* dated licence set and the site's trading-hours calendar. One national configuration produces unlawful sales in one province and blocked lawful ones in another.

Producer obligations. Packaging material and mass per article feed a statutory return [1], which makes a physical attribute of the pack into a reporting obligation and makes own-label a category with extra duties, because for own-label the retailer is the brand owner.

The pattern across all four: the code list is owned by somebody outside the business, it changes on their schedule rather than yours, and the change is an *event* your estate has to ingest. A classification maintained by annual spreadsheet is a compliance finding waiting for an inspection.

Ownership, lineage and the gate that actually works

The article master is the retailer's most-integrated object: read by the till, the scales, the label printer, the website, the ERP, the warehouse system, the forecasting engine, the space planner, the tax engine and every supplier portal. Which is why **it has many writers, no owner and no acceptance criteria.**

The pattern is consistent enough to state as a proposition. The buyer creates the record to place an order, with the minimum fields the order screen enforces. Marketing adds copy later, or does not. Space planning needs dimensions nobody captured. Tax needs a code that defaulted to standard-rated. Compliance needs allergen and packaging data nobody asked for. The website needs images. Each team patches what it needs in its own system, and the estate acquires four partially correct versions of every article.

The usual fixes fail for a reason worth naming. A governance committee with no authority over the order screen changes nothing, because the only enforcement point that works is creation — and the buyer will not accept sixty mandatory fields to raise one order. An MDM tool deployed over unchanged processes becomes a fifth copy.

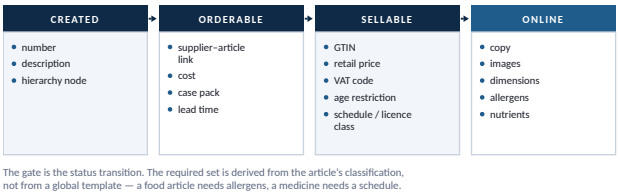


Figure 10.4 Staged mandatory sets tied to lifecycle status. The buyer will not accept sixty mandatory fields to raise one order, and does not have to — the gate is the status transition, not the create screen.

What constrains it is a staged gate. An article reaches *orderable* with the order-critical fields; it cannot reach *sellable* without the till fields; it cannot reach *online* without copy and images; and it cannot reach *sellable at all* without the regulatory fields its classification demands. The status transition is the gate, the required set is derived from the classification rather than from a global template, and supplier-sourced data arrives through a defined channel with rejection on failure — because the supplier knows the pack dimensions and the buyer does not.

Lineage is the other half. For every attribute that matters: where it came from, who asserted it, when the estate learned it, and which version of which rule derived it if it was derived. Two questions settle whether the lineage is real. Can you reproduce last month's gross margin using the cost version that was in force then, rather than today's? And can you say which forecast version and which parameter set produced a particular order? If the answer to either is no, the estate is storing outputs without inputs.

The second question is the one store operations will ask you, in those words. "Why did the system order forty-eight of this?" is the commonest operational question in retail, and it is answerable only if the position, the forecast version and the parameter version in force at computation time can all be recovered. That is a bitemporal requirement on the planning data, not a logging preference, and it is almost always discovered a year after go-live when somebody tries to tune the engine and finds there is nothing to tune against.

Milkwood's first restart came from exactly this. Three cost figures existed for every article — the supplier-article cost, the last receipt cost and the standard cost used for valuation — and the margin report used whichever one its author had found first. None of the three was wrong. Nobody had written down which one was authoritative for which question, which is a one-page deliverable that nobody had been asked to produce.

Retention: the longest clock wins, and it is not one clock

Retention in a retailer is not a number. It is a schedule per data class, and the inputs conflict.

Tax records have their own period. Promotional-competition records must be kept at least three years and produced to the Commission at the promoter's own expense [10]. Card data must be minimised under PCI DSS, which pushes the other way [11]. POPIA says personal information may not be kept longer than is necessary for the purpose. And pharmacy runs on a clinical clock that has nothing to do with the commercial one: Ontario, for contrast, requires records relating to a patient's care to be kept at least ten years from the last professional service, **or until ten years after the patient reached or would have reached eighteen, whichever is longer** [12]. That rule makes the patient's date of birth a *retention input*, not optional demographic data — and it means retention has to be a per-jurisdiction policy attached to the record type rather than a global setting.

The honest design is a retention class on every data object, a stated legal basis for each class, and an exception path for a lawful hold — because a shrink investigation extends the retention of specific CCTV footage, and a blanket rule will delete the evidence.

Time in a retailer

This is the long section, and it is long because almost every reporting failure in a retailer is a time failure wearing something else's clothes.

Effective dating on price and assortment

Three dated statements carry most of the load.

Price is covered in chapter 9 as an entity. Here the requirement is evidential and precise: effective-from and effective-to are **timestamps, not dates**. A price change applied at 06:00 before the store opens and one applied at 14:00 mid-trade are different facts with different reconciliation consequences, and section 23 of the Consumer Protection Act binds the retailer to what was displayed at a place at a time [13].

Assortment is three dated statements that retailers routinely collapse into one boolean. *Assorted* is the commercial decision that this store should carry this SKU. *Planogrammed* is the space decision that it has a facing. *Orderable* is the operational state that it can be replenished. A SKU can be assorted and not planogrammed, waiting for the next range review; planogrammed and not orderable, because the supplier is out; or orderable and not assorted, because somebody did a bulk buy. One boolean makes the range review invisible and the phantom-range problem undiagnosable.

Hierarchy membership is dated for a harder reason. Oracle's merchandising documentation states that when an item is reclassified, "its sales history, purchase orders, inventory, and planning information are reclassified too" [14]. That is a choice, and it has a twin: a reclassification that carries history rewrites last year's numbers so this year's comparison works and last year's published report no longer reconciles; one that does not carry history preserves the report and destroys the comparison. Both are defensible. The retailer must choose, must choose the same way every time, and in practice different systems in the same estate choose differently. Find out which does which; it is a two-day question with a five-year consequence.

The general pattern is two timelines. Every dated master statement is a version with a **business validity period** — when the fact was true in the world — and a **record timestamp** — when the estate learned it. Retail needs the second specifically to answer "what did the replenishment engine know when it ordered" and "which forecast was in force when the buy was committed".

The retail calendar, and the week that is not there

Retailers do not use calendar months.

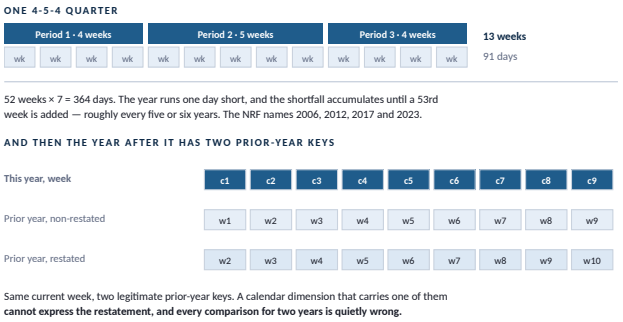


Figure 10.5 The 4-5-4 quarter, the missing day, and the restatement that breaks a naive calendar dimension. A prior-year key has to exist twice: once restated, once not.

The dominant convention is the **4-5-4 calendar**: each quarter holds three "months" of four, five and four weeks, so the year is 52 weeks and 364 days, with the extra day accumulating until a 53rd week is added — about every five or six years, and only when four or more days remain in January after the 52 weeks are laid out [15]. The National Retail Federation names 2006, 2012, 2017 and 2023 as 53-week years, and gives the first week of 2017 as 29 January to 4 February on the non-restated calendar [15]. The retail year begins in late January or early February.

The purpose is comparability. Comparable months hold the same number of Saturdays and Sundays, and holidays line up, so like days compare with like days; the convention dates to the 1930s, when Gregorian monthly reporting produced period-on-period noise purely from weekend counts [15].

The 53rd-week restatement is the part that breaks data models. The NRF restates a 53-week year in the following year by pushing each week back one and skipping the year's first week, so the holidays stay aligned — the restated 2017 first week runs 5 to 11 February rather than 29 January to 4 February — and it publishes both restated and non-restated versions [15]. The alternative is to leave the year unrestated and ignore the 53rd week in comparisons [15].

Either choice is fine. What is not fine is a calendar dimension carrying one set of prior-year keys, because then the restatement cannot be expressed and every comparison in the 53-week year and the year after it is quietly wrong. The requirement is a calendar dimension carrying, per day: fiscal year, quarter, period, week, day of fiscal week, **and an explicit prior-year-comparable day key populated differently for the restated and non-restated bases**, plus public-holiday, trading-day and season flags.

> South African retailers' financial year-ends do not cluster where the American convention puts them, and the Gregorian date matters here in a way a pure 4-5-4 dimension often drops. Easter moves between fiscal periods and shifts a material trading week. The pay cycle and the social-grant dates mean the *day of month* is a driver alongside the day of fiscal week. Keep the Gregorian attributes on the calendar dimension. The specific South African year-end pattern is something I could not source and would not state as fact.

Three dates, and then two more

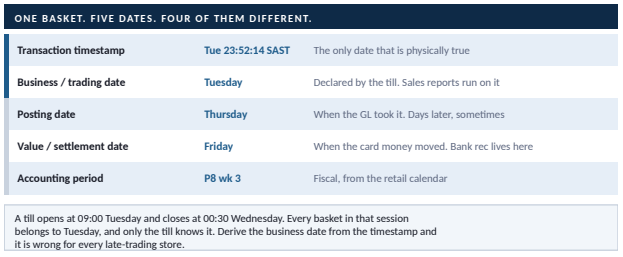


Figure 10.6 Five dates on one basket, four of which differ. Four reporting questions run on four of these columns, and collapsing them is why store reconciliations fail.

Four queries, four date columns, one transaction. Sales reports run on business date. Bank reconciliation runs on value date. The general ledger runs on posting date. The trading statement runs on fiscal period. Collapse them into one and the store reconciliation fails and nobody can explain why.

One South African convenience is worth naming because it will not survive expansion. South Africa has no daylight saving and one time zone, which removes a whole class of defect. A Canadian estate spans six time zones with daylight saving in most of them, so it must store timestamps with an offset and must be able to express a 23-hour and a 25-hour trading day. A model built in South Africa and lifted to Canada breaks here first.

Late-arriving store data

Several hundred sites, each closing tills at a different time, each uploading over links of varying quality, some buffering through a power or network outage. The consequence is simple and most reporting architectures refuse to accept it: **yesterday's sales number is not final today, and may not be final this week.**

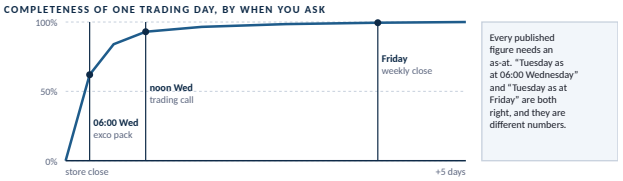


Figure 10.7 Yesterday's trade, arriving. The curve is illustrative, but its shape is the point: completeness is a measured state with an as-at, not a belief about last night's batch.

Four facts the model needs. **Completeness** as a measured, published state: per site, per business date, a reconciliation of expected till sessions to received till sessions — not a belief. An **arrival timestamp** on every transaction batch, distinct from the business date. A **restated-on** marker on every aggregate, so a number that moved can be shown to have moved. And an **as-at convention** on every published figure.

The anti-pattern is overwriting the daily aggregate as data arrives, with no prior value and no completeness marker, so the trading number changes between the morning and afternoon packs with no explanation available. In South Africa the load-shedding dimension makes this structural rather than occasional: stores trade offline and buffer, then upload in bursts, and some tills trade cash-only with no card network. Degraded-mode transactions must be identifiable, because their tender mix and basket composition are not comparable to normal trade.

The restatement problem

A restatement is a change to a previously published figure. Retail generates them routinely and legitimately, and there are eight common causes.

1. **Late-arriving data**, as above.
2. **A hierarchy reorganisation** — last year's departmental sales change because the department changed.
3. **Stores moved** between regions, formats or legal entities.
4. **The 53rd week**, and the choice of restated or non-restated basis.
5. **Acquisitions and disposals**, where like-for-like must exclude the acquired estate until it has a comparable base.
6. **Corrections** — a mis-scanned count, a duplicated transaction batch, a wrongly dated price change.
7. **Accounting policy changes** — a change in the markdown provision basis or in supplier-income recognition restates gross margin.
8. **Currency**, for a group trading across borders.

Only two of those eight are errors. The other six are the business changing shape, which it does continuously, and an architecture that treats restatement as an exception will be surprised six times a year.

The requirement follows: every published aggregate must be reproducible **as at the date it was published** and **on the current basis**. That means the warehouse holds atomic facts plus versioned dimensions plus versioned structures, and reporting always states its basis. One mutable `SALES_BY_DEPT` table satisfies neither half.

Like-for-like is an exclusion set with a policy

Like-for-like — comparable-store, same-store — growth is the most scrutinised number a listed retailer publishes, and it is **defined by exclusions**. It is therefore a modelled set, not a calculation on a report.

SITE EVENT	IN THE SET?	WHY — AND IT IS A POLICY, NOT A F
Store opened mid-base period	OUT	no comparable base
Closed for refit 9 weeks	OUT	policy: >6 weeks dark
Extended by 340 m²	OUT	policy: >15% floor change
Relocated 400 m down the road	?	policy must say
Dispensary added	?	policy must say
Converted franchise → corporate	OUT	ownership change
Online orders picked in store	?	channel attribution rule
Traded both periods, unchanged	IN	the comparable set

Like-for-like is an exclusion set with a policy, and the policy is data. Hold it or the number cannot be audited.

Figure 10.8 Like-for-like, as it is actually decided. Three of these eight rows have no right answer — only a policy — and the policy has to live in the model if the published number is to be auditable.

A store enters the comparable set only if it traded through the whole of both periods, which immediately requires, per site: opening and closing dates and every non-trading period — refit, flood, fire, civil unrest, extended outage; **material change events** such as an extension, a downsizing, a relocation, a format conversion, the addition or removal of a dispensary or liquor section, or a change from franchise to corporate; the retailer's own **policy thresholds** for how much floor-space change disqualifies a store and how long a closure does; and a **channel treatment** rule saying whether online is in, and whether click-and-collect belongs to the fulfilling store, the collecting store or neither.

Each of those is an entity or a dated attribute on the site, plus a **comparable-set membership** fact per site per period pair carrying the reason for inclusion or exclusion. A retailer computing like-for-like in a spreadsheet from a store list cannot tell an auditor why a store was included, cannot restate when the policy changes, and cannot reconcile the published number to the warehouse.

Reference and master data, and the five distribution channels

Six reference sets carry the estate: the article master, the supplier master, the location master, the price file, the hierarchy tables and the barcode and case-code set. Chapter 9 defined what they are. Here is what goes wrong with each as *data*.

The article master is covered above: many writers, no owner, a staged gate as the only fix that works.

The supplier master has four recurring problems. Dual identity, because the same legal entity is a supplier in procurement, a creditor in finance, a trading partner in EDI and a counterparty in the contract repository, usually with four keys. Regulated attributes with expiry dates — VAT registration, tax clearance, B-BBEE certificate or affidavit, food-safety certification, a wholesale licence for medicines — each needing a validity period and a block-on-expiry rule. Bank-detail change control, which is the highest-value fraud target in the estate. And the absence of a supplier *hierarchy*, which is why "what is our total spend with this group" takes a week to answer. Add to that the South African compliance load: the Grocery Retail Market Inquiry recommended uniform, objectively justified trade terms [16], and B-BBEE status must be held **as at the transaction date** rather than currently, or a prior year's scorecard cannot be reconstructed.

The location master fails in four familiar ways: duplicated across the till system, the merchandising system, the warehouse system and the website with four different store numbers; trading hours held as text; licences held as scanned PDFs with no expiry; and no geography hierarchy beyond a free-text province, which makes municipal analysis impossible even though trading hours and zoning are municipal.

The price file is the distribution artefact, distinct from the price master in the way a payload is distinct from a record.

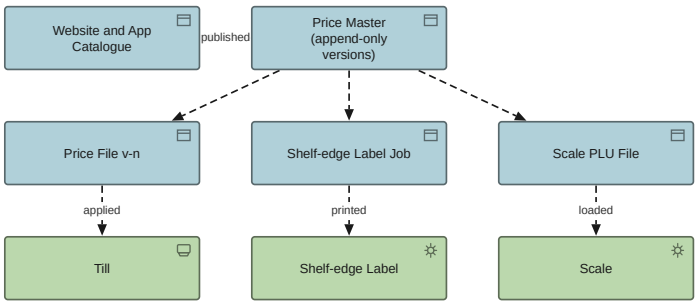


Figure 10.9 The price master and four of its downstream channels, each with its own lag. The edge that is missing from almost every real estate is the one running back — a per-endpoint acknowledgement that the version was applied, which is what makes a displayed-price defence possible.

The architecture needs four things of it: a complete base file plus deltas; a version and an effective time, so a till can be asked which version it is running; an **acknowledgement** from every endpoint, so the group knows which stores have the new prices; and a **reconciliation** comparing what the till charged to what the master said. That last control is what makes section 23 defensible [13]. And the shelf-edge label is a separate channel with its own lag — the gap between a till price change and a reprinted label is precisely the two-displayed-prices scenario the Act punishes — so label print status per SKU per site is a compliance fact, not a store-operations detail.

The hierarchy tables are four tables, not one flattened view: a hierarchy with a type and an owner; a hierarchy node with a parent and effective dates; an article-to-node assignment, itself dated; and a node mapping for the cross-hierarchy equivalences reporting needs. Denormalised flat views are built *from* these for query speed. They are never the system of record.

Barcodes and case codes are the set architects most often treat as one field, and there are four distinctions to hold. First, **symbology is not data**: EAN-13, UPC-A, ITF-14, GS1-128, GS1 DataBar and GS1 DataMatrix are ways of printing, and the same GTIN appears in several of them. Store the data, record the symbology separately, and do not build logic on the shape of the print. Second, **consumer unit, trade unit and logistics unit are three identifiers for one product** — the each scans at the till, the case at goods receiving, the pallet on the dock — with conversions between them that are dated properties of the supplier-article link, because suppliers change case packs mid-season. GS1 has ruled that GTIN-8 is not to be used for groupings [21], so a small-pack code is never a case code. Third, **codes that carry more than a GTIN** — GS1-128 and DataMatrix with their application identifiers — bring batch, expiry, serial and weight into the scan, which the till and the warehouse system must be able to read *and store*. That is mandatory in pharmacy for recall traceability, and it is the direction of travel everywhere else. Fourth, **own-label makes the retailer the brand owner**, and therefore the GTIN allocator, with the allocation-rule obligations and the producer obligations that follow.

Classification, consent and POPIA

The loyalty programme is the retailer's largest holding of personal information and its most purpose-ambiguous.

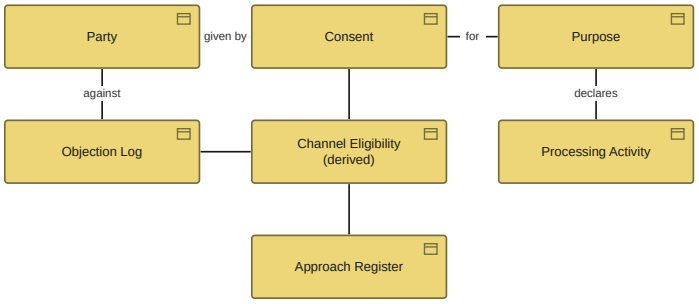


Figure 10.10 Consent as an entity with a purpose, and channel eligibility as a derivation over consent, objections and the once-only approach register. A stored flag cannot carry any of this.

Purpose is an attribute of a consent, and consent is an entity. The Information Regulator's position is explicit: using loyalty data for marketing outside the purpose it was collected for is further processing, and its worked example — sharing a loyalty member's data with group companies for marketing — requires written consent [17]. A group with a shared programme across banners therefore needs the sharing purpose consented at enrolment, or the data cannot cross the banner boundary.

The direct-marketing rules turn that into a specification. For non-customers, consent must be obtained using **Form 4 or a substantially similar form**, free of charge, recording the goods or services marketed and **the communication method the person chooses, which the responsible party must then use**; telephone consent requires reading out Form 4 and recording the call; and a responsible party may approach a person **once only**, and only if they have not previously refused [17]. The burden of proof sits with the retailer. For existing customers, the section 69(3) exemption applies only if the contact details were obtained in the context of a sale, the marketing is of the party's **own similar** products, and the customer was given a free and easy opportunity to object both at collection and in every message [17]. The Regulator's own illustration is that funeral insurance is not similar to clothing [17].

So, as data: a consent record with form version, capture channel, chosen method, scope, timestamp, evidence reference including the call recording, and withdrawal events. A **single-approach register** so the once-only rule can be honoured. An **objection log** distinct from the consent record. And a **channel-eligibility derivation** per party that resolves consent, section 69(3) eligibility and the retailer's own preferences into one answerable yes-or-no per channel per campaign — computed, not stored as a flag. Most retailers have the flag and not the evidence, and have suppression in the campaign tool rather than as a shared service.

CCTV and biometrics. Store CCTV processes personal information, and where it identifies individuals it edges into biometric information, which POPIA treats as special personal information and which, used as a unique identifier for a purpose other than collection, engages the prior-authorisation regime [18]. The data artefact is a **surveillance asset register**: camera, location, field of view, purpose, lawful basis, retention period, access list, and whether any analytics on the feed produces biometric templates. I could not source retail-specific guidance from the Regulator on in-store facial recognition or shelf-camera analytics, and an architect should check the prior-authorisation register before designing one.

And the group's legal structure becomes a data-governance object. Each subsidiary must register its own information officer, registration must happen before taking up duties and be updated at least annually, and the access manual must agree with what the Regulator was told [19]. A group with a separate legal entity per banner, per property company and per franchise vehicle has a registration obligation per entity.

The pharmacy boundary is a hard stop. Patient dispensing data is health information, and Good Pharmacy Practice limits access to those who need it for the patient's care, requires written consent for disclosure, requires the disclosure itself to be recorded on the patient's record, and names the identifiers to strip before patient data may be collated for commercial purposes [20]. The loyalty programme may know that a basket included a dispensary transaction and its value. It must not know what was dispensed.

The analytics layer, and the object everybody draws too small

The grain of the sales fact settles the design: one row per transaction line item. Any coarser grain — per basket, or per SKU-store-day — destroys basket analysis and makes promotional attribution impossible. The dimensions that grain needs are article at every level, location, calendar with both fiscal and Gregorian attributes, customer where identified, operator, promotion and offer, tender type, channel, price type and tax code. The conformed facts are units, gross value, discount value **by funding source**, net value, VAT, cost of goods at the cost basis in force, and gross margin.

That last clause is load-bearing. **Cost must be the cost version in force at the time**, so the cost dimension is dated and margin is computed at load against that version — not recomputed later against current cost, which silently restates every historical margin.

Basket analysis is what the line-item grain is for, and it is the reason the grain cannot be compromised later. It answers affinity — what is bought together — cannibalisation, the incrementality of a promotion, the basket-building effect of a category, the recognisable shapes of trip, and customer-level repertoire and switching. Three structural requirements are routinely missed. The basket must be identifiable **as a unit across channels**, including where one order splits across fulfilments or where click-and-collect creates two records. **The anonymous basket must participate**, because an analysis restricted to loyalty members measures a self-selected minority and most South African grocery baskets are unidentified. And **promotion context must be on the line, not inferred later**: which offer fired, at what funded cost. Reconstructing that from price differences is unreliable, and it gets worse as mechanics get more complex.

Inventory position history is the hardest table in the warehouse, because it is a state over time rather than an event stream. A full daily snapshot is simple and enormous: sixty thousand SKUs across six hundred locations is thirty-six million rows a day even where nothing moved. A change-only snapshot with validity periods is far smaller and harder to query. A movement log with derived position is smallest and most honest and most expensive to query. The pattern that works is a movement log *plus* immutable period-end snapshots *plus* a rolling recent-days snapshot: the snapshots serve

reporting and get signed off, the log serves investigation and reconciliation. Position must be held by **status bucket**, not as one quantity, and any snapshot reported to the market or used in a statutory return must be immutable.

The volume problem

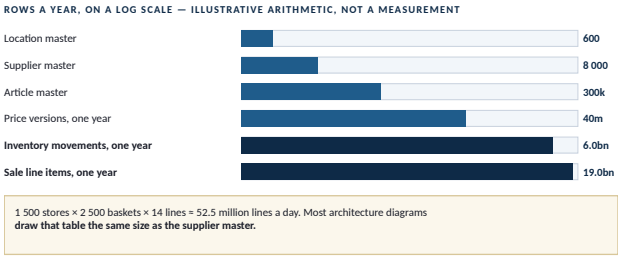


Figure 10.11 The estate's objects on a log scale. The line-item table is four orders of magnitude larger than the article master, and on most architecture diagrams the two boxes are the same size.

A large retailer's sale-line-item table is the biggest object in the estate by an order of magnitude, and most architecture diagrams draw it as one box the same size as the supplier master. The arithmetic is worth doing in front of a sceptical audience: 1 500 stores, 2 500 baskets per store per day, 14 lines per basket is about 52.5 million line items a day, 1.6 billion a month, roughly 19 billion a year — before online, before discount applications, before tender lines, and before the inventory movement log, which in a replenishment-heavy grocery estate is of comparable size. At seven years of retention that is hundreds of billions of rows. Those figures are illustrative arithmetic, not a measurement of any retailer; no South African retailer publishes its line-item volumes.

Five consequences belong in the architecture rather than in a footnote.

- **Partition by business date, not timestamp**, or partition pruning fails for every late-trading store.
- **Retention is a schedule per data class** with conflicting legal inputs, not one number for "transactions".
- **Aggregates must be reproducible from the atoms**, with the basis stated.
- **Query patterns diverge.** Operational reporting wants recent fine grain, planning wants long coarse history, data science wants long fine history for a narrow slice. A columnar store for the long tail plus a serving layer for the recent window is the honest answer.
- **The card-data boundary cuts straight through this table.** Storing tender detail alongside line items drags the entire sales history into PCI DSS scope [11] — an expensive mistake made once and lived with for a decade.

What a single version of the truth has to mean

It cannot mean one number. A listed retailer publishes a trading statement, and the moment that number is public it acquires a basis: a period definition, a hierarchy version, a store set, a cost basis, a completeness as-at and a restatement position. Change any one of those and the number legitimately changes.

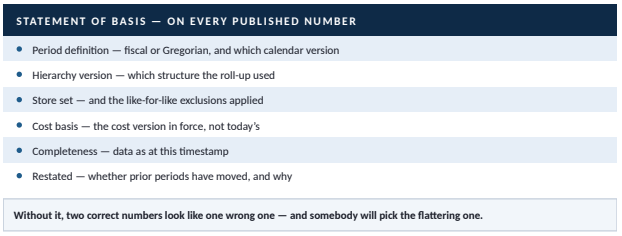


Figure 10.12 What has to travel with a published number. A single version of the truth is one set of atomic facts and versioned structures, plus this, on everything derived from them.

So the useful definition is narrower and much more achievable. **One set of atomic facts, one set of versioned dimensions and structures, and a stated basis on every number derived from them.** Two numbers that differ are then not a data-quality problem; they are two bases, and the statement of basis says which.

That is the test to take into the first week. Ask for yesterday's sales by department, then ask the same question again tomorrow. If the answer moved and nobody can tell you why, the estate has no information model — it has a warehouse with opinions in it. Chapter 11 turns to the applications that have to produce all of this, and asks which of them a retailer actually needs.

> Three sentences to take out of this chapter. Time is the hard part, and the calendar is part of the data model rather than part of the report. An identifier maintained by somebody else is a feed, not a field. And the biggest object in the estate is the one nobody draws.

THE CHAPTER IN TEN LINES

- An attribute earns its place when a named role needs it, somebody owns its value, a blank has a defined meaning and the estate can say when it changed.
- The barcode is not a key: a SKU carries several GTINs over time, GTINs get reissued, and the GS1 prefix is not a country of origin.
- Normalise every trade-item identifier to fourteen digits, right-aligned and stored as a string, at every system boundary.
- A South African pharmacy item master carries GTIN and NAPPI for the same product, maintained by different bodies on different cadences — and the retailer owns the mapping nobody else maintains.
- The article master has many writers and no owner; the only enforcement that works is a staged mandatory field set tied to the lifecycle status transition.
- The 4-5-4 calendar makes the year 364 days, so a 53rd week arrives every five or six years and the calendar dimension needs two prior-year keys, restated and not.
- Business date, transaction timestamp, posting date, value date and fiscal period are five different columns on one basket.
- Yesterday's sales number is not final today; completeness is a measured state and every published figure needs an as-at.
- Like-for-like is an exclusion set with a policy, and if the model cannot hold the policy the number cannot be audited.
- The sale-line-item table is the biggest object in the estate by orders of magnitude, and a single version of the truth means one set of atoms plus a stated basis, not one number.

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CHAPTER 11

The conceptual application model

Twenty-five boxes, named by what they do and not by who sells them — the five that are almost always missing, and why they are the same five in every retailer.

The model you draw before anyone has chosen a vendor

Chapter 12 names the systems a retailer runs, with their trade names on them. This chapter is the layer above: the applications a retailer *needs*, named by what they do, with nobody's product on the page.

It is the artefact you draw when you are designing a target rather than describing an estate, and it is the one most often skipped, because an estate is visible and a target is an argument. The cost of skipping it arrives later — in a procurement where three suppliers answer three different questions, or in a merchandising replacement whose scope is a list of systems rather than a list of capabilities.

Retail has a particular reason to need it, and it is worth naming before anything else. The industry has **no shared reference model**. Banking has BIAN, insurance has ACORD. Retail's nearest equivalent is the set of data models built by the Association for Retail Technology Standards, founded in 1991, taken over by the National Retail Federation in 1998, and since a 2017 agreement managed jointly by the NRF and the Object Management Group [7]. The Operational Data Model is still described on OMG's pages [8], and it is genuinely good on the hard parts — it models a merchandise hierarchy *function*, so different business processes get different hierarchies over the same items [8]. But it carries no current version marker, it is absent from OMG's formal retail specification list, and the device and receipt programme that sits on top of it is far more active than the models beneath it. Do not send anybody to fetch it; ask OMG whether it exists in usable form, and plan on the answer being no.

So every retailer draws its own, usually by copying the shape of whatever suite it last bought. The twenty-five boxes in this chapter are an attempt at the artefact the industry does not publish: a synthesis of the system classes a retailer actually runs, not a standard, and they should be argued with rather than adopted.

Three artefacts that get confused

A great deal of wasted meeting time comes from mixing these up.

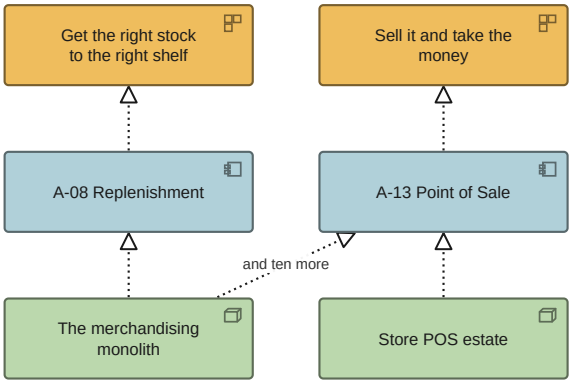


Figure 11.1 Three layers, realised upward, and the relationship that matters: one system realising several conceptual capabilities. The merchandising monolith is the case in point, and drawing it on its own page is what makes a replacement scopeable.

A **business capability** is something the retailer can do: range a category, get stock to a shelf, take money at a till, dispense a prescription lawfully. It is independent of how, and it changes about once a decade.

A **conceptual application capability** is a coherent lump of application function that realises part of one or more business capabilities. "Replenishment" is one. It is independent of who supplies it, and it changes when the industry changes — two-dimensional barcodes arguably added one, and the shift to in-store fulfilment certainly changed one.

A **system** is a thing with a licence, a version, a support contract and somebody's logo on it. It changes when a contract ends.

The relationship between the layers is realisation, and it is many-to-many in both directions. One conceptual capability is realised by three systems; one system realises eleven conceptual capabilities, which is exactly what a merchandising monolith is. Draw the three layers on separate pages and the conversation about replacement becomes tractable. Draw them on one page and you have produced something nobody can use.

Milkwood Retail Group — invented, South African, about 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm and an online business — is the worked example at the end.

What goes in a box, and what does not

Four tests, and a rule about size.

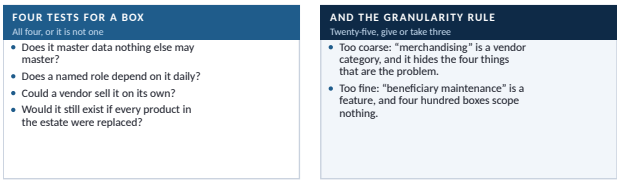


Figure 11.2 Four tests and a size rule. The mastering test does most of the work, because in retail the arguments are about who owns the data rather than about who runs the function.

The first test does most of the work, because in retail the mastering question is where the arguments are. Price is mastered by A-15 and nowhere else, which immediately tells you that a promotions engine holding its own copy of the base retail is a finding rather than a design. The article is mastered by A-01, which tells you that a product-information tool calling itself the item master is a category error — it has no concept of cost, inventory or a purchase order.

The granularity rule is doing more work than it looks. Too coarse and the model says nothing: a box called "merchandising" is a vendor category, and it hides the four things inside it that are the actual problem — the article record, the commercial ledger, the inventory ledger and the transaction engine. Too fine and you have drawn a service catalogue, and a model with four hundred boxes cannot be used to scope anything.

Twenty-five is the right order of magnitude for a multi-format group. An independent grocer needs about nine of them. A listed multi-format group needs all twenty-five and several instances of four.

Three things that are not application capabilities

These three appear on first drafts, and each makes the model worse.

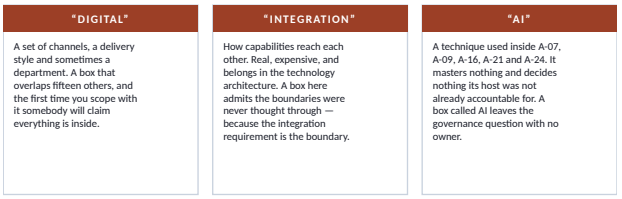


Figure 11.3 Three boxes that appear on first drafts and make the model worse. Each overlaps everything, masters nothing and leaves a governance question with no owner.

> If a supplier's architecture diagram has boxes called Digital, Integration and AI, you are looking at a marketing artefact. Ask which of the twenty-five each product actually *masters*, and watch the answer change shape.

The twenty-five

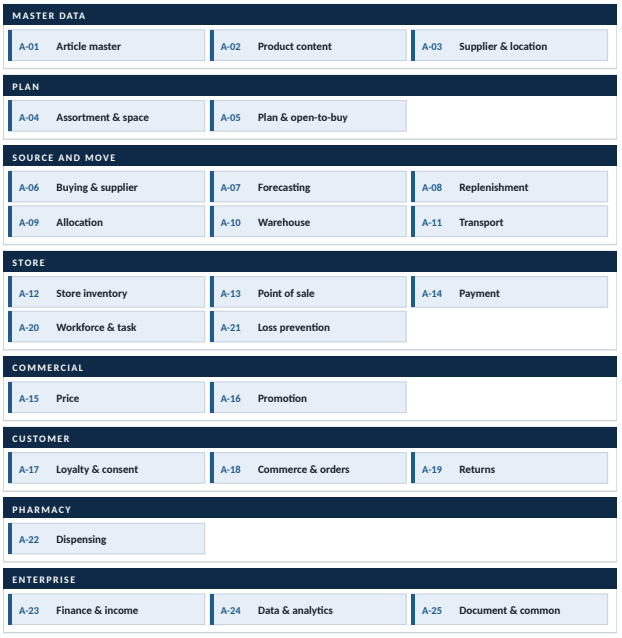


Figure 11.4 The twenty-five, in eight groups. Group names are a reading aid, not architecture — the capability and its mastering statement are the model.

Master data (A-01 to A-03). A-01 masters the article at its three levels, the merchandise hierarchy, the supplier-article link, the cost and the authorised retail. A-02 masters the descriptive and regulatory content the article record cannot hold — long copy, images, attributes by category, allergens and nutrients, translations. The division that works is that A-01 owns the commercial and operational record and A-02 owns the descriptive one; the division that does not work is A-02 as the article master, because a content tool has no concept of a cost, an inventory position or a purchase order. A-03 masters the supplier party, the supply source, the trading relationship and the location set, down to the selling area and the dated licence — and it is drawn as one box rather than two because the failure mode is identical in both halves: a legal entity with no site beneath it, and a site with no licence attached to it.

A-01 is also the hardest box in the model to replace, for a reason that has nothing to do with its own complexity.

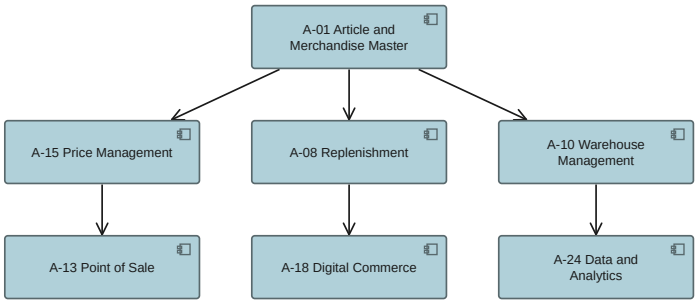


Figure 11.5 The article master serving the estate. Six consumers shown, and a real retailer has twelve; the nightly 300-field extract that feeds them all is the reason A-01 is the hardest box in the model to replace.

Everything downstream depends on it, through an interface estate that accumulated one generation at a time. The same article record is typically published as a nightly flat file, read directly from the database by two reporting tools, and exposed as an interface for the digital estate — three field sets, three latencies, three consumers each believing theirs is authoritative. When somebody proposes replacing the merchandising system, that fan is the project, and the article master itself is the easy part.

Plan (A-04, A-05). A-04 decides what to sell where and how much shelf it gets: store clusters, category roles, range architecture, the listing and delisting decision, the planogram, the facing and the derived shelf capacity. Shelf capacity is the number the rest of the estate needs, because it caps the replenishment order. A-05 is the money side — sales, margin, stock and intake by hierarchy level and period, and the open-to-buy that stops the business committing more than it can sell.

Source and move (A-06 to A-11). A-06 turns a buying decision into a commitment a supplier can act on, and carries the message traffic that makes it real — the order, the acknowledgement, the despatch advice, the receipt advice and the invoice, over whichever of the four or five channels that particular supplier actually uses. It is drawn as one box because the message layer without the order state machine is a gateway rather than a capability. A-07 produces expected demand; A-08 turns demand and a position into an order.

Those two are separate boxes on purpose, and the reason is worth a sentence. A forecast is a prediction with an as-at date that can be assessed afterwards; an order is a commitment that cannot. Conflate them and the planner's override lands on the order instead of on the forecast, so the forecast never learns and nobody can measure accuracy at the grain that drives the ordering. That single modelling decision is the difference between a forecasting capability and a forecasting argument.

A-09 is the one-time distribution of a fixed quantity, with size curves and pack constraints, and it is not replenishment: allocation is capacity-constrained distribution under fixed supply, replenishment is demand-driven pull against open supply. Their inputs, their failure modes and their audit questions all differ. A-10 runs the building at bin and batch grain, which is a strictly finer grain than A-01 holds — the interface between them is a summarisation, the two will drift, and the daily reconciliation is a named process in a well-run distribution centre and an unowned mystery in a badly run one. A-11 moves goods between nodes and produces the delivery event that the store's receipt is matched against, which is the one output of the transport capability that the rest of the estate genuinely needs.

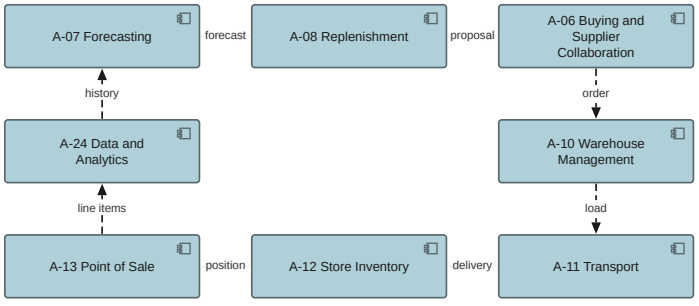


Figure 11.6 The source-to-shelf-to-insight loop, as conceptual applications. The closing edge — history back into forecasting — is the one that makes availability self-correcting, and it is the one most often broken by a forecast trained on sales rather than demand.

Store (A-12, A-13, A-14, A-20, A-21). A-12 holds the store's own position, and the whole argument of retail inventory architecture is whether that sentence is true: in the centralised model the merchandising system holds the authoritative store position and the store system captures movement

events; in the devolved model the store holds it and publishes. Most real estates are a confused hybrid in which the centre believes it holds the position, the store acts on its own, and the two are reconciled at stock-take, at which point the difference is called shrinkage and written off. Draw the box, then ask which model this retailer is actually running. A-13 records the transaction, resolves price and promotion against a local copy, enforces age and licence restrictions, handles tender, and keeps trading when the line is down — which in South Africa is a first-order requirement rather than a resilience nicety. A-14 is the terminal estate, the authorisation path and the settlement file, and it owns the card-data boundary, which is the single largest payments decision a retailer makes. A-20 converts a central instruction into work a named person does, with evidence. A-21 works the exceptions and decomposes the shrink number into causes that different owners can act on.

Commercial (A-15, A-16). A-15 masters every price with its type, scope and effective times, and distributes it to five endpoints with acknowledgement. A-16 holds the promotion, the offer rule, the funding and the discount application that records what actually fired.

Customer (A-17, A-18, A-19). A-17 masters the member, the balance, the consent and the suppression set, and resolves identity across channels. The points balance is a financial liability that has to reconcile to a provision, which makes loyalty an unexpected finance integration; and the suppression set has to be a shared service rather than a feature of the campaign tool, or the group will market to somebody who objected through a different banner. A-18 is the online catalogue, the cart, the order as an object with a life, and the sourcing decision against available stock — and that last part is the half most often missing. A-19 is the return with its legal basis, its disposition and its refund path; the disposition decision is where the money is, because a returned unit put straight back on the shelf is worth full retail and one sent to a returns centre is worth considerably less.

Pharmacy (A-22). Patient, prescriber, prescription, dispensing act, substitution, batch, scheduled-substance register, clinical check and reimbursement claim, priced against a regulated price the retailer does not set. It masters all of those alone, and nothing else in the estate may be authoritative for any of them.

Enterprise (A-23, A-24, A-25). A-23 is the general ledger, accounts payable, the stock ledger that bridges merchandising to the GL, and the trading terms, accruals and claims that decide reported gross margin. A-24 moves and conforms data, holds history, and serves the marts, models and the supplier-facing data products that are becoming a product in their own right. A-25 keeps the document of record with its retention class, produces and proves outbound communication, and carries the common services every other box assumes — workforce identity, entitlement, authority limits and segregation of duties.

The table

ID	Capability	Purpose	Primary user roles	Data mastered	Typical chapter 12 system
A-01	Article and merchandise master data	Hold what we sell, what it costs and what it may be sold for	Article data steward, buyer, buying assistant	Article at three levels, hierarchy node, supplier-article link, cost, authorised retail	Merchandising system or retail ERP
A-02	Product information and content	Hold the descriptive and regulatory content the article record cannot	Content manager, category marketer, compliance analyst	Copy, images, attributes by category, allergens and nutrients, translations	PIM and digital asset management
A-03	Supplier, partner and location master	Know who we buy from, out of which door, and where we trade	Supplier master steward, store operations, compliance	Supplier party, supply source, trading relationship, site, selling area, dated licence	Vendor master, MDM, store master
A-04	Assortment, range and space planning	Decide what each store carries and how much shelf it gets	Category manager, space planner, range planner	Range authorisation, listing decision, planogram, facing, shelf capacity	Assortment planning, space planning
A-05	Merchandise financial planning and open-to-buy	Set the sales, margin, stock and intake plan, and cap commitment	Merchandise planner, merchandiser, trading director	Plan by level and period, open-to-buy, WSSI	MFP suite — or a spreadsheet
A-06	Buying, purchase orders and supplier collaboration	Turn a buying decision into a commitment a supplier acts on	Buyer, supply planner, import clerk, supplier	Purchase order and line, order state, acknowledgement, despatch advice, claim	PO module, EDI gateway, supplier portal
A-07	Demand forecasting	Say what customers will want, and record who disagreed	Demand planner, forecast analyst	Forecast version with as-at, override with reason, causal set, accuracy	Forecasting engine
A-08	Replenishment	Turn demand and a position into an order, every day, everywhere	Replenishment analyst, store manager	Replenishment rule version, proposal with its inputs, exception	Auto-replenishment, computer-assisted ordering
A-09	Allocation and size curve	Distribute a fixed quantity across stores before the season tells you anything	Allocator, merchandiser	Allocation and line, size curve, store grade, hold-back	Allocation system
A-10	Warehouse management	Run the building and hold stock at bin and batch grain	DC manager, picker, inbound clerk	Stock by bin, pallet and batch; task; receipt; despatch and SSCC	WMS
A-11	Transport and delivery execution	Move goods between nodes and prove they arrived	Transport planner, driver, 3PL	Load, trip, route, delivery event, temperature log, freight cost	TMS, carrier platform
A-12	Store inventory and counting	Hold the store's own position, and know how wrong it is	Store manager, stock controller, receiving clerk	Store on-hand by status, receipt, adjustment with reason, transfer, count variance	Store inventory / back office
A-13	Point of sale and the till journal	Take the sale, enforce what may be sold, and keep trading offline	Cashier, supervisor, store manager	Basket, line item, tender line, discount application, operator session, electronic journal	POS estate
A-14	Payment acceptance and settlement	Authorise, settle and reconcile, and contain the card-data boundary	Payments manager, treasury, store cash office	Terminal estate, authorisation, settlement file, chargeback, token	PED estate, switch, acquirer
A-15	Price management and distribution	Hold every price with its scope and dates, and prove what was displayed	Pricing manager, buyer, store operations	Price version by type, zone, channel and effective time; price file version; label state	Pricing engine, price-file distribution, ESL
A-16	Promotion and offer execution	Hold the offer, the funding and the record of what actually fired	Promotions planner, category manager, supplier	Promotion, offer rule, eligibility set, discount application with funding split	Promotions engine
A-17	Customer, loyalty and consent	Know who the shopper is, what we owe them and what they agreed to	Loyalty manager, CRM analyst, information officer	Member, identifier, points balance, consent with purpose, objection log, resolved identity	Loyalty platform, CRM, CDP
A-18	Digital commerce and order management	Sell online and run the order through to collection or delivery	Digital trading manager, fulfilment lead, picker	Online catalogue placement, cart, customer order and state, sourcing decision	E-commerce platform, OMS, picking app
A-19	Returns and reverse logistics	Take it back on the right legal basis and put it in the right place	Store supervisor, returns centre, customer service	Return with legal basis, return line, disposition, refund, return-to-vendor claim	Returns module, reverse logistics
A-20	Workforce, task and store execution	Staff the store to demand, and confirm that central instructions were carried out	Store manager, scheduler, area manager	Roster, clock event, credential validity, task, completion evidence	WFM, time and attendance, task management
A-21	Loss prevention and exception management	Decompose the shrink number and work the cases	Loss prevention manager, investigator, store manager	Exception, case, shrink attribution by cause, surveillance asset register	Exception-based reporting, case management
A-22	Pharmacy dispensing and the clinical record	Dispense lawfully, price to the regulated price, claim, and keep the record	Responsible pharmacist, pharmacist's assistant, patient	Patient, prescription, prescription item, dispensing act, batch, scheduled-substance register, claim	Dispensing system
A-23	Finance, stock ledger and supplier income	Close the books, bridge the stock ledger, and earn the back margin defensibly	Financial controller, accounts payable, commercial finance	Journal, chart of accounts, stock-ledger movement, trading term, accrual, claim, settlement	ERP finance, deals / rebate system
A-24	Data integration, history and analytics	Move and conform data, hold history, serve marts, models and supplier data products	Data engineer, data steward, analyst, data scientist	Conformed dimensions, retail calendar, atomic history, derived measures, features	Warehouse or lakehouse stack
A-25	Document, evidence and the common layer	Keep the document of record and the entitlements every other box assumes	Records manager, identity administrator, risk manager	Document of record, retention class, proof of delivery, identity, entitlement, authority limit	ECM, communication management, workforce IAM

The five boundaries that are actually contested

Twenty of the twenty-five boundaries are uncontroversial. Five are argued about in every retailer, and an architect who settles them in the first fortnight has done most of the useful work.

A-01 against A-02. Who owns an attribute that is both commercial and descriptive — pack dimensions, country of origin, the allergen set? The rule that holds: if the attribute changes the *behaviour* of another capability, it belongs to A-01; if it only changes what a human reads, it belongs to A-02. Dimensions drive shelf capacity, so they are A-01's, even though the website needs them too.

A-15 against A-16. Who decides what the customer pays? In the better-architected estates A-16 is the authority for the price at the till, with A-15 holding the base retail and A-16 holding everything that modifies it. In the worse ones both hold both. Either is workable; the unworkable arrangement is the one nobody has written down, because the day a multi-buy and a markdown collide, two engines produce two answers and the Consumer Protection Act settles it in the customer's favour [1].

A-12 against A-10. The distribution centre holds stock at bin and batch grain; the store holds it at article grain; and somewhere a summarisation runs between them. Which is authoritative for the *in-transit* quantity is the question that exposes whether anybody is reconciling, because in-transit stock that is counted at both ends is the commonest reason a group's total position is wrong by a number nobody can explain.

A-18 against A-13. When a store fulfils an online order, the store is simultaneously selling to a walk-in customer and picking for a remote one. Who holds the reservation? Answering "the order management system" means A-12 must publish a near-real-time available position; answering "the point of sale" means A-18 promises against a stale feed and oversells. South African grocery has largely solved this operationally, with substitution and refund, rather than architecturally, with accuracy. That is a legitimate design choice and it should be written down as one rather than discovered.

A-23 against A-01. The stock ledger is a subsidiary ledger, not a reporting artefact, and it lives at the boundary between merchandising and finance. Whether a department is valued on the retail inventory method or on cost decides what a markdown *is* in the data — under the retail method a markdown reduces inventory value the moment it is applied, with no physical movement at all. An architect who treats that as a configuration detail will mis-scope the finance integration by a wide margin.

The patterns

Five patterns run through the twenty-five, and knowing which one a capability follows tells you more about programme risk than anything else on the page.

Usually one. A-01, A-03, A-15, A-23 and A-24 should exist once. When they exist more than once, the duplication *is* the retailer's central data problem rather than an inconvenience. A-23 is the only one that reliably is singular, because finance will not tolerate two ledgers. A-01 is routinely not singular in a multi-format group, and for a defensible reason: a grocery merchandising model and an apparel one are genuinely incompatible, so the group runs a merchandising system per *merchandise type* rather than per banner. Say that out loud in the model rather than treating it as debt.

Usually several. A-12, A-13, A-18 and A-22 come in multiples — by banner, by format, by acquired business, by country. The question is never "how many" but "which one is authoritative for what", and the answer should be written down somewhere other than in three people's heads.

Usually bought. A-07, A-09, A-10, A-11, A-14 and A-21 are bought almost everywhere, because the mathematics, the device certifications or the scheme rules make building uneconomic. Buying them is correct, and the architectural work is the boundary rather than the build. Two of them deserve a warning, though. A-10 bought at the same time as A-23, and implemented together, is the shape of the best-documented failure in South African retail technology [6]. And A-07 bought without A-24 behind it will be configured on sales history that was never corrected for out-of-stocks, which guarantees that every line the retailer failed on stays under-forecast.

Usually built, or heavily extended. A-18 drifts towards build in every retailer that competes on delivery promise, because the sourcing decision is the differentiator. A-20 tends to be built because the task set is the retailer's own operating model. And A-15's *distribution* half is almost always local, because the endpoint estate is local.

The build-and-buy line is usually drawn in the wrong place. The question is not which capabilities are differentiating; individually almost none of them are. It is **which capabilities carry a local specification that changes faster than a vendor's release train.** A-15 does, because the two-displayed-prices rule is a South African statutory exposure [1] and the shelf-label estate is yours. A-22 does, because the medicine schedule is amended by gazette [2] and the Single Exit Price adjusted on the state's timetable — there were **two** adjustments in 2026, an annual 1.47% in January and an extraordinary 2.88% effective 1 October [3], so a system that treats "the 2026 SEP" as one annual load was wrong from the first of October. A capability with a domestic specification that can change in a gazette needs either a supplier with a local product team or an internal build. There is no third answer, and a roadmap commitment is not one.

Usually missing. Five, and they are the same five in most retailers.



Figure 11.7 The five most commonly missing, and the reason they are the same five everywhere. Absence is silent because none of them stands between a transaction and a ledger.

The common factor is the most useful sentence in this chapter. None of those five sits on the path of a financial transaction. Nothing bounces when they are absent. No journal fails, no settlement breaks, no customer complains the same day. So the gap stays invisible until gross margin becomes unexplainable, or a website describes last year's formulation, or a price change reaches the till and never reaches the shelf — and by then a spreadsheet has quietly become the system of record, with one person who understands it.

The same twenty-five, three different retailers

A format does not need a different model. It needs a different *subset*, with the weight in different places, and pretending otherwise is how a grocery template gets imposed on an apparel business.

CAPABILITY		GROCERY	FASHION	PHARMACY
A-01	Article and merchandise master data	●	●	●
A-02	Product information and content	◑	●	◑
A-03	Supplier, partner and location master	●	●	●
A-04	Assortment, range and space planning	●	●	◑
A-05	Merchandise financial planning and open-to-buy	◑	●	◑
A-06	Buying, purchase orders and supplier collaboration	●	●	●
A-07	Demand forecasting	●	◑	◑
A-08	Replenishment	●	○	●
A-09	Allocation and size curve	○	●	○
A-10	Warehouse management	●	●	●
A-11	Transport and delivery execution	●	◑	◑
A-12	Store inventory and counting	●	●	●
A-13	Point of sale and the till journal	●	●	●
A-14	Payment acceptance and settlement	●	●	●
A-15	Price management and distribution	●	◑	○
A-16	Promotion and offer execution	●	◑	○
A-17	Customer, loyalty and consent	●	●	●
A-18	Digital commerce and order management	●	●	◑
A-19	Returns and reverse logistics	◑	●	◑
A-20	Workforce, task and store execution	●	◑	●
A-21	Loss prevention and exception management	●	●	●
A-22	Pharmacy dispensing and the clinical record	○	○	●
A-23	Finance, stock ledger and supplier income	●	○	◑
A-24	Data integration, history and analytics	●	●	●
A-25	Document, evidence and the common layer	◑	◑	●

● core, and it is where the format competes ▸ needed, but light ○ rarely worth a box

Figure 11.8 The same twenty-five, weighted three ways. The judgement in this table is mine, drawn from the system-class research rather than from a survey — use it as a starting position to argue with, not as a finding.

Grocery lives on A-07 and A-08. Volume is the architecture, price is zoned and changes weekly, space is the binding constraint so A-04 carries real weight, and A-23's supplier income half is material to reported margin. A-09 allocation barely exists, because supply is open-ended.

Fashion inverts most of that. A-09 is the capability the business is won or lost on, with size curves and pack constraints; A-05 is a weekly grid a merchandiser lives inside; A-02 matters because the customer buys from a photograph. A-08 replenishment is close to absent, because there is no repeat buy. And A-23's supplier income half is thin, because there is little back margin to misstate.

Pharmacy adds A-22 and takes away the things a pharmacist is not allowed to do. A-15 and A-16 are largely bypassed for scheduled lines, because the retailer cannot price, promote or mark down a medicine: the Single Exit Price is the only price at which a manufacturer may sell to anyone other than the state, with a capped dispensing fee layered on top [5]. A-20 gains a hard edge, because the till must be able to refuse an act for which no credentialed person is present. And A-25 matters more here than anywhere else, because retention runs on a clinical clock rather than a commercial one.

Cash-and-carry is the quiet fourth case. It needs A-13 to produce a compliant full tax invoice above R5,000, including the business customer's name, address and VAT number [4] — a capability most retail point-of-sale deployments lack until the first business customer complains. It also needs A-17 to hold a trade account rather than a loyalty membership, and A-15 to carry a volume-break price that is party-based rather than zone-based.

There is a fifth case that is not a format at all, and it changes the model more than any of them: **the retailer that does not own its stores.** This is not an edge case in South Africa. Pick n Pay reported 992 company-owned stores alongside 620 franchise stores at 1 March 2026 [9]; SPAR is not a franchise at all but a voluntary-trading group whose members own their stores, join by signing a membership agreement with a Guild, and **may stock from any supplier they choose** [10]. In a voluntary-trading or heavily franchised estate the group does not have an inventory position, an assortment or a price to master for much of its turnover. A-12, A-04 and A-15 stop being central controls and become *offered services*, A-06 points outward to a member rather than inward to a store, and the integration becomes business-to-business rather than intra-enterprise. The twenty-five boxes survive; what changes is who is on the other side of each one, and that is a bigger architectural fact than the format mix.

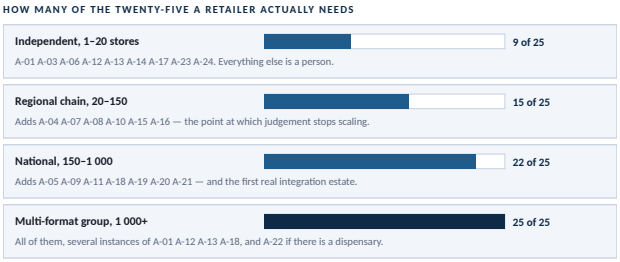


Figure 11.9 Count by size band. The interesting band is the regional chain: it is where judgement stops scaling and forecasting, replenishment and price distribution have to become systems.

Size changes the count rather than the shape. An independent needs about nine of the twenty-five; everything else is a person with judgement, and that works until roughly twenty stores. The interesting band is the regional chain, because that is where judgement stops scaling and A-07, A-08 and A-15 have to become systems. The jump to a national estate adds the planning, allocation and execution-confirmation capabilities, and it is the point at which the integration estate becomes an architecture problem in its own right.

How to use it

Three uses. A model not used for at least one of them is decoration.

As a target skeleton. Draw a boundary round the boxes a programme touches and you have the most honest one-page scope statement anybody will produce. It survives procurement because it is in business language. "This programme touches A-01, A-06, A-10, A-12 and A-23; it does not touch A-13, A-15 or A-22" is a sentence an executive can hold a supplier to — and it is also, incidentally, the sentence the defining South African retail failure case would have benefited from. SPAR's own stated remediation after its SAP programme disrupted the KwaZulu-Natal distribution hub was **"separating the ERP from the warehouse system and implementing them independently"** [6]. On this model that is a boundary between A-23 and A-10, and it is visible on one page before anybody signs anything. It is the most transferable sentence in South African retail technology, and it is a scoping statement expressed in capabilities.

As a package-fit assessment. Take the twenty-five into a demonstration and score each one, once, per product.

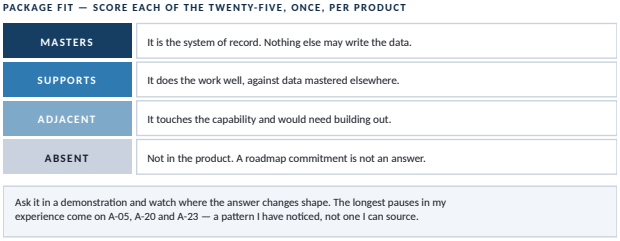


Figure 11.10 A four-level package-fit scale. Score each of the twenty-five once per product, in the demonstration, and a shortlist resolves itself into arithmetic.

The silences are informative. Score honestly and a three-product shortlist usually resolves itself, because the overlap and the gaps both become arithmetic rather than opinion.

As the basis for the application portfolio. Every real system in chapter 12 maps to one or more of the twenty-five, and every capability maps to one or more systems. Where a capability has three systems, you have a consolidation question. Where a system realises eleven, you have a decomposition question and a replacement that will take years. Where a capability has none, you have either a deliberate gap or a spreadsheet. And the mapping itself is the portfolio's spine: once it exists, cost, criticality, lifecycle and risk hang off capabilities rather than off products, which means the portfolio survives the next procurement.

A practical note on producing it. The model is built in a room, not at a desk, and the right room holds a buying controller, a supply-chain planner, a store operations manager, the pricing lead, the financial controller and — if there is a dispensary — the responsible pharmacist. Put the twenty-five on a wall with nothing but their names and their mastering statements, and ask one question per box: *who owns this today, and where does the data live?* An afternoon of that produces a better portfolio baseline than a month of interviews, because the disagreements happen in front of each other rather than in sequence.

What the model deliberately does not contain. No integration, because that is the technology architecture, and because the boundary between two boxes *is* the integration requirement. No organisation, because the model outlives the structure and because naming a department on a capability invites the department to defend it. No instance counts, because those belong to the portfolio view, where they can change without the target changing. No deployment model, because whether A-07 runs in somebody's cloud is a sourcing decision and not an architectural one. And no sequencing, because a target with a roadmap drawn on it stops being a target and becomes a plan nobody revisits.

Milkwood against the model

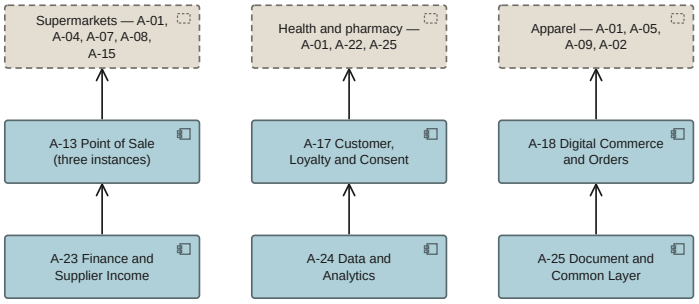


Figure 11.11 Milkwood mapped: three trading formats with different needs, above a common layer that serves all of them. Two instances of the article master are defensible; three instances of the point of sale are not, and the model is what makes the difference arguable.

Milkwood is invented, and its shape is the point. Three trading formats with genuinely different needs below the till, and a common layer underneath them.

It runs **two instances of A-01**, one for food and one for apparel, because the two merchandise models do not fit in one. That is defensible. It runs **three instances of A-13**, which is not — two of them arrived with acquisitions and the third was a pharmacy decision, and the group has three versions of what a basket is. It has **no A-05** at all; the apparel merchandiser maintains the plan in a workbook she built in 2019. It has **no A-20**, so when a price change goes out nobody can say which stores relabelled. And its **A-23 supplier income** half is a shared drive.

It does have a single A-17, which is better than most groups manage, and a single A-24. But A-17's consent model was built for one banner and the group has since started marketing across all three — which, on this page, is visible as one box serving three formats with one consent scope, and that is a conversation worth having before the Regulator has it.

The useful output of the exercise was not the diagram. It was four sentences: A-13 is a consolidation programme, A-05 is a buy, A-20 is a build, and A-23's income half is the one that has to be fixed first, because it is the only one of the four that moves the reported number. Three years of roadmap, agreed in a morning, because the argument was about boxes rather than about products.

Where this goes wrong

Five failure modes, and the first is by far the commonest.

Drawing the estate and calling it a model. Somebody produces a diagram of what exists, relabels the boxes with generic names, and presents it as a target. The tell is that the box count matches the system count. A conceptual model that happens to look exactly like the current estate has described rather than designed, and it will never be used to argue for anything.

One box per supplier product. A suite ships eleven modules, so the model gets eleven boxes with the suite's own names generically rewritten. The supplier's packaging becomes the retailer's architecture, and the next procurement is scored against the incumbent's shape.

Letting a monolith be one box. The merchandising system realises A-01, A-03, A-06, A-12 in part, A-15 in part and A-23's stock-ledger half. Draw it as one box called "merchandising" and the model has hidden the exact thing a replacement programme needs to decompose. Draw the capabilities and then show, on a separate page, that one system realises six of them. That second page is what makes the replacement conversation honest.

Two boxes mastering the same data with no stated authority. A-15 and A-16 both holding base retail. A-01 and A-02 both holding dimensions. A-12 and A-10 both holding a position for the same stock. Each of those is survivable if the model says which one wins. None of them is survivable if it does not, and the symptom is always the same: two correct numbers, a monthly reconciliation nobody owns, and an argument that resurfaces every quarter.

The mastering statement is therefore the part of the model to write first and to defend hardest. It is one line per box, it takes an afternoon, and it is the only thing on the page that a project can actually be held to. Everything else — the grouping, the ordering, the colours, the count — is presentation.

Modelling the group and ignoring the ownership of the stores. This is the retail-specific one, and it is the mistake an architect arriving from banking or insurance is most likely to make. Every box on the page quietly assumes that the retailer can instruct the location at the other end of it. In a franchised or voluntary-trading estate that assumption is false for a material share of turnover, and it is false as a matter of contract rather than of capability. The model does not change, but the question "who is on the other side of this box" has to be asked of all twenty-five, and the answers will not be the same for the company-owned estate and the member estate. A target architecture that assumes central control of price, range and inventory across a members' estate is not ambitious. It is wrong, and the first member meeting will say so.

> The model is twenty-five boxes, a mastering statement for each, and nothing else. If you can hand it to a supplier and get back an honest answer about where each box lives in their product, it has already paid for the afternoon it took. Chapter 12 puts the real names back on.

THE CHAPTER IN TEN LINES

- A business capability, a conceptual application capability and a system are three different artefacts; draw them on three pages, not one.
- A box earns its place when it masters data nothing else may master, a named role depends on it daily, a vendor could sell it alone, and it survives replacing every product.
- Twenty-five is the right order of magnitude for a multi-format group; an independent grocer needs about nine.
- “Digital”, “integration” and “AI” are not application capabilities — they overlap everything and master nothing.
- Five capabilities should exist once, four usually come in multiples, six are almost always bought, and three drift towards build.
- Build where a domestic specification changes faster than a vendor's release train — price distribution and pharmacy dispensing both qualify.
- The five most commonly missing are planning and open-to-buy, product content, order management, store task confirmation and supplier income.
- They are missing for one reason: none of them sits between a transaction and a ledger, so nothing fails loudly when they are absent.
- Grocery, fashion and pharmacy need genuinely different subsets of the same twenty-five, and a grocery template imposed on apparel is a predictable failure.
- Use the model to scope a programme, to score a package and to build the portfolio — and if it looks exactly like the current estate, it describes rather than designs.

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CHAPTER 12

The application landscape

This is the chapter you will come back to. It names the thirty-one systems a retailer actually runs, says what each is authoritative for, where each one hurts and what it is called in the trade — then the integration as it really is, the batch clock, and the legacy question, with the sourced claims and the practitioner consensus kept apart.

Read this before you read anything else

A bank's unit of record is the account. An insurer's is the policy. A retailer's is **the article at a location**, and the only two numbers that matter about it are how many there are and what they cost the customer. Almost every hard architecture problem in retail reduces to one of three things: an inventory position two systems disagree about, a price three systems disagree about, or an item identifier that means something different in every system holding it.

Nothing crashes when those numbers diverge. There is no double-entry discipline forcing a reconciliation. Margin leaks quietly and the business finds out at stock take. That is why retail estates look undisciplined to an architect arriving from financial services. They are not. They are built in an industry where the feedback loop on a data error is twelve months long.

Two more framings. **The estate is physically distributed in a way no other industry's is.** Pick n Pay reported 2,261 stores at 1 March 2026 across South Africa and the rest of Africa [12]; Shoprite reported 2,839 Supermarkets RSA stores at 28 June 2026 [13]; Clicks reported 990 stores and 780 pharmacies at its 2025 year end [14]. Each has its own tills, its own network, its own power supply, its own stock position and its own staff who will work around a system that fights them. Any design assuming a reliable network to every node is wrong, and in South Africa it is wrong twice over.

Retail time is not calendar time. The retail week runs Monday to Sunday or Sunday to Saturday depending on the chain; the retail month has four or five of those weeks; the retail year occasionally has 53 [11]. The overnight cycle has to finish before the first store opens. In banking the batch window constrains one system. In retail it is the heartbeat of the whole estate.

> THE WORKED EXAMPLE
>
> Where this chapter needs a concrete retailer it uses **Milkwood Retail Group**: roughly 420 supermarkets, a 180-store pharmacy and health chain, an apparel format, a cash-and-carry arm and an online business. **Milkwood is invented**, nothing about it is a case study, and no number attached to it is sourced. The multi-format shape is deliberate, because grocery, pharmacy and apparel have genuinely different architectures below the POS.

First question: who owns the store?

Ask this before you draw anything, because the answer relocates authority for price, assortment and stock — and South Africa has all three answers inside its top five grocers.

Corporate-owned. The group owns the store, the stock and the price, and everything in this chapter is the group's system of record. Shoprite's Supermarkets RSA estate and Clicks' 990 stores are of this type [13][14].

Franchised. The franchisee owns the business; the group owns the brand, the supply chain and usually the systems. Pick n Pay reported **992 company-owned and 620 franchise stores** at 1 March 2026, with franchisees sourcing inventory through the central distribution network [12]. Architecturally the franchise store is a *customer*, not a location — which is exactly how Oracle's model lets a "store" represent a customer location such as a franchisee [2]. Price recommendation versus price mandate becomes a contractual question, and stock is sold rather than transferred.

Voluntary trading. SPAR stores are **independently owned**, and SPAR describes the model as explicitly *not* a franchise: SPAR acts as wholesaler and distributor and earns its main income from selling goods to the stores; retailers join by signing a membership agreement with a non-profit Guild; and **retailers may stock their stores from any supplier**, so each store has its own range [15]. Who sets the shelf price is not in the public material and I could not source it.

THREE OWNERSHIP MODELS, THREE DIFFERENT SYSTEMS OF RECORD

Model	The group masters	What it gets instead
Corporate-owned	Price, range, stock, labour, the shelf	Nothing missing. Every class in this chapter is the group's
Franchised	Brand, supply chain, usually the systems	A store that is a customer, not a location. Price becomes recommendation; stock is sold, not transferred
Voluntary trading	The wholesale catalogue and the membership standard	A wholesale order book, and whatever the member chooses to share. Members may buy elsewhere

Shoprite Supermarkets RSA and Clicks are corporate. Pick n Pay runs 992 company-owned and 620 franchise. SPAR is voluntary trading, and says so.

Figure 12.1 Who owns the store decides what the group has to master. South Africa runs all three models inside its top five grocers, and the third one removes price, range and inventory from the group's control for much of its turnover.

The consequence is immediate and most imported reference material misses it. **In a voluntary-trading or heavily franchised estate the group does not have an inventory position, a price or an assortment to master** for a large part of its turnover. It has a wholesale order book and whatever the member chooses to share. Replenishment, range, pricing and loyalty become *offered services* rather than central controls, and the integration is business-to-business rather than intra-enterprise.

The map

Expect multiple instances of merchandising, store inventory, POS and e-commerce — by banner, by country, by acquired business, by format. Expect one ERP, and a shadow merchandise plan in Excel.

#	Class	System of record for	Called in the trade
1	Merchandising / article master	Article, hierarchy, cost, retail price, supplier-article link, inventory position	"the merch system", MMS, RMS
2	Assortment and range planning	Range by cluster, listing and delisting decision	category planning
3	Space planning	Planogram, bay layout, facings, shelf capacity	space planning, "the planos"
4	Merchandise financial planning	Sales, margin, stock and intake plan, open-to-buy	MFP, OTB, WSSI
5	Buying and purchase orders	Purchase order, order line, delivery schedule, order cost	"the PO system", buying
6	Supplier collaboration and EDI	Message state, acknowledgement, dispute	EDI, VAN, supplier portal
7	Demand forecasting	Base forecast, promotional uplift, override	forecasting, DF
8	Replenishment	Proposed order, safety stock, order-up-to level	auto-replen, CAO
9	Allocation	Store-level quantity for a fixed buy	allocation, distro
10	Warehouse management	Stock by bin and pallet, task, pick, receipt	WMS
11	Transport management	Load, trip, route, delivery event, freight cost	TMS
12	Store inventory	Store on-hand, receipt, adjustment, count	SIM, back-office
13	Point of sale	The sale — basket, line, tender — and the till journal	POS, "the tills", lanes
14	Payment acceptance	Terminal, authorisation, settlement file	PED, PIN pad, acquirer
15	Promotions and pricing	Promotion, offer rule, price-change event, markdown	promo engine, pricing
16	Electronic shelf labels	Label-to-article binding, label state	ESL
17	Loyalty and CRM	Member, card, points balance, offer response	loyalty, "the card"
18	Customer data platform	Resolved customer identity, segment, audience	CDP
19	E-commerce platform	Online catalogue, cart, online order	"the site", digital
20	Order management	Customer order, sourcing decision, fulfilment state	OMS, DOM
21	Marketplace	Third-party seller, seller listing, seller order	marketplace, 3P
22	Returns	Return authorisation, disposition, refund	RMA, reverse logistics
23	Workforce management	Schedule, clock event, leave, labour cost	WFM, T&A
24	Store task management	Task, completion evidence, compliance score	task management
25	Loss prevention	Exception, case, shrink attribution	LP, exception-based reporting
26	Master data and product information	Golden record, stewardship workflow, rich content	MDM, PIM
27	Data platform	Conformed history, dimensions, marts, features	warehouse, "the DW"
28	ERP and finance	GL, AP, AR, fixed assets, statutory accounts	ERP, finance
29	Supplier income and rebates	Trading term, accrual, claim, settlement	commercial income, deals
30	Pharmacy dispensing	Patient, prescription, dispensing event, register	dispensary system
31	Integration layer	Nothing. That is the problem.	the bus, middleware, SFTP

Those thirty-one do not form one estate. They form three, running at different speeds and owned by different parts of the business, and an architect who draws them as one diagram with one governance model has already lost the argument.

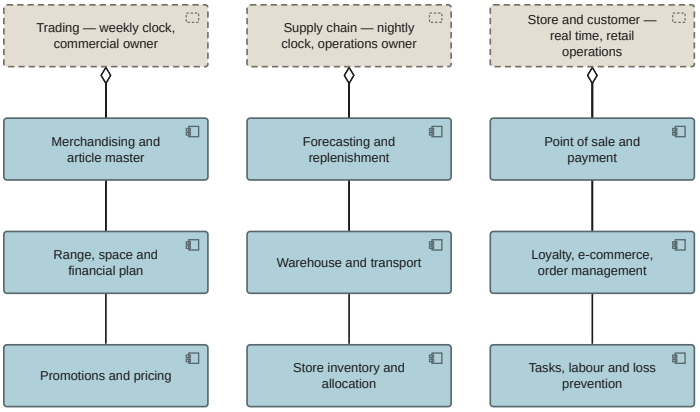


Figure 12.2 Three clock speeds in one estate. The groupings are not organisation units — they are the rate at which each part changes and the function that owns it. A single governance model across all three is the commonest early mistake.

Merchandising and the master-data core

Retail's equivalent of core banking, with the same definitional ambiguity: it is four things in one box and nobody agrees which.

FOUR READINGS OF ONE SYSTEM, AND THE COST OF BELIEVING ONLY ONE

Reading	"Merchandising" is…	If you believe only this
The article master	The item record, its hierarchy, its attributes, its barcodes	You under-scope: cost, price and supplier terms live here too, and are the harder half
The commercial ledger	Cost, retail price, margin, supplier terms, intake	You discover the inventory ledger is a different subsystem with its own period close
The inventory ledger	On-hand, on-order, in-transit by article and location, valued	You discover stores run their own stock system and it does not agree
The transaction engine	Orders, receipts, transfers, adjustments, invoice match	You discover planning, allocation and replenishment are satellites writing back into it
All four	The monolith	Correct for most retail estates, and the reason replacement programmes run five years

The retail inventory method against the cost method is a data-model difference, configured per department. Establish it in week one.

Figure 12.3 "Merchandising" is a question with four answers, and in a real estate it is all four at once. Believing only one of them is how a replacement programme is under-scoped by a factor of three.

Properly scoped it does article creation and lifecycle with effective dating; hierarchy maintenance and reclassification; supplier and supplier-site maintenance; the supplier-article relationship with cost, case pack, lead time and minimum order quantity; cost management across standard, weighted average, last and landed cost; retail price management by price zone with effective dating; purchase order creation through closure; receipt, return-to-vendor, transfer, adjustment and count posting; inventory valuation; the stock ledger and period close; invoice matching; and the outbound feeds telling every other system what an article is and what it costs.

It masters the article identifier, the merchandise and organisational hierarchies, the supplier-article relationship, the cost, the authorised retail price, the inventory position by article and location, and the stock-ledger history. **It does badly** anything a planner wants to model before committing, anything customer-facing, anything event-driven, rich content for a website, and any calculation needing a re-run on a changed assumption.

One structural fork most architects miss. Oracle's documentation exposes a department-level profit calculation type of *Direct Cost* or *Retail Inventory*, with a matching open-to-buy calculation type [1]. That is not a configuration detail. The **retail inventory method** values stock at retail and derives cost by a cumulative mark-on, so the stock ledger is driven by *price-change events* as much as by movement events — a markdown reduces inventory value the moment it is applied, with no physical movement at all. The **cost method** values stock at cost and treats price as a selling-side attribute. Which method a department uses decides what a markdown *is* in the data, and therefore what "gross margin" means in every downstream report. Find out, per department, in week one.

A department is not a label. Oracle's department record carries both calculation types, purchase type (normal, consignment or concession), markup method and percentages, expected annual sales, outlier tolerances and buyer assignment; tax regions attach at department level and default down to items [1]. It bundles accounting policy, tax default, buying authority, budget and statistical parameter — move an article between departments and you move all of it. And here is the trap: Oracle states that when an item is reclassified, "its sales history, purchase orders, inventory, and planning information are reclassified too" [1]. A reclassification that *does* carry history rewrites last year's numbers so this year's comparison works and last year's published report no longer reconciles. One that *does not* preserves the report and destroys the comparison. Both are defensible; the retailer must choose, must choose the same way every time, and different systems in the same estate choose differently.

TWO TRAPS IN THE VOCABULARY, AND BOTH COME FROM SAP ESTATES

SAP builds a **merchandise category hierarchy** with an unlimited number of levels, numbered bottom-up, where branches may have different depths. Two node types exist: merchandise categories, which hold articles, and hierarchy levels, which never can — "articles are always assigned to the merchandise categories, and never to the hierarchy levels" [3]. SAP also supplies a **reference article** used as a creation template that is never sent to POS, so it can never be operational [3].

And the sharper one: a characteristic inherited from a hierarchy level is **informational only** — it supports analysis and reporting and has no operational effect in the wider retail system [3]. An attribute that looks like a product property may be a reporting tag with nothing behind it, and a project that builds logic on it is building on sand.

How it fails. The article master open to direct edit by 400 users with no stewardship workflow. Cost held in three places with no stated authority. The hierarchy reclassified without carrying history. Price maintained here *and* in the promotions engine *and* in the POS. Custom code in the posting layer carrying fifteen years of special cases nobody dares touch. A nightly 300-field article extract to twelve consumers, each using nine fields and all breaking when a field widens.

The three planning systems above it

Assortment and range planning decides what to sell where: store clustering by size, demographic, trading pattern, climate and format; category role definition; range architecture, meaning how many articles a category gets split by price tier, brand, pack size and own-label; scenario modelling; space-constraint feedback; and listing and delisting workflow with supplier notification. **It masters the article-to-cluster authorisation and the listing decision** — and that authorisation is what the rest of the estate needs and most often cannot get, because replenishment must not order an unranged article into a store and the website must not offer it for collection there. In practice the range is held as a status flag in merchandising *and* as a plan in the range tool, and the two diverge within a month of the review ending. It does badly local exception handling, mid-season change, and anything where the constraint is space rather than money. And there is a South African wrinkle with teeth: the Competition Commission's Grocery Retail Market Inquiry found that national chains receive rebates for shelf access and category management that wholesalers serving independents generally cannot access, and that some had no efficiency justification; its remedy is that suppliers must offer **uniform trade terms**, objectively justified, clearly communicated and not volume-based [10]. So range review now needs a listing audit trail that survives an inquiry and an explicit link from each listing to its trading term.

Space planning turns the range into shelf. A **planogram** specifies where products sit [49]; in practice it is a dataset — a fixture of known dimensions, a sequence of articles, facings, depth, a stacking rule, and a resulting shelf capacity. It masters the fixture, the planogram, the article position and the **shelf capacity**, and that last number is what the rest of the estate needs, because space-aware replenishment uses it to set presentation stock and to cap the order at what the shelf will physically hold. It does badly: keeping up. Planograms are authored centrally against an idealised fixture and real stores have the fixture they have, so the biggest failure is that the store does not match the planogram and nobody measures the gap — compliance measurement traditionally means a merchandiser with a clipboard, which is sampled, late and arguable. Image recognition and electronic shelf labels change that by making the article-to-bay binding a maintained object rather than a design intent.

Merchandise financial planning sets top-down targets for sales, margin, inventory and intake by hierarchy level and period; **open-to-buy** is the derived constraint on how much more may be committed — planned closing stock less opening stock, on-order and in-transit, adjusted for planned sales and markdowns [35]. In the United Kingdom and South African fashion trade the artefact is usually a **WSSI**, a weekly sales-stock-and-intake grid maintained by a merchandiser; the clearest vendor framing is of a pre-season and in-season plan reconciling top-down targets with bottom-up assortment plans [36]. Open-to-buy is the control that stops the business buying more than it can sell, and it is a cross-system calculation a buyer must trust mid-negotiation: planned sales from the plan, actual sales from the data platform, actual stock from merchandising (which may disagree with the stores), on-order from purchase orders, planned markdown from pricing, and the calculation types configured at department level, which decide whether the arithmetic runs on cost or retail [1]. Planning is naturally department-by-month; buying is article-by-week. Every retailer bridges that gap and most bridge it in Excel. **"The planning tool is a spreadsheet" is not a legacy finding in retail. It is the normal state.**

Buying, and how an order reaches a supplier

Purchase order management turns a buying decision into a commitment: order creation manually, from an allocation, from a replenishment proposal or a contract call-off; approval against open-to-buy and authority limits; amendment with supplier re-acknowledgement; delivery scheduling and appointment booking at the distribution centre; order-cost derivation including deals, freight and up-charges; expected-landed-cost calculation for imports; and receipt and invoice matching with discrepancy workflow. It masters the purchase order, the committed quantity and cost, and a state machine more elaborate than architects expect: created, approved, transmitted, acknowledged, amended, part-received, fully received, closed, cancelled, plus a parallel set of shipment states in import trade.

Two splits to get right. Everything above is **goods for resale**, and it lives in merchandising, never in ERP; **goods not for resale** — shelving, uniforms, IT, services — runs through ERP procurement with a different approval model, a different supplier master and a different matching rule. Two purchase-to-pay processes, two supplier masters, one payables team, and architects routinely scope "procurement" and find they have described the smaller half. And **import orders are a different system behaviour, not a variation**: long lead times, letters of credit, customs entry, duty at import, freight apportionment, partial shipment, and a landed cost that must be trued up. A fashion retailer importing from Asia runs a materially different buying system from a grocer buying locally, which is usually why the group has two merchandising instances. It fails when orders are amended by telephone and not in the system, when case quantity is wrong so the order is an order of magnitude out, and when deals are applied at invoice rather than at order, so order cost and invoice cost differ systematically and nobody can reconcile intake margin.

Supplier collaboration and EDI is how the order, the despatch and the invoice actually move, and the honest answer is: more slowly and in flatter files than anyone expects. In the EDIFACT world — Europe, the Commonwealth, South Africa — the retail message set is **ORDERS** (purchase order), **ORDRSP** (acknowledgement), **DESADV** (despatch advice, the electronic packing note, carrying serial shipping container codes), **RECADV**, **INVOIC**, **INVRPT** (inventory report), **SLSRPT** (sales report), **PRICAT** (price and sales catalogue), **DELFOR** and **REMADV** [51]. In the North American X12

world the practitioner names are 850 purchase order, 855 acknowledgement, 860 change, 856 advance ship notice, 810 invoice, 852 product activity and 846 inventory advice [50]. Chapter 13 treats which ones matter and warns about X12's licensing position.

Two carry most of the weight. **DESADV, or the 856,** is what makes receipt-by-scanning possible: a hierarchical despatch advice keyed on serial shipping container code [6] lets the centre scan a pallet label and receive the whole content in one event, with discrepancies by exception. Without it, receipt is a line-by-line count, and the difference in labour cost and accuracy is large. **PRICAT** is the message meant to solve article-data onboarding, and it is the least adopted of the set, because it requires the supplier to maintain data to the retailer's standard.

And that is the real finding. EDI in a large retailer is a minority of supplier traffic. The majority is a spreadsheet emailed to a buyer's assistant, a delimited file dropped on SFTP at an agreed time with an agreed filename, a fixed-width file from a 1990s interface nobody has touched, a portal where a supplier types the data in by hand, and a small but growing REST estate — each with its own error handling, usually "the file did not arrive and nobody noticed until Thursday". No South African retailer publishes an interface inventory, so the proportions are practitioner consensus; settle it for your own client by counting. One trend has an architectural tail: supplier data sharing is becoming a product, and Pick n Pay disclosed launching **Boxer B-Inside** in FY26 to extend supplier insight and create a new revenue stream [12]. That gives the data platform an external, commercial, contractually committed consumer.

Forecasting, replenishment and allocation

Demand forecasting produces expected unit demand per article per location per period. Four things make it structurally harder than it looks. **Intermittency**: at article-store-day grain most series are mostly zeros and classical time-series methods assume otherwise, so the grain at which a forecast is *statistically* meaningful is coarser than the grain at which it is *operationally* needed, and reconciling the two is where accuracy is won or lost. **Censored history**: sales are not demand, an out-of-stock article sold zero because there was nothing to sell, and training on sales without correcting for availability makes out-of-stocks self-perpetuating — most forecasting failures in grocery trace to exactly this. **Promotions dominate the variance**, and uplift depends on the mechanic, the price point, the position in the leaflet, the competitor's simultaneous activity and whether the end-of-aisle display was actually built, so the promotional forecast is partly an *execution* forecast. And **cannibalisation and halo**: promoting one line suppresses its substitutes and may lift its complements, so forecasting lines independently over-states the category by construction. It masters the forecast and, importantly, the **override** — who, when, why, by how much. Override capture is the difference between a forecasting capability and a forecasting argument. It fails with no availability correction, with promotional uplift as a flat multiplier set once in 2019, and with overrides applied to the replenishment proposal rather than the forecast, so the model never learns.

Replenishment turns a forecast and a stock position into an order: the highest-volume decision engine in the estate, article by location by day. Order-point and order-up-to-level calculation; safety stock from demand and lead-time variability against a service level; review-cycle and order-calendar handling, because a store ordered three times a week has different mathematics from one ordered daily; case rounding and order multiples; shelf-capacity and presentation-stock constraints; minimum shelf life for fresh; promotional pre-build and post-promotion run-down; and the routing choice between centre and supplier-direct-to-store. The vocabulary: *auto-replen* generally, **computer-assisted ordering** in grocery — the honest name, because the store keeps an override — *vendor-managed inventory*, and *continuous replenishment*. Fresh is a separate engine or at least a separate parameter set, because shelf life changes the objective from "do not run out" to "do not run out and do not throw away". **The two-echelon problem** is the one to name early: optimise store orders in isolation and the centre sees the sum of many case-rounded, calendar-constrained orders, lumpier than consumer demand, so retailers that do not model both echelons together carry safety stock twice and still run out.

Two South African factors change the mathematics. Long inbound lead times combined with port variability widen lead-time variance and therefore safety stock. And the pay cycle: Stitch's platform data for the first quarter of 2026 shows over 80% more transactions on the first of the month than on the twenty-second, with peak volume on the twenty-fifth [22]. That is payments data for digitally active consumers rather than grocery footfall and must be cited as such, but the pattern behind it — salary and grant dates concentrating demand into a few days — is well known here. It means the **day-of-month profile matters as much as the day-of-week profile**, which imported forecasting configurations do not model at all.

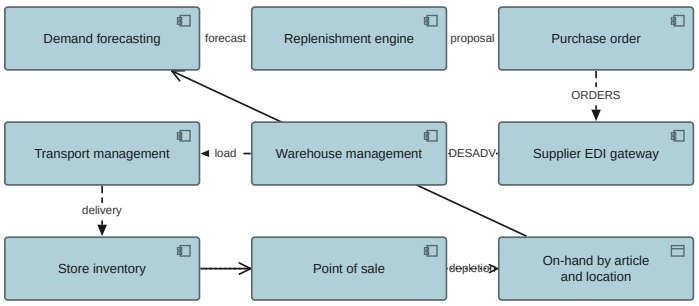


Figure 12.4 Forecast to shelf, and back. Read the top row left to right, then down the right-hand column, then the bottom row right to left. The loop closes because the on-hand position feeds the next night's replenishment run — which is why an inaccurate store position is self-perpetuating.

Allocation is a separate class, and the distinction is the thing to get right. Replenishment answers "the shelf is low, send more" and assumes supply is repeatable. **Allocation answers "we have bought 40,000 units and will never buy more — how many to each store?"** and assumes supply is fixed and the decision is distributive: allocation against a fixed quantity, store-level quantity from need weighted by store grade, size-curve application, pack constraint handling, hold-back and warehouse retention, phased allocation across drops. **Size curves and packs are what architects miss.** In apparel the decision is not how many but how many of each size, and the profile differs by store. A **size curve** is the proportional distribution of demand across the size range for a style at a store or cluster; a **prepack** is a supplier-assembled carton with a fixed size ratio [34]. Buy in prepacks and the allocation is constrained to integer multiples of that ratio, so the optimisation is combinatorial rather than proportional and the gap between the ideal curve and the

achievable pack mix is a measurable margin cost [34]. It does badly: reacting, because an allocation is a one-shot decision made before the season tells you anything, and the corrective mechanisms are inter-store transfer, which is expensive, and markdown.

Warehouse and transport

Warehouse management runs the building, and it is the only system in the estate whose primary job is directing human and machine movement in real time: inbound appointment and dock scheduling; receipt against an advance ship notice or blind; quality inspection and quarantine; put-away by rule — velocity, temperature, weight, hazard, batch; bin management; stock by batch, lot, expiry and serial; pick-face replenishment from reserve; wave planning; pick confirmation by voice, scan, light or radio terminal; pack and despatch with serial shipping container code generation and GS1-128 label printing [6] [7]; load building; cycle counting; reverse logistics; cross-dock; value-added services such as ticketing and garment-on-hanger; and labour management against engineered standards [33].

It masters stock at the bin, pallet and batch grain — strictly finer than merchandising holds. Merchandising wants one net number per article per warehouse; the warehouse system holds hundreds of rows to produce it. The interface is a *summarisation*, the two will drift, and the reconciliation is a named daily process in a well-run centre and an unowned mystery in a badly run one. Three archetypes behave differently enough to be different configurations: a **stockholding ambient centre** receives, holds and picks store orders; a **flow-through centre** receives deliveries already store-allocated and moves them across the dock without put-away, so if the inbound is late the outbound is late because there is no buffer; and a **cold-chain centre** has temperature-regimed zones, picks first-expired-first-out rather than first-in-first-out, and carries a traceability obligation. A fourth is now common: the **e-commerce fulfilment centre or dark store**, picking units to a customer order rather than cases to a store order, which makes it a warehouse-and-order-management hybrid. It fails when the two positions differ by a number nobody can explain and the difference is written off at stock take; when put-away rules reflect a product mix five years old; when cycle counting is suspended at peak and never resumed; and when batch and expiry are captured at receipt but not at pick, so first-expired-first-out is aspirational.

Transport management plans and executes movement, is often the least-integrated system in the estate and frequently a third party's: load planning and consolidation; route optimisation; fleet and driver scheduling against legal driving-hours constraints; carrier selection and rate management; delivery execution with proof of delivery; in-transit tracking; temperature monitoring for cold chain; backhaul loading on the return leg, which is real money; freight-cost allocation to landed cost; and damage and shortage claims. It masters the load, the trip, the delivery event and the freight cost — and **the delivery event is the one other systems need**, because it tells store inventory stock is arriving, closes the despatch, and is what the receipt is matched against. Store delivery in South Africa carries risk and cost factors imported models under-weight: long distances between metros, hijacking risk on certain corridors, variable last-mile access in dense township retail. Pick n Pay disclosed in FY26 both a materially improved logistics contract and a strategic supply chain partnership with DP World [12] — so for at least one major grocer the direction is to outsource more, which makes the integration boundary a *contractual* one and raises the question of who owns the delivery-event data. One retailer's disclosure is not a sector pattern and I would not write it as one.

Store inventory management

The store's own stock system, and the class where you will find the widest gap between what the estate believes and what is true. Receipt with discrepancy capture; return to centre and to vendor; inter-store transfer; adjustment with reason code for damage, waste, internal use, theft write-off and found stock; markdown capture; counts across cycle, perpetual and full stock takes; direct store delivery receipt where a supplier delivers to the store and invoices centrally; price-change and label confirmation; shelf-edge label printing; and the store's ordering override.

It masters on-hand by article at that store — and whether that is true is the whole argument. In the **centralised model** merchandising holds the authoritative position and the store system captures movement events. In the **devolved model** the store holds the position and publishes to the centre. Most real estates are a confused hybrid: the centre *believes* it holds the position, the store *acts on* its own, and the two are reconciled at stock take, where the difference is called shrinkage and written off.

> **THE SINGLE MOST CONSEQUENTIAL UNEXAMINED ASSUMPTION IN RETAIL ARCHITECTURE**
>
> Whether the centre or the store is authoritative for store on-hand. It drives replenishment, online order promising, and the shrink number the store manager is measured on. Nobody will volunteer the answer because nobody has written it down. Ask, then test it: compare the two numbers for twenty articles at three stores on one morning.

Why store inventory is structurally inaccurate, and none of these is anyone's fault: receipt counted at case level while sales deplete at unit level; a wrong case quantity; unrecorded waste, especially fresh; internal consumption; staff and customer theft; mis-scans where the cashier scans one of two visually similar products twice; variable-measure items from the in-store scale that deplete a value rather than a unit; and transfers sent but never received. Each is small. Together they produce an article-level accuracy rate well below what anyone would guess — worse still where reason codes exist but go unused so adjustments are all "other", where negative on-hand is tolerated, and where there is no article-level accuracy measure at all, only the aggregate shrink percentage, which hides compensating errors.

Point of sale

The one system every customer touches and the one most likely to be twenty years old. Item entry by scan, by lookup, by price-embedded barcode and by weight from an integrated scale; price and promotion resolution against a local price file and rule set; tender handling across cash, card, voucher, gift card, account, loyalty points, mobile wallet, social-grant cards and split tenders; loyalty identification and accrual; age-restriction prompts; tax calculation and the compliant receipt; void, refund, exchange and no-sale with authorisation levels; cash management; supervisor override; electronic journal capture; and **offline operation**. It masters the transaction — basket header, lines, tenders, discounts, loyalty linkage and operator — and the electronic journal, which is the evidential record of what happened at that till and the input to nearly all analytics.

The inheritance worth knowing. Peripheral integration — scanner, printer, scale, cash drawer, PIN pad, line display — is standardised by **UnifiedPOS**, which originated in the retail standards body and is now the Object Management Group's; the current formal release is 1.15.1, adopted December 2024 [45]. So **the peripheral layer is the most stable part of the POS stack and the application layer the least**, which is why replacements keep the hardware and why a hardware refresh and a software refresh are separable programmes.

Three South African specifics. The **till slip is a tax document** — a value-added-tax vendor must produce a valid tax invoice, with an abridged form permitted below a threshold; I could not source the current threshold and content list to a standard this book will stand behind, so no figure appears here. There is **no fiscal-device mandate**, unlike several African and European jurisdictions: South Africa does not impose certified fiscal printers or real-time reporting on general retail — stated as an absence, and absences are hard to source positively, so confirm it. And **age-restricted sales are a time-and-place problem, not an age prompt**: liquor trading days and hours are set by provincial legislation and municipal by-law, and tobacco sale to under-eighteens is prohibited under the Tobacco Products Control Act 83 of 1993 as amended [37]. The same article must be sellable at one store at 19:00 and blocked at another, the rule set has many owners, and this is routinely under-built.

> **"WE HAVE OFFLINE MODE" IS FOUR CLAIMS, AND MOST RETAILERS HAVE TWO**
>
> **Offline item lookup and pricing** — the till holds a local price file and rule set and prices any basket without a call. Most estates have this, and it is why the price file is distributed to every till rather than queried. **Offline transaction logging** — the till writes the journal locally and forwards on recovery with duplicate suppression. Usually present. **Offline tender acceptance** — the hard one, because a card authorisation needs a round trip to the acquirer; the mechanisms are floor limits, stand-in authorisation and store-and-forward, each trading fraud risk for availability, and the parameters are a commercial decision disguised as a technical one. **Offline loyalty and gift card** — accrual can be deferred; redemption cannot safely be, because the balance is unknown.
>
> A store with the first two and not the third **cannot trade when the line is down**, because in a largely card market most baskets cannot be tendered. Test this specifically. A store that cannot take money is closed, and the cost is not the lost basket — it is the queue abandoning, the perishables degrading, the labour paid for no output and the customer who does not come back from the rival 400 metres away. A branch that cannot transact sends the customer to the app. A store that cannot transact sends the customer to a competitor.

The classical architecture put a **store server** in the back office holding the price file, distributing to tills, aggregating the journal and acting as the single integration point. It made the store autonomous and made the estate 2,000 servers to patch and secure in rooms with variable power and no IT staff. The direction of travel is to remove it, with tills caching enough to trade standalone [46] — fewer things to manage, more dependence on the link, which here is precisely the dependence you do not want. Because the power crisis set the rules. Between roughly 2019 and 2024 South Africa ran systematic rotational outages; Shoprite's diesel spend to keep stores trading was reported at R1.3bn for the period reported in September 2023 [32] and R754m in the period reported in August 2024 [31] — historic peak figures, not a run rate, so date them whenever you use them. Telecommunications failed alongside: the regulator's inquiry recorded operator submissions that network availability fell below 93% at stage two for some operators, that most backup systems need nine to twelve hours to recharge and above stage four did not get it, and that tower companies recorded a 250% increase in generator loss during stage six in 2022 [29]. Card authorisation therefore failed for reasons entirely outside the retailer's estate, repeatedly and predictably. Outages were suspended for extended periods from 2024 [30], but the positions they forced are the right ones for any distributed retail estate: **the store trades on local capability, the link is an optimisation rather than a dependency, power and network are both assumed to fail, and the mitigation is itself an asset needing monitoring and protection** — because generators, batteries and cabling are theft targets. Write that into the principles, because somebody will propose designing it back out on cost grounds within two years.

Payment acceptance and the card estate

Distinct from POS and usually owned by a different team, often Treasury rather than IT. Terminal estate management with key injection and configuration distribution; tender routing to acquirer; authorisation; chip-and-PIN and contactless processing; tokenisation; refund and reversal; offline and stand-in authorisation with floor limits; batch settlement and reconciliation to bank statement; multi-acquirer and least-cost routing; chargebacks; gift-card and voucher processing; wallet acceptance; and card-data scope control. It masters the tender authorisation, the settlement file and the terminal estate — and **the terminal estate exists twice**, once in the retailer's asset records and once in each acquirer's and scheme's records as merchant and terminal identifiers. These drift, and the drift shows up as unreconciled settlement and as terminals the retailer pays for and does not have.

One South African fact reshapes the economics. Interchange is set by the central bank rather than negotiated: the Reserve Bank's National Payment System Department has published a position paper on interchange and a subsequent policy paper for comment [38]. So the largest single component of acceptance cost is a regulated number the retailer cannot negotiate, which means **tender steering and routing matter more than merchant-discount negotiation** — and the data to do that, cost per transaction by card type, scheme and acquirer, must be captured at the transaction, which many estates do not do.

And the tender mix is moving somewhere UK and US material does not predict. Stitch's 2026 consumer report, covering roughly 3,000 digitally active South African consumers plus platform data, found wallets at 57.5% combined adoption in the prior year, buy-now-pay-later at 38.9%, and bank-initiated methods at 38.6% combined; on its own platform Apple Pay became the most-used e-commerce method by volume, overtaking card [22]. Those are e-commerce and digitally-active figures, not in-store grocery figures, and must be cited as such — but the direction is clear enough to design for: **an architecture built around card-present chip-and-PIN needs a route for bank-initiated and wallet tenders that is not a card rail.** And the question that is under-asked: does cardholder data ever touch the retailer's own network? A point-to-point-encrypted terminal estate with tokenisation at the acquirer removes most of the store estate from card-security scope; an architecture passing card data through the POS puts every till in it. Across a 2,000-store estate that is the largest single payments-architecture decision a retailer makes.

Promotions and pricing engines

Where commercial intent becomes executable rules: promotion definition with mechanic, funding, period, participating articles and locations, and limits; offer construction across price-point, percentage, multi-buy, mix-and-match, threshold, gift-with-purchase, loyalty-exclusive and personalised forms; a planning calendar with slot allocation and conflict detection; margin simulation; funding and supplier-contribution capture; price-change scheduling and

approval; markdown planning; competitor price capture and rule-based matching; price-file build and distribution; and post-promotion evaluation. It masters the promotion, the offer rule and the price-change event. **In the better-architected estates the promotions engine, not merchandising, is the authority for what price the customer pays**, with merchandising holding the base retail and the engine holding everything that modifies it. In the worse ones both hold both.

Three distinctions to hold. **Promotion versus markdown**: a promotion is time-boxed, usually supplier-funded in whole or part, with the base price unchanged underneath it; a **markdown** is a permanent reduction funded by the retailer, with an immediate inventory-value consequence under the retail method. Model both as "a price record with dates" and you lose the funding attribution and the accounting treatment — and then nobody can answer how much of the margin erosion was promotional and how much was markdown, which is the most important question in merchandise performance. **Multi-buy is not a price**: "three for R50", "buy two get one free", "spend R500 get R50 off" are *basket-level rules* evaluated at the till against the whole basket with a defined priority and exclusivity, which is why the engine is a separate class. And **the promotion calendar is a scarce resource**: leaflet pages, end-of-aisle positions and home-page tiles are finite, allocating articles to them is a negotiation with suppliers paying for the space, and the conflict rules — two promotions on one article, a promotion on an article being delisted, a promotion that breaks a price ladder against the retailer's own private label — need machine checking, because humans will not catch them at volume.

> **PRICE IS THE MOST DANGEROUS OBJECT IN A RETAIL ESTATE, AND THE REASON IS LEGAL**
>
> Section 23(6)(b) of the Consumer Protection Act provides that where more than one price is displayed at the same time, the supplier **must not charge more than the lower or lowest of those prices** [9][8]. A shelf label that disagrees with the till is not an operational irritation. It is a statutory obligation to sell at the lower number.
>
> The escape is narrow. Section 23(9) relieves the supplier of a displayed price containing an "inadvertent and obvious error" only once the supplier has **both** corrected the display **and** taken reasonable steps to inform affected consumers of the error and the correct price [8]. The Consumer Goods and Services Ombud's Advisory Note 2 of 1 March 2021 sets out what reasonable steps look like — withdraw the product, fix the label, then resell; for in-store notices, put them up at the affected stores and keep dated photographic proof — and notes that a consumer who has already acted as if buying, for example by taking the item to the till, may still pay the error price [8]. Section 23(10) relieves the supplier where an unauthorised person altered or obscured the price [8]. The Note is explicit that sections 23 and 30 appear to conflict on errors and that only the Tribunal or the courts can resolve it [8].
>
> **So the architecture must produce evidence, not just correctness.** A provable record of what was displayed, where and when. A correction workflow whose output is proof. And a reconciliation between the price in the POS, the price on the label and the price in the advertising.

Electronic shelf labels

Battery- or field-powered e-paper displays at the shelf edge, bound to an article, updated over a wireless protocol from a store controller or cloud service, so a price change reaches the shelf in minutes rather than in a labour-intensive weekly relabel. What matters: label-to-article binding and re-binding; price and message rendering with template control; **update confirmation per label**; battery and health monitoring across a large estate; flash and locate functions, so a label can light up to guide an online picker to the bay; and an **auditable record of what was displayed and when**.

That last one is why this is a compliance capability and not only a labour saving. A paper estate cannot prove what the shelf said at 14:37 last Tuesday. An electronic estate can, per label, with a timestamp — which turns a disputed complaint into a query. It creates a sharper risk in exchange: **the label-to-article binding becomes a data object that can be wrong at scale**, and a mis-bound label mis-prices the shelf at every store that received the same update. Pick n Pay piloted the technology and its Faerie Glen hypermarket was reported as the first hypermarket in the country to deploy Pricer labels [28]; I could not source current penetration across the South African estate and would treat the technology as **deploying rather than deployed** here as at 2026. It fails when the binding is maintained by hand so a planogram reset orphans hundreds of labels; when update confirmation goes unmonitored, so failed labels sit showing an old price — exactly the scenario section 23(6)(b) punishes; and when labels update from a different source than the POS price file, so the two can disagree by design.

Loyalty, CRM and customer data platforms

Loyalty is the retailer's only reliable means of attaching a transaction to a person: enrolment and identifier issue; identification at till and online; accrual and redemption with expiry; tiering; personalised and segment offers with redemption tracking; partner earn and burn across fuel, airline and bank; household grouping; consent and preference management; and member service. It masters the member, the identifier, the balance and the offer-response history — and **the points balance is a financial liability** that must reconcile to a provision in the accounts, which makes loyalty an unexpected finance-integration concern. South African penetration is high by international standards: Clicks reported 12.6 million loyalty members at its 2025 year end, with 745,500 Baby Club and 1.6 million Seniors Club members [14], and Pick n Pay operates Smart Shopper and Boxer Rewards Club with points redeemable for fuel at participating stations [12]. What no major retailer here discloses is the proportion of sales identified to a member, which is the only figure that matters architecturally.

A **customer data platform** sits above loyalty and resolves identity across channels: deterministic resolution on loyalty number, card token, email and telephone; probabilistic on device and behaviour; profile assembly; audience definition; activation; and consent propagation. **It masters the resolved identity — the claim that these identifiers are one person — and nothing else.** Everything in it is a copy, which is both its value and its risk. It earns its place when the retailer has genuinely separate identity domains to reconcile: an in-store member, an online account, an app install, a marketplace buyer and a pharmacy patient. It does not earn its place as a second data warehouse, which is what it usually becomes.

POPIA IS SHARPER THAN GDPR IN ONE RESPECT, AND IT LANDS HERE

The Information Regulator's Guidance Note on Direct Marketing sets out that electronic direct marketing is prohibited unless the data subject has consented or is an existing customer; that the responsible party bears the burden of proving consent; that consent must be requested using **Form 4 or a substantially similar form**, free of charge, naming the goods or services and letting the person choose their preferred contact method, which the marketer must then follow; that a responsible party may approach a person **only once** to request consent, and only if they have not previously refused; and that silence is not consent [20]. The definition of electronic communication expressly includes telephone calls, email, SMS, automatic calling machines, fax, push notifications, social-media direct messages and cookies [20]. The existing-customer exception applies only if **all** four of its conditions are met, including that the marketing promotes the responsible party's own *similar* products [20].

Read as architecture that is a specification: consent captured per channel per purpose with the Form 4 provenance retained; recorded evidence, including call recordings; a **suppression store authoritative across every outbound channel including partners**; a "similar products" determination per campaign that someone can defend; and a once-only rule enforced across the estate, which means the consent *request* is itself an event to be logged. Most South African retail loyalty estates have the consent flag and not the evidence, and have suppression in the campaign tool rather than as a shared service. That is the gap.

E-commerce, order management and fulfilment

Three classes the trade talks about as one and that fail separately. **The e-commerce platform** owns the online catalogue, search and browse, cart, checkout, customer account and content, and it masters the *online* product presentation — enriched content, images, attributes, reviews — which is why "the website has the wrong description" is a content problem rather than a web problem. **The order management system** owns the customer order as an object with a life: created, paid, sourced, allocated, picked, packed, dispatched, delivered, returned. It makes the **sourcing decision** — which node fulfils which line — against available-to-promise inventory, node capacity, cost to serve and promise. This is the class most often missing from an estate that has a website, and its absence shows up as a platform doing a bad job of promising against a nightly stock feed. **Fulfilment** is physical execution: in-store picking with an app and a route through the store, a dark store, a dedicated centre, or a courier integration.

Available-to-promise is the hard problem. Promising stock in a store that is simultaneously selling it to walk-in customers needs a near-real-time store position — which is structurally inaccurate — a safety buffer that is per-article and per-store rather than a flat percentage, and a substitution policy for when the pick fails. South African grocery has solved this operationally with substitution-and-refund rather than architecturally with accuracy. That is a legitimate design choice and it should be written down as one.

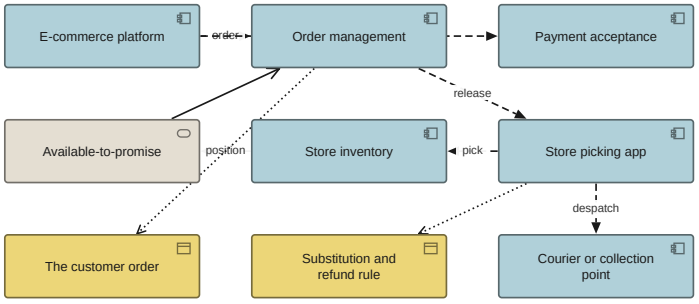


Figure 12.5 The three classes the trade calls “online”. The order management system is the one most often missing, and its absence shows up as a website promising stock against last night's feed while a walk-in customer buys the last unit.

The figures matter because they move the constraint. Online retail here was estimated at about R130bn in 2025 and forecast at about R159bn for 2026, reaching about 10.7% of total retail at constant prices in 2025 [23]. Checkers Sixty60 was 10.3% of Shoprite's South African sales in FY2026 and Woolworths Dash 7.3% of Woolworths' South African Food sales [23]; Sixty60 turnover was R25.5bn for the year to 28 June 2026, up 34.5% [13]. Pick n Pay reported online sales up 32.7% on a comparable basis, with more than 620 stores fulfilling orders and over 30,000 products offered with delivery within 60 minutes [12]. Mr Price reported more than 55% of online orders collected in store in FY2026 [23]. Three readings. **In-store fulfilment is the dominant South African model**, not dedicated centres — so store systems, store labour and store inventory accuracy are the binding constraints on e-commerce growth, not the website. **The sixty-minute promise is a systems promise**: order-to-pick-start must be minutes, which rules out a batch interface anywhere on the path. And **click-and-collect is material in non-food**, which is a different order-management flow with a different failure mode, uncollected orders, and a different store task.

Marketplace, and returns

Marketplace is where the retailer lists third-party sellers' products alongside its own, and architecturally it is a different business wearing the retailer's brand: seller onboarding and vetting; catalogue ingestion and mapping to the retailer's taxonomy; listing policy enforcement; order routing to seller; service-level enforcement; commission calculation and seller settlement; seller-side returns; and delisting. It masters the seller, the seller listing and the seller order. **It must not master the article** — mapping third-party listings into the retailer's own article master pollutes it with records that have no cost, no supplier-article relationship and no inventory, so keep the catalogue separate and join on the trade item number where one exists. Two obligations get underestimated: the retailer is liable to the customer for the experience regardless of who shipped it; and the retailer becomes a payment intermediary holding money for sellers, which raises a question about whether the arrangement is a payment service. I could not settle that from public material and would put it to the regulator rather than guess.

Returns is the class most likely to be a process rather than a system: authorisation or over-the-counter acceptance; reason capture; disposition; refund to original tender or alternative; inventory reinstatement at the correct location and value; return-to-vendor claim; and fraud detection on return behaviour. **The disposition decision is the one with money in it.** Back on the shelf is worth full retail; to a returns centre is worth retail less handling; back to the vendor is worth cost; destroyed is worth nothing and may cost to dispose of. Retailers that do not make this decision systematically default to "back on the shelf" and carry unsellable stock as good. Statutory rights set the floor — the Consumer Protection Act gives consumers rights to return goods in defined circumstances [9] — so a returns policy is a floor set by statute rather than a commercial choice. In pharmacy, returned medicines generally cannot re-enter stock at all.

Workforce, tasks and loss prevention

Workforce management is the second-largest controllable cost after cost of goods and routinely the least-integrated system: demand-driven labour forecasting, converting a sales or transaction forecast into hours by department and fifteen- or thirty-minute interval; schedule generation against skills, availability, contract terms and legal constraints; self-service for availability and shift swap; clock capture; exception handling; leave; overtime interpretation; payroll export; and productivity reporting against the schedule. **The forecast-to-hours conversion is where it breaks**: the labour model needs the *same* demand forecast the replenishment engine uses, at a finer time grain, and in practice gets a separate, coarser, differently-owned forecast — so the store is stocked for one demand picture and staffed for another. The compliance surface is the Basic Conditions of Employment Act, the sectoral determination for wholesale and retail, and bargaining-council agreements, which means **award interpretation is reference data that changes by gazette** and a system with hard-coded rules will be non-compliant within two years. Biometric clocking is special personal information under POPIA, which is a basis question rather than a procurement question.

Store task management is thin, newish and disproportionately valuable: task creation from a system event — a price change to apply, a planogram reset, a recall to execute, an order to pick, a count to perform — or from a central instruction; assignment by role and store; completion capture with evidence such as a photograph or scan; escalation; and compliance reporting. Here is why it matters more than its size suggests. **Every system in this chapter produces work for a human in a store, and almost none of them can tell whether that work was done.** The price change was distributed — was the label changed? The planogram was published — was the bay reset? The recall was issued — was the stock pulled? Task management is the **execution-confirmation layer for the entire estate**, and without it every central system's measure of its own success is an assumption. It fails when volume is so high that stores triage by ignoring, when completion is self-reported for want of evidence capture, when six systems create tasks with no deduplication, and when there is no link from a task back to the event that caused it, so the central system never learns its instruction failed.

Loss prevention runs exception-based reporting over transaction data — voids, no-sales, refunds without receipt, manual overrides, discount abuse, operator outliers — plus video analytics with transaction overlay, article surveillance and tag management, shrink attribution, case management, and organised-crime pattern detection across stores. **Shrink decomposes into four causes with very different responses**: external theft, internal theft, process error and waste. That decomposition is only possible with a stock-take variance analysed against adjustment and waste records, which is why retailers with poor reason-code discipline cannot say what their shrink is *made of* — only how big it is.

> **THERE IS NO PUBLISHED SOUTH AFRICAN SHRINK FIGURE**
>
> Retail crime and shrinkage are consistently named as material pressures on South African retailers, organised retail crime specifically, but the trade and security commentary available carries **no quantified figures and cites no underlying data**. The likeliest primary sources would be the Consumer Goods Council, individual annual reports — where waste and shrink are occasionally mentioned and rarely quantified — or academic work on physical stock loss.
>
> So: **any shrink percentage quoted to you must carry its source and date, and the United States benchmark may be used only if it is labelled as a United States figure.** And a South African retailer cannot benchmark its own shrink against its market. Say that out loud, because somebody will ask for the comparison and the honest answer is that it does not exist.

Master data, product information and the data platform

Master data management governs the golden record: stewardship workflow, matching and merging, survivorship rules, data-quality rules with measurement, hierarchy management, and distribution to consumers. The candidate domains are article, supplier, location and customer, and in practice retailers attempt article first, succeed partially, and never reach the others. **Product information management** manages rich, marketing-facing content: descriptions, images and digital assets, attributes by category, nutritional and allergen data, ingredients, country of origin, dimensions and weights, regulatory statements and translations; its primary consumer is the website and marketplace listing and its primary supplier is the supplier. **The division of labour that works**: merchandising masters the *commercial and operational* article record; the content system masters the *descriptive* record; master data management governs the process by which both are created and holds the cross-references. The division that does not work is content-as-article-master, because a content system has no concept of cost, inventory or a purchase order.

The data platform ingests transaction logs, stock movements, orders, forecasts, plans and loyalty events; builds conformed dimensions for article, location, calendar, supplier and customer; handles slowly-changing hierarchies; serves marts for trading, supply chain, finance, store operations and customer; holds features for forecasting and personalisation; and increasingly publishes external data products for suppliers. Three retail-specific decisions, all commonly got wrong. **The grain of the sales fact**: transaction-line grain is the only grain supporting basket analysis, loyalty attribution and promotional evaluation, and it is enormous — a large South African grocer generates many millions of lines a day, where article-store-day aggregates are a hundredth of the volume and answer most questions. Most estates build both, and the two disagree because of late-arriving transactions and voids. **The calendar dimension is not generic**: it must carry the retail week, the period, the quarter, the retail year, the 53rd-week flag and the comparable-period mapping including the restatement convention [11]. This is the single most common thing a competent general-purpose data engineer builds badly, because nothing about it is wrong in any other industry. And **hierarchy history**: whether the warehouse restates history on reclassification, and whether that matches merchandising's behaviour. State the answer in the dimension design, because the default is a type-one dimension that restates everything silently.

ERP, finance and supplier income

In most retailers **ERP is not the merchandising system.** It is the general ledger, payables and receivables, fixed assets, cash management and statutory reporting, plus goods-not-for-resale procurement. The integration points that matter: the merchandising stock ledger posts to the general ledger, usually as a summarised journal, and reconciling the two is a named monthly control; matched supplier invoices post to payables; POS sales post as a daily journal per store with tender analysis, and reconciling that journal to banked cash and to acquirer settlement is the retailer's highest-volume financial control; supplier income accruals post from the deals system; payroll posts from workforce management. **The two-ledger problem**: merchandising holds inventory at article-location grain on a retail or cost basis and the general ledger holds it as one balance per entity per account, so the bridge is the stock ledger — and the stock ledger is where markdowns, shrink, waste and intake margin become accounting entries. An architect who treats it as a reporting artefact rather than a subsidiary ledger will mis-scope the finance integration by a wide margin.

Supplier income is the highest-risk commercial system in a retailer and frequently the least systematised. It covers everything moving money from supplier to retailer outside the invoice price: volume rebates and growth incentives, settlement discounts, listing allowances and new-line fees, promotional funding and co-operative advertising, shelf-access and category-management payments, refurbishment contributions, markdown and waste support, and year-end over-riders. Why it is dangerous: the amounts are material to reported profit; they are earned over a period against conditions; they are often *estimated* and accrued before they are claimed; and the estimate is made by the commercial team whose performance it flatters. **Tesco's 2014 accounting misstatement arose precisely here** — the recognition of commercial income relative to the period in which it was earned and the costs it related to, with a resulting overstatement of profit [24]. The transferable lesson: **commercial income must be keyed to the thing that earns it** — a promotion, a volume achieved, a line listed — with a system-held condition, a system-calculated entitlement and a documented claim-to-cash trail. Keyed to nothing, it is an accrual based on judgement, and an accrual based on judgement is where misstatement lives. In South Africa there is a second reason to systematise it: after the Grocery Retail Market Inquiry remedy, a rebate system that cannot list every term offered to every supplier, state the objective justification for each differential and show that terms are not volume-based is a compliance exposure, not a data-quality niggle [10].

Pharmacy dispensing and its regulatory separation

The most regulated system in a retail estate, and the one an architect must not treat as another department. In a South African pharmacy retailer the dispensary is a **separately licensed professional practice operating inside a shop**, and almost every architectural instinct from the rest of the estate is wrong inside its walls.

The dispensing system does patient registration; prescriber capture and validation; prescription capture including item, strength, dose, quantity, directions and repeats; clinical checking for interaction, duplication, dose range, allergy and contra-indication; generic substitution offer and recording; label generation; the scheduled-substance register; repeat management; medical-scheme eligibility and claim submission with real-time adjudication; patient medication record maintenance; stock control of dispensary stock separately from front shop; and the statutory registers. **It masters the patient, the prescription, the dispensing event and the register, and it must master them alone.** No other system in the estate may be authoritative for any of them.

The frame. The Pharmacy Act 53 of 1974 establishes the South African Pharmacy Council, and the Council's **Good Pharmacy Practice** rules are published under section 35A of that Act [16]. The Medicines and Related Substances Act 101 of 1965 and its General Regulations govern scheduling, registration, labelling, record-keeping and prescription requirements, classifying medicines into schedules that determine who may sell, who may prescribe, whether a prescription is required, whether it may be repeated, and what register entries are mandatory [17].

Pricing is national and statutory, and this is the fact that breaks an imported design. The **single exit price** is the only price at which a manufacturer may sell a medicine to anyone other than the state; final sellers charge that price to the patient or scheme; declared prices for registered medicines are published by the Department of Health [18][19]. Wholesalers add no mark-up and instead receive a logistics fee taken as a portion of the exit price [18]. The **dispensing fee** is capped annually as a flat amount plus a percentage of the exit price across four price bands, with the percentage falling as the price rises and separate limits for pharmacists and licensed dispensing practitioners [18]. Sellers must disclose both and may not exceed the gazetted maximum [18].

> **WHAT THAT MEANS ARCHITECTURALLY, AND IT IS UNLIKE ANYTHING ELSE IN RETAIL**
>
> **The pharmacy retailer cannot price medicines.** For a scheduled line there is no price zone, no promotion, no markdown, no competitor matching and no margin optimisation.
>
> What there is instead: a **national price feed** ingested on a published schedule and applied exactly; a **fee calculation** that is a banded formula whose parameters change by Government Gazette; and a **disclosure obligation on the receipt**. The pricing engine must be bypassed entirely for scheduled lines, and the article master must carry the schedule as a controlling attribute rather than a reporting tag.
>
> A design that treats medicines as articles with a price is wrong at the first adjustment — and in 2026 there were **two**, an annual increase effective in January and an extraordinary one effective 1 October. An architect who heard "the annual increase" and built for one event a year was wrong twice in twelve months.

The separation requirements are physical as well as logical, and they are specified. Good Pharmacy Practice requires that a pharmacy inside another business be clearly identified and demarcated with **permanent, floor-to-ceiling barriers** enclosing the waiting, clinic, semi-private and private areas; that it have a single compliant entry and exit; and that the pharmacist have unfettered access around the clock [16]. Another business operating inside a pharmacy requires Council approval. **Only registered pharmacy personnel may have direct access to medicine**, and keys or access devices must remain on the person of a pharmacist [16]. And the confidentiality rules constrain the data architecture directly: access to patient information is limited to those who need it for the patient's care or in the public interest; anyone with access must understand it is confidential and must not disclose it without the patient's **written** consent; any disclosure must be recorded on the patient's record; anonymised patient data may be collated for research or commercial purposes only with names, addresses, telephone numbers and postal codes removed; and software relied on for warnings or interaction checking **must be the latest version available** [16].

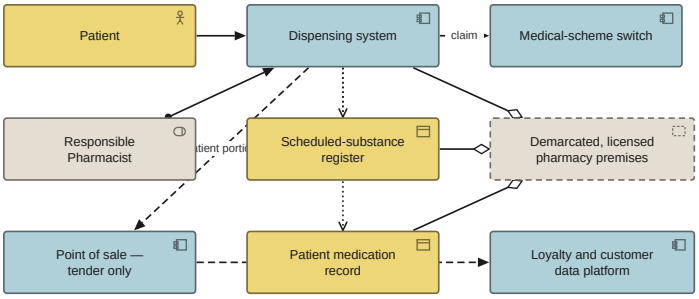


Figure 12.6 The dispensary boundary is a wall, not a lane. Good Pharmacy Practice specifies permanent floor-to-ceiling demarcation and pharmacist-only access to medicine; POPIA makes the dispensing detail special personal information. The loyalty programme may know the basket value and nothing else.

Put that beside POPIA, under which health information is special personal information with a stricter processing basis, and a hard boundary falls out. Patient and prescription data is access-controlled to pharmacy personnel — not store staff, not head-office merchandising, not marketing and not the customer data platform. The loyalty programme may know a basket included a dispensary transaction and its value; it must not know what was dispensed, absent a separate consented basis. The data platform receives dispensary data either pseudonymised or into a segregated zone with its own access model, and the anonymisation standard names the identifiers to strip [16]. Audit logging on the patient medication record is a professional-conduct requirement, not an IT nicety.

Two more structural constraints. Regulations govern who may own a pharmacy and how it is licensed [48], and the Constitutional Court settled a long-running dispute on the meaning of beneficial interest in pharmacy ownership in 2023 [47]. **The legal entity owning a pharmacy may differ from the retail group entity owning the store it sits in**, which makes the dispensary a separate location, a separate stock-holding entity, a separate tax and posting path, and potentially a separate licence holder in the location master — and an estate that models the dispensary as a department of the store cannot produce the entity-level records the licence requires. And "latest version available" [16] means the drug-interaction database vendor's release cadence sits on the retailer's compliance critical path: a dispensing estate including stores that have not taken a content update in weeks is non-compliant on the plain words of the rule, which is a real consequence for offline-capable stores in a market where connectivity fails.

It fails when the dispensary is integrated to POS for tendering only, so dispensary and store stock are reconciled by hand; when the price file is loaded manually and late; when the fee bands are hard-coded, so a gazette change becomes a software release; when patient data flows into the group warehouse because a generic extract was pointed at the whole store database; when the register is kept in the system *and* on paper because nobody is sure the electronic version satisfies the regulation — a genuinely contested point I could not settle from the Regulations; and when interaction software is out of version, which is an explicit breach.

Integration, as it really is

In descending order of volume in a large retailer. The component list is directly observable in any interface inventory; the ordering is practitioner consensus.

WHAT IS ACTUALLY CARRYING THE TRAFFIC, MOST TO LEAST

1 · Flat files on SFTP	Article extracts, price files, sales polling, stock snapshots, GL journals. Cron, sentinel file, "retry then page someone".
2 · Store polling	Vendor-supplied push of price and promotion down, pull of sales and stock up. The most reliability-engineered path in the estate.
3 · EDI to partners	AS2, SFTP or a value-added network — so the retailer usually does not own its own message archive.
4 · Direct database	Reporting tools and satellites reading the merchandising schema, and in older estates writing to it. The reason it cannot be replaced.
5 · Bus or iPaaS	Stock enquiry, loyalty lookup, order status, payment authorisation. Introduced for one programme, then partially adopted.
6 · REST APIs	Concentrated in the digital estate, increasingly supplier-facing.
7 · Event streaming	Transaction events to the data platform, inventory events to order management.

Component list observable in any interface inventory; the ordering is practitioner consensus.

Figure 12.7 The integration estate is a sediment, not a layer, and the ordering is by volume rather than by modernity. Every generation added a style without retiring the previous one, so the same article master arrives three ways with three latencies.

1. **Scheduled flat files over SFTP.** The backbone. Article master extracts, price files, sales polling, stock snapshots, general-ledger journals, supplier catalogues. Fixed-width or delimited, named by convention, triggered by cron, with a sentinel file and an error-handling strategy of "retry then page someone".
2. **Store polling and distribution.** A dedicated, often vendor-supplied mechanism pushing price and promotion files down to stores and pulling sales and stock up. The most reliability-engineered path in the estate, because it has to be.

3. **EDI** to trading partners, over AS2, SFTP or a value-added network — which means the retailer usually does not own its own message archive, and that matters when a dispute needs proof of what was sent and when.
4. **Database-level integration.** Direct reads of the merchandising database by reporting tools and satellite applications, and in older estates direct writes. Undocumented, untraceable, and the main reason the merchandising system cannot be replaced.
5. **An integration platform or bus**, carrying the newer synchronous traffic: stock enquiries, loyalty lookups, order status, payment authorisation. Usually introduced for one programme and then partially adopted.
6. **REST APIs**, concentrated in the digital estate and increasingly supplier-facing.
7. **Event streaming**, in the newest layers: transaction events to the data platform, inventory events to order management.

The honest state. A large retailer's integration estate is not a layer, it is a sediment. Each generation of programme added a style without retiring the previous one, so the same article master is published as a nightly file, read directly from the database and exposed as an API, with three field sets and three latencies — and three consumers each believing theirs is authoritative. This is not primarily a failure of discipline. It is the arithmetic of a 2,000-store estate where every change must be tested against every store and the cheapest safe change is always to add rather than replace.

Two patterns worth insisting on, and they are cheap. Normalise identifiers at the boundary — trade item numbers to fourteen digits, right-aligned, zero-padded, stored as a string [4][5]; location to an internal key with a maintained global location number cross-reference [7]; supplier to a legal-entity key with a site sub-key. That removes the largest category of retail integration defect, which is an identifier that lost a leading zero in transit. And **separate the reference-data path from the transaction path**, engineering the reference path for *confirmed* delivery rather than best effort. The price file is the case in point: it is not enough to send it, the retailer must know it arrived and was applied, **per till and per label**, because the legal consequence of failure is immediate [8][9].

The batch clock and the trading day

Retail time governs retail architecture, and this is the section architects from banking and insurance consistently under-weight.

The calendar first. The National Retail Federation publishes the 4-5-4 calendar as a **voluntary guide**: each year divides into months on a four-week, five-week, four-week pattern, so comparable months contain the same number of Saturdays and Sundays and holidays stay aligned [11]. It came out of informal industry discussion in the 1930s and into wide use in the 1940s, because as weekends grew as a share of sales the varying number of weekends per calendar month made month-on-month comparison meaningless [11]. A 52-week year is 364 days, so a 53rd week is needed roughly every five or six years, on no fixed schedule; the rule is that if four or more days remain in January after the 52-week layout, a 53rd week is added **for sales reporting only** [11]. For comparability a 53-week year is typically restated the following year by shifting each week back one and ignoring the first, or the 53rd week is just left out of the comparison [11]. The South African wrinkle is that nobody shares a year end: Pick n Pay's FY26 was a 52-week period ended 1 March 2026, reported on a 52-against-52-week comparable basis because the prior year was not comparable [12]; Shoprite's year ended 28 June 2026 [13]; Clicks reports to the end of August [14]. Sector comparison therefore requires period alignment, and the data platform's calendar must support more than one retail year definition if the retailer benchmarks. Chapter 13 treats the calendar as a standards artefact.

Then the nightly cycle, which is the estate's heartbeat. The dependency order is the point.

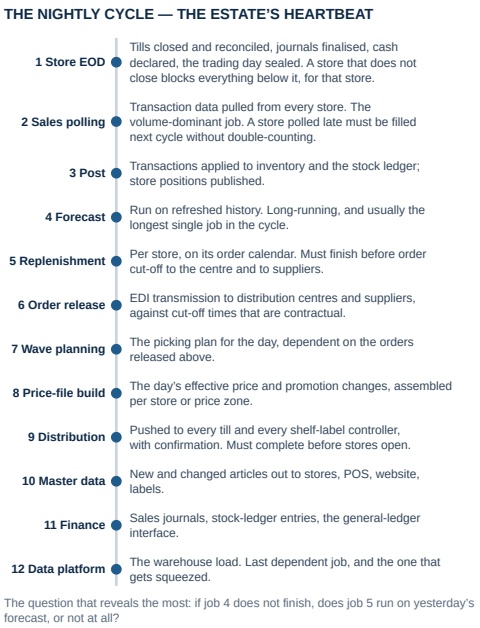


Figure 12.8 The overnight cycle, in dependency order. The dependency is the point: every job below a failure runs on stale input or does not run at all, and the whole chain has to finish before the first store opens. Ask what happens when job 4 overruns.

The critical-path properties to establish in your first month. What is the cycle's total duration and its slack to store opening? Which jobs are on the critical path and which are merely slow? What is the recovery procedure for each — re-run, skip, or run degraded? And the one that reveals the most: **what happens if the forecast run does not complete? Does replenishment run on yesterday's forecast, or not run at all?** In most retailers that answer is a convention held in an operator's head rather than a designed behaviour, and finding it out is one of the more valuable things you will do early. Above

the nightly clock sit the weekly trading review and promotion changeover, the periodic close with its stock-ledger reconciliation, the statutory close, and the stock-take weekend, which freezes movement and runs a heavier batch.

And then peak, which here is not the UK or US shape. The festive peak runs from Black Friday to early January with December dominating the year. **Back to school** in January is a second genuine peak, concentrated in stationery, uniform and school shoes, and because the school year starts in January it collides directly with the post-festive trough in food. Easter moves. Month-end and the pay-and-grant cycle produce a *monthly* peak sharper than in most markets [22]. **The change freeze is the architectural consequence**, and it is wider than IT change: software releases, infrastructure change, structural master-data change — hierarchy reclassification especially — range change, planogram resets and store refits. No South African retailer publishes its change calendar, so confirm the window in week one rather than assuming it. Two second-order effects are worth stating anyway. **The freeze compresses the delivery year into roughly nine months**, so a plan assuming twelve is wrong by a quarter. And **the freeze accumulates change**, so the January and February releases are the largest and riskiest of the year, immediately after the period in which the estate was least exercised.

The legacy question

What an architect actually finds. Most of this is **practitioner consensus rather than sourced fact**, it is labelled as such, and it is here because it is what the first eight weeks look like.

The merchandising system nobody can replace. Implemented between 1995 and 2008, modified continuously, carrying the accumulated commercial logic of twenty-five years of decisions nobody wrote down. Not replaceable in one step, because every other system reads it — often directly at database level — the modifications are undocumented and their authors have gone, and the business cannot articulate the rules well enough for anyone to rebuild them. The route out, where one exists, is to strangle it: move price out, move planning out, move inventory out. That takes years and needs exactly the integration discipline the estate lacks. What settles the general claim is a published replacement case study with duration and cost; what settles it for one retailer is an interface inventory and a count of direct database consumers.

The planning tool that is a spreadsheet. The financial plan, the weekly grid, the open-to-buy and usually the allocation live in workbooks owned by individuals, with formulas nobody has audited, feeding decisions worth hundreds of millions. Not always wrong — the spreadsheet is faster and the merchandiser is accountable for the number — but unmanaged.

The POS that is twenty years old. It works, it has been hardened by two decades of store reality, and it trades offline. Its promotion logic is in code so new mechanics are slow, its journal retention is short, and nobody will fund replacing it until something external forces the issue. The architectural reason it survives *is* sourced: peripheral integration is standardised [45], so hardware and application layers evolve independently and the application layer has no customer-visible reason to change.

And the archetypal finding: the systems that each think they own the price.

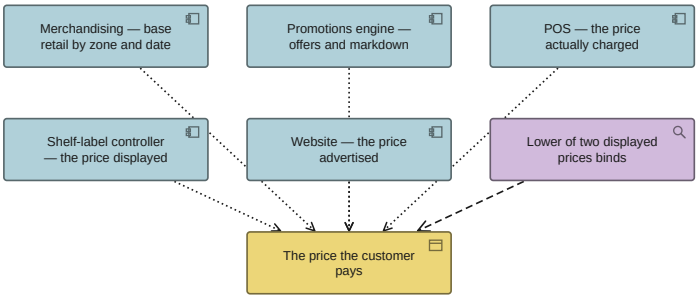


Figure 12.9 Five stores of one concept. Each was correct in isolation: merchandising needs base retail for the stock ledger, the engine needs basket rules it cannot express as an article price, the till needs a local file to trade offline. The failure is that none is declared authoritative — and the statute binds the retailer to the lower of any two that are displayed.

Merchandising holds regular retail by article, zone and effective date, because the margin and the stock ledger need it there [1]. The promotions engine holds promotional and markdown prices and the basket-level rules, which cannot be expressed as an article price. The POS holds the price it will actually charge, in a local file, because it must trade offline. Add the shelf-label controller and the website and it is **five stores of one concept, with five update paths and five latencies**. Each was correct in isolation. The failure is that none is declared authoritative and no reconciliation exists — so "what price will this article ring at, at this store, tomorrow" has no systemic answer, and the retailer is bound by the lower of whatever two of them display [8] [9].

The South African anchor, and what it cost

SPAR replaced its enterprise system with SAP, beginning at the **KwaZulu-Natal distribution centre**. **The sources disagree on when it went live**: ITWeb dates it to February 2023 [39], TechCentral to October 2022 [40], and Business Day describes the S/4HANA rollout as begun in early 2023 [41]. That disagreement is itself instructive — this is the best-documented case in the country and the basic date is contested, so cite whichever source you use and do not resolve it.

What happened is not contested. Go-live and integration issues significantly disrupted distribution in the region; stock deliveries to retailers' stores were disrupted, reducing service levels and retailer loyalty [39]. The KZN logistics hub was incapacitated, and SPAR supplied KZN stores from the Eastern Cape, North Rand and South Rand distribution centres, with **fuel and distribution costs up 12.8%** [40]. The numbers differ by source, so carry the source. For the year to 30 September 2023: an estimated **R1.6bn loss of group turnover**, about **R720m of lost profit for the KZN region**, and a **R94m write-off** of SAP asset under construction tied to a change in rollout approach for foreign regions [39]. TechCentral puts SAP-related losses at R1.8bn in the period to end-September 2023, with profit after tax of R0.4bn, **a decrease of 80% on the previous year** [40]. The Register, citing SPAR's 2024 report

and Bloomberg, reports about R2bn of lost sales in the year to 30 September 2024, an estimated R2bn in FY2023, and an **implementation cost estimated at R1.8bn** [42].

TWO SOURCED CASES, AND WHAT EACH ONE TEACHES

Fact	Figure, as reported	Source, and the caveat
Go-live	February 2023 / October 2022 / early 2023	ITWeb / TechCentral / Business Day. Three dates. Do not resolve them
Lost turnover FY23	R1.6bn group; −R720m KZN profit	ITWeb, on the year to 30 Sep 2023
SAP-related losses	R1.8bn; PAT R0.4bn, down 80%	TechCentral, same period — a different framing of the same year
Lost sales FY24	−R2bn	The Register, citing SPAR's 2024 report and Bloomberg
Implementation cost	Estimated R1.8bn	Bloomberg via The Register, Sept 2023. An estimate, not a disclosure
Write-off	R94m asset under construction	ITWeb. Tied to a change in rollout approach abroad
Franchisee claim	R168.7m lodged 28 Jan 2026	Business Day. Allegations; no court findings reported
Pick n Pay POS data	Planned <6 months to Nov 2021; live Jan 2022	SAP's own newsroom. 13+ interfaces; team "pushed its internal resources too hard"

In retail the cost of a systems failure is measured in sales, not in licences, because the supply chain stops.

Figure 12.10 The two South African anchors, with their sources attached because the figures differ. SPAR's lost sales exceeded the estimated implementation cost in a single year. Pick n Pay's is the more useful case, because the vendor's own write-up admits the overrun and names its cause as internal capacity.

Put those together and you have the sentence that should change how you brief a steering committee. **The lost sales exceeded the implementation cost, in a single year.** In retail, the cost of a systems failure is measured in sales, not in licences, because the supply chain stops — and a grocery supply chain that stops loses perishable stock and customers simultaneously. That is the fundamental difference from a bank core replacement, where the failure mode is service disruption and regulatory exposure rather than immediate revenue loss.

The consequences ran on. Three directors had left by the time of the trading update reported on 30 September 2024 [39]. On **28 January 2026** a franchisee family owning 46 stores lodged a summons in the Durban High Court for about **R168.7m** plus interest — R142.9m for lost gross profit and margin across 2023 to 2025 and R25.8m tied to SPAR's rebate and over-rider schemes [41]. Those are allegations, no court findings have been reported, and the case is live. But note what the claim shows: in a voluntary-trading or franchised estate **the blast radius of a go-live extends to parties with a contractual claim**, and the rebate entitlement failed when the distribution system did.

> **SPAR'S OWN REMEDIATION LINE, AND IT IS THE MOST TRANSFERABLE SENTENCE IN THIS CHAPTER**
>
> SPAR conducted a reassessment covering stabilising the KZN implementation, reassessing whether the warehouse management system is fit for purpose, and reviewing how the rollout could continue at reduced risk — and stated: **"This will include separating the ERP from the warehouse system and implementing them independently"** [40].
>
> No end date was given. The implementation partner is not named in any source retrieved, and SPAR described SAP as having remained a supportive partner throughout.
>
> Read it as a design rule rather than a news item. **An enterprise system and a warehouse system should not go live together.** One changes how the business records a transaction; the other changes how a human moves a pallet. Together, nobody can tell which one broke, and the recovery path for each is different.

The counterpoint, which is also South African and also sourced

Failure cases are easy to over-learn from, so set one honest success beside it — and the most useful one is honest precisely because it admits what went wrong. Pick n Pay's own vendor-published case study records a move **from SAP POS DM, first deployed in 2004, to SAP CAR with POS data transfer and audit**, with an implementation partner supplying an accelerator. **The migration was planned to take just under six months, ending November 2021, and went live in January 2022 — later than initially planned** — with more than thirteen interfaces to implement. And the article states that the team **pushed its internal resources too hard and had to revise its timelines** [43]. Separately, the vendor's newsroom records Pick n Pay deploying SAP CAR across more than 2,000 stores in South Africa plus five neighbouring countries, processing "more than 30 billion transactions across our stores per year" [44].

That is a vendor-published admission of internal-capacity-driven slippage on a six-month project, and it is worth more than any number of unblemished case studies. Two readings. **Thirteen interfaces is the real scope of a POS data-layer migration**, not the platform. And **internal capacity, not vendor capability, is the constraint that moved the date** — which is the thing a resource plan built from a vendor's estimate never accounts for.

The three deliverables that make everything afterwards possible

1. **A named authority per data concept** — article, cost, price, on-hand, member, patient — with the reconciliation for each written next to it.
2. **An interface inventory, with the direct-database-read consumers flagged.** That flag is the list of reasons the merchandising system cannot be replaced.
3. **A list of the spreadsheets that make commitments.**

None of the three is glamorous. Produce them in your first six weeks and you will know more about the estate than anyone who has worked in it for five years, because nobody has ever written them down in one place.

THE CHAPTER IN TEN LINES

- A retailer's unit of record is the article at a location, and almost every hard architecture problem here is an inventory position two systems disagree about, a price three systems disagree about, or an identifier that means something different in each system holding it.
- Ask who owns the store before you draw anything: in a voluntary-trading or heavily franchised estate — SPAR is the first, Pick n Pay runs 620 franchise stores alongside 992 company-owned — the group has no price, range or inventory position to master for much of its turnover.
- "Merchandising" is four systems in one box, and the department-level choice between the retail inventory method and the cost method is a data-model fork that decides what a markdown is and what gross margin means downstream.
- Store on-hand is structurally inaccurate, and whether the centre or the store is authoritative for it is the single most consequential unexamined assumption in a retail estate — it drives replenishment, online promising and the shrink number.
- A store that cannot take money is closed, so offline tender acceptance is the capability to test; offline pricing and offline journalling are the two most retailers actually have.
- Price is the most dangerous object in the estate because section 23(6)(b) of the Consumer Protection Act binds the retailer to the lower of two displayed prices — so the architecture must produce evidence of what was displayed, where and when, not just correctness.
- A South African pharmacy retailer cannot price medicines: the single exit price and a banded, gazetted dispensing fee bypass the pricing engine entirely, and in 2026 there were two adjustments rather than one.
- The integration estate is a sediment rather than a layer, and the two cheap disciplines are normalising identifiers at every boundary and engineering the reference-data path for confirmed delivery per till and per label.
- SPAR's SAP programme lost R1.6bn–R2bn of turnover against an implementation estimated at R1.8bn, so the lost sales exceeded the implementation cost in a single year — and the sources disagree on when it even went live.
- SPAR's own remediation line is the most transferable sentence in the chapter: separate the ERP from the warehouse system and implement them independently. Pick n Pay's vendor-published counterpoint is the honest one, because it names internal capacity as the reason the date moved.

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- **44.** SAP Africa newsroom, "Pick n Pay sparks retail revolution with powerful SAP deployment" (vendor-published case material) https://news.sap.com/africa/2023/04/pick-n-pay-sparks-retail-revolution-with-powerful-sap-deployment/
- **45.** Object Management Group, "About the Unified POS Specification version 1.15.1" (adopted December 2024) https://www.omg.org/spec/UPOS/1.15.1/About-UPOS
- **46.** Extenda Retail, "Providing speed and resiliency in point of sales" (vendor source) https://www.extendaretail.com/blog/pos/providing-speed-and-resiliency-in-point-of-sales/
- **47.** Independent Community Pharmacy Association v Clicks Group Ltd and Others (CCT 11/22) [2023] ZACC 10 (28 March 2023) https://www.saflii.org/za/cases/ZACC/2023/10.html
- **48.** South African Pharmacy Council, "Regulations relating to the Ownership and Licensing of Pharmacies" (2003) https://pharmcouncil.co.za/media/default/documents/Ownership_and_licencing_of_pharmacies_(2003).pdf
- **49.** Wikipedia, "Planogram" https://en.wikipedia.org/wiki/Planogram

- **50.** SPS Commerce, “EDI transactions and usage” (vendor source) https://www.spscommerce.com/blog/edi-documents/
- **51.** Stacksync, “EDIFACT message types” https://www.stacksync.com/edi/edifact

CHAPTER 13

The reference architecture

Banking has several maintained answers to "what are the words". Retail has one standards body that genuinely binds the estate, one whose data models have gone quiet while its device programme is thriving, a calendar, a process taxonomy and a set of vendor models that are marketing artefacts with real content inside them. There is no retail BIAN, and saying so is more useful than pretending.

What a reference architecture is actually for

An architect arriving in retail from banking expects to find a BIAN. There is nothing like it, and the useful response is to say so plainly rather than dress a vendor model up as an industry one.

That matters because a reference architecture earns its keep in only three ways, none of which is "a design". **It settles vocabulary before an argument starts** — when the trading director says "range", the supply chain director says "assortment" and the data team says "listing", a published taxonomy lets you point at a numbered thing instead of negotiating; the value is the numbering, not the content. **It bounds a scope you cannot see the edge of**, because "does the package cover this?" is answerable against a list and unanswerable against a conversation. And **it is an argument you do not have to win from first principles**: saying "the standard separates identification from the data carrier" ends a debate that would otherwise run three weeks.

Everything else is misuse, and three misuses are common enough to name.

WHAT A REFERENCE ARCHITECTURE IS FOR, AND WHAT IT IS NOT FOR

Use it to…	Because
Settle vocabulary	A numbered taxonomy ends a terminology argument in one sentence. The value is the numbering, not the content
Bound a scope	"Does the package cover this?" is answerable against a list and unanswerable against a conversation
Win a settled argument	"The standard separates identification from the carrier" saves three weeks of debate
Adopt as target state	A reference model is decomposed for comparability; a target state is decomposed for delivery. Adopting one inherits somebody else's boundaries
Treat a vendor's as neutral	The dominant misuse in retail, because the best models in the sector are vendors' and there is no industry alternative
Let it stand in for thinking	The tell is a capability map with no volumes, no latencies, no costs and no named systems. It decides nothing

If a model cannot be used to refuse a request, it is decoration.

Figure 13.1 Three honest uses and three misuses. The misuses are not careless — each is what happens when a model built for comparability is asked to do a job that needs commitment.

The first is **adopting one as a target state**. A reference model describes what *could* exist, decomposed for comparability; a target state describes what *will* exist, decomposed for delivery. They look the same on a slide. A programme that adopts a reference model as its target inherits a boundary set drawn by somebody who was not solving its problem.

The second is **using a vendor's model as though it were neutral**. In retail this is the dominant misuse, because the most detailed reference architectures in the sector are published by vendors, there being no industry body maintaining an equivalent.

The third, and the most expensive, is **letting the model stand in for thinking**. The tell is a capability map with no numbers on it: no volumes, no latencies, no cost, no ownership, no named system. It is complete, it is defensible, and it decides nothing. If a model cannot be used to refuse a request, it is decoration.

> **THE WORKED EXAMPLE**
>
> Where this chapter needs a concrete retailer it uses **Milkwood Retail Group** — roughly 420 supermarkets, a 180-store pharmacy chain, an apparel format, a cash-and-carry arm and an online business. **Milkwood is invented**, and nothing attributed to it is sourced.

ARTS, the NRF and the OMG Retail Domain Task Force

This section exists because commentary treats ARTS as live and it is not, and simultaneously treats UnifiedPOS as dead and it is not. **The asymmetry is the whole story**, and getting it right puts you ahead of most consultants in the room.

The lineage. ARTS — the Association for Retail Technology Standards — was founded in **1991** as a share group for retail chief information officers and **incorporated in 1993**. The **National Retail Federation acquired it in 1998**, and since **2017 the Object Management Group and the NRF have managed and developed the retail standards jointly, as the Retail Domain Task Force** [6][5]. So "ARTS" is a historical name, and writing "ARTS publishes…" in the present tense is wrong.

What the family contained, because the names turn up in legacy schema and old request-for-proposal boilerplate: the **ARTS Data Model**, whose one recoverable hard version marker is a conformance certification for the **4.01 Data Model dated 16 January 2006** [6]; the **Operational Data Model**, covering

store-level transactional data — inventory, pricing, sales reporting, workforce [5]; the **Data Warehouse Model**, complementing it and defining key performance indicators, cited at over 40 [5]; **ARTS XML**, formerly IXRetail, cited at eleven schemas in one place and over twenty message sets in another, a discrepancy worth noting and not reconciling [6][5]; **UnifiedPOS**, now OMG's [6]; plus **template requests for proposal** and **retail business process models** [6][5].

Now the part that settles the question. OMG's formal retail specification catalogue currently holds **five** items [1]:

Specification	Acronym	Version	Status	Published
Digital Receipt API	DRAPI	1.0	Formal	June 2025
Pedigree and Provenance Model and Notation	PPMN	1.0	Formal	September 2026
Unified POS RCSD	UPOS	1.16 beta	Beta	April 2020
ARTS XML Video Analytics	VIDEOANALYTICS	1.0	Formal	January 2019
Web Services for Point of Sale	WS-POS	1.3.1 beta 2	Beta	March 2019

The Operational Data Model and the Data Warehouse Model are not on that list. They are described on OMG's pages, with no version numbers and no dates [4][5], and they are absent from the formal catalogue [1].

OMG'S RETAIL WORK: ALIVE ON THE LEFT, DORMANT BELOW

Artefact	Status as at October 2026
UnifiedPOS 1.15.1	Formal, adopted December 2024, superseding 1.15. RAND terms. 2024 contributors: Aptos, Diebold Nixdorf, Digimarc, GS1, Microsoft, Toshiba. Not legacy
Digital Receipt API 1.0	Formal, June 2025. Non-assert terms. Copyright across six named firms plus GS1. Machine-readable files published
Live RFPs	Three in progress: digital receipt API, UnifiedPOS fiscal API, UnifiedPOS v2 model and printer API. Plus a point-release revision task force
Video analytics, WS-POS	In the formal catalogue, but January 2019 and March 2019 respectively, the latter still beta
PPMN 1.0	Formal, September 2026. Generic provenance modelling, filed under retail. No retailers among its copyright holders
Operational Data Model	Described on OMG pages. No version. No date. NOT in the formal specification catalogue. Availability unknown
Data Warehouse Model	Same: described, undated, absent from the catalogue. "Over 40 KPIs" is a reporting starter kit, not a design
ARTS Data Model	The only hard version marker anywhere is a conformance certification for 4.01, dated 16 January 2006

Do not tell a reader to go and get the ARTS ODM. Tell them to ask OMG whether it exists in usable form, and to record the answer with the date.

Figure 13.2 The asymmetry is the story. The device-and-receipt programme is genuinely active with real industrial participation; the data models are absent from the formal catalogue, undated where they are described, and of unknown availability. Most commentary gets both halves the wrong way round.

What is alive

Separately from the catalogue table, the current formal UnifiedPOS release is **1.15.1, adopted December 2024**, superseding 1.15, published with machine-readable schema files alongside the specification, under a **RAND** intellectual-property mode, with the 2024 contributing companies listed as **Aptos, Diebold Nixdorf Global Solutions B.V., Digimarc, GS1, Microsoft and Toshiba** [2]. **Do not call UnifiedPOS legacy.** It is the one piece of the ARTS inheritance that is demonstrably maintained, and that contributor list is a who's-who of the live store-systems market.

The **Digital Receipt API 1.0** became formal in **June 2025**, developed by a digital-receipt subcommittee under a contract project for a national ministry, with copyright notices from Aptos, Diebold Nixdorf, Fiscal Solutions, GK Software, GS1, Lexmark and Microsoft, under a non-assert intellectual-property mode [3]. The task force also lists three live requests for proposal — a digital receipt API, a UnifiedPOS fiscal API, and a UnifiedPOS version 2 model and printer API — plus a revision task force on a later point release [4].

That is a **narrow, device-and-receipt standards programme with real industrial participation**, and it gives an architect two usable questions. In any point-of-sale or self-checkout procurement, ask **which UnifiedPOS version the peripheral drivers implement**, because the answer decides whether swapping a scanner, scale or printer is a configuration change or a development project. And if digital receipts are on the roadmap, ask whether the vendor implements the formal specification or its own format, because a receipt is evidence and evidence outlives a vendor relationship. If you meet "OPOS drivers" in an older estate — OPOS being the Windows-specific ancestor UnifiedPOS generalised — the community maintains the common control objects publicly [8], and the right question is still which UPOS version the device claims.

What is dormant, and what to say about it

The data models are the part that did not survive into the formal catalogue. They are described but undated, the task force's mailing list is member-only, and **whether the Operational Data Model and the Data Warehouse Model are obtainable at all today — at what version, on what terms, and whether they have been revised since the ARTS era — could not be established** [1][4].

DO NOT SEND ANYONE TO "GET THE ARTS ODM"

That instruction appears in architecture documents and consultancy templates, and it may well send a colleague on a week-long hunt for a document that is not available. **The correct instruction is: ask OMG whether the Operational Data Model and the Data Warehouse Model exist in a usable form, at what version and on what terms — then record the answer in your decision log, with the date.**

If the answer is yes you have gained a documented data model and lost nothing. If it is no you have saved a week and have an evidenced reason to choose something else. Either way the question is cheap and the assumption is not.

Where the ARTS material is genuinely useful is archaeology, which is not a small thing. A retailer's data warehouse or point-of-sale log schema built between the late 1990s and the mid-2010s may well use ARTS-derived naming, and in that case the data model is the documentation nobody kept. **Where it is not useful**: not as a target data architecture, because the Operational Data Model's own description is a store-systems scope rather than an enterprise one, and "over 40 key performance indicators" is a reporting starter kit rather than a dimensional design; and not as an integration standard for anything but devices and receipts, because the XML message sets lost to GS1 EDI, X12 and bespoke flat files.

One entry deserves a caveat rather than enthusiasm. The newest formal retail specification, **PPMN 1.0**, published September 2026, describes a repository of elements capturing the lineage, custody and ownership of entities of interest; its copyright holders are research institutes, consultancies and a university, and **its contributor list contains no retailers** [7]. It is tagged retail and plainly relevant to traceability, but its stated purpose is generic provenance modelling. Characterise it as a general provenance standard filed under retail, not as a retail traceability standard.

Retail has no BIAN, and here is the evidence

Say it directly, because an architect who spends a month looking for the retail BIAN has wasted a month.

THE RETAIL BIAN DOES NOT EXIST — HERE IS WHAT DOES

You expect	Banking has	Retail has
A maintained service landscape	Service domains, versioned, with release notes	Nothing equivalent. The nearest body's data models are absent from its own catalogue
A control-record concept	An ownership semantic per domain	No equivalent. Process taxonomies are silent on ownership
A semantic API set	Published and certifiable	No equivalent
Certification	Against a maintained landscape	Peripheral conformance only, at the device layer
A neutral vocabulary	Several, overlapping	GS1 keys, a process taxonomy, and vendor models

Why: retail regulation does not compel a shared vocabulary, format variance exceeds banking's segment variance, and the vendors had no incentive to fund comparability.

Figure 13.3 What a banking architect expects to find, and what is actually there. The right response is to say so plainly: in retail you choose a vocabulary rather than inherit one, and that choice should be an explicit decision rather than a default.

There is no retail equivalent of a maintained service-domain landscape, no retail control-record concept, no retail semantic application-programming-interface set, no certification scheme, and no retail body publishing release notes with a metrics table. The only industry-body candidate is the ARTS lineage, and the previous section establishes what that is now [1][4]. Commercial retail business reference architectures do exist from research firms [10] — but those are **subscription research products, not industry standards**, their contents and licence terms are not public, and they should be characterised as such.

Why retail has no BIAN is instructive rather than an excuse, and this is analysis rather than sourced fact. **Retail's regulatory surface does not force a common vocabulary**: banking's shared models exist because payments, prudential and reporting regulation compel banks to describe the same things the same way, whereas retail's binding constraints are product identification, card security and consumer law, none of which requires a shared capability decomposition. **Retail's variance by format exceeds banking's variance by segment** — a grocer, a pharmacy chain, an apparel vertical and a building-materials big-box share a checkout and very little else. And **retail software consolidated into a few large vendors earlier**, and those vendors had no incentive to fund a neutral model that would make their suites comparable, which is precisely what BIAN does to banking suites and precisely why banks funded it.

So the formulation transfers from banking with one clause changed. **TOGAF tells you how to do architecture; ArchiMate is how you draw it; and in retail there is no industry body telling you what the things are called, so you choose.** That is the most consequential early decision in a retail architecture practice: adopt a vendor's vocabulary and get depth at the price of a taxonomy encoding that vendor's module boundaries; adopt a process taxonomy and get neutrality at the price of no data-ownership semantics; or build your own and get fit at the price of months of elapsed time. My own position, as practitioner judgement rather than finding: **build a thin, stable, two-level capability model in-house — forty to sixty capabilities, not four hundred — and map vendor models onto it rather than adopting them.** It then survives vendor changes, which in retail arrive every few years.

ONE ARCHIMATE NOTE SPECIFIC TO RETAIL

Retail's heavy use of **physical** concepts — stores, distribution centres, lanes, scanners, shelf edges, labels, pallets — means the technology and physical layers earn their keep here in a way they rarely do in banking. A retail landscape model showing only business and application layers has hidden the half of the estate that actually breaks: the device estate, the in-store network, the edge compute at the lane, and the dependency of a whole store on a scanner driver version. Draw the devices.

GS1 — the only standards body that genuinely binds a retailer

If this book teaches one standards body properly it must be this one, because **GS1 identifiers are the primary keys of the retail data estate** and almost every integration failure, inventory discrepancy and reporting dispute in a retailer traces to someone treating them as display strings.

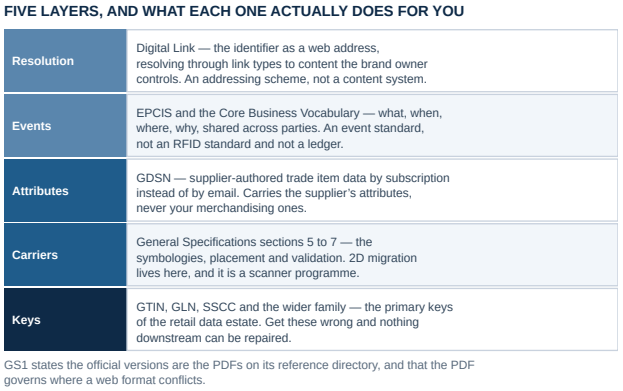


Figure 13.4 The GS1 stack, bottom-up. Only the bottom two layers are genuinely binding on a retailer. Everything above them is optional, valuable and dependent on getting the bottom two right first.

The keys

The ones to be able to name and distinguish are **GTIN** for a trade item, **GLN** for a party or a location, and **SSCC** for a specific logistic unit — this pallet, this carton. A wider family covers assets, service relationships, documents, consignments, shipments, coupons and models, all defined in section 1 of the General Specifications [9]. Four architectural points, each a real failure mode.

A GTIN identifies a trade item at a given packaging level, not a product. A consumer unit, an inner pack and a case are three different numbers with a hierarchy between them. Model "product" as one row and lose the hierarchy, and you cannot reconcile a distribution-centre receipt against a store sale — the commonest single cause of inventory variance. **A GLN identifies a party or a location, and retailers routinely conflate the two**: store, distribution centre, legal entity, ship-to, bill-to and remit-to are distinct, and trading partners discover this at the first invoice dispute. **An SSCC identifies one physical logistic unit once**, and reuse is the defect that makes receiving automation impossible. And **application identifiers are the grammar, not an afterthought** — section 3 defines the identifier set and section 4 the application rules [9], batch, expiry, serial and weight arrive as prefixed elements inside a single carrier, and a till or warehouse system that parses the trade item number and discards the rest throws away exactly the traceability data the business is about to be asked for.

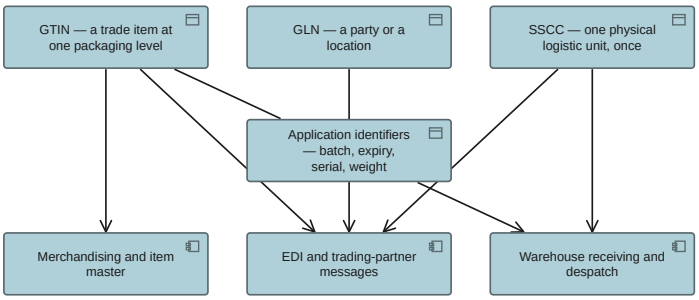


Figure 13.5 The three keys an architect must be able to separate, and who needs each. The failure mode in each case is the same shape: treating a key as a display string rather than as the primary key it is.

The General Specifications

GS1's foundational standard for how identification keys, data attributes and barcodes are used. The current release is **26.0, ratified January 2026**, in nine sections: identification, application standards, application identifier definitions, application rules, data carriers, barcode placement, validation rules, application standard profiles, and a glossary [9]. Change is governed through numbered change notifications, each release carrying a log naming them and the sections they affect; release 26.0's log includes radio-frequency and near-field communication as additional data carriers [9]. One governance detail is a real instruction: GS1 states that the official versions are the **PDFs on its online reference directory**, that web representations are for convenience, and that **if a web format conflicts with the PDF, the PDF governs** [9]. So a retailer's internal standard must cite the PDF release, not a page. And it is an identification and data-carrier standard, not a product information model — it has no opinion on assortment, merchandising hierarchy or attribute governance.

The cost is in the identifiers rather than the document. GS1 South Africa publishes a tiered fee table in rand with a once-off initial fee and, above the single-item level, an annual renewal due in March: a single trade item number at R176 with no renewal; ten at R1,964 then R433 a year; a hundred at R6,823 then R2,409; a thousand at R8,215 then R4,709; ten thousand at R9,051 then R9,628; and a hundred thousand at R9,888 then R28,878 [16]. Three readings. **The curve is flat on the initial fee and steep on renewal**, so prefix capacity is a long-term commitment rather than a procurement line. **A single number costs R176 with no renewal**, which is why small suppliers buy single numbers and why a retailer's item master fills with identifiers

belonging to no coherent company prefix. And **location numbers are priced identically to item numbers** [16], so there is no economic reason not to assign them to every location properly — and most retailers still have not.

GDSN, EPCIS and Digital Link

The Global Data Synchronisation Network is the mechanism by which a supplier publishes trade item master data once into a certified data pool and retailers subscribe to it, with a global registry as the directory. Its business problem is familiar: a grocery retailer's item master is assembled from supplier emails, and every attribute is entered twice and disagrees. It moves that to a subscription. Two cautions. **GS1's own standards listing shows a maintenance release dated November 2021** with later versions shown as future or legacy [10] — four years stale, on a page that does not describe data pools, the registry, certification or access terms, so **the current release, the number of certified pools and the joining terms could not be established**. And the deeper caution is architectural: the network carries supplier-authored *trade item* data, not the retailer's merchandising attributes, hierarchy, cost structure, ranging decisions or selling attributes. **A retailer treating it as its product information system has outsourced its item master to its suppliers' data quality.**

EPCIS is the standard for creating and sharing **visibility event data** across enterprises — a shared view of physical or digital objects in a business context [11]. Its naming is a trap that costs retailers real capability: "EPC" is historical, and **EPCIS requires neither electronic product codes nor radio-frequency identification, and since version 1.1 does not require instance-level identification** [11]. Architects reject it because they think it is a radio-frequency standard. It is an event standard. Ratifications run 1.0 in April 2007, the companion **Core Business Vocabulary** at 1.0 in October 2010, 1.1 of both in May 2014, 1.2 of both in September 2016, and **2.0 of both in June 2022** [11]. That companion supplies the controlled values — business steps, dispositions, transaction types — without which two implementations exchange well-formed events that mean nothing to each other. Its live retail uses are traceability and recall, cold chain, controlled-substance pedigree, and node-to-node visibility. **Where not to use it**: not as a message standard for commercial documents, which is EDI's job, and not as an inventory system — it records that an event happened and does not hold a balance.

GS1 Digital Link is a standardised way to encode identifiers as web addresses, so one scan both identifies the item and resolves to online information [12]. An application either constructs the address itself or queries a conformant resolver, which returns the **link types** associated with that item — and the link type is the mechanism that matters, because the same scan can resolve to product information, instructions for use, a certificate, a recall notice or a traceability record, and **the brand owner can change what sits behind a link type without changing the code on the pack** [12].

> **FOUR VERSION STRINGS ARE IN PLAY AT ONCE**
>
> GS1 now publishes Digital Link as component standards: **URI Syntax 1.5.0, June 2024**, and **GS1-Conformant Resolver 1.0.0, February 2024**, with **version 1.1.3** applying to anything the two components do not cover [12]. GS1's reference directory additionally shows a **legacy Digital Link standard version 1.1.4, published January 2026**, and dates the original ratification to August 2018 [13].
>
> So a document citing "GS1 Digital Link version X" without naming the component is imprecise, and two people can both be right and mean different things. **Name the component.** The access terms, licensing and resolver conformance arrangements are not stated on GS1's own pages and could not be established here.

GS1 itself draws a sector split that runs straight through a pharmacy retailer. In **retail**, the goal is one multipurpose two-dimensional barcode per product that works at the till *and* gives access to digital content. In **healthcare**, products keep their existing linear or data-matrix code and scanning applications either build the address themselves or use a resolver, so **the address is not encoded in a two-dimensional barcode in that sector** [12]. For Milkwood's 180 pharmacy stores that means the front shop follows one model and the dispensary another, inside one building, with one item master. And where not to use it: it is an addressing scheme, not a content management system. The content behind the link, its governance, its localisation and its accuracy are the retailer's or brand owner's problem, and a programme without a content owner produces scannable codes that resolve to nothing.

The two-dimensional migration, and Sunrise 2027 precisely

This is the single most consequential standards change in retail, and also the one most often overstated.

Where it came from. GS1 US issued a press release on **12 January 2022** describing retail moving from the twelve-digit code to two-dimensional barcodes on packaging by 2027, calling the timeline one set by the retail industry in collaboration with GS1 US. It records a test kit piloted with three unnamed national retailers and collaboration with OMG, states that guidance will be issued on **both** dual marking and single marking, and **does not say which one is required by 2027** [15].

What GS1 US's own current page says. A global transition from one-dimensional to two-dimensional codes such as QR codes and data-matrix codes, with the focus on making sure point-of-sale systems can read them. It calls 2027 a major milestone and **gives no specific day or month**. The model is a voluntary crawl-walk-run: the retailer's first phase is to install scanners capable of reading a two-dimensional code and extracting the trade item number at the till, and later retailer phases are explicitly optional. And the sentence that is the honest centre of the topic: **"it is not necessary to implement all the capabilities of 2D by that date"** [14].

WHAT GETS SAID, AND WHAT THE SOURCE ACTUALLY SAYS

Commonly said	GS1 US's own page
"GS1 requires 2D by 2027"	No enforcement mechanism appears in GS1's own material. It is an industry capability ambition
"The deadline is 1 January 2027"	The page calls 2027 a major milestone and gives no specific day or month
"Everything has to be in place"	"It is not necessary to implement all the capabilities of 2D by that date"
"It is a packaging programme"	The retailer's phase one is scanners that read 2D and extract the GTIN at the till. Later phases are explicitly optional
"It applies globally"	The retrievable material is GS1 US. The global and South African positions were not established
What is actually true	The carrier is changing, the standards infrastructure is in place, and the binding constraint is the scanner and POS estate

A reader who tells a steering committee "GS1 requires 2D by 2027" will be contradicted by GS1's own page, and will lose the room.

Figure 13.6 Sunrise 2027, precisely. The left column is what circulates in steering-committee papers; the right is what GS1 US's own page says. The sourced material is GS1 US — the global and South African positions could not be established and should not be assumed from it.

> **NEVER WRITE THAT GS1 REQUIRES 2D BY 2027**
>
> There is no enforcement mechanism in GS1's own material, no stated day or month, an explicitly voluntary phasing model, and an explicit statement that full capability is not required by the date [14]. **A reader who tells a steering committee "GS1 requires 2D by 2027" will be contradicted by GS1's own page**, and will lose the room for everything else they say that day.
>
> Two further precisions. The sourced material is **GS1 US**; the global position and the South African position could not be established and should not be assumed from the North American timetable — settle them with GS1 South Africa directly. And "Sunrise 2027" is an **industry capability ambition**, which is a real and useful thing to plan against, not a mandate with a legal date.

Three things are true at once. The direction is not in doubt: the carrier is changing from a twelve- or thirteen-digit number to a structure that can carry batch, expiry, serial and a web address, the General Specifications have already added further carrier types [9], and the Digital Link components were re-released in 2024 [12]. The date is soft. And **the binding constraint is not packaging — it is the scanner and point-of-sale estate**, which is where the work sits. That phase-one ask is specific and testable: can every lane, self-checkout, handheld and receiving scanner read a two-dimensional symbol and extract a trade item number, and does every downstream system tolerate the additional prefixed elements arriving with it? That is a hardware refresh cycle, a driver and UnifiedPOS version question [2], an application change, a schema change, and a set of exception behaviours for codes that resolve to nothing — bounded, in a large grocery estate, by scanner replacement economics.

And now the part that outlives the date, which is why this section is here at all.

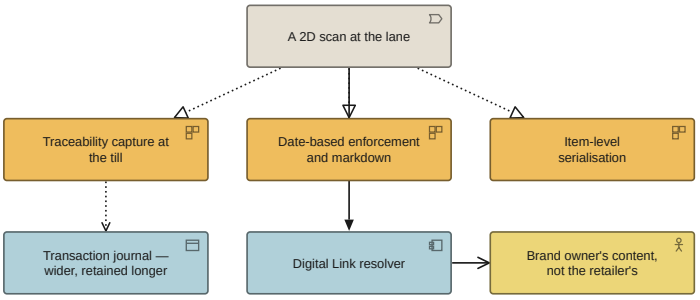


Figure 13.7 The four consequences that outlive the date. The last one is the one nobody has planned for: because the link resolves to the brand owner's content and the brand owner can change it without changing the pack, the scan in your aisle is a touchpoint you do not control.

The point of sale becomes a traceability capture point. When batch and expiry arrive in the scan, the transaction log can hold them — changing what the journal stores, how long it is retained and what a recall costs to execute. A retailer that can answer "which baskets contained batch 2719" has a materially different recall from one that can answer "which stores received it".

Date-based enforcement at the lane becomes possible. If the code carries an expiry date the till can refuse an out-of-date sale or apply a date-driven markdown without a separate reduced-to-clear label. That is a change to the pricing and promotions engine, not to the scanner.

Item-level serialisation becomes possible in some categories, changing returns, warranty and loss prevention — and introducing a volume problem the data platform has not been sized for.

And the one nobody has thought about: the scan becomes a consumer touchpoint the retailer does not control. Because Digital Link resolves to the brand owner's content, and the brand owner can change what sits behind a link type without changing the pack [12], a shopper standing in Milkwood's aisle can scan a product and be taken to content the brand owner chose — potentially an offer, a direct-to-consumer channel, or a comparison the retailer

would rather they did not see. That is a governance question with commercial teeth, and no retailer has an answer yet because the question has not arrived in the aisle.

EDI, and which messages actually matter

The point of this section is to kill one illusion: that a modern retailer's supplier integration is an application-programming-interface estate. It is not. It is EDI, it is decades old, and the parts that are not EDI are flat files.

Two grammars. GS1's EDI family comprises **EANCOM**, a fully compliant subset of the United Nations EDIFACT standard and the European and Commonwealth retail grammar a South African retailer will actually meet; **GS1 XML**; and **GS1 UN/XML**, profiles of four cross-industry standards for order, order response, invoice and despatch advice [17]. Maintenance releases are typically published every two years and code lists can be updated up to four times a year [17] — an architectural fact in itself, because the translator carries a version matrix and the code-list cycle is four times the message-structure cycle. The current EANCOM edition in general retail use could not be established here.

The other grammar is ANSI X12, and its licensing posture is the practically important fact about it. X12 states that it manages the exclusive copyright to all its standards, publications and products, that all its work products are copyrighted, and that access runs through a licensing programme with tiered commercial, internal and developer categories; the standard is published once a year in January [18]. **A retailer or integrator embedding implementation guides in its own documentation, or shipping segment definitions inside a product, needs a licence.** And the versions in common retail practice are far older than the current release — the standard moves annually and the market does not.

SEVEN MESSAGES, AND WHAT BREAKS WITHOUT EACH

Message	What it carries	What breaks without it
ORDERS / 850	The purchase order	Nothing. Everyone implements this one
ORDRSP / 855	The supplier's commitment	Your expected-receipt position is your own order, not theirs. Availability is lost in that gap
DESADV / 856	The advance ship notice, keyed on logistic unit codes	No cross-dock, no scan-receive. Receipt becomes a line count. One with a bad code hierarchy is worse than none
INVOIC / 810	The invoice	Three-way match fails. Mismatches are almost always unit-of-measure or packaging-level errors
INVRPT, SLSRPT / 852	Your sales and stock, sent to the supplier	No vendor-managed inventory — but note this discloses trading data to a party that also supplies competitors
PRICAT / 832	The supplier's item and price catalogue	Article onboarding reverts to spreadsheets. Running this and GDSN gives two masters for one attribute set
997 and 824	Envelope acknowledgment, then content acceptance	You cannot tell a lost message from a rejected one — the commonest EDI operational failure

The integration backlog in a retailer is measured in trading partners, not in interfaces: the cost is the n × m mapping estate, not the standard.

Figure 13.8 The messages that actually matter. The EDIFACT and X12 names are given as practitioner mapping rather than sourced equivalence — only the invoice, the application advice and the functional acknowledgment titles were confirmed against X12's own listing.

Two messages deserve emphasis. **The despatch advice is the single highest-value message in retail EDI**, because it determines whether a distribution centre can cross-dock, scan-receive against a logistic unit code and put away without counting — and one that does not carry a valid code hierarchy matching the physical pallet labels is *worse* than none, because receiving staff stop trusting the system. And **the product activity message reverses the direction of the relationship**: the retailer sends the supplier its own sales and inventory position, which is what makes vendor-managed inventory and collaborative replenishment possible, and which is a disclosure of trading data to a party that also supplies competitors. Treat it as a commercial decision, not an integration pattern.

And underneath all of it, a working retailer runs on flat files on a file-transfer server; EDI brokered through a service provider rather than direct connections, so the message logs live with a third party; spreadsheets as the supplier onboarding and new-line interface, including for suppliers large enough to know better; and **a translator with a mapping per trading partner**, because every retailer publishes its own implementation guide and no two agree on optional segments. That last one is the actual reason retail EDI is expensive: not the standard, the $n \times m$ mapping estate. So **the integration backlog in a retailer is measured in trading partners, not interfaces.** Replacing a merchandising system does not mean rebuilding twelve interfaces. It means re-certifying several hundred trading-partner maps.

The retail calendar as an architecture artefact

This is the section that most repays a sceptical reader's attention, because it looks like a finance topic and is not.

What it is. The National Retail Federation publishes the **4-5-4 calendar** as a guide ensuring sales comparability between years by dividing the year into months on a four-week, five-week, four-week pattern, so the same number of Saturdays and Sundays fall in comparable months and holidays line up. The federation calls it **a voluntary guide for the retail industry**, derived in the 1930s from informal inter-industry discussion and in use among stores from the 1940s [19]. A 52-week year is 364 days, so a **53rd week is added roughly every five or six years**; the rule is to add it if four or more days remain in January after the 52nd week, and the federation cites 2006, 2012, 2017 and 2023 as 53-week years. For comparability a 53-week year is restated the following year by shifting each week back one and ignoring the first week of the fiscal year; the alternative is to leave it unrestated and ignore the 53rd week [19]. The calendars are downloadable with no stated fee and no login requirement [19].



Figure 13.9 The retail calendar as a schema object. The 53rd week is not a reporting event — it shifts every seasonal profile by a week and breaks the week-to-week comparison key, which is why the restatement convention belongs in the time dimension rather than in a report.

Why that is architecture. Because the retail calendar is **a dimension in every data model in the business, a partition key in every large fact table, and a parameter in every planning and forecasting engine** — and because the 53rd week is a schema event, not a reporting event.

- **Every time dimension must carry the retail calendar alongside the Gregorian one**, with fiscal week, period, quarter, year, and the week-over-week and year-over-year comparison keys. A warehouse with only a Gregorian date dimension cannot answer the only question retail asks, which is "versus last year, like for like".
- **Like-for-like depends on the comparison key, not on arithmetic.** When the 53rd week appears the mapping from this year's week to last year's breaks, and whichever restatement convention the business chose must be encoded **in the dimension, not applied in a report**. Encoding it in reports guarantees two reports disagree.
- **Forecasting and replenishment engines are calibrated on fiscal weeks.** A 53-week year shifts every seasonal profile by a week. Engines taking the calendar as configuration survive this; ones assuming 52 weeks do not.
- **It is a cross-system contract.** Merchandising, planning, point-of-sale reporting, finance and the warehouse must all agree what week one is. In an estate assembled from a global enterprise system, a local till application and a third-party planning tool they frequently do not, and the resulting one-week offset is extremely hard to find because every individual system is internally consistent.

> **DO NOT ASSUME THE FEDERATION'S DATES APPLY IN SOUTH AFRICA**
>
> Whether the 4-5-4 calendar or a local variant is the convention among South African retailers, and how it interacts with a June, August or February year end, **could not be established**; it is settled by each retailer's own annual report, which states its period definition.
>
> Practitioner expectation, presented as expectation: South African retailers commonly run a 4-4-5, 4-5-4 or 52/53-week calendar aligned to their own year end rather than to the federation's February start. **Teach the pattern; do not assert the dates.** And the calendar is not a statutory reporting calendar and does not align to value-added-tax periods or payroll — so a retailer runs two calendars and must reconcile them. A design pretending there is one has pushed the reconciliation into spreadsheets.

APQC's Retail Process Classification Framework

The nearest thing retail has to a neutral process taxonomy. APQC maintains a cross-industry process classification framework and a set of industry variants covering twenty or so sectors including retail, consumer products, banking, healthcare provider and property and casualty insurance [20]. The retail variant documented in its spreadsheet release at **version 6.1.1, published 28 April 2023**, describes **twelve enterprise-level categories including process groups and over a thousand processes and associated activities**, listed as public content with no stated price or membership requirement on that page — while also pointing to a newer version elsewhere [21].

Two cautions, both real. APQC's industry index shows several other sectors at version 8.0 with 2026 dates while showing no version against retail [20], and **the current retail version and date could not be established** — so do not print a version number without re-checking it. And **APQC's licensing terms for non-member commercial use could not be established** either [20]. The framework is widely used inside consulting deliverables, and the licence question is the one most often not asked.

What it is good for: a process inventory a retailer does not have, faster than any workshop; a neutral vocabulary for scoping; and benchmarking, which is APQC's actual business and the reason the taxonomy is numbered. **Where not to use it**: it is a *process* classification, not an application architecture, not a capability model in the TOGAF sense and emphatically not a data model; it has no ownership semantics, so unlike a banking control record nothing in it says who owns the data; and there is **no pharmacy framework**, the nearest options being healthcare provider and life sciences [20], so a retail-pharmacy business must stitch two taxonomies together itself.

PCI DSS and EMV — contractual, not statutory

Two conformance obligations that bind the estate rather than the industry, and both are commonly misdescribed.

The Payment Card Industry Data Security Standard is published by a council formed by the card schemes. It is **a contractual obligation flowing from the merchant agreement and the scheme rules, not a statute**. That distinction matters in a room: nobody is prosecuted for non-compliance, but an acquirer can impose assessments, raise fees, require a forensic investigation or decline to continue acquiring — and for a retailer that is a trading risk rather than a legal one. The council's document library features **version 4.0.1** as current [22].

STATE NO VERSION DATE YOU CANNOT SOURCE

A detailed public timeline circulates for this standard — an errata release month, a retirement date for the previous version, and dates on which future-dated requirements became mandatory. **The research behind this chapter could not verify any of those dates against the council's own site**: the standards page gives none of them and the document library table did not render a publication date [22][23]. So this chapter states the current version and nothing else, and an architect should do the same — re-verify against the council's own document library and bulletins before putting a date in a board paper, and put the date you checked beside it.

On EMV, state nothing at all. No primary source was retrieved in this research round, so **no specification version, book number or mandate date appears in this book.** Settle them from EMVCo directly.

What *can* be said safely is the part that changes a design. **Card-security scope is decided by the architecture, not by the security team.** Three decisions move the burden by an order of magnitude: **whether the till ever sees a card number** — point-to-point encryption and a pin pad that never exposes card data to the application collapse scope dramatically; **whether the store network is segmented** from the corporate network and from guest wireless; and **where tokenisation happens and who holds the token vault**, because the vault is the exit cost when the acquirer or gateway changes. Note too that the requirements concerning payment page scripts and page-change detection land on the **e-commerce team, not the store team**, a boundary retailers routinely get wrong.

And the safe architectural statement about the card-present conformance regime is a planning one. It is scheme-mandated with deadlines the retailer does not control, and its consequence is that **the pin pad and its kernel are certified assets with their own release cycle, independent of the point-of-sale application**. So a point-of-sale upgrade and a terminal certification are two programmes with two approval bodies, and conflating them is a classic retail planning error.

Vendor reference architectures, and how to use one without being captured

The honest framing: **in retail, the most detailed and most usable reference architectures are published by vendors**, because no industry body maintains an equivalent. That is both a real asset and a real trap.

Oracle lists, alongside its products, three artefacts that are explicitly architectural rather than functional — a retail reference model, a retail data store and a retail intelligence foundation [24]. The reference model is the most substantial vendor reference architecture in the sector, and **its contents, format, version and availability could not be established**, so this book describes none of them.

SAP's retail industry page names its business suite for retail and a fashion management product alongside functional descriptions, and three further named items presented in the same visual frame [25]. One of those three is a pricing platform belonging to a separate company, which SAP's own partner pages confirm [26].

A VENDOR'S INDUSTRY PAGE IS A MARKETPLACE, NOT A PRODUCT LIST

That is the lesson, and it is not about one vendor's honesty. **An industry page is constructed as a partner ecosystem catalogue with the vendor's own products mixed in**, at the same visual weight, in the same typeface, under the same logo. At least one product on a major vendor's retail page belongs to a third party [25][26], and the owners of two others could not be established.

So **check every name on a vendor industry page against that vendor's own partner listing before you put it in a landscape document.** The commonest source of mis-attribution in a retail estate is a slide built from an industry page.

Microsoft is a named contributing company on the current UnifiedPOS specification [2] and a copyright holder on the formal digital receipt specification [3], which is genuine sourced evidence of active retail standards participation. But **no current Microsoft retail reference architecture artefact could be verified**, so this book names none. If somebody hands you one, check its date and status before building on it.

A fourth source is local: in the South African market one large integrator publishes an intelligent-retail-enterprise capability model with six named areas and a customer-to-checkout framework [27]. Treat an integrator's model exactly as a vendor's — useful taxonomy, self-interested boundaries, a bill of materials at the bottom — and note that this one sits inside a listed telecommunications group [27], which is relevant when assessing independence.

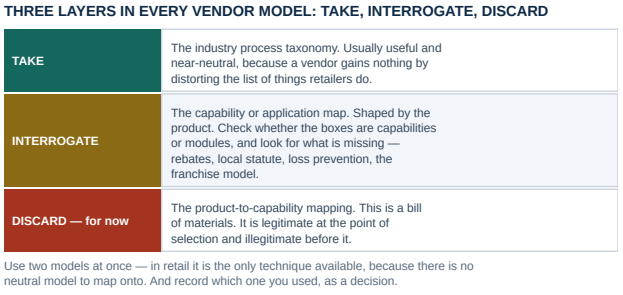


Figure 13.10 The anatomy of a vendor reference architecture, and what to do with each layer. The test for the middle layer is whether any capability spans two products or is split across two — if none does, it is a price list in architecture notation.

The method, offered as method rather than fact. Separate the three layers: a vendor reference architecture holds an industry process taxonomy, a capability or application map, and a product-to-capability mapping; the first is usually useful and near-neutral, the second is shaped by the product, the third is a bill of materials — so take the first, interrogate the second, and discard the third until after selection. **Test whether the capability boundaries**

are the module boundaries: if every box maps one-to-one to a licensable module it is a price list in architecture notation, and the tell is that no capability spans two products and none is split across two. **Look for what is missing**, because a vendor omits what it does not sell — in retail the recurring omissions are supplier income and rebates, local statutory and fiscal requirements, loss prevention, and the whole franchise or buying-group operating model, which in a market where SPAR is voluntary-trading and Pick n Pay runs hundreds of franchise stores is not a minor gap. **Use two at once**, because mapping one retailer onto two vendors' models is the cheapest way to find the real boundaries, and in retail it is the *only* technique available since there is no neutral model to map onto. And **record which one you used, as an architecture decision**, because the vocabulary outlives the vendor relationship and the next programme inherits it without knowing where it came from.

How to choose, and how to tell when a model is standing in for thinking

Put the chapter together and the synthesis is short. **GS1 gives you the keys and the carriers** — the trade item, location and logistic-unit numbers, the General Specifications, the carrier, the synchronisation network for supplier-authored attributes, the event standard and the resolution scheme. It is the only layer that is genuinely binding, genuinely maintained and genuinely external to your vendors, so get it right first because everything downstream inherits it. **EDI gives you the commercial grammar**, with a flat-file substrate underneath, and that is where the integration cost lives. **The retail calendar gives you time**, as a dimension and a cross-system contract rather than a finance convention. **The ARTS inheritance gives you the device layer and your own archaeology** — a maintained peripheral standard, a formal receipt specification, and a data model that is documentation for schema you inherited rather than a target. **APQC gives you a process inventory**: numbered, neutral, benchmarkable and silent on ownership. **Vendor reference models give you depth at the price of their boundaries**, so use two. **Card security and the scheme conformance regime give you non-negotiable constraints** the architecture can make cheap or expensive by where it puts data. **And there is no BIAN**, so you will choose a vocabulary rather than inherit one.

That leaves the judgement, and it is worth stating as a test rather than a principle. **A reference architecture is standing in for thinking when it cannot be used to refuse anything.** Three symptoms, in increasing order of expense. **The model has no numbers on it** — no volumes, no latencies, no transaction counts, no costs; a capability map of a retailer that does not say the transaction log is millions of lines a day is describing a different retailer. **Nobody can name the system behind a box**: if "Demand Planning" maps to no application, no owner and no licence, it is a wish. And **every box is green**, which means the model was produced to be approved rather than used.

The counter-test is simpler. Take the model into a real decision — a vendor shortlist, a programme scope, an integration dispute — and see whether it changes the answer. If it does, it is architecture. If the decision would have gone the same way without it, you have a diagram.

And the sentence to carry out of this chapter, because it survives every vendor change and every standards revision: **the architecture decision that outlives everything else in retail is the one about identifiers and the calendar.** Get the trade item number, the location number, the logistic unit code and the retail week right, and a retailer can change merchandising vendor, point-of-sale vendor, planning vendor and data platform in turn without losing the ability to say what it sold, where, to whom, against last year. Get them wrong and every system downstream inherits the error, no package fixes it, and no amount of money spent on merchandising software will make the numbers agree.

THE CHAPTER IN TEN LINES

- A reference architecture settles vocabulary, bounds a scope and ends a settled argument. It is not a target state, it is not neutral when a vendor published it, and if it cannot be used to refuse a request it is decoration.
- The ARTS asymmetry is the story: the device-and-receipt programme is alive — UnifiedPOS 1.15.1 adopted December 2024 under RAND with Aptos, Diebold Nixdorf, Digimarc, GS1, Microsoft and Toshiba contributing, and a formal Digital Receipt API from June 2025 — so do not call UnifiedPOS legacy.
- The ARTS Operational Data Model and Data Warehouse Model are absent from OMG's formal retail catalogue, undated where they are described, and of unknown availability. Do not send anyone to get them; tell them to ask OMG and record the answer.
- Retail has no BIAN: no maintained service landscape, no control-record concept, no semantic API set and no certification scheme. You choose a vocabulary rather than inherit one, so make the choice explicitly.
- GS1 is the only layer that genuinely binds — GTIN, GLN and SSCC are the primary keys of the retail data estate, and the General Specifications' official version is the PDF on the reference directory, which governs where a web format conflicts.
- Sunrise 2027 is an industry capability ambition, not a mandate with a legal date: GS1 US's own page gives no day or month and says it is not necessary to implement all 2D capabilities by that date. Never write that GS1 requires 2D by 2027, and note the source is GS1 US rather than global.
- The durable 2D content is architectural — the point of sale becomes a traceability capture point, date-based enforcement at the lane becomes possible, and the scan becomes a consumer touchpoint the retailer does not control because the Digital Link resolves to the brand owner's content.
- Four Digital Link version strings are in play at once, so naming a version without naming the component is imprecise; and X12's licensing posture means embedding implementation guides in your own documentation needs a licence.
- The 4-5-4 calendar is an architecture artefact rather than a finance convention: the 53rd week is a schema event, and the restatement convention belongs in the time dimension rather than in a report.
- Card security is contractual rather than statutory and its scope is decided by the architecture — state no version date you cannot source, and state no EMV version, book number or mandate date at all.

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CHAPTER 14

Technology providers

Somebody will ask you which merchandising system to buy before you know enough to answer. This chapter will not tell you. It teaches what kind of thing each vendor sells, which attributions the market routinely gets wrong, and why in South Africa the honest answer to "what does this retailer run" is usually that nobody has published it.

Read this before you read anything else

In your first month at a retailer somebody will ask which merchandising system to buy, or which point-of-sale platform. You will not know; nobody three weeks into the industry does. What you *can* learn quickly is what **kind** of thing each vendor sells, which question that category exists to answer, and what it costs architecturally to say yes.

> FRAMING, AND IT IS NOT NEGOTIABLE
>
> **Every company named in this chapter is an example of who plays in a category. Nothing here is a recommendation. Nothing is an endorsement. It is not a shortlist and it must not be used as one.**
>
> Inclusion means a vendor was recognisable enough to illustrate a category *and* had a source behind the claim made about it. Omission means nothing — several of the strongest products in retail are missing because no source could be found for anything useful to say about them. There are no rankings, scores or quadrants. In one category below, **two vendors each claim on their own sites to be number one in overlapping markets**, which is exactly why this chapter ranks nothing and attributes everything.
>
> Where a vendor's own site is the source, scale claims are that vendor's claims, dated and attributed. Treat them as data about market *perception*, not as measurement.

The research is dated October 2026, and here is what rots first

Knowing which parts of this chapter to distrust first is more useful than the chapter itself.

SHELF LIFE · RESEARCH DATED OCTOBER 2026

What	Why, and the date that moved it
The 2D barcode timetable	Roughly fourteen months out, and the definition of what is required keeps being softened in guidance rather than hardened
Retail software ownership	Six ownership or naming changes in six categories in about two years. That is the tempo, not the exception
The dormant retail data models	Absent from the formal catalogue today, undated where described. Could resolve either way
Card-security versions	Version and retirement dates move, and could not be verified against the council's own site
The SPAR litigation	A franchisee claim lodged 28 January 2026. Live, and the figures will move
SA acquirer consolidation	The consolidating merchant-services group is still buying. Reads differently in a year

What does not rot: the twenty-six categories, and the question each exists to answer.

Figure 14.1 What goes stale first. Distrust this chapter from the top of the list down, and put the date you re-checked next to every ownership and product name that reaches a board paper.

Ownership rots fastest, and it is an architectural attribute rather than a trivium. A platform under private-equity margin pressure behaves differently in a roadmap conversation than a founder-led, listed or conglomerate-owned one. A platform owned by a hardware company has a reason to bundle hardware with software. Record the owner in your decision record, with the date you checked it.

And in retail software the tempo is faster than in banking. In roughly the two years this research covers, **six ownership or naming changes landed in six different categories**: a space and planning vendor's rebrand, an electronic-shelf-label company's second name change in two years, a workforce vendor renamed and then acquired, a data-integration vendor absorbed by a customer-relationship platform, a supply-chain network taken by a logistics software group, and a store-systems vendor delisted under a new majority owner. That is the rate to expect, not the exception.

What does not rot is the shape of the problem. The twenty-six categories below have been recognisably the same for twenty years, and the question each exists to answer has not changed at all.

The corrections — get these right, because the market gets them wrong

Each was checked against the vendor's or body's own material. **Getting one wrong in a board paper is worse than omitting the vendor entirely**, because every one of these is a wrong line in somebody's live architecture repository, and most got there from a vendor slide.

IN CIRCULATION, AND WRONG

Commonly written	Actually
Blue Yonder, JDA, Blackstone's	Panasonic's — $7.1bn, closed 17 Sep 2021
Greenplum, open source	Broadcom's, closed-source since May 2024
NCR, the one company	Split 16 Oct 2023 — NCR Voyix and NCR Atleos
"NCR" in a South African sentence	The National Credit Regulator. Disambiguate every time
Reflexis, antuit.ai	Zebra's — 2020 ($575m) and 2021, now Workcloud
Revionics, independent	Aptos's
dunnhumby, an agency	Tesco's — a competitor's subsidiary
Nielsen, in retail data	NielsenIQ — Advent's since March 2021
Infor, a software vendor	Koch Industries' — two-thirds Feb 2017
Informatica, independent	Salesforce's, complete November 2025
e2open, a product	WiseTech Global's, and a portfolio
Ceridian / Dayforce	Dayforce since 1 Feb 2024, Thoma Bravo's Feb 2026
"SAP POS" in a grocery estate	May be GK code — GK is Fujitsu's, resold by SAP
Kerridge Commercial Systems	Trades as Klipboard
SES-imagotag / VusionGroup	Vusion since 7 January 2026
Oracle Retail, Oracle-built	Retek + ProfitLogic + 360Commerce + Micros
Toshiba retail, Toshiba-built	The former IBM retail store business. Brand: ELERA
GAAP, a grocery POS	Lesaka via Adumo — and a restaurant product
A vendor's industry page	A marketplace. Some products on it are partners'

Getting one wrong in a board paper is worse than omitting the vendor entirely.

Figure 14.2 Attributions that are wrong in current commentary. Each was checked against the vendor's or body's own material. These are not trivia — they are wrong lines in somebody's live architecture repository, and most got there from a vendor slide.

- **Blue Yonder is Panasonic's, "JDA" is a former name rather than a sister product, and neither Blackstone nor New Mountain owns it now.** Panasonic announced the acquisition on **23 April 2021** [1] and bought the company from Blackstone and New Mountain Capital for **$7.1bn**, closing **17 September 2021** [2].
- **Greenplum is Broadcom's and has not been open source since May 2024.** The open-source successor is **Apache Cloudberry (incubating)** [3][4].
- **"NCR" no longer exists as one company.** The distribution completed at five in the afternoon, New York time, on **16 October 2023**: **NCR Voyix** kept retail, restaurant and digital banking; **NCR Atleos** took automated teller machines, self-service banking, payments and network [5]. **And in a South African retail sentence "NCR" also means the National Credit Regulator.** Write "NCR Voyix" or do not write it.
- **Reflexis and antuit.ai are Zebra's** — Reflexis in 2020 for **$575m**, antuit.ai in 2021, under the **Workcloud** brand launched 2023 [6]. Zebra's current Workcloud page **does not mention Reflexis at all** [7], so a retailer running "Reflexis" runs a product whose vendor no longer markets the name.
- **Revionics is Aptos's.** Its own About page is titled "About Revionics: An Aptos Company" [8]. The acquisition date and consideration could not be established.
- **dunnhumby is Tesco's, not an independent agency** — 53% in 2001, 84% in 2006, later the remainder [9]. **A retailer buying dunnhumby services is buying from a competitor's subsidiary**: an architecture and commercial fact, not a gossip item.
- **NielsenIQ is not Nielsen.** Spun off January 2021, acquired by **Advent International in March 2021**, merged with GfK in 2023, and holder of **Brandbank** [10]. "Nielsen" in a retail data sentence almost always means NielsenIQ.
- **Infor is Koch Industries' — $2.68bn for two-thirds in February 2017**, minority bought out **February 2020** at an $11bn valuation [11]. A conglomerate's software arm behaves differently from a listed software vendor.
- **Informatica is Salesforce's**, complete **November 2025** [12]. **e2open is WiseTech Global's**, announced May 2025 [13] — and is itself a stack of acquisitions, so it names a portfolio rather than a product.
- **Dayforce is the former Ceridian and is Thoma Bravo's**: renamed **1 February 2024**, acquisition of **$12.3bn** completed **February 2026** [14].
- **GK Software is Fujitsu's, and SAP resells GK's software.** GK delisted in 2023 when **Fujitsu ND Solutions AG** became majority owner, and **SAP has distributed GK products since 2009, growing to five, globally from 2013** [15]. **So "SAP POS" in a grocery estate may be GK code under an SAP contract.**
- **Kerridge Commercial Systems trades as Klipboard** [16]. Whether the IQ Retail line the South African market knows survives under that name could not be established.
- **SES-imagotag is VusionGroup is Vusion** — 2014, 2024, and **Vusion on 7 January 2026** [17]. Three names in two years.
- **Oracle Retail is Retek plus ProfitLogic plus 360Commerce plus Micros — acquired, not built.** Oracle outbid SAP for **Retek at $11.25 a share, about $650m**, against SAP's $8.50 [18]. The other three transactions could not be dated and are not stated.
- **Toshiba Global Commerce Solutions is the former IBM retail store business** and its software brand is **ELERA**; its own site still lists a legacy group of IBM-era names [19]. The transaction date, price and stake could not be established.
- **GAAP Point-of-Sale sits inside the Lesaka group via Adumo, and it is a restaurant product rather than a grocery one** [20]. It is frequently named in South African conversation as though it were a grocery platform. It is not.
- **And the one that catches everyone: a vendor's industry page is a marketplace, not a product list.** SAP's retail industry page presents at least one third party's pricing platform in the same visual frame as SAP's own products, which SAP's own partner pages confirm [21][22], and the owners of two further names on that page could not be established. Check every name against the vendor's own partner listing before it enters a landscape document.

HOW TO USE A CORRECTIONS LIST

Not as trivia. When you inherit an architecture repository, spot-check the vendor and ownership fields against each vendor's own site, with a date. You will find three or four wrong, and finding them is cheap credibility.

Part one: the categories, and who plays in each

Vendors sell suites, built so you cannot see where one capability stops and the next begins. Invisible boundaries are harder to argue with and harder to leave. So decompose any suite onto capabilities before comparing anything.

WHERE EACH VENDOR'S OWN SITE PLACES IT

	SAP	Oracle Retail	Blue Yonder	Manhattan	Aptos	Zebra	Toshiba
Merchandising and retail ERP	•	•	•		•		
Assortment, space, financial plan		•	•	•			
Forecast, replenishment, allocation		•	•	•	•	•	
Price and promotion optimisation		•			•		
Warehouse and transport		•	•	•			
Store inventory and point of sale	GK	•	•	•	•	•	•
Commerce and order management		•	•	•	•		
Customer, workforce, loss prevention		•	•	•	•	•	•

Blanks mean the vendor's own material did not name it in this research round, not that the capability is absent. "GK" marks software SAP resells rather than owns.

Figure 14.3 A readable subset of the market, by capability rather than by vendor, built only from what each vendor's own material names. Read it by rows: a vendor in one row is a product decision, and a vendor in four or more is a concentration decision.

Read that figure by rows. A vendor in one row is a product decision. **A vendor in four or more rows is a concentration decision** — a different kind of risk, a different approval path, and a different conversation about continuity.

And before the categories, the three shapes a "retail suite" actually comes in, because they are not substitutes and a shortlist mixing them is comparing different things.

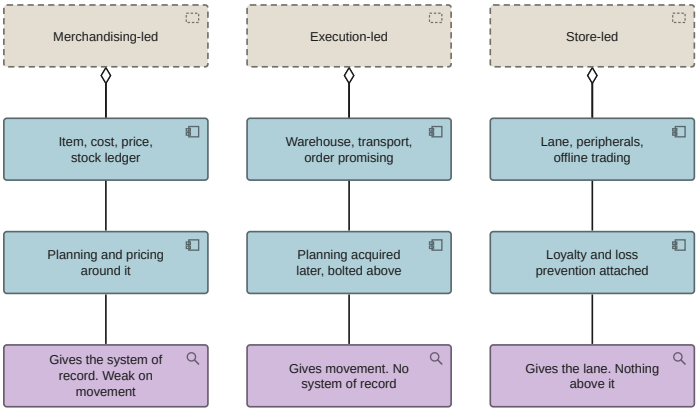


Figure 14.4 Three shapes sold as "a retail suite", and they are not substitutes. A shortlist mixing them is comparing different things, and the integration surface you inherit depends entirely on which shape you started from.

Merchandising and retail ERP

What it is for. The system of record for the item, the location, the cost, the retail price, the on-hand and on-order position, the purchase order, the receipt, the stock ledger and the supplier invoice match. **Separate three axes before comparing anything**: the **cost method** — retail inventory method against weighted average against standard cost, a data-model difference rather than a configuration flag; **grocery against fashion against hardlines**, because size, colour, style, seasonality and perishability are different data models; and **corporate chain against franchise or buying group**, because in the last of those the retailer does not own the stores and the merchandising system is a wholesale system with a retail front.

Merchandising and retail ERP — examples, not recommendations

- **SAP** — the current retail industry page names SAP Business Suite for retail and Fashion Management [21]; the products documented in South African grocery are **SAP ERP / SAP Retail** and **SAP Customer Activity Repository** [23][24], and **S/4HANA** is the platform named in the SPAR litigation [25].
- **Oracle Retail** — its merchandising group names Merchandising Foundation, Merchandising, Sales Audit, Invoice Matching, Allocation and Brand Compliance [26], and the lineage runs through **Retek** [18].
- **Aptos** — Merchandising alongside Point of Sale, Order Management, Store Fulfillment, Allocation/Forecasting & Replenishment, CRM, Sales Audit and Analytics, with self-reported claims of "hundreds" of client retailers and "40 years" in retail [27]. Its current owner is a Goldman Sachs affiliate, bought from Apax funds [28]; the completion date and consideration could not be established.
- **Infor — Koch Industries'** [11]. Its current retail product naming could not be established, so no Infor retail product is named here.
- **Klipboard**, formerly Kerridge Commercial Systems, claims 55,000-plus customers in 70-plus countries, with South African products listed as ERP One, ERP Go, OnRent and Manufacturing [16].
- **Arch Retail / Spinnaker Software** (South Africa) sells Arch Retail in store and Arch Enterprise centrally, plus mobile, accounting, loyalty, e-store, replenishment, fuel and self-checkout modules, across supermarkets, wholesalers, liquor, butcheries, bakeries, convenience, forecourt and DIY [29] — a precise description of the independent and buying-group segment that dominates South African store counts.

Assortment, space and merchandise financial planning

Three categories that get bought together and fail separately. **Assortment** decides what to carry where — clusters, localised range, breadth against depth, option counts. **Space** turns that into shelf: planograms, fixtures, facings, days of supply at shelf, and the reset process, with the architecturally under-appreciated point that **the planogram is a data contract**, because shelf capacity is an input to replenishment and an unsynchronised planogram estate produces phantom capacity and systematic availability failure. **Merchandise financial planning** is the money plan, and its distinguishing question is **whether the plan and the forecast live in the same object** — a tool that cannot reconcile to the demand forecast produces two numbers and a monthly argument.

Assortment, space and planning — examples, not recommendations

- **Oracle Retail** names Assortment Planning and Category Management with customer decision tree and demand transference as the underlying analytics [26] — that pair is what separates assortment *optimisation* from assortment *administration*, and it is worth understanding before you pay for it. It also names Merchandise Financial Planning explicitly [26].
- **Blue Yonder** covers assortment, space and the financial plan under Retail Planning and Category Management [30]; its space lineage is the **Intactix** acquisition of 2000 at $20.5m [2].
- **SymphonyAI** names Assortment Intelligence, Merchandising Intelligence and Store Intelligence within a connected retail platform, citing named grocery customers [31].
- **DotActiv** (South Africa) is the local specialist: a category management consultancy from **1990**, with the DotActiv brand founded in **2001**, five integrated packages covering localised assortment, data-driven planograms, heat-mapped floor plans and promotion planning — and **New Clicks Group Ltd** named as having rated it a top-ten IT supplier for three consecutive years [32]. That is a rare sourceable South African vendor-to-retailer attribution.
- **First Insight** sells something different inside the same decision: a platform gathering consumer product-testing data, adopted by **Woolworths South Africa's womenswear division**, announced **15 May 2023** [33].

Demand forecasting, replenishment and allocation

This is where retail value is created or destroyed, because availability and waste are both outputs of it and they trade against each other. Allocation is a separate problem — the constraint is supply rather than demand and the objective is sell-through rather than service level — and the question that discriminates is **whether it can allocate on a size curve and a store cluster simultaneously, with a single-store override that does not break the rest**.

Forecasting, replenishment and allocation — examples, not recommendations

- **RELEX Solutions** is the category's best-known pure-play. **Its founding, ownership, customer counts and current module naming could not be established, so this book states no RELEX figures at all.**
- **Blue Yonder** — Supply Chain Planning, with a forecasting lineage running through **E3 (2001)**, **Manugistics (2006, over $200m)** and **i2 Technologies (announced 2008 at $346m, completed January 2010)** [2][30].
- **Oracle Retail** — Retail Demand Forecasting, Inventory Planning Optimization, Retail Supply Chain Management and Retail Allocation [26].
- **Manhattan Associates** — ActivePlanning, with demand forecasting, allocation and replenishment as named components [34].
- **Zebra** — demand intelligence as one of four Workcloud modules [7], which is the **antuit.ai** acquisition of 2021 [6]. A hardware company selling forecasting is a structural fact worth recording.
- **o9 Solutions** — a platform built on an enterprise knowledge graph with a sense-model-simulate-decide-execute-learn workflow [35]. Founding date, funding and retail customers could not be established.

Price and promotion optimisation

Three separable jobs that get conflated and should not be: **base price** — everyday price architecture, zone pricing, price families, competitor positioning; **promotion** — mechanic design, funding, forecast lift, cannibalisation and halo; and **markdown** — clearance optimisation against a sell-through-by-date objective. Personalised offers belong with loyalty.

Pricing and promotion — examples, not recommendations

- **Revionics, an Aptos company**, sells exactly that split — base price, markdown and promotions [8].
- **Oracle Retail Lifecycle Pricing Optimization** [26]. **SymphonyAI Merchandising Intelligence**, covering promotional optimisation, pricing and margin management [31].
- **dunnhumby**, through the former KSS Retail rebranded as its price and promotion line in 2015, sells pricing and promotions optimisation — **and is Tesco's** [9], which a competing retailer must weigh deliberately.
- **Competera Pricing Platform** belongs to **Competera UK Limited** and appears on SAP's partner pages as well as being surfaced on SAP's retail industry page [22][21].

THE SOUTH AFRICAN PHARMACY CAVEAT, REPEATED BECAUSE IT CHANGES A BUSINESS CASE

Under the single exit price regime, **dispensary medicine pricing is out of scope for every tool in this category**. For a retailer whose pharmacy arm is a material share of turnover, the addressable scope of a price-optimisation purchase is the front shop only. Ask the vendor to price accordingly.

Warehouse and transport management

Warehouse software is receiving, put-away, slotting, picking, packing, despatch, internal replenishment, labour management, yard and automation integration; the retail-specific demands are **store-friendly picking** in aisle sequence, **mixed-case and each-picking for online**, and **cross-docking against an advance ship notice**. Transport is route planning, load building, carrier selection and tendering, freight audit, telematics, proof of delivery and, increasingly, last-mile courier orchestration.

Warehouse and transport — examples, not recommendations

- **Manhattan Associates** — warehouse management plus labour and yard management, on a platform with a low-code extension layer, and transportation management with dispatch and modelling [34].
- **Blue Yonder** — warehouse and transportation management [30], whose execution lineage is the **RedPrairie** merger of 2012 [2]. That is the single most useful lineage fact when assessing Blue Yonder's planning-against-execution balance.
- **Oracle Retail Warehouse Management** [26].
- **e2open**, now **WiseTech Global's** [13], holds BluJay and INTTRA in the transport and trade space.

THE WARNING THE SPAR CASE EARNS, AND IT BELONGS IN THIS CATEGORY

When SPAR reassessed its programme it stated the remediation as **"separating the ERP from the warehouse system and implementing them independently"** [57]. Put that question into the warehouse selection itself: *will we go live separately from the enterprise system?* If the answer is no, the answer to the next question — how will we know which one broke — is also no.

Store inventory, point of sale and store systems

Store inventory is the store's own view: back-door receiving, counts, adjustments and shrink, transfers, returns to vendor, and the available-to-promise position the website is quoting against. **It is the least accurate data in a retailer and the most exposed once e-commerce quotes against it**, and three decisions determine whether omnichannel works: where the single inventory position lives, what safety buffer applies per location and channel, and how fast a sale at the lane decrements the figure the website is reading. **Point of sale** is the transaction and the device estate around it, and **offline capability is the defining non-functional requirement** — harder in South Africa than in most markets.

Store inventory, point of sale and store systems — examples, not recommendations

- **Toshiba Global Commerce Solutions**, the former IBM retail store business — software under **ELERA**, hardware under TCx and MxP, and a legacy group still listing the IBM-era names [19]. It self-reports as number one worldwide for electronic point-of-sale installations and number two for self-checkout, with 220,000-plus units shipped in 2025 and 2.6 million-plus installed, citing a May 2026 analyst assessment [19]. **Those are the vendor's own claims, dated.**
- **NCR Voyix** — a commerce platform with retail applications across point of sale, checkout, payments, loyalty, loss prevention, store operations and corporate planning, plus an edge capability for local operation when connectivity drops [36]. It self-reports as "the number one global point-of-sale software provider", footnoted to the first quarter of 2026 [36]. **Two vendors therefore claim number one in overlapping categories on their own sites. Attribute both and rank neither.**
- **Diebold Nixdorf** — a retail software suite with a cloud platform, self-service, point-of-sale and engagement software, plus a self-service hardware line [37].
- **GK Software**, **Fujitsu's**, sells a store platform covering point of sale, back office, merchandise management, device control, promotions, vouchers, label printing, open scale and monitoring [15]; **SAP has distributed GK products since 2009** [15], which is why "SAP POS" in a grocery estate may be GK code.
- **Oracle Xstore Point of Service** with hardware and payment [26]; **Aptos ONE** [27]; **Manhattan Point of Sale** within unified commerce [34]; **Zebra Workcloud inventory** [7]; **Blue Yonder Merchandise Operations** [30].

Payment acceptance, and the South African distinction

Everything between the customer's card or telephone and the retailer's bank account: terminal estate and certification, gateway, tokenisation, routing, settlement, reconciliation, chargebacks, alternative payment methods, and the card-security scope decisions chapter 13 described. The discriminating questions are **whether the till ever sees a card number** and **who holds the token vault when you change acquirer**.

Payment acceptance — South African examples first, because the local market is genuinely distinct

- **Yoco** self-reports 200,000 small businesses, over $2bn processed a year and over $100m raised, selling card machines, a counter point-of-sale product, tap-to-pay on a telephone, online payments and merchant capital [38].
- **iKhokha**, **founded 2012**, self-reports 380-plus employees, selling card readers, tap-on-phone, a payments and invoicing suite, a cash advance product and its own point-of-sale application; **its own About page says it is a subsidiary of a parent it names only in an image**, and that parent's identity could not be established [39]. Do not name it.
- **Lesaka Technologies** self-reports more than two million active consumers, 130,000-plus merchants, 750-plus enterprise clients and 4,000-plus staff across five countries [40]; **GAAP Point-of-Sale carries an "an Adumo company" mark and a Lesaka unified-brand banner, and is a restaurant product** [20]. Lesaka's acquisition dates and considerations, and whether several other local names sit inside the group, could not be established.
- The switching and acquiring layer behind South African retail card acceptance could not be sourced for any named retailer, and **no reliable count or market share exists for any platform in this market. Do not estimate one.**

Electronic shelf labels, and what you are also buying

The business cases are price-change labour, price integrity at the shelf — which in South Africa is a statutory exposure rather than a tidiness concern — and increasingly retail media and in-store task direction. The architectural consequence is that **the shelf edge becomes a system with an item-location data dependency, a network dependency and a device-management problem across thousands of endpoints per store**.

Electronic shelf labels — examples, not recommendations

- **Vusion**, formerly VusionGroup, formerly SES-imagotag, formerly Store Electronic Systems — renamed in 2014, 2024 and **Vusion on 7 January 2026**; BOE Technology held 79.94% from 2018, reduced to about 25% by May 2025. Paris-listed since 2006, with reported revenue of €1.011bn in 2024 and €1.5bn in 2025, about 1,200 employees, 650 million labels sold across 75,000 stores in 2025, and named European and North American grocery customers [17].
- **Pricer AB**, established **1991**, uses an optical wireless system and sells labels, shelf-edge displays and a cloud platform, self-reporting over 380 million labels delivered in more than 80 countries [41].
- **Hanshow**, **founded 2012**, sells several label families plus an inventory robot, a planogram camera and a smart cart, self-reporting deployment in over 55,000 stores across 70-plus countries [42].

> **THE PATTERN HERE IS A GENUINE STRUCTURAL OBSERVATION**
>
> **The shelf-label vendors are becoming retail-media and in-store-data companies.** One describes itself as supplying retail devices, a retail data platform and retail media systems [17].
>
> So a retailer buying labels on a labour business case is also buying a platform with an advertising revenue model attached. That may be welcome. It should be decided rather than discovered, and it belongs in the contract, not in a later conversation about who owns the screen at the shelf edge.

E-commerce, order management and marketplace

Three categories, three different questions. For **e-commerce platforms** the axes are suite against composable, and whether the platform owns the inventory and order lifecycle or defers to an order management system. For **order management** the question is whether the sourcing decision optimises cost to serve or merely finds stock — **this is the system that makes omnichannel economics work or not**. For **marketplace** it is who is the seller of record for tax and consumer-law purposes, per transaction.

Commerce and orders — examples, not recommendations

- **E-commerce platforms: none was verified in this research round, so none is named.** The category's well-known names are easy to find and their current positioning changes quickly; go to each vendor's own site. The one sourced adjacent fact is that **Manhattan sells a storefront capability inside its unified commerce range** [34], which is an example of supply-chain vendors moving into the front end.
- **Order management** — **Manhattan** order management with enterprise promise-and-fulfil, store inventory and fulfilment, and digital self-service [34]; **Blue Yonder** order management and commerce plus returns management, built out through the Yantriks acquisition of 2020 and, on the returns side, three acquisitions in 2023 and 2025 [2][30]; **Oracle** order management [26]; **Aptos** order management and store fulfilment [27].
- **Marketplace** — **Mirakl**, founded in Paris and Boston, with marketplace, connect, payout, advertising and agentic-commerce products, and marketplaces reported at over 450 companies [43]. **Takealot** in South Africa is the market-defining example of a retailer that *is* a marketplace: takealot.com began in 2011, its marketplace has operated since **2014**, the group supports over 12,000 entrepreneurs with warehousing, logistics and seller funding, operates 150-plus pickup points including some at Pick n Pay locations, and comprises Takealot, Mr D and a fulfilment arm [44]. **Takealot's ownership could not be verified against a primary filing and is therefore not asserted here.**

Loyalty, customer data platforms, workforce, tasks and loss prevention

Four categories an architect meets as one conversation about the store and the customer, and that behave quite differently.

Loyalty is where retail's identity graph lives, and the consequential decision is **whether the loyalty identifier is resolvable at the lane in real time while offline**. A programme whose offer engine requires connectivity cannot apply a personalised offer during an outage, which here means it cannot apply it reliably. The design answers are offer pre-distribution to the store, a deterministic offer file, or accepting that personalisation is online-only — and the choice should be explicit. **Customer data platforms**: the real job is joining the anonymous web session to the loyalty-identified basket, and the questions are whether the platform *masters* identity — creating a second customer master — and where the consent record lives, since under POPIA the consent record and the processing record must be reconcilable. **Workforce management** is demand-driven labour forecasting, scheduling, time and attendance, compliance and the payroll handoff, and here the compliance surface is **bargaining-council agreements, sectoral determinations and the Basic Conditions of Employment Act** rather than United States-style predictive-scheduling law; which councils bind South African retail could not be established, and **this is where a global workforce product most predictably generates local configuration work**. **Task management** is a separate category and not a feature, because it is where every other system's output lands, and the architectural test is whether task origination is pluggable. **Loss prevention** splits into exception-based reporting over transaction data, computer vision at the lane and on the floor, and article surveillance at the exit.

Customer, workforce and store execution — examples, not recommendations

- **Loyalty** — **Eagle Eye Solutions Group**, an AIM-listed company whose platform covers real-time loyalty, omnichannel promotions, smart checkout and predictive offers, with named grocery and pharmacy customers in the United Kingdom and North America [45]. *Its customer list includes a "Woolworths Group" not confirmed to be Woolworths South Africa — do not use it as a local reference.* Also **Oracle Retail Loyalty and Engagement** [26], **Aptos CRM** [27], **Toshiba ELERA Loyalty and Promotions** [19], and **dunnhumby**, which is **Tesco's** [9].
- **Customer data platforms — unverified in their entirety, so none is named.** This category has consolidated heavily and at least one ownership mapping practitioners commonly repeat is out of date.
- **Workforce** — **Zebra Workcloud workforce optimisation** [7], which is the **Reflexis** acquisition of 2020 at $575m [6]; **Blue Yonder Workforce and Labor Management** [30], from the RedPrairie lineage [2]; **Dayforce**, the former **Ceridian**, now **Thoma Bravo's** [14]; **Manhattan Labor Management** for the distribution-centre side [34]. **GK Software** had a January 2020 partnership embedding a workforce product into its point-of-sale platform [15] — workforce management reaching into the till.
- **Task management** — **Zebra Workcloud enterprise collaboration** [7], again the Reflexis lineage; **NCR Voyix store operations** [36]; **SymphonyAI Store Intelligence** with a named category-reset agent [31].
- **Loss prevention** — **Everseen**, a vision platform for checkout loss, shelf loss and operational intelligence [46]; **Toshiba ELERA Security Suite, Loss Prevention and Produce Recognition** [19], the last being the specific answer to weighed-item fraud at self-checkout; **NCR Voyix** loss prevention [36]. The most interesting design on the South African public record is not a product: **Shoprite's radio-frequency self-checkout**, where the label handles both item identification and article surveillance, with each paid item's identifier sent to a cloud service that alerts only when an unpaid item passes the gates [47].

Master data, data platforms, supplier collaboration and rebates

Master data and product information: the dominant retail sub-problem is the item, because it is the widest, most contested and most externally sourced object in the business, and the discriminating question is **whether the global synchronisation network is a source into your master-data platform or a competitor to it**. **Data platforms**: retail's distinguishing characteristic is volume at the lowest grain — Pick n Pay states more than 30 billion transactions a year across its stores [24] — so decisions are driven by cost per query and cost per terabyte rather than by features, basket-level grain is non-negotiable, forecast and plan versioning is a first-class requirement because a retailer must be able to reproduce the plan it approved, and the retail calendar belongs in the time dimension. **Supplier collaboration and EDI**: the only question worth asking is **who does the onboarding**, because a network vendor charging per connection has an incentive to onboard and a translator vendor does not — and in a retailer with several hundred suppliers the cost of the

category is almost entirely onboarding and re-certification. **Rebate and supplier-income management** is, in grocery, frequently the difference between gross and net margin, and the least systematised major money flow in retail.

Data and supplier-facing — examples, not recommendations

- **Master data** — **Stibo Systems** sells an enterprise master-data platform with product information management, onboarding and syndication, plus further domains for customer, supplier, location and sustainability [48]. **Informatica**, now **Salesforce's**, lists master data management among its portfolio [12]; its current retail product naming could not be established. **NIQ Brandbank** collects and distributes product and brand content for fast-moving consumer goods [10], competing with and overlapping both product information management and the synchronisation network.
- **Data platforms** — **Greenplum**, treated in full below. **Snowflake** is named by Blue Yonder as underlying its current analytics branding, on a hyperscaler's AI services [2]. **Oracle** sells a retail data store and a retail intelligence foundation as named retail-specific artefacts [26]. **SAP** names a business data cloud in its retail suite [21], and its retail-specific analytical component in documented South African use is the customer activity repository with point-of-sale data transfer and audit [23][24]. **Apache Cloudberry** (**incubating**) is the open-source massively-parallel option with Greenplum compatibility [4].
- **Supplier collaboration and EDI** — **SPS Commerce**, a cloud supply-chain network with EDI plus testing and certification, fulfilment, analytics, invoicing, assortment and revenue-recovery products, self-reporting 46,000-plus subscribing customers [49]. **e2open**, now **WiseTech Global's** [13]. **Blue Yonder** gained adjacent capability with **One Network Enterprises, August 2024, $839m** [2]. **Oracle Supply Chain Collaboration** [26].
- **Rebates and supplier income — unverified in its entirety, so no product is named.** The honest statement of the category: supplier income in a large grocer is commonly managed in spreadsheets by a small team, the agreements are not machine-readable and the accrual is an estimate — practitioner consensus, not a finding. Two sourced facts establish why it matters: the SPAR franchisee claim apportions **R25.8m of a R168.7m claim specifically to rebate and over-rider schemes** [25], and a network vendor sells a product literally called revenue recovery [49], which exists because supplier income leaks systematically.

Pharmacy dispensing systems

The dispensary: prescription capture, patient medication record, interaction and allergy checking, generic substitution, scheduling and controlled-substance control, labelling, claim submission and adjudication to medical schemes, dispensing-fee calculation, and the clinic workflows attached. **It is a clinical system in a retail store, with a different regulator, a different identifier scheme and a different audit expectation.** The discriminating question is the one chapter 12 set up: **how is the interaction database kept at "the latest version available" in a store that was offline for a week?**

Pharmacy dispensing — one sourced South African example, and a warning

- **BCX Unisolv Evolve** is described by BCX as a cloud-enabled pharmacy management platform, the next version of a system BCX says has been in use for over thirty years, serving independent pharmacies, corporate groups and hospital and public-sector pharmacies; cloud-deployed with **offline operation during connectivity outages**, open interfaces to medical aids, suppliers, enterprise systems and loyalty, **automated updates for regulatory changes**, role-based access and encryption, and stated built-in compliance with POPIA, the Pharmacy Council's rules and card-security requirements. BCX claims it is trusted by over 50% of pharmacies in Southern Africa — **a market-share claim, attributed and not independently verified** [50].
- Note how directly that feature list maps to the regulatory rules: offline operation, automated regulatory content updates and open medical-aid interfaces are the three things the practice rules and the claims model force.
- **No other dispensing system is named in this book.** Other South African systems, Canadian systems and United States systems were all unverified. In Canada a dispensing system must additionally be conformed **per province**. **Getting a pharmacy system attribution wrong is worse than most**, because the buyers are a small community who will all know.

Blue Yonder, in detail

It gets its own treatment because it is the single vendor most likely to appear across planning, execution and store simultaneously, and because **its lineage and ownership are the most commonly mis-stated facts in retail software**.



Figure 14.5 Blue Yonder is seven acquisitions and a name it bought. Every one of them is still visible in the product set, which is the honest explanation for both its breadth and its integration seams — and the reason "JDA" is a former name rather than a sister product.

The six-word version: Blue Yonder is Panasonic's. The longer version is that it is Arthur plus Intactix plus E3 plus Manugistics plus i2 plus RedPrairie plus a German artificial-intelligence company whose name it took — and every one of those seven acquisitions is still visible in the product set, which is the honest explanation for both its breadth and its integration seams [2]. **What it covers today**, per its own solutions page: supply chain planning; retail planning and category management; merchandise operations; order management and commerce; returns management; warehouse management; transportation management; workforce and labour management; sustainable supply chain management; a supply chain command centre; and mobility and automation [30].

> NAMING CAUTION
>
> **"Luminate"** was the platform brand through roughly 2020 to 2024 and still appears across third-party review sites and job advertisements. The current solutions page does not use it [30], and whether it has been formally retired could not be established. **Write "Blue Yonder" and the solution area, not a platform brand name**, because the platform brand has changed twice in six years.

Where it sits against the alternatives, as structural comparison rather than ranking. It is the only vendor in this chapter credibly appearing in planning, execution, store and commerce at once. The alternatives cluster: Oracle Retail and SAP compete across merchandising, planning and store but are weaker in execution-grade warehouse and transport software; Manhattan competes hard in execution and omnichannel and has moved into planning but does not sell merchandising; RELEX, o9 and SymphonyAI compete in planning without execution or store; Aptos, Toshiba, NCR Voyix, Diebold Nixdorf and GK compete in store without planning. **So the real decision is not a feature comparison. It is a concentration decision**: one vendor across four domains buys integration you do not have to build, and a single relationship you cannot walk away from one domain at a time.

Two facts about the risk profile. In **November 2024 the company disclosed a ransomware attack**; a group claimed to have stolen 680 gigabytes of data, and by **12 December 2024** most affected customers had been restored [2]. For a retailer whose warehouse management or workforce scheduling runs on a vendor-operated service that is a sourced illustration of concentration risk — and the right response is not to criticise the vendor but to ask what your continuity plan for a software-as-a-service supply-chain outage actually is, because most retailers do not have one. And **the owner is a Japanese electronics conglomerate rather than a software company** [1][2], which changes roadmap behaviour, bundling incentives and the likelihood that hardware and software get sold together.

Greenplum, in detail

It gets its own treatment because **its licensing status changed materially in 2024, and a reader who repeats the pre-2024 position will tell a client something false**.



Figure 14.6 Greenplum's twenty-one years, and the two dates that matter now: closed-source in May 2024 without announcement, and an open-source successor still incubating. The manner of the closure is the architecturally relevant fact.

What the closure looked like is the architecturally relevant part. By the successor project's own account, in May 2024 nearly all of Greenplum's public repositories were archived and made read-only, the chat workspace was deleted, and the mailing lists went silent — done, in that project's words, **"without any announcements"** [4]. Whatever one makes of the decision, the *manner* of it is the lesson: **an open-source dependency can become unmaintained without notice**, and a data platform strategy that assumed community continuity had no warning.

The successor is **Apache Cloudberry**, created by original Greenplum developers, evolved from the open-source database but on a newer PostgreSQL kernel and aiming to stay Greenplum-compatible. It was voted into the Apache Incubator on **12 October 2024**, repositories transferred **5 November 2024**, and it **remains incubating** [4]; whether it has since graduated could not be established.

The architecture matters because it sets the migration cost. Greenplum is massively parallel processing on PostgreSQL: a master node, a standby master and segment nodes, with catalogue on the master and data on the segments, each segment a modified PostgreSQL instance with a primary and a mirror on different hosts. Version 7.0.0, September 2023, is based on PostgreSQL 12.12 [3].

The honest read for an architect who finds it in an estate. It is **not a legacy platform to be migrated on principle**. It is a capable warehouse with a changed commercial position, and the decision is commercial and continuity rather than technical. Its genuine strengths in retail are PostgreSQL compatibility — SQL, drivers, extensions and a hiring pool — in-database analytics on a cluster you control, predictable cost on owned hardware when the workload is a decade of basket-line data scanned repeatedly, and on-premises residency, which for health-adjacent data is sometimes a requirement rather than a preference. Its structural disadvantages against the cloud warehouses are equally real: compute and storage are coupled, so a seasonal peak means provisioning for the peak year-round; cluster operations including failover and mirroring are yours; and the roadmap now sits inside a closed-source commercial suite [3].

So ask three questions: the forward licence cost and term; the migration cost to either a cloud warehouse or the open-source successor, with the compatibility claim being what makes the second option cheap *if* it holds; and the exposure if the owner changes terms again, given it has now done so once without announcement. **Practitioner consensus rather than finding**: the migration cost in a large retail estate is dominated not by moving data but by rewriting in-database analytics and stored procedures with no direct equivalent, and by the extract and reporting estate pointed at the cluster. Do not put a number on it.

The South African picture, honestly

The rule here is strict. If it is not in an annual report, a vendor press release or credible trade press, say so rather than repeat it. South African retail technology attribution is mostly market folklore, and the unsourceable list is longer than the sourceable one.

SOURCED, AND NOT SOURCED

Retailer or vendor	What the public record actually shows
Shoprite	SAP ERP enterprise-wide, SAP Retail rollout completed 2018, 1,819 stores and 30,355 users in twelve months — per SAP. GK OmniPOS across ~2,500 stores and ~30,000 tills — per GK
Pick n Pay	SAP CAR across 2,000+ stores, 30bn+ transactions a year. POS DM to CAR/DTA planned <6 months, live January 2022, 13+ interfaces, internal resources pushed too hard — per SAP
Massmart	Delisted, publishes nothing. Walmart's own Global Integrated Fulfilment platform behind the Africa app — per trade press
Woolworths SA	One partnership, womenswear, announced 15 May 2023. Nothing else. And "Woolworths" in a search returns an Australian group
Clicks, Dis-Chem, TFG, Mr Price, Pepkor, Lewis, Cashbuild	No technology attribution sourceable for any of them. Mr Price's investor pages name no operational vendors
Oracle Retail, Blue Yonder, RELEX, Manhattan, Aptos, Toshiba, NCR Voyix, Diebold Nixdorf, Vusion, Pricer, Hanshow, SPS Commerce, Stibo	Not one South African retail customer sourceable for any of them. State no RELEX figures at all
Systems integrators	Effectively unsourceable. Two exceptions, both named by the software vendor rather than the retailer

The absence is the finding. The only reason Shoprite's and Pick n Pay's estates are known is that SAP and GK published case studies.

Figure 14.7 What is on the South African public record, and what is not. The right conclusion from the second half is that this market does not disclose — so a vendor's reference list in your shortlist will be its global one.

Shoprite is the best-documented estate in the country, and it is documented by its vendors. SAP's own Africa newsroom states Shoprite uses SAP ERP enterprise-wide, has used SAP human-resource and finance software since the late 1990s, and completed the rollout of SAP Retail to all stores and distribution centres **in 2018**; it reports 2,791 corporate-owned stores, that **1,819 stores, 30,355 users and nine distribution centres moved onto SAP within twelve months**, and roughly 16,000 employees on the system at any time [23]. On the store side, GK Software's own case study records **GK OmniPOS** deployed across the estate using GK's standardised SAP integration method, and a **cashless radio-frequency self-checkout** built for a new apparel banner, the first store opening in Cape Town on **30 March 2023** and nine operating by June 2023, against an estate of about **2,500 stores and about 30,000 tills** in the modernisation programme [47]. **Treat the quantified delivery claims — self-checkout in six months, SAP integration in eight weeks — as vendor-reported, and note the scope: one new banner, nine stores.**

Pick n Pay is documented more usefully, because the documentation admits an overrun. SAP's newsroom records the customer activity repository deployed across more than 2,000 stores in South Africa plus five neighbouring countries, processing more than 30 billion transactions a year [24]. The second article is the valuable one: a move **from SAP POS DM, first deployed in 2004, to the customer activity repository with point-of-sale data transfer and audit**, **planned at just under six months ending November 2021** and **live in January 2022, later than initially planned**, with **more than thirteen interfaces** — and the team **pushed its internal resources too hard and had to revise its timelines** [51]. That is a vendor-published admission of internal-capacity-driven slippage, and it is the single most honest data point in South African retail technology. Pick n Pay also selected a hyperscaler as strategic cloud provider in August 2022 [52], scope unestablished.

Massmart is delisted and publishes nothing. Walmart bought out the remaining 47% stake and took it off the exchange; the sourced account dates completion of the full acquisition to 2022 and the delisting is commonly dated 2023 — the useful fact is the silence either way. What *is* sourced is that the **Walmart Africa application runs on Walmart's own Global Integrated Fulfilment platform**, with the parent embedding its global stack more deeply after taking full ownership, and Walmart-branded stores opened at Clearwater Mall in Roodepoort and in Fourways on Black Friday, 28 November 2025 [53]. **That is the clearest sourced example of a multinational parent replacing a local estate with its own platform** — and for Massmart the build-versus-buy question was answered by the shareholder.

Woolworths South Africa has exactly one sourced platform decision: a partnership with First Insight announced 15 May 2023, applied in womenswear [33]. **Its merchandising, planning, point-of-sale and data platforms are not on the public record from any source retrieved.** And note the name trap: **"Woolworths" in a search returns Woolworths Group, which is Australian and irrelevant here.** Letting that material into a South African document is an easy and embarrassing error.

> **SAY THE ABSENCE OUT LOUD, BECAUSE THE ALTERNATIVE IS FOLKLORE**
>
> **No South African retailer's technology estate is publicly sourceable except Shoprite's and Pick n Pay's, and those only because SAP and GK published case studies.** Clicks, Dis-Chem, TFG, Mr Price, Pepkor, Lewis, Cashbuild and SPAR's non-SAP estate could not be sourced at all. Mr Price's investor-relations page names no operational technology vendors. Shoprite's own newsroom contains nothing on the technology behind its online grocery business.
>
> And more pointedly: **not one South African retail customer was sourceable for Oracle Retail, Blue Yonder, RELEX, Manhattan, Aptos, Toshiba, NCR Voyix, Diebold Nixdorf, Vusion, Pricer, Hanshow, SPS Commerce or Stibo.** That is not evidence they are absent. It is evidence that this market does not disclose — which means **a vendor's reference list in your shortlist will be its global one.** Insist on named references in your own market, format and size band, and insist on speaking to them.
>
> The one sourceable local vendor-to-retailer attribution found anywhere is New Clicks Group rating DotActiv a top-ten IT supplier for three consecutive years [32] — thin, vendor-published, and the only one there is.

On systems integrators, attributions here are effectively unsourceable. What is sourced: a Canadian-headquartered SAP retail implementation partner named by SAP on Pick n Pay's work [24][51]; GK Software itself delivering the Shoprite store programme with its own method [47]; and BCX, a Telkom

Group subsidiary, publishing a retail capability model while **naming no retail clients and no case studies** [54]. Named South African retail implementations by any global consultancy or large local integrator could not be sourced for any of them. One structural observation: the two best-documented programmes were delivered **by the software vendor itself** or **by a specialist boutique brought in by the software vendor**, not by a global integrator [47][24]. Practitioner consensus rather than finding — product-specialist boutiques outperform generalists on packaged delivery, and generalists win on programme scale and change management, so a sensible buyer splits the two roles deliberately.

What a merchandising or point-of-sale replacement costs, and how long it takes

The most useful thing this chapter can do is destroy your intuition — and in retail, unlike banking, **the public record is dominated by failures, because that is what triggers disclosure**.

WHAT THE PUBLIC RECORD ACTUALLY ESTABLISHES ABOUT COST AND DURATION

Case	The figure	The scope caveat
SPAR · SAP S/4HANA	R1.6bn–R2bn lost turnover; implementation estimated R1.8bn	One distribution centre's go-live. Lost sales exceeded implementation cost in a year
SPAR · consequences	~R720m lost regional profit; R94m write-off; PAT down 80%; three directors gone	Go-live date given as Feb 2023, Oct 2022 and early 2023 by three sources
SPAR · contagion	Fuel and distribution up 12.8%; R168.7m franchisee claim, 28 Jan 2026	Allegations, no findings reported, case live. Third-party damage is a buying-group risk
Pick n Pay · POS data	Planned under six months to Nov 2021; live Jan 2022; 13+ interfaces	Vendor-published, and it admits internal resources were pushed too hard
Shoprite · SAP Retail	1,819 stores, 30,355 users, nine DCs in twelve months; rollout completed 2018	Vendor-published. Measures a rollout, not a selection-to-steady-state arc
Shoprite · GK store	Self-checkout built in six months; SAP integration in eight weeks	Vendor-reported, and the scope is nine stores of one new banner
Target Canada	$2.1bn lost; 133 stores opened and closed inside 25 months	Told as supply chain; actually item master, ordering authority and an immovable date

No South African retailer has disclosed the cost or duration of a merchandising or POS replacement that worked. The only local costed figure is an estimate of a failure.

Figure 14.8 The sourced anchors, with scope attached. These calibrate expectations; none of them is a benchmark, because a vendor case study measures from a convenient start and a failure is disclosed for reasons a success is not.

SPAR is the anchor, and chapter 12 sets out the figures. The short version: a distribution-centre go-live whose date three credible sources give differently [58][57][25]; R1.6bn–R2bn of lost turnover against an implementation estimated at R1.8bn, so **the lost sales exceeded the implementation cost in a single year** [58][56]; about R720m of lost regional profit and a R94m write-off [58]; profit after tax down 80% and fuel and distribution costs up 12.8% while three other centres covered the gap [57]; three directors gone [58]; and a R168.7m franchisee claim lodged 28 January 2026 [25]. Four lessons, and they are this section's spine.

1. **The go-live scope was a distribution centre, and the damage landed on independent retailers who did not choose the system.** In a franchise or buying-group model the blast radius extends to parties with a contractual claim, now being tested in court. **Model third-party damage as a programme risk**, not only as a service-level one.
2. **The enterprise system and the warehouse system should not go live together.** SPAR's own stated remediation is to separate them and implement them independently [57]. Repeat it verbatim.
3. **In retail the cost of a failure is measured in sales, not licences**, because the supply chain stops and a grocery supply chain that stops loses perishable stock and customers at once. That is the fundamental difference from a bank core replacement, where the failure mode is service disruption rather than immediate revenue loss.
4. **The damage lasted years, not weeks** — the claim alleges problems persisting to at least September 2025 from an early-2023 go-live [25]. **Stabilisation is a programme phase with its own budget, and nobody plans it.**

Target Canada is the international anchor, and it is usually told wrong. Supply problems were blamed on new inventory software and on not giving staff time to resolve its issues; the parent refused to delay the launch rather than keep paying rent on unopened stores; and Canadian stores **lacked local authority to order their own merchandise**, so border-town stores stocked the wrong regional sports apparel. The chain lost **$2.1bn**; the first stores opened 5 March 2013, 133 were operating by January 2015, restructuring began 15 January 2015 and all had closed by **12 April 2015** [55]. **It is usually told as a supply-chain story and it is actually an item-master and governance story**: centralised ordering authority, plus unvalidated item data, plus an immovable launch date. The widely repeated item-master error rates could not be sourced and are not quoted.

Asda is an active case worth watching. A January 2025 round-up cites two 2024 headlines, one describing a stock discrepancy of tens of millions during an enterprise migration and one a decision not to proceed with a mass rollout of a store conversion; neither carries sales figures, and the product, programme name, budget and status could not be established [56]. The practitioner note is that it is a **separation** from a parent's systems rather than a replacement — Massmart's shape in reverse — and **separation programmes are systematically under-estimated because the as-is is undocumented by definition**.

DO NOT USE THE LIDL STORY

A story circulates about a cancelled multi-year enterprise programme at a European discounter, abandoned after very large spend over an inventory-valuation mismatch. **Two attempts to retrieve a primary or near-primary source failed, so the figure, the duration and the project name do not appear in this book, and should not appear in yours.**

The underlying point is sound without the case: inventory valuation method is a property of the data model, and discovering mid-programme that the package implements the other one is a re-platform, not a change request. Make the point. Drop the anecdote.

On duration, print the caveat with every figure. Public dates establish *elapsed market reality*, not how long any single implementation took, and a vendor case study measures from a convenient start. **Calibrate expectations; do not benchmark.** And note what is missing from the entire public record: **no South African retailer has disclosed the cost or duration of a merchandising or point-of-sale replacement that worked.** The only costed local figure in this chapter is a secondary estimate of a failure's implementation spend.

The cost lines that are not the licence, which is the smallest of them. **Item and supplier master data remediation**, the retail equivalent of a bank's account-data migration and, on the Target Canada evidence, the most consequential [55]. **Trading-partner re-certification** — not interfaces, partners; several hundred maps, each with a partner's own implementation guide. **The device and peripheral estate**, because a point-of-sale replacement is a hardware programme with a software component. **Payment terminal re-certification**, on a separate approval body's calendar. **Store training at scale**, where 30,355 users within twelve months is the sourced measure of what that means [23]. **Parallel distribution capacity** — SPAR's sourced cost of running one region from three other centres was a 12.8% rise in fuel and distribution costs [57], **and this is the retail-specific line banking does not have**. **Lost sales during stabilisation. Third-party damage exposure** in a franchise or buying-group model. **Stabilisation itself**, as a funded phase with a defined exit test. And **deconversion and exit cost**, asked at selection and written into the contract.

THE RETAIL FAILURE SIGNATURE, AS PRACTITIONER CONSENSUS

Go-live into a peak trading period. The enterprise system and the warehouse system together. An immovable external date — a lease, a launch, a parent-company separation deadline. Item master data accepted on trust from suppliers. And no funded stabilisation phase.

Three of those five are visible in the sourced SPAR and Target Canada records. **The software is almost never the cause.**

Build, buy or configure

The question is not build or buy. **It is where on the spectrum each capability sits**, and the answer differs per capability inside one retailer.

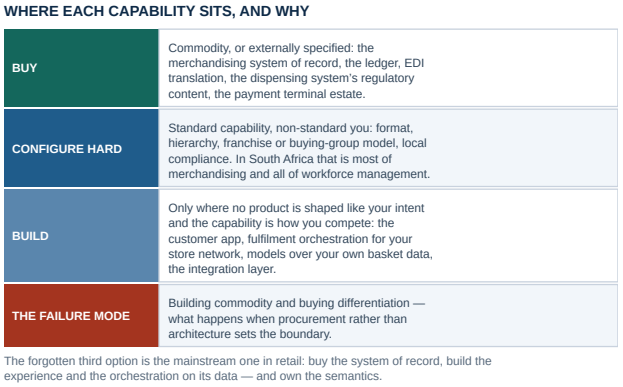


Figure 14.9 Four positions on the spectrum, with the question each answers and the failure that follows from getting it the wrong way round. The South African evidence puts almost every sourced large-retailer decision in the first two bands.

What the South African evidence shows is that every sourced large-retailer decision is buy-and-configure, heavily: Shoprite on SAP with GK at the store [23][47]; Pick n Pay on SAP's customer activity repository with a hyperscaler as strategic cloud provider [24][52]; SPAR on S/4HANA [25]; Woolworths buying a specialist product for one merchandising decision in one division [33]. **The one visible build is a parent's, not a local decision**: Massmart runs Walmart's internally built fulfilment platform [53].

That is a materially different picture from South African banking, where the companion volume records a bank self-building a core, another building proprietary risk and pricing engines, and a third's chief technology officer rejecting the core-banking category outright. The reason, as analysis: **retail's differentiating capabilities — range, price, availability, store experience — are operating-model and data capabilities rather than software capabilities.** A retailer can out-execute a competitor on the same enterprise instance; a bank cannot out-execute a competitor on product definition if its core will not define the product. So **retail's build energy goes into the channels, the fulfilment orchestration and the data estate**, and almost never into the merchandising system of record.

The heuristic that fits. Buy where the capability is commodity or externally specified: the merchandising system of record, the ledger, EDI translation, the dispensing system's regulatory content, the payment terminal estate. **Configure heavily** where the capability is standard but your format, hierarchy, franchise model or local compliance is not — which here is most of merchandising and all of workforce management. **Build** only where no product is shaped like your intent and the capability is how you compete: the customer application, the fulfilment orchestration logic for your specific store network, the analytical models over your own basket data, and the integration layer holding the estate together. **The failure mode** arrives by the same

route as in banking: **building commodity and buying differentiation**, which is what happens when procurement rather than architecture sets the boundary.

And the forgotten third option is, in retail, the mainstream one. Buy the system of record; build the experience and the orchestration on its data — which is what a transactional data layer is in the two best-documented South African estates [51][47], and the same shape as banking's "buy the ledger, build the product layer". The cost is identical: **you now own the integration and the semantics**, and you must say out loud where a second item master or a second inventory position is not allowed to appear.

The pure-play against the incumbent chain

They are **not the same problem**, and conflating them is the commonest error a reader will import from conference material.

TWO DIFFERENT PROBLEMS, OFTEN COMPARED AS ONE

Dimension	The pure-play	The incumbent chain
Store estate	None. No tills, lanes, offline trading, article surveillance, shelf labels, store rosters or cash	Thousands of stores, each a node that must keep trading when the link drops
Inventory	Few locations. Availability is a simpler question	Every store a stock location, structurally inaccurate, quoted against by the website
Range risk	Externalised to third-party sellers	Owned, with a buying calendar and an open-to-buy
Identity	One channel, one customer identity from day one	Loyalty, online account, app, marketplace and patient — and a base that cannot re-onboard
Item data	Digital-native, no decades of undocumented variation	An item master encoding twenty-five years of decisions nobody wrote down
Real constraint	Last-mile cost, seller quality, catalogue matching, and being an intermediary for tax and consumer law	It cannot choose its starting point. Phased rollout with a funded stabilisation plan is the only honest pattern

"Operate like a pure-play" is not advice, it is a category error. The thing that takes years is not the software — it is the supply chain and the people who depend on it.

Figure 14.10 The pure-play and the incumbent chain are not the same problem. The transferable lessons are in the right-hand column of the last two rows; the headcount, the timeline and the absence of stores are not transferable at all.

The pure-play, locally, is Takealot: a marketplace since 2014, over 12,000 third-party sellers, 150-plus pickup points including some at a competing grocer's stores, fast delivery in four metropolitan areas, and group companies covering retail, food delivery and fulfilment [44]. Its structural advantages are all absences: **no store estate** — no tills, no lanes, no offline-trading problem, no article surveillance, no shelf labels, no store scheduling, no cash; **inventory in few locations**; **a marketplace model that externalises range and inventory risk to sellers**; one channel and one customer identity from the start; and a data estate with no decades of undocumented product variation. Its real constraints are last-mile cost, seller quality and catalogue matching, and the fact that **being an intermediary changes its tax and consumer-law position per transaction**.

The incumbent chain carries every one of the things the pure-play does not: a device estate in thousands of stores; scheme certifications; cash; a franchise or buying-group estate; a supplier base integrated by EDI over decades; an item master encoding product variation nobody documented; a regulated dispensary in the pharmacy formats; a workforce under bargaining-council agreements; and **a customer base that cannot be asked to re-onboard**. It cannot choose its starting point, so its available patterns are **phased rollout by region or banner with an explicit stabilisation plan** — what SPAR is now doing, having learned the hard way — and **buy-and-configure at the system of record with building concentrated in channel and data**, which is what Shoprite and Pick n Pay demonstrably do.

> **THE HONEST READ FOR AN INCUMBENT ARCHITECT**
>
> "Operate like a pure-play" is not advice. It is a category error.
>
> The transferable lessons are **one inventory position, one customer identity, product definition as configuration, and a fulfilment decision that optimises cost to serve rather than finding stock**. Not the headcount, not the timeline and not the absence of stores. And the transferable lesson from the incumbents is the one SPAR paid for: **the thing that takes years is not the software. It is the supply chain and the people who depend on it.**

And once more, because it matters

This chapter is a map of what plays in each category. It is not a recommendation, a ranking, an endorsement or a shortlist. Named vendors are examples chosen for recognisability; omission is not a judgement and inclusion is not a blessing. There are no scores and no quadrants, and where a vendor claims a rank it is attributed to that vendor and dated — in one category two vendors claim to be number one in overlapping markets, which is exactly why nothing here is ranked. Reference architectures and standards are described and pointed at, never reproduced.

The research behind this chapter is dated October 2026. Re-verify every ownership and product attribution before it reaches a board paper, and put the date you checked next to it.

What goes stale first, restated. The two-dimensional barcode timetable, because the definition keeps being softened in guidance and the date is close. Retail software ownership, because six changes landed in six categories in roughly two years. The availability of the dormant retail data models, because it is unclear today and could resolve either way. The card-security standard's version and dates. The SPAR litigation, because it is live. And South African acquirer consolidation, because the consolidating group is still buying.

And the sentence to leave you with. By the time three vendors are in a procurement, the architecture has already been chosen — data model, extension model, release cadence, integration style, cloud posture, the skills market you hire from, and the exit cost you will one day pay. Negotiation moves price

and dates and very little else. So write the capability model and the identifier standard first. Fix the non-negotiable constraints before the market scan. Shortlist for the shape of the problem and the format of the retailer rather than the size of the vendor. Take references in your own jurisdiction, format and size band, and insist on speaking to them. **Separate the enterprise and warehouse go-lives. Fund the stabilisation phase.** And record the decision with the alternatives and the constraints that excluded them, naming the reversibility — because an option that cannot be undone belongs at board level, not in a procurement scorecard.

THE CHAPTER IN TEN LINES

- Every company named here is an example of who plays in a category — not a recommendation, not an endorsement, not a shortlist — and the research is dated October 2026, so re-verify every ownership and product name before it reaches a board paper.
- Blue Yonder is Panasonic's at $7.1bn, closed 17 September 2021; “JDA” is a former name, and the lineage runs JDA plus i2 plus Manugistics plus RedPrairie, which is why the product set has seams.
- Greenplum is Broadcom's and has been closed-source since May 2024 — done without announcement — with Apache Cloudberry the incubating open-source successor. It is not a platform to migrate on principle; the decision is commercial and continuity, not technical.
- “NCR” split on 16 October 2023 into NCR Voyix and NCR Atleos, and in a South African retail sentence it also means the National Credit Regulator. Disambiguate every time.
- Attributions to get right: Reflexis and antuit.ai are Zebra's; Revionics is Aptos's; dunnhumby is Tesco's, so buying its services is buying from a competitor's subsidiary; NielsenIQ is not Nielsen; Infor is Koch Industries'; Informatica is Salesforce's; e2open is WiseTech Global's; Dayforce is the former Ceridian and is Thoma Bravo's.
- GK Software is Fujitsu's and SAP resells it, so “SAP POS” in a grocery estate may be GK code; Oracle Retail is Retek plus ProfitLogic plus 360Commerce plus Micros, acquired rather than built; Toshiba Global Commerce is the former IBM retail store business under the ELERA brand; Kerridge trades as Klipboard; SES-imagotag became VusionGroup and then Vusion.
- A vendor's industry page is a marketplace rather than a product list — some products on it belong to partners — and a vendor appearing in four or more capability rows is a concentration decision rather than a product decision.
- No South African retailer's technology estate is publicly sourceable except Shoprite's and Pick n Pay's, and those only via SAP and GK case studies. Not one SA customer was sourceable for thirteen named global vendors, and state no RELEX figures at all. The absence is the finding.
- SPAR lost R1.6bn–R2bn of turnover against an implementation estimated at R1.8bn, so in retail the cost of a failure is measured in sales rather than licences — and Pick n Pay's own vendor-published case study admits the team pushed its internal resources too hard and revised its timelines.
- The sourced decisions here are buy-and-configure almost without exception, with build energy going into channel, fulfilment orchestration and the data estate; and “operate like a pure-play” is a category error, because the incumbent cannot choose its starting point.

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CHAPTER 15

Regulation, risk and what the regulator inspects

No retail regulator asks how your systems are built. An inspector walks into one store during trade and scans the shelf-edge labels. This chapter is about the constraints that are not negotiable, where each one lands in the model, and why almost none of them can be satisfied retrospectively.

The inspector is in the store

A banking supervisor arrives with a request list and sits in a meeting room. A retail inspector arrives in a store, unannounced, during trade, and starts scanning shelf-edge labels.

That difference is the whole chapter. Retail regulation is *operational*. It reaches the till, the shelf label, the deli scale, the chiller, the dispensary door and the packaging on an own-label tin. It is enforced by people who walk into one site out of hundreds and form a view of the group from what that site was doing on a Tuesday afternoon. No quarterly return stands between the architecture and the finding.

So these are not a compliance backlog. They are design inputs, and several cannot be satisfied at all by a system built without them. Three properties recur.

The obligation attaches to a site, not to the enterprise. A liquor licence, a pharmacy licence, a certificate of acceptability for food premises, a verified scale: all per premises, with their own numbers, conditions and expiry dates. The location master is a regulatory object, and a flat store table with a type column cannot hold it.

The clock starts from an event the retailer does not control. Six months from delivery, seven days from receipt, thirty days from a cancellation — each needing an authoritative timestamp for something that happened elsewhere, where a carrier's proof of delivery is often the only evidence.

The defence is a record that had to exist at the time. Almost nothing here can be remediated retrospectively: either the system wrote the evidence when the thing happened or there is no defence. That is the sentence to repeat in a steering committee.

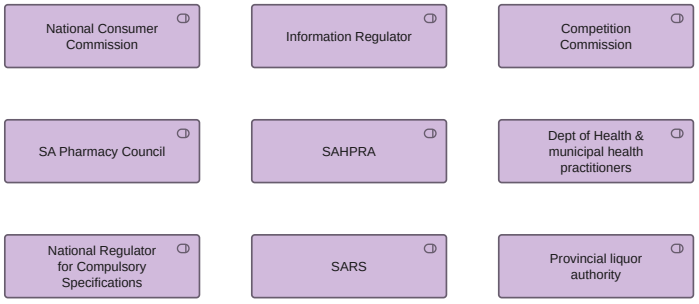


Figure 15.1 Nine enforcers, no single return. Each is a Stakeholder with its own concern, its own instrument and its own inspection habit — and several inspect in the store rather than in the boardroom. There is no consolidated retail regulator and therefore no consolidated compliance system; what holds the set together is an obligations register keyed to instrument, obligation, owner, evidence and system.

Nine enforcers, and that is before the municipal environmental health practitioner and, for a listed group, the JSE. No single regulator, no single return, and therefore no single compliance system.

The Consumer Protection Act, which binds the systems most directly

The Consumer Protection Act 68 of 2008 came into force on **1 April 2011** [2]. Of everything here it reaches furthest into a retailer's systems, and it is most often treated as a policy matter rather than an architectural one.

Scope is narrower than people assume and wider in two places that matter. Section 5(2)(b) excludes a transaction where the juristic consumer's asset value or annual turnover equals or exceeds a threshold set at **R2 million**, unchanged since the Act took effect, and the credit agreement itself is out where the National Credit Act applies — though the goods bought on credit stay under the CPA [2]. Then the overrides: **a franchisee is always a consumer regardless of size**, and sections 60 and 61 on product safety and liability apply to goods even in otherwise-exempt transactions [2]. So a group with a franchise estate and a cash-and-carry arm sells to consumers, to exempt juristic buyers and to statutory consumers at once. Customer type is a legal attribute, not a segmentation.

Enforcement runs through the **National Consumer Commission**, with the National Consumer Tribunal adjudicating and accredited industry ombuds alongside — among them the Consumer Goods and Services Ombud, whose industry code was prescribed in Gazette 38637 of 30 March 2015 [14] and whose levy is compulsory for suppliers in the sector [15]. The Commission's principal tool is the compliance notice and it uses it in volume: 45 notices to consumer-goods suppliers in a tranche announced on 2 July 2025 [3], and 62 in a later tranche [4].

Section 23, and why a stale shelf label is a legal exposure

Section 23 is the most operationally invasive provision in South African retail law. From the Act's text [1]: a retailer must display a price for goods offered for sale, adequately displayed if it is in rand and affixed to the goods, the packaging, the shelving or the display; a supplier **may not charge more than the displayed price**, and **where more than one price is concurrently displayed, may not charge more than the lower or lowest of them**. A price that

fully covers and obscures another becomes the displayed price, and a reduction placard makes the displayed price the shown price *less* that reduction. "Price" includes a **unit price**, which is where the Act meets the legal-metrology format rules later in this chapter. And none of it applies where the price is set by or under a public regulation — which is how single-exit-price medicines sit outside it.

Read the two-prices rule again slowly, because it converts a familiar operational nuisance into a statutory liability. A shelf-edge label that silently failed to update is not a data-quality defect with a customer-service consequence. It is a second concurrently displayed price, the lower one binds, and the group's own till is charging above a displayed price at every store where the label did not print, until someone notices.

The escape is deliberately narrow. Section 23(9) releases the supplier from an **inadvertent and obvious** error only once it has **corrected the display** *and* **taken reasonable steps to inform consumers who may have seen the erroneous price** [1]. Both limbs are data. A retailer that corrected the label and told nobody has not made out the defence, and cannot prove the correction time either way without a record of it.

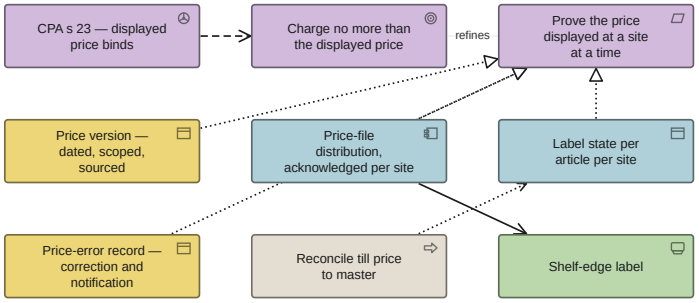


Figure 15.2 Section 23 from driver to control. The Act is a Driver and cannot be realised by anything; the specific duty is a Requirement that a system can satisfy or fail. Three elements realise it — the dated price version, the confirmed distribution of the price file, and the label state per site — and the price-error record is what makes the narrow s 23(9) defence provable. A model with the Driver and no Realization edges is decoration.

What that forces: **price as a dated, scoped, sourced entity**, append-only, with effective *timestamps* — a change applied at 06:00 before opening and one at 14:00 mid-trade are different facts; **price-file distribution with per-site acknowledgement**, because sending the file is not the obligation and knowing it was applied is; **label print status per article per site**; a **till-versus-master reconciliation**; and a **price-error record** carrying both the correction and the notification.

Three return clocks, running at once

Sections 55 and 56 set the implied warranty: goods must be safe, of good quality, in good working order, free of defects, usable and durable for a reasonable period. Within **six months** of delivery the consumer may return goods failing that standard — and the choice between repair, replacement and refund **is the consumer's**, at the supplier's risk and expense [6].

That is the clause retail systems get wrong. It is not a returns policy with configurable options. It is a per-sale-line statutory clock with the election held by someone outside the business. A returns engine that offers repair first, or routes by category, or applies a thirty-day store policy, is implementing a policy the statute overrides.

Two other clocks run alongside it with different anchors: five business days under section 16 to rescind a transaction resulting from direct marketing, from the later of conclusion or delivery [5], and seven days from receipt for a distance sale under the Electronic Communications and Transactions Act [31].

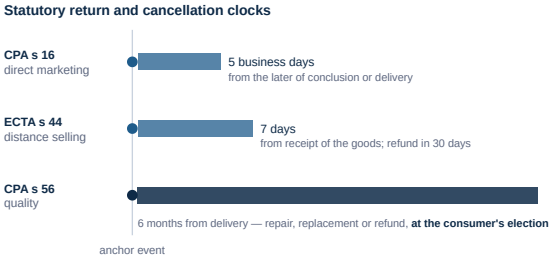


Figure 15.3 Three statutory clocks, three anchors, running at once. A consumer may rely on whichever is most favourable, so the return record needs a legal basis with its own anchor date rather than one reason code. The bars are not to scale; the anchors are the point.

A consumer may rely on whichever is most favourable, so the model needs a **legal basis on every return**, each with its own anchor date and clock, plus the link to the original sale line. One `return_reason` list cannot carry it: "faulty" is a reason, section 56 is a basis, and the same reason arises under three bases with three different outcomes. Then section 61 on product liability, which applies along the supply chain and survives the exemptions [2] — for an own-label range that means **batch traceability from receipt to sale**, which for food and medicines means batch capture at goods-receiving and at the till, not only in the distribution centre.

Lay-by is not a sale and not credit

Section 62 applies where a supplier agrees to sell goods, accept payment in instalments and hold the goods until the full price is paid. A system routing lay-by through either the sales path or the credit path gets the statute wrong [8]. Each instalment **remains the consumer's property** until delivery, and

the goods remain at the supplier's risk until then. If the supplier cannot deliver after full payment, and the shortfall was not beyond its control, **the refund is double the amount paid** — with a failure to handle routine business matters diligently expressly not beyond its control. On early termination, or failure to pay within **60 business days of the anticipated completion date**, a penalty may be charged — but **not where non-payment resulted from death or hospitalisation**, and only if its fact and extent were disclosed before the agreement was made.

So lay-by needs its own entity: agreement and anticipated completion dates, the instalment schedule, instalments held as **consumer-owned funds** rather than revenue, goods reserved at a named location with the retailer bearing risk, the disclosed penalty terms with proof of pre-agreement disclosure, and a sixty-business-day clock. Recognising the instalments as revenue is an accounting error and a statutory one; leaving the reserved goods in the store's sellable bucket is a second, and it is the one that oversells a promotion weekend. The maximum penalty is prescribed by regulation 34 of the Regulations; I could not retrieve the figure from a source I trust, so this book prints none.

Promotional competitions are a records obligation

Section 36 with regulation 11 makes a competition a prescribed-record exercise rather than a marketing one [9]. The cost of an electronic entry, **including all subsequent electronic communication about that entry**, is capped at **R1.50**; prizes of R1.00 or less fall outside the definition; and there must be **independent oversight by an accountant, registered auditor, attorney or advocate**. The prescribed record set runs from the rules and the prize list through the marketing materials with a log of where and when each ran to signed prize-receipt acknowledgements carrying winner identity numbers — kept **at least three years** and produced to the Commission forthwith, at the promoter's own expense, on written request.

The 2026 amendment regulations, and the date to plan to

This is the newest material change in South African retail compliance and the one most likely to be mis-planned, because the legal date and the operational dates are not the same.

The amendment regulations operationalising the national opt-out registry under section 11(6) were gazetted as **Government Notice R. 7380 in Government Gazette 54521 on 15 April 2026**, and they **take effect on publication, being 15 April 2026** [10][11]. They oblige **registration** with the Commission, renewable annually, with **no contact with any consumer for direct marketing unless registered**; **cleansing of databases at least monthly** against the registry; a **central** block, so a consumer registering once covers all marketers; and **identification** in every communication, with dissemination barred from any platform where the originator cannot be identified [10][11]. Compliance does not displace POPIA, and section 69 in particular [11].

> **TWO DATES, AND THE SOURCES DO NOT AGREE ON WHICH GOVERNS**
>
> The regulations took effect on publication, 15 April 2026 [10], and one commentary describes the obligations as already mandatory [11]. The Commission announced the registry's launch on **7 October 2026**, with **Phase 1, marketer registration, from 15 September 2026 to December 2026, Phase 2, cleansing, from December 2026 to April 2027**, and **consumer pre-emptive blocking only from May 2027** [12]; it cites spam-call volumes above 17 billion in six months as the driver [13]. The legal position and the operational position are different things. **Plan to the earlier date.** A programme planning to May 2027 is planning to the date the consumer-facing function switches on, not to the date the obligation began — and the registration obligation is not phased in any way a retailer can rely on.

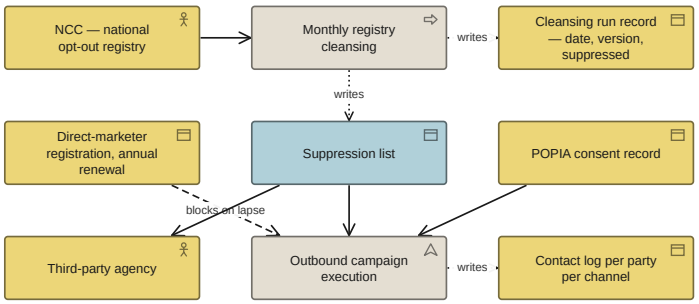


Figure 15.4 The suppression list has to outrank the channels, not sit inside one of them. Monthly cleansing against the registry is a provable obligation, so the cleansing run needs its own audit record. Note that the CPA pre-emptive block and the POPIA consent record are separate objects with separate legal effects — one flag satisfies neither.

It lands on the loyalty and customer estate: a **registration record** with a renewal date and a control that blocks outbound campaigns when it lapses; a **cleansing job with a per-run audit record** — run date, registry version, records suppressed — because an un-logged monthly job proves nothing; a **suppression list authoritative over every outbound channel**, third-party agencies and in-app push included; and a **contact log** per party per channel per campaign. The CPA suppression list and the POPIA consent record are **two objects with different legal effects**, and a single `marketing_opt_in` flag satisfies neither.

POPIA, with the retail bite

The Protection of Personal Information Act 4 of 2013 commenced substantially on 1 July 2020 and became enforceable on **1 July 2021**. These are the retail-specific points.

Loyalty data and purpose limitation. The loyalty programme is the retailer's largest holding of personal information and its most purpose-ambiguous. The Regulator's position is explicit: using loyalty data for direct marketing outside the original purpose is further processing under section 15, and its own worked example — sharing a member's data with group companies for marketing — **requires written consent** [16]. Read that against a multi-

banner group building a single customer view by joining banner datasets: without that purpose consented at enrolment, the join is a breach dressed as an architecture decision.

Direct marketing and section 69. The Regulator's Guidance Note sets out the reading to build to [16]. For non-customers, consent before electronic direct marketing; the first message may itself request it; the party may approach a person **once only**, and only if they have not previously refused. Consent must use **Form 4 or a substantially similar form**, recording the goods marketed and **the communication method the person chooses — which the party must then use**; telephone consent requires reading out Form 4 and recording the call; and the **burden of proof** sits with the responsible party. For existing customers the section 69(3) test applies: the contact detail was obtained in the context of a sale, the marketing is of the party's own similar products, and the customer had a free and simple opportunity to object at collection and on every communication. The Regulator also treats **telephone calls as electronic communications**, which brings the whole outbound call centre inside the regime. The Note carries no date on its face; commentary places publication in December 2024 [16].

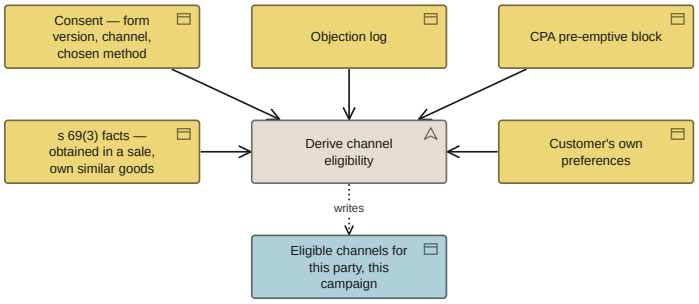


Figure 15.5 Eligibility to contact is a derivation, not a flag. Four dated facts resolve into one answerable yes or no per channel per campaign, computed at send time. Stored as `marketing_opt_in` it will be wrong, and wrong in a way nobody can reconstruct when the Regulator asks which consent the message relied on.

CCTV and biometrics in stores. In-store CCTV processes personal information, and where it is used to identify individuals it edges into biometric information — special personal information under section 26, and where used as a unique identifier for a purpose other than collection, subject to **prior authorisation** under sections 57 and 58 [17]. I could not source retail-specific guidance on in-store facial recognition or shelf-camera analytics, nor establish whether prior authorisation has ever been granted or refused for a retail deployment, so this book states the obligation and not the practice. It forces a **surveillance asset register**: camera, location, purpose, lawful basis, retention, access list, and whether analytics on the feed produces a biometric template — with the evidential hold on footage used in a shrink investigation recorded against that footage, or the blanket retention rule deletes the evidence.

Automated decisioning. Section 71 prohibits a decision with legal consequences, or affecting the data subject substantially, based solely on automated processing intended to create a profile. In retail it bites on the store-card credit decision, automated fraud blocking of an account or refund, personalised pricing, and an automated refusal to accept a return — each needing a human-review path, a representation mechanism, a renderable reason and the retained inputs that make the reason true.

Cross-border transfer and the information officer. Section 72 permits transfer abroad only on a listed ground, which reaches every marketing platform, offshore analytics environment, group data lake and supplier portal hosted outside the country. And the duty architects most often miss: **each subsidiary must register its own information officer and deputies**, before taking up duties, with the PAIA manual matching what the Regulator was told [18]. A group with a legal entity per banner, per property company and per franchise vehicle has an obligation per entity — which makes the legal-entity structure a data-governance object.

Pharmacy and health products

This is the most architecture-relevant regulation in retail and most reference material skips it. A South African pharmacy is a **separately licensed professional practice operating inside a shop**, and nearly every instinct from the rest of the estate is wrong inside its walls.

The Pharmacy Act 53 of 1974 establishes the South African Pharmacy Council and requires every pharmacy to be recorded with Council as well as licensed. The **Regulations relating to the Ownership and Licensing of Pharmacies, GNR.553 of 25 April 2003, commenced 2 May 2003** [19]. They opened ownership beyond pharmacists on conditions that are unexpectedly structural: a community pharmacy may be owned by any person who **is not an authorised prescriber**, does not hold an interest on behalf of such a person, and **holds no interest in a manufacturing pharmacy** — and for a body corporate those conditions apply to its **shareholders, directors, trustees, beneficiaries or members** [19]. Licensing requires proof of ability to meet Good Pharmacy Practice standards, **a licence per premises with conditions**, and **notification to Council within 30 days of issue and before any services are provided** [19].

Two consequences follow, and the second surprises people. A **pharmacy licence entity per site**, with conditions, Council reference, dates and the responsible-pharmacist designation, plus a block preventing the dispensary trading when either lapses. And an **ownership-interest register** structured enough to test the prescriber and manufacturing prohibitions against shareholders and directors — which makes the group's own shareholder register a compliance dataset.

The Medicines and Related Substances Act 101 of 1965 governs registration, manufacture, import, distribution, sale and advertising, and establishes the schedules; the South African Health Products Regulatory Authority replaced the Medicines Control Council as regulator. Substances are scheduled **0 to 8**: Schedule 0 in open retail, Schedule 1 and 2 through a pharmacy or authorised person, Schedule 3 and above on prescription, Schedule 5 and 6 with register and storage obligations. The Authority publishes consolidated schedules, most recently those of **August 2025** [20], and amends them by gazette repeatedly — which makes **schedule a dated attribute of the article, not a constant**. Section 22C(1)(b) licences are required to wholesale or distribute,

so a group running its own pharmaceutical distribution centre needs one for that site. And since **Notice 669 of 2024** the Council prohibits the sale of tobacco and e-cigarettes in pharmacies [25] — a site-category sales block, not a merchandising preference.

The schedule therefore drives POS behaviour: whether the item may be sold at this site at all, whether a pharmacist must be present, whether a prescription is required, whether a register entry is generated, and whether a quantity limit applies. A gazette amendment silently changes what a till may sell, so schedule-change ingestion is a process with an owner.

The Single Exit Price, and the year there were two of them

Section 22G and the Regulations relating to a Transparent Pricing System for Medicines and Scheduled Substances, 2005, create the Single Exit Price: **the only price at which a manufacturer may sell a medicine to any person other than the State**, and the price the final seller charges the patient or scheme [21]. Every private buyer pays the same price regardless of volume, so bulk purchasing cannot negotiate it down, and wholesalers add no mark-up [21]. This is the clearest case in retail of a price the retailer does not set: no price zone, no promotion, no markdown, no margin optimisation on a scheduled medicine, and the pricing engine bypassed entirely for those lines.

> **THERE WERE TWO SEP ADJUSTMENTS IN 2026, AND MOST SOURCES GIVE ONE**
>
> The annual adjustment for 2026 was **1.47%**, down from 5.25% in 2025, effective in **January 2026** [21][23]. An **extraordinary second adjustment of 2.88% took effect on 1 October 2026**, applying uniformly to all medicines registered and listed on the national pricing database **as at 30 September 2026**, including related pack sizes, in the private sector only; the Department of Health announced it on 30 September 2026 and described it as a response to exceptional domestic and geopolitical pricing pressures [22][23]. The announcement of the October increase does not itself mention that it is the second for the year [22]. **Any source giving "the 2026 SEP increase" as a single figure is wrong.** I could not source the gazette notice numbers for either adjustment.

The dispensing fee is capped annually, differs for pharmacists and for licensed dispensing practitioners, and is a flat amount plus a percentage of the exit price across four bands, the percentage falling as the price rises [21]. The bands published on 26 September 2025 in Gazette 51352 are these [24]:

SEP band	Pharmacist dispensing fee
Below R159.00	R23.13 + 46% of SEP
R159.02 to below R423.55	R42.91 + 33% of SEP
R423.55 to below R1 530.72	R112.59 + 15% of SEP
R1 530.73 and above	R270.54 + 5% of SEP

Treat that as the September 2025 position and check it: a 2026 notice exists — **GN 7599, dated 19 June 2026** — whose band values I could not retrieve [26], so the 2025 bands may have been superseded.

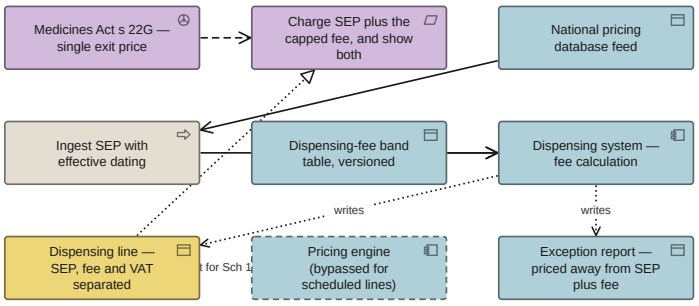


Figure 15.6 A price the retailer does not set. The exit price arrives as a national feed and must be applied exactly; the dispensing fee is a banded formula whose parameters change by gazette. Both are dated artefacts — which is what the two 2026 adjustments proved. The pricing engine is bypassed entirely for a scheduled line, and the exception report is the control that proves it.

What that forces is specific and testable: **a SEP ingestion process** with effective dating, because an adjustment re-prices thousands of lines on one date; **the dispensing-fee calculation as a versioned band table with effective dates**, never as constants in code; **SEP, dispensing fee and VAT separated on the dispensing line**; and **an exception report** for any line priced away from the exit price plus the capped fee. A system that treated "the 2026 SEP" as a single annual load was wrong from 1 October.

> **NORTH AMERICAN ASIDE — PHARMACY PRICING**
>
> Canada has no single exit price: formulary and generic prices are negotiated province by province and reimbursement is adjudicated in real time against the payer's formulary. So instead of one regulated national price with a capped fee, a Canadian estate needs payer-specific price and fee resolution at the point of dispensing, with a claims round-trip inside the transaction. A system built for the single exit price will not price a Canadian script, and one built for Canadian adjudication will over-engineer a South African dispensary.

Good Pharmacy Practice, and a rule that turns a database into a compliance obligation

The Rules relating to Good Pharmacy Practice, published under section 35A of the Pharmacy Act, set the professional standards a recorded pharmacy must meet [27]. Three of them carry architectural weight.

Rule 1.2.11.4(f): where software provides warnings or interaction checks, the latest version must be used [27]. Read that as an architect. A drug-interaction content database stops being a reference dataset and becomes a **regulated currency obligation on the compliance critical path**. Not patched quarterly. Latest. And the consequence falls on exactly the parts of the estate least able to bear it: a dispensary on an intermittent rural link, a store trading through a power outage on local capability, a dispensing system that holds its clinical content locally because it must keep working when the line is down. The design has to deliver content currency to an offline-capable endpoint and *prove* it delivered, per site, per version — the same confirmed-delivery pattern as the price file, with a professional-conduct consequence instead of a consumer-law one.

Temperature. Thermolabile products must have storage temperatures recorded at least twice daily at intervals of not less than seven hours and reviewed daily, with annual mapping and annual calibration of the recording instruments [27]. The chiller is a monitored asset with a calibration certificate, not a fixture.

Records. Recording of medicines and scheduled substances follows Regulation 11 of the General Regulations under the Medicines Act [27]. The rules on dispensing records and the Schedule 5 and 6 register did not render in the text available to me, so this book states no retention period for a dispensing record — only the design rule: pharmacy retention runs on a clinical clock, not a commercial one, and must not inherit the transaction retention schedule.

Food labelling and safety

The parent statute is the **Foodstuffs, Cosmetics and Disinfectants Act 54 of 1972**. Enforcement is by **environmental health practitioners of local authorities** and the national Department of Health — which is why the inspection happens in the store and findings arrive one municipality at a time. Food premises need a **certificate of acceptability** per premises under the general hygiene regulations, R638 [28]; I could not confirm R638's gazette number or commencement date.

R146 is the labelling law in force, published as **Government Notice R.146 in Government Gazette 32975 on 1 March 2010**, with one promulgated amendment, **R1091 of 19 November 2010** [29][30]. Its mandatory particulars are the architecture: the name of the foodstuff; the manufacturer, importer or seller; the ingredient list in descending order of mass; storage conditions; net contents in SI units; **country of origin**; **batch identification**; and **date marking** using *best before*, *use by* or *sell by*, supported by a shelf-life study [29]. Allergens sit in regulation 43 and the list is **eight categories** — egg, cow's milk, crustaceans and molluscs, fish, peanuts, soybeans, tree nuts and significant gluten-containing cereals; **sulphites are not on it** [29]. Nutrition information is **not compulsory on all prepackaged food** — it is triggered principally by a claim — and where a claim is made the analysis must come from an accredited laboratory and be **re-verified every three years** [29].

> **R3337 IS STILL A DRAFT, AND THIS BOOK WILL NOT PRINT ITS NUMBERS**
>
> The draft labelling and advertising regulations **R3337 were published for comment in Gazette 48460 on 21 April 2023** [31][32]. **As at October 2026 R3337 remains a draft and is not in force. R146 remains the law.** A July 2026 review records the Department of Health as reviewing it and expected to engage NEDLAC in May 2026, with no official public update, and warns that regulations passed since R3337 would conflict with it in its current form [33]. An October 2026 opinion piece says only "**if** the regulations are now before NEDLAC" [34]. Treat 21 April 2023 and Gazette 48460 as best supported; one summary instead gives "late 2023" [32]. R3337 would require warning labels on products above thresholds for added sugar, sodium and saturated fat, assessed against a nutrient profiling model in Annexure 8 [31][33]. **The Annexure 8 thresholds and the transition period are unverified and are not printed here.** Nor should a programme plan to them.

The practical instruction for an architecture being built now is three sentences. Build the nutrient-value fields per article today, because R146 needs them the moment a claim is made. Keep warning-label status as a **derived indicator** rather than a stored flag, so it recomputes when a threshold is finally gazetted. And do not commit pack artwork until the final regulations are published — because a front-of-pack warning changes pack face dimensions, which changes facings, which changes planograms, for thousands of articles at once.

Two instruments limit what is *in* the food rather than what is on the label, which makes them **supplier-specification** data. The sodium regulations, **R214, signed 18 March 2013 and published 20 March 2013**, set maximum total sodium per 100 g for listed categories — bread, cereals, fat spreads, savoury snacks, processed meats, soup powders and others — with target dates in the regulation's own table; regulation 4 was substituted by Notice 1071 in Gazette 41164 of 6 October 2017 [35][36]. I could not verify the limits or the table dates; read the gazette. The trans fat regulations were signed in **February 2011** and cap trans fat at **2 g per 100 g** of any oil or fat, with a "trans fat-free" claim requiring less than 1 g per 100 g [37].

The regulation number is itself contested: the World Health Organization's own feature calls it "Regulation 249" [37] while legal databases catalogue it as **R127 of 2011** [38]. Cite both, or cite the gazette.

So the article master needs **sodium per 100 g** and **trans fat per 100 g of fat** as specification attributes with a supplier declaration and a verification date, plus a category-to-limit mapping so a breach is detectable at listing rather than at inspection. For an own-label range the retailer is the manufacturer and carries the primary obligation. And under the **Agricultural Product Standards Act 119 of 1990**, which governs grading, packing and display [39], fresh produce carries a **grade axis behaving like a variant axis**: a model treating grade as a quality note can neither price nor report Class 1 and Class 2 apples.

Tobacco: the Act in force, and a Bill that is not

The Act in force is the **Tobacco Products Control Act 83 of 1993**, effective **1 February 1994**, as amended through 2008 [40]. Its subordinate instruments are what bind a store — chief among them **R.976 of 2000**, in force 1 January 2001, prescribing point-of-sale signs including availability and price signs, notices that sales to minors are prohibited, and health warnings [40].

THE BILL'S STATUS, AND A STALE MARKER THAT IS WIDELY REPEATED

The **Tobacco Products and Electronic Delivery Systems Control Bill [B33-2022]** was introduced in the National Assembly on 9 December 2022. The Parliamentary Monitoring Group's bill page still shows it as **"Lapsed" on 21 May 2024** under National Assembly rule 333(2) [41]. **That marker is stale.** The Bill lapsed at the end of the Sixth Parliament and was revived in the Seventh. The Portfolio Committee on Health **adopted the motion of desirability on 24 June 2026**, ten in favour and one against, with the chairperson framing the vote as a mandate to improve the Bill rather than approval of its text [42]; and **clause-by-clause deliberation began on 12 August 2026**, covering the preamble and definitions only, with the Committee recognising a distinction between combustible and non-combustible products and continuing its work on the remaining sections [43]. So as at mid-October 2026 the Bill is **in clause-by-clause committee deliberation — not passed, not assented to, not in force**. Do not plan to it as law, and do not repeat the lapse date.

What is sound today is modest: a **tobacco flag and age restriction on the article**, the prescribed point-of-sale signage as a site compliance record, and the Council's pharmacy prohibition as a site-category sales block. What is worth anticipating is that the Bill as introduced is commonly described as providing for plain packaging and a point-of-sale display ban — though the Committee has signalled amendment, so the final provisions are unknown. A display ban moves product behind a gantry with no customer-facing facing, which breaks facing-derived shelf capacity. Plain packaging collapses brand differentiation on pack, which removes the visual pick confirmation store replenishment relies on. Both are space changes driven by a labelling statute.

Liquor, and what it binds at the till

The **Liquor Act 59 of 2003** is a national statute concerned principally with manufacturing and distribution. The **retail sale of liquor is licensed provincially**, under nine provincial liquor acts — the Western Cape Liquor Act 4 of 2008 is representative [45] — administered by provincial authorities, with municipalities regulating zoning and local trading conditions.

The **National Liquor Norms and Standards** include clause 4.14 on uniform trading hours, which sets a table of **minimum trading-hour standards that apply in municipal areas where by-laws are not enforced** [44]. That is the trap: the national instrument sets a floor for a residual case, while the actual hours differ by province, by municipality and by licence type. I could not source a current national table, and an architect should not expect one to exist.

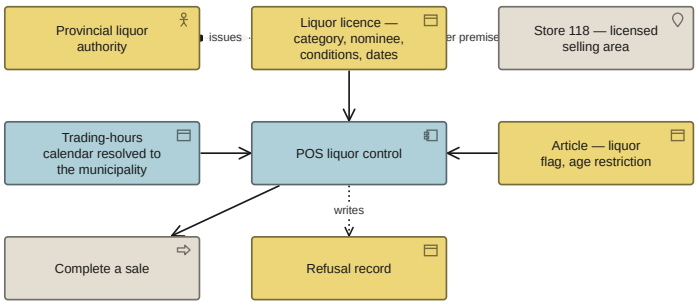


Figure 15.7 The licence is per premises and the hours are municipal, so the control has to be too. A single national trading-hours configuration fails in both directions at once: unlawful sales in one province, lawful ones blocked in another. The refusal is itself a record, because an inspector asks what the till did when it should have said no.

So: a **licence entity per site per licence type**, carrying category, authority, number, nominee, conditions and validity dates; a **trading-hours calendar resolved to the site's municipality**, public holidays included; a liquor flag on the article; and a **POS control that refuses a sale outside permitted hours or outside the licensed selling area, and records the refusal**. A single national trading-hours configuration is the commonest failure here, and it fails in both directions at once — unlawful sales in one province while lawful ones are blocked in another.

Competition: exclusive leases, trade terms and buyer power

The Grocery Retail Market Inquiry was initiated by Commission notice on 30 October 2015, with the **Final Report published 25 November 2019** [46]. Its headline finding was on **exclusive lease agreements**: long-term exclusivity in shopping centres entrenches the national chains and blocks smaller retailers from malls, with initial terms generally ten years and renewals extending some contracts beyond thirty. It recommended that chains stop enforcing exclusivity against SMMEs and other grocery retailers in non-urban centres immediately, that new leases contain none, and that remaining enforcement end at the next lease extension or within five years — about 25 November 2024 [46]. Whether all enforcement has ceased, and whether legislation followed, I could not establish as at October 2026. On supplier terms it recommended that trade terms be offered **uniformly**, each with an **objective justification**, and **volume not a basis for qualifying** [46] — so a trading term has to be comparable across suppliers: structured fields and a stored justification, not free text in a contract.

Buyer power. Section 8(4) prohibits a dominant firm in a designated sector from imposing unfair prices or trading conditions on SME or historically disadvantaged suppliers, and from avoiding buying from them to circumvent the prohibition [47]. The **Buyer Power Regulations were issued in Government Gazette 43018 on 13 February 2020** and the Commission's **Enforcement Guidelines are dated 12 May 2020**, with the Commission's own position that there is **no grace period**; the Guidelines give no commencement date for section 8(4) itself [47]. The designated sectors include **grocery wholesale and retail** [47].

Four numbers are worth memorising. Above **35%** market share is a rebuttable presumption of dominance and above **45%** it is not rebuttable — but **a buyer with a material share, typically 15% or more, may still be dominant on buyer-power grounds** [47]. HDP protection applies only where the supplier provides **less than 20%** of the buyer's purchases of the relevant good, and the Commission **screens price differences at 3%** [47]. Annexure 2

lists practices considered unfair in grocery, including **payment later than 30 days from delivery**, short-notice cancellation of perishables and charges unrelated to the sale — with listing and promotional charges unfair unless clearly agreed in the contract and reasonably costed [47]. Separate guidelines exist for SMEs in the same sectors [48].

Then the part that makes this the most specific data requirement in South African competition law: **the onus reverses**. Once the Commission has a prima facie case, the respondent must show the price or condition was not unfair [47].

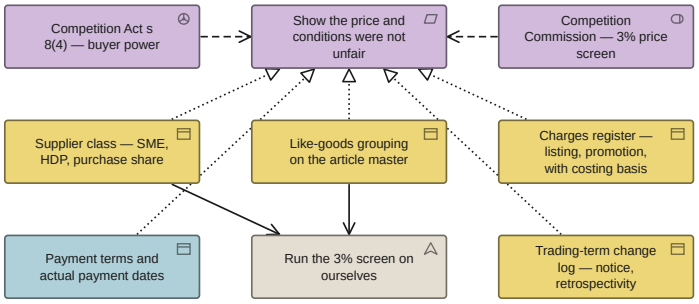


Figure 15.8 The onus reverses, so the dataset is the defence. Once the Commission has a prima facie case the retailer must show the price or condition was not unfair — which means producing the price paid to every supplier of like goods, with each supplier's size and ownership status, dated. The like-goods grouping is a classification the retailer has to assert and defend; nobody else will supply it.

To discharge that, a retailer must produce on demand, for each comparable good, the price paid to **every** supplier of like goods with each supplier's size and ownership status, over the period. Almost no retailer maintains that deliberately. It needs supplier classification by the dual employment-and-turnover test, dated and evidenced; a **like-goods grouping** on the article master, because that is a classification the retailer must assert and defend; payment terms and **actual payment dates** per invoice, so the thirty-day benchmark is measurable; a **charges register** per supplier per charge type with the contract reference and costing basis; and a **3%-screen report run on itself**, because the Commission's screen is public.

Tax, credit, measurement and the card estate

VAT. The Value-Added Tax Act 89 of 1991 at a standard rate of 15%, with basic foodstuffs **zero-rated** under Part A of Schedule 2 [49]. The boundaries are the architecture problem, because zero-rating is defined at a granularity finer than any commercial category: brown bread is zero-rated, brown bread rolls are not; milk is, flavoured milk is not. **There is no hierarchy under which the VAT rate is inheritable**, which is why regulated classification is a separate hierarchy from the merchandise one. I could not verify the exact current contents of Part A, and SARS has issued rulings because the boundaries are contested.

The till slip is a document, not a report. Per SARS a **full tax invoice is required above R5 000**, an **abridged invoice at R5 000 or less**, and **none at R50 or less** — though a till slip showing the VAT is still needed to support an input-tax claim — and the invoice must be issued **within 21 days** of the supply [50]. The prescribed particulars, effective from **8 January 2016**, run from the words "Tax Invoice" through the supplier's VAT number to the recipient's details where the recipient is a vendor [50].

So: a **VAT code per article** as a governed, auditable attribute with a change history and an approval trail — a wrongly standard-rated staple over-collects from the poorest customers and a wrongly zero-rated line under-declares to SARS. **VAT computed and stored per line**, never per basket at a blended rate. And a till-slip engine able to produce a compliant full invoice above R5 000, which requires the POS to *capture* a business customer's name, address and VAT number — a capability most retail POS deployments lack until the first cash-and-carry customer complains. One contrast for a multi-market design: South Africa needs **one VAT code per article**, North America **a taxability matrix of article category by jurisdiction**.

Credit. The National Credit Act 34 of 2005, regulated by the National Credit Regulator — and in a South African retail sentence "NCR" means that Regulator, not the former point-of-sale manufacturer that split into two companies in 2023. The Act reaches store cards, instalment agreements on furniture and appliances, and the apparel credit book, and the threshold for compulsory registration as a credit provider was **reduced to nil with effect from 11 November 2016** [51]; the gazette number is inferred rather than stated in the source I had. What bites is the affordability assessment against prescribed expense norms, with reckless-lending exposure defended only by the retained assessment, declines included. The retail-specific points are three. The credit decision happens **at a till or in an app, in minutes**. The **same party is simultaneously a loyalty member and a credit consumer**, with different lawful bases and retention rules. And **lay-by is not credit** — it is CPA section 62 — so a system routing lay-by through the credit engine applies the wrong statute to it.

Measurement. The Legal Metrology Act 9 of 2014, administered by the National Regulator for Compulsory Specifications, governs measuring instruments used for prescribed purposes and the declared quantity of prepackaged goods. Its commencement is contested: gov.za records **1 August 2014 per Government Gazette 37887** [52], while the Act's own section 43 provides for commencement by proclamation with no date in the text [53]. Note the uncertainty and move on, because the **Legal Metrology Regulations**, published as **Government Notice 877 in Government Gazette 41854 on 24 August 2018**, are what bind the store [54]. Three of them reach a retailer directly. **Anyone buying prepackaged goods for resale must take steps to ensure the actual quantity matches the stated quantity** — a direct obligation for goods somebody else packed. **Price displays must use the required unit-price format.** And an instrument without valid verification must be verified before use, with a **re-installed instrument treated as unverified until verified in its new position**.

So the scale — at the deli, the butchery and the produce counter — is an **asset with a verification lifecycle**: an instrument register per site with model-approval number, verification date, next-verification date and installation position, and a **re-verification trigger on relocation**, the control nobody builds, and a store refit moves scales. Add a **non-conforming segregation state** in the inventory status buckets, because the regulations require non-conforming goods to be kept physically separate.

Distance selling. The Electronic Communications and Transactions Act 25 of 2002 requires, in section 43, prescribed information on the website — identity and legal status, physical address, a description of the goods, the full price including transport and taxes, dispatch times and the return and refund policy; I could not render the exact enumeration, so read the Act. Section 44 is clearer: a consumer may cancel without reason and without penalty **within seven days after receipt of the goods**, the only charge is the **direct cost of returning them**, and a **full refund is due within thirty days** [31]. That needs a **disclosure-page version record** with effective dates, a **delivery-receipt timestamp** treated as an authoritative fact, and a measured thirty-day refund service level.

PCI DSS is different in kind: imposed by the card schemes through acquirer contracts. It is **not South African law**, and that is the point — no regulator to appeal to, no commencement proclamation, no national derogation. **Version 4.0.1 was published on 11 June 2024**; **v4.0 was retired on 31 December 2024**; and the **future-dated v4 requirements became effective on 31 March 2025** [55]. Three structural consequences, all expensive to retrofit. The primary account number **is not a key, a join column or a search field**, and sensitive authentication data must not be retained after authorisation at all. **Tender detail must be physically separable from sales line items**, or the entire multi-year sales history falls inside the audit scope — the single most expensive avoidable decision in a retail data architecture. And **retention is a per-data-class schedule**, because card data carries a shorter mandated retention than the transaction class.

Packaging, empowerment and the board

Extended producer responsibility. The EPR Regulations, 2020, made under section 69(1) of the National Environmental Management: Waste Act 59 of 2008 and published as **GN 1184 in Gazette 43879 on 5 November 2020**, as twice amended, with regulation 14 setting **commencement on 5 May 2021** [56].

Here is the part that catches retailers. A "producer" includes a **brand owner** — a person who makes or sells an identified product under a brand label. **A retailer with an own-label range is therefore a producer**, with registration, scheme membership, an annual submission of **quantities placed on the market, collected, diverted from landfill, exported and landfilled**, fees applied **proportionally based on products placed on the market**, and penalties reaching fifteen years' imprisonment, a fine, or both [56].

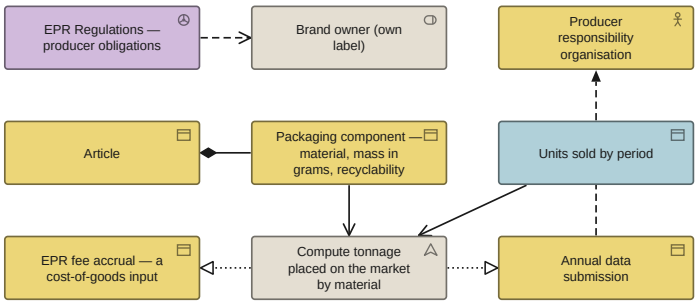


Figure 15.9 The obligation that cannot be back-filled. An own-label range makes the retailer a brand owner and therefore a producer. The fee and the return both run on packaging mass by material, so component mass has to be captured when the article is created — re-collecting it later from every supplier for every own-label line is a programme, not a data fix.

That is the obligation in this chapter that **cannot be back-filled**. Fee and return both run on packaging mass by material, so the article master needs, per article: packaging components, material type per component against the scheme's governed list, **mass per component in grams**, recycled content and recyclability classification. Multiplied by units sold, that is the return. A retailer that did not capture component mass at article creation must re-collect it from every supplier for every own-label line, which is a programme, not a data fix.

B-BBEE. The Codes of Good Practice make the supplier record a compliance dataset, and the requirement architects miss is that procurement spend must be classified by the supplier's attributes **as at the transaction date** — so B-BBEE status is a dated version, and a retailer storing only the current certificate cannot reconstruct a prior year's scorecard. Proposed amendments to the generic Codes were gazetted on **29 January 2026** for comment and were not final at the time of the commentary I relied on [57]; whether they have since been gazetted finally I could not establish.

King IV. The King IV Report on Corporate Governance for South Africa, 2016, is effective for financial years beginning on or after **1 April 2017**, applied on an apply-and-explain basis and incorporated by the JSE Listings Requirements — which is how it becomes binding in practice for a listed retailer [58]. **Principle 12** makes the **governing body**, not management and not IT, accountable for technology and information: the governance framework, the value delivered by technology investment, information assets including security and privacy, technology risk including continuity and third-party risk, the ethical use of data, and disclosure in the integrated report. King IV separates "technology" from "information" deliberately, because governing data as an asset is a different job from governing the platforms it sits on. I could not verify the exact wording and numbering of Principle 12's recommended practices; read the Report.

King IV is what turns the rest of this chapter into a governance artefact rather than a backlog. It forces an **information asset register** with named owners — in a retailer, the article master, the transaction log, the loyalty database and the dispensing records, four holdings with four owners and four regimes — and an **obligations register** keyed to instrument, obligation, owner, evidence and system, from which the disclosure pack is produced rather than written from scratch each year. A board cannot apply Principle 12 to a data estate it cannot see described, which is the strongest internal argument an enterprise architect in a listed South African retailer has.

What an inspector actually asks to see

Strip away the statutes and the inspection is a short list of questions, each of which a system either answers from data or does not. Test an estate against this list.

1. **"Show me the price you displayed for this article in this store on this date"** — price versions with effective timestamps, scoped to site and channel — and **"which shelf labels did not print last night?"**
2. **"Show me the error you corrected and who you told"** — the price-error record, both limbs — and **"this return, the basis you accepted it on, and the original sale line."**
3. **"Show me your monthly registry-cleansing runs for the last year,"** with registry versions, and **"the consent you rely on to have sent this message on this channel."**
4. **"Show me this pharmacy's licence, its responsible pharmacist on that date, and the version of the interaction-checking software the dispensary was running."** Three records, one of which almost nobody keeps.
5. **"Show me the exit price and the dispensing fee applied to this line."**
6. **"Show me the verification certificate for this scale and the date it was last moved."**
7. **"Show me the price you paid every supplier of this comparable good last year, with their size and ownership status."** The one most retailers cannot produce.
8. **"Show me your packaging tonnage by material placed on the market, reconciled to units sold,"** and **"your obligations register, with the system that satisfies each obligation."** The King IV answer, and the artefact an architect should own.

Notice what is not on that list: a policy document, a framework, a control matrix. Every question is answered by a record that either existed at the time or did not — which is why these constraints belong in the model rather than in an appendix to it.

THE CHAPTER IN TEN LINES

- Retail regulation is operational: it reaches the till, the shelf label, the scale, the chiller and the dispensary, and it is enforced in the store during trade rather than through a quarterly return.
- Most obligations attach to a site rather than to the enterprise, which makes the location master a regulatory object carrying dated licences with conditions.
- CPA s 56 gives the consumer six months from delivery with the consumer electing repair, replacement or refund at the supplier's risk and expense — a per-sale-line statutory clock, not a configurable returns policy.
- CPA s 23(6) makes the lower of two concurrently displayed prices binding, so a shelf label that silently failed to update is a legal exposure rather than a data-quality issue, and the s 23(9) defence has two limbs and both are data.
- The 2026 CPA amendment regulations took effect on publication, 15 April 2026, while the Commission's registry phases run to May 2027 — the legal and operational dates differ and a programme should plan to the earlier one.
- POPIA's retail bite is loyalty purpose limitation, s 69 consent with its Form 4 evidence, in-store biometrics, s 71 automated decisions and a per-subsidiary information-officer registration.
- A pharmacy is a separately licensed practice inside a shop: licence per premises, ownership conditions tested against shareholders and directors, schedule as a dated article attribute, and a price the retailer does not set.
- There were two single-exit-price adjustments in 2026 — 1.47% from January and an extraordinary 2.88% from 1 October on medicines listed as at 30 September 2026 — so the price feed and the fee band table must both be dated artefacts.
- Good Pharmacy Practice rule 1.2.11.4(f) requires the latest version of interaction-checking software, turning a content database into a regulated currency obligation with real consequences for offline stores.
- R146 of 2010 remains the food labelling law; the R3337 draft of April 2023 is still a draft, its Annexure 8 thresholds and transition period are unverified, and nothing should be planned to them.
- The tobacco Bill [B33-2022] is in clause-by-clause committee deliberation, not passed; PMG's "lapsed 21 May 2024" marker is stale and the Act in force remains Act 83 of 1993.
- The buyer-power onus reversal is the most specific data requirement in South African competition law: the price paid to every supplier of like goods, with each supplier's size and ownership status, dated.
- Packaging component mass captured at article creation is the EPR obligation that cannot be back-filled, and PCI DSS — contractual rather than statutory — is why tender detail must be physically separable from sales lines.
- King IV Principle 12 makes the board accountable for technology and information, which is what turns an obligations register from a compliance chore into the architect's strongest internal argument.

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CHAPTER 16

Modelling retail in ArchiMate

Forty-two elements, a few hundred thousand articles and no dictionary. Retail's difficulty is not vocabulary — it is instance volume and time, and ArchiMate models neither. This chapter is the mapping, with the reasoning behind every choice, because the reasoning is what you will need for the concept it missed.

Forty-two elements, four hundred thousand articles

An architect arriving in a retailer wants a dictionary. Article equals this element, promotion equals that one. There is no such dictionary, and the reason is not that retail has too many words. It has fewer than banking. The reason is that retail's difficulty is **instance volume and time**, and ArchiMate models neither.

A mid-sized South African grocer carries a few hundred thousand articles across a few hundred sites, each with a price that changed at a particular minute, a facing count on a particular planogram version, and a stock position that is nine different numbers depending on who is asking. None of that is expressible in the notation. What the notation is good at is saying what *kinds* of thing exist, what behaviour acts on them, and which system is responsible — and that turns out to be exactly the argument a retailer cannot otherwise have, because the merchandising system, the promotions engine and the point of sale each believe they own the price.

So three things happen to the vocabulary.

Many concepts collapse onto one element. Shopper, loyalty member, patient, credit account holder, franchisee, supplier, buyer, planner, allocator, responsible pharmacist: all Role. The notation is telling you something true — these are positions held against something, for a period, and the differences live in the data.

Many concepts are not elements at all. Facings, case pack, lead time, cover, like-for-like, sell-through, open-to-buy, shrink, margin: attributes and derivations. Drawn as elements they produce a diagram that is unarguably complete and that nobody will read.

A few need a relationship rather than an element. Stock moving from a distribution centre to a store is a Flow. One promotion firing on three lines of one basket in a particular order is an Association whose cardinality and sequence ArchiMate cannot express.

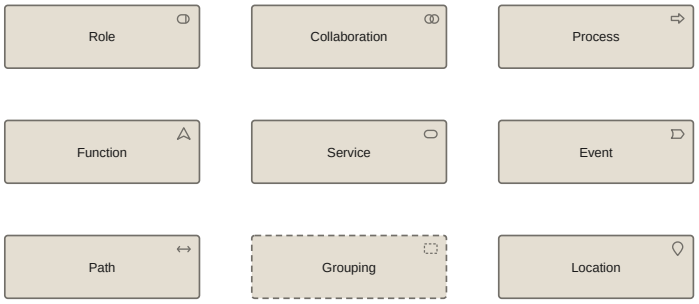


Figure 16.1 The nine elements that carry no layer. Role, Collaboration, Process, Function, Service, Event, Path, Grouping and Location are drawn in one neutral colour, so a Process looks identical whether it is business or application behaviour. Context tells you the layer — what the element is assigned from, what it realises, what it accesses. Retail needs the neutrality, because repricing an article is partly a buyer's judgement and partly a batch job.

> **THE RULE THAT PREVENTS MOST OF THE DAMAGE**
>
> ArchiMate models **types**, not instances, and it has no clock. There is no element for article 6001234567890, no element for Store 118's Tuesday price, no cardinality, no effective dating, no quantity. If you need to say "one style has one or more options, each with one or more SKUs, each with a dated price per zone", you need a data model, and this book's conceptual data model chapter exists for that. An architect who makes ArchiMate do entity-relationship work produces something that fails as both.

One more thing before the decisions. Nine elements carry no layer at all, so a Process looks identical whether it is business or application behaviour. Colour no longer tells you the layer; context does — what the element is assigned from, what it realises, what it accesses. That is uncomfortable for a fortnight and then useful, because retail is full of behaviour that sits across the seam. Repricing an article is partly a buyer exercising judgement and partly a batch job building a file, and a notation that forces you to pick one is lying for you.

Five decisions that settle most of a retail model

Nearly every argument in a retail model reduces to one of five questions. Learn each test and the rest follow by analogy.

Is the article a Product?

No. This is the first thing to get right and the one most reference models get wrong.

A **Product** in ArchiMate is a coherent set of services and objects offered as a whole, with a contract attached. The test is whether there are *terms*. A loyalty programme is a Product: it has terms, a membership, an earn-and-burn mechanic and a liability. So is a store card, a delivery subscription, an extended

guarantee, and a lay-by agreement. Each is something the customer enters into.

An **article** is a Business Object. It is a thing the retailer holds information about, buys, prices, ranges, counts, puts on a shelf and sells. Nobody signs anything to buy a tin of beans.

Why it matters: a Product carries a contract and a service bundle, so drawing four hundred thousand articles as Products implies four hundred thousand contracts and tells you nothing about the only structure that matters — the style, option and SKU hierarchy below, and the several classification hierarchies above. Model the article as an object and the merchandise structure becomes expressible. Model it as a Product and you have built a catalogue of nouns.

Is it a service or a process?

A **Service** is behaviour visible from outside, with a consumer who would notice its absence. A **Process** is the sequence inside that produces it. A **Function** is behaviour with no fixed sequence.

Retail gets this wrong in a particular way, because departments are named after functions — Replenishment, Pricing, Space, Loss Prevention — and those names arrive in the model as processes. "Replenishment" has no sequence; it is a function that runs nightly across hundreds of thousands of article-location pairs. "Release the store order to the distribution centre before the 21:00 cut-off" does have a sequence, and it is a process. The service, the thing the store actually consumes, is stock arriving on the dock in time to be on the shelf before trade.

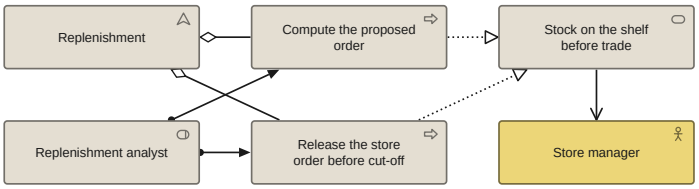


Figure 16.2 Service or process. "Replenishment" is the department's name and has no sequence, so it is a Function aggregating the processes beneath it. Each process has a sequence and a cut-off. The store manager consumes one service — stock on the shelf before trade — and does not care which engine computed the order quantity, which is exactly why that is a process and not a service. The Role is assigned to the processes, never to the service.

The test has two parts: would someone outside the boundary ask for this by name, and could you reorganise it completely without them caring? If both, it is a service, and the sequence behind it is a process. A store manager asks for stock on the shelf. She does not care which engine computed the order quantity, which is why that is a process and not a service.

Is a channel an interface, a location or an actor?

"Channel" conflates four things in retail, and the model has to separate them: the **party** (the shopper, a franchisee, a seller on the marketplace), the **place** (the store, the distribution centre, a pick-up point), the **access point** (the till, the app, the website, the call centre, the handheld), and the **banner** (the fascia the customer thinks they are shopping).

The party is a Business Actor in a Role. The place is a **Location**. The access point is a Business Interface with an Application Component behind it, and the physical device is a **Node**. The banner is a **Grouping**, because a banner is not a thing that does anything — it is a set of sites, articles, prices and terms that travel together.

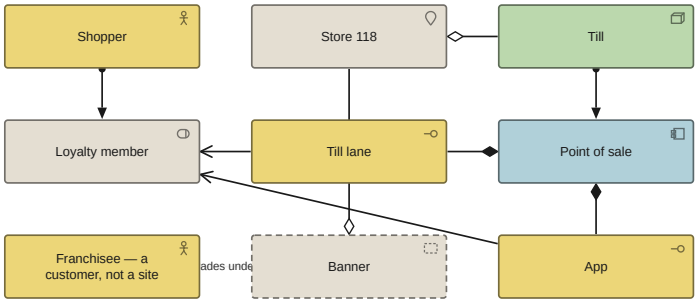


Figure 16.3 "Channel" is four things. The party is an Actor in a Role, the place is a Location, the access point is a Business Interface with a component behind it and a Node under it, and the banner is a Grouping — because a banner does nothing; it is a set of sites, prices and terms that travel together. Note the franchisee: drawn as a Business Actor the group serves, not as a site the group controls, because in a franchise estate there is often no price or stock position to master.

The test for interface versus actor is whether the thing can decide. A franchisee decides what to range; a till cannot decide anything. That matters more in retail than elsewhere because of the ownership question: in a franchise or voluntary-trading estate, **the group may not have a price, an assortment or an inventory position to master at all** for a large part of its turnover. Drawing the franchise store as a Location inside your own estate asserts control you do not have. Draw it as a Business Actor in a customer Role, served by group services, and the architecture tells the truth.

Is it behaviour or a thing?

A promotion is behaviour. The promotion plan is a thing. A price change is an event; the price version is a thing. A stock count is behaviour; the count and its variance are things. Dispensing is behaviour; the dispensing record is a thing, and a regulated one.

The test: can it be filed? If you can imagine retrieving it, showing it to an inspector and asking whether it was correct, it is an object. If it only happens, it is behaviour, and what you file is the record it wrote.

Retail models fail here in a consistent direction. They draw the artefact and lose the behaviour, so the model has a Promotion box and nothing that explains why a basket satisfying three offers can be discounted in several valid orders producing different totals.

Is it an element at all?

An element earns its place if behaviour acts on it, something depends on it, or someone owns it. If a concept is a property of another concept, write it in the data model and keep it out of the picture.

Facings, shelf capacity, case pack, order multiple, lead time, safety stock, forecast accuracy, sell-through, cover, open-to-buy, markdown depth, shrink rate, like-for-like: attributes and derivations, every one. Stock on hand is the sharpest case — it is not one number but nine, each a filter and a sum over an append-only movement log, so what the model should carry is the **movement** and the **status bucket**, not a quantity.

> THE COMPLETENESS TRAP
>
> There is a kind of retail model, produced around month four, that contains every noun in the merchandising system's data dictionary. Eleven hundred elements, it validates, and it answers no question anyone asked. The test for an element is not "is it real" but "does a decision change if it is here".

The mapping

Here is the mapping itself, in four parts. The third column is the one that matters: it is the reasoning, and the reasoning is what you will use on the concept this table does not contain.

Merchandise, space and price

Retail concept	ArchiMate element	Reasoning
Style / generic article	Business Object	A design or recipe identity with a cost base and a buy decision. Not stockable, not sellable, and it still needs to exist or apparel cannot be planned.
Option	Business Object, composed into the style	Style plus the axis the customer chooses before size. The unit of range planning and of the product page.
SKU / article / variant	Business Object, composed into the option	The stockable, sellable, barcoded atom. One bin, one count, one price point, one sale line.
GTIN, case code	Not elements — identifiers in the data model	A SKU legitimately carries several, and a code can be reused after a sunset period. The barcode identifies the trade item, not the merchandise record.
Merchandise, financial, space, online, GS1 and regulated hierarchies	Grouping per hierarchy, with a dated assignment in the data model	Five or more hierarchies, each correct for its purpose. Picking one as canonical is the error.
Price	Business Object, versioned	Dated, scoped, typed, sourced, with a reason. As an attribute of the article it is wrong within a week.
Price change	Event	The thing with a time, which the price version records.
Markdown	Process writing a new price version, plus a revaluation	Permanent, and under the retail inventory method it changes stock value with no physical movement.
Single exit price	Business Object, realised by an external Data Object	A price the retailer does not set, ingested on a published schedule.
Planogram	Business Object; the assignment to sites is dated data	A dated SKU-to-position plan with a derived capacity. Not a PDF.
Fixture, bay, gondola	Equipment, in a technology or physical view	Physical holding structures with dimensions and a temperature class.
Shelf space	Resource	The retailer's scarcest asset, consumed by courses of action. See the capability section.
Facings, capacity, depth	Not elements — derivations	Facings times depth times stack. Drawn as elements they overwhelm the view.

Supply, stock and the stores

Retail concept	ArchiMate element	Reasoning
Supplier	Business Actor	The legal entity, with its registration, tax and empowerment attributes.
Trading relationship	Business Object	The dated commercial arrangement: terms, rebates, payment days, returns policy. Separate from the party, or a terms change rewrites history.
Supply source	Location	A factory, depot or agent's warehouse, with its own lead time and order multiple. Dual sourcing is unsayable without it.
Purchase order, order line, delivery, receipt, claim	Business Objects, written by Processes	All five, because the relationships are many-to-many in every direction and `received_qty` on the line destroys lead-time measurement.
Stock movement	Business Object, append-only	The durable fact. Signed, typed, located, timestamped, reason-coded.
Inventory position	Not an element — a derivation over movements	Nine named positions, each a filter and a sum. Store one and the channels cannot be adjudicated when they disagree.
Stock in a distribution centre	Material, where the view is physical	ArchiMate's Material is for physical matter, and a logistics view earns it. In a business view the movement is the object.
Store, distribution centre, dark store, depot	Location; Facility where the building itself is the subject	A site is a regulatory entity carrying dated licences, not just an operational one.
Selling area — dispensary, liquor, butchery	Location, nested	Separately licensed and separately counted, which is why it is not an attribute.
Virtual location — in transit, quarantine, web stock	Location	Every movement needs both ends. Without these, receipts come from nowhere.
Till, handheld, label controller	Node, with the application on it	The physical thing. The POS application is a separate element.
Scale at the deli counter	Equipment	An asset with a verification lifecycle and a legal consequence.
Store network link	Communication Network; Path where the logical route is the point	The store's dependence on the link is a first-order South African design concern.

Customer, sale and promotion

Retail concept	ArchiMate element	Reasoning
Party — person or organisation	Business Actor	Identity, not behaviour. One actor, many roles, modelled once.
Shopper, loyalty member, patient, account holder	Role	Four positions with four lawful bases and four retention rules. The anonymous basket is still a sale, so the transaction cannot require a customer key.
Loyalty programme, store card, delivery subscription	Product	Terms, membership, mechanic, liability. This is what Product is for.
Transaction / basket	Business Object; the sale is a Process	The filed thing the customer disputes, and the behaviour that produced it.
Line item, tender line, tax line, discount application	Business Objects composed into the transaction	All four, or the cash-up does not reconcile and nobody can say why.
Till slip	Business Object, realised by an Artifact	A prescribed document with a reproduction obligation, not a report.
Promotion plan	Business Object	The commercial intent: period, funding, approval, expected uplift.
Offer / mechanic	Business Object, read by the pricing process	The condition-and-reward rule a basket can satisfy.
Discount application	Business Object, written by the process	Which offer fired, on which line, for how much, with whose money.
Coupon, voucher, gift card	Business Object	An instrument with its own identity, issue, expiry and redemption record.
Consent, objection	Business Objects	Dated, scoped, with evidence. Not a flag.

Retail concept	ArchiMate element	Reasoning
A statute — the CPA, POPIA, the Medicines Act	Driver, with the specific duty as a Requirement	A Driver cannot be realised by a system; a Requirement can. That Realization edge is the trace an inspector asks for.
A licence — liquor, pharmacy, certificate of acceptability	Business Object held against a Location	Per premises, dated, with conditions. The site record is where compliance lives.
Regulator	Stakeholder, and Business Actor where it receives or inspects something	Both: a stakeholder with concerns, and a counterparty who walks into a store.
Capability	Capability, with an owner and a maturity	An ability the retailer has. Not an activity and not a department.
Value stream	Value Stream, with stages	The chief operating officer's view of the same facts.
Shelf space, working capital, trading hours	Resource	Finite things the enterprise holds, consumed by what it chooses to do.
Roadmap	Work Packages, Deliverables and Plateaus	Plateaus are the only place the model says what is true at the end of each year.

Three relationships that carry most of the meaning

Eleven relationships exist and three of them do nearly all the work in a retail model. Getting these right matters more than getting any single element right, because a wrong relationship makes a true statement unprovable.

Realization says "this satisfies that". It runs upward: a Data Object realises a Business Object, a Process realises a Service, a Deliverable realises a Plateau, an Application Component realises a Requirement. In a retailer the realisation edges are where the arguments get settled. Four Data Objects realising one article is how you show that the merchandise system, the product information system, the price file and the online catalogue are four copies of one thing. A price-file distribution component realising a Consumer Protection Act requirement is how you show what breaks if it is replaced. A model with no Realization edges from systems to requirements is decoration, and it will not survive its first compliance review.

Serving says "this is used by that". It runs across the boundary between a provider and a consumer, and it is the relationship people reach for when they mean Realization. The test: Serving implies the consumer could in principle use something else, Realization implies identity of purpose. The replenishment engine *serves* the store order process. The store order process *realises* the stock-on-the-shelf service. A forecasting engine serves replenishment; it does not realise it.

Assignment says "this performs that". It runs from active structure to behaviour, and from a Node to the software deployed on it. A cashier Role is assigned to the sale process. A registered person is assigned to the dispensing act — and in a pharmacy that edge is a statutory control rather than a convenience, because the act is attributable to a named person with a current registration. A till Node is assigned the point-of-sale component. What Assignment never means is a reporting line, and what it never runs to is a Service: assign the Role to the process, let the process realise the service.

Two more earn their keep and are routinely confused with each other. **Access** is behaviour touching an object — the only way to say that a process reads the price file and writes the discount application, which in retail is most of what matters. **Flow** is value or information moving between two pieces of behaviour or two actors, and it is the right edge for stock moving between locations, for a return going to a supplier and for an order going out over a trading network. Where a reviewer objects that Flow should be Triggering, the test is whether anything is actually transferred: triggering is about sequence, flow is about the thing that moves.

The article, and the hierarchy it hangs from

This is the piece a retail model lives or dies on, and it is worth drawing before anything else.

One style, five options, eight sizes each: forty SKUs. Grocery degenerates that cleanly — one style, one option, one SKU — which is exactly why grocery-born systems cannot be retrofitted to apparel and apparel-born systems are gratuitous in grocery. A model that cannot express both levels has already chosen a format.

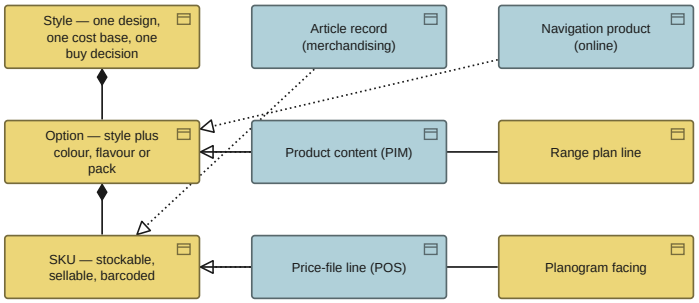


Figure 16.4 One style, five options, eight sizes each: forty SKUs. Composition rather than aggregation, because an option does not exist outside its style and a SKU does not exist outside its option — aggregation would imply a SKU could be shared, which is the mistake that produces one table with a level column. The four Data Objects realising the same Business Objects are the figure's real argument: they are copies of one thing, and drawn as unrelated boxes the duplicate-mastering conversation cannot happen.

Three choices in that figure will be challenged.

Composition, not aggregation, from style to option to SKU. An option does not exist outside its style and a SKU does not exist outside its option. Aggregation would imply a SKU could be shared between styles, which is exactly the mistake that produces a `PRODUCT` table with a level column.

The Data Objects are realisations, not copies. The merchandise system holds the article record, the product information system holds the content, the price file holds what the till needs, the online catalogue holds the navigation grouping. Four Data Objects realising one Business Object is the sentence that makes the duplicate-mastering argument possible. Draw them as four unrelated boxes and the argument cannot be had.

Size is not the same axis as colour, and the notation cannot say so. Colour drives photography, assortment breadth and markdown behaviour. Size drives the size curve, allocation ratios and broken-size analysis. An out-of-stock size is a lost sale on the *option*, because the customer may buy the next size up. ArchiMate gives you one composition edge for both, so the distinction has to live in the data model and in the view's notes.

Then the hierarchies, which is where most retail models quietly become wrong.

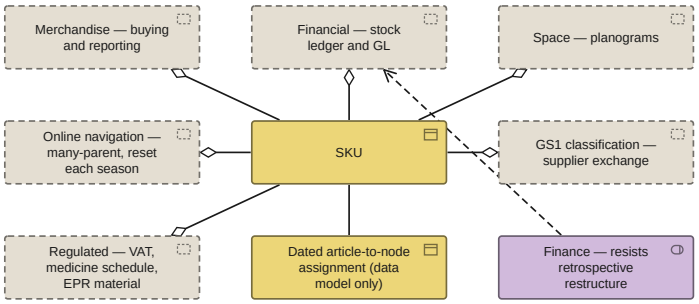


Figure 16.5 Six hierarchies, all correct, none canonical. Drawing each as a Grouping makes the plurality visible and ends the "what category is this" argument by showing it as a question about purpose rather than fact. The hierarchy *nodes* are deliberately absent: the pattern that works is a dated article-to-node assignment, and ArchiMate cannot date an assignment — so it lives in the data model and the view says so.

A retailer does not have *a* product hierarchy. It has at least six, they disagree, and each is correct for its purpose: merchandise for buying and reporting, financial for the stock ledger and the general ledger, space for planograms, online navigation for browsing, the GS1 classification for supplier exchange, and regulated classification for VAT rate, medicine schedule and packaging material. They disagree legitimately. Merchandise hierarchies are strictly one-parent and get restructured when buying teams reorganise; online navigation is many-parent and gets restructured every season; financial hierarchies cannot be restructured retrospectively without restating the stock ledger; and zero-rated VAT status lines up with no commercial category at all.

Drawing each as a **Grouping** does two useful things. It makes the plurality visible, which ends the "what is the category of this product" argument by revealing it as a question about purpose rather than fact. And it keeps the hierarchy *nodes* out of the picture, which they must be, because the useful pattern is a dated article-to-node assignment and ArchiMate cannot date an assignment.

Price, which the notation cannot date

A price is a dated, scoped, typed, sourced number with a reason. Nine price types coexist — cost in several flavours, regular, promotional, markdown, zone, recommended, regulated, member and unit — and two of them can be correct at the same moment for the same article in different places.

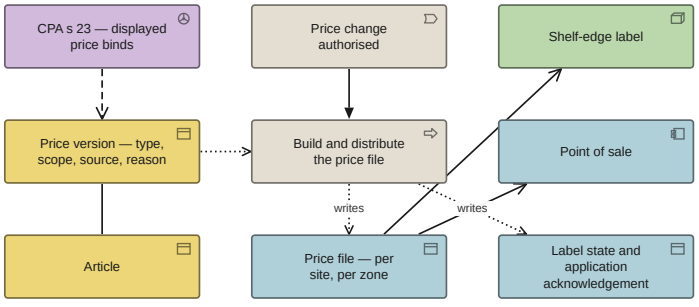


Figure 16.6 Price is an object with versions; the change is an Event. Separating them is what lets the model say a price was authorised at head office, built into a file overnight, and applied at a till and a shelf label at different times — which is the whole of the price-integrity problem. The label is a Node with its own state, because a label that silently failed to update is a second concurrently displayed price and the lower one binds. Effective dating is not in the picture and cannot be.

The figure carries four arguments.

Price is a Business Object, not an attribute. Because effective dating is a legal requirement rather than a convenience: the Consumer Protection Act binds the retailer to the displayed price and releases it from an error only on proof of correction and consumer notification, so the retailer must be able to state what price was displayed, where, when [1].

The price change is an Event and the price file is a Data Object. Separating them is what lets the model say that a price was authorised at head office, built into a file overnight, and applied at a till and a shelf label at different times — which is the whole of the price-integrity problem.

The shelf-edge label is a Node with its own state. Not a detail. A label that silently failed to update is a second concurrently displayed price, and the lower one binds [1].

Effective dating is not in the picture. It cannot be. The notation has no clock, so the dated scope of a price lives in the data model, and the view's note says so. This is the single most important honest limitation to state out loud in a retail modelling review, because a reviewer who discovers it unaided concludes the model is wrong rather than bounded.

The store, the till and what the link does

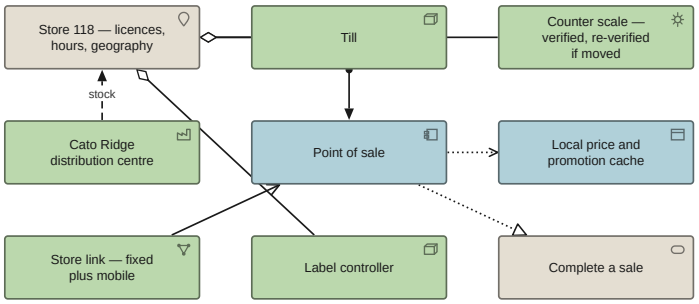


Figure 16.7 The store is a Location, the till is a Node, the point of sale is an Application Component and the scale is Equipment. Four elements, because the questions people ask — how many tills are on the current release, how many have a verified scale beside them, how many can trade with the link down — are questions about Nodes and Equipment. The local cache is drawn deliberately: the store's ability to complete a sale must not depend on anything outside the store.

Read that figure as a set of choices.

The store is a Location, not an Application Component and not a Business Actor. A Location is where behaviour and objects can be, which is precisely what a store is. It carries the licences, the trading hours and the geography.

The till is a Node and the point of sale is an Application Component. Two elements, because the estate has thousands of the first and a handful of versions of the second, and the questions people ask — how many are on the current release, how many have a verified scale beside them, how many can trade when the link is down — are questions about Nodes.

The scale is Equipment. It is a regulated instrument with a verification lifecycle, and a store refit that moves it makes it unverified until verified in its new position.

The network is a Communication Network, and it is drawn as a dependency rather than a decoration. Here is the design rule that makes retail different from banking: a branch that cannot transact sends the customer to the app; a store that cannot transact sends the customer to the competitor four hundred metres away. So **the store's ability to complete a sale must not depend on any system outside the store.** Everything else may degrade. Drawing the till's dependence on a central service without drawing its local price cache is how an architecture quietly loses that property.

The sale, which is behaviour with four objects inside it

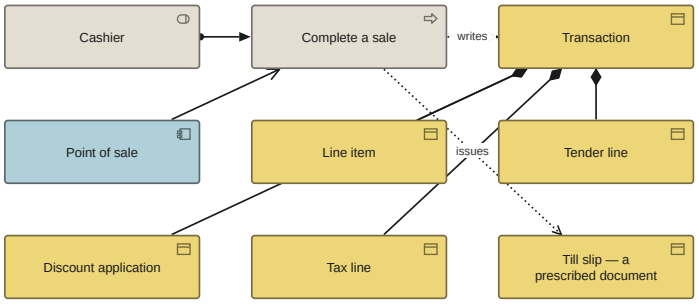


Figure 16.8 The sale is behaviour with four objects inside the transaction. Tender is not a column — a split across two cards and some cash is three tender lines, and tender type decides fee, settlement timing and reconciliation path. Tax is per line, because a South African basket mixes standard-rated and zero-rated items. And the discount application is stored rather than derived, because offers interact and finance has to know which offer cost what.

The sale is a Process. The transaction is a Business Object with line items, tender lines, tax lines and discount applications composed into it. The cashier is a Role assigned to the process. The point of sale serves it. The till slip is an object realised by an Artifact, because it is a prescribed document with its own particulars and a reproduction obligation [4].

Four choices worth defending.

Tender is not a column. A split tender across two cards and some cash is three tender lines, and tender type determines fee, settlement timing, fraud exposure and reconciliation path. In a market where cash is the majority of payment volume, tender type is a first-class dimension of the sale.

Tax is per line. A single South African basket mixes standard-rated and zero-rated items, and VAT at a blended basket rate cannot be reconstructed or defended.

The discount application is stored, not derived. Offers interact; a basket satisfying three offers can be discounted in several valid orders producing different totals; the engine picks one; and finance needs to know which offer cost what so the supplier can be billed for its share. Store only the net line price and none of that is recoverable.

The invariant belongs in the view's notes. Sum of line nets, plus tax, less discounts, equals sum of tenders plus change plus rounding. ArchiMate cannot express it, and a model of a sale that does not state it has left out the only thing that makes the sale checkable.

The supplier, the buyer and the planner

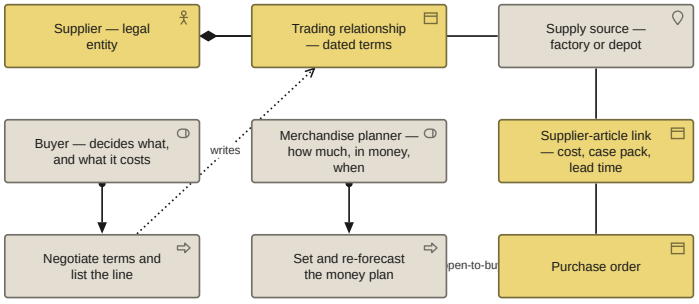


Figure 16.9 Supplier party, trading relationship and supply source are three elements. One legal entity supplies through three factories with different lead times, and one SKU is dual-sourced with the split changing weekly. And buying and planning are two Roles with two mandates: buying decides *what*, planning decides *how much, in money, and when*. A model that merges them produces a design in which nobody owns open-to-buy. Assignment here is "performs", never "reports to".

Two separations, both of which practitioners will recognise immediately and most models miss.

Supplier party, trading relationship and supply source are three elements. One legal entity supplies through three factories with different lead times; one SKU is dual-sourced and the split changes weekly. A single supplier box with a lead-time attribute carries none of it, and it is also why the competition regulator's price-comparability question cannot be answered.

Buying and planning are two Roles with two mandates. Buying decides *what*; planning decides *how much, in money, and when*. They are separate jobs with separate systems and frequent conflict. A buyer who plans over-buys; a planner who buys buys safe. A model that merges them has mis-modelled the organisation, and will produce a system design in which nobody owns open-to-buy.

Note what Assignment is doing in that figure and what it is not. Assignment runs from a Role to the behaviour that Role performs. It is not a reporting line. ArchiMate has no organisational hierarchy, deliberately, and the planner reporting to the trading director is a fact about March that the model should not carry.

A promotion is behaviour, not a thing

Open ten retail models and nine have a box labelled Promotion. It is usually an Application Component, sometimes a Business Object, and it is the reason nobody can explain the group's own margin.

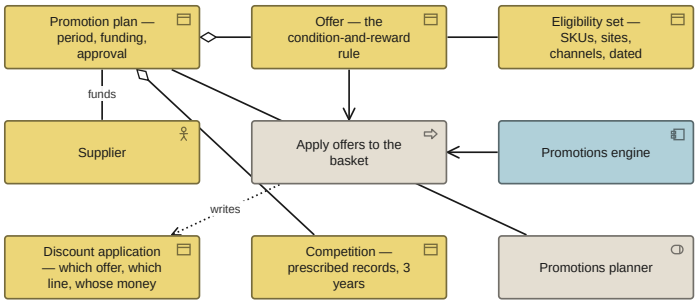


Figure 16.10 The plan is commercial intent; the offer is an executable rule; the discount application is what actually happened. Funding is an Association to a supplier rather than an attribute, because retailer-funded and supplier-funded promotions with identical customer-facing mechanics are two different financial events. Eligibility is a dated set. And a promotional competition is a separate object with a records lifecycle, not a mechanic.

Here is the shape that works. The **promotion plan** is an object: name, period, categories, funding source, approval, expected uplift. The **offer** is a separate object, because the plan is commercial intent and the offer is an executable rule. **Applying offers to the basket** is a Process. The **discount application** is what that process writes, and it carries the funding split — which is the edge that lets a retailer bill a supplier for its share and explain the margin afterwards.

Then the three things the figure deliberately shows that a single Promotion box cannot.

Funding is an Association to a supplier, not an attribute. Retailer-funded and supplier-funded promotions with identical customer-facing mechanics are two different financial events.

Eligibility is a dated set, not a list on the promotion. Which SKUs qualify, at which sites, in which channels, from when.

A promotional competition is a different animal. In South Africa it is regulated separately, with prescribed records and a three-year retention obligation, which makes it an object with a compliance lifecycle rather than a mechanic [2].

The pharmacy, which is a bounded piece of the estate

Nothing else in a retail model behaves like this. A South African pharmacy is a separately licensed professional practice operating inside a shop, with its own regulator, its own price it does not set, its own records and a confidentiality boundary that the rest of the estate may not cross.

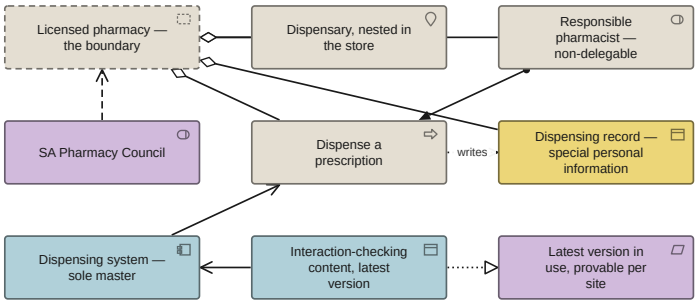


Figure 16.11 The pharmacy boundary is drawn as a wall, not a lane. Inside: the dispensary as a nested Location, the dispensing system as sole master of patient, prescription and dispensing record, and a responsible pharmacist whose accountability is personal and non-delegable. Across the boundary, four edges are the whole design — the till learns the value and not the contents, the exit price arrives from outside, the interaction content must be current and provable, and nothing leaves unpseudonymised.

The boundary is drawn as a **Grouping**, and it is drawn as a wall rather than a lane. Inside it: the dispensary as a nested Location, the dispensing system as the sole master of patient, prescription and dispensing record, the responsible pharmacist as a Role whose accountability is personal and cannot be delegated to a system, and the dispensing act as a Process assigned to a registered person. Outside it: the front shop, the loyalty programme, the customer data platform, the group data warehouse.

Four edges across that boundary are the whole design.

The till tenders the dispensary transaction and learns its value, not its contents. The loyalty programme may know a basket included a dispensary line. Absent a separate consented basis it may not know what was dispensed.

The single exit price arrives from outside and is applied exactly. The group pricing engine has no authority here at all [5].

The interaction-checking content must be current, because Good Pharmacy Practice requires the latest version of software providing warnings or interaction checks [3]. That is a Realization edge from a Data Object to a Requirement, and it is the one architects are surprised by.

Nothing flows outward unpseudonymised. The data platform receives dispensary data into a segregated zone with its own access model, or not at all.

Draw the dispensary as a department of the store and every one of those four statements becomes unsayable. That single modelling choice is the difference between a model a pharmacy director will work with and one she will dismiss in the first five minutes.

A regulation, from driver to control

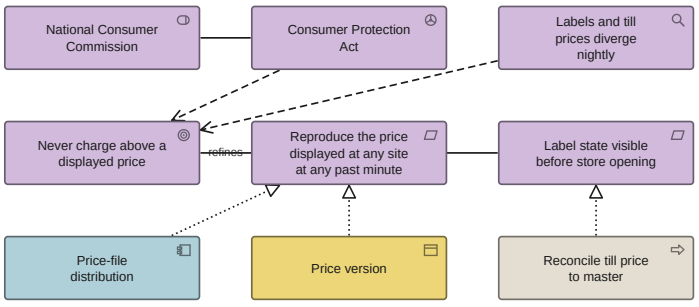


Figure 16.12 A statute is a Driver and cannot be realised by anything; the duty is a Requirement and can. Writing "CPA compliance" as a Requirement is the commonest error in this layer because it makes the model untestable. A Requirement is a sentence a system satisfies or fails. The edges that matter are the three Realizations at the bottom, because they answer the question nothing else in a retailer answers: if this system is replaced, which obligations lose their realisation?

The chain is Stakeholder, Driver, Assessment, Goal, Requirement — and then the part people skip: Realization from a core element up to the Requirement.

The discipline each element enforces is the point. Writing "CPA compliance" as a Requirement is the commonest error in this layer and it makes the model untestable, because the Act is a Driver and cannot be realised by anything. A Goal is a direction. A **Requirement is a sentence a system can satisfy or fail**: "the price displayed at any site at any past minute is reproducible from stored records", or "no liquor sale completes outside the hours on that site's licence".

Then the realisation. An Application Component realises it. A Process realises it. A Data Object realises it. Now the question that justifies the whole motivation layer becomes answerable: if the promotions engine is replaced in 2028, which regulatory requirements lose their realisation, and what has to take them over? No other artefact in a retailer answers that — not the control register, not the application inventory, not the programme plan.

Capabilities, value streams and the scarcest resource

Every retailer has a capability map. Most are wallpaper. The difference is whether anything hangs off each box.

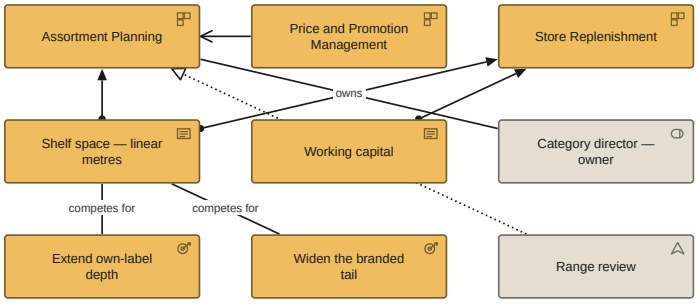


Figure 16.13 Shelf space is a Resource, and that is the figure's point. Retail has a hard, physical, externally fixed constraint on how much business it can write: linear metres are finite, a range review is a zero-sum allocation of them, and every listing displaces something. Modelling space as a Resource assigned to capabilities and competed for by courses of action carries the constraint without pretending to be a space-planning tool. Working capital behaves the same way.

A **Capability** is an ability the enterprise possesses, with an owner. It is realised by core behaviour and enabled by **Resources**. "Assortment Planning" is a capability; "sign off the autumn range review" is a process; "Category Management" is a department and belongs nowhere in the model.

The figure's most useful edge is the Resource. Retail has something most industries do not: a hard, physical, externally fixed constraint on how much business it can do, and it is **shelf space**. Linear metres are finite, a range review is a zero-sum allocation of them, and every listing decision displaces something. Modelling space as a Resource assigned to capabilities and consumed by courses of action is how an architecture model carries that constraint without pretending to be a space-planning tool. Working capital and trading hours behave the same way.

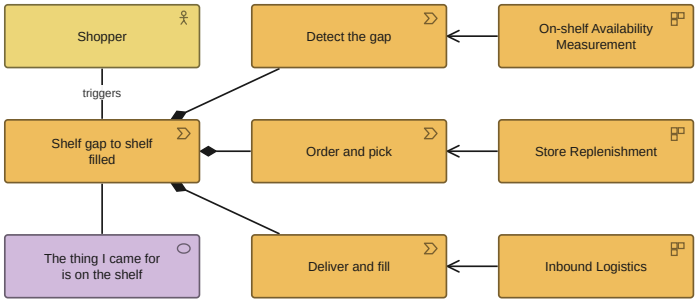


Figure 16.14 The same facts, arranged for the chief operating officer. Three stages from a trigger a store would recognise to a value a customer would: the thing I came for is on the shelf. Each stage is served by a capability, which is the edge that lets availability be argued as cost to serve on one page. “Supply chain” is not a value stream; it is a set of capabilities. The outcome is.

A **Value Stream** is stages from a customer trigger to a value the customer would recognise, served by capabilities. The useful ones in retail are short and cross every organisational boundary: shelf gap to shelf filled, order to collection, prescription to dispensed, supplier listed to first delivery. The test is whether a customer or a store would recognise the trigger and the outcome. "Supply chain" is not a value stream; it is a set of capabilities. "The thing I came for is on the shelf" is the outcome of one.

The roadmap, and what is true on 31 December

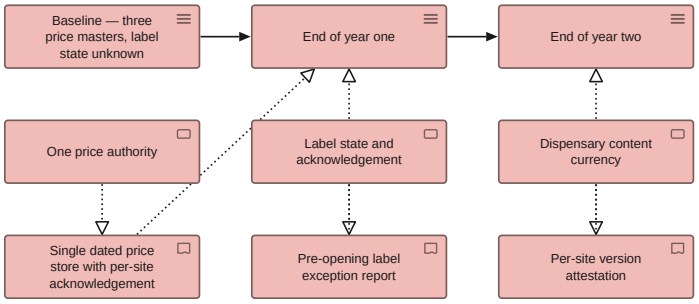


Figure 16.15 Plateaus are the only place the model says what is true on 31 December. Three states, two years, three work packages and the deliverables that move the retailer between them. Every plateau carries exit criteria you could test — one price authority with per-site application confirmed, label state visible before opening, dispensary content currency attested per site. And the view should carry the peak change freeze, because the delivery year is nine months, not twelve.

A **Work Package** is work with a start, an end and a budget. A **Deliverable** is what it produces. A **Plateau** is a relatively stable state of the enterprise, and the only construct in the notation that says what is true on a date.

One rule: every plateau has exit criteria you could test. Not "foundations laid". Something like: one price authority with per-site application confirmed for every store, label state visible before store opening, and the dispensary's content currency attested per site. If a plateau's criteria cannot be tested, it is a date with ambition attached, and the programme board will discover that in month fourteen.

Retail adds one constraint to the roadmap that other industries do not have, and it belongs on the view: **the change freeze**. A retailer freezes the production estate across peak, commonly from mid-November to mid-January for a grocer and longer for a general merchandiser, and the freeze covers more than software — master-data restructuring, range change, planogram resets and store refits as well. So the delivery year is roughly nine months, and a plan that assumes twelve is wrong by a quarter. Put the freeze on the plateau view as a period, and the conversation about sequence becomes honest.

Viewpoints that earn their keep in a retailer

A viewpoint is a view for a named audience with a named question. These are the ones asked for repeatedly, and the last three are missing from the standard set and should not be.

Viewpoint	What it shows	Who asks	What it settles
Capability map with maturity	Capabilities, owners, supporting applications, maturity	Executive committee, chief financial officer	Where the next few hundred million rand goes
Value stream	Stages from trigger to value, served by capabilities	Chief operating officer, retail operations	Cost to serve and turnaround, argued on one page
Application co-operation	Components, services and dependencies	Chief information officer, integration lead	Coupling, duplication, and who really masters what
Information structure	Business object to data object to artifact, with the system holding each	Chief data officer, merchandise systems	Granularity and lineage, and whether a number can be produced at all
Technology and deployment	Nodes, networks, facilities, what runs where	Store systems, operations	Estate scale, patch state, and what happens when the link drops
Requirements realisation	A requirement traced to the component, process or data object that satisfies it	Compliance, internal audit, the inspector	Which system satisfies which rule, and what breaks if it is switched off
Implementation and migration	Plateaus, work packages, deliverables, and the freeze	Programme board, finance	Sequence, and what is true at the end of each year
Price authority view	Every system that can set or change a price, with the scope it may change and where that is enforced	Pricing, merchandise, legal	How many price masters exist, which is authoritative, and whether the label and the till agree
Format divergence view	What is genuinely shared across grocery, pharmacy and apparel, and what is not	Group executive, architecture	Whether one merchandising stack is a strategy or a wish
Site authority view	Every licence, certificate and verification held per site, with the control that enforces it	Risk, store operations	Whether a site is lawfully trading, and whether the system knows

The price authority view is the one that most quickly earns an architecture function the right to be consulted. In most retailers the honest answer is three masters and no stated authority, nobody has drawn it, and several people urgently need it.

The honest limits

ArchiMate is good at what a retailer is made of and bad at five things a retailer runs on. Say so early, and say what you use instead, because the alternative is a reviewer finding the gap and concluding the model is wrong rather than bounded.

No cardinality. The notation cannot say that one style has one or more options, that one delivery carries lines from three orders, that one receipt covers part of one line, or that a membership has one primary card and up to three family cards. Every one of those is load-bearing. Put them in a conceptual data model — crow's-foot notation when cardinality is the point of the picture — and reference it from the view. This book does exactly that: the multiplicity argument about article, option and SKU is settled in the data-model chapter, not here.

No quantity and no money. There is no element for 240 facings, a R19.99 price point, a twelve-unit case or a 96% service level. A limit, a rate, a case pack and a margin are attributes, and a model that draws them becomes unreadable before it becomes complete. This is why shelf space appears above as a Resource rather than as a number of metres.

No time, and especially no time of day. This is the big one for retail, and it is worse than the usual complaint about effective dating. A retailer's architecture is organised around a clock: the overnight cycle, in which store end-of-day gates sales polling, which gates posting, which gates the forecast, which gates replenishment, which must finish before the supplier cut-off, while the price file must be built, distributed and applied before the stores open. The order and the slack are the architecture. ArchiMate can draw Triggering edges between those processes and cannot say that the forecast run takes four hours and has forty minutes of slack. So this book draws the overnight cycle as a **timing diagram** instead, and for an architect arriving from banking or insurance it is the single most useful picture in the book.

No reporting line. Assignment from an Actor to a Role is not "reports to". The org chart changes next March and the model should not describe anything that does not survive the change. Model Roles and the behaviour assigned to them, and leave the hierarchy in the human resources system.

No swimlanes, and therefore no hand-offs. In retail the hand-off is frequently the failure mode: the promotion whose funding is not confirmed before the leaflet locks, the price change that reaches the till and not the label, the online order picked against stock that is not there. ArchiMate has no lane, so a process view shows the sequence and hides who was holding the work. This book therefore draws level-2 process models in **BPMN 2.0**, with a pool, lanes for the roles and systems, and explicit gateways, and the process chapter owns and argues that decision. Two notations, each doing what it is good at, beats one used dishonestly.

> FOUR TRAPS, EACH OF WHICH PRODUCES A MODEL THAT VALIDATES
>
> **Modelling the org chart.** Business Actors named after departments, composed into a hierarchy, with processes named after the departments that run them. Quick, recognisable, worthless, and it teaches the organisation that architecture is documentation.
>
> **Modelling the system menu.** Application functions named after screens: Article Maintenance, Price Change Entry, Stock Enquiry. That is what a model built from a user manual looks like. The test for an application service is whether something outside the component depends on it. Nobody outside depends on Stock Enquiry; five systems depend on "available to sell at this site, now".
>
> **Drawing the article as a process.** A chain of boxes labelled Create, Range, Trade, Markdown, Delist, with arrows. Those are **events** in the life of a Business Object, each triggering different behaviour, in no fixed order, often concurrently and sometimes retroactively. Drawing them as a sequence encodes the false assumption that markdown follows ranging, and that assumption later shows up as a system that cannot reinstate a delisted line.
>
> **One box for merchandising.** The expensive one. "The merch system" is at least four things — the article master, the commercial ledger, the inventory ledger and the transaction engine — and whether they are separable decides whether modernisation is a sequence of affordable moves or one unaffordable replacement. The decomposition is not documentation. It is the highest-value hour an architect spends inside a retailer.

THE CHAPTER IN TEN LINES

- Retail's modelling difficulty is instance volume and time rather than vocabulary, and ArchiMate models types with no clock — so build a conceptual data model alongside it instead of asking the notation to be one.
- The article is a Business Object, not a Product; Product is for the things with terms — the loyalty programme, the store card, the delivery subscription, the lay-by agreement.
- Five decisions settle most arguments: article or product, service or process, channel as interface or location or actor, behaviour or thing, and element or attribute.
- Realization says “this satisfies that”, Serving says “this is used by that”, and Assignment says “this performs that” — and Assignment is never a reporting line.
- Style, option and SKU compose downward, and four Data Objects realising the same article is how the duplicate-mastering argument becomes possible to have.
- A retailer has at least six disagreeing hierarchies; draw each as a Grouping and keep the dated article-to-node assignment in the data model.
- Price is a versioned Business Object, the price change is an Event and the shelf label is a Node with its own state — and effective dating cannot appear in the picture at all.
- The store is a Location, the till a Node, the point of sale an Application Component and the deli scale Equipment with a verification lifecycle.
- A promotion is behaviour: the plan is intent, the offer is a rule, and the discount application is the stored fact that says which offer cost whose money.
- Draw the pharmacy as a Grouping — a wall, not a lane — with the dispensing system as sole master, the responsible pharmacist's accountability personal, and four controlled edges across the boundary.
- A statute is a Driver, the duty is a Requirement, and the Realization edges from components, processes and data objects upward are the only artefact that says what breaks when a system is retired.
- Shelf space is a Resource, which is retail's equivalent of a bank's capital: finite, zero-sum at a range review, and competed for by courses of action.
- Plateaus carry testable exit criteria, and the retail roadmap also carries the peak change freeze — so the delivery year is nine months, not twelve.
- ArchiMate has no cardinality, no quantity, no time of day, no reporting line and no swimlane — so cardinality goes to a crow's-foot model, the overnight cycle to a timing diagram, and level-2 processes to BPMN.

SOURCES FOR THIS CHAPTER

- **1.** Consumer Protection Act 68 of 2008, s 23 — disclosure of price of goods or services https://www.acts.co.za/consumer-protection/23_disclosure_of_price_of_goods_or_services
- **2.** CPA Regulations, reg 11 — promotional competitions https://www.acts.co.za/consumer-protection/r293_11__promotional_competitions
- **3.** Rules relating to Good Pharmacy Practice, consolidated https://www.saflii.org/za/legis/consol_reg/rrtgpp362
- **4.** SARS, Tax Invoices https://www.sars.gov.za/businesses-and-employers/tax-invoices/
- **5.** How medicine pricing works in SA, Daily Maverick / Spotlight, 19 February 2026 https://www.dailymaverick.co.za/article/2026-02-19-how-medicine-pricing-works-in-sa-and-how-it-might-change-in-future/

CHAPTER 17

Milkwood Retail Group, end to end

One invented South African multi-format retailer — supermarkets, pharmacy, apparel, cash-and-carry and online — modelled from motivation through to technology, a view at a time. The argument for every choice is shown rather than asserted, and three views are kept in their wrong first drafts, because those are the ones that teach.

A warning before we start

Milkwood Retail Group does not exist. It is invented for this book. Nothing in this chapter is a case study, no number here describes a real company, and the system names are made up. What is real is the *shape*: the formats, the conflicts between them, the places the model goes wrong on the first attempt, and the arguments you will have to win. Those are drawn from the research behind this volume and from the published record of South African retail, which is thinner than it should be.

The point of a worked example is not to admire a finished model. It is to watch a model being argued into existence. So two of the views in this chapter are shown wrong first, and one is shown wrong twice, because those are the ones that teach.

The group in one page

Milkwood is a JSE-listed multi-format retailer with its head office in Century City. Five businesses, one balance sheet, and almost nothing shared below the till.

Format	Shape	What makes it different
Milkwood Supermarkets	420 stores: 316 company-owned, 104 franchised	Food, fresh, liquor in 210 of them. The franchise estate is a third of the store count and the group does not set its prices
Milkwood Health	180 stores, 142 with licensed dispensaries	A separately licensed professional practice inside each shop, on a price the group does not set
Thread	168 apparel stores	Style, option and SKU; the size curve; markdown as the primary margin lever
Milkwood Cash & Carry	38 depots serving about 2,400 independent traders and spazas	A wholesale business whose customers are businesses, needing full tax invoices above R5 000
Milkwood Now	milkwood.co.za plus an app, picked in 46 supermarkets	One-hour delivery and click-and-collect, competing for the same stock the store is selling

The cast, because an architecture argument is always between named people:

- **Lerato Mahlangu**, Group Chief Operating Officer. Wants availability and labour cost.
- **Dirk Steenkamp**, Group Chief Information Officer. Has been asked for a single merchandising platform and does not believe in it.
- **Yusuf Adams**, Lead Enterprise Architect. New. Six weeks in at the start of this chapter.
- **Ntombi Zwane**, Merchandise Director, Supermarkets. Owns range and supplier terms.
- **Hennie Loots**, Head of Pricing. Owns the shelf price and knows where the bodies are.
- **Dr Ayesha Moosa**, Superintendent Pharmacist, Milkwood Health. Personally accountable for 142 dispensaries.
- **Pieter de Waal**, Supply Chain Director. Four distribution centres and a fifth he wants.
- **Zanele Dube**, Head of Customer and Loyalty. Owns Milkwood Rewards and its eleven million members.
- **Rajesh Naidoo**, Head of Commercial Finance. Owns supplier income, which nobody can reconcile.
- **Kobus Venter**, Head of Store Systems. Owns 4,100 tills and the thing that keeps them trading.

The systems, as the group names them. Harvest is the grocery merchandising system, implemented in 1999 and modified continuously since. Loom is the apparel merchandising and range-planning system. Scrip is the pharmacy dispensing system. Tally is the point-of-sale application, one version across all formats. Lantern is the promotions and pricing engine. Tide does forecasting and replenishment. Keel runs the warehouses. Gantry does space planning. Trellis is the e-commerce platform and order management. Moya is loyalty and customer data. Quay is the supplier portal and trading gateway. Abacus is the group ledger. Kelp is the data platform.

The motivation view: why this group is changing

Yusuf's first view is not a landscape. It is the answer to "why are we spending money".

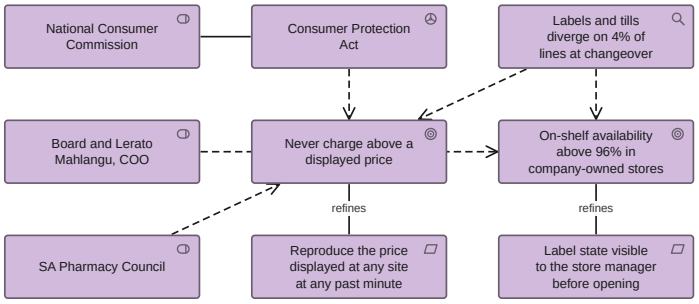


Figure 17.1 Milkwood's motivation view: why the money is being spent. The Act is a Driver and the Commission a Stakeholder, separately, because a Driver cannot be realised and a Commission walks into stores. The Assessment carries the honest number rather than the diplomatic phrase. And availability and price integrity are two Goals influenced by one Assessment, which is what let one programme replace two. Milkwood is invented.

Three things about that figure are worth defending, because each was challenged in the room.

The Consumer Protection Act is a Driver and the National Consumer Commission is a Stakeholder. Both, separately. The Act cannot be realised by a system; the duty under it can. And the Commission is not an abstraction — it issues compliance notices in tranches of dozens, and it does so after walking into stores [1].

The Assessment is the honest sentence, not the diplomatic one. "Labels and till prices diverge in about 4% of lines on a promotion changeover weekend, and nobody can say which stores" is an assessment. "Price integrity requires attention" is a slogan. Lerato approved the figure in the assessment before she approved the programme, which is the correct order.

Availability and price integrity are separate Goals with a shared root cause. They came to the board as two problems and one budget. Drawing them as two Goals influenced by the same Assessment is what let Dirk argue for one programme rather than two, and it is the single most useful thing the motivation view did all year.

One more thing about this view, and it is procedural rather than notational. Yusuf drew it in week six, before he had drawn a single application. The brief he had been given was "give us a view of the landscape", and a landscape drawn before anybody has said what hurts is an inventory that will then be defended because it took four months. The motivation view took nine days, it was wrong in two places that Lerato corrected in the first meeting, and it is what made the landscape worth drawing later.

The strategy view, first draft — and why it was wrong

Here is what Yusuf drew in week seven, and it survived two workshops before somebody killed it.

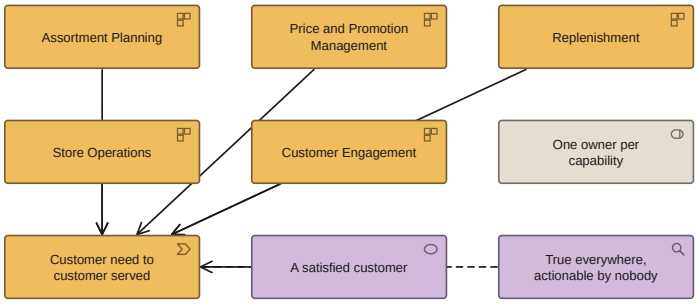


Figure 17.2 The first draft, and it survived two workshops. One capability map, one value stream, an owner against each box. The merchandise director killed it in eleven minutes: “Assortment Planning” is three different abilities with three different planning grains, and “Price and Promotion Management” is a capability the group does not possess in 142 dispensaries or 104 franchise stores. A capability the enterprise does not have is a wish with an owner.

It looks reasonable. One capability map for the group: Assortment Planning, Price and Promotion Management, Replenishment, Store Operations, Customer Engagement. One value stream. Owners against each box. It validates, it fits on a page, and the executive committee nodded at it.

Ntombi Zwane killed it in eleven minutes, and her objection was not about the picture.

"Assortment Planning" is not one capability, because the planning grain is different per format. Supermarkets plan category money and SKU-store units. Thread plans option money and a size curve. Milkwood Health plans dispensing volume and a front shop. These are not maturity differences. They are different abilities, requiring different data, different systems and different people, and a single capability box asserts that one platform could serve all three.

"Price and Promotion Management" is worse, because for two of the five formats the group does not set the price. In 142 dispensaries the price is the single exit price plus a capped dispensing fee, and the group's pricing engine has no authority at all [2]. In 104 franchise stores the group recommends and the franchisee decides. A capability the enterprise does not possess for a third of its stores is not a capability; it is a wish with an owner.

And the single value stream hid the thing that actually hurts. One stream from "customer need" to "customer served" is true of every retailer on earth and tells Lerato nothing.

> **WHAT THE FIRST DRAFT GOT RIGHT**
>
> It was still worth drawing. A wrong capability map that provokes an eleven-minute correction from the merchandise director is worth more than a right one nobody reads, and it bought Yusuf the next two hours of Ntombi's time. Draw the thing that can be argued with.

The strategy view, second draft

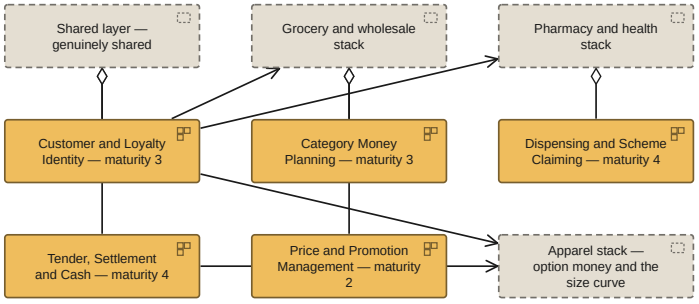


Figure 17.3 The second draft: one shared layer, three format stacks. Shared after argument — customer identity, tender and settlement, the ledger, people, the site master and the store technology estate. Not shared — planning, assortment, space, replenishment and pricing, because the grain differs and in two formats the price authority is not the group's. Maturity is scored against named evidence rather than opinion, and against the capability rather than the system.

The second draft makes one structural change and one judgement.

The structural change is the **Grouping**: a shared layer that genuinely is shared, and three format-family stacks that genuinely are not. Shared, after argument: customer and loyalty identity, tender and settlement, the ledger and statutory reporting, people and rostering, the site and property master, and the in-store technology estate. Not shared: merchandise planning, assortment, space, replenishment and pricing, because the grain differs and the price authority differs.

That is not a compromise. It is the research position stated plainly — a pharmacy and an apparel chain share almost nothing below the point of sale, and groups that have run one merchandise system across grocery, apparel and pharmacy have ended up with a grocery system plus two sets of workarounds. Dirk had been arguing it for two years without a picture. The picture ended the argument in one meeting.

The judgement is **maturity, scored and owned**. Five levels, assessed against evidence rather than opinion, with the evidence named on the view: Price and Promotion Management scores 2 of 5 because the group cannot answer what price was displayed at a site at a past minute; Store Replenishment scores 4 because forecast accuracy is measured at article-location grain and acted on; Supplier Income Management scores 1 because the agreements are not machine-readable and the accrual is an estimate Rajesh re-negotiates annually.

> **TWO RULES FOR A MATURITY SCORE THAT WILL SURVIVE CONTACT**
>
> State the evidence, not the level. "Level 2" invites an argument about the scale; "we cannot reproduce a displayed price" invites an argument about the fact, which is the argument you want. And score the capability, never the system. A good system inside a capability nobody owns still scores low, and that distinction is what stops a maturity assessment turning into a vendor review.

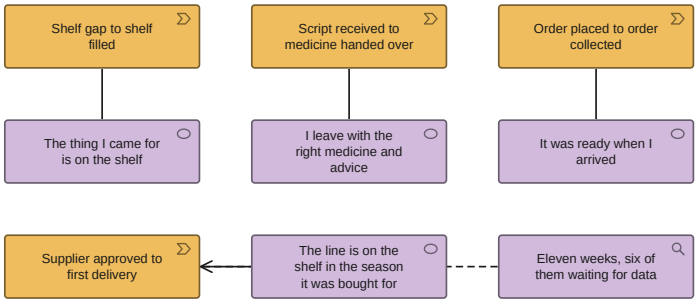


Figure 17.4 Four value streams Lerato can act on. Each has a trigger a store would recognise and an outcome a customer would, and each crosses every organisational boundary. The fourth is there because the merchandise director asked for it: listing a new supplier takes eleven weeks at Milkwood and six of those are waiting for data.

Then the value streams, and the reason they replaced the single stream of the first draft: Lerato can act on them. Four streams, each with a trigger a store would recognise and an outcome a customer would. Shelf gap to shelf filled. Script received to medicine handed over. Order placed to order collected.

Supplier approved to first delivery. The last one is there because Ntombi asked for it: listing a new supplier takes eleven weeks at Milkwood and six of those weeks are waiting for data.

The business layer

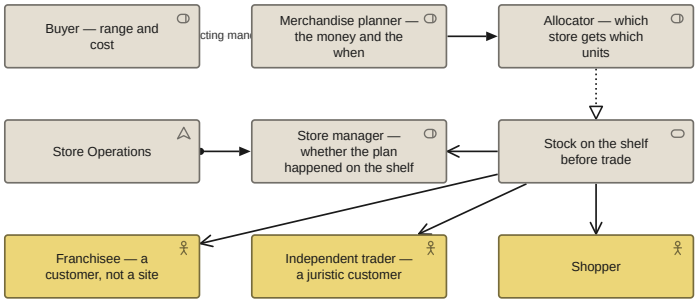


Figure 17.5 The business layer, with the roles named as the industry names them. Buyer and planner are two Roles with conflicting mandates even though they sit in one department. The franchisee is a Business Actor in a customer Role, not a Location — Milkwood sells stock to it, recommends a price and sees whatever the agreement compels. The independent trader is a juristic customer, which is why that format's till must issue a full tax invoice.

The business view is where the roles get named properly, and naming them properly is most of the value. Note four choices.

Buyer and merchandise planner are two Roles. Buying decides what and what it costs; planning decides how much, in money, and when. At Milkwood they sit in the same department and report to Ntombi, and they still have different mandates and conflicting incentives.

The franchisee is a Business Actor in a customer Role, not a Location. This is the choice that makes the franchise estate expressible. Milkwood sells stock to its franchisees, it does not transfer stock to them; it recommends prices, it does not set them; and its visibility of their shelf depends on what the franchise agreement compels. Drawing those 104 stores as Locations inside the estate would assert a control the group does not have, and every downstream model — replenishment, pricing, availability reporting — would inherit the lie.

The independent trader buying from Cash & Carry is a third kind of customer, and a juristic one, which is why that format needs a point of sale that can capture a business customer's name, address and VAT number and issue a full tax invoice above R5 000 [3]. Milkwood discovered this when its largest spaza customer asked for twelve months of reissued invoices.

Store operations is a Function, not a process. It has no sequence. The trading day has a sequence and belongs in a level-2 process model with lanes, which is where the book draws it.

The supplier side, and the dataset a regulator will ask for

Yusuf's fifth view is the one Rajesh Naidoo had been waiting four years for, and it has no systems on it at all.

Milkwood buys from about 3,700 suppliers. Behind that number sit four entities the group had been treating as one: the **supplier party**, which is a legal entity with a registration number, a VAT number, banking details and an empowerment status; the **trading relationship**, which is a dated commercial arrangement with terms, rebates, payment days and a returns policy; the **supply source**, which is the factory or depot goods actually come from, with its own lead time and order multiple; and the **supplier-article link**, with the cost price, case pack and conversion that an order is placed against. Harvest has one vendor record and a lead-time field.

Two consequences, and the second one is the expensive one.

Dual sourcing is unsayable. Pieter de Waal's team dual-sources about 400 lines and the split moves weekly. With no supply-source level, an open order cannot say which source it is against, so lead-time reliability cannot be computed per source, so the forecast cannot learn, so the safety stock is set by argument.

The competition regulator's question cannot be answered. Under the buyer-power provisions the onus reverses: once the Commission has a prima facie case, the retailer must show the price and trading conditions were not unfair to a small or historically disadvantaged supplier, which means producing the price paid to *every* supplier of like goods, with each supplier's size and ownership status, across the period [10]. Milkwood has none of the four things that question needs — a dated supplier classification, a defensible like-goods grouping on the article master, actual payment dates per invoice against the thirty-day benchmark, and a charges register for listing and promotional fees with the contract reference and costing basis. Rajesh's team can produce a rebate accrual. It cannot produce a price-comparability dataset, and nobody had told the board that the absence is a legal exposure rather than a reporting inconvenience.

That is also, incidentally, the same data the supplier-income accrual needs. One dataset, two reasons — which is how a compliance requirement and a commercial finance requirement ended up in the same work package.

The article and the price

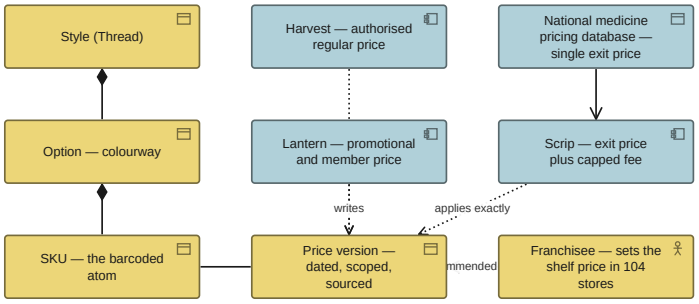


Figure 17.6 Two format shapes side by side, and four price authorities in one picture. Thread needs style, option and SKU; supermarkets degenerate to one level, which is why the instinct to "extend the grocery system to clothing" dies here — the missing level is not a field. Then the authorities: Harvest holds the authorised regular price, Lantern writes prices Harvest never authorised, the national database holds the exit price, and in 104 stores the franchisee holds the actual shelf price.

This is the view Yusuf drew third and should have drawn first.

Two format shapes sit side by side. Thread needs style, option and SKU, because one jacket is one buy decision, five colourways and eight sizes. Supermarkets degenerate that to one level, which is why Harvest has no concept of an option and why Thread runs on Loom. The group's instinct had been to "extend Harvest to clothing". Drawing the two shapes next to each other is what killed that idea, because the missing level is not a field.

Then price, and the figure makes four separate price authorities visible in one picture.

1. **Harvest** holds the authorised regular retail price for supermarket and cash-and-carry lines.
2. **Lantern** holds promotional prices and member prices, and — this is the finding — it can and does write prices that Harvest never authorised.
3. **The national medicine pricing database** holds the single exit price for scheduled lines, which Scrip ingests and applies exactly, with the capped dispensing fee computed from a banded table that changed twice in 2026 [2].
4. **The franchisee** holds the actual shelf price in 104 stores, and the group holds a recommendation.

Four authorities, one article master, and the Consumer Protection Act binding the group to whichever price was displayed [1]. That picture is why the programme got funded.

One more thing lives on this view and it surprises everybody who has not worked in a pharmacy retailer. **A Milkwood Health article carries two identifier schemes for the same physical product.** The GTIN is what gets received at the Midrand distribution centre, scanned at the shelf and rung up at the till. The NAPPI code is what gets claimed and reimbursed. They are maintained by different bodies on different cadences, they do not map one to one, and **the retailer owns a mapping that nobody else maintains for it**. Get that mapping wrong and the till sells a box the scheme will not pay for, which surfaces weeks later as a rejected claim nobody can reconstruct. Draw it as two identifier attributes on one Business Object with an explicit mapping object between them, name an owner, and put the mapping's error rate on a report — because in the first draft of Milkwood's data model it was a column in a spreadsheet maintained by one person in Dr Moosa's team.

The application landscape, first draft

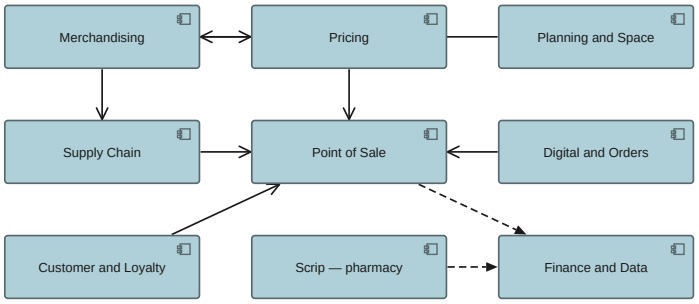


Figure 17.7 The landscape the chief information officer put in a board pack, and the expensive error in it. One box labelled Pricing, with three price masters behind it. A landscape organised by layer and labelled by capability hides every duplicate master in the estate, every time, because duplication is invisible at that level of abstraction. The second, quieter error: the dispensing system drawn as one departmental component among twenty-two.

Twenty-two boxes, grouped by layer, arrows between them. Everybody recognised their system. The chief information officer put it in a board pack. And it is wrong in one specific, expensive way.

There is a single box labelled **Pricing**, and there are three price masters behind it.

That is not a drawing error. It is the exact mistake that keeps a retailer from fixing its price integrity, because the diagram answers the question "do we have a pricing system" — yes — instead of the question anybody actually needs answered, which is "how many systems can change a price, and which

one wins". A landscape organised by layer and labelled by capability will hide every duplicate master in the estate, every time, because duplication is invisible at that level of abstraction.

The second error in the same figure is quieter. Scrip appears as one application component among twenty-two, which makes it look like a departmental system. It is not. It is the sole lawful master of a regulated record class, inside a boundary the rest of the estate may not cross.

The application landscape, second draft

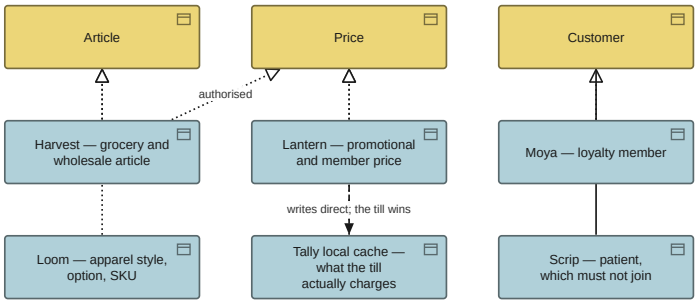


Figure 17.8 The second draft draws the mastering, not the systems. Three article masters, one of them accidental; three price masters where the till wins because the till is where the money moves; four customer identities with four lawful bases, one of which must never join the others. Nine days, four interviews and an afternoon in a store. The first draft asked what systems exist; this one asks who may change each thing and who wins.

The second draft changes one thing: it draws the **mastering** rather than the systems, and it draws it per data object.

Read the figure as three findings.

The article is mastered three times. Harvest for grocery and cash-and-carry, Loom for apparel, and Trellis holds an online catalogue that has quietly become authoritative for product content because nobody else would own it. Three masters, one of them accidental.

The price is mastered three times, and nobody is authoritative. Harvest holds the authorised regular price. Lantern holds promotional and member prices and writes to the till directly. Tally's local cache holds what the till actually charges, and when the three disagree the till wins, because the till is where the money moves. Hennie Loots has known this for six years. Nobody had drawn it.

The customer is mastered four times. Moya holds the loyalty member, Trellis holds the online account, the store-card book holds a credit consumer under the National Credit Act, and Scrip holds a patient whose record is special personal information and must not join to any of the others without a separate lawful basis. Four identities, four lawful bases, four retention rules, and a customer data platform project that had been scoped as an integration exercise.

> **WHAT THE SECOND DRAFT COST**
>
> Nine days, four interviews and one afternoon in a store. It is not a bigger diagram than the first draft; it is a differently organised one. The first draft asked "what systems do we have". The second asked "for each thing we care about, who may change it and who wins". That question is the whole difference between an inventory and an architecture.

The store and the channel

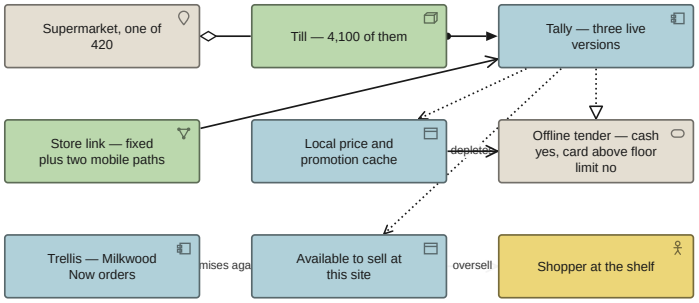


Figure 17.9 The store slice, and the four things the head of store systems insisted on. Till as Node and Tally as component, because every question he is asked is about Nodes. The local cache on the view, because the store's ability to complete a sale must not depend on anything outside the store. Offline tender drawn separately from offline pricing, because Milkwood has three of the four capabilities that matter. And Milkwood Now drawn as competing for the same inventory position the shelf is selling.

Four choices in the store view, each of which Kobus Venter insisted on.

The till is a Node and Tally is an Application Component. There are 4,100 of the first and three live versions of the second. Every question Kobus is asked is about Nodes: how many are on the current release, how many have a verified scale beside them, how many trade when the link drops.

The local price and promotion cache is on the view. Deliberately. Milkwood's design rule, written after load-shedding and kept afterwards, is that **the store's ability to complete a sale must not depend on any system outside the store.** Everything else may degrade. A view that shows the till depending on a central price service and does not show the cache has described a system that closes when the line drops.

Offline tender is drawn separately from offline pricing, because they are different capabilities and Milkwood has only three of the four that matter. It can look up and price offline. It can log the journal offline and forward. It can accept cash. It cannot authorise a card above the floor limit, which in a market where cards carry most of the *value* of payments means a store on a dead link can trade in volume and not in value.

Cash handling is a capability on the view, not a legacy footnote. Cash is the majority of South African payment volume even where it is a minority of value, and the monthly pay-and-grant cycle produces an amplitude that no weekly model captures: Milkwood's busiest hour of the month is on the first and its quietest is around the twenty-second. That shows up on the technology view as till capacity and on the business view as rostering, and it shows up here as cash declaration, cash-up, float management, safe control and cash-in-transit scheduling across 806 sites — which is a set of capabilities with systems behind them, and the first draft of the view had none of them.

Milkwood Now is drawn as competing for the same stock. The app promises a one-hour delivery against available-to-sell at a supermarket that is simultaneously selling that stock to the person standing in front of the shelf. Drawing the two demands on one inventory position is how the oversell problem became a design conversation rather than a weekly argument between Zanele and the store operations team.

The pharmacy, and the boundary around it

Milkwood's third wrong model was the pharmacy, and it was wrong in the dullest possible way: Scrip was drawn as an application serving a department of the store, and the dispensary was an attribute of the site.

Dr Moosa's objection was short. She is personally accountable, under the Pharmacy Act, for whether 142 dispensaries may lawfully operate. That accountability cannot be delegated to a manager or a system. A model in which her dispensary is an attribute cannot represent the thing she is responsible for.

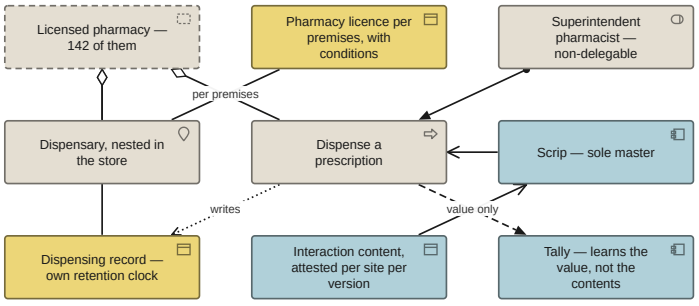


Figure 17.10 Milkwood's third wrong model, corrected. Scrip had been drawn as an application serving a department of the store, with the dispensary as an attribute of the site — a model in which the superintendent pharmacist's personal, non-delegable accountability for 142 dispensaries cannot be represented at all. The Grouping is a wall, not a lane. Four controlled edges cross it, and drawing them switched off a four-year-old extract that had been pulling patient records into the group warehouse.

The corrected view draws the boundary as a **Grouping** — a wall rather than a lane — and it changed four things in the programme.

The dispensary became a nested Location with its own licence record. A licence per premises, with conditions, the Council recording reference, and the responsible-pharmacist designation, plus a control that prevents the dispensary trading when either lapses [4].

Scrip became the sole master of patient, prescription and dispensing record, and the generic extract that had been feeding Kelp from the whole store database was switched off in week three. It had been running for four years.

The interaction-checking content became a tracked, attested artefact. Good Pharmacy Practice requires the latest version of software providing warnings or interaction checks [5]. Milkwood has seventeen dispensaries on links that fail weekly, which means content currency has to be delivered to an offline-capable endpoint and *proved*, per site, per version. The programme had a work package for it within a month of the view being drawn.

The till learns the value and not the contents. Milkwood Rewards may know a basket included a dispensary line and what it cost. It may not know what was dispensed, absent a separate consented basis. That one edge removed a personalisation feature from the roadmap and probably saved the group a regulatory finding.

The data view

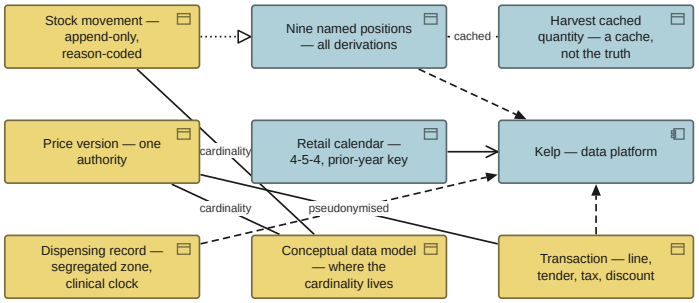


Figure 17.11 The data view, carrying three statements and one admission. The movement log is the system of record for stock and every position is a derivation — Harvest's cached quantity is a cache being treated as truth, which is why the website and the store back-office disagreed every Saturday. The dated price version has one authority, which is a target rather than a baseline. And the dispensing record sits in its own zone with its own clock. None of the cardinality is here, and the note says so.

The data view is where the article and price structure meets the systems that hold it, and Milkwood's version carries three deliberate statements.

The movement log is the system of record for stock, and every position is a derivation. Nine named positions, each a filter and a sum over append-only movements. Harvest holds a cached quantity per article-location, which is fine as a cache and was being treated as truth, which is why Trellis and the store back-office disagreed every Saturday.

The dated price version is a first-class object with a single authority. That is the target, not the baseline, and the view says so.

The dispensing record sits in its own zone with its own access model and its own retention clock. Pharmacy retention runs on a clinical clock, not a commercial one, and a deletion request under POPIA collides with a statutory retention obligation — the statute wins, but only if the obligation is recorded against the record.

One thing the view cannot say, and the note on it says so: none of the cardinality is here. One style has one or more options; one delivery carries lines from three orders; one membership has one primary card and up to three family cards. Those live in the conceptual data model, in crow's-foot notation, and the view points at it.

The technology view

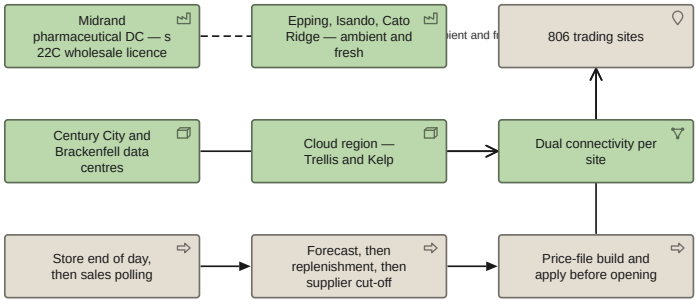


Figure 17.12 Four distribution centres, 806 sites, two data centres and a cloud region — and three questions nobody else answers. Midrand holds a wholesale licence under the Medicines Act, which makes it a different kind of facility with its own inspection exposure. The store link is a dependency, not a decoration: dual connectivity and power on the till and network layer, forced by load-shedding and kept because power and network are both assumed to fail. And the batch chain is here, with the honest note that the notation cannot express its slack.

Four distribution centres, 806 sites, 4,100 tills, two data centres and a cloud region. The technology view earns its place by answering three questions nobody else can.

Where does the regulated distribution happen? The Midrand pharmaceutical distribution centre holds a wholesale licence under the Medicines Act, which makes it a different kind of facility from Epping, Isando and Cato Ridge, with its own licence record and its own inspection exposure.

What does the store depend on? The store link is drawn as a Communication Network with a dependency, not as a line. Dual connectivity — fixed plus two mobile paths — and uninterruptible power on the till and network layer rather than only on the back-office server. That configuration was forced by load-shedding, and Milkwood's architecture principles now keep it on the grounds that power and network are both assumed to fail. Somebody will propose designing it back out on cost grounds within two years.

Which calendar is authoritative? Milkwood runs a 4-5-4 retail calendar with a June year end, and three systems derive it independently — which is how Thread's Sunday-start week and Supermarkets' Monday-start week put the same Monday's sales into different retail weeks in different reports. The calendar is a single Data Object on this view with every system reading it, and that is a target rather than a baseline. It also has to carry the prior-year comparable day key explicitly, because a fifty-three-week year cannot be compared without one and every like-for-like number in the results announcement depends on it.

What is the batch window? The view carries the overnight dependency chain as Triggering edges and the honest note that ArchiMate cannot express duration or slack. Milkwood's cycle has four hours of slack to store opening on a Tuesday and eleven minutes on the Monday after a promotion changeover, and that single fact constrains every programme in the group. The book draws the cycle itself as a timing diagram for exactly this reason.

Gaps, and the roadmap in two plateaus

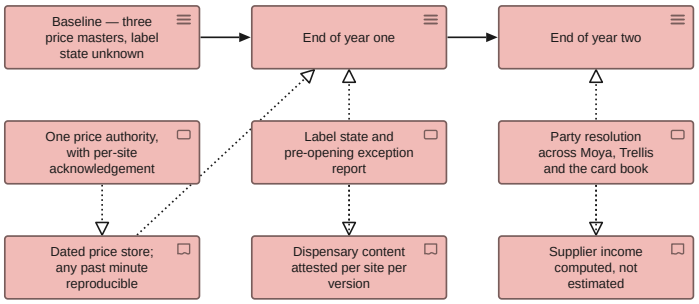


Figure 17.13 Two plateaus with exit criteria a sceptical chief financial officer can test. Year one: one dated price store with per-site acknowledgement, a pre-opening label exception report, and content currency attested per dispensary — testable by picking a store and a past minute and reproducing the displayed price. Year two: one resolved party across three systems with the patient deliberately not joined, and supplier income computed rather than estimated. Harvest is strangled, not replaced, and the view carries the peak freeze.

Six gaps survived the analysis, ranked by what they cost rather than by how hard they are. Ranking by cost rather than by effort is what stopped the roadmap becoming a list of the easy things.

Gap	What it costs	Owner
No single price authority	Statutory exposure on every unit sold at a price the master never authorised, at unknown scale	Hennie Loots
No label state before store opening	A section 23 complaint cannot be answered from data, and nobody knows which stores are exposed today	Hennie Loots
Three article masters, one of them accidental	Every listing done three times; online content drifts; range reporting cannot be reconciled across formats	Ntombi Zwane
Four customer identities, no resolved party	Personalisation unprovable, erasure requests unservable, and a patient record one careless join away from a marketing audience	Zanele Dube
Supplier income in spreadsheets	The number most likely to cause a restatement is an estimate renegotiated annually	Rajesh Naidoo
No attested content currency in 17 dispensaries	A Good Pharmacy Practice breach that is provable against the group rather than arguable	Dr Ayesha Moosa

The roadmap has two plateaus and the exit criteria are testable, which is the only rule that matters.

End of year one. One dated price store with a single authority for company-owned supermarket and pharmacy front-shop lines, per-site application acknowledgement on every price file, a pre-opening label exception report in every store manager's hands, and content currency attested per dispensary. Testable: pick a store and a past minute, and reproduce the displayed price.

End of year two. One party identifier resolved across Moya, Trellis and the store-card book, with Scrip's patient deliberately *not* joined; supplier income agreements machine-readable and the accrual computed rather than estimated; and the online available-to-sell published as a service rather than computed in the channel.

Two things are deliberately not on the roadmap, and saying so out loud was Dirk's contribution.

Harvest is not being replaced in either plateau. It is being strangled: price out first, then the inventory ledger, then planning. The argument for that sequence is not technical preference. It is the South African failure case, and it is worth stating to a board in its own words. SPAR's distribution centre go-live — dated variously to October 2022, February 2023 and "early 2023", and the sources do not agree [6][7][8] — is reported to have cost an estimated R1.6 billion of group turnover and about R720 million of KwaZulu-Natal profit in the year to 30 September 2023, with profit after tax down 80% and an implementation estimated at around R1.8 billion [6][7]. **The lost sales exceeded the implementation cost in a single year.** Three directors had left by the August 2024 trading update [9], and in January 2026 a franchisee owning 46 stores sued for about R168.7 million [8]. And SPAR's own remediation line is the most transferable sentence in this book: the group said it would include **"separating the ERP from the warehouse system and implementing them independently"** [7]. Milkwood's roadmap does exactly that, and cites it.

The merchandising platform consolidation is not on the roadmap at all, because the second-draft strategy view established that the three formats do not share a planning grain. What is on the roadmap is the shared layer: one party, one price authority, one tender and settlement path, one calendar.

> **THE RETAIL DELIVERY YEAR IS NINE MONTHS**
>
> Milkwood freezes the production estate from mid-November to mid-January, and the freeze covers more than software — master-data restructuring, range change, planogram resets and store refits as well. So a two-year roadmap has eighteen months of delivery in it, the January and February releases are the largest and riskiest of the year, and a plan built on a twelve-month year is wrong by a quarter before it starts. Put the freeze on the plateau view as a period, and the sequencing conversation becomes honest.

What the model was for

Nine months after Yusuf started, the model had done four things, none of which was documentation.

It ended the single-merchandising-platform argument with a picture rather than a memo. It made three price masters visible to a board that had been told there was one pricing system. It stopped a data extract that had been pulling patient records into a group warehouse for four years. And it gave the programme a plateau whose exit criteria a sceptical chief financial officer could test.

That is the test for a worked model. Not whether it validates, and not whether it is complete. Whether anybody's decision changed because it existed. If the answer is no after six months, the problem is not the notation.

THE CHAPTER IN TEN LINES

- Milkwood Retail Group is invented; what is real is the shape — five formats, one balance sheet and almost nothing shared below the till.
- Draw the motivation view before the landscape: a landscape produced before anybody has said what hurts becomes an inventory that is then defended because it took four months.
- The first strategy view was wrong because “Assortment Planning” is three abilities with three planning grains, and “Price and Promotion Management” is a capability the group does not possess in its dispensaries or its franchise stores.
- The second draft separates a genuinely shared layer from three format stacks, and scores maturity against named evidence rather than against opinion.
- The franchisee is a Business Actor in a customer Role, not a Location — and that single choice stops every downstream model inheriting a control the group does not have.
- A pharmacy article carries two identifier schemes for the same product, GTIN for receiving and selling and NAPPI for claiming, maintained by different bodies and mapped by nobody but the retailer.
- The first application landscape hid three price masters behind one box labelled Pricing; organising the second draft around mastering rather than systems is what made duplication visible.
- Four customer identities with four lawful bases — loyalty member, online account, credit consumer and patient — and the patient must never join the others without a separate basis.
- The pharmacy was modelled wrong twice over as a department with the dispensary as a site attribute; drawing the boundary as a Grouping switched off a four-year-old extract of patient records into the group warehouse.
- The buyer-power onus reversal means the group must produce a price-comparability dataset it does not maintain — and that is the same data the supplier-income accrual needs.
- Harvest is strangled rather than replaced, and the argument for that sequence is the published South African failure case, whose own remediation line was to separate the ERP from the warehouse system and implement them independently.
- The retail delivery year is nine months, because the peak freeze covers master data, range, planograms and refits as well as software — so put the freeze on the plateau view.
- The test of a worked model is not whether it validates but whether anybody's decision changed because it existed.

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CHAPTER 18

Your first ninety days

Ninety days is not enough to understand a retailer, but it is enough to be useful — if you get into a store and a distribution centre before you open the application inventory, ask each specialist the one question only they can answer, and draw something small enough that the people who do the work will correct it.

Ninety days is not enough, and it is enough to be useful

You will not understand a retailer in ninety days. Nobody does. People who have been inside one for twenty years do not understand the whole of it, which is why they are so generous with the parts they do.

What ninety days is enough for is this: to know what the business is for, to know what hurts, to have drawn one true thing that the people who do the work have corrected, and to have earned the right to be in the room when the next decision is made. That is the whole ambition of this chapter, and achieving it has less to do with modelling than with the order in which you do things.

The order matters because of one specific trap. An architect arriving in a retailer is handed a mandate that sounds like "give us a view of the landscape". It is the wrong first task and it is almost irresistible, because it is the thing you know how to do, it needs no political capital, and it produces something visible in a fortnight. Resist it. A landscape diagram produced before anybody has told you what hurts is a diagram nobody asked a question of, and it sets the price of architecture at the cost of documentation. Worse, in a retailer it will be organised by system and labelled by capability, which hides every duplicate master in the estate — and duplicate mastering is the problem you were actually hired to find.

Retail has one more reason to care about order, and it is the thing that makes this different from a bank or an insurer. **The people who know how the business works are standing in a store or a distribution centre.** Not in head office. The buyer knows the range, the planner knows the money and the store manager knows whether any of it happened. If your first fortnight is spent in meeting rooms in a business park, you will learn the vocabulary and not the mechanism, and you will spend month two unlearning.

Before week one: what is already public

Two evenings of reading will put you ahead of most people who have been there a year, because almost nobody inside a retailer reads what the retailer publishes about itself.

The annual report and the results presentation. Read the segment results first: which formats make money, which do not, and what the chief executive says has to change. Then the trading statement language — like-for-like growth, internal selling-price inflation, trading density per square metre, space growth — because those four numbers are what the board actually manages, and every one of them is a derivation your data model has to support. Then the analyst questions at the end of the results call, which are the most honest document a retailer produces.

The year-end date, because it is not January. South African retailers do not share a financial year. Three of the large listed groups report to the end of February, the end of June and the end of August respectively, and each runs a 4-5-4 retail calendar on top of that, with a fifty-third week roughly every five to six years [1]. Know your employer's year end before you walk in, because every plan, target, promotion period and like-for-like comparison hangs off it.

The Competition Commission's grocery retail market inquiry report. Published 25 November 2019, and still the best public description of how South African grocery actually works — exclusive leases, trade terms, buyer groups, spaza shops and the economics of a mall [2]. It is the single most useful document for an outsider, and nobody will mention it to you.

And the honest limit on what is knowable. No South African retailer's technology estate is publicly sourceable except where a vendor has published a case study, and there are only two of those. Market folklore about who runs what circulates freely and is mostly wrong. So the two published cases are worth reading precisely because they are the exception, including the one that admits its own overrun: a point-of-sale programme planned at just under six months went live two months late with more than thirteen interfaces, and the retailer's own account says the team pushed its internal resources too hard and had to revise its timelines [3].

Week one: get into a store, then a distribution centre

Do this in the first week, before you have read the application inventory. Not a tour. A shift.

Stand at a till for two hours, on a Saturday or the first of the month. You are looking for four things: how many price disputes happen and how the cashier resolves them; how often an age-restricted or scheduled item stops the sale; what the operator does when the authorisation takes too long; and how much of the basket is cash. The first of the month matters because South African retail has a monthly amplitude that no weekly model captures — the busiest hour of the month is on the first, and the till capacity, the rostering and the cash-handling design all follow from it.

Walk the back of the store with the receiving clerk. Ask to see yesterday's deliveries booked in, and ask what happens when a delivery is short. In most retailers the answer is that there is no time to check at the dock and the claim is not followed up, which means a short delivery quietly becomes shrink. That is administrative shrink with a systems cause, and you have just found it in week one.

Find the shelf-edge labels that did not print. Walk an aisle after a promotion changeover and compare labels to the till. If the store manager has a printed exception report in her hand, the estate is healthier than average. If she tells you her staff walk the aisles checking, you have found the finding — because where two prices are displayed concurrently, the lower one binds [4], which makes a label that silently failed to update a legal exposure rather than a housekeeping task.

Then a distribution centre, on a night shift if they will let you. Watch a wave being picked, ask the manager what the worst thing a store can do to him is, and ask what happens when the order file arrives late. The answer to the last question is the single most useful thing you can know about the

overnight cycle, and nobody in head office will give it to you accurately.

And if the group has a pharmacy format, spend an hour in a dispensary. Stand on the public side of the counter and watch the boundary being enforced. Then ask the pharmacist two questions: what does the system stop you doing that you need to do, and when did the interaction-checking content last update? The second question matters because the rules require the latest version of software providing warnings or interaction checks [5], and a dispensary on a poor link is where that obligation breaks.

Eight documents, asked for by name

Ask for these on day one. Asking by name is itself a signal: it says you know what a retailer has, which shortens the period during which people explain retailing to you.

Document	Who has it	What it tells you
The price-file distribution and application report	Store systems, or pricing	Whether the group knows its prices were applied, per site, per till, per label. If it does not exist, you have found artefact one
Last month's overnight batch failure log, with the critical path	Operations	The truest document about the estate. Nothing else tells you what actually breaks, or how much slack there is before the stores open
The change-freeze calendar	The programme office or change management	The delivery year, which is about nine months. Get this before you commit to any date
The last full stock-take variance report	Finance or loss prevention	Shrink by store and category, and critically how much of the adjustment is reason-coded. Uncoded adjustments mean variance cannot be analysed
The licence register — liquor, pharmacy, certificates of acceptability, scale verifications	Risk, store operations, and often nowhere	Whether the group can say which sites are lawfully trading today. Usually a spreadsheet
The interface inventory, with a count of direct database consumers	Integration	Why the merchandising system cannot be replaced. The number of direct readers is the number you want
The supplier trading-terms schedule	Commercial finance	Whether terms are data or free text in contracts. This is both a competition-law question and the supplier-income question
The trading calendar and the period-close timetable	Finance	The 4-5-4 structure, the year end, and which systems derive the calendar independently

Two notes on reading them. The batch failure log is the one people are surprised you want; ask anyway, because the critical path through the overnight cycle is the constraint every programme in the group will hit. And the licence register is the one that most often does not exist in any usable form, which makes it the easiest credibility win available to you.

Who to meet, in what order

Order by what each person can tell you that nobody else can, and start furthest from your own discipline.

1. **A store manager, in her store.** Ask: what do you do by hand every week that the system should do? Then ask how many systems she logs into for one job. The answer is usually eight or more.
2. **The receiving clerk.** Covered above. Fifteen minutes, and it reframes shrink.
3. **The head of pricing.** Ask: how many systems in this group can change a price, and which one wins at the till? Nobody has drawn this. The answer is usually three and the till.
4. **The replenishment analyst.** Ask: when a promoted line sells out on day two, whose fault is it? The answer explains the whole relationship between the promotional calendar, the forecast and availability, and the failure it describes — promoted volume not ordered — is the commonest availability failure in grocery.
5. **A buyer, and separately a merchandise planner.** Two meetings, never one. The buyer will tell you what is in the range; the planner will tell you whether the money allows it. If they describe the same job, somebody has mis-designed the organisation.
6. **The supplier income accountant.** Ask: can you reconstruct last year's rebate entitlement from data? The pause before the answer is the finding. This is reliably the highest-value unautomated process in a retailer.
7. **The responsible or superintendent pharmacist, if there is a pharmacy format.** Ask what she is personally accountable for. The answer will reorganise your model.
8. **The space planner.** Ask how often the planogram matches the fixture. Then ask who measures compliance. Usually nobody, which means shelf capacity — an input to every replenishment calculation — is an estimate.
9. **The head of store systems.** By now you can ask in the business's language: how many tills are on the current release, how many can take a card when the line is down, and how many have a scale whose verification is current [6]?
10. **The integration architect, last.** Ask for the count of direct database readers on the merchandising system. That number is the replacement cost of the estate, expressed honestly.

> **THE ONE QUESTION TO ASK EVERYONE**
>
> "What do you do by hand every week, and what would have to be true for you to stop?" Ask all ten. The overlaps are your architecture backlog, ranked by how many people independently named the same thing.

What to draw first, and why it must be small

Your first model is not a landscape. It is one slice, narrow enough to finish and check, chosen because somebody has already told you it hurts.

In a retailer the best first slice is almost always **the price change**: from the buyer's decision to the shelf-edge label and the till, with the acknowledgement or its absence drawn in. It is small, it crosses six systems, it has a hard legal consequence, and everybody from the store manager to the chief financial officer understands why it matters.

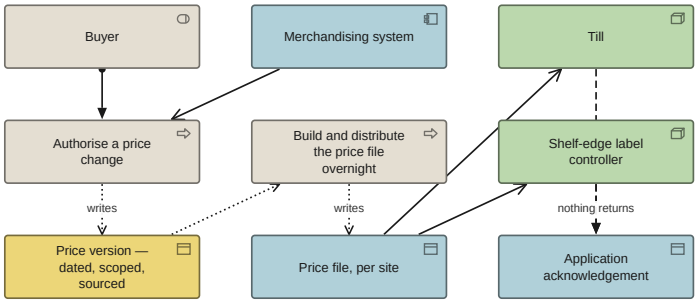


Figure 18.1 The first slice: a price change from the buyer's decision to the shelf label. Nine elements, three days, and one edge that is a finding — the acknowledgement that comes back from the tills and does not come back from the label controller. That absence is a sentence somebody has to answer, and it is why this slice earns its keep where a landscape does not. Take it to the head of pricing and to a store manager and let them correct it.

Eleven elements. Drawn in three days, wrong in two places, and that is the point — you take it to Hennie in pricing and to a store manager and let them correct it. A model corrected by the head of pricing is worth ten that have not been, and the correction is the relationship.

Three rules for the first slice.

It must contain an edge that is a finding. Not a description. In the figure it is the missing acknowledgement from the label controller: that absence is a sentence somebody has to answer. If your first model contains no edge anybody would argue with, you have drawn an inventory.

It must be finishable in a week. An unfinished model is indistinguishable from no model, and the first one has to exist before you know enough to make it good.

It must name no vendor you cannot source. This matters more in South African retail than people realise. Put the group's own internal system names on your boxes, not a vendor's, unless you have seen the contract.

What to refuse in month one

Four things, all of which will be asked of you, and all better refused politely with a reason and an alternative.

A full current-state landscape. Offer a slice and a date for the next slice. The honest argument is that a landscape built before you know the questions will be organised around the wrong things, and will then be defended because it took four months.

A target-state architecture. You cannot have one in month one because you do not know what constrains the group. Offer three or four principles instead, each with a decision it has already caused — and in a South African retailer one of those principles writes itself: the store trades on local capability, and the link is an optimisation rather than a dependency.

A merchandising-replacement recommendation. Everybody will ask, and everybody has an opinion. Say you will have a view in month five, and that the first thing you will do is decompose what the group calls "the merch system", because the decomposition changes the question. It is at least four things — the article master, the commercial ledger, the inventory ledger and the transaction engine — and whether they are separable decides whether modernisation is a sequence of affordable moves or one unaffordable replacement.

A diagram for a board pack in week two. The hardest to refuse, because it comes from somebody senior being helpful. The problem is not the board pack. It is that a diagram drawn to be reassuring in week two becomes the thing your function is known for. Offer one page in month three, built from a model, containing something the board does not already know.

Six numbers that tell you whether the estate is healthy

You can form a defensible view of a retail estate in a month from six numbers, none of which is a technology metric. Ask for each by name; if nobody can produce one, that is the finding.

1. **The percentage of price files acknowledged as applied, per site, last month.** Most groups cannot produce it. A group that can is better run than its peers.
2. **The number of shelf-edge labels that failed to print last night, and who reads that report.**
3. **The slack between the end of the overnight cycle and store opening, on a normal Tuesday and on the Monday after a promotion changeover.** The second number is the one that constrains every programme in the group.
4. **The proportion of stock adjustments that carry a reason code.** Below about half, shrink cannot be decomposed into its causes, which means loss prevention is working from months-cold stock-take variance.

5. **The number of systems that can change a price, and the number that master an article.** Three and three is normal. One of the article masters is usually accidental.
6. **Which of the four offline capabilities the till estate actually has** — offline lookup and pricing, offline journalling, offline card tender, offline loyalty. Most estates have the first two and not the third, which means a store on a dead link can trade in volume and not in value.

The three artefacts that buy you credibility

Credibility is bought with artefacts other people need and nobody is producing. These three qualify in nearly every retailer, and all three are achievable in ninety days.

One: the price authority map. Every system that can set or change a price, the scope it may change, and where that is enforced. Nobody has drawn it, several people urgently need it, and it is the artefact most likely to change a funding decision in your first quarter.

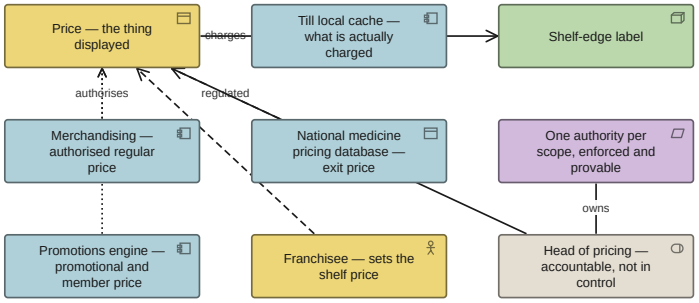


Figure 18.2 The price authority map — the artefact most likely to change a funding decision in your first quarter. Every system that can set or change a price, the scope it may change, and where that is enforced. Three internal masters, one regulated feed the group must apply exactly, and a franchise estate where the group holds a recommendation rather than a price. The head of pricing is accountable for all of it and controls one box.

Two: the site authority register, as data. Every licence, certificate and instrument verification held per site, with the control that enforces it and the date it expires — liquor licence and its trading hours, pharmacy licence and responsible pharmacist, certificate of acceptability, scale verifications [6]. It usually exists as four spreadsheets owned by four people. Turning it into one dated dataset with a control answers a question the board is asked and cannot currently answer: which of our sites is lawfully trading today?

Three: the overnight timing diagram. Not a swimlane. A timing diagram of the twelve-step cycle from store end-of-day to the data-platform load, with durations and slack marked, and the price-file distribution's deadline against store opening drawn as a hard line. For an architect arriving from banking or insurance this is the single most clarifying picture in retail, and in most groups the critical path exists only in an operator's head.

The politics, honestly

The merchandising replacement everybody has an opinion about. You will be asked within a fortnight. Some of the opinions are informed and all of them are strongly held. The useful move is to decline the question and reframe it, and the reframing has public evidence behind it. The best-documented South African programme failure was a distribution-centre go-live whose lost sales exceeded its implementation cost in a single year, whose damage ran for years rather than weeks, and which landed on independent retailers who had not chosen the system and have since sued [7][8][9]. The retailer's own stated remediation is the most transferable sentence available to you: it would include separating the enterprise resource planning system from the warehouse system and implementing them independently [8]. Quote it. It converts an argument about preference into an argument about sequence, which is the argument you can win.

The buyer who will not be told what to range. The buyer's authority over range is real, commercially justified and defended. An architect who arrives with a data-governance framework that implies the buyer needs approval to list a line will lose, and should. The way in is not governance, it is the thing the buyer cannot currently get: total supplier income by supplier, landed cost that is actually landed, and a range review that does not have to be rebuilt from scratch each cycle. Solve one of those and the master-data conversation becomes possible.

November, when nobody will talk to you. The trading calendar is not a scheduling inconvenience, it is a political fact. From roughly mid-November to mid-January a grocer is in peak and in change freeze, and the freeze covers master-data restructuring, range change, planogram resets and store refits as well as software. Nobody senior will give you an hour, no decision will be taken, and anything you start will be parked. Plan your ninety days to end before it, or to start after it. If your first day is 1 November, spend the first six weeks in stores — which is the one thing that gets *more* useful at peak, not less.

And the thing nobody will say out loud. There is no published South African shrink benchmark. A retailer cannot compare its own shrink to the industry, which means the number is argued about internally with no external reference. If you are shown a benchmark, ask where it came from; if it is the American retail federation's figure, it is a United States number and must be labelled as one.

What week thirteen looks like



Figure 18.3 Ninety days, in four overlapping moves. The order is the whole method: stores before systems, documents and people before drawings, one small slice before any landscape, and a ranked statement of what hurts instead of a target state. The bar along the bottom is what you decline, politely, with a reason and an alternative.

By week thirteen you should be able to hand over four things and say one sentence.

The four things: a first slice that has been corrected by the people who do the work; one of the three credibility artefacts, finished; a ranked list of what hurts, with the number of people who independently named each item; and three or four architecture principles, each with a decision it has already caused.

The sentence is the ambition of the whole chapter: *here is what this group cannot currently answer, and here is the smallest change that would let it*. If you can say that, with evidence, to the chief operating officer and the chief financial officer in the same meeting, you have done the ninety days properly — and you will be in the room for the next decision, which was the point.

What you will not have is a model. You will have four views, two of them wrong, one artefact and a backlog. That is the correct state at week thirteen, and anybody who tells you a retailer of this size can be modelled in a quarter is selling something.

Then the method. Volume 2 takes the notation and the vocabulary of this book and puts them inside a repository: a populated metamodel for an invented retailer, with the views, the properties, the governance and the maintenance cycle that turns a set of diagrams into an architecture a group can actually run on. This volume teaches you what retail is. The next one teaches you how to keep a model of it true.

THE CHAPTER IN TEN LINES

- A landscape diagram produced before anybody has told you what hurts sets the price of architecture at the cost of documentation — and in a retailer it will hide every duplicate master in the estate.
- The people who know how the business works are standing in a store or a distribution centre, so spend week one at a till, on a receiving dock, in an aisle after a promotion changeover and on a DC night shift.
- Read what the group publishes about itself first — segment results, the trading-statement derivations, the year end and the 4-5-4 calendar — and read the grocery retail market inquiry report, which nobody will mention to you.
- Ask for eight documents by name; the overnight batch failure log and the price-file application report are the two people are surprised you want, and the licence register is the one that usually does not exist.
- Meet the store manager, the receiving clerk and the pricing, replenishment, buying, planning, supplier-income, pharmacy and space people before any technologist, and ask everyone what they do by hand every week.
- Draw the price change first: small, six systems, a hard legal consequence, and an edge that is a finding rather than a description.
- Refuse the full landscape, the target state, the merchandising verdict and the week-two board diagram — each politely, with a reason and an alternative.
- Six numbers tell you whether the estate is healthy, and none is a technology metric; the slack between the end of the overnight cycle and store opening is the one that constrains every programme in the group.
- The three artefacts that buy credibility are the price authority map, the site authority register as data, and the overnight timing diagram with its slack marked.
- Reframe the merchandising replacement with the published failure case and its own remediation line — separate the ERP from the warehouse system and implement them independently — which turns an argument about preference into one about sequence.
- The buyer's authority over range is real and defended; the way in is total supplier income, true landed cost and a range review that does not have to be rebuilt each cycle.
- Nobody will talk to you from mid-November to mid-January, so plan the ninety days around the peak freeze — and if you must start in November, spend it in stores.
- At week thirteen you have four views, two of them wrong, one finished artefact and a ranked backlog — and that is the correct state.

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- **2.** Competition Commission, Grocery Retail Market Inquiry — Final Report summary, 25 November 2019 https://www.compcom.co.za/wp-content/uploads/2019/12/Grocery-Retail-Market-Inquiry-SUMMARY_.pdf
- **3.** Pick n Pay modernises its POS, SAP News Africa, July 2023 https://news.sap.com/africa/2023/07/pick-n-pay-modernises-its-pos/
- **4.** Consumer Protection Act 68 of 2008, s 23 — disclosure of price of goods or services https://www.acts.co.za/consumer-protection/23_disclosure_of_price_of_goods_or_services
- **5.** Rules relating to Good Pharmacy Practice, consolidated https://www.saflii.org/za/legis/consol_reg/rrtgpp362
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- **7.** SPAR loses R1.6bn due to failed SAP project, ITWeb https://www.itweb.co.za/article/spar-loses-r16bn-due-to-failed-sap-project/Pero3qZ35JyvQb6m
- **8.** SPAR taking the pain of a botched SAP project, TechCentral https://techcentral.co.za/spar-taking-pain-botched-sap-project/240082/
- **9.** Giannacopoulos family sues SPAR for R170m over SAP fiasco, Business Day, 28 January 2026 https://www.businessday.co.za/companies/2026-01-28-giannacopoulos-family-sues-spar-for-r170m-over-sap-fiasco/

APPENDIX A

The cross-maps

The reference pages an architect actually uses. Five dimensions defined with local IDs, then seven maps that join them — all hinged on business process, because retail has no industry reference model to hinge them on instead.

What these are, and what they are not

Everything here hangs off **business process**, because process is the only dimension the business will argue about and therefore the only safe hub. Nobody fights over a capability map; everybody fights over who signs off a range review. Put process at the centre and every other dimension has something to attach to.

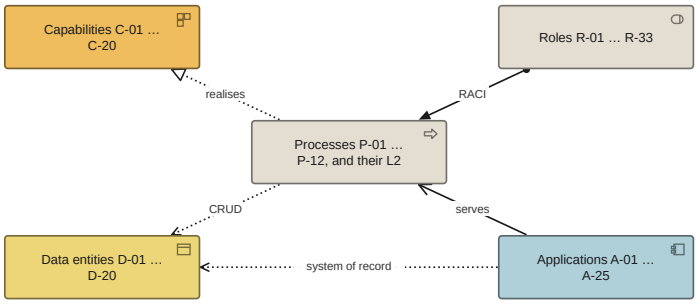


Figure 19.1 Process as the hub, and each cross-map as the ArchiMate relationship it really is: a process realises a capability, a role is assigned to it, it accesses the data, an application serves it. The one map that does not touch process — application to data — is the Access edge along the bottom, and it is the one that settles the system of record.

Retail makes that choice easier than banking does, for an uncomfortable reason: **there is no industry reference model to inherit.** Retail has GS1 for identifiers [3], a device-and-receipt programme at OMG whose data models are not in the formal catalogue [2], and a process taxonomy whose licence terms are not stated [1]. The ID series is yours — a freedom, and a trap, because nothing external will tell you it has drifted.

Four warnings before the tables.

A worked starting set for Milkwood Retail Group, not a universal reference. Milkwood is the invented South African multi-format group carried through this book: 420 supermarkets, a 180-store pharmacy chain, an apparel format, a cash-and-carry arm and an online business.

The IDs are local, illustrative, and L1 and L2 only. `P-06`, `C-09` and `A-16` mean something inside this book and nothing outside it — not APQC identifiers [1], not GS1 keys [3]. A real cross-map comes out of workshops with the named role in the room, and where my assignment is contestable the row says so.

Format breaks the map, which is the point of a multi-format example. A row true for the supermarket is often false for the dispensary and half-true for apparel. Price in a supermarket is a commercial decision; in a dispensary it is a gazetted input [10].

Ownership breaks it harder. In a franchised or voluntary-trading estate the group may have **no price, no assortment and no inventory position to master** for much of its turnover: SPAR's members own their stores and set their own range [5], and Pick n Pay reported 992 company-owned stores alongside 620 franchise at 1 March 2026 [6]. A blank cell there is a *contractual* fact.

Adapt it in this order. Processes first. Then roles, the cheapest to verify. Then data, then applications, and capabilities last.

Half one — the definitions

Business processes

Chapter 8's L1 inventory, verbatim, with its triggers and outcomes. **The owners are mine** — chapter 8 carries an honest note instead — and they are the first thing to reconcile with an operations leader, because in most retailers two people claim P-07 and nobody claims P-02.

Two of chapter 8's observations carry into the maps. **Four of the twelve are cycles rather than triggered flows** — P-01, P-03, P-05 and P-12 — and forcing a swimlane onto a cycle produces neither a process model nor a schedule. And **P-03 is deliberately not modelled at L2**, which is why the L2 table has fourteen rows covering eleven processes.

ID	L1 process	Trigger → the outcome that ends it	Owner
P-01	Review the range and authorise an assortment	Review slot → an authorised range per cluster, dated	Category manager, with space and supply chain. A **cycle**
P-02	Onboard a supplier and list an article	A new supplier or line → a payable supplier and a scannable article	**Four owners**: buyer, master data, finance, compliance
P-03	Plan the money and hold open-to-buy	A planning period → a plan and a spend constraint	Merchandise planner. A **cycle**, usually a spreadsheet
P-04	Order goods, receive them and pay for them	A proposal or a buy → goods on hand, a matched invoice, a payable	Supply planner, DC manager, accounts payable
P-05	Forecast and replenish	The nightly run → a released order and stock on a shelf	Demand planning and supply chain. A **nightly cycle** with a contractual cut-off
P-06	Allocate a fixed buy	A fixed quantity → a store-level plan and the movements that execute it	Allocator. Apparel and seasonal
P-07	Change a price and take a promotion live	A booked promotion or cost change → the price at every till and label, on time, with evidence	Pricing manager, with category management and stores [4]
P-08	Execute a markdown	Terminal stock or a delist → stock cleared, inventory revalued	Buyer, store, finance. Three, by category
P-09	Serve a customer at the point of sale	A basket presented → a sale, a settled tender, depleted stock, a compliant receipt	Store manager. In pharmacy a **separately licensed sibling** [7]
P-10	Fulfil a customer order across channels	A checkout → goods with the customer and a settled payment	E-commerce operations; store manager for the pick
P-11	Take a return, refund and dispose	Goods come back → a refund, a disposition, correct stock	**Two owners**: store in store, digital online
P-12	Count the stock and state the inventory	The count calendar → a corrected position, a variance, a shrink number	Store or DC manager; finance; audit. A **cycle**

ID	L2 process, as chapter 8 models it	What happens
P-01.1	Review the range and authorise an assortment	Space feasibility drawn before commercial sign-off, the opposite of how it usually happens
P-02.1	Onboard a supplier and list an article	Nine hand-offs between a commercial agreement and a scannable article
P-04.1	Purchase order to receipt and payable	The receipt step changes shape entirely depending on whether a shipping notice arrived
P-05.1	The nightly replenishment run	One gateway carrying three constraints owned by three functions
P-06.1	Allocate a seasonal buy	Supply is fixed, so the decision is distributive
P-07.1	A price change decided and built	One gateway standing for two checks with different owners
P-07.2	A price change distributed and confirmed	Three exits on purpose: applied, failed, not yet answered
P-08.1	Execute a markdown	An accounting decision disguised as a commercial one [9]
P-09.1	A sale at the till	Loyalty identified first, because personalised offers must resolve before the basket does
P-09.2	A prescription dispensed	The pharmacy boundary drawn as a wall rather than a lane [7][10]
P-10.1	An online order to a picked basket	Order management in the middle lane, because it is what most estates lack
P-10.2	Substitution, handover and the payment that follows	The short-pick refund, and which system may calculate it
P-11.1	A return and a refund	The first gateway is a statute, not a policy [4]
P-12.1	Cycle count and stock take	Blind counting and the movement freeze are what make the variance mean anything

Business capabilities

Twenty capabilities, each of which stays true if you switch off every computer. **Built for a retailer, with local IDs.**

There is no capability called data management, and the banking volume has one. A retailer's data problem is not an absent capability; it is that six legitimate classification structures disagree — merchandise, financial, reporting, space, organisational and price zone — each with a real owner [9]. Naming it lets everybody agree it matters and nobody own a hierarchy, so information governance sits inside C-02 and C-20.

ID	Capability	What the business must be able to do
C-01	Range and assortment management	Decide what to sell, where, from when
C-02	Article and product information	Hold one defensible record of every thing it sells
C-03	Supplier management and trading terms	Know who it buys from, on what terms, at what cost
C-04	Merchandise planning and open-to-buy	Decide how much money may be committed, and commit it
C-05	Demand forecasting	Say how much will sell, where, next week
C-06	Replenishment and inventory management	Keep the shelf full without dead stock; know what it holds
C-07	Allocation and distribution	Decide which units of a fixed buy go where
C-08	Space and shelf management	Decide what sits where, and know whether the store did it
C-09	Price management	Set a lawful price; land it at the till and label at once
C-10	Promotion and markdown management	Run a funded offer; cut permanently when stock will not clear
C-11	Warehousing, transport and goods flow	Receive, store, pick and move goods at a known cost
C-12	Store trading operations	Trade, bank and close daily — and trade when the power goes
C-13	Payment acceptance and cash handling	Take every tender offered, and account for it
C-14	Workforce and credential management	Staff the floor; prove the qualified were present
C-15	Customer identification, loyalty and consent	Know who shopped; use their data only as agreed
C-16	Digital trading and order fulfilment	Sell online against real stock; deliver the order
C-17	Returns, refunds and reverse logistics	Take goods back lawfully; restate stock; recover from suppliers
C-18	Dispensing and regulated supply	Supply a medicine lawfully, at the regulated price
C-19	Loss prevention and shrink management	Measure what went missing, decompose it, reduce it
C-20	Merchandise accounting and supplier income	Value the stock; collect what suppliers owe; file returns

Roles

Chapter 5 names fifty-two jobs in eleven families, each with the decision it owns and what it complains about. These thirty-three appear in the maps, under its own titles. **R-14 covers every fresh department manager** — butchery, bakery, deli — because that is one process with four product sets. The pharmacy and front-shop manager, digital product owner, marketplace operations, CRM analyst, trade marketing, category finance, accounts payable, store controller, internal audit, shrink analyst, investigator, property, store development and the seven technology roles **are not rows here**: in the maps they appear only as consulted or informed.

And the omission that matters: **chapter 5's master data manager is a technology role**, and there is no business role that owns the article master. That is the finding behind C-02's coverage rating, not a gap in this table.

ID	Role	What the person does
R-01	Buyer / category manager	Picks suppliers, range, cost price and trading terms
R-02	Merchandise planner	Plans sales and receipts in money; owns the open-to-buy
R-03	Range / assortment planner	Decides the assortment per cluster, including the tail
R-04	Allocator / distributor	Decides which store gets which units in which sizes
R-05	Replenishment analyst	Maintains forecast parameters and order rules at vast scale
R-06	Space planner	Builds planograms: which SKU, which bay, how many facings
R-07	Supply chain planner (inbound)	Turns the receipt plan into orders; manages lead times
R-08	Demand planner	Owns the forecast and its accuracy
R-09	Distribution centre manager	Runs receiving, put-away, picking, despatch, the yard
R-10	Transport / fleet planner	Builds routes and loads; owns the cost per drop
R-11	Cold-chain / food-safety manager	Owns temperature discipline from supplier to shelf
R-12	Regional / area manager	Runs a portfolio of stores on sales, shrink, labour, standards
R-13	Store manager	Runs the store result, roster, cash, stock, standards
R-14	Department manager (fresh)	Orders, prepares, prices to clear, records waste
R-15	Replenisher / shelf-filler	Fills shelves, faces up, reports gaps
R-16	Cashier / till operator	Scans, takes tender, verifies age, enrols loyalty
R-17	Front-end supervisor	Authorises overrides, voids and refunds. The control point
R-18	Receiving clerk	Books deliveries in; records shortages and damages
R-19	Stock controller	Runs counts, investigates variances, posts adjustments
R-20	Online picker (store-based)	Picks online orders alongside trade; decides substitutions
R-21	Responsible pharmacist	Accountable for lawful operation. Cannot delegate it [7]
R-22	Dispensing pharmacist	Dispenses, screens interactions, counsels, substitutes
R-23	Pharmacist's assistant	Works inside a scope the system must enforce
R-24	Medical-scheme claims clerk	Reconciles claims to remittances; works rejections
R-25	E-commerce trading manager	Runs the online range, price, search, promotion
R-26	Fulfilment operations manager	Owns pick, pack, slot capacity, on-time delivery
R-27	Loyalty programme manager	Owns membership, identification rate, the liability
R-28	Pricing manager	Owns the price file: architecture, zones, rounding
R-29	Promotions planner	Builds the promotional calendar; locks the leaflet
R-30	Markdown / clearance manager	Decides markdown timing, depth and scope
R-31	Supplier income accountant	Accrues, claims and reconciles rebates and allowances
R-32	Inventory accountant	Values stock, books shrink and markdown, reconciles
R-33	Head of loss prevention	Owns total shrink and its decomposition

Conceptual data entities

Chapter 9's fifteen, by ID and under its own names, plus the three it admits as borderline — planogram, shrink event and the supplier income agreement — and two from its second ring, allocation and return. No attributes, no keys, no system names.

Two of chapter 9's cautions carry into the maps. **Shrink event is a classification rather than a new fact**, and it is here because loss prevention will not accept a reason code on a movement. And **markdown is a typed price event rather than an entity**, so it has no D number.

ID	Entity	The question it answers
D-01	Article	What we sell, at one of three levels — style, option, SKU. The argument the model lives or dies on
D-02	Hierarchy node	Whose number this is, under which of the six structures, in which version
D-03	Assortment	Whether this store should carry this, and whether it is ranged, planogrammed or orderable
D-04	Price	What may be charged, where, through which channel, from when. A version, not an attribute [4]
D-05	Promotion	What was offered, who funded it, and what actually fired in the basket
D-06	Location	Where it is, who owns the stock there, and what may lawfully be sold there
D-07	Supplier	Who we buy from, on what terms, out of which door
D-08	Purchase order	What we have committed to, and whether it has arrived
D-09	Inventory movement	Why the position changed, and who says so. Append-only, signed, reason-coded
D-10	Party	Who this person or organisation is, whatever they hold
D-11	Loyalty membership	What they agreed to, and what we owe them
D-12	Transaction	What happened at the till, and whether it reconciles
D-13	Employee	Who did this, and whether they were qualified to [7]
D-14	Forecast	What we expected, as at when, and who changed it
D-15	Dispensing record	Who supplied what to whom, on whose authority, from which batch [7]
D-16	Planogram	A dated SKU-to-position assignment, with the shelf capacity derived from it
D-17	Allocation	A finite quantity pushed to many locations, with a method and a size curve
D-18	Shrink event	The decomposition of book against counted, by cause and by owner
D-19	Return	Goods back under a stated legal basis, each basis carrying its own clock [4]
D-20	Supplier income agreement	The entitlement, and the proof the retailer must keep that it earned it [8]

Conceptual applications

Chapter 11's twenty-five, by ID and under its own names. That chapter carries each one's purpose, its primary user roles, the data it masters and the product category it maps to in chapter 12.

ID	Application capability	Domain
A-01	Article and merchandise master data	Master data
A-02	Product information and content	Master data
A-03	Supplier, partner and location master	Master data
A-04	Assortment, range and space planning	Planning
A-05	Merchandise financial planning and open-to-buy	Planning
A-06	Buying, purchase orders and supplier collaboration	Trading
A-07	Demand forecasting	Planning
A-08	Replenishment	Supply chain
A-09	Allocation and size curve	Supply chain
A-10	Warehouse management	Supply chain
A-11	Transport and delivery execution	Supply chain
A-12	Store inventory and counting	Store
A-13	Point of sale and the till journal	Store
A-14	Payment acceptance and settlement	Store
A-15	Price management and distribution	Commercial
A-16	Promotion and offer execution	Commercial
A-17	Customer, loyalty and consent	Customer
A-18	Digital commerce and order management	Digital
A-19	Returns and reverse logistics	Digital and store
A-20	Workforce, task and store execution	Store
A-21	Loss prevention and exception management	Control
A-22	Pharmacy dispensing and the clinical record	Regulated
A-23	Finance, stock ledger and supplier income	Finance
A-24	Data integration, history and analytics	Data
A-25	Document, evidence and the common layer	Common

Half two — the maps

Seven maps, as lists of pairs rather than wide matrices: a twenty-by-twenty-five grid does not fit a page, and a list of pairs survives being pasted into a repository.

Capability to process

Read it both ways: a capability exercised by no process is an aspiration, and a process exercising seven is probably two.

Capability	Processes that exercise it
C-01	P-01.1
C-02	P-02.1, and **no process maintains it afterwards**
C-03	P-02.1, P-04.1
C-04	P-03, which chapter 8 leaves **unmodelled at L2**
C-05	P-05.1
C-06	P-05.1, P-04.1, P-12.1
C-07	P-06.1
C-08	P-01.1 as a feasibility check only
C-09	P-07.1, P-07.2
C-10	P-07.1, P-08.1
C-11	P-04.1, P-10.2
C-12	P-09.1, P-12.1
C-13	P-09.1
C-14	**None.** Neither rostering nor credential checking is an L1 process
C-15	P-09.1, P-10.1
C-16	P-10.1, P-10.2
C-17	P-11.1
C-18	P-09.2 only — **modelled at L2 and absent from the L1 inventory**
C-19	P-12.1, and P-11.1 as refund fraud
C-20	P-04.1, P-08.1, P-12.1

Three findings fall out, and chapter 8 predicts all three in its own warning box.

C-14 is exercised by no process at all. Rostering is invisible, and checking that a registered pharmacist is present before a dispensing act appears nowhere, though the act is unlawful without one [7]. Chapter 8 names "trade the day at a store" as a process almost always missing, and this is the hole it leaves.

C-18 has no L1 process either. Dispensing is modelled at L2 as P-09's licensed sibling and is not in the twelve — exactly chapter 8's complaint that pharmacy gets filed as a department rather than a separately licensed professional practice.

C-02 is created once and never maintained. The article is born in P-02.1 and no process touches its data again. That row explains every item master an architect will open: hundreds of users with edit rights, cost in three places, dimensions wrong often enough that planograms do not fit [9].

Process to role, as RACI

One accountable role per process: that is the whole discipline. Where I could not honestly name one the row says so.

Process	Accountable	Responsible	Consulted	Informed
P-01	R-01	R-01, R-03, R-06	R-02, R-08	R-13, R-28
P-02	**Nobody.** R-01 owns the decision, master data the record, finance the payment, compliance the vetting	R-01, R-31	R-06, R-28	R-13, R-18
P-03	R-02	R-02	R-01, R-03	R-12, R-32
P-04	R-07 to the receipt; R-09 at the dock	R-07, R-09, R-18	R-01, R-11	R-31, R-32
P-05	R-05 for the engine; R-08 for the forecast. **Two accountables**	R-05, R-08	R-01, R-13	R-09, R-15
P-06	R-04	R-04, R-02	R-03, R-13	R-09, R-12
P-07	R-28	R-28, R-29	R-01, R-30	R-12, R-13, R-16
P-08	R-30 for apparel; R-14 for fresh; R-01 for grocery delists. **Three, by category**	R-30, R-13	R-02, R-32	R-12, R-28
P-09	R-13 at the till; R-21 for the dispensing sibling, personally and not delegably [7]	R-16, R-17, R-22	R-12, R-33	R-28, R-32
P-10	R-26 end to end; R-13 for the pick. **Contested at the store door**	R-20, R-25, R-26	R-13, R-16	R-10, R-27
P-11	R-13 in store; R-26 online. **Two, with two policies**	R-16, R-17	R-19, R-33	R-01, R-32
P-12	R-19	R-19, R-13	R-32, R-33	R-01, R-12

P-02 has no accountable role. Four functions own a piece each and nobody owns the end-to-end date. That is why the commonest failure in retail happens at all: the line is listed, the leaflet is printed, and the article does not scan on day one. Fixing it is an accountability decision, not a systems one.

P-05 has two accountables, because forecast accuracy is R-08's number and availability R-05's. **P-09 has two by law**, one of which cannot be delegated. And **P-11 has two with two policies**, which is why a garment bought online and returned in store is the hardest four minutes in retail.

Process to data, as CRUD

Which entities each process touches, and how.

Process	Entities touched
P-01	D-03 **CU**, D-01 RU, D-02 RU, D-16 R, D-14 R
P-02	D-07 **CU**, D-01 **C**, D-20 **C**, D-02 RU, D-04 C, D-06 R
P-03	D-14 R, D-01 R, D-02 R, D-12 R. **Creates nothing**
P-04	D-08 **CU**, D-09 **C**, D-07 R, D-20 CU as a claim, D-01 R
P-05	D-14 **CU**, D-08 C, D-09 R, D-03 R, D-16 R, D-01 R
P-06	D-17 **CU**, D-08 C, D-09 C, D-01 R, D-06 R
P-07	D-04 **CU**, D-05 **CU**, D-01 R, D-06 R, D-16 R
P-08	D-04 **U**, D-09 C as revaluation, D-01 R, D-06 R
P-09	D-12 **C**, D-09 **C**, D-15 **C**, D-13 RU, D-10 RU, D-11 RU
P-10	D-12 **C**, D-09 CU, D-10 RU, D-11 R, D-03 R, D-06 R
P-11	D-19 **C**, D-12 R, D-09 CU, D-20 C as a claim, D-10 R
P-12	D-18 **C**, D-09 **C**, D-01 R, D-06 R, D-12 R

Bold marks the natural *creator* of an entity, and that column repays reading alone. Three rows are worth arguing about.

P-03 creates nothing. The money plan and the open-to-buy are not among chapter 9's fifteen entities, because they are constraints rather than facts — and the consequence is the oldest complaint in merchandising, that the unit plan and the money plan never reconcile. Chapter 11 gives the function an application, chapter 9 no entity, chapter 8 no L2 model. Three sensible decisions, one hole.

Nothing creates D-16, Planogram. Three processes read it and none writes it, which is the data-side statement of C-08's finding: the shelf plan is an input everywhere and an output nowhere.

D-09, Inventory movement, is created by eight of twelve processes and updated by none. That is the correct shape, and the most useful thing to tell a retailer whose system of record is a `qty_on_hand` column: every one of chapter 9's nine stock positions is a filter and a sum over this entity.

Process to application

Process	Applications
P-01	A-01, A-04, A-05, A-06
P-02	A-01, A-02, A-03, A-06, A-23
P-03	A-05, A-24
P-04	A-01, A-06, A-10, A-11, A-12, A-23
P-05	A-01, A-07, A-08, A-12, A-24
P-06	A-01, A-05, A-09, A-10, A-12
P-07	A-01, A-13, A-15, A-16, A-20
P-08	A-01, A-13, A-15, A-16, A-23
P-09	A-12, A-13, A-14, A-17, A-22
P-10	A-11, A-12, A-13, A-17, A-18
P-11	A-12, A-13, A-18, A-19, A-21
P-12	A-01, A-10, A-12, A-21, A-23

Count the applications per process and the procurement lesson appears. **Order-to-receipt touches six, and the price change five with a statutory consequence if any is late** [4]. **A-01 appears in eight of twelve rows and A-12 in seven**, which is why replacing the merchandising system is a programme rather than a project [9].

And **P-09 should stop a reader.** A-22 is in chapter 11's list, which is right — but it is a separate regulated estate with its own vendor, its own content-currency obligation and patient data that must not reach A-24 [7]. One box among twenty-five is correct at this grain and dangerous at the next.

Application to data, with the system of record

The most useful table here. **Bold means system of record** — it creates the entity and owns its lifecycle.

Application	System of record for	Also reads
A-01	**D-01, D-02**, and D-04 as authorised retail	D-03, D-07, D-09
A-02	nothing; masters copy, images, allergens and nutrients	D-01, D-02
A-03	**D-06, D-07**	D-01, D-13
A-04	**D-03, D-16**	D-01, D-02, D-06
A-05	nothing; masters the plan and the open-to-buy, **for which chapter 9 has no entity**	D-01, D-02, D-12
A-06	**D-08**	D-01, D-06, D-07
A-07	**D-14**	D-05, D-09, D-12
A-08	nothing; masters the replenishment rule version and the proposal	D-01, D-03, D-09
A-09	**D-17**	D-01, D-06, D-09
A-10	**D-09** inside the building	D-06, D-08
A-11	nothing; masters load, trip, delivery and the temperature log	D-06, D-08
A-12	**D-09** at store level. **Contested with A-01 and A-10**	D-01, D-04
A-13	**D-12**	D-01, D-04, D-05, D-10
A-14	nothing; masters terminal, authorisation, settlement and token	D-12
A-15	**D-04**. Contested with A-01	D-01, D-06
A-16	**D-05**	D-01, D-02, D-04
A-17	**D-10, D-11**, with the consent and objection log	D-01, D-12
A-18	nothing; masters the customer order and its sourcing decision	D-01, D-03, D-04, D-09
A-19	**D-19**	D-09, D-12, D-20
A-20	**D-13**	D-06, D-10
A-21	**D-18**	D-09, D-12, D-13
A-22	**D-15**	D-01, D-04, D-10
A-23	**D-20**	D-04, D-08, D-09
A-24	history of everything; authoritative for nothing current	all
A-25	nothing; masters the document of record, the retention class, identity and entitlement	D-10, D-13, D-19

Four rows deserve an argument.

D-09, Inventory movement, has three candidate masters — A-01 for the group position, A-10 for the distribution centre's, A-12 for the store's — and the honest answer is "all three, differently". That is the commonest reason an online order is accepted against stock a walk-in customer has just bought.

D-04, Price, is mastered by A-15 and by A-01, and cached by A-12, A-13 and A-18. Where two prices are displayed concurrently the supplier may not charge more than the lower [4], so a label that fails to update is a legal exposure. The fix is a reference-data path engineered for *confirmed* delivery, per till and per label — which needs A-20, and A-20 is where most of this goes wrong.

A-04 masters D-16 and cannot join to it. The article-to-bay relationship is re-derived inside the space-planning tool rather than mastered [9], so planogram compliance rests on a key that does not exist.

A-05 masters something the conceptual data model does not contain. That is not an error in either chapter: it is the signature of a function that is real, consequential and unmodelled, and the first thing to check when an application map and a data model disagree about what exists.

Application to role

Chapter 11 names its own user roles. Where one is not among the thirty-three, the cell names the job in words.

Application	Roles
A-01	R-01; and an article data steward, which chapter 5 carries only as a technology role
A-02	a content manager and a compliance analyst. **No business role on this list**
A-03	R-13; and a supplier master steward
A-04	R-03, R-06, R-01
A-05	R-02; R-01 as a consumer
A-06	R-07, R-01
A-07	R-08
A-08	R-05, R-13
A-09	R-04
A-10	R-09, R-18
A-11	R-10, R-11
A-12	R-13, R-18, R-19
A-13	R-16, R-17, R-13
A-14	R-17; and a payments and acquiring manager
A-15	R-28, R-01, R-13
A-16	R-29, R-01
A-17	R-27; and a CRM analyst and an information officer
A-18	R-25, R-26, R-20
A-19	R-17, R-13
A-20	R-13, R-12
A-21	R-33
A-22	R-21, R-22, R-23, R-24
A-23	R-32, R-31
A-24	R-08, R-33; and a data engineer and a data steward
A-25	a records manager and an identity administrator. **No business role on this list**

Read the role column instead and look for roles appearing once. **R-11, the cold-chain manager, appears only in A-11's temperature log**, which exists and is not joined to stock, so a write-off cannot be evidenced to an excursion [7]. **R-14 appears nowhere**, though the fresh department orders, produces, prices to clear and records waste — a modelling failure to correct in the first workshop. And **A-02 and A-25 have no business role at all**: both are everybody's input and nobody's job.

Capability to application — the coverage map

The map that goes in the board pack, assessed against the Milkwood estate chapter 11 describes: **two instances of A-01**, food and apparel, which is defensible; **three of A-13**, which is not, so the group holds three versions of what a basket is; **no A-05** and **no A-20**; one A-17 whose consent scope was built for one banner and now covers three; and **A-23's supplier-income half on a shared drive**. **Full** means one application serves the capability and masters its data, **partial** served but not mastered, **gap** nothing serves it.

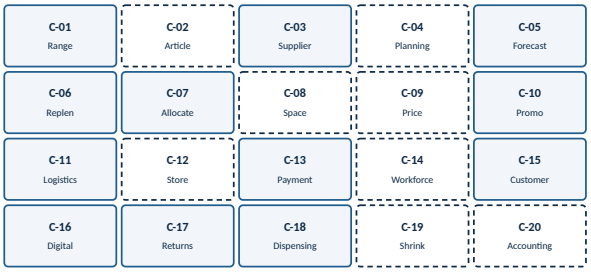


Figure 19.2 The coverage map as one picture, against the invented Milkwood Retail Group estate chapter 11 describes. Dashed cells are gaps, tinted cells are partial, and nothing is solid — because nothing is rated full. That is the finding, not the eight gaps.

Capability	Applications	Milkwood coverage
C-01	A-04, A-01	Partial. Cash-and-carry members range themselves [5]
C-02	A-01, A-02	**Gap.** Two A-01 instances, no business owner for either
C-03	A-03, A-23	Partial. One supplier master; terms on A-23's shared drive
C-04	A-05	**Gap.** No A-05, and chapter 8 does not model the process either
C-05	A-07	Partial. Sound in the supermarket; apparel has no plan to hang it on
C-06	A-08, A-12, A-01, A-10	Partial. D-09 mastered three times, so availability disagrees by channel
C-07	A-09	Partial. National size curves; prepacks found at the pick face
C-08	A-04	**Gap.** Planograms produced and never confirmed, because confirmation needs A-20
C-09	A-15, A-01, A-13, A-18	**Gap.** Three A-13 instances, three price caches, no reconciliation [4]
C-10	A-16	Partial. Markdown taken as promotion, so no revaluation
C-11	A-10, A-11	Partial. Temperature logged and never joined to stock
C-12	A-12, A-13, A-20	**Gap.** Three versions of a basket and no task execution
C-13	A-14	Partial. Terminal register and the acquirer's differ continuously
C-14	A-20	**Gap.** No A-20, so nobody can say which stores relabelled [7]
C-15	A-17	Partial. One A-17, with a consent scope built for one banner and three now marketed to
C-16	A-18	Partial. Available-to-promise computed in the channel, not published
C-17	A-19	Partial. Legal basis captured; disposition decided at the desk
C-18	A-22	Partial. Compliant in the dispensary; the group cannot see it, and must not see all of it [7]
C-19	A-21	**Gap.** Till exceptions only; shrink measured at stock take, causes months cold
C-20	A-23	**Gap** on the supplier-income half. The ledger half closes

Eight gaps, twelve partials, nothing rated full — and **the nothing-rated-full is the finding**. Duplication, not absence, stops a capability having one owner: A-01 twice, A-13 three times, D-09 mastered three ways, D-04 held in five places. That reads as a healthy estate on an application list.

The useful output is the ordering, and chapter 11's own is four sentences: A-13 is a consolidation programme, A-05 a buy, A-20 a build, and A-23's income half first because it is the only one that moves the reported number. Read against the capability map that holds, in three bands.

Fund first, where the downside is quantified and nobody will resist. C-20, for the reason chapter 11 gives. C-09, because the exposure is statutory, immediate and per shelf label [4]. C-14, because A-20 is what would let anybody prove a label changed, so it is the precondition of fixing C-09 and C-08 at once.

Second, the organisational fights. C-04, because buying A-05 means somebody must own an open-to-buy no process approves. C-02, because two article masters is a decision somebody must unmake. C-12, because consolidating three A-13 instances means agreeing what a basket is across three formats.

Last, the one nobody will sponsor. C-19 — the cheapest return in retail, its beneficiary the gross margin rather than a department, and no South African shrink figure published to benchmark against.

Keeping these maps alive

One owner per dimension, named. Processes to an operations director, capabilities to strategy, data to the head of data, applications to the lead architect, roles to human resources — who will produce a better list than yours.

Never change the hub. Renumbering a process invalidates every map here. Add `P-13`; do not renumber, and do not reuse a retired ID.

Review on a trigger, not a date, and never in the freeze. Quarterly will be skipped by the second quarter. Before an architecture board, before a package evaluation, after a reorganisation, after an instrument lands. A South African change freeze runs mid-November to mid-January.

Publish the gaps with the maps. A cross-map circulated without its gaps reads as a healthy estate, and the next reader will believe it. The gaps are the only part of this appendix anybody will act on.

THE CHAPTER IN TEN LINES

- Hinge every cross-map on business process. It is the only dimension the business will contest, and therefore the only one that ever gets corrected.
- Five dimensions, five ID series — processes P, capabilities C, roles R, data entities D, applications A. Keep them local, version them, and never renumber.
- Retail has no BIAN to borrow identifiers from, so the series is yours and nothing external will ever tell you it has drifted. Write down the day you built it.
- Two capabilities have no L1 process at all: trading the day at a store, and dispensing. Chapter 8 predicts both, and both are places where the accountability is real and unwritten.
- The money plan has an application in chapter 11, no entity in chapter 9 and no L2 model in chapter 8. Three sensible decisions, one hole, and it is why the unit plan and the money plan never reconcile.
- Inventory movement has three candidate masters and price is held in five places with nothing reconciling them. In South Africa the second is a statutory exposure, because the lower displayed price binds.
- In a franchised or voluntary-trading estate a blank cell is a contractual fact, not a gap. Say which it is before you publish the map.
- This is a worked starting set for an invented retailer. Rebuild it in workshops with the named role in the room, and expect a third of the role assignments to change in the first twenty minutes.

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APPENDIX B

Glossary

Most retail vocabulary is ordinary English used with a precise and different meaning. These are the words, in plain language, with the South African ones flagged, the North American divergences marked, and the dangerous ambiguities spelled out.

Most retail vocabulary is ordinary English used with a precise and different meaning, and the precision is where an architect gets caught. A *store* is four things with four boundaries. *Stock* has nine answers and the right one depends on who is asking. An *article* in a SAP estate and a *SKU* in a North American one are the same object with different names, except in apparel, where they are not. A *price* is at least seven numbers for the same item on the same day. And a *lay-by* in South Africa is neither a sale nor a credit agreement, which makes recognising its instalments as revenue both an accounting error and a statutory one.

This appendix is the list, written for somebody who has never worked in retail. One or two sentences each. Where a word means different things in different parts of a retailer, both meanings are given and the ambiguity is marked, because that ambiguity is the defect source. Where a term is **South African and has no close overseas equivalent** — trading density, trading profit, lay-by, spaza shop, voluntary trading, retailer loyalty, single exit price, NAPPI — it is flagged. Where an **overseas term means something different here**, that is flagged too: "NCR" in a South African retail sentence usually means the National Credit Regulator, and "CPI" usually means a competitor price index.

The North American divergences the research identified are marked where they land, and the strongest of them by a long way is pharmacy. South Africa has one national regulated price per medicine and a four-band national dispensing fee; Canada has no national exit price, per-province public-plan fee caps and mark-up formulas, "usual and customary" fees charged to private plans, billable professional services, and confidential generic rebates. **A dispensing system built for a single exit price will not price a Canadian script, and one built for Canadian adjudication will over-engineer a South African dispensary.** The others worth knowing before a conversation: the SKU-versus-article vocabulary split, the twelve-against-thirteen-digit identifier split, shrink benchmarking, the retail versus cost method of inventory accounting, and the statutory-versus-policy basis of a return.

Three conventions. Terms in **bold** inside a definition are defined elsewhere in the list. Where a figure appears it is as published at October 2026. And where the research behind this book could not source something, the entry says so rather than guessing — **shrink** is the clearest case, because no South African figure is published at all.

The terms

Term	What it means
1P / 3P	First-party inventory, which the platform owns, and third-party inventory, which a seller owns. A marketplace needs a seller entity that nothing else in the model touches.
4-5-4 calendar	The National Retail Federation's retail calendar: months of four, five and four weeks, so comparable periods hold the same number of weekends. **NRF calls it a voluntary guide**, not a standard, and South African retailers have three different year-ends.
53rd week	The extra trading week added to a 52-week retail year roughly every five or six years. **It breaks every growth comparison**, and the calendar dimension must carry a prior-year-comparable day key on both a restated and a non-restated basis.
Account sales	Sales made on the retailer's own credit facility. **In South Africa they bring the National Credit Act with them**, including a mandated affordability assessment, and the same person is then simultaneously a loyalty member and a credit consumer with different lawful bases and retention rules.
Adjacency	The deliberate placing of related categories next to each other in a store, so that a shopper buying one is prompted to buy the other.
Adjustment	A posted correction to an inventory position. It should always carry a reason code; where it does not, shrink cannot be attributed to a cause.
Advance shipping notice	An electronic notification that a delivery is on its way, with what is on it. Without one, a goods receipt is a line count by hand.
Age-restricted item	An item that may be sold only to a buyer above a stated age, and often only at stated hours — liquor and tobacco in South Africa. The till must prompt, and must also be able to refuse outside permitted hours.
Allergen declaration	The mandatory statement of allergenic ingredients. **South Africa's R146 names eight categories** — egg, cow's milk, crustaceans and molluscs, fish, peanuts, soybeans, tree nuts and significant gluten-containing cereals. **Sulphites are not on that list**, and precautionary "may contain" labelling is not a substitute for allergen control.
Allocation	**Two meanings.** In merchandising, the decision of which units of a finite buy go to which store, in what size curve, on what date — a push, used in apparel and seasonal goods. In inventory, a quantity that is physically present but committed to a customer order and therefore not available to sell. Say which one you mean.
Allocation engine	The application that turns a fixed buy into store-level quantities by size. Distinct from a replenishment engine, and a group running supermarkets and clothing needs both.
Allowances	Discounts plus markdowns — the gap between the markup set when an item was ticketed and the margin actually achieved.
Analogue	A historic item used as a proxy when forecasting a new line that has no sales history. Also called a like item.
Anchor tenant	The large tenant whose presence draws shoppers to a shopping centre. A supermarket is usually the anchor, which is why its lease terms differ from everyone else's.
Application identifier	The prefix inside a GS1 data carrier that says what the following data element is — batch, expiry, serial, weight. **A till or warehouse system that parses only the GTIN and discards the rest throws away the traceability data the business is about to be asked for.**
Arrival timestamp	When a transaction batch reached the centre, as distinct from the business date it belongs to. Needed because **yesterday's sales number is not final today**, and may not be final this week.
Article	**Two meanings.** In a SAP or European estate, the merchandise-managed item: what the buyer buys, the system prices and the shelf holds — and in grocery it is the same thing a North American would call a SKU. In apparel the word is used loosely for any level of the product structure, which is why "how many articles do you carry" gets three answers in one room. SAP additionally distinguishes a *single article*, a *generic article with variants* and a *structured article*.
ARTS	The Association for Retail Technology Standards, founded 1991, acquired by the National Retail Federation in 1998, and managed jointly with the Object Management Group as the Retail Domain Task Force since 2017. **A historical name**: do not write "ARTS publishes" in the present tense. Its data models are absent from OMG's formal catalogue and may not be obtainable at all.
ARTS ODM	The ARTS Operational Data Model, covering store-level transactional data. Described on OMG's pages with no version and no date. **Useful as archaeology for a schema you inherited, and not available as a target architecture.**
Assortment	The range actually carried by a specific store or cluster, as distinct from the range the retailer offers overall. Three dated statements hide inside it: assorted (the commercial decision), planogrammed (it has a shelf facing) and orderable (it can be replenished).
Attach rate	The proportion of transactions that include an added product or service.
Attribute	A descriptive property of an item used for planning, search or reporting. In SAP, note that a characteristic assigned to a hierarchy level may be informational only, with no operational effect — logic built on one will fail.
Availability	The proportion of lines present and saleable on the shelf when a shopper wants them. Also called on-shelf availability, and it is not the same as system stock.
Available to promise	The stock position the retailer is willing to sell online: physical, less reservations, less unsellable, sometimes plus inbound inside a promise horizon. Computed in the channel it will be wrong; published as a service it can be right.
Average transaction value	Net sales divided by transactions. The basket.
Average unit cost	Cost of sales divided by units sold.
Average unit retail	Net sales divided by units sold. Together with units per transaction it decomposes the basket.
B-BBEE	Broad-Based Black Economic Empowerment. For a retailer it is a supplier-master obligation: a dated, evidenced classification per supplier, feeding spend reporting by classification. Most retailers hold it in a procurement system separate from the merchandising supplier master, producing two supplier masters that disagree about which entities exist.
Back stock	Stock held in the store but not on the shelf. Its existence is why a gap on the shelf is not the same thing as an out-of-stock.
Backhaul	Using an outbound delivery vehicle's return leg to collect from a supplier. The cheapest freight in retail and the hardest to schedule.
Banner	The trading brand over the door. One group routinely runs several banners over one supply chain, and the banner is not the same thing as the legal entity or the format.

Baseline	The sales that would have happened without the promotion. Everything about promotional evaluation depends on it and nobody can measure it directly.
Basket	**Two meanings.** The physical or notional set of items in one transaction; and the measure, average transaction value. Also, in competitive pricing, a defined set of lines whose total is compared with a rival's.
Basket abandonment	An online basket not converted to an order.
Batch clock	The overnight cycle, in dependency order: store end of day, sales polling, sales and stock posting, the forecast run, the replenishment run, order release and transmission, distribution-centre wave planning, the price-file build, price-file distribution and application, master-data distribution, finance posting, and the data-platform load. **Establish the total duration, the slack to store opening, and what happens if the forecast run does not finish — in most retailers that last answer is a convention in an operator's head rather than a designed behaviour.**
Bay	A standard unit of shelving width, and the unit a planogram is drawn against. The join between an article and a bay usually exists only inside the space-planning tool.
Bi-temporal	Holding both the date from which a fact was true in the world and the date the system learned it. Retail needs the second to answer what the replenishment engine knew when it ordered, and which forecast was in force when the buy was committed.
Bias	Persistent directional forecast error. Worse than random error, because it compounds into stock.
Blue Yonder	A supply-chain and retail planning vendor, **owned by Panasonic** since September 2021. "JDA" is a former name, not a sister product, and the lineage runs JDA plus i2 plus Manugistics plus RedPrairie.
BNPL	Buy now pay later — short-term interest-free instalment credit offered at checkout.
BOGOF	Buy one get one free. A basket-level rule, not a price.
Book stock	What the system believes is physically present. The counterpart of counted stock, and the difference between them is shrink.
BOP / EOP	Beginning and end of period inventory. The two anchors of the open-to-buy arithmetic.
BOPIS	Buy online, pick up in store. Also called click and collect.
Bracketing	Ordering several sizes intending to return most of them. The behaviour that makes online apparel return rates what they are.
Breakage	Loyalty value earned and never redeemed. It reduces the liability and flatters the programme's economics.
Business date	**The retailer's declared trading day for a site**, which is not the same as the transaction timestamp. A till open at 23:50 and closed at 00:30 trades across midnight and every transaction in that session belongs to one business date, which the till declares. Derive it from the timestamp and it is wrong for every late-trading store.
Buying	Choosing the suppliers, the range and the cost price. **Not the same job as planning**, and a system that merges them has mis-modelled the organisation.
Buying group	Independent retailers pooling their purchasing power to obtain better terms. **South African relevance:** the grocery inquiry found buying groups genuinely benefit independents, and that store-level rebates available to national chains are not available to the wholesalers serving them.
Cannibalisation	Sales of one line lost because another was promoted. Uplift without cannibalisation is not incremental.
CAR / DTA	SAP's Customer Activity Repository and its point-of-sale Data Transfer and Audit component — a **point-of-sale data platform, not a till application**. A retailer running these has not told you what software its tills run.
Case	The shipping-level pack a retailer buys in, with its own barcode and its own quantity. Also called an outer or a traded unit.
Case code	The trade word for the barcode on a case, as opposed to the one on the consumer unit. Scanning one and expecting the other is a daily error at a goods-in dock.
Case pick	Picking whole cases rather than individual units. Cheaper per unit and useless for a small store.
Cash and carry	A wholesale format where the customer collects the goods and pays immediately. **A significant South African channel**, serving spaza shops, caterers and small traders, and its turnover is invoiced sales rather than till sales.
Cash conversion cycle	Inventory days plus receivable days less payable days. In grocery it is negative, which means suppliers fund the working capital — and that is the business model, not a quirk.
Cash sales	Sales settled immediately by any non-credit tender. The term survives from a time when that meant notes.
Cashback	Loyalty value returned as money. **The most-preferred South African mechanic**, and it creates a real liability on the balance sheet rather than a points balance.
Catchment	The geographic area a store draws its trade from. It is the basis of a new-store forecast and of the like-for-like argument when a second store opens nearby.
Category	A shopper-defined grouping of substitutable products, and the unit of commercial management. It is not the same as a merchandise department, because shoppers do not shop the way buyers buy.
Certificate of Acceptability	The local-authority certificate every South African food premises needs under the food hygiene regulations. One per premises, which makes it an attribute of the site record.
Change freeze	The period during which no change is made to the production estate. In South African retail it typically runs from mid-November to mid-January and covers software, infrastructure, master-data structure, range, planograms and refits. **It makes the delivery year nine months long**, and it makes the January and February releases the largest and riskiest of the year.
Characteristic (SAP)	An attribute used to create or describe variants of a generic article. Variant-creating characteristics have operational effect; characteristics inherited from a hierarchy level are informational only.
Claim	The money a retailer is owed because a delivery was short, damaged or wrong. Claims raised and never chased are a measurable, recurring loss.
Clearance	The final markdown stage, intended to exit the stock rather than to optimise its margin.
Clearance price	The last price in a markdown sequence, set to exit the stock.

Click and collect	Buying online and collecting in store. **In South Africa the collection share is high** — one listed apparel group reported over 55% in-store collection — which makes the uncollected-order problem a real one.
Cluster	A group of stores planned together because they are alike on size, demographics, climate or trade pattern. Clusters are usually defined once and then never revisited, which is a standing finding.
Co-op advertising	A supplier contribution to the retailer's advertising of the supplier's product. One of the forms of supplier income.
Cold chain	The temperature-controlled path from supplier to shelf. In a pharmacy its records are a regulated obligation, not a quality nicety.
Colourway	A specific colour execution of an apparel style. It drives photography, assortment breadth and markdown behaviour, which is why colour and size are not interchangeable variant axes.
Comparable-set membership	A stored fact per site per period pair saying whether the store is in the like-for-like base, and why. The alternative is a spreadsheet nobody can audit.
Competitor price index	**In retail this is what "CPI" means** — the retailer's prices measured against a competitor's basket. Not the consumer price index, and in a sentence about inflation the ambiguity is real.
Consent	**Dated, purpose-bounded permission to process somebody's data**, and not a boolean. Under POPIA, using loyalty data outside the purpose it was collected for is further processing, and the Information Regulator has said that sharing a member's data with group companies for marketing needs written consent. A retailer also has to store *how* a contact detail was obtained and *which* product category the relationship arose from, because the existing-customer exemption depends on both.
Consignment stock	Stock sitting on the retailer's shelf that the supplier still owns. Common in concessions, magazines and some pharmacy lines, and it must not be counted as the retailer's inventory.
Convenience format	A small store with a narrow range, long hours and a higher price zone. Often attached to a forecourt.
Conversion rate	Transactions divided by footfall. Measurable in store only where footfall is counted, which is fewer stores than you expect.
Cooling-off	A right to cancel without reason. **Three different clocks run concurrently in South Africa** and a consumer may rely on whichever is most favourable: five business days from the later of conclusion or delivery for a direct-marketing sale, seven days from receipt for an electronic transaction, and six months from delivery for a quality failure.
Core range	The lines every store must carry, whatever its local assortment. The counterpart of the tail.
Corporate store	A store owned and operated by the group. The only kind of store where the group straightforwardly owns the price, the assortment and the stock.
Cost method	Tracking inventory at cost by specific identification, average, first-in-first-out or last-in-first-out. South African grocers use it; the retail method's institutional home is the North American department store. **Expect an imported package to assume the other one.**
Cost price	The price invoiced by the supplier before any supplier income is applied. One of at least five numbers that all get called "the cost".
Coupon	A redeemable instrument conferring a discount. Its redemption is an event with its own reconciliation.
Cover	See *Weeks of supply*.
CPA	The Consumer Protection Act 68 of 2008, in force 1 April 2011. **The spine of a South African retailer's customer-facing controls**, and operational in a way most financial regulation is not: it reaches the shelf-edge label, the till slip and the returns desk.
CPA section 23(6)(b)	The two-prices rule: **where more than one price is displayed concurrently, the supplier may not charge more than the lower**. A shelf label that silently fails to update is therefore a legal exposure, not a data-quality issue. The escape in section 23(9) is a two-limb defence — correct the display *and* take reasonable steps to tell affected consumers — and both limbs are data.
CPA section 56	The six-month return right: goods failing the quality standard may be returned within six months of delivery for repair, replacement or refund, **with the consumer electing** which, at the supplier's risk and expense, **whatever the retailer's policy says**. **North American divergence:** United States and Canadian practice is dominated by retailer policy plus implied warranties, so that estate needs a configurable policy engine where the South African one needs a per-sale-line statutory clock.
Cross-dock	Receiving and despatching goods without putting them away. Fast, cheap, and intolerant of a bad advance shipping notice.
Customer	**Three or four things, and collapsing them is the most expensive modelling decision in retail.** A *party* is a person or organisation the retailer knows about. A *customer* is a party with at least one purchase — a derived, dated role. A *loyalty member* is a party enrolled in a programme, with a membership number, a tier, a balance, a consent record and a terms version: a contractual relationship. An *account holder* has a credit facility, which is a regulated relationship. And in a pharmacy a *patient* has a dispensing history, which is a regulated health-information relationship. **A transaction cannot require a customer key**, because the unidentified basket is still a transaction and still has to reconcile.
Customer data platform	The system that resolves a customer identity across channels and builds segments and audiences from it. Distinct from loyalty, which holds the contractual relationship.
Customer fulfilment centre	A dedicated warehouse for online grocery orders, as opposed to picking from a store.
Cycle count	A partial count of selected lines on a rolling schedule, as opposed to a full stock take.
Dark store	A facility built or converted to store format and used only for online fulfilment, with no walk-in trade.
Days inventory outstanding	The average number of days stock is held. One of the three legs of the cash conversion cycle.
Days payable outstanding	The average number of days taken to pay suppliers. Also called creditor days, and in retail it is a strategy rather than an accident.
Deduction	A claim netted off a supplier payment rather than invoiced. Disputed more often than not, because the evidence sits somewhere else.
Delisted line	A line removed from the range. For the supplier it is a contractual event; for the retailer it is the start of a run-down, and a delist with no run-down plan strands stock that can then only be marked down.

Demand sensing	Adjusting a short-term forecast from very recent actuals.
Department	**Two meanings.** In the merchandise hierarchy, the top reporting level, where the commercially material attributes live. In a store, a physical area with its own manager, stock responsibility and often its own licence — the dispensary, the butchery, the liquor section.
Digital Link	A GS1 standard for encoding an identifier as a web URI, so one scan both identifies the item and resolves to content. **The content behind the link belongs to the brand owner, not the retailer** — which is a governance question no retailer considers until a competitor's offer appears behind a scan in its own aisle.
Digital Receipt API	A formal OMG specification published June 2025, developed by a Japanese industry council under a government contract project. Evidence that the device-and-receipt half of retail standards is alive.
Direct import	Buying from an overseas supplier without a local intermediary. It lengthens lead time, moves the risk and makes landed cost the number that matters.
Discontinued line	A line the retailer has stopped buying but may still be selling through. Different from delisted, which is the contractual act.
Discount	A temporary reduction applied at the point of sale, leaving the base price unchanged underneath it.
Discount format	A limited-assortment grocery store trading on price, with a narrow range, high private-label share and low operating cost.
Dispensing fee	**The regulated fee a South African pharmacy may charge per prescription**, as a flat amount plus a percentage of the single exit price, across four bands with the percentage falling as the price rises. Capped annually by gazette, and different for a licensed dispensing doctor. **Implement it as a versioned band table with effective dates, never as constants. North American divergence:** Canada has no national fee — each province caps its own for its public plan, and private plans are charged a "usual and customary" fee that varies by chain.
Dispensing item	A medicine identified by a national medicine code, whose price in South Africa is set by regulation rather than by the retailer.
Dispensing record	**The regulated act of supply**, and structurally unlike a sale line: the dispensed product may differ from the prescribed one through generic substitution; the batch is mandatory because a recall must reach a patient; the act is attributable to a named registered person; retention runs on a clinical clock rather than a commercial one; the price is set by regulation; it is special personal information; and **it cannot be deleted on request**, because a statutory retention obligation beats an erasure request — but only if the obligation is recorded against the record.
Distribution centre	A warehouse that supplies stores. A regional distribution centre supplies other distribution centres.
DIY format	Building materials and home improvement. A distinct format with trade and retail customers, bulk goods, and a delivery model closer to builders' merchanting than to grocery.
Drop	A single delivery. Drop density — the number of drops per route area — is what makes last-mile economics work or not.
Dropship	The supplier ships direct to the customer or the store, bypassing the retailer's distribution centre.
Drug Identification Number	**North American only.** Health Canada's eight-digit identifier for every drug product approved for sale in Canada, required on the label. **The DIN, not the GTIN, is the key for formulary matching, provincial plan adjudication and recall** — so an architect inheriting a Canadian pharmacy estate must treat it as a first-class article attribute with its own lifecycle and reference feed. South Africa has no equivalent.
DSD	Direct store delivery — the supplier delivers to the store rather than to a distribution centre. It bypasses the retailer's receiving controls, which is why DSD lines over-represent in shrink.
Each pick	Picking individual consumer units rather than cases. The expensive mode, and the one online fulfilment runs on.
EAN / EAN-13	The 13-digit international article number encoded in the standard retail barcode outside North America. **"EAN" is a legacy name for GTIN-13.**
EANCOM	GS1's retail subset of UN/EDIFACT, and the electronic data interchange grammar a South African grocer or apparel retailer will actually meet.
Earn and burn	The accrual and redemption sides of a loyalty programme. They have different system paths and different offline tolerances: accrual can be deferred, redemption cannot be, safely.
ECTA	The Electronic Communications and Transactions Act 25 of 2002. Section 43 prescribes what an online seller must disclose; **section 44 gives a seven-day cancellation from receipt**, with only the direct cost of returning the goods chargeable and a refund due within thirty days.
EDLP	Every day low price — a stable low-price strategy with little promotion. The opposite of high-low, and a different systems problem: less promotional machinery, more price-file discipline.
Effective dating	Recording the date from which a change applies, independently of when it was captured. The minimum requirement on price, assortment and hierarchy membership.
Electronic shelf label	A digital shelf-edge display updated from a controller rather than printed. **It turns a weekly manual price audit into a systems-integration problem, which is better — and it makes the label-to-article binding a controlled object with a legal consequence.**
End cap	The display at the end of a gondola. Premium promotional space, and usually sold to a supplier.
End of day	The store process that closes and seals the trading day. **It gates everything downstream**: a store that does not close blocks every overnight job for that store, and a store closed before polling completes loses the day's sales to the centre until the next cycle.
Entry price point	The cheapest option in a category. Also called the opening price point, and it anchors the shopper's sense of the whole range.
EPCIS	A GS1 standard for sharing **visibility event data** — what, when, where, why. **It is not an RFID standard and has not required instance-level identification since version 1.1**, which is why architects reject it for the wrong reason. It records that an event happened; it does not hold a balance.
EPR	Extended producer responsibility for packaging, under regulations commencing 5 May 2021. **A retailer with an own-label range is a brand owner and therefore a producer**, and the fee is proportional to products placed on the market **by weight and material** — which means packaging components, material type, mass in grams, recycled content and recyclability must be on the article record. **This is the obligation that cannot be back-filled**: a retailer that did not capture component mass at article creation must re-collect it from every supplier for every own-label line.
ERP (in retail)	The finance and ledger estate. Expect **one** of it, and a shadow merchandise-planning system in spreadsheets beside it.
Exception-based reporting	Loss-prevention analytics over till and operator data, looking for the pattern — the store, the SKU, the till, the operator, the supplier, the process.
Exception report	A report listing only the anomalies that need investigating. Without one, internal audit tests by sampling.

Exchange	Replacing returned goods with others rather than refunding. A different event from a refund, with a different inventory consequence.
Exclusivity clause	A lease term barring the landlord from letting space in the same centre to a defined competitor. **South African specific:** the Competition Commission's grocery inquiry recommended these be phased out, and a lease register that cannot answer "where do we still have exclusivity" is a compliance problem.
Facings	The number of units of one SKU presented to the shopper across the front of the shelf. The facing count, not the shelf depth, is what a planogram specifies and what shelf capacity derives from.
Fascia	The signage over a store. Used loosely as a synonym for banner.
FEFO / FIFO	First-expired-first-out and first-in-first-out rotation rules. FEFO is what fresh and pharmacy need; FIFO is what most systems implement.
Fill rate	The proportion of ordered lines or units a supplier actually supplied.
Fixture	A physical shelving, rail or display unit. Fixture data decays after every store refit, which is why planograms stop fitting.
Floor limit	A value below which a card transaction may be accepted without an online authorisation, with the retailer taking the risk. One of the three mechanisms that let a store take card payments while offline.
Floor plan	The layout of fixtures, aisles and departments in a store.
Flow-through	Pre-allocated stock passing through a distribution centre to stores without being stored.
Footfall	The count of people entering a store. Also called traffic.
Forecast error	The difference between forecast and actual demand. Measured at the wrong grain it flatters; measured at the order grain it is actionable.
Forecourt	The retail site attached to a fuel station. A distinct format with a higher price zone, longer hours and a licence of its own.
Format	A defined store type with a size, range and price envelope — hypermarket, supermarket, convenience, forecourt, cash-and-carry. Format matters more than banner, because it determines the architecture below the till.
Franchise store	An independently owned store trading under the brand for a fee and a royalty. **Architecturally the franchise store is a customer, not a location**: stock is sold to it rather than transferred, price becomes a recommendation rather than a mandate, and the group's visibility of the shelf is whatever the franchise agreement compels.
Fulfilled by the platform	An arrangement in which a marketplace stores and ships a third-party seller's stock for a fee.
Funded promotion	A promotion to which the supplier contributes. **The distinction from an unfunded one is the single most important fact about any promotion**, and the one most often not held as data.
GDSN	The GS1 Global Data Synchronisation Network: suppliers publish trade-item master data once into a certified data pool and retailers subscribe. It replaces the supplier spreadsheet, and **it carries supplier-authored data only** — a retailer that treats it as its product information system has outsourced its item master to its suppliers' data quality.
GK Software	A store-systems vendor, **owned by Fujitsu**, whose software SAP has resold since 2009. So “SAP POS” in a grocery estate may be GK code under an SAP contract.
GLA / gross lettable area	The leased area of a store, which is larger than the trading area. Rent is calculated on it; trading density is not.
GLN	Global Location Number — the GS1 identifier for a party or a location. **Store, distribution centre, legal entity, ship-to, bill-to and remit-to are distinct GLNs**, and trading partners address documents to them rather than to the retailer's internal store number.
GMV	Gross merchandise value — the total value transacted on a platform, which is **not** the platform's revenue. Its revenue is the take rate on that value.
Golden source	The single authoritative version of an entity's data. Also called master data, and in retail the hard part is not building one but getting a hierarchy's owner to accept somebody else's.
Gondola	A double-sided free-standing shelving run.
Good Pharmacy Practice	The rules made under section 35A of the South African Pharmacy Act, and **the rules that actually specify a dispensing system**. Rule 1.2.11.4(f) requires the **latest version** of interaction-checking software, which turns a content database into a regulated currency obligation on the compliance critical path. Labels must be in at least one official language. Cold-chain temperatures must be recorded twice daily with annual mapping and calibration.
Goods received note	The record of a receipt against a purchase order, and the document the three-way match depends on.
Gross margin return on inventory	Gross margin rand divided by average inventory at cost. The measure that catches a line with good margin and no turn.
GS1	The global standards body for identification. **The only external standard that is a primary key in a retailer's data estate.** Its keys are GTIN, GLN and SSCC; its General Specifications govern the data carriers.
GS1-128	A barcode symbology used on logistics labels and anywhere batch, date or weight must travel with the identifier.
GS1 company prefix	The licensed number range from which a company allocates its own GTINs. **South Africa's range is 600–601**, and it tells you where the company is registered, not where the product was made.
GS1 DataBar	A compact omnidirectional symbology for coupons and fresh items with expiry data.
GS1 DataMatrix	A two-dimensional symbology used in pharmacy and healthcare.
GS1 General Specifications	GS1's foundational standard for how keys, attributes and barcodes are used. **Release 26.0 was ratified January 2026.** GS1's own governance instruction matters: the official version is the reference PDF, and where another format conflicts the PDF governs.
GTIN	Global Trade Item Number — the GS1 identifier for a trade item **at one packaging level**. A consumer unit, an inner and a case are three GTINs with a hierarchy between them, not three names for one product.

GTIN-12	A twelve-digit GTIN — the number historically called a UPC, and the dominant retail identifier in the United States and Canada.
GTIN-13	A thirteen-digit GTIN — the number historically called an EAN, and the dominant identifier in the rest of the world. Usually stored as fourteen digits with a leading zero.
GTIN-14	A fourteen-digit GTIN for a package containing several units, built from the inner GTIN plus an **indicator digit** of 1 to 8 and a new check digit. **Normalise every GTIN to fourteen digits, zero-padded, stored as a string**, and the commonest retail data defect disappears.
GTIN-8	An eight-digit GTIN allocated on application for very small retail packs where a full barcode will not fit. Not for use on groupings.
Guild	The member body in a voluntary-trading group. Members sign a membership agreement with the guild and agree to its standards; there is no royalty.
Halo	Incremental sales of complementary lines caused by a promotion on another.
Hierarchy (merchandise)	The fixed classification tree from division down to the item, used for buying authority, budget, margin policy and reporting. Oracle's canonical shape is Company, Division, Group, Department, Class, Subclass; SAP's is a merchandise category hierarchy with an unlimited number of levels, numbered from the bottom. A retailer has at least six hierarchies and they disagree, which is correct.
High-low	A strategy of higher base prices with frequent deep promotions. It needs a promotions engine, a leaflet calendar and a funding negotiation; every day low price does not.
Hypermarket	A very large store combining a full grocery range with general merchandise. The format where the merchandise hierarchy has to span food and non-food in one estate.
Identification rate	The share of sales linked to an identified customer. **In South African grocery the identified basket is a minority of transactions**, which is why the model cannot assume one.
IFRS 16 right-of-use asset	The capitalised value of a leased store on the balance sheet. It is why South African retailers report **trading profit** as well as operating profit — the lease accounting moved the numbers and management kept managing to the old shape.
In duplum	The South African rule capping post-default charges on a credit agreement at the unpaid principal at the time of default. Relevant wherever a retailer runs its own credit book.
In transit	Stock shipped and not yet received. Owned, not available, and at somebody's risk depending on the shipping terms — and counted at both ends by any model without an in-transit location.
Incrementality	Uplift net of cannibalisation, halo and pantry loading. The only honest measure of a promotion, and almost never available.
Indicator digit	The leading digit of a GTIN-14 that says which packaging level this is. GS1 calls it the logistic variant, and it allows up to eight levels for one product.
Informal trade	Unregistered or semi-registered small retail. Also called traditional trade, and in South Africa it is roughly a fifth of grocery.
Initial markup	The markup set when an item is first ticketed.
Inner	The intermediate pack between the consumer unit and the case, with its own GTIN and quantity.
Instant savings	A loyalty discount applied at the till in real time rather than accrued as points. A member price, by another name.
Integration layer	What a retailer has instead of an integration architecture. In descending order of volume: scheduled flat files over SFTP, store polling and distribution, electronic data interchange, direct database reads of the merchandising system, an enterprise service bus adopted for one programme, REST interfaces in the digital estate, and event streaming in the newest layers. **It masters nothing, which is the problem**, and it is a sediment rather than a layer because every generation added a style without retiring the previous one.
Internal selling-price inflation	The retailer's own weighted shelf-price movement, which is what a South African retailer reports to the market rather than the national inflation rate.
Item master	The authoritative record of every item the retailer sells and its attributes. In most retailers it is open to direct edit by hundreds of users with no stewardship workflow, which is the root of most downstream failures.
ITF-14	The case-level barcode symbology, printed on corrugated board, deliberately tolerant of poor print quality.
Jaws	The gap between sales growth and cost growth. Positive jaws is the whole of a retail turnaround story.
Known value item	A line whose price shoppers remember and compare — bread, milk, a two-litre cola. Priced for perception rather than margin.
Landed cost	Purchase price plus freight, duty, clearing and inbound handling. The only cost number that means anything on a direct import.
Last mile	The final delivery leg to the customer, and the dominant cost in online retail.
Lay-by	**A specifically South African regulated mechanic, with no close overseas equivalent.** The retailer holds goods while the customer pays in instalments. Under section 62 of the Consumer Protection Act each instalment **remains the consumer's property** and **the goods remain at the retailer's risk** until delivery; if the retailer cannot deliver after full payment through its own fault the refund is **double** what was paid. **It is not a sale and not a credit agreement**, so recognising the instalments as revenue is both an accounting and a statutory error, and routing a lay-by through the credit engine applies the wrong statute.
Lead time	Elapsed time from placing an order to the stock being available to sell.
Leaflet	The printed promotional vehicle. **Its print deadline is what locks the promotional calendar**, which is why the calendar often locks before the supplier funding is confirmed.
Legal metrology	The regulation of measuring instruments and declared quantities, administered in South Africa by the National Regulator for Compulsory Specifications. **A scale is an asset with a verification lifecycle**, a re-installed instrument is unverified until verified in its new position, and **the duty to check a declared quantity falls on anybody buying prepackaged goods for resale** — the retailer, for goods somebody else packed.
Like-for-like	Growth measured only across stores trading in both periods. **It is defined by its exclusions, which makes it a modelled set rather than a calculation**: opening and closing dates, non-trading periods, refits, relocations, format conversions, the addition or removal of a dispensary, and the retailer's own policy thresholds. If the model cannot hold the policy, the number cannot be audited.
Line	**Two meanings.** Commercially, a product the retailer sells — loosely a SKU or an option, as in "a thousand new lines". On a transaction or an order, one row: a line item on a till slip, an order line on a purchase order. Both senses appear in the same sentence often enough to matter.

Line item	One product line within a transaction. The highest-volume row in a retailer's data estate.
Linear metre	A metre of shelf frontage, and the denominator of shelf productivity.
Liquor licence	A provincial licence to sell liquor, with a category, a nominee, a licensed area within the store and permitted trading days and hours set provincially and municipally. **A single national trading-hours configuration is the commonest failure**, and it produces unlawful sales in one province while blocking lawful ones in another.
Listing fee	Payment for the right to be ranged. Also called a slotting allowance, and in South Africa it must be objectively justified and consistently offered.
Loyalty programme	A scheme that identifies customers in exchange for savings or rewards. South African programmes run to tens of millions of members, which makes the consent and retention position a first-order data obligation rather than a marketing detail.
Maintained markup	Ticket price less cost less allowances — the margin actually achieved, as opposed to the one set at ticketing.
MAPE	Mean absolute percentage error, the standard forecast-accuracy measure. Useful split between promotional and non-promotional lines, and misleading when it is not.
Margin	Sales less cost of sales, divided by sales. Also called gross margin or GP. In retail it cannot usually be decomposed into cost, price, mix, markdown, shrink and supplier income, which is why everybody argues about it.
Markdown	**A permanent reduction to the base price**, funded by the retailer, with an immediate inventory-value consequence under the retail method of accounting. Distinct from a promotion: *does the regular price come back?* If no, it is a markdown. Systems that model both as "a price record with dates" lose the funding attribution and the accounting treatment.
Markdown support	A supplier contribution to clearance, or to a price fall. Also called price protection.
Marketplace	A platform selling third-party sellers' inventory alongside, or instead of, its own.
Markup	Price less cost, divided by cost. Not the same arithmetic as margin, and confusing the two is a rite of passage.
Master data management	The discipline and the system class that holds a golden record and a stewardship workflow. Its retail sibling, product information management, holds the rich content a website needs and a merchandising system does not.
Mechanic	The structural form of an offer — price cut, multibuy, threshold, mix-and-match. The promotions engine is organised around mechanics, not prices.
Merchandise hierarchy	See *Hierarchy (merchandise)*.
Merchandising system	What the trade calls "the merch system", and **it is four things in one box**: the article master, the commercial ledger of cost and price, the inventory ledger, and the transaction engine for orders, receipts and adjustments. Correct for most estates, and the reason replacement projects run five years.
Micro-fulfilment centre	Small automated fulfilment capacity, often in a store back room.
Min-max	A reorder point and an order-up-to level held per SKU per location. The simplest replenishment policy and the one most retailers actually run.
Mix-and-match	A multibuy across a defined set of lines rather than one line.
Multibuy	A quantity-based offer such as three for R50. **Not a price**: it is a basket-level rule evaluated against the whole basket, with a priority and an exclusivity relative to other offers.
NAPPI	The South African identifier for pharmaceutical, surgical and healthcare consumable products, used in electronic reimbursement since the early 1990s. **MediKredit owns the trademark and maintains the file; the public-domain file is free.** A South African pharmacy's item master therefore carries **two identifier schemes for the same product** — GTIN for receiving and selling, NAPPI for claiming — maintained by different bodies, on different cadences, not mapping one to one. **The retailer owns the mapping and nobody else will maintain it.**
National brand	A supplier-branded product, as opposed to the retailer's own brand.
National Credit Act	The South African credit statute, with the National Credit Regulator as regulator. The registration threshold for a credit provider has been **nil since 11 November 2016**, so anybody trading as a credit provider must register. For a retailer it reaches store cards, instalment agreements and the clothing account book — and the decision happens at a till or in an app in minutes.
NCPDP	The United States pharmacy standards body, publisher of the Telecommunication Standard for claim adjudication and SCRIPT for electronic prescribing. **Member-gated**, so a vendor claiming conformance may be citing a specification the buyer cannot read. **North American divergence:** the United States dispensing estate is integration-standard-bound; the South African one is scheme-relationship-bound.
NCR	**Disambiguate this every time.** In a South African retail sentence "NCR" usually means the **National Credit Regulator**. As a technology company it no longer exists: it split on 16 October 2023 into **NCR Voyix** (retail, restaurant and digital banking) and **NCR Atleos** (ATMs and self-service banking). Write the full name or do not write it.
Negative stock	A system on-hand below zero. Always a process defect, and always a signal rather than a state.
Net net cost	Cost price after every discount and all supplier income. The number a buyer negotiates towards and the finance system rarely holds.
Net sales	Gross sales less returns.
No sale	Opening the cash drawer without completing a sale. A shrink exception signal, and one of the first things an exception report should count.
NPD	New product development, and by extension a newly introduced line. A new line has no sales history, so it must be forecast from an analogue.
Objection log	A record of every occasion on which a person asked not to be marketed to. **Distinct from a consent record and from the statutory suppression list**, with a different legal effect — and one `marketing_opt_in` flag satisfies none of the three.
Offline mode	**Four separable capabilities, and most retailers have two.** Offline item lookup and pricing from a local file; offline transaction logging with forwarding on reconnect; **offline tender acceptance**, which is the hard one because a card authorisation needs a round trip; and offline loyalty, where accrual can be deferred and redemption cannot. **Test the third specifically** rather than accepting "we have offline mode".
Omnichannel	Coordinated trading across store, online and app with shared stock, shared price and a shared view of the customer. Shared stock is the hard one.

On-demand delivery	Same-hour or same-day delivery, usually picked from a store.
On order	Purchase-order quantity committed but not yet received. Unreliable on-order data is the commonest reason an open-to-buy figure is wrong.
Open-to-buy	Planned receipts less commitments already on order — the buying budget, **held in money, by category, by period, not in units**. The stock ledger records units of a SKU at a location, so reconciling the two is where most merchandise-planning architecture lives and fails.
Opt-out registry	The South African national direct-marketing registry administered by the National Consumer Commission under the 2026 amendment regulations. Marketers must register and **cleanse their databases at least monthly**; a consumer's block applies to every marketer at once. Registration opened 15 September 2026, cleansing from December 2026, consumer blocking from May 2027.
Option	**Two meanings, and the second is the one that matters.** In general speech, a choice. In apparel and general-merchandise planning, a style in one colour — style plus the axis the customer chooses before size. It is the grain of range planning, of merchandise financial planning in fashion, and of the photoshoot. In grocery the level collapses and the word is not used.
Order management system	The application that takes a customer order, decides which node will source it, releases it, and tracks it to delivery. Also called distributed order management.
Order multiple	The quantity increment in which a SKU can be ordered. Case rounding against a small store over-orders slow lines, permanently.
ORDERS / DESADV / INVOIC	The EANCOM purchase order, despatch advice and invoice messages. Their X12 counterparts are 850, 856 and 810.
OTIF	On time, in full — the combined measure of supplier delivery reliability. Suppliers deliver short and most systems have no memory of it.
Outer	See *Case*.
Overrider	A retrospective payment based on purchase volume. Also called a rebate.
Pack	A grouping of consumer units sold as one, with its own GTIN. Distinguish it from an inner and a case, which are logistics levels rather than selling levels.
Pallet	The bottom logistics level, identified for a class of goods by a GTIN-14 and for a specific physical pallet by an SSCC.
Pantry loading	Forward buying by shoppers during a promotion, which depresses the demand that follows it.
PCI DSS	The Payment Card Industry Data Security Standard. **Not law** — imposed through the acquirer's contract, so there is no regulator to appeal to. **Version 4.0.1 was published 11 June 2024, v4.0 retired 31 December 2024, and the future-dated v4 requirements became effective 31 March 2025.** The architecture decides the scope: whether the till ever sees a card number, whether the store network is segmented, and who holds the token vault.
Perpetual inventory	A continuously updated on-hand record per SKU per location, as opposed to a count-based periodic record.
Perpetual versus periodic	Continuously updated inventory records against count-based ones.
Phantom inventory	Stock the system holds that does not physically exist. It sends an online picker to an empty shelf and suppresses the replenishment order that would have fixed it.
Pick face	The shelf position from which stock is taken. Its capacity caps the replenishment order, which is why a wrong planogram over-orders into a shelf that cannot hold the stock.
Pick path	The sequence a picker walks, optimised to minimise travel.
Pick rate	Units or lines picked per hour. The denominator of warehouse and online fulfilment cost.
Planned receipts	Planned sales plus planned closing stock plus planned markdowns, less opening stock. The first half of the open-to-buy arithmetic.
Planning	Deciding how much to buy, in money, by period. **Not the same job as buying.** A buyer who plans over-buys; a planner who buys buys safe.
Planogram	A drawing specifying which SKU sits where on a defined piece of shelving, with how many facings and how deep. Usefully defined as a dated SKU-to-position assignment with a derived capacity.
PLU	Price look-up code — a short number keyed at the till for an item with no barcode, typically loose produce.
Point of sale	The till application and the estate around it: lanes, scanners, scales, label printers, payment devices. **Its journal is the system of record for the transaction**, and its retention period is usually shorter than anybody assumes.
Point-to-point encryption	Encrypting card data at the pinpad so the till never sees a card number. **The single decision that collapses a retailer's card-security scope most.**
Points	An accrued unit of loyalty value, redeemable later. The balance is a liability.
POPIA	The Protection of Personal Information Act 4 of 2013, enforceable from 1 July 2021, with the Information Regulator as authority. Its retail bite is on loyalty data, direct marketing, in-store cameras and biometrics, automated decisioning and cross-border transfer.
Posting date	The date a transaction was recorded in the financial ledger — later than the business date, sometimes by days. **Sales reports run on business date, bank reconciliations on value date, the ledger on posting date and the trading statement on fiscal period. Four queries, four dates, one transaction.**
Price	**At least seven things, and the same article has all of them on the same day.** Cost, in several flavours; the regular or base retail by price zone; a promotional price; a markdown price; a clearance price; a competitor-matched price; and, in a South African pharmacy, a **regulated price** the retailer has no discretion over at all. **A price is a version with effective-from and effective-to timestamps, not an attribute of an article.**
Price architecture	The deliberate structure of price points within a category — entry, mid, premium — rather than a set of independent prices.
Price change	The execution of a new price to every till and every shelf label. In South Africa it has a statutory consequence if it lands late or partially.
Price elasticity	The responsiveness of demand to a change in price.

Price ending	The convention governing the final digits of a price. A rounding rule, and a surprisingly contested one.
Price file	The effective-dated set of prices by SKU, by zone, by date, built and distributed to stores. **The most dangerous object in a retail estate**, because it is high-volume, time-critical, legally binding at the shelf, and held in five systems with nothing reconciling them.
Price point	A specific shelf price used as a commercial anchor.
Price zone	A group of stores sharing a price position, so that the retailer need not maintain a price per store. **It is a fourth hierarchy over locations** and it agrees with none of the others. South African practice typically distinguishes metro, rural, convenience, forecourt and township, with the convenience and forecourt formats priced materially higher — though no retailer publishes its zones.
Primary distribution	The supplier-to-distribution-centre leg. Secondary distribution is the distribution-centre-to-store leg.
Private label	Product sold under the retailer's own brand. Also called own brand or house brand. **It makes the retailer the manufacturer for labelling, composition and packaging-responsibility purposes**, which is a larger consequence than the margin.
Promotion	**A time-bound offer with a defined mechanic, scope and funding, under which the base price is unchanged.** Distinct from a markdown, and the distinction carries both a funding attribution and an accounting treatment.
Promotional calendar	The locked schedule of promotions by period and channel. It locks on the leaflet deadline.
Promotions engine	The application holding offer definitions and basket-level rules, with priority and exclusivity between them. It exists as a separate system class **because multibuys and threshold offers cannot be expressed as a price on an article**.
Put-away	Moving received stock to its storage location in a warehouse.
R146	The South African food labelling regulations currently in force, published 1 March 2010. Mandatory particulars include ingredient list, storage conditions, net contents in SI units, **country of origin, batch identification and date marking**. Country of origin is a declared attribute and **cannot be derived from the barcode prefix**.
R3337	The draft replacement food labelling regulations, published for comment 21 April 2023. **Still a draft as at October 2026, and R146 remains the law.** It would add front-of-pack warning labels above thresholds for added sugar, sodium and saturated fat, assessed against a nutrient-profile model, with a logo covering about a quarter of the front of pack and a bar on claims and child-directed marketing on warned products. **The thresholds and the transition period are unverified and are not printed in this book.** Build the nutrient fields now, keep the warning status as a derived indicator, and do not commit artwork.
Range	**Two meanings.** The set of lines a retailer offers; and, in apparel, a seasonal collection. The first is a data object, the second is a commercial event.
Reason code	A mandatory classification on an adjustment, void, refund or write-off. **The single cheapest improvement available in most retailers**: without it, shrink cannot be decomposed and a write-off's tax treatment cannot be derived.
Rebate	A retrospective payment from a supplier based on volume. The largest component of supplier income, and the part attributable to unsold stock must be deferred.
Reference article (SAP)	An article used as a template when creating others. It is never sent to the point of sale, so it cannot be operational — and a project that treats it as a live article will discover this at a till.
Refit	A store refurbishment, usually changing space and layout. It invalidates the fixture survey, which invalidates the planograms.
Refund	Returning money for returned goods. **The principal till-fraud vector**, which is why the authority rules around it are a control rather than a convenience.
Regulated price	A price set by or under public regulation, which the retailer may not alter. In South Africa the single exit price of a medicine is the example, and it takes the dispensary out of the scope of every price-optimisation tool.
Reorder point	The on-hand level that triggers an order.
Replenishment	Automatically reordering to maintain availability against a forecast — a pull. **Not the same as allocation**, which is a push of a fixed quantity, and the two engines will disagree about the same store on the same day.
Repricing	See *Price change*.
Restatement	A change to a previously published figure. Retail generates them routinely and legitimately: late-arriving store data, hierarchy reorganisation, store transfers between regions, the 53rd week, acquisitions and disposals, corrections, accounting-policy changes and currency. **Every published aggregate must be reproducible as at the date it was published and on the current basis**, which means atomic facts plus versioned dimensions.
Restricted-circulation number	A GS1 prefix range reserved for identifiers that are valid only inside one company — prefixes beginning 02 and 20 to 29. In-store scales generate them, so **a material share of a grocer's turnover passes through identifiers that match nothing a supplier holds**.
Retail Domain Task Force	The OMG and National Retail Federation body that now owns the former ARTS standards. Its live work is device and receipt standards; its data models are dormant.
Retail media	Advertising sold by a retailer against its own audience and its own data. **No South African retailer publishes retail media revenue**, so any figure in circulation is an estimate.
Retail method	Tracking inventory at selling price and converting to cost using a cumulative mark-on. **Its consequence is structural**: a markdown reduces inventory value the moment it is applied, with no physical movement at all, so the stock ledger is driven by price-change events as much as by movements. Oracle exposes the choice as a department-level setting, so one group can run both.
Retailer loyalty (SPAR usage)	The share of a member's purchases bought through the group's distribution centres. **A South African term of art that has nothing to do with customer loyalty**, and in a voluntary-trading group it is the number that matters most.
Return rate	Return units divided by sales units. In online apparel it is the number that decides whether the channel is profitable.
RTV	Return to vendor — sending stock back to the supplier for credit.
Safety stock	Buffer stock held against variability in demand and in lead time. The dial that trades availability against working capital.
Sales per employee	The labour productivity measure. Sales per hour worked is the sharper version.

Sales per linear metre	Net sales divided by shelf frontage. The space-productivity measure a space planner works to.
Sales per square metre	Net sales divided by trading area. **In South Africa it is called trading density** and it is the market's standard measure of space productivity.
Sales per store	Net sales divided by store count. The format measure, and the one that exposes space growth that bought no sales.
Sales polling	The overnight job that pulls transaction data from every store to the centre. The volume-dominant job, and the one where a missing store has to be filled next cycle without double-counting.
Scheduled substance	A medicine classified Schedule 0 to 8 under the South African Medicines Act, with escalating control. **The schedule is a dated attribute, not a constant**: it is amended by gazette, and an amendment silently changes what a till may sell.
Seasonality	The recurring within-year pattern in demand. In South Africa it is cut across by a monthly pay-and-grant cycle that most imported forecasting configurations do not expect.
Sell-through	Units sold as a percentage of opening units on hand. **Apparel's central measure; grocery barely uses it**, because grocery replenishes rather than selling through a finite buy.
Service level	The target probability of not running out of stock. Expressed as a percentage, and the input nobody can defend.
Settlement discount	A discount for paying early. Economically it is interest, and it should be priced as such.
Shelf edge / shelf-edge label	The price and product label on the front of the shelf. **In South Africa it is a legal instrument**: where it shows a lower price than the till, the lower price binds.
Ship from store	Fulfilling an online order from store stock for delivery to the customer.
Short delivery	A delivery smaller than ordered. If the claim is not raised and chased, the shortfall becomes shrink.
Shrink	Book stock less counted stock — the loss nobody can account for. **It is a decomposition, not a number**: external theft, internal theft, unclaimed supplier shortage, administrative error, scan error, waste, expiry, damage, process loss in the bakery and butchery, and an honest residual. **There is no published South African shrink figure**, so a South African retailer cannot benchmark its own, and the United States retail-federation benchmark may only be used if it is labelled as a United States figure.
Shrink rate	Shrink at retail or at cost as a percentage of sales.
Single exit price	**The only price at which a manufacturer may sell a medicine in South Africa to anyone other than the State**, and the price the pharmacy charges the patient or scheme. All private buyers pay the same price regardless of volume, so bulk purchasing cannot negotiate it down; wholesalers add no mark-up and are paid a logistics fee out of the exit price instead. Annual adjustments are capped by gazette. **There were two adjustments in 2026** — 1.47% in January and an extraordinary 2.88% from 1 October — so any source quoting “the 2026 SEP increase” as one figure is wrong.
Size curve	The planned distribution of units across sizes for an option. Getting it wrong by store is the commonest cause of a broken size break and an early markdown.
SKU	Stock-keeping unit — the smallest unit distinguished for stock purposes: one barcode, one bin, one count, one price point. **In North American usage it means precisely the sellable unit; in South African usage it is used loosely for any level**, so when counting, state the level.
Slot	A bookable delivery or collection window with finite capacity. When the app promises more slots than the store has labour for, both the fulfilment manager and the store manager are right.
Slotting	Assigning SKUs to warehouse locations by velocity and handling profile. Rarely re-optimised as velocity changes.
Slowly changing dimension	A warehouse technique for keeping history on a changing attribute. It matters here because **the merchandise hierarchy restates history on reclassification and the warehouse hierarchy usually does not**, so a reclassified article's history moves in one system and stays in the other, permanently.
Space growth	The year-on-year change in trading area. The denominator of the question every retail investor asks: did the new space earn anything?
Space payment	Payment from a supplier for shelf position or facings.
Space plan	The allocation of selling space between categories in a store, above the level of individual planograms.
Spaza shop	A small informal outlet, typically in a township and run from a home or a container. A major South African channel in its own right, and a customer of the cash-and-carry format.
Split tender	One transaction settled across more than one means of payment. Three cards is three tender lines, and a model with one tender column cannot represent it.
SSCC	Serial Shipping Container Code — eighteen digits identifying **one specific physical logistic unit**: this pallet, this cage, this parcel. Not a class of goods. Reusing one makes receipt automation impossible.
Stand-in authorisation	A card authorisation decision taken locally — by the terminal or a store controller — when the acquirer cannot be reached. The parameters are a commercial decision disguised as a technical one.
Stock	**At least nine meanings in one retailer, and the question is always “who is asking?”** Physical or book stock; available to sell; on order; in transit; reserved or allocated; damaged, quarantined or expired; soft-held in an online basket; customer-owned but retailer-held, as with a lay-by; and supplier-owned consignment. A pharmacy adds a tenth: bulk stock under dispensing, measured in a different unit from the purchase unit. **Stock is not a column. The durable fact is the inventory movement, and every position above is a filter and a sum over those.**
Stock take	A full physical count of a location, usually over a weekend with trading frozen. The variance is partly a measure of integration quality, and almost nobody reads it that way.
Stock-to-sales ratio	Opening inventory divided by the period's sales.
Stock turn	Cost of sales divided by average inventory. How many times a year the shelf empties and refills.
Store	**Three or four meanings, and they do not share a boundary.** A selling location; a stock-holding location; a tax and licensing location; and a labour location. In some systems a "store" is also a *virtual* selling location such as a website, or a *customer* location such as a franchisee — which is exactly how a franchise store should be

	modelled.
Store-and-forward	Queuing transactions locally while offline and sending them when the link returns, with duplicate suppression on recovery.
Store card	A credit facility usable only at the issuing retailer.
Store reset	A coordinated range and planogram change executed across many stores at once.
Store server	The back-office computer that historically held the price file, distributed to tills, aggregated the journal and integrated with the centre. Removing it reduces two thousand servers to patch and increases dependence on the link — **which in South Africa is precisely the dependence you do not want**, so a retailer that removes it must over-invest in till-level autonomy.
Structured article (SAP)	An article made up of other articles — a display, a prepack, a set. It behaves differently in ordering and in inventory from its components.
Style	An apparel design across all its colours and sizes: one cut, one construction, one cost base, one buy decision. **Not stockable and not sellable.**
Subcategory	A level below category. Successively: class and subclass, where items finally attach.
Substitution	Replacing an unavailable picked item with an alternative. It needs a policy, the customer's consent, and exactly one place that calculates the refund — not two.
Sunrise 2027	An industry capability ambition to have retail point-of-sale estates able to read two-dimensional barcodes. **It is not a mandate and has no legal date.** GS1 US's own page gives no day or month, runs a voluntary crawl-walk-run model, and says it is not necessary to implement all two-dimensional capabilities by then. **Never write that GS1 requires 2D by 2027.** The durable content is the consequence: the till becomes a traceability capture point, and the scan becomes a consumer touchpoint the retailer does not control.
Supplier income	Money from suppliers that is not a reduction in the invoice price — rebates, listing fees, co-op advertising, space payments, markdown support. Also called commercial income or back margin. **The most argued-over number in a retailer's profit and loss, and the one most likely to cause a restatement**, because it is accrued on an estimate made by the team whose performance it flatters.
System of record	The authoritative source for a given data domain. In retail the question is almost never “which system” but “which system for which part”: quantity and value of inventory frequently have different masters.
System-wide sales	Total sales through a franchise network, including the sales the franchisor never books as revenue. **A South African reader will meet this in every franchise or voluntary-trading result announcement**, and confusing it with turnover overstates the group by a wide margin.
Tail	The slow-selling lines that make up most of a retailer's item count and little of its sales. Nobody can measure what the tail costs, which is why it survives every range review.
Take rate	Platform commission as a percentage of gross merchandise value.
Task management	The application that pushes work to a store and collects evidence that it was done. Its absence is why a price change can be distributed and never confirmed.
Tax invoice	The till slip as a tax document. South African practice requires a **full tax invoice above R5,000**, an abridged one at R5,000 or less, and none at R50 or less, issued within twenty-one days. A till that cannot produce a compliant full invoice blocks the business a cash-and-carry depends on.
Tender	The means of payment used for a transaction. Also used for the *tender line*, one row per means of payment.
Tender type	The classification of a means of payment: cash, debit, credit, wallet, voucher, gift card, loyalty points, account. **In South Africa cash is 56% of payment volume and 21% of value**, and more than 80% more transactions happen on the first of the month than on the twenty-second — so cash handling and payday amplitude are first-order design concerns rather than legacy footnotes.
Terminal stock	Stock remaining at the end of a season with no further full-price demand.
Three-way match	Reconciling purchase order, receipt and invoice before paying. The first-pass rate is a direct measure of receiving discipline.
Threshold offer	A reward triggered by total basket spend — spend R500, get R50 off. A basket-level rule.
Tokenisation	Replacing a card number with a surrogate. **The token vault is the exit cost** when the acquirer or gateway changes, so who holds it is an architecture decision with a commercial tail.
Tote	The container an order is picked into.
Trade spend	The supplier's total investment behind a retailer, front and back margin combined.
Trading area	The floor area used for selling, and the denominator of trading density. Smaller than gross lettable area.
Trading date	See *Business date*.
Trading density	**South African usage: sales per square metre per year.** The market's measure of space productivity, quoted in rand per square metre.
Trading margin	Trading profit as a percentage of turnover.
Trading profit	**South African usage: the primary operating measure a listed retailer reports**, usually before capital items, and the number management actually manages to. It is not operating profit under IFRS and the reconciliation matters.
Trading terms agreement	The effective-dated contract defining rebates, allowances, settlement discounts and the conditions attached to each. Usually not in the supplier master, usually in a spreadsheet, and the gap is where supplier income goes wrong.
Transfer	A stock movement between stores, or from store to distribution centre. In-season transfers often cost more than the margin they recover.
Turnover	Revenue as reported. **For a wholesaler or a voluntary-trading group it is invoiced sales to members, not till sales to shoppers**, which is a different and much smaller number than the brand's visible trade.
Turnover rent	Rent calculated as a percentage of store sales above a threshold. It makes the landlord a partner in the store's trade and makes store-level sales a contractual figure.
Ullage	Breakage and leakage loss, chiefly in liquor and beverages.

Unfunded promotion	A promotion the retailer pays for out of its own margin. Knowing which promotions were unfunded is the only way to answer how much margin erosion was chosen.
UnifiedPOS	OMG's platform-neutral specification for connecting point-of-sale peripherals to till applications. **It is alive and it is not legacy**: version 1.15.1 was adopted in December 2024 under reasonable-and-non-discriminatory terms, with Aptos, Diebold Nixdorf, Digimarc, GS1, Microsoft and Toshiba contributing. **Ask any till vendor which UPOS version its drivers implement**: the answer decides whether swapping a scanner is configuration or development.
Units per transaction	Units sold divided by transactions.
UPC / UPC-A	The 12-digit North American barcode. **A legacy name for GTIN-12.**
Uplift	Sales above baseline during a promotion. Gross, before cannibalisation is deducted.
Usage clause	A lease term restricting what a tenant, or another tenant, may sell.
Value-added network	A third-party service that routes electronic data interchange traffic between trading partners. Still common, still billed per character in places.
Value date	When money actually moves on a card or electronic funds transfer tender. Different again from the business date and the posting date, and bank reconciliation lives here.
Value-only article (SAP)	An object supporting inventory management on a value basis at merchandise-category level rather than by unit. Useful where unit-level control is not worth the cost; dangerous if anybody expects a unit count from it.
Van sales	Selling and delivering from a vehicle. The route to market for informal trade in South Africa.
Variable-weight item	An item priced at the scale by mass, with a barcode generated in store. Typically under a GS1 restricted-circulation prefix, so the identifier is valid only inside that retailer and matches nothing a supplier holds.
Variance	The difference between counted and book stock, before it is explained.
Variant	A version of a product differing in one attribute — size, colour, flavour, pack size. In SAP, the variant is the sellable unit beneath a generic article.
VAT zero-rating	The South African regime under which a list of basic foodstuffs carries no value-added tax. **The boundaries are defined finer than any commercial category** — brown bread is zero-rated, brown bread rolls are not; raw vegetables are, prepared salads are not — and **there is no hierarchy under which the rate is inheritable**, which makes regulated classification a separate hierarchy from the merchandise one.
Vendor-managed inventory	An arrangement in which the supplier determines replenishment quantities. It moves the forecast to the party with less visibility of the shelf.
Virtual location	A location with no address: in transit, quarantine, web stock, a third-party logistics provider, a supplier's consignment pool. **Mandatory, because every inventory movement needs both ends** — without them receipts come from nowhere, write-offs go nowhere, and in-transit stock is owned by no one and counted at both ends.
Void	The cancellation of a line or a transaction before completion. A legitimate necessity and a shrink vector at once, which is why the authority to void is the front-end supervisor's.
Voluntary trading	**A South African top-five grocery model, and not a franchise.** Independent retailers buy through a shared wholesaler under a shared brand, with no royalty; members own their stores, sign a guild membership agreement, **and may stock from any supplier**. For an architect the consequence is immediate: for much of the group's turnover there is no central price, no central assortment and no central inventory position to master, only a wholesale order book and whatever the member chooses to share.
Warehouse management system	The application holding stock by bin and pallet, and the tasks that move it. It masters the distribution centre's inventory position, which is one of three in the estate.
Waste	Stock lost to expiry, damage or a temperature excursion. Distinct from theft, and in fresh it is the dominant loss.
Waste rate	Waste value as a percentage of sales, usually quoted for fresh.
Weeks of supply	On-hand units divided by average weekly unit sales. Forward weeks of supply uses the *forecast* instead, which is the more useful number and the less available one.
Weighed item	See *Variable-weight item*.
Wet stock	Fuel held in underground tanks at a forecourt, reconciled by dip against throughput rather than by counting units.
Workforce management	Scheduling, time and attendance. In retail it is driven by a forecast of transactions by hour, so a bad forecast produces either queues or idle payroll.
Write-off	Removing stock from inventory at zero recovery. Its VAT treatment differs from a sale's, which makes the reason code a tax attribute.

THE CHAPTER IN TEN LINES

- Retail's vocabulary problem is not jargon, it is ordinary words used precisely: store, stock, price, article, line, option, allocation and customer each mean several things.
- Stock is not a column. It has at least nine meanings in one retailer, and every one of them is a filter and a sum over inventory movements.
- Price is at least seven numbers for the same article on the same day, and a price record is a version with effective timestamps rather than an attribute.
- Buying decides what; planning decides how much, in money, and when. A system that merges the two roles has mis-modelled the organisation and removed a control.
- A promotion is time-boxed with the base price intact underneath; a markdown is permanent and revalues the stock. Model both as "a price with dates" and the funding attribution and the accounting treatment are both lost.
- Several of the most load-bearing words are South African and have no overseas equivalent: trading density, trading profit, lay-by, voluntary trading, retailer loyalty, single exit price, NAPPI.
- The pharmacy vocabulary diverges hardest from North America, because South Africa regulates one national price and Canada regulates per province and per payer.
- Where a term carries a figure, the figure is as at October 2026. Where nothing could be sourced — South African shrink, most retailers' technology estates — the entry says so.

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APPENDIX C

Standards, bodies and where to find them

A working register of the identification standards, reference models, conformance obligations and regulators a retail architect has to cite: what each one is, who owns it, whether you can read it without paying, and what to read first.

An architect in retail spends a great deal of time hunting for the authoritative text of something, and in South Africa more time still establishing which version is operative. This is the register. Checked October 2026.

Three warnings. **Retail has no BIAN** — no maintained industry reference architecture with a release cadence, a certification scheme and a published vocabulary. There is one standards body that genuinely binds the estate, a live device-and-receipt programme sitting on dormant data models, a voluntary calendar, a process taxonomy, and vendor models shaped like their price lists [1][2][7]. Choosing a vocabulary is an architecture decision, not an inheritance. **Availability is not a licence**: several artefacts here download freely with no stated usage terms, so read the access column as "can I obtain this". And **almost everything South African is free because it is gazetted** — the difficulty is currency. Instruments are amended by notice rather than reissued, and two of the most consequential items in retail are a three-year-old draft and a dispensing-fee notice whose values could not be retrieved at all [18][26].

Identification and interchange — the standards that bind

Item	What it is	Access and cost	Read first
GS1	The identification system, and the only external standard that is a primary key in a retailer's data. **GTIN** identifies a trade item at one packaging level, **GLN** a party or location, **SSCC** one logistic unit once. Around them sit the General Specifications (**release 26.0, January 2026**), **GDSN**, **EPCIS** and **Digital Link** [1].	On `ref.gs1.org` with no stated fee. **The costed part is the identifiers.** The reference PDF governs where formats conflict [1].	Section 1, for the keys; then Section 3, the application identifiers — the grammar a till must parse once 2D codes arrive.
GS1 South Africa	The local member organisation, which **trades as the Consumer Goods Council of South Africa**. The prefix range is **600–601** [3].	**Priced in rand, and the shape of the table is the finding.** One GTIN: R176 once, no renewal. A 100,000-item prefix: R9,888 once and **R28,878 a year**. **GLNs cost the same as GTINs** [4].	The fee table [4]. Prefix capacity is a long-term commitment, not a procurement line — and there is no economic reason not to GLN every location.
GS1 Digital Link, and 2027	GS1 keys encoded as web URIs, so one scan identifies and resolves. Four version strings run at once, so citing "version X" without the component is imprecise [5].	Published; **terms and resolver conformance were not retrievable** [5].	GS1 US's Sunrise 2027 page [6], and its own sentence: **"it is not necessary to implement all the capabilities of 2D by that date."** A capability ambition with no day, no month and no enforcement. **Never write that GS1 requires 2D by 2027.** The global and South African positions are unverified.
EDI — EANCOM and X12	The commercial grammar of supplier integration, and not an API estate. **EANCOM is a subset of UN/EDIFACT**, the family a South African retailer meets; X12 is the North American one. Code lists change up to four times a year [8].	Commercial. In practice your translator's version matrix and your partners' guidelines govern.	Your own trading-partner map, and the despatch advice — the pallet code in it makes receipt-by-scanning possible.

Reference models, process frameworks and the calendar

Item	What it is	Access and cost	Read first
OMG's Retail Domain Task Force, and what ARTS became	**"ARTS" is a historical name.** Founded 1991, **acquired by the NRF in 1998**, **jointly managed by OMG and the NRF as the RDTF since 2017** [2]. The asymmetry is the story: **the device and receipt layer is alive** — UnifiedPOS 1.15.1 adopted December 2024 under RAND, with Aptos, Diebold Nixdorf, GS1, Microsoft and Toshiba contributing, and a formal **Digital Receipt API in June 2025** [9][10].	Specifications on omg.org; the task-force list is member-only [12].	The retail specification category [11]. Then ask a till vendor which UPOS version its drivers implement.
The ARTS data models specifically	The Operational Data Model and Data Warehouse Model are described on OMG's pages **with no version and no date, and are absent from the formal catalogue** [11][12]. The only hard marker anywhere is an **ARTS 4.01 conformance certification dated 16 January 2006** [2].	**Unknown, and that is the honest answer.**	Nothing, because I cannot tell you where to get them. **Do not send anybody off to "get the ARTS ODM"**; ask OMG whether it exists in usable form. It is good for **archaeology**: a till-log schema built on ARTS naming has no other documentation.
The National Retail Federation	The United States trade body, former owner of ARTS, publisher of the **4-5-4 calendar** — which NRF itself calls **"a voluntary guide"**. A **53rd week** is added when four or more days remain in January; NRF names 2006, 2012, 2017 and 2023 [13].	**Free.**	The 4-5-4 FAQ, then the restatement rule — the part that makes it a data-model requirement, because the calendar dimension needs a prior-year-comparable day key on both bases.
APQC's Retail PCF	A numbered process hierarchy for benchmarking. The Excel release documented at **6.1.1, 28 April 2023**: **twelve enterprise categories and over 1,000 processes** [7].	**Unclear. Do not assert it is free.** Retail carries no version on APQC's own index while four other industries show 2026 releases at 8.0, and **non-member commercial terms are not stated** [14].	The Excel edition, for the identifier scheme. Two limits: no data model, so it cannot settle a system boundary, and **no pharmacy framework** [14].
The Open Group	TOGAF the method, ArchiMate the notation. In banking it also publishes ArchiMate expressions of BIAN; **no retail equivalent was verified** [7].	Standards free on registration; certification paid.	ArchiMate's technology and physical layers, which earn their keep in retail as they rarely do in banking. Stores, lanes, scanners and pallets are the half of the estate that breaks.

Cards and payments

Item	What it is	Access and cost	Read first
PCI Security Standards Council	Publisher of PCI DSS. **It is not law** — it is imposed through the acquirer's contract, so there is no regulator to appeal to and no national derogation. **v4.0.1 published 11 June 2024**; **v4.0 retired 31 December 2024**; future-dated v4 requirements effective **31 March 2025** [15].	**Standards free.** Assessor qualification commercial.	Requirement 3, on stored account data. **The architecture decides the scope**: whether the till ever sees a card number, whether the store network is segmented, and who holds the token vault.
EMVCo	The body behind chip, contactless, three-domain-secure and payment tokenisation.	Downloadable subject to terms; certification commercial.	The specification index, and **check the versions yourself** — no EMVCo primary source was retrievable for this book, so **no EMV version or mandate date appears in it**. The safe point: the pinpad kernel is a certified asset with its own release cycle, so a till upgrade and a terminal certification are two programmes.

The South African statute book and its regulators

Item	What it is	Access and cost	Read first
National Consumer Commission	Enforcer of the Consumer Protection Act 68 of 2008, in force **1 April 2011**. It issues compliance notices in tranches, and now administers the **national opt-out registry**: marketer registration from **15 September 2026**, cleansing from **December 2026**, consumer blocking from **May 2027** [16] [17].	**Free.**	**Section 23** on price display: where two prices are displayed concurrently the supplier may not charge more than the lower, and the s 23(9) escape is a **two-limb defence** — correct the display *and* tell affected consumers. Then **section 56**: six months, with the **consumer electing** the remedy.
Consumer Goods and Services Ombud	The accredited industry ombud and binding code under CPA s 82, prescribed in **Gazette 38637 of 30 March 2015**. Participation and the levy are compulsory in the sector [16].	**Free** to read; the levy is not optional.	Advisory Note 2 on pricing errors (1 March 2021) — the only practical guidance on "reasonable steps" under s 23(9): withdraw, fix the label, resell, keep dated proof.
Consumer Goods Council of South Africa	The industry body — **and also GS1 South Africa**, one organisation wearing two hats [3].	Mixed; member material sits behind the public pages.	Its barcode guidance [3], which kills a persistent myth: the prefix shows where the company is *registered*, not where the product was made.
Competition Commission	Investigator under the Competition Act 89 of 1998. The **Grocery Retail Market Inquiry** (final report **25 November 2019**) covered exclusive leases, buyer power and uniform trade terms; **section 8(4) buyer power** has Regulations in **Gazette 43018 of 13 February 2020** and Guidelines of **12 May 2020** [18][19].	**Free.**	The Buyer Power Guidelines [19], for the **reversed onus**: the retailer must show a price or condition is *not* unfair, which means producing the price paid to every supplier of like goods with each one's SME and historically-disadvantaged status.
Information Regulator	The authority under POPIA, Act 4 of 2013 — commenced 1 July 2020, enforceable **1 July 2021** [20].	**Free.**	The **Guidance Note on Direct Marketing**, whose s 69(3) existing-customer test is a data test: the detail obtained in the context of a sale, the marketing of the retailer's own *similar* products, and an objection opportunity at collection and on every message. Then **section 71**.
SARS	The revenue service, and for a retailer the Value-Added Tax Act 89 of 1991: 15% standard rate and a **zero-rated basket** defined finer than any commercial category — brown bread zero-rated, brown bread rolls not [21].	**Free.**	The tax-invoice guidance: **full invoice above R5,000**, abridged below, none required at R50 or less [21]. A till that cannot produce a compliant invoice above R5,000 blocks cash-and-carry trade. And **no hierarchy makes the VAT rate inheritable**.
NRCS, for legal metrology	The National Regulator for Compulsory Specifications, under the **Legal Metrology Act 9 of 2014**, commenced **1 August 2014 per Gazette 37887**. Regulations: **Notice 877, Gazette 41854, 24 August 2018** [22].	**Free.**	Regulations 13 and 27. Regulation 13 puts the duty to check declared quantity on **anyone buying prepackaged goods for resale**. Regulation 27 sets the required **unit-price format**. And a **re-installed instrument is unverified until verified in its new position**.
Statistics South Africa	The national statistics office. For retail, the monthly **retail trade sales** series P6242.1, and the Quarterly Labour Force Survey [23].	**Free.**	P6242.1, the only neutral market measure a South African architect can cite. Careful with labour: the Q1 2026 QLFS release carries no trade-industry row.

Pharmacy — a second regulated business inside the store

Item	What it is	Access and cost	Read first
South African Pharmacy Council	Established under the **Pharmacy Act 53 of 1974**; regulates pharmacists, support personnel and **premises**. Publishes the **Good Pharmacy Practice rules under section 35A** — the rules that actually specify a dispensing system [24].	**Free.**	Rule **1.2.11.4(f)**: where software is relied on for interactions, **it must be the latest version available**. That turns a content database into a regulated currency obligation on the compliance critical path. Then rule 1.1.2, labelling in at least one official language, and the twice-daily cold-chain recording.
SAHPRA	The health products regulator under the Medicines and Related Substances Act 101 of 1965. Registers medicines, runs scheduling, licenses wholesale and distribution under **s 22C(1)(b)** [25].	**Free.**	The **Consolidated Schedules**, most recently **August 2025**. Schedules run 0 to 8 and are amended by gazette repeatedly, which makes **schedule a dated attribute of an article, not a constant** — and an amendment silently changes what a till may sell.
Department of Health	Sets the **Single Exit Price** under s 22G: the only price at which a manufacturer may sell a medicine privately, with capped annual adjustments and a capped dispensing fee on top [26].	**Free**, but **currency is the problem.**	The adjustment notices. **There were two adjustments in 2026** — **1.47%** in January and an extraordinary **2.88%** from **1 October 2026** — so any source giving "the 2026 SEP increase" as one figure is wrong. The bands in circulation are the **26 September 2025** position; **GN 7599 of 19 June 2026** exists and its values could not be retrieved.
MediKredit, for NAPPI	**NAPPI** identifies pharmaceutical, surgical and healthcare consumable products across electronic reimbursement, in use since the early 1990s. **MediKredit owns the trademark and maintains the file** [27].	**The public-domain file is free** and always has been; anything beyond it is a subscription.	The public file layout — and the fact that makes it architecture: a South African retail pharmacy's item master carries **two identifier schemes for the same product**, GTIN for selling and NAPPI for claiming, on different cadences and not mapping one to one. **The retailer owns that mapping.**

What will date fastest

In order. **The 2D barcode timetable**, because the definition keeps softening and the date is close. **Software ownership**, because retail vendor attributions changed in six categories in roughly two years. **The ARTS data models' availability**, because it could resolve either way. **PCI DSS**, because versions and retirement dates move and the council's own pages do not carry them. And **the South African instruments in motion** — the draft food labelling regulations, the Tobacco Bill in committee, the direct-marketing registry's phasing, and the dispensing-fee bands.

Two things will not date. **GS1's keys are the primary keys of the estate.** Get GTIN, GLN, SSCC and the retail week right and a retailer can change merchandising vendor, till vendor, planning vendor and data platform in turn and still say what it sold, where, to whom, against last year. Get them

wrong and no package fixes it. And **retail has no BIAN**, so you will choose a vocabulary rather than inherit one: write down which model and which version, and make it survive the next vendor change.

THE CHAPTER IN TEN LINES

- Retail has no BIAN. There is one binding standards body — GS1 — and after that you choose a vocabulary rather than inherit one. Record the choice as an architecture decision.
- GS1's keys are the primary keys of the data estate. GTIN identifies a trade item at one packaging level, GLN a party or location, SSCC one logistic unit once.
- Sunrise 2027 is an industry capability ambition with no day, no month and no enforcement mechanism. GS1 US's own page says it is not necessary to implement all 2D capabilities by that date.
- "ARTS" is a historical name. OMG's Retail Domain Task Force runs a live device-and-receipt programme, and the ARTS data models are absent from the formal catalogue. Do not send anybody to fetch them.
- UnifiedPOS is alive and it is OMG's: version 1.15.1, adopted December 2024. Ask any point-of-sale vendor which UPOS version its peripheral drivers implement.
- The 4-5-4 calendar is a voluntary NRF guide and an architecture artefact. The 53rd-week restatement is what makes it a data-model requirement rather than a reporting convention.
- PCI DSS is contractual, not statutory, and the architecture decides the scope. Where the card number flows, and who holds the token vault, move the burden by an order of magnitude.
- Almost everything South African is free because it is gazetted. The problem is currency: the food labelling regulations are a three-year-old draft, and the 2026 dispensing-fee band values could not be retrieved at all.

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APPENDIX D

References

Every source this book rests on, de-duplicated across four research passes and grouped by theme — then an honest list of what could not be sourced at all.

Everything asserted in this book either carries a source or is marked as the author's judgement. This appendix collects the sources in one place, de-duplicated across the four research passes behind the book, grouped by theme and numbered continuously. Where one source supports several chapters it appears once.

When the research was done. The underlying research was compiled and verified on **16 October 2026**, in four passes: the business of retail and the South African market; the systems, the master data and the end-to-end processes; the reference architectures, the standards and the technology providers; and the conceptual data model and the regulatory environment. The four passes cited 60, 75, 106 and 113 sources. De-duplicated across passes that is **337 distinct sources**, which is the list below. Every market figure, ratio, version number and commencement date is as published at that date.

The source hierarchy used. Primary sources first: statutes and gazetted regulations, standards bodies' own publications, regulators' own documents, official statistics and audited company results. Then near-primary: exchange announcements, standards-body specification pages, vendor technical documentation and industry-body publications. Then trade press, used for facts a primary source should have carried and did not — which in South African retail is most of what can be learned about a retailer's technology, because retailers do not publish it. Vendor marketing is used only to record what a vendor claims, is attributed to the vendor, and is dated.

Three source problems specific to retail in South Africa. The first is that **the technology estate of a South African retailer is almost entirely off the public record.** Two exceptions exist, and both exist only because a vendor published a case study: Shoprite's SAP and GK estate, and Pick n Pay's SAP point-of-sale data platform. Not one South African customer was sourceable for Oracle Retail, Blue Yonder, RELEX, Manhattan, Aptos, Toshiba, NCR Voyix, Vusion, Pricer, Hanshow, SPS Commerce or Stibo. Where a platform attribution appears in this book it is attributed to the outlet or the vendor that published it, never asserted as the retailer's own disclosure. The second is that **the commercially decisive numbers are not published at all** — price-zone structures, shrink, change-freeze windows, return rates, retail media revenue, and the adoption rate of anything. The third is that **retail's standards landscape has an availability problem rather than a cost problem**: the General Specifications are free to read and the identifiers are not, the process framework downloads without stated terms, and the industry data models may not exist in usable form.

What will date fastest, in order. *The two-dimensional barcode timetable*, because the definition keeps softening and the date is close. *Software ownership and product naming* — twenty-two attributions in the research's own corrections register are commonly got wrong, and several more are one transaction from joining them. *The South African instruments in motion*: the draft food labelling regulations, the Tobacco Bill in clause-by-clause committee, the direct-marketing registry's phased commencement through to May 2027, and the dispensing-fee bands. *Market results*, because every turnover, margin and store count is a point in time and the point has passed. *Standards versions* — the General Specifications, Digital Link's four component versions, PCI DSS and the Retail process framework will all move.

What will not date. The mechanisms. That a retailer is the inventory between buying and selling, and that its hardest problems are inventory-integrity problems. That GS1's keys and the retail calendar are the two decisions every downstream system inherits. That stock is a derivation over movements and not a column. That a price is a version with a timestamp, and that in South Africa the lower displayed price binds. That a promotion and a markdown are different objects with different funding and different accounting. That the store must be able to complete a sale with nothing outside it working. That a franchise or voluntary-trading estate gives the group no price, no assortment and no inventory position to master. And that a pharmacy inside a retailer is a second regulated business with its own identifier scheme, its own records and a price the retailer does not set. Those outlast any statute, platform or version number in the list below.

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What could not be sourced

An honest list of gaps is worth more than a confident guess. These are the items the research behind this book registered as unverified, in the order of how much they would have improved it. **None of them appears in the book as a fact.** Each is settled by the source named.

South African shrink, in any form. The single biggest gap. No South African retailer discloses shrink separately in published results, and no South African industry survey was located. The only benchmark available anywhere is the United States National Retail Security Survey, which is used in this book **only** where it is labelled as a United States figure. The consequence is worth saying out loud: **a South African retailer cannot benchmark its own shrink**, and that is a real strategic disadvantage and an argument for building internal measurement properly. Settled by a Consumer Goods Council of South Africa loss-prevention survey, a Trade Intelligence or Deloitte South Africa retail report, or the retailers' own integrated reports.

Whether the ARTS data models exist in usable form. The Operational Data Model and the Data Warehouse Model are described on the Object Management Group's pages with no version number and no date, and they are absent from OMG's formal retail specification catalogue. The only hard version marker recoverable anywhere is a conformance certification dated January 2006. Whether they are obtainable at all, at what version and on what terms could not be established. **This book therefore does not tell a reader to go and get them.** Settled by an OMG membership enquiry.

GS1 South Africa's position on Sunrise 2027. All the sourced material on the two-dimensional barcode transition is **GS1 US**. The global GS1 page could not be retrieved, and GS1 South Africa's own position, whether any South African retailer has made a public two-dimensional commitment, and whether local point-of-sale estates are being specified to read them, are all unverified. Do not assume the North American timetable applies locally. Settled by GS1 South Africa directly.

The R3337 Annexure 8 nutrient thresholds and transition period. The most commercially consequential open item in the set, because the thresholds decide which products carry a front-of-pack warning label, and the warning label decides pack artwork, promotion eligibility and advertising copy. The draft has been a draft since April 2023. **The thresholds are not printed in this book.** Settled by the draft text in Gazette 48460.

The current dispensing-fee bands. The four bands in circulation are the position published on 26 September 2025 in Gazette 51352. A 2026 notice exists — Government Notice 7599 of 19 June 2026 — and its band values and effective date could not be retrieved; a public price registry shows fee data last updated 27 August 2026 against prices valid at 26 May 2026. The bands may be superseded. **Implement the band table as a dated, versioned artefact and never as constants.** Settled by Government Notice 7599 in the Gazette.

PCI DSS publication and retirement dates against the council's own site. The council's standards page carries none of the dates, and its document library table rendered no publication date in the research round. The dates used in this book come from the council's own blog announcement; a second commonly cited timeline comes from a vendor glossary and is not relied on. Any version-5 timeline is unknown. Settled by the council's document library date column and its bulletins.

EMV versions, book numbers and mandate dates — entirely. No EMVCo primary source was retrieved. **No EMV version or date appears anywhere in this book.** Settled by EMVCo's own specification pages.

South African price-zone practice. The zone structures of South African grocers are commercially sensitive and unpublished. The description in this book of metro, rural, convenience, forecourt and township zones with a materially higher convenience and forecourt position is practitioner expectation, and it is marked as such. Settled by a retailer's own zone definition, which will not be published; the Competition Commission's grocery inquiry material is the most likely public proxy.

Adoption rates for electronic data interchange and electronic shelf labels. South African advance-shipping-notice and electronic data interchange adoption is unmeasured, and so is electronic shelf label penetration across the national estate. Both would be settled by GS1 South Africa or Consumer Goods Council supply-chain benchmarking, if it were published.

South African return rates and retail media revenue. Only low-quality aggregator sources were located for online and in-store return rates in apparel and general merchandise, none of them citing anything. **No South African retailer publishes retail media revenue**, and the one confirmed fact is that a listed grocer launched a retail media business. Both are likely unobtainable until disclosed.

Every South African retailer's technology estate, with two exceptions. The merchandising, planning, point-of-sale and data platforms of Woolworths, Clicks, Dis-Chem, TFG, Mr Price, Pepkor, Lewis, Cashbuild and SPAR's non-SAP estate were not sourceable. Nor was the technology behind Checkers Sixty60, which Shoprite's own newsroom does not describe; nor which till software Pick n Pay runs, since the sourced platform is a point-of-sale *data* layer rather than a till application; nor a single named South African retail implementation by any of the global or local systems integrators; nor the implementation partner on SPAR's SAP programme. **Massmart delisted in 2023 and publishes nothing**, which is itself a useful fact. **No count or market share of South African retailers running a given platform exists, and this book does not estimate one.**

And four items the research registered as disagreements rather than gaps, which the book states rather than resolves. SPAR's SAP go-live is dated February 2023 by one outlet, October 2022 by a second and early 2023 by a third. The trans-fat regulation is catalogued as R127 of 2011 by legal databases and called "Regulation 249" by the World Health Organization. The Tobacco Bill is shown as lapsed on a parliamentary monitoring page and as live in

Parliament's own media statements, and the monitoring page is the stale one. And the Legal Metrology Act's commencement is recorded as 1 August 2014 by the gazette record while the Act's own section 43 leaves it to proclamation with no date in the text.

THE CHAPTER IN TEN LINES

- The research behind this volume was compiled and verified on 16 October 2026; four passes citing 60, 75, 106 and 113 sources de-duplicate to 337 distinct sources.
- Almost nothing about the technology inside a South African retailer is on the public record. The two exceptions exist only because a vendor published a case study.
- Where a platform attribution appears in this book it is attributed to the outlet or vendor that published it, never asserted as the retailer's own disclosure.
- There is no published South African shrink figure of any kind, so a South African retailer cannot benchmark its own. That absence is itself a finding.
- No EMV version or date appears in this book, because no EMVCo primary source could be retrieved. An omission is better than a guess.
- The ARTS data models may not exist in obtainable form. The book says so rather than sending a reader after them.

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